

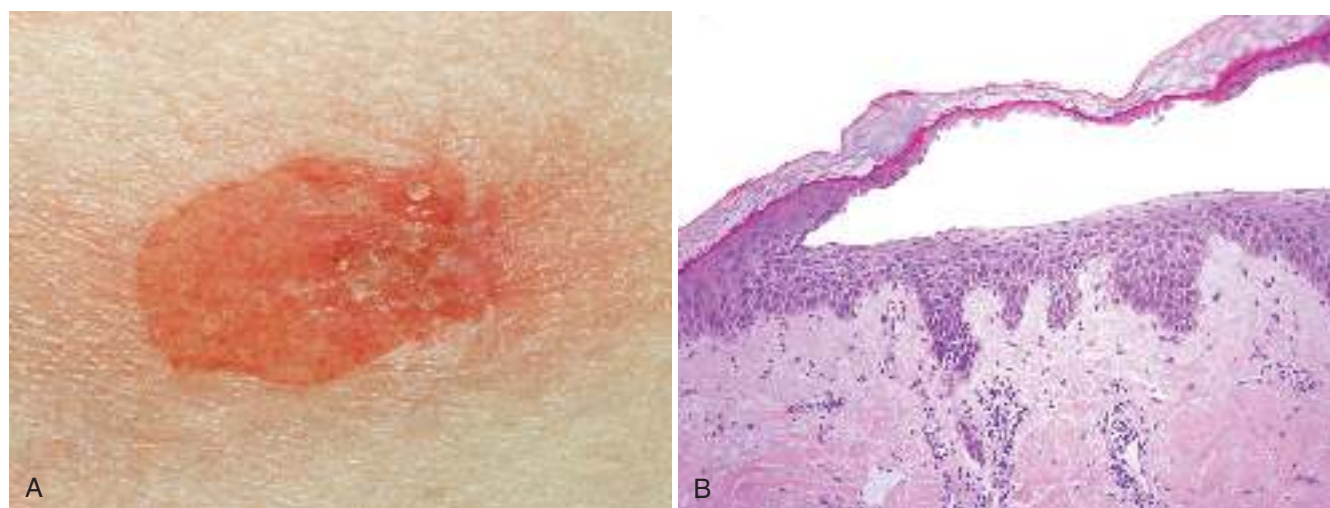


Figure 25.30 Pemphigus foliaceus. (A) The delicate superficial (subcorneal) blisters are much less erosive than those seen in pemphigus vulgaris. (B) Subcorneal separation of the epidermis is seen.

erythema and crusting; these represent superficial erosions at sites of blister rupture (Fig. 25.30A).

- *Pemphigus erythematosus* is considered to be a localized, less severe form of pemphigus foliaceus that may selectively involve the malar area of the face in a lupus erythematosus-like fashion.
- *Paraneoplastic pemphigus* occurs in association with various malignancies, most commonly non-Hodgkin lymphoma.

Pathogenesis

All forms of pemphigus are autoimmune diseases caused by IgG autoantibodies against desmogleins that disrupt intercellular adhesions and result in the formation of blisters. By direct immunofluorescence, lesions show a characteristic net-like pattern of intercellular IgG deposits. IgG is usually seen at all levels of the epidermis in pemphigus vulgaris, but tends to be more superficial in pemphigus foliaceus (Fig. 25.31). The distribution of desmoglein 1 and 3 in the epidermis and whether there are autoantibodies to one or both proteins appear to explain the position and severity of the blisters (see Fig. 25.28). The antibodies cause these lesions primarily by disrupting the intercellular adhesive function of the desmosomes; they may also act indirectly by activating intercellular proteases. Paraneoplastic pemphigus arises most often in the setting of lymphoid neoplasms and is also caused by autoantibodies that recognize desmogleins or other proteins involved in intercellular adhesion.

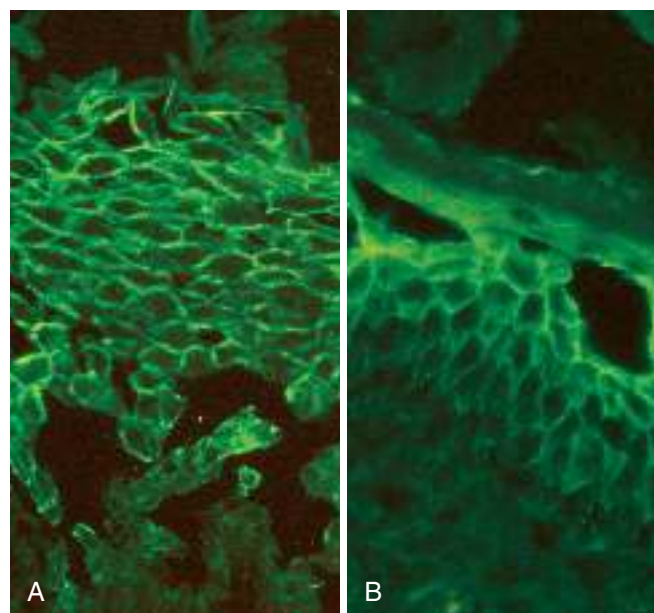


Figure 25.31 Direct immunofluorescence staining for immunoglobulin of epidermis involved by pemphigus. (A) In pemphigus vulgaris there is deposition of immunoglobulin along the plasma membranes of keratinocytes in a reticular or fishnet-like pattern accompanied by suprabasal loss of cell-to-cell adhesion (acantholysis). (B) In pemphigus foliaceus the immunoglobulin deposits and acantholysis are more superficial.

MORPHOLOGY

The common histologic feature in all forms of pemphigus is **acantholysis**, the dissolution of the intercellular bridges that connect squamous epithelial cells. Acantholytic cells dissociate from one another, lose their polyhedral shape, and become rounded. In pemphigus vulgaris (see Fig. 25.29B) and pemphigus vegetans, acantholysis selectively involves the cells immediately above the

basal cell layer, producing a suprabasal blister. In the vegetans variant, there is also overlying epidermal hyperplasia. The single layer of intact basal cells that forms the blister base in pemphigus vulgaris has been likened to a row of tombstones. In pemphigus foliaceus, blisters form by similar mechanisms but are found in the superficial epidermis at the level of the stratum granulosum (see Fig. 25.30B). Variable superficial dermal infiltration by lymphocytes, macrophages, and eosinophils accompanies each type of pemphigus.

The mainstay of treatment in all forms of pemphigus is immunosuppressive agents, which decrease the titers of the pathogenic antibodies.

Bullous Pemphigoid

Generally affecting elderly individuals, bullous pemphigoid shows a wide range of clinical presentations. Sites of involvement include the inner aspects of the thighs, flexor surfaces of the forearms, axillae, groin, and lower abdomen. Oral lesions are present in 10% to 15% of affected individuals, usually appearing after the cutaneous lesions. Some patients present with urticarial plaques and severe pruritus.

MORPHOLOGY

The lesions are tense bullae filled with clear fluid involving erythematous or normal-appearing skin (Fig. 25.32A). The bullae are usually less than 2 cm in diameter but occasionally may reach 4 to 8 cm in diameter. They do not rupture easily, unlike the blisters seen in pemphigus, and heal without scarring unless they

become infected secondarily. The separation of bullous pemphigoid from pemphigus is based on the identification of **subepidermal, nonacantholytic** blisters. Early lesions show a superficial and sometimes deep perivascular infiltrate of lymphocytes and variable numbers of eosinophils, occasional neutrophils, superficial dermal edema, and basal cell layer vacuolization (Fig. 25.32B). Eosinophils are typically present directly beneath the epidermal basal cell layer. The vacuolated basal cell layer eventually lifts away, allowing space for a fluid-filled blister to form.

Pathogenesis

Bullous pemphigoid is caused by autoantibodies that bind to proteins that are required for adherence of basal keratinocytes to the basement membrane. Most antibody deposition occurs in a continuous linear pattern at the dermoepidermal junction (Fig. 25.33A), where specialized structures called hemidesmosomes link basal keratinocytes to the underlying basement membrane (Fig. 25.33B). The so-called bullous pemphigoid antigens (BPAGs) are components of hemidesmosomes

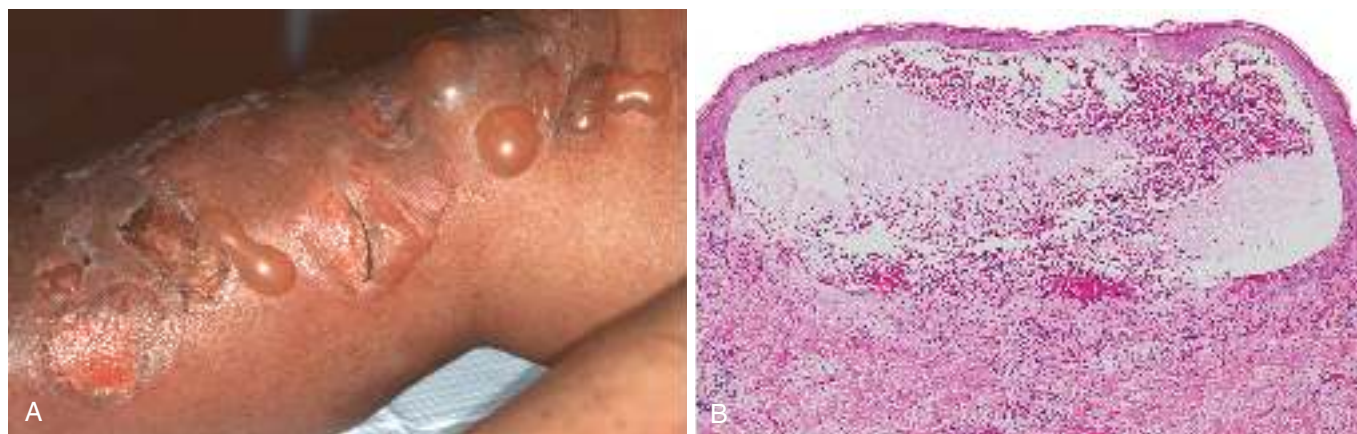


Figure 25.32 Bullous pemphigoid. (A) Bullae consist of tense blisters that usually fail to rupture, as their roof consists of intact epidermis. Ulcers form if the blisters rupture. (B) Intact subepidermal blister associated with eosinophils, lymphocytes, and occasional neutrophils.

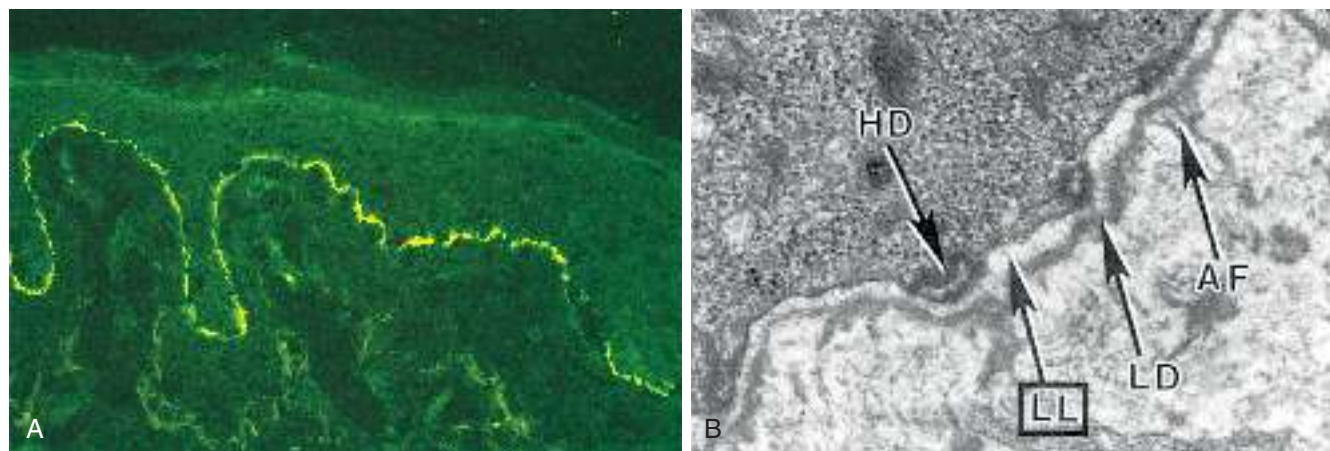


Figure 25.33 (A) Linear deposition of complement along the dermoepidermal junction in bullous pemphigoid. (B) Electron micrograph showing the ultrastructural features of the dermoepidermal junction. The bullous pemphigoid antigen (BPAG) is located in the basal portion of basal keratinocytes in association with hemidesmosomes (HD), which attach the epidermis to the lamina lucida (LL) of the basement membrane. AF, Anchoring fibrils; LD, lamina densa. (See also Fig. 25.31.)

Figure 25.34 Dermatitis herpetiformis. (A) Lesions consist of groups of intact and eroded erythematous blisters (seen here on elbows and arms). (B) Selective deposition of IgA autoantibody at the tips of dermal papillae is characteristic. (C) The blisters are associated with the accumulation of neutrophils (microabscesses) at the tips of dermal papillae. (B, Courtesy Dr. Victor G. Prieto, Houston, Tex.)

(see Fig. 25.29). Antibodies against one such component called BPAG2 are proven to cause blistering. Pathogenic autoantibodies also activate complement, leading to the recruitment of neutrophils and eosinophils, inflammation, and disruption of epidermal attachments.

Dermatitis Herpetiformis

Dermatitis herpetiformis is a rare disorder characterized by urticaria and grouped vesicles. The disease affects predominantly males, most often in the third and fourth decades of life. In some cases it occurs in association with intestinal celiac disease (Chapter 17). The plaques and vesicles are extremely pruritic.

Pathogenesis

The association of dermatitis herpetiformis with celiac disease provides a clue to its pathogenesis. Genetically predisposed individuals develop IgA antibodies to dietary gluten (derived from the wheat protein gliadin). The antibodies cross-react with reticulin, a component of the anchoring fibrils that tether the epidermal basement membrane to the superficial dermis. The resultant injury and inflammation produce a subepidermal blister. In some people with dermatitis herpetiformis and gluten-sensitive enteropathy, both disorders respond to a gluten-free diet.

MORPHOLOGY

The lesions are bilateral, symmetric, and grouped, involving preferentially the extensor surfaces, elbows, knees, upper back, and buttocks (Fig. 25.34C). Fibrin and neutrophils accumulate selectively at the **tips of dermal papillae**, forming small microabscesses (Fig. 25.34A). The basal cells overlying these microabscesses show vacuolization and focal dermoepidermal separation that ultimately coalesce to form a true **subepidermal blister**. By direct immunofluorescence, dermatitis herpetiformis shows discontinuous, **granular deposits of IgA** that selectively localize in the tips of dermal papillae (Fig. 25.34B).

Noninflammatory Blistering Disorders

Epidermolysis Bullosa and Porphyrria

Some disorders characterized by vesicles and bullae are mediated by inherited or, in some cases, acquired defects involving structural proteins that maintain the integrity of the skin. Two such disorders are epidermolysis bullosa and porphyria.

Epidermolysis Bullosa. Epidermolysis bullosa is a blanket term for a group of disorders caused by inherited defects

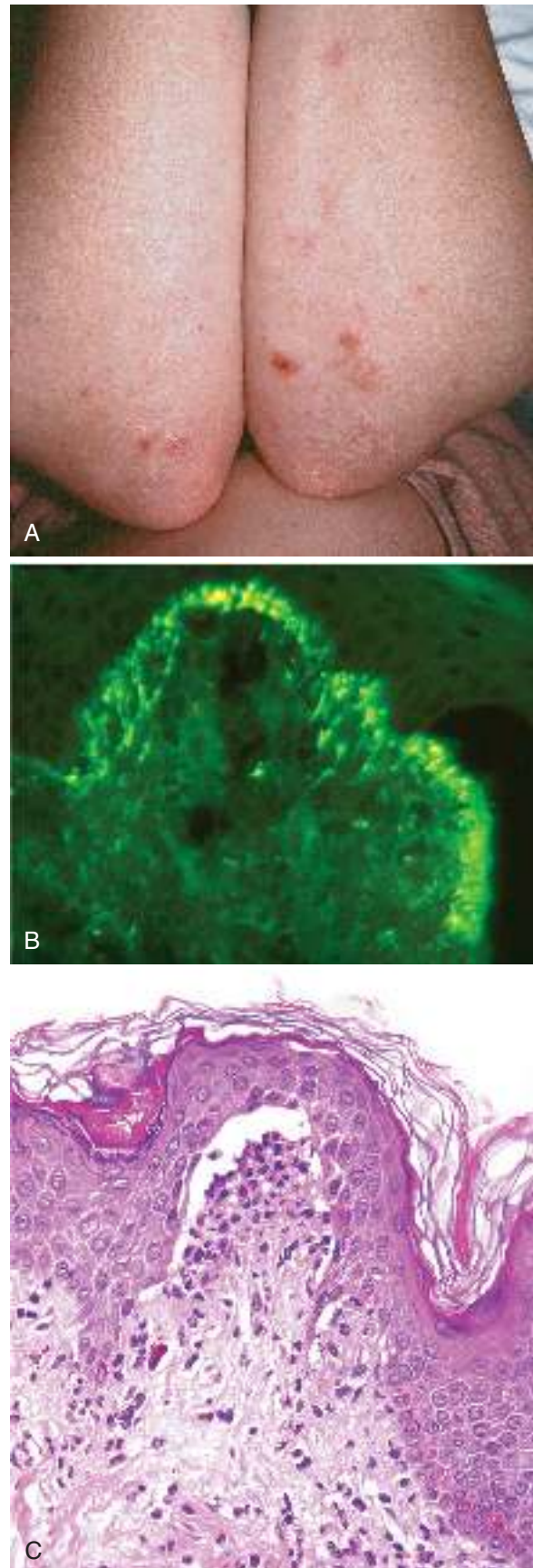




Figure 25.35 Epidermolysis bullosa. (A) Junctional epidermolysis bullosa showing typical erosions in flexural creases. (B) Subepidermal blister at the level of the lamina lucida. There is no associated inflammation.

in structural proteins that lend mechanical stability to the skin. The common feature is a proclivity to form blisters at sites of pressure, rubbing, or trauma, at or soon after birth. The histologic changes in all forms are so subtle that electron microscopy may be required to differentiate among the various types.

- In the *simplex type*, defects of the basal cell layer of the epidermis almost always result from mutations in the genes encoding keratin 14 or keratin 5. These two proteins normally pair with one another to make a functional keratin fiber, thus explaining the similar phenotype resulting from mutations in either gene. The mutated proteins have a dominant negative activity, and as a result the disorder shows an autosomal dominant mode of inheritance. This is the most common type of epidermolysis bullosa, encompassing 75% to 85% of cases.
- In the *junctional type*, blisters occur in otherwise histologically normal skin at precisely the level of the lamina lucida (Fig. 25.35; see Fig. 25.28). Most cases are caused by autosomal recessive defects in one of the subunits of laminin, a multicomponent protein located in the lamina lucida that binds to both hemidesmosomes and anchoring filaments. Some of the remaining cases are caused by mutations in BPAG2, the same protein that is targeted by autoantibodies in bullous pemphigoid.
- In the scarring *dystrophic types*, blisters develop beneath the lamina densa in association with rudimentary or defective anchoring fibrils. Dystrophic epidermolysis bullosa usually results from mutations in the *COL7A1* gene, which encodes type VII collagen (Chapter 3), a major component of the basement membrane anchoring fibrils. Depending on the mutation, the disorder may follow an autosomal dominant or autosomal recessive mode of inheritance.
- *Mixed types*, marked by defects at several levels, are also recognized.

Porphyria. Porphyria refers to a group of uncommon inborn or acquired disturbances of porphyrin metabolism. Porphyrins are pigments that are normally present in hemoglobin, myoglobin, and cytochromes. The classification of porphyria is based on both clinical and biochemical features. The five major types are (1) congenital erythropoietic porphyria, (2) erythrohepatic protoporphyria, (3) acute intermittent

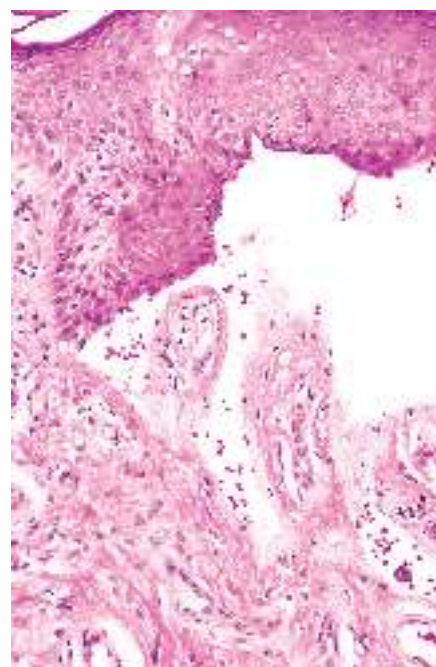


Figure 25.36 Porphyria. A noninflammatory blister at the dermoepidermal junction; note the seemingly rigid dermal papillae at the base that contain abnormal superficial vessels.

porphyria, (4) porphyria cutanea tarda, and (5) mixed porphyria. Cutaneous manifestations consist of urticaria and vesicles associated with scarring that are exacerbated by exposure to sunlight. The vesicles are subepidermal in location, and the adjacent dermis contains vessels with walls that are thickened by glassy deposits of serum proteins, including immunoglobulins (Fig. 25.36). The pathogenesis of these alterations is not understood.

KEY CONCEPTS

BLISTERING DISORDERS

- Blistering disorders are classified based on the level of epidermal separation.

- These disorders are often caused by autoantibodies specific for epithelial or basement membrane proteins that lead to unmooring of keratinocytes (acantholysis).
- Pemphigus is associated with IgG autoantibodies to various intercellular desmogleins, resulting in bullae that are either subcorneal (pemphigus foliaceus) or suprabasilar (pemphigus vulgaris).
- Bullous pemphigoid is associated with IgG autoantibodies to basement membrane proteins and produces a subepidermal blister.
- Dermatitis herpetiformis is associated with IgA autoantibodies to fibrils that bind the epidermal basement membrane to the dermis and also produces subepidermal blisters.
- Noninflammatory blistering disorders include inherited defects in proteins that stabilize the epidermis (e.g., epidermolysis bullosa), as well as defects in porphyrin synthesis (the porphyrias) that lead to sun-induced skin damage through uncertain mechanisms.

DISORDERS OF EPIDERMAL APPENDAGES

Acne Vulgaris

Virtually universal in the middle to late teenage years, acne vulgaris affects males and females, with males tending to have more severe disease. Acne is seen in all races but is usually milder in people of Asian descent. It may be induced or exacerbated by drugs (corticosteroids, adrenocorticotrophic hormone, testosterone, gonadotropins, contraceptives, trimethadione, iodides, and bromides), occupational exposures (cutting oils, chlorinated hydrocarbons, and coal tars), and conditions that favor occlusion of sebaceous glands, such as heavy clothing, cosmetics, and tropical climates. Some families seem to be particularly prone to acne, suggesting a hereditary component.

Acne is divided into noninflammatory and inflammatory types, although both types may coexist. Noninflammatory acne may take the form of open and closed comedones.

- *Open comedones* are small follicular papules containing a central black keratin plug. This color is the result of oxidation of melanin pigment (not dirt).
- *Closed comedones* are follicular papules without a visible central plug. Because the keratin plug is trapped beneath the epidermal surface, these lesions are potential sources of follicle rupture and inflammation.

Pathogenesis

The pathogenesis of acne is incompletely understood and is likely multifactorial. At least four factors contribute to its development: (1) keratinization of the lower portion of the follicular infundibulum and development of a keratin plug that blocks outflow of sebum to the skin surface; (2) hypertrophy of sebaceous glands during puberty under the influence of androgens; (3) lipase-synthesizing bacteria (*Propionibacterium acnes*) colonizing the upper and mid-portion of the hair follicle, converting lipids within sebum to proinflammatory fatty acids; and (4) secondary inflammation of the involved follicle. Androgens were first implicated in times past when it was noted that young

castrated males generally did not develop the condition (a questionable tradeoff). Elimination of *P. acnes* is the rationale for administration of antibiotics to individuals with inflammatory acne. The synthetic vitamin A derivative 13-*cis*-retinoic acid (isotretinoin) brings about remarkable improvement in some cases of severe acne through its strong antisebaceous action.

MORPHOLOGY

Inflammatory acne is marked by erythematous papules, nodules, and pustules (Fig. 25.37A). Severe variants (e.g., **acne conglobata**) result in sinus tract formation and dermal scarring. Depending on the stage of the disease, open or closed comedones, papules, pustules, or deep inflammatory nodules may develop. **Open comedones** have large, patulous orifices, whereas the orifices of **closed comedones** are identifiable only microscopically (Fig. 25.37B, C). Variable infiltrates of lymphocytes and macrophages are present in and around affected follicles, and extensive acute inflammation accompanies follicle rupture. Dermal abscesses may form in association with rupture (Fig. 25.37B) and lead to scarring.

Rosacea

Rosacea is a common disease of middle age and beyond, affecting up to 3% of the U.S. population, with a predilection for females. Four stages are recognized: (1) flushing episodes (pre-rosacea), (2) persistent erythema and telangiectasia, (3) pustules and papules, and (4) rhinophyma (permanent thickening of the nasal skin by confluent erythematous papules and prominent follicles).

Pathogenesis

A clue to the cause comes from observations showing that individuals with rosacea have high cutaneous levels of the antimicrobial peptide cathelicidin, an important mediator of the cutaneous innate immune response. The cathelicidin peptides present in affected individuals are qualitatively distinct from those seen in individuals without rosacea as a result of alternative processing by proteases such as kallikrein 5 (also known as stratum corneum tryptic enzyme). Injection of cathelicidin peptides from patients into mice induces some of the cutaneous changes seen in rosacea, including inflammation and vascular dilation. In addition, it has been noted that activation of TLR2 upregulates kallikrein 5 expression in keratinocytes, suggesting that factors that stimulate TLR2 are involved. Several microbial triggers have been proposed, but none are proven.

MORPHOLOGY

Rosacea is characterized by a nonspecific perifollicular infiltrate composed of lymphocytes surrounded by dermal edema and telangiectasia. In the pustular phase, neutrophils may colonize the follicles, and follicle rupture may occur and cause a granulomatous dermal response. The development of rhinophyma is associated with hypertrophy of sebaceous glands and follicular plugging by keratotic debris.

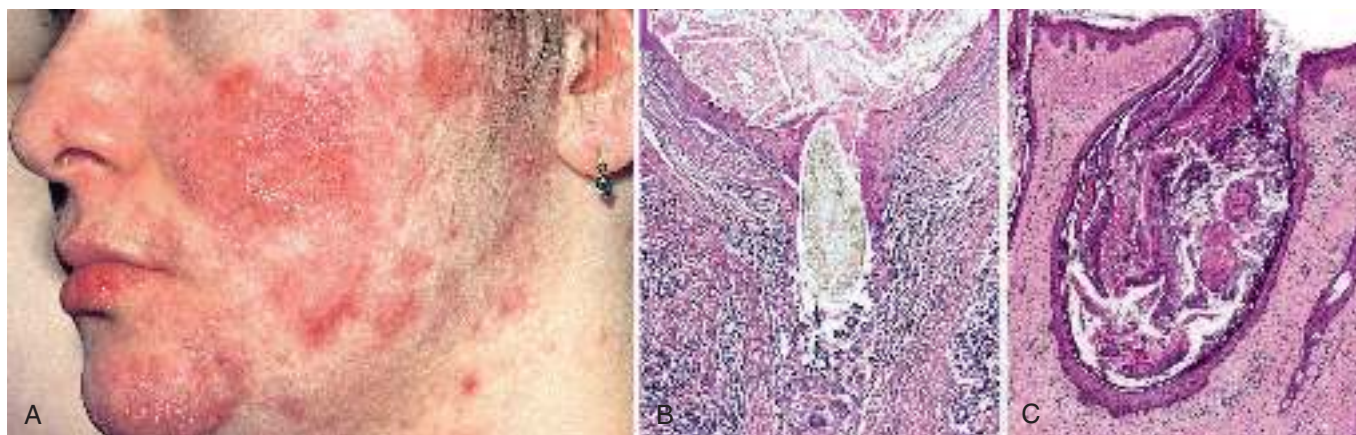


Figure 25.37 Acne. (A) Inflammatory acne associated with erythematous papules and pustules. (B) A hair shaft pierces the follicular epithelium, eliciting inflammation and fibrosis. (C) An open comedone.

PANNICULITIS

Erythema Nodosum and Erythema Induratum

Panniculitis is an inflammatory reaction in the subcutaneous adipose tissue that may preferentially affect (1) the lobules of fat or (2) the connective tissue that separates fat into lobules. Panniculitis often involves the lower legs. *Erythema nodosum* is the most common form and usually has a subacute presentation. A second somewhat distinctive form, *erythema induratum*, also merits brief discussion.

- *Erythema nodosum* presents as poorly defined, exquisitely tender, erythematous plaques and nodules that may be more readily palpated than seen. Its occurrence is often associated with infections (β -hemolytic streptococcal infection and tuberculosis and, less commonly, coccidioidomycosis, histoplasmosis, and leprosy), drug administration (sulfonamides and oral contraceptives), sarcoidosis, inflammatory bowel disease, and certain malignant neoplasms, but many times a cause cannot be identified. Fever and malaise may accompany the cutaneous signs. It is thought to be caused by a delayed hypersensitivity reaction to microbial or drug-related antigens. In some cases, immune complexes have been implicated, but in many cases the pathogenesis is mysterious. Over the course of weeks, lesions usually flatten and become bruise-like, leaving no residual clinical scars, while new lesions develop. Biopsy of a deep wedge of tissue to generously sample the subcutis is usually required for histologic diagnosis.
- *Erythema induratum* is an uncommon type of panniculitis that affects primarily adolescents and menopausal women. Although the cause is not known, most observers regard this as a primary vasculitis of deep vessels supplying the fat lobules of the subcutis; the associated vascular compromise leads to fat necrosis and inflammation. *Erythema induratum* presents as an erythematous, slightly tender nodule that usually goes on to ulcerate. Originally considered a hypersensitivity response to tuberculosis, *erythema induratum* today most commonly occurs without an associated underlying disease.

MORPHOLOGY

The histopathology of **erythema nodosum** is distinctive. In early lesions the connective tissue septa are widened by edema, fibrin exudation, and neutrophilic infiltration. Later, infiltration by lymphocytes, histiocytes, multinucleated giant cells, and occasional eosinophils is associated with septal fibrosis. Vasculitis is not present. In **erythema induratum**, on the other hand, granulomatous inflammation and zones of caseous necrosis involve the fat lobule. Early lesions show necrotizing vasculitis affecting small- to medium-sized arteries and veins in the deep dermis and subcutis.

Other Types of Panniculitis

Many other types of panniculitis have also been described, a few of which merit brief mention.

- *Weber-Christian disease (relapsing febrile nodular panniculitis)* is a rare form of lobular, nonvasculitic panniculitis seen in children and adults. It is marked by crops of erythematous plaques or nodules, predominantly on the lower extremities, created by deep-seated foci of inflammation containing aggregates of foamy macrophages admixed with lymphocytes, neutrophils, and giant cells.
- *Factitial panniculitis* is a form of secondary panniculitis caused by self-inflicted trauma or injection of foreign or toxic substances.
- Rare types of *T-cell lymphoma* home to fat lobules, producing fat necrosis and superimposed inflammation that mimics panniculitis.
- *Systemic lupus erythematosus* may occasionally cause inflammation of the subcutis and an associated panniculitis.

INFECTION

Despite its ample protective mechanisms, the exposure of the skin to frequent trauma and the external environment leaves it vulnerable to infection by microorganisms, parasites, and insects. We have already discussed the possible role of bacteria in the pathogenesis of common acne, and the

dermatoses resulting from viruses are too numerous to list. In the setting of the immunocompromised individual, ordinarily trivial cutaneous infections may become life threatening. Many disorders, such as herpes simplex and herpes zoster, viral exanthems, deep fungal infections, and immune reactions in skin provoked by infectious agents, are discussed in Chapter 8. Here we cover a representative sampling of common infections whose primary clinical manifestations are in the skin.

Verrucae (Warts)

Verrucae are squamoproliferative lesions caused by human papillomaviruses (HPVs). They are common lesions of children and adolescents, although they may be encountered at any age. Transmission of disease usually involves direct contact between individuals or autoinoculation. Verrucae are generally self-limited, regressing spontaneously within 6 months to 2 years.

Pathogenesis

More than 150 types of papillomavirus have been identified, many of them capable of producing warts in humans. The clinical variants of warts are often associated with distinct HPV subtypes. For example, anogenital warts are caused predominantly by HPV types 6 and 11. HPV type 16 has been associated with in situ squamous cell carcinoma of the genitalia and with bowenoid papulosis; the latter is a genital

lesion of young adults with the histologic appearance of carcinoma in situ, but which usually regresses spontaneously (see also Chapter 21). The relationship of HPV subtypes 5 and 8 to squamous cell carcinomas, particularly in individuals affected by the rare condition epidermodysplasia verruciformis, was mentioned earlier. These patients develop multiple flat warts that contain HPV genomes, some of which progress to carcinoma. Viral typing can be accomplished by either in situ hybridization (Fig. 25.38D) or polymerase chain reaction.

HPVs that are associated with a high risk of cancer produce E6 proteins that abolish p53 function (Chapter 7). By contrast, HPV subtypes 5 and 8 produce variant E6 proteins that do not affect p53, explaining why these forms of HPV have low oncogenic potential. By contrast, the E6 proteins of low-risk HPVs interfere with Notch signaling, which is required for the normal maturation of keratinocytes; this effect likely contributes to the epidermal hyperplasia that characterizes warts.

MORPHOLOGY

The classification of verrucae is based largely on appearance and location. **Verruca vulgaris** is the most common type of wart. The lesions of verruca vulgaris may occur anywhere but are most often found on the hands, particularly on the dorsal surfaces and periungual areas, where they appear as gray-white to tan, flat to

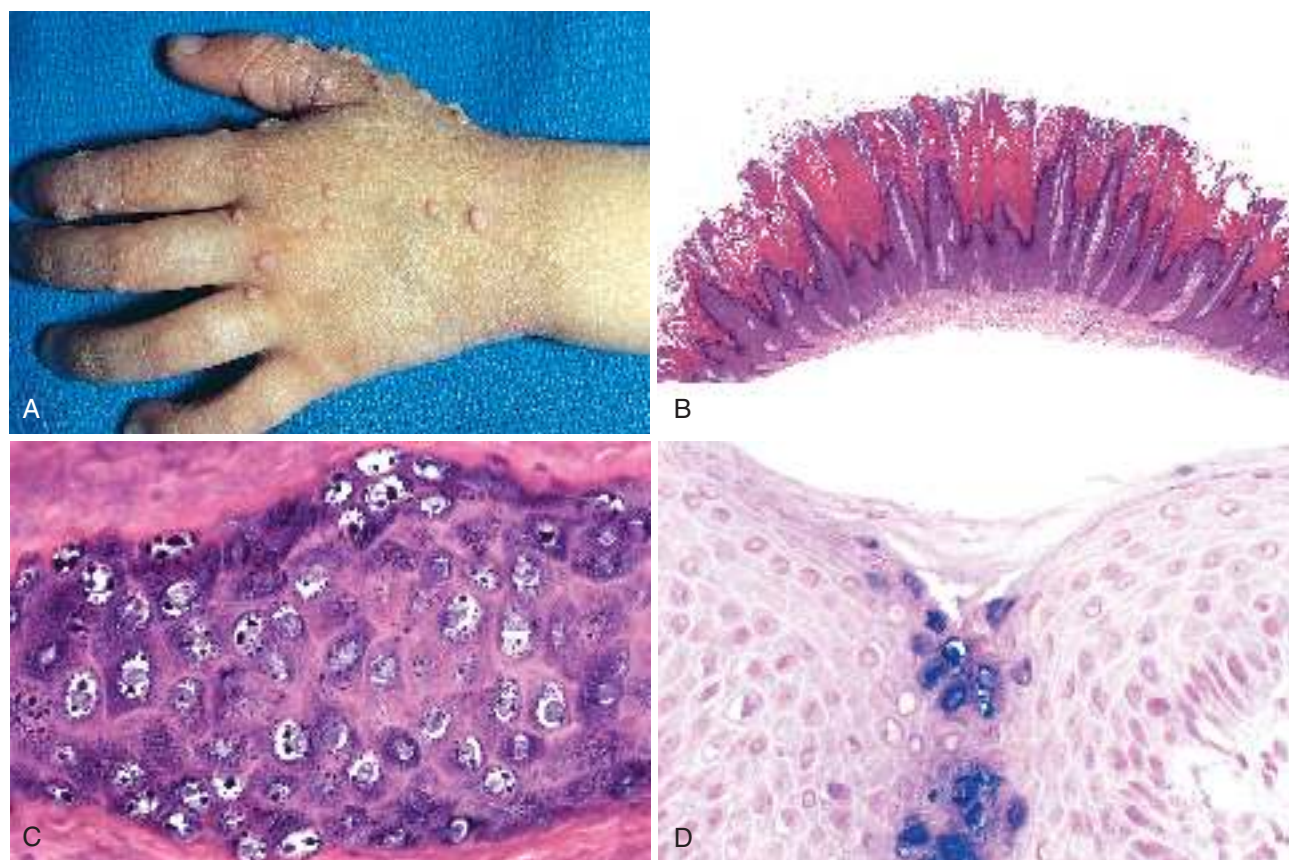


Figure 25.38 Verruca vulgaris. (A) Multiple papules with rough pebble-like surfaces. Low-power (B) and high-power (C) lesions showing papillomatous epidermal hyperplasia and cytopathic alterations including nuclear pallor and prominent keratohyaline granules. (D) In situ hybridization demonstrating human papillomavirus DNA within epidermal cells.

convex, 0.1- to 1-cm papules with a rough, pebble-like surface (Fig. 25.38A). **Verruca plana**, or **flat wart**, is common on the face or the dorsal surfaces of the hands. The warts are slightly elevated, flat, smooth, tan papules that are generally smaller than verruca vulgaris. **Verruca plantaris** and **verruca palmaris** occur on the soles and palms, respectively. These rough, scaly lesions may reach 1 to 2 cm in diameter, may sometimes coalesce, and may be confused with ordinary calluses. **Condyloma acuminatum** (**venereal wart**) occurs on the penis, female genitalia, urethra, perianal areas, and rectum. Venereal warts appear as soft, tan, cauliflower-like masses that occasionally reach many centimeters in diameter.

Histologic features common to verrucae include epidermal hyperplasia that is often undulant or spire-like in character, termed **verrucous** or **papillomatous epidermal hyperplasia** (Fig. 25.38B), and **cytoplasmic vacuolization (koilocytosis)** involving the more superficial epidermal layers, producing haloes of pallor surrounding infected nuclei. Electron microscopy of these zones reveals numerous HPV virions within nuclei. Infected cells may also demonstrate prominent, condensed keratohyaline granules and jagged eosinophilic intracytoplasmic keratin aggregates as a result of viral cytopathic effects (Fig. 25.38C). These cellular alterations are not as prominent in condylomas; hence, their diagnosis is based primarily on hyperplastic papillary architecture containing wedge-shaped zones of koilocytosis.

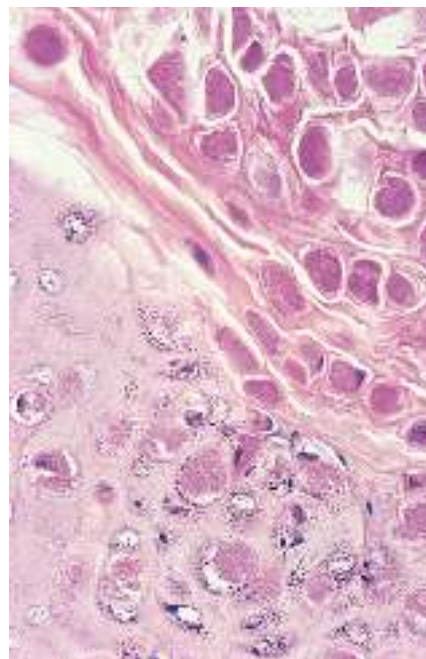


Figure 25.39 Molluscum contagiosum. A focus of verrucous epidermal hyperplasia contains numerous cells with ellipsoid cytoplasmic inclusions (molluscum bodies) within the stratum granulosum and stratum corneum.

Molluscum Contagiosum

Molluscum contagiosum is a common, self-limited viral disease of the skin caused by a poxvirus. The virus is characteristically brick shaped, has a dumbbell-shaped DNA core, and measures 300 nm in maximal dimension, making it one of the largest pathogenic poxviruses in humans and one of the largest viruses in nature. Infection is usually spread by direct contact, particularly among children and young adults.

MORPHOLOGY

Multiple lesions may occur on the skin and mucous membranes, with a predilection for the trunk and anogenital areas. Individual lesions are firm, often pruritic, pink to skin-colored umbilicated papules, generally ranging in diameter from 0.2 to 0.4 cm. Rarely, “giant” forms occur measuring up to 2 cm in diameter. A curd-like material can be expressed from the central umbilication. Smearing this material onto a glass slide and staining with Giemsa often shows diagnostic molluscum bodies.

On microscopic examination, lesions show cup-like verrucous epidermal hyperplasia. The diagnostically specific structure is the **molluscum body**, which is a large (up to 35 μm), ellipsoid, homogeneous, cytoplasmic inclusion that is found in the cells of the stratum granulosum and the stratum corneum (Fig. 25.39). In hematoxylin and eosin stains, the inclusions are eosinophilic in the stratum granulosum and acquire a pale blue hue in the stratum corneum. Numerous virions are present within molluscum bodies.

Impetigo

Impetigo is a common superficial bacterial infection of skin. It is highly contagious and is frequently seen in otherwise healthy children as well as occasionally in adults in poor health. The infection usually involves exposed skin, particularly that of the face and hands. Two forms exist, classically referred to as *impetigo contagiosa* and *impetigo bullosa*; they differ from each other simply by the size of the pustules. Over the past several decades a remarkable shift in etiology has been observed. Whereas in the past impetigo contagiosa was almost exclusively caused by group A β -hemolytic streptococci and impetigo bullosa by *Staphylococcus aureus*, both are now usually caused by *S. aureus*.

Pathogenesis

Bacteria in the epidermis evoke an innate immune response that causes epidermal injury, leading to local serous exudate and formation of a scale crust (scab). The pathogenesis of blister formation in impetigo is related to bacterial production of a toxin that specifically cleaves desmoglein 1, the protein responsible for cell-to-cell adhesion within the uppermost epidermal layers. Recall that in pemphigus foliaceus, which has a similar plane of blister formation, desmoglein 1 is compromised not by a toxin but by an autoantibody (see Fig. 25.28). Because there is virtually no involvement of the dermis, once the bacteria are eliminated the lesions heal without scarring.

MORPHOLOGY

Impetigo presents as an erythematous macule, but multiple small pustules rapidly supervene. As pustules break, shallow erosions

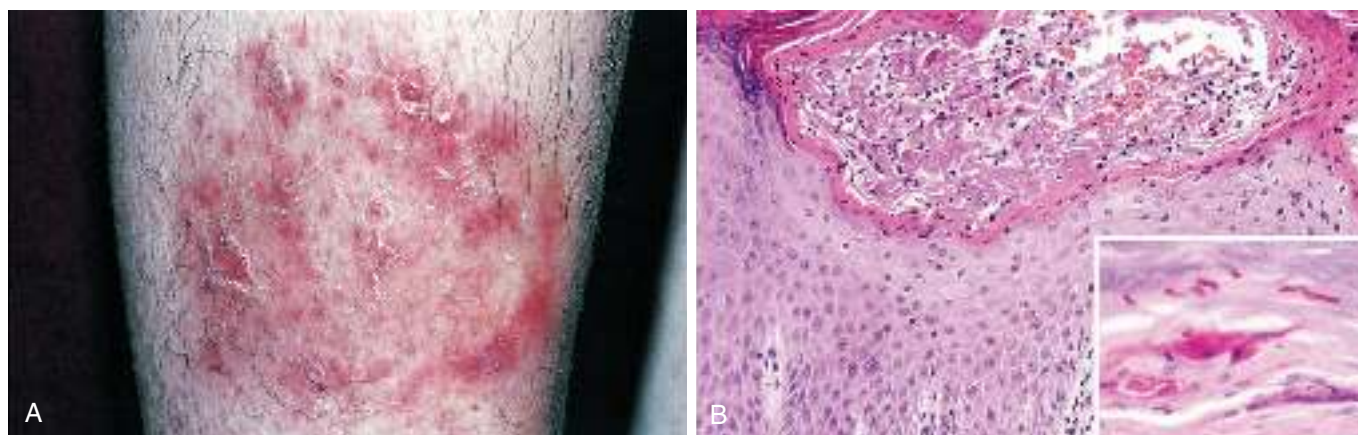


Figure 25.40 Tinea. (A) Characteristic plaque of tinea corporis. (B) Routine histology shows a mild eczematous (spongiotic) dermatitis and focal neutrophilic abscesses. Periodic acid–Schiff stain (inset) reveals deep red hyphae within the stratum corneum.

form, covered with drying serum, giving the characteristic appearance of **honey-colored crust**. If the crust is not removed, new lesions form about the periphery, and extensive epidermal damage may ensue. A bullous form of impetigo mainly occurs in children.

The characteristic microscopic feature of impetigo is **accumulation of neutrophils beneath the stratum corneum**, often producing a subcorneal pustule containing serum proteins and inflammatory cells. Special stains reveal the presence of bacteria in these foci. Nonspecific, reactive epidermal alterations and superficial dermal inflammation accompany these findings. Rupture of pustules releases serum, neutrophils, and cellular debris, which layer out and dry to form the characteristic crust.

Superficial Fungal Infections

As opposed to deep fungal infections of the skin, where the dermis or subcutis is primarily involved, superficial fungal infections of the skin are confined to the stratum corneum and are caused primarily by dermatophytes. These organisms grow in the soil and on animals and produce a number of diverse lesions with characteristic distributions, as follows:

- *Tinea capitis* usually occurs in children and is only rarely seen in infants and adults. It is a dermatophytosis of the scalp characterized by asymptomatic, patchy skin lesions associated with mild erythema, crust formation, scaling, and frequent hair loss.
- *Tinea barbae* is a dermatophyte infection of the beard area that affects adult men; it is relatively uncommon.
- *Tinea corporis*, on the other hand, is a common superficial fungal infection of skin that affects persons of all ages, but particularly children. Predisposing factors include excessive heat and humidity, exposure to infected animals, and chronic dermatophytosis of the feet or nails. The most common type of tinea corporis is an expanding, round, slightly erythematous plaque with an elevated scaling border (Fig. 25.40A).
- *Tinea cruris* occurs most frequently in the inguinal areas of obese men during warm weather. Heat, friction, and maceration all predispose to its development. The infection

usually first appears on the upper inner thighs as moist, red patches with raised scaly borders.

- *Tinea pedis* (athlete's foot) affects 30% to 40% of the population at some time in their lives. There is diffuse erythema and scaling, often initially localized to the web spaces. Most of the inflammatory reaction, however, appears to be the result of bacterial superinfection and is not directly related to the primary dermatophytosis. Spread to (or primary infection of the nails) is referred to as *onychomycosis*. This produces discoloration, thickening, and deformity of the nail plate.
- *Tinea versicolor* usually occurs on the upper trunk and is highly distinctive in appearance. Caused by *Malassezia furfur* (a yeast, not a dermatophyte), the lesions consist of groups of macules of varied size and color with a fine peripheral scale.

MORPHOLOGY

The histologic features of dermatophytoses are variable, depending on the properties of the organism, the host response, and the degree of bacterial superinfection. There may be mild eczematous dermatitis associated with intraepidermal neutrophils (Fig. 25.40B). Due to cell walls rich in mucopolysaccharides, fungi stain bright pink to red with periodic acid–Schiff stain. They are found in the anucleate cornified layer of lesional skin, hair, or nails (Fig. 25.40B, inset). Culture of material scraped from these areas usually permits identification of the offending species.

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Bones, Joints, and Soft Tissue Tumors

26

Andrew Horvai

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Bone**BASIC STRUCTURE AND FUNCTION OF BONE**

The adult human skeleton is composed of 206 bones that account for ~12% of body weight. Bone functions include mechanical support, force transmission, internal organ protection, and mineral homeostasis, and it also serves

as the major site of hematopoiesis during postnatal life. Bone consists of extracellular matrix and several cell types.

Matrix

The extracellular component of bone, the matrix, is composed of osteoid (35%) and minerals (65%). The latter,

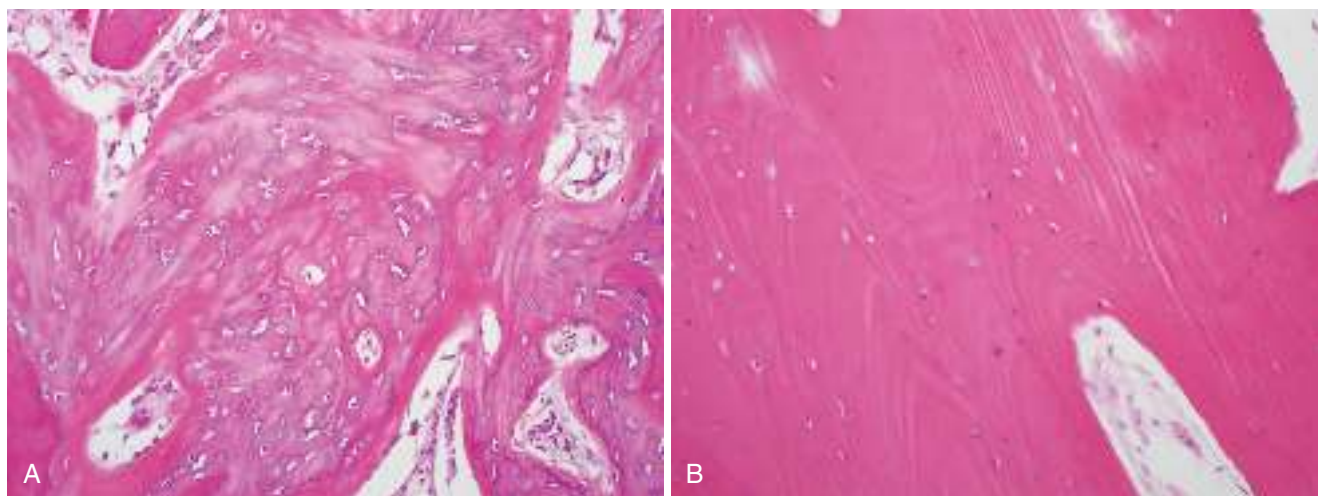


Figure 26.1 (A) Woven bone. (B) Lamellar bone contrasts with the more cellular, disorganized appearance of woven bone.

primarily hydroxyapatite [$\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$], give bone its hardness and serves as a repository for 99% of the calcium and 85% of the phosphorus in the body.

Osteoid consists of type I collagen and smaller amounts of glycosaminoglycans and other proteins. One of these proteins, osteopontin (also called osteocalcin), is produced by osteoblasts and contributes to the regulation of bone formation, mineralization, and calcium homeostasis. Serum osteopontin levels are used as a specific marker of osteoblast activity. Bone maturation and metabolism are also sensitive to cytokines and growth factors and are thus regulated by diverse inputs, which include locally and systemically produced factors, as well as mechanical force.

Bone matrix can be woven or lamellar (Fig. 26.1). Woven bone is produced rapidly (e.g., during fetal development or fracture repair), but the haphazard arrangement of collagen fibers imparts less structural integrity than the parallel collagen fibers of lamellar bone. Woven bone is always abnormal in adults, but its presence is not specific for any particular disease. Long bones are composed of a dense outer cortex and a central medulla. The latter is supported by bony trabeculae interspersed with marrow, which may be fatty (white) or hematopoietic (red).

Cells

The cellular components of bone include osteoblasts, osteocytes, and osteoclasts.

- *Osteoblasts* on the surface of the osteoid matrix synthesize, transport, and assemble matrix and regulate mineralization (Fig. 26.2A). Osteoblast activity is tightly regulated by hormonal and local mediators. Quiescent osteoblasts, which can be recognized by a decrease in cytoplasmic volume, may remain on the trabecular surface or become embedded within the matrix as osteocytes.
- *Osteocytes* are interconnected by an intricate network of dendritic cytoplasmic processes through tunnels (canaliculi) within the matrix. Osteocytes help control calcium and phosphate levels in the microenvironment, detect mechanical forces, and translate those forces into biologic activity—a process called mechanotransduction.

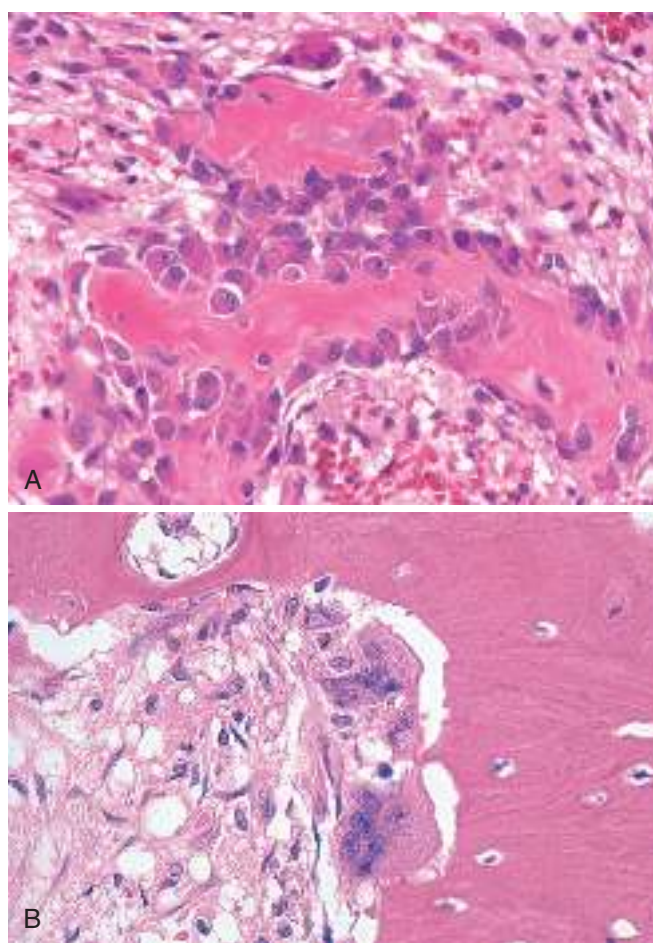


Figure 26.2 (A) Active osteoblasts synthesizing bone matrix. The surrounding spindle cells represent osteoprogenitor cells. (B) Two osteoclasts resorbing bone.

- *Osteoclasts* are specialized multinucleated macrophages that are derived from circulating monocytes and resorb bone (Fig. 26.2B). Surface integrin proteins allow osteoclasts to attach to the matrix and create a sealed extracellular trench (resorption pit). Secretion of acid and neutral

proteases, predominantly matrix metalloproteases (MMPs), into the pit results in dissolution of inorganic and organic bone components.

Development

Most bones that form during embryogenesis develop from a cartilage mold via endochondral ossification. The cartilage mold (anlagen) is synthesized by mesenchymal precursor cells. A central medullary canal within the anlagen is created by chondroblasts at approximately 8 weeks of gestation. Simultaneously, osteoblasts begin to deposit the cortex beneath the nascent periosteum of the midshaft (diaphysis). This forms a primary center of ossification resulting in radial bone growth. At the longitudinal ends (epiphysis), endochondral ossification forms secondary centers of ossification. Eventually, plates of cartilage anlage become entrapped between the expanding centers of ossification forming physes or growth plates (Fig. 26.3). Chondrocytes within the growth plates undergo sequential proliferation, hypertrophy, and apoptosis. Matrix mineralizes during apoptosis and is invaded by capillaries, providing the nutrients for activation of osteoblasts and osteoid synthesis. Most calcified cartilage matrix is ultimately resorbed leaving only strut-shaped remnants that serve as scaffolding for bone deposits known as primary spongiosa, the earliest bone trabeculae (see Fig. 26.3). Over time, this process produces longitudinal bone growth.

Flat bones, for example the cranium, are formed by intramembranous ossification, in which a dense layer of mesenchyme is directly ossified by osteoblasts without a cartilage anlagen. Bones enlarge by deposition of new bone on a preexisting surface, a process called appositional growth.

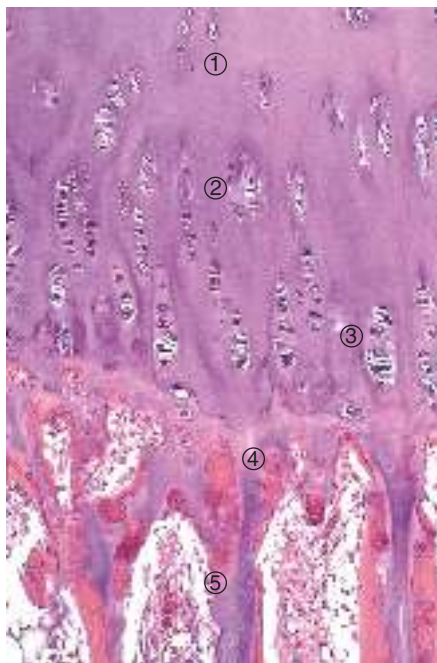


Figure 26.3 Active growth plate with ongoing endochondral ossification. 1, Reserve zone. 2, Zone of proliferation. 3, Zone of hypertrophy. 4, Zone of mineralization. 5, Primary spongiosa.

Local and systemic factors that regulate bone development include the following:

- Growth hormone (GH) secreted by the anterior pituitary gland induces and maintains chondrocyte proliferation.
- Thyroid hormone secreted by the thyroid gland acts on proliferating chondrocytes to induce hypertrophy.
- Indian hedgehog (Ihh) is secreted locally by prehypertrophic chondrocytes and coordinates chondrocyte proliferation and differentiation with osteoblast proliferation.
- Parathyroid hormone-related protein (PTHrP) produced by perichondrial stromal cells and early proliferating chondrocytes activates the PTH receptor to maintain chondrocyte proliferation.
- Wnt growth factors are expressed in the growth plate-proliferating zone and, via Frizzled and LRP5/6 receptors, activate β -catenin to promote chondrocyte proliferation and maturation.
- SOX9 is a transcription factor expressed by proliferating, but not hypertrophic, chondrocytes that is essential for differentiation of chondrocyte precursors.
- RUNX2 is a transcription factor expressed in early hypertrophic chondrocytes and immature mesenchymal cells that controls terminal chondrocyte and osteoblast differentiation.
- Fibroblast growth factors (FGFs) are secreted by a variety of mesenchymal cells. FGFs (most notably FGF-3) act on hypertrophic chondrocytes to inhibit proliferation and promote differentiation.
- Bone morphogenic proteins (BMPs), members of the TGF- β family, are expressed at various stages of chondrocyte development and have diverse effects on chondrocyte proliferation and hypertrophy at the growth plate.

Homeostasis and Remodeling

The adult skeleton appears static but actually undergoes continuous change via a tightly regulated process known as remodeling. This process, which turns over approximately 10% of the skeleton each year, repairs damage, and may change the shape of bones in response to mechanical forces. Remodeling occurs within the *bone (or basic) multicellular unit (BMU)*, which consists of a unit of coupled osteoblast and osteoclast activity on the bone surface. Osteoclast attachment, bone resorption, osteoblast attachment and proliferation, and, finally, matrix synthesis occur sequentially at the BMU.

Events at the BMU are regulated by cell-to-cell interactions and cytokines, and several signaling pathways (Fig. 26.4). One pathway involves three factors: (1) the transmembrane receptor activator for NF- κ B (RANK), which is expressed on osteoclast precursors; (2) RANK ligand (RANKL), which is expressed on osteoblasts and marrow stromal cells; and (3) osteoprotegerin (OPG), a secreted decoy receptor made by osteoblasts and several other types of cells that binds RANKL and thus prevents its interaction with RANK. When stimulated by RANKL, RANK signaling activates NF- κ B, which is essential for the generation and survival of osteoclasts. A second important pathway involves macrophage colony-stimulating factor (M-CSF), a factor produced by osteoblasts that is also crucial for the generation of osteoclasts. Finally, WNT proteins produced by osteoprogenitor cells bind to LRP5 and LRP6 on osteoblasts to activate β -catenin signaling and osteoprotegerin (OPG) synthesis (Fig. 26.5).

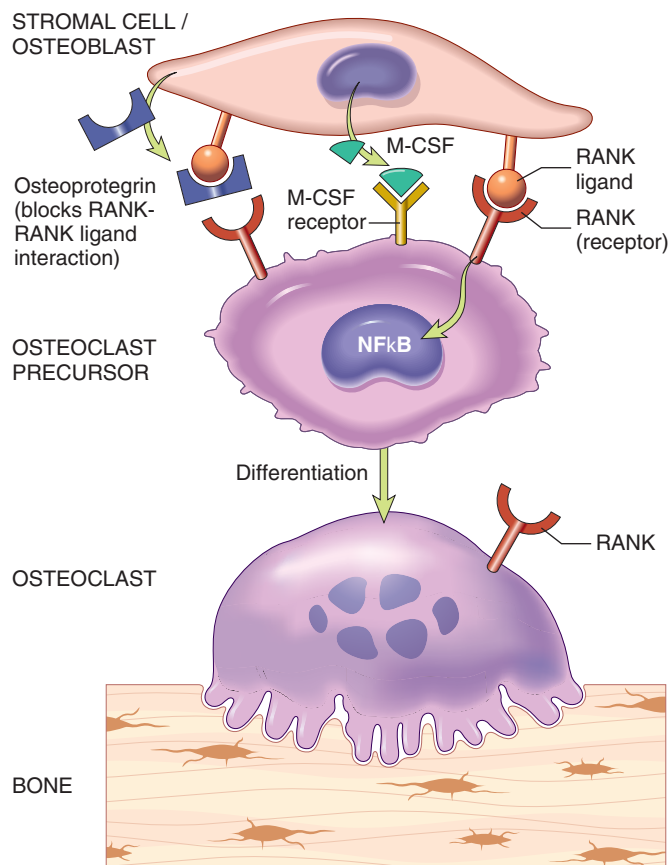


Figure 26.4 Paracrine molecular mechanisms that regulate osteoclast formation and function. Osteoclasts are derived from the same mononuclear cells that differentiate into macrophages. Osteoblast/stromal cell membrane-associated RANKL binds to its receptor RANK located on the cell surface of osteoclast precursors in concert with macrophage colony-stimulating factor (M-CSF). RANK activation induces precursor cells to become functional osteoclasts. Osteoblasts and stromal cells also secrete osteoprotegerin (OPG), which acts as a decoy receptor for RANKL, preventing it from binding the RANK on osteoclast precursors. Consequently, OPG prevents bone resorption by inhibiting osteoclast differentiation.

Conversely, osteocytes produce sclerostin, which inhibits WNT/ β -catenin signaling and promotes bone formation. The importance of these pathways is emphasized by rare germline mutations in *OPG*, *RANK*, *RANKL*, *LRP5*, and *sclerostin* genes, which severely disrupt bone metabolism and produce congenital bone disorders (described later).

The balance between bone formation and resorption is modulated by RANK and WNT signaling. For example, because OPG and RANKL oppose one another, bone resorption or formation can be favored by increasing or decreasing the RANK-to-OPG ratio, respectively. Systemic factors affecting this balance include hormones (parathyroid hormone, estrogen, testosterone, and glucocorticoids), vitamin D, inflammatory cytokines (e.g., IL-1), and growth factors (e.g., bone morphogenetic factors). Each of these acts by altering RANK/NF- κ B and WNT/ β -catenin signaling. Parathyroid hormone, IL-1, and glucocorticoids promote osteoclast differentiation and bone turnover, while bone morphogenic proteins and sex hormones generally block osteoclast differentiation or activity by promoting OPG expression, alterations that favor bone deposition.

Another level of control involves paracrine signaling between osteoblasts and osteoclasts. Matrix breakdown by osteoclasts liberates and activates growth factors, cytokines, and enzymes (e.g., collagenase), some of which stimulate osteoblasts. Thus, substances that initiate bone deposition are released into the microenvironment during bone resorption (see Fig. 26.5).

Peak bone mass is achieved in early adulthood after the cessation of skeletal growth and is determined by factors that include vitamin D receptor and *LRP5/6* polymorphisms, nutrition, physical activity, age, and hormonal status. Beginning in the fourth decade, however, resorption exceeds formation, resulting in a steady decline in skeletal mass.

DEVELOPMENTAL DISORDERS OF BONE AND CARTILAGE

Developmental abnormalities of the skeleton often stem from inherited mutations and become apparent during the earliest stages of bone formation. In contrast, acquired diseases usually appear in adulthood. The spectrum of disorders of bone development is broad, and there is no standard approach to their classification. Here we will categorize the major diseases according to their pathogenesis.

Developmental anomalies can result from localized disruption of the migration and condensation of mesenchyme (dysostosis) or global disorganization of bone and/or cartilage (dysplasia). Dysostoses may occur in isolation or as part of more complex syndromes, and are caused by defects in mesenchymal condensation and differentiation into cartilage anlage. The most common forms include complete absence of a bone or entire digit (aplasia), extra bones or digits (supernumerary digit), and abnormal fusion of bones (e.g., syndactyly, craniosynostosis). Genetic alterations that affect genes encoding transcription factors (especially homeobox genes), cytokines, and cytokine receptors are especially common among dysostoses. In contrast, dysplasias arise from mutations in genes that control development or remodeling of the entire skeleton. It is important to note that while the term dysplasia in this context implies abnormal growth, it is not a precursor of neoplasia, as is the case with dysplasias of epithelial cells (Chapter 7).

More than 350 skeletal dysostoses and dysplasias are recognized, most of which are extremely rare. The classification has evolved from clinical and radiographic descriptions to one that includes the causative genetic defects. Table 26.1 lists some of the better characterized developmental abnormalities and their associated defective genes. The relationships between specific mutations and phenotypes are complex; different point mutations in a single gene (e.g., *COL2A1*) can result in distinct phenotypes, while mutations in disparate genes (e.g., *LRP5*, *RANKL*) can lead to similar phenotypes.

Defects in Nuclear Proteins and Transcription Factors

Defects in nuclear proteins and transcription factors, especially homeobox proteins, result in disorganized mesenchymal condensation and differentiation of osteoblasts and chondrocytes, leading to abnormal bone development.

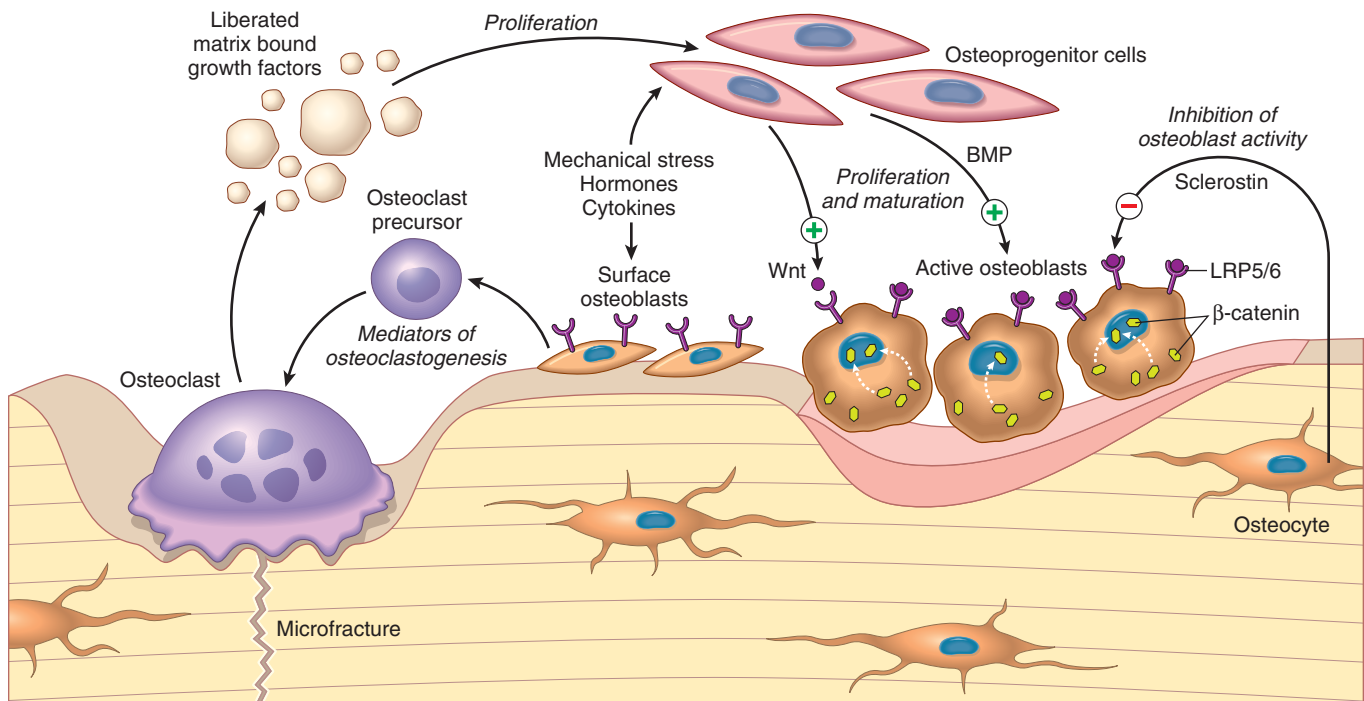


Figure 26.5 Bone cells and their interrelated activities. Hormones, cytokines, growth factors, and signal-transducing molecules are instrumental in bone formation and maturation and allow communication between osteoblasts and osteoclasts. Bone resorption and formation in remodeling are coupled processes controlled by systemic factors and local cytokines, some of which are deposited in the bone matrix. *BMP*, Bone morphogenetic protein; *LRP5/6*, LDL receptor–related proteins 5 and 6.

- *Brachydactyly types D and E* are caused by mutations in the homeobox *HOXD13* gene and are characterized by shortening of the terminal phalanges of the thumb and big toe, respectively.
- Loss-of-function mutations in *RUNX2* result in *cleidocranial dysplasia*, an autosomal dominant disorder characterized by patent fontanelles, delayed closure of cranial sutures, Wormian bones (extra bones that occur within a cranial suture), delayed eruption of secondary teeth, primitive clavicles, and short stature.

Defects in Hormones and Signal Transduction Proteins

Achondroplasia is the most common skeletal dysplasia and a major cause of dwarfism. It is an autosomal dominant disorder caused by gain-of-function mutations in the FGF receptor 3 (*FGFR3*) gene, ~90% of which stem from new mutations in the paternal allele. FGF-mediated *FGFR3* activation normally inhibits endochondral growth. This effect is exaggerated by *FGFR3* gain-of-function mutations. The retarded cartilage growth results in shortened proximal extremities, an enlarged head with bulging forehead, and depression of the root of the nose despite a trunk of relatively normal length. These skeletal abnormalities are not usually associated with changes in longevity, intelligence, or reproductive status.

Thanatophoric dysplasia is the most common lethal form of dwarfism. It affects ~1 in every 20,000 live births and is caused by *FGFR3* gain-of-function mutations distinct from those that cause achondroplasia, which appear to cause greater increases in *FGFR3* signaling than those that produce

achondroplasia, and thus a more severe phenotype. Affected individuals have disproportionately short (micromelic) limbs, frontal bossing, relative macrocephaly, a small chest cavity, and a bell-shaped abdomen. The underdeveloped thoracic cavity leads to respiratory insufficiency, and these individuals frequently die at birth or soon after. Histologic examination of the growth plate reveals reduced chondrocyte proliferation and disorganization within the zone of proliferation.

Defects in Extracellular Structural Proteins

Mutations in the major bone and cartilage collagens (types I, II, IX, X, and XI) give rise to highly variable presentations ranging from lethal disease to premature osteoarthritis (OA).

Type I Collagen Diseases (Osteogenesis Imperfecta)

Osteogenesis imperfecta (OI), or brittle bone disease, the most common inherited disorder of connective tissue, is a phenotypically heterogeneous disorder caused by deficiencies in type I collagen synthesis. OI principally affects bone, but it also impacts other tissues rich in type I collagen (joints, eyes, ears, skin, and teeth). It is caused by mutations in genes encoding the $\alpha 1$ and $\alpha 2$ chains of type I collagen. Many of the over 800 mutations identified lead to replacement of a glycine residue within the triple-helical domain with another amino acid. Collagen synthesis and extracellular transport require triple-helix formation, and these mutations result in misfolding of collagen polypeptides and defective assembly of higher-order collagen chains. Mutant collagens also interfere with assembly of wild-type collagen chains, that is, they exert a dominant negative effect, thereby explaining the autosomal dominant inheritance pattern.

Table 26.1 Diseases of the Skeleton With Identified Genetic Defects

Disorder	Gene Symbol	Affected Molecule	Clinical Phenotype
Defects in Transcription Factors Producing Abnormalities in Mesenchymal Condensation and Related Cell Differentiation			
Brachydactyly types D and E	<i>HOXD13</i>	Transcription factor	Short, broad terminal phalanges of first digits
Camptomelic dysplasia	<i>SOX9</i>	Transcription factor	Sex reversal, abnormal skeletal development
Cleidocranial dysplasia	<i>RUNX2</i>	Transcription factor	Abnormal clavicles, Wormian bones, supernumerary teeth
Holt-Oram syndrome	<i>TBX5</i>	Transcription factor	Congenital abnormalities, forelimb anomalies
Nail-patella syndrome	<i>LMX1B</i>	Transcription factor	Hypoplastic nails, hypoplastic or aplastic patellas, dislocated radial head, progressive nephropathy
Waardenburg syndrome types 1 and 3	<i>PAX3</i>	Transcription factor	Hearing loss, abnormal pigmentation, craniofacial abnormalities
Defects in Hormones and Signal Transduction Proteins Producing Abnormal Proliferation or Maturation of Osteoblasts, Osteoclasts, or Chondrocytes			
Achondroplasia	<i>FGFR3</i>	Receptor	Short stature, rhizomelic shortening of limbs, frontal bossing, midface deficiency
Hypochondroplasia	<i>FGFR3</i>	Receptor	Disproportionately short stature, micromelia, relative macrocephaly
Osteopetrosis, autosomal dominant	<i>LRP5</i>	Receptor	Increased bone density, hearing loss, skeletal fragility
Osteopetrosis, infantile form	<i>RANKL</i>	Receptor ligand	Increased bone density
Osteoporosis-pseudoglioma syndrome	<i>LRP5</i>	Receptor	Congenital or infant-onset loss of vision, skeletal fragility
Thanatophoric dysplasia	<i>FGFR3</i>	Receptor	Severe limb shortening and bowing, frontal bossing, depressed nasal bridge
Defects in Extracellular Structural Proteins			
Achondrogenesis type 2	<i>COL2A1</i>	Type II collagen	Short trunk
Metaphyseal dysplasia, Schmid type	<i>COL10A1</i>	Type X collagen	Mildly short stature
Osteogenesis imperfecta types 1 to 4	<i>COL1A1</i> , <i>COL1A2</i>	Type I collagen	Bone fragility
Defects in Metabolic Enzymes and Transporters			
Osteopetrosis with renal tubular acidosis	<i>CA2</i>	Carbonic anhydrase	Increased bone density, fragility, renal tubular acidosis
Osteopetrosis, late onset type 2	<i>CLCN7</i>	Chloride channel	Increased bone density, fragility

Modified from Mundlos S, Olsen BR: Heritable diseases of the skeleton. Part I: Molecular insights into skeletal development—transcription factors and signaling pathways, *FASEB J* 11(2):125–132, 1997; Mundlos S, Olsen BR: Heritable diseases of the skeleton. Part II: Molecular insights into skeletal development—matrix components and their homeostasis, *FASEB J* 11(4):227–233, 1997; Superti-Furga A, Bonafé L, Rimoin DL: Molecular-pathogenetic classification of genetic disorders of the skeleton, *Am J Med Genet* 106(4):262–293, 2001; Krakow D, Rimoin DL: The skeletal dysplasias, *Genet Med* 12(6):327–341, 2010.

The fundamental abnormality in OI is too little bone, resulting in extreme skeletal fragility. Other findings include blue sclerae caused by decreased collagen content, making the sclera translucent and allowing partial visualization of the underlying choroid; hearing loss related to both a sensorineural deficit and impeded conduction due to abnormalities in the bones of the middle and inner ear; and dental imperfections (small, misshapen, and blue-yellow teeth) secondary to dentin deficiency.

OI is separated into four major clinical subtypes of varying severity (Table 26.2). Mutations that result in decreased synthesis of qualitatively normal collagen are associated with mild skeletal abnormalities. More severe or lethal phenotypes are associated with mutant collagens that interfere with triple helix formation. The type 2 variant, which is uniformly fatal in utero or during the perinatal period, is characterized by extraordinary bone fragility and multiple intrauterine fractures (Fig. 26.6). In contrast, individuals with type 1 OI have a normal life span but experience childhood fractures that decrease in frequency following puberty.

Defects in Metabolic Pathways (Enzymes, Ion Channels, and Transporters)

Osteopetrosis

Osteopetrosis comprises a group of rare genetic diseases characterized by reduced bone resorption due to deficient osteoclast development or function, which leads to diffuse, symmetric skeletal sclerosis. Although the term osteopetrosis implies that the bones are stonelike, they are actually brittle and fracture easily. Osteopetrosis is classified into variants based on both the mode of inheritance and the severity of clinical findings.

Pathogenesis

Most of the mutations underlying osteopetrosis interfere with acidification of the osteoclast resorption pit, which is required for the dissolution of calcium hydroxyapatite within the matrix. For example, *Albers-Schönberg disease*, a mild autosomal dominant form of osteopetrosis, is caused by mutation of *CLCN7*, which encodes a proton-chloride

Table 26.2 Subtypes of Osteogenesis Imperfecta

Subtype	Collagen Defect	Inheritance	Major Clinical Features	Prognosis
I	Decreased synthesis of pro- $\alpha 1(I)$ chain Abnormal pro- $\alpha 1(I)$ or pro- $\alpha 2(I)$ chains	Autosomal dominant	Postnatal fractures, blue sclera Normal stature Skeletal fragility Dentinogenesis imperfecta Hearing impairment Joint laxity Blue sclerae	Compatible with survival
II	Abnormally short pro- $\alpha 1(I)$ chain Unstable triple helix Abnormal or insufficient pro- $\alpha 2(I)$	Most autosomal recessive Some autosomal dominant New mutations	Death in utero or within days of birth Skeletal deformity with excessive fragility and multiple fractures Blue sclera	Perinatal lethal
III	Altered structure of pro-peptides of pro- $\alpha 2(I)$ Impaired formation of triple helix	Autosomal dominant (75%) Autosomal recessive (25%)	Compatible with survival Growth retardation Multiple fractures Progressive kyphoscoliosis Blue sclera at birth that become white Hearing impairment Dentinogenesis imperfecta	Progressive, deforming
IV	Short pro- $\alpha 2(I)$ chain Unstable triple helix	Autosomal dominant	Postnatal fractures, normal sclerae Moderate skeletal fragility Short stature Sometimes dentinogenesis imperfecta	Compatible with survival



Figure 26.6 Skeletal radiograph of a fetus with lethal type 2 osteogenesis imperfecta. Note the numerous fractures of virtually all bones, resulting in accordion-like shortening of the limbs.

exchanger on the osteoclast surface that is required for resorption pit acidification. Similarly, most cases of autosomal recessive osteopetrosis are caused by a mutation of *TCIRG1*, which encodes a subunit of the osteoclast vacuolar H^+ -ATPase that is also necessary for acidification of the resorption pit. Other causes of autosomal recessive osteopetrosis include defects in carbonic anhydrase 2 (CA2), which, like all carbonic anhydrase isozymes, generates protons and bicarbonate from carbon dioxide and water. CA2 facilitates resorption pit acidification by osteoclasts and urinary acidification by renal tubular epithelial cells. Osteopetrosis due to CA2 mutations is, therefore, accompanied by renal tubular acidosis. Mutations in *IKBKG*, which encodes NEMO, the regulatory subunit of the inhibitor of kappaB kinase (IKK) complex that is involved in NF- κ B activation, is a cause of osteopetrosis that is not due to defective acidification. NEMO, which is X-linked, is required for osteoclastogenesis and osteoclast survival, which are regulated by NF- κ B. Because NF- κ B serves many other functions, *IKBKG* mutations result in a multisystem disorder called X-linked anhidrotic ectodermal dysplasia with immunodeficiency that includes osteopetrosis.

MORPHOLOGY

Due to deficient osteoclast activity, bones involved by osteopetrosis lack a medullary canal, and the ends of long bones are bulbous (Erlenmeyer flask deformity) and misshapen (Fig. 26.7). The neural foramina are small and can compress exiting nerves. The primary spongiosa, which is normally removed during growth, persists and fills the medullary cavity, leaving no room for the hematopoietic marrow and preventing the formation of mature trabeculae (Fig. 26.8). Deposited bone is not remodeled and tends to be woven rather than lamellar. Depending on the underlying genetic defect, the number of osteoclasts may be normal, increased, or decreased.



Figure 26.7 Radiograph of the upper extremity in an individual with osteopetrosis. The bones are diffusely sclerotic, and the distal metaphyses of the ulna and radius are poorly formed.

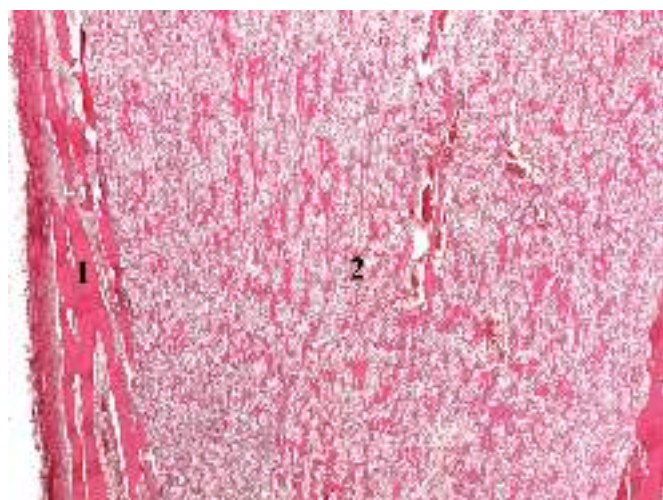


Figure 26.8 Section of proximal tibial diaphysis from a fetus with osteopetrosis. The cortex (1) is present, but the medullary cavity (2) is filled with primary spongiosa, which replaces the hematopoietic elements.

Clinical Features

Severe infantile osteopetrosis is an autosomal recessive disorder that usually becomes evident in utero or soon after birth. Fracture, anemia, and hydrocephaly are often seen, resulting in postpartum mortality. Affected individuals who survive into their infancy have cranial nerve defects (optic

atrophy, deafness, and facial paralysis) and repeated—often fatal—infections because of leukopenia due to reduced marrow space and decreased hematopoiesis. The compensatory extramedullary hematopoiesis can lead to prominent hepatosplenomegaly. Autosomal dominant forms are milder and may not be detected until adolescence or adulthood when discovered on x-ray studies, either incidentally or because of repeated fractures.

Osteopetrosis was the first genetic bone disease treated with hematopoietic stem cell transplantation, which is effective because osteoclasts are derived from hematopoietic precursors. The normal osteoclasts produced from donor stem cells reverse many of the skeletal abnormalities.

Diseases Associated With Defects in Degradation of Macromolecules

Mucopolysaccharidoses

The mucopolysaccharidoses, discussed in Chapter 5, are lysosomal storage diseases. They are caused by deficiencies in enzymes, primarily acid hydrolases, that degrade dermatan sulfate, heparan sulfate, and keratan sulfate. Mesenchymal cells, particularly chondrocytes, degrade extracellular matrix mucopolysaccharides. In these diseases, mucopolysaccharides accumulate within chondrocytes and induce apoptosis. Extracellular mucopolysaccharide accumulation leads to structural defects in articular cartilage. Consequently, many of the skeletal manifestations of the mucopolysaccharidoses result from abnormalities in the cartilage anlage, growth plates, costal cartilages, and articular surfaces. Affected individuals are frequently of short stature and have chest wall abnormalities and malformed bones.

KEY CONCEPTS

DEVELOPMENTAL DISORDERS OF BONE AND CARTILAGE

Abnormalities in a single bone or a localized group of bones are called dysostoses and arise from defects in mesenchyme migration and condensation. They manifest as absent, supernumerary, or abnormally fused bones. Global disorganization of bone and/or cartilage is termed dysplasia. Developmental abnormalities can be categorized by the associated genetic defect.

- **Transcription factors:** Homeobox genes, such as *HOXD13*, are frequently mutated in brachydactyly syndromes.
- **Signal transduction molecules:** *FGFR3* mutations are responsible for achondroplasia and thanatophoric dysplasia, both of which manifest as dwarfism.
- **Structural proteins:** Mutations in the genes for type I collagen underlie most types of osteogenesis imperfecta (brittle bone disease), characterized by defective bone formation and skeletal fragility.
- **Metabolic enzymes and transporters:** Mutations that interfere with osteoclast acidification of the resorption pit or osteoclastogenesis cause osteopetrosis, in which bones are hard but brittle. Depending on the gene mutated, extraskelatal manifestations may also be present.

METABOLIC DISEASES OF BONE

Osteopenia and Osteoporosis

Osteopenia refers to decreased bone mass; osteoporosis is defined as osteopenia that is severe enough to significantly increase the risk of fracture. Radiographically, osteoporosis is considered bone mass at least 2.5 standard deviations below mean peak bone mass in young adults. Osteopenia is 1 to 2.5 standard deviations below the mean. The disorder may be localized to a certain bone or region, as in disuse osteoporosis of a limb, or may involve the entire skeleton as a manifestation of a metabolic bone disease. Generalized osteoporosis may be primary or secondary to a variety of conditions (Table 26.3). Each year, approximately 1 million Americans experience an osteoporosis-related fracture with an estimated cost of more than \$14 billion. The following discussion relates largely to these forms of osteoporosis.

Pathogenesis

Peak bone mass is achieved during young adulthood. Its magnitude is determined largely by hereditary factors, especially polymorphisms in the genes that influence bone metabolism (Fig. 26.9). Physical activity, muscle strength, diet, and hormonal state also make important contributions.

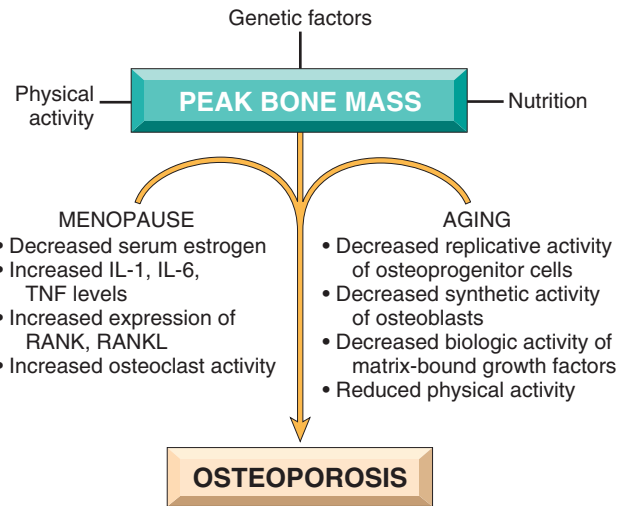


Figure 26.9 Pathophysiology of postmenopausal and senile osteoporosis (see text).

Once maximal skeletal mass is attained, bone resorption slightly exceeds formation, resulting in age-related bone loss that averages 0.7% per year. Both sexes are affected equally, but the process is more rapid on average in Caucasians than in those of African descent.

Although numerous factors affect bone mass, the most common forms of osteoporosis are senile and postmenopausal.

- *Age-related changes* include a reduced proliferative and biosynthetic capacity, and attenuated response to growth factors of osteoblasts, resulting in a diminished ability to make bone. This form of osteoporosis, known as *senile osteoporosis*, is categorized as a low-turnover variant.
- *Reduced physical activity* increases the rate of bone loss in experimental animals and humans, because mechanical forces normally stimulate bone remodeling. Bone loss in an immobilized or paralyzed extremity or in astronauts experiencing reduced gravitational forces for prolonged periods, and, conversely, increased bone density in athletes, exemplify the importance of physical forces in bone maintenance. When considering physical activity, the effect of load magnitude on bone density is greater than load repetition, explaining why resistance exercises, such as weight training, increase bone mass more effectively than endurance activities such as bicycling. The decreased physical activity that is associated with normal aging contributes to senile osteoporosis.
- *Genetic factors* such as single-gene defects are rare causes of osteoporosis. However, polymorphisms in certain genes may contribute to variation in peak bone density within populations. The most strongly linked genes identified by genome-wide association studies include *RANK*, *RANKL*, and *OPG*, all of which encode key osteoclast regulators; the HLA locus (for unknown reasons); and the estrogen receptor gene (discussed later).
- *Calcium nutritional state* contributes to peak bone mass. Adolescent girls (more than boys) tend to have insufficient calcium intake in the diet. If it occurs during a period of rapid bone growth, calcium deficiency reduces peak bone mass and increases the risk of osteoporosis. Relative

Table 26.3 Categories of Generalized Osteoporosis

Primary
Idiopathic
Postmenopausal
Senile
Secondary
Endocrine Disorders
Addison disease
Diabetes, type I
Hyperparathyroidism
Hyperthyroidism
Hypothyroidism
Pituitary tumors
Neoplasia
Carcinomatosis
Multiple myeloma
Gastrointestinal
Hepatic insufficiency
Malabsorption
Malnutrition
Vitamin C, D deficiencies
Drugs
Alcohol
Anticoagulants
Anticonvulsants
Chemotherapy
Corticosteroids
Miscellaneous
Anemia
Homocystinuria
Immobilization
Osteogenesis imperfecta
Pulmonary disease

deficiencies of calcium and vitamin D and elevated PTH levels may also contribute to development of senile osteoporosis.

- **Hormonal influences.** In the decade after menopause, up to 2% of cortical bone and 9% of cancellous bone may be lost each year. Estrogen deficiency plays the major role in this phenomenon, and close to 40% of postmenopausal women are affected by *postmenopausal osteoporosis*. Although decreased estrogen increases both bone formation and resorption, the latter dominates, resulting in high-turnover osteoporosis. Estrogen loss leads to increased secretion of inflammatory cytokines, such as IL-6, TNF, and IL-1, by innate immune cells in the blood and marrow through unknown mechanisms. The resulting increases and decreases in RANKL and OPG, respectively, stimulate osteoclast recruitment and activity.

MORPHOLOGY

The hallmark of osteoporosis is histologically normal bone that is decreased in quantity. The entire skeleton is affected (Fig. 26.10), but certain bones tend to be more severely impacted. Postmenopausal osteoporosis affects mainly bones or portions of bones that have increased surface area, such as the cancellous compartment of vertebral bodies. The trabecular plates become perforated and thinned, and lose their interconnections (Fig. 26.11), leading to microfractures and eventually to vertebral collapse. In senile osteoporosis, the cortex is thinned by subperiosteal and endosteal resorption, and the Haversian systems are widened.

Clinical Features

The clinical manifestations of osteoporosis depend on which bones are involved. Vertebral fractures that frequently occur in the thoracic and lumbar regions are painful, and, when multiple, can cause significant loss of height and deformities such as lumbar lordosis and kyphoscoliosis. Fractures of the femoral neck, pelvis, or spine lead to immobilization and complications such as pulmonary embolism and pneumonia, resulting in 40,000 to 50,000 deaths per year.

Osteoporosis cannot be reliably detected in plain radiographs until 30% to 40% of the bone mass is lost. The best estimates of bone loss, aside from biopsy (which is rarely performed), are specialized radiographic imaging techniques,

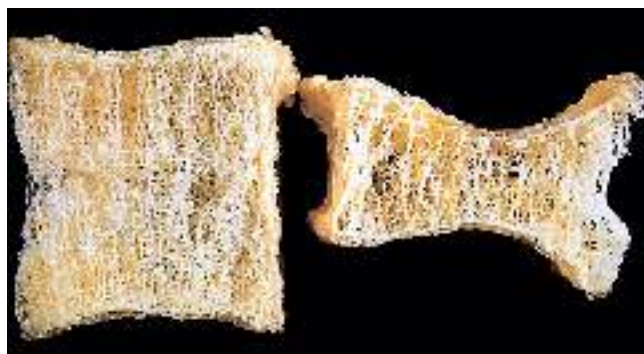


Figure 26.10 Osteoporotic vertebral body (right) shortened by compression fractures compared with a normal vertebral body (left). Note that the osteoporotic vertebra has a characteristic loss of horizontal trabeculae and thickened vertical trabeculae.

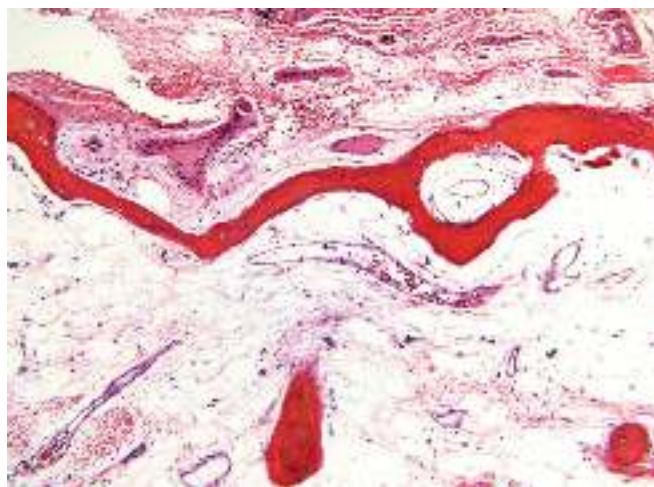


Figure 26.11 In advanced osteoporosis, both the trabecular bone of the medulla (bottom) and the cortical bone (top) are markedly thinned.

such as dual-energy x-ray absorptiometry and quantitative computed tomography, both of which measure bone density.

Preventive and therapeutic management of osteoporosis includes exercise, appropriate calcium and vitamin D intake, and pharmacologic agents, most commonly bisphosphonates, which reduce osteoclast activity and induce their apoptosis. Denosumab, an anti-RANKL antibody, has shown promise in treating some forms of postmenopausal osteoporosis. Although menopausal hormone therapy has been used to prevent osteoporosis and associated fractures, complications, particularly deep vein thrombosis and stroke, have prompted search for more selective estrogen receptor modulators.

Osteomalacia and Rickets

Osteomalacia and rickets are manifestations of impaired mineralization of bone matrix. This contrasts with osteoporosis, in which mineralization of bone is normal and bone mass is decreased. Most examples of undermineralized matrix result from abnormal vitamin D metabolism or vitamin D deficiency (detailed in Chapter 9). Rickets refers to the disorder in children, in whom it interferes with the deposition of bone in the growth plates. Osteomalacia is the adult counterpart, in which bone formed during remodeling is undermineralized and predisposed to fractures.

Hyperparathyroidism

Hyperparathyroidism causes increased bone resorption. As discussed in Chapter 24, parathyroid hormone (PTH) has a central role in calcium homeostasis through the following effects:

- *Activation of osteoclasts*, increasing bone resorption and calcium mobilization. PTH mediates the effect indirectly by increasing RANKL expression on osteoblasts.
- *Increasing calcium resorption* by the renal tubules
- *Increasing urinary phosphate excretion*
- *Increasing synthesis of active vitamin D, 1,25(OH)₂-D*, by the kidneys, thereby enhancing intestinal calcium absorption and mobilizing bone calcium by inducing RANKL expression on osteoblasts

The net result of PTH action is elevated serum calcium, which normally inhibits PTH production. Excessive or inappropriate PTH release may stem from autonomous parathyroid secretion (*primary hyperparathyroidism*) or may be the result of renal disease (*secondary hyperparathyroidism*) (see Chapter 24). In either setting, hyperparathyroidism produces changes throughout the skeleton as a consequence of unrestrained osteoclast activity. Elevated PTH is responsible for bone changes in primary hyperparathyroidism, but additional factors contribute in secondary hyperparathyroidism. In chronic renal insufficiency, $1,25\text{-(OH)}_2\text{-vitamin D}$ synthesis is reduced due to decreased α_1 -hydroxylase activity, which stems from loss of renal function and the suppressive effects of hyperphosphatemia on α_1 -hydroxylase. Inadequate vitamin D ultimately limits intestinal calcium absorption. Secondary hyperparathyroidism can also be complicated by metabolic acidosis and aluminum deposition in bone. With time, loss of bone mass increases susceptibility to fractures, bone deformation, and joint problems.

MORPHOLOGY

Symptomatic, untreated primary hyperparathyroidism manifests with three interrelated skeletal abnormalities: osteoporosis, brown tumors, and osteitis fibrosa cystica. The **osteoporosis** is generalized but is most severe in the phalanges, vertebrae, and proximal femur. Hyperparathyroidism most prominently enhances osteoclast activity within cortical bone (subperiosteal and endosteal surfaces), but cancellous bone can also be affected. At these sites, osteoclasts may tunnel into and dissect centrally along the length of the trabeculae, leaving adjacent marrow spaces to be replaced by fibrovascular tissue, producing **dissecting osteitis** (Fig. 26.12). Radiographically, dissecting osteitis is seen as decreased bone density or osteoporosis.

Bone loss in hyperparathyroidism predisposes to microfractures, secondary hemorrhage, macrophage recruitment, and ingrowth of reparative fibrous tissue to create a mass lesion called **brown tumor** (Fig. 26.13). The brown color reflects vascularity, hemorrhage, and hemosiderin deposition. Cystic degeneration of brown

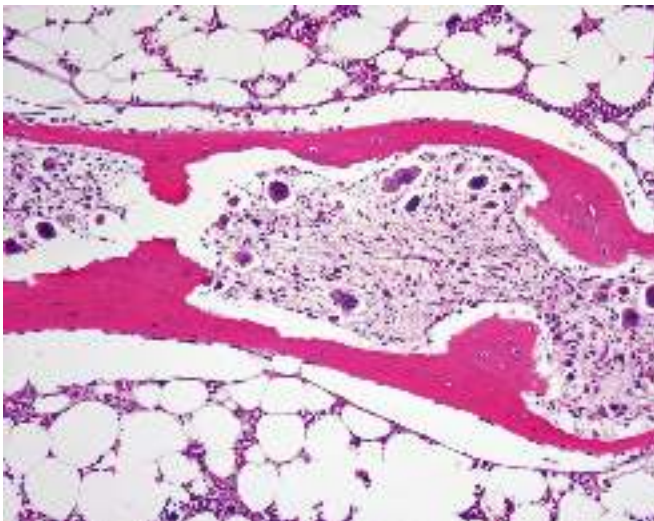


Figure 26.12 Hyperparathyroidism with osteoclasts boring into the center of the trabeculum (dissecting osteitis).



Figure 26.13 Resected rib, harboring an expansile brown tumor adjacent to the costal cartilage.

tumors is common. The combination of increased bone cell activity, peritrabecular fibrosis, and cystic brown tumors is the hallmark of severe hyperparathyroidism and is known as **generalized osteitis fibrosa cystica (von Recklinghausen disease of bone)**.

Osteitis fibrosa cystica is now uncommon because hyperparathyroidism is usually diagnosed on routine blood tests and treated at an early stage. Although secondary hyperparathyroidism can cause similar changes, the process is usually less severe, and skeletal abnormalities tend to be milder. Bony changes regress or disappear completely when hyperparathyroidism is controlled.

Renal Osteodystrophy

The term **renal osteodystrophy** describes the collective skeletal changes that occur in chronic renal disease, including those associated with dialysis. The manifestations include many of the entities described earlier including (1) osteopenia/osteoporosis, (2) osteomalacia, (3) secondary hyperparathyroidism, and (4) growth retardation. As medical advances have prolonged the lives of individuals with renal disease, the impact of this disease on skeletal homeostasis has assumed greater clinical importance. Histologic bone changes in individuals with end-stage renal failure can be divided into three major types of disorders:

- *High-turnover osteodystrophy* is characterized by increased bone resorption and bone formation, with the former predominating.
- *Low-turnover or aplastic disease* is manifested by adynamic bone (little osteoclastic and osteoblastic activity) and, less commonly, osteomalacia.
- *Mixed pattern disease* with areas of high turnover and low turnover.

Pathogenesis

Kidney disease causes skeletal abnormalities through three mechanisms (Fig. 26.14):

- *Tubular dysfunction* most commonly leading to renal tubular acidosis. The associated systemic acidosis dissolves hydroxyapatite, resulting in matrix demineralization and osteomalacia.
- *Secondary hyperparathyroidism*, due to reduced phosphate excretion, chronic hyperphosphatemia, and hypocalcemia.

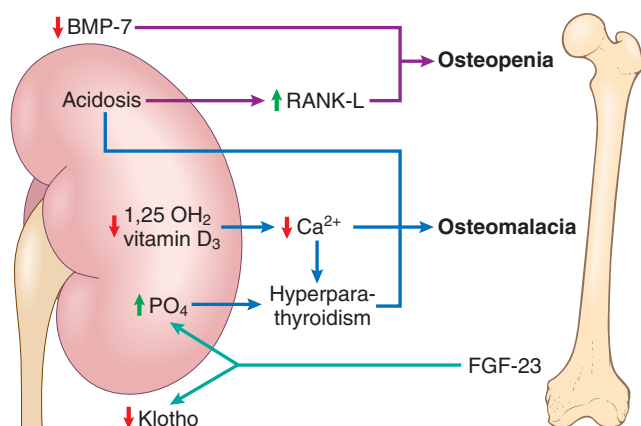


Figure 26.14 Mechanisms of renal osteodystrophy involve homeostasis and endocrine signaling between bone and kidney.

As discussed earlier, the resulting metabolic state is not completely analogous to primary hyperparathyroidism, and bone volume, turnover, and mineralization can vary independently.

- *Decreased biosynthetic function*, including reduced renal vitamin D hydroxylation to generate 1,25-OH₂-vitamin D₃. This results in hypocalcemia that contributes to secondary hyperparathyroidism. A hormonal feedback loop between kidney and bone that regulates calcium and phosphate homeostasis involves the secreted proteins BMP-7 and FGF-23 and the membrane protein Klotho. BMP-7, produced by renal tubular cells, induces osteoblast differentiation and proliferation, whereas FGF-23, made by osteocytes, acts on the kidney to regulate Klotho-dependent renal phosphate homeostasis and vitamin D hydroxylation. Chronic renal failure disrupts this signaling axis and contributes to osteopenia and osteomalacia.

KEY CONCEPTS

METABOLIC DISEASES OF BONE

- Osteopenia and osteoporosis are conditions in which bone is normally mineralized but decreased in quantity. Osteoporosis is defined as bone loss sufficient to increase fracture risk and is associated with significant morbidity and mortality from fractures. Multiple factors including peak bone mass, age, activity, genetics, nutrition, and hormonal influences contribute to its pathogenesis.
- Osteomalacia is defined by the presence of bone that is insufficiently mineralized. In the developing skeleton, this results in rickets.
- Hyperparathyroidism arises from either autonomous or compensatory hypersecretion of PTH and can lead to osteoporosis, brown tumors, and osteitis fibrosa cystica. In high income countries, where early diagnosis is the norm, these manifestations are rare.
- Renal osteodystrophy is marked by a constellation of bone abnormalities (osteopenia, osteomalacia, hyperparathyroidism, and growth retardation) that occur as a consequence of chronic renal failure. The bone changes stem from decreased tubular, glomerular, and hormonal renal functions.

Paget Disease (Osteitis Deformans)

Paget disease is a disorder marked by increased, but disordered and structurally unsound, bone mass. It develops in three sequential phases: (1) an initial osteolytic stage; (2) a mixed osteoclastic-osteoblastic stage; and (3) a burned-out quiescent osteosclerotic stage in which osteoblast activity predominates (Fig. 26.15).

Paget disease often presents in late adulthood and becomes progressively more common with increasing age. An intriguing aspect is the striking geographic variation in its prevalence. It is relatively common in Caucasians in England, France, Austria, regions of Germany, Australia, New Zealand, and the United States, and is rare in native populations of Scandinavia, China, Japan, and Africa. Its

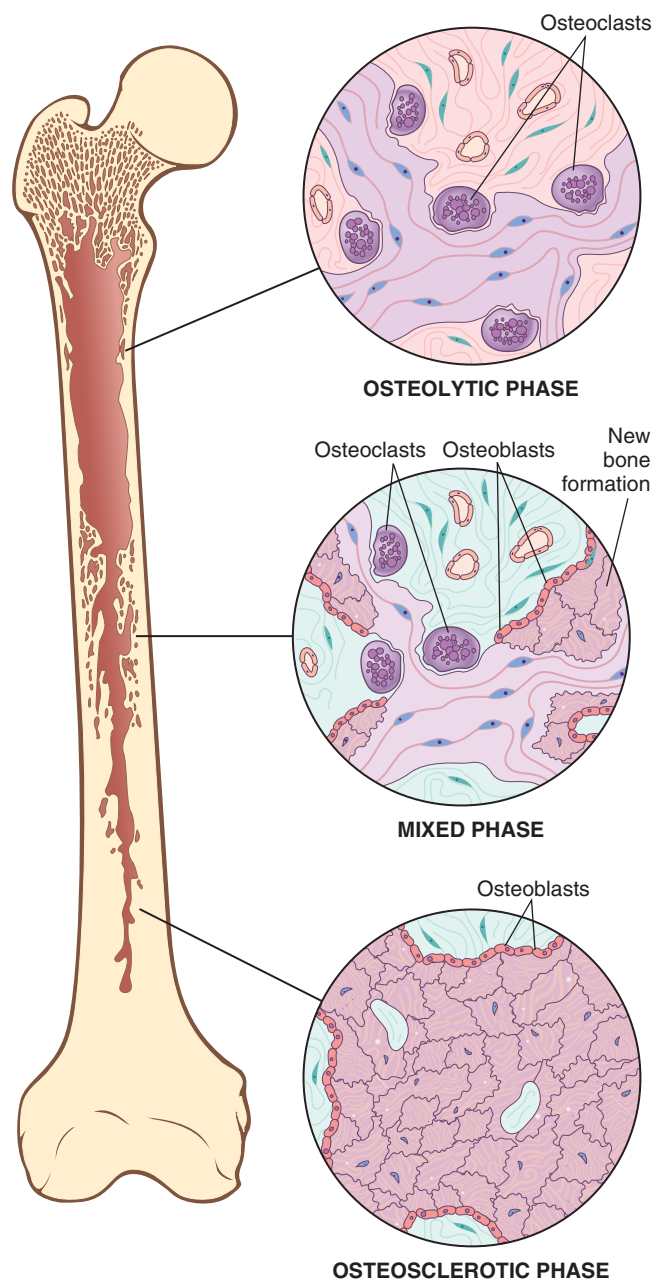


Figure 26.15 Diagrammatic representation of Paget disease of bone demonstrating the three phases of disease evolution.

exact incidence is difficult to determine because many affected individuals are asymptomatic; it is estimated that 1% of the US population older than 40 years of age is affected. The prevalence in England is 2.5% for men and 1.6% for women 55 years of age or older. A decrease in new cases has been observed in some countries over the past 30 years.

Pathogenesis

The cause of Paget disease remains uncertain, but current evidence suggests both genetic and environmental contributions. Approximately 50% of cases of familial Paget disease and 10% of sporadic cases are associated with mutations in the *SQSTM1* gene that increase NF- κ B activity and thus enhance osteoclast activity. Activating *RANK* mutations and inactivating *OPG* mutations account for some cases of juvenile Paget disease. In vitro studies suggest that chronic infection of osteoclast precursors by measles or other RNA viruses may also play a role.

MORPHOLOGY

Paget disease shows remarkable histologic variation over time and across sites. Its hallmark is a mosaic pattern of lamellar bone that develops in the sclerotic phase. This jigsaw puzzle–like appearance is produced by unusually prominent **cement lines**, which join haphazardly oriented units of lamellar bone (Fig. 26.16). The features are less specific during other phases of disease, but include waves of osteoclastic activity and numerous resorption pits in the lytic phase. The osteoclasts are abnormally large and have many more than the normal 10 to 12 nuclei; sometimes 100 nuclei are present. Osteoclasts persist in the mixed phase, but many of the bone surfaces are also lined by plump osteoblasts. The marrow adjacent to the bone-forming surface is replaced by loose connective tissue that contains osteoprogenitor cells and numerous blood vessels. The newly formed bone may be woven or lamellar, but eventually it is all remodeled into lamellar bone. As the mosaic pattern takes hold and cell activity decreases, periosseous fibrovascular tissue recedes and is replaced by normal marrow. In the end, the coarsely thickened trabeculae and soft and porous cortices lack structural stability and make the bone vulnerable to fractures.

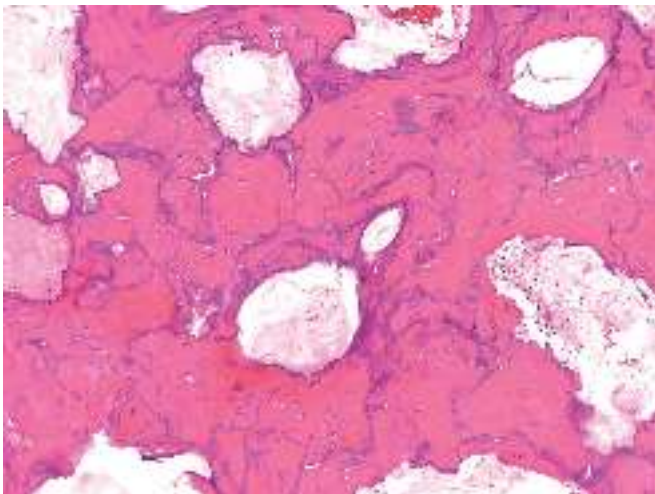


Figure 26.16 Mosaic pattern of lamellar bone pathognomonic in Paget disease.

Clinical Features

Clinical findings are extremely variable and depend on the extent and site of the disease. Paget disease is monostotic in about 15% of cases and polyostotic in the remainder. The axial skeleton or proximal femur is involved in up to 80% of cases. Most cases are asymptomatic and are discovered as an incidental radiographic finding. Pain localized to the affected bone may be present due to microfractures or bone overgrowth that compresses spinal and cranial nerve roots. Enlargement of the craniofacial skeleton may produce leontiasis ossea (lion face) and a cranium so heavy that it is difficult for the person to hold the head erect. The weakened Pagetic bone may lead to invagination of the skull base (platybasia) and compression of the posterior fossa. Weight bearing causes anterior bowing of the femurs and tibiae and distorts the femoral heads, resulting in the development of severe secondary osteoarthritis. Chalk stick–type fractures are common and usually involve the long bones of the lower extremities. Vertebral compression fractures can result in spinal cord injury and kyphosis. Rarely, the hypervascularity of Pagetic bone warms the overlying skin, and in severe polyostotic disease the increased blood flow can act as an arteriovenous shunt, leading to high-output heart failure or exacerbation of underlying cardiac disease.

The most dreaded complication of Paget disease is sarcoma, which occurs in less than 1% of all individuals with Paget disease, and in 5% to 10% of those with severe polyostotic disease. The sarcomas, usually osteosarcoma or fibrosarcoma, arise in Paget lesions in the long bones, pelvis, skull, and spine.

The diagnosis of Paget disease can be made radiographically. Pagetic bone is typically enlarged with thick, coarsened cortices and medulla (Fig. 26.17). Active disease has a wedge-shaped lytic leading edge that may progress along the length of the bone at a rate of 1 cm per year. Many affected individuals have elevated serum alkaline phosphatase levels, but serum calcium and phosphate levels are normal.

In the absence of malignant transformation, Paget disease is usually not serious or life-threatening. Most affected individuals have mild symptoms that are readily suppressed by treatment with calcitonin and bisphosphonates.

FRACTURES

Fractures are defined as loss of bone integrity. They are some of the most common pathologic conditions affecting bone. The following qualifiers describe fracture types and affect treatment:

- *Simple*: the overlying skin is intact.
- *Compound*: the bone communicates with the skin surface.
- *Comminuted*: the bone is fragmented.
- *Displaced*: the ends of the bone at the fracture site are not aligned.
- *Stress*: a slowly developing fracture that follows a period of increased physical activity in which the bone is subjected to repetitive loads.
- *Greenstick*: extending only partially through the bone, common in infants when bones are soft.



Figure 26.17 Severe Paget disease. The tibia is bowed. The affected portion is enlarged and sclerotic, and it exhibits irregular thickening of both cortical and cancellous bone.

- *Pathologic*: involving bone weakened by an underlying disease process, such as a tumor.

Healing of Fractures

Bone has a remarkable capacity for repair. This process involves regulated expression of a multitude of genes and can be separated into overlapping stages. Immediately after fracture, rupture of blood vessels results in a hematoma that fills and surrounds the area of injury (Fig. 26.18). The clot provides a fibrin mesh that seals the fracture site and provides a framework for the inflammatory cell influx, fibroblast ingrowth, and capillary proliferation that characterize granulation tissue. Release of PDGF, TGF- β , FGF, and other growth factors by degranulated platelets and inflammatory cells activates osteoprogenitor cells in the periosteum, medullary cavity, and surrounding soft tissues to stimulate osteoclastic and osteoblastic activity. Uncalcified tissue known as *soft tissue callus* or *procallus* forms, providing some anchorage but not structural rigidity for weight bearing.

Within 2 weeks of injury, the activated osteoprogenitor cells deposit subperiosteal trabeculae of woven bone oriented perpendicular to the cortical axis and within the medullary cavity. These processes transform the procallus into *bony callus*, which reaches maximal girth at the end of the second or third week and helps stabilize the fracture site. Activated soft tissue mesenchymal cells may also differentiate into chondrocytes that produce fibrocartilage and hyaline cartilage. Endochondral ossification creates a contiguous

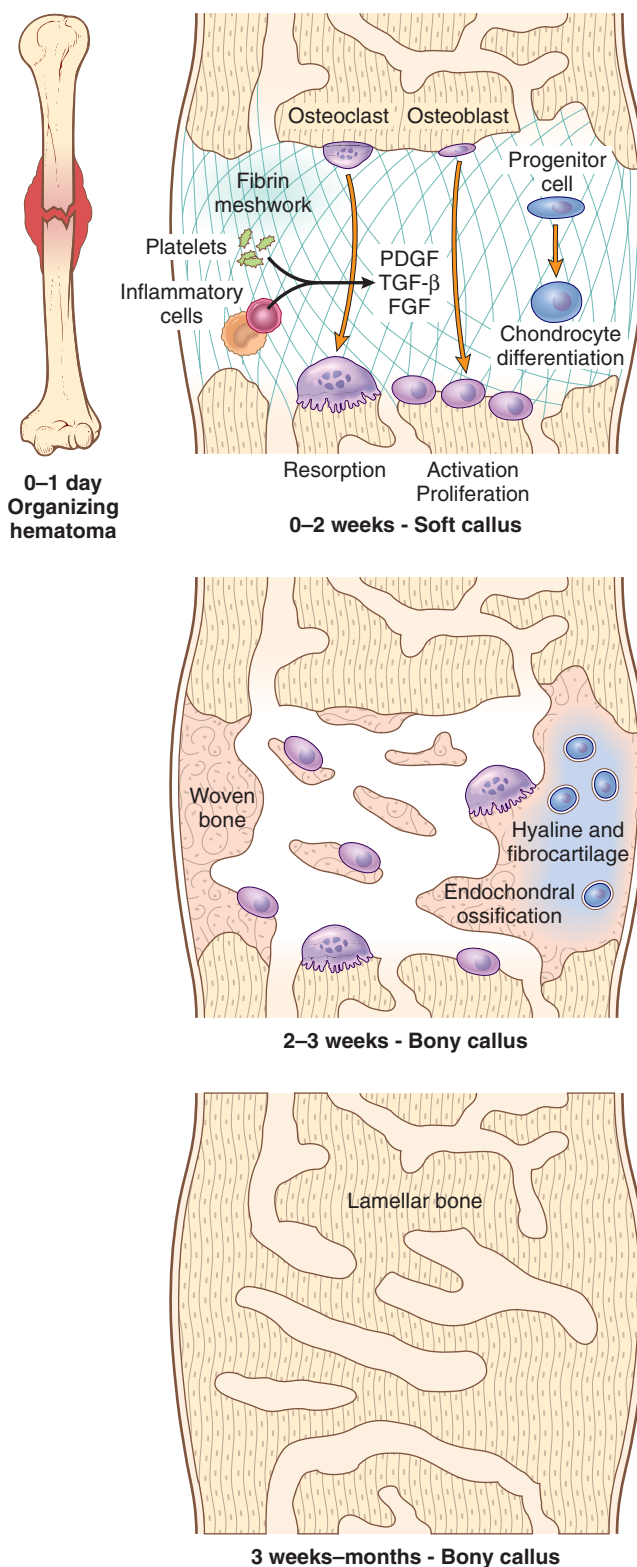


Figure 26.18 The reaction to a fracture begins with an organizing hematoma. Within 2 weeks, the two ends of the bone are bridged by a fibrin meshwork in which osteoclasts, osteoblasts, and chondrocytes differentiate from precursors. These cells produce cartilage and bone matrix, which, with adequate immobilization, remodels into normal lamellar bone.

network of bone and newly deposited bone trabeculae in the medulla and beneath the periosteum. As a result, fractured bone ends are bridged and, with progressive mineralization, the stiffness and strength of the callus increases to allow weight bearing.

In the early stages of callus formation, excess fibrous tissue, cartilage, and woven bone is produced. The portions that are not subjected to physical stress are resorbed as the callus matures, which both reduces the size of the healing bone and recreates lamellar bone. The healing process is completed by restoration of the medullary cavity.

In children and young adults, near-perfect union is the norm, although some deformity typically persists after healing of displaced and comminuted fractures. In older adults, fractures often occur in the background of other bone disorders (e.g., osteoporosis and osteomalacia). In such settings, surgical immobilization is often needed for adequate repair. Other factors may also interfere with healing. Inadequate immobilization, which permits movement of the callus and interferes with normal maturation, can result in *delayed union* or *nonunion*. If nonunion persists, the malformed callus undergoes cystic degeneration and the luminal surface can become lined by synovial-like cells, creating a false joint or *pseudoarthrosis*. Infection of the fracture site, especially common in open fractures, is another serious obstacle to healing, as are malnutrition and skeletal dysplasia.

OSTEONECROSIS (AVASCULAR NECROSIS)

Infarction of bone and marrow is relatively common. It can be limited to the medullary cavity or involve both the medulla and cortex. Fractures and corticosteroid administration are the two most common causes, but many other conditions also predispose to osteonecrosis, including alcohol abuse, bisphosphonate therapy, connective tissue disease, chronic pancreatitis, Gaucher disease, pregnancy, radiation therapy, sickle cell crisis (Chapter 14), tumors, and dysbarism (e.g., decompression sickness).

MORPHOLOGY

Medullary infarcts are geographic in shape and involve both trabecular bone and marrow. Collateral blood flow usually limits cortical involvement. In subchondral infarcts, a triangular or wedge-shaped segment with the subchondral bone plate as its base undergoes necrosis; the overlying articular cartilage remains viable due to nutrients within synovial fluid. Microscopically, dead bone is characterized by empty lacunae surrounded by necrotic adipocytes. The released fatty acids bind calcium and form insoluble calcium soaps. The remaining trabeculae act as scaffolding for deposition of new bone, while osteoclasts resorb necrotic trabeculae. The slow pace of substitution in subchondral infarcts (Fig. 26.19) results in collapse of the necrotic bone, fracture, and sloughing of the articular cartilage.

Clinical Features

Symptoms depend on location and extent of infarction. Typically, subchondral infarcts cause pain that is initially associated with activity but becomes constant as secondary



Figure 26.19 Femoral head with a subchondral, wedge-shaped pale area of osteonecrosis. The space between the overlying articular cartilage and bone is caused by trabecular compression fractures without repair.

changes supervene. Subchondral infarcts often collapse, resulting in secondary osteoarthritis. Medullary infarcts are usually small and clinically silent.

OSTEOMYELITIS

Osteomyelitis denotes inflammation of bone and marrow, virtually always secondary to infection. Osteomyelitis frequently manifests as a primary solitary focus of disease, but can also be a complication of any systemic infection. All types of organisms, including viruses, parasites, fungi, and bacteria, can produce osteomyelitis, but certain pyogenic bacteria and mycobacteria are the most common culprits. In the United States, unusual infections in immigrants from lower income countries and opportunistic infections in immunosuppressed individuals have made diagnosis and treatment of osteomyelitis challenging.

Pyogenic Osteomyelitis

Pyogenic osteomyelitis is almost always caused by bacterial infection. Organisms may reach the bone by hematogenous spread, extension from a contiguous site, or direct implantation (e.g., in traumatic injury). In otherwise healthy children, osteomyelitis is most often due to hematogenous spread and involves long bones. The initiating bacteremia may stem from seemingly trivial mucosal injuries, such as may occur during defecation or vigorous chewing of hard foods, or from minor skin infections. In adults, osteomyelitis more often occurs as a complication of open fractures, surgical procedures, and infections of the feet in the setting of diabetes. Osteomyelitis is particularly worrisome in infants, in whom epiphyseal infection can spread to a joint, resulting in septic or suppurative arthritis, articular cartilage destruction, and permanent disability.

Staphylococcus aureus is responsible for 80% to 90% of culture-positive pyogenic osteomyelitis. Bacterial cell wall proteins bind to bone matrix components, such as collagen, which facilitates adherence to bone. *Escherichia coli*,

Pseudomonas, and *Klebsiella* are more frequently isolated from individuals with genitourinary tract infections or intravenous drug users. Mixed bacterial infections typically reflect direct spread or inoculation during surgery or in open fractures. *Haemophilus influenzae* and group B streptococci are common in neonates, and sickle cell disease predisposes to *Salmonella* infection. No specific organism is identified in nearly 50% of patients.

The location of bone infections is influenced by the osseous vascular circulation, which varies with age. In neonates, metaphyseal vessels penetrate the growth plate, resulting in frequent infection of the metaphysis, epiphysis, or both. In older children, involvement of the metaphysis is typical. The epiphyses and subchondral regions are more commonly involved in adults, after growth plate closure, in which merger of metaphyseal and epiphyseal vessels provides a route for bacterial spread.

MORPHOLOGY

Changes associated with osteomyelitis vary temporally and by location. In the acute phase, bacteria proliferate and recruit neutrophils to the site. Necrosis of bone cells and marrow ensues within 48 hours. Bacteria and the associated inflammatory response then spread longitudinally and access the Haversian systems to reach the periosteum. Because the periosteum is loosely attached to the cortex in children, subperiosteal abscesses may form and extend for long distances. Associated periosteal lifting further impairs blood supply and contributes to the necrosis. Soft tissue abscesses may also form after periosteal rupture, and these can channel to the skin as draining sinuses. Dead bone, or **sequestrum**, can crumble and release fragments into the sinus tract.

As the inflammatory process evolves, chronic inflammatory cells recruited during the first week release cytokines that stimulate bone resorption, fibrous tissue ingrowth, and peripheral deposition of reactive bone. This new bone can form a living shell, or **involucrum**, around the devitalized, infected bone (Fig. 26.20).



Figure 26.20 Resected femur in a person with draining osteomyelitis. The drainage tract in the subperiosteal shell of viable new bone (involucrum) reveals the inner native necrotic cortex (sequestrum).

Clinical Features

Hematogenous osteomyelitis may present acutely as a systemic illness with malaise, fever, chills, leukocytosis, and intense throbbing pain over the infected bone. In other instances, the presentation is subtle, with only unexplained fever (infants) or localized pain (adults). The characteristic radiographic finding of a lytic focus of bone destruction surrounded by sclerosis strongly suggests osteomyelitis. Blood cultures can be positive before treatment, but biopsy and bone cultures are required to identify the pathogen in most instances. The combination of antibiotics and surgical drainage is usually curative.

Acute osteomyelitis fails to resolve and persists as a chronic infection in 5% to 25% of cases. These are typically associated with delayed diagnosis, extensive bone necrosis, inadequate antibiotic therapy or surgical debridement, or weakened host defenses. Chronic infections may be punctuated by spontaneous flares that may occur after years of dormancy. Other complications of chronic osteomyelitis include pathologic fracture, secondary amyloidosis, endocarditis, and sepsis.

Mycobacterial Osteomyelitis

Mycobacterial osteomyelitis tends to be more destructive and resistant to control than pyogenic osteomyelitis.

Mycobacterial osteomyelitis was previously limited to lower income countries, but the incidence has risen worldwide due to immigration and increased numbers of immunocompromised individuals. Risk of mycobacterial osteomyelitis is also increased in those with pulmonary or extrapulmonary tuberculosis, up to 3% of whom have osseous infection. The infection may persist for years before diagnosis and typically presents with localized pain, low-grade fever, chills, or weight loss. Infection is usually solitary in immunocompetent patients but can disseminate in the immunocompromised.

Mycobacteria are usually blood borne and originate from a focus of active visceral disease during the initial stages of primary infection. Bone involvement may occur by direct extension (e.g., from a pulmonary focus into a rib or from tracheobronchial nodes into adjacent vertebrae) or following spread via blood vessels and lymphatics. The histologic findings of granulomatous inflammation and caseous necrosis are typical of tuberculosis (Chapter 8).

Forty percent of mycobacterial osteomyelitis cases involve the spine (*Pott disease*). Infection breaks through intervertebral discs to involve multiple vertebrae and surrounding soft tissues. Discs and vertebral destruction result in compression fractures that culminate in scoliosis, kyphosis, and neurologic deficits. Tuberculous osteomyelitis may also cause tuberculous arthritis, sinus tract formation, psoas abscess, and amyloidosis.

Skeletal Syphilis

Syphilis (*Treponema pallidum*) and yaws (*Treponema pertenue*) can involve bone. Although syphilis incidence is increasing, bone involvement remains rare because diagnosis and treatment typically occur before this develops.

In congenital syphilis, bone lesions appear in the fifth month of gestation and are fully developed at birth. The spirochetes concentrate at sites of active endochondral ossification and within the periosteum to cause osteochondritis and periostitis, respectively. The characteristic *saber shin* is produced by reactive periosteal bone deposition on the medial and anterior surfaces of the tibia.

In acquired syphilis, bone disease begins early in the tertiary stage, usually 2 to 5 years after the initial infection. Bones of the nose, palate, skull, and extremities, especially the long tubular bones such as the tibia, are involved most frequently.

MORPHOLOGY

Spirochetes can be detected with silver stains or by immunohistochemistry. Edematous granulation tissue with numerous plasma cells and necrotic bone characterizes syphilitic bone disease. **Obliterative endarteritis** typically accompanies caseous necrosis as part of gummas, which may be present in congenital or acquired syphilis (Chapter 8).

BONE TUMORS AND TUMOR-LIKE LESIONS

Primary bone tumors are rare and are vastly outnumbered by metastases and hematopoietic tumors. Nevertheless, the

dismal survival of only 50% and disfiguring surgery often required for treatment make management of bone malignancies challenging. Therapy for the ~2400 new bone sarcomas diagnosed each year in the United States aims to optimize survival and maintain function of affected body parts.

Most bone neoplasms have a propensity for the long bones of the extremities. The age groups and anatomic sites affected are typical of specific tumor types. For example, osteosarcoma incidence peaks during adolescence and most frequently involves the knee. In contrast, chondrosarcoma affects the pelvis and proximal extremities of older adults.

Benign bone tumors are often asymptomatic and identified incidentally. Others, however, produce pain, cause a slow-growing mass, or produce a pathologic fracture. Radiologic imaging defines tumor location and can also detect features that narrow the differential diagnosis, but biopsy is necessary for definitive diagnosis in almost all cases.

Bone tumors are classified according to the normal cell types recapitulated or matrix produced. Lesions that do not have normal tissue counterparts are grouped according to clinicopathologic features (Table 26.4). After exclusion of hematopoietic neoplasms, the most common primary bone cancers are osteosarcoma, chondrosarcoma, and Ewing sarcoma.

Bone-Forming Tumors

Tumors in this category produce unmineralized osteoid or mineralized woven bone.

Table 26.4 Classification of Major Nonhematopoietic Primary Bone Tumors

Category and Fraction (%)	Behavior	Tumor Type	Common Locations	Age (years)	Morphology
Cartilage forming (40)	Benign	Osteochondroma	Metaphysis of long bones	10–30	Bony excrescence with cartilage cap
		Chondroma	Small bones of hands and feet	30–50	Circumscribed hyaline cartilage nodule in medulla
		Chondroblastoma	Epiphysis of long bones	10–20	Circumscribed, pericellular calcification
		Chondromyxoid fibroma	Tibia, pelvis	20–30	Collagenous to myxoid matrix, stellate cells
	Malignant	Chondrosarcoma (conventional)	Pelvis, shoulder	40–60	Extends from medulla through cortex into soft tissue, chondrocytes with increased cellularity and atypia
Bone forming (32)	Benign	Osteoid osteoma	Metaphysis of long bones	10–20	Cortical, interlacing microtrabeculae of woven bone
		Osteoblastoma	Vertebral column	10–20	Posterior elements of vertebra, histology similar to osteoid osteoma
	Malignant	Osteosarcoma	Metaphysis of distal femur, proximal tibia	10–20	Extends from medulla to lift periosteum, malignant cells producing woven bone
Unknown origin (19)	Benign	Giant cell tumor	Epiphysis of long bones	20–40	Destroys medulla and cortex, sheets of osteoclasts
		Aneurysmal bone cyst	Proximal tibia, distal femur, vertebra	10–20	Vertebral body, hemorrhagic spaces separated by cellular, fibrous septae
	Malignant	Ewing sarcoma	Diaphysis of long bones	10–20	Sheets of primitive small round cells
		Adamantinoma	Tibia	30–40	Cortical, fibrous bone matrix with epithelial islands
Notochordal (5)	Malignant	Chordoma	Clivus, sacrum	30–60	Destroys medulla and cortex, foamy cells in myxoid matrix

Modified from Unni KK, Inwards CY: *Dahlin's Bone Tumors*, ed 6, Philadelphia, 2010, Lippincott-Williams & Wilkins, p 5; by permission of Mayo Foundation.

Osteoid Osteoma and Osteoblastoma

Osteoid osteoma and osteoblastoma are benign bone-producing tumors with similar histologic features but different sizes, sites of origin, and symptoms; malignant transformation is rare. By definition, osteoid osteomas are less than 2 cm in diameter. They are most common in young men in their teens and 20s and have a predilection for the appendicular skeleton, with about 50% involving the cortex of the femur or tibia. A thick rim of reactive cortical bone may be the only radiographic clue. Despite their small size, osteoid osteomas present with severe nocturnal pain that is probably caused by prostaglandin E_2 (PGE_2) produced by the proliferating osteoblasts and is relieved by aspirin and other NSAIDs. Osteoblastomas are larger than 2 cm, typically involve the posterior spine (laminae and pedicles), cause pain that is unresponsive to aspirin, and do not induce reactive cortical bone. Osteoid osteoma can be treated by radiofrequency ablation, but osteoblastoma usually requires curettage or en bloc excision.

MORPHOLOGY

Osteoid osteoma and osteoblastoma are round-to-oval, well-circumscribed masses of hemorrhagic, gritty tan tissue. Microscopic examination demonstrates randomly interconnected trabeculae of woven bone rimmed by a single layer of prominent osteoblasts (Fig. 26.21) and surrounded by loose connective tissue with many dilated and congested capillaries. The relatively small size, well-defined margins, and benign cytologic features of the neoplastic osteoblasts help distinguish these tumors from osteosarcoma. Reactive bone encircles the actual neoplasm, or nidus, which forms a small radiolucent core that may be centrally mineralized (Fig. 26.22).

Osteosarcoma

Osteosarcoma is the most common primary malignant tumor of bone and accounts for approximately 20% of bone cancers. The age distribution of osteosarcoma is bimodal, with 75% occurring before 20 years of age. A smaller peak occurs in older adults, in whom it is frequently associated with predisposing conditions such as Paget disease, bone

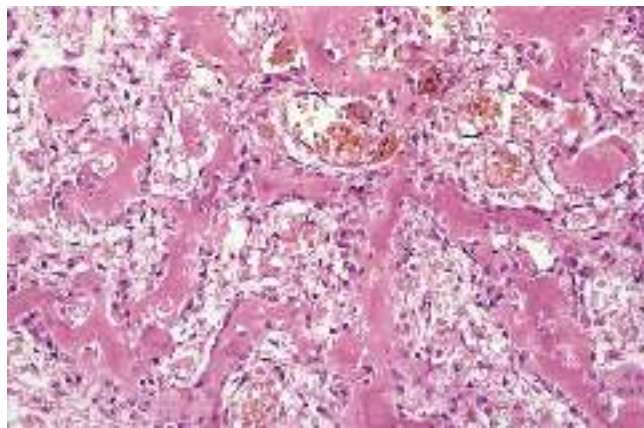


Figure 26.21 Osteoid osteoma composed of haphazardly interconnecting trabeculae of woven bone that are rimmed by prominent osteoblasts. The intertrabecular spaces are filled by vascularized loose connective tissue.

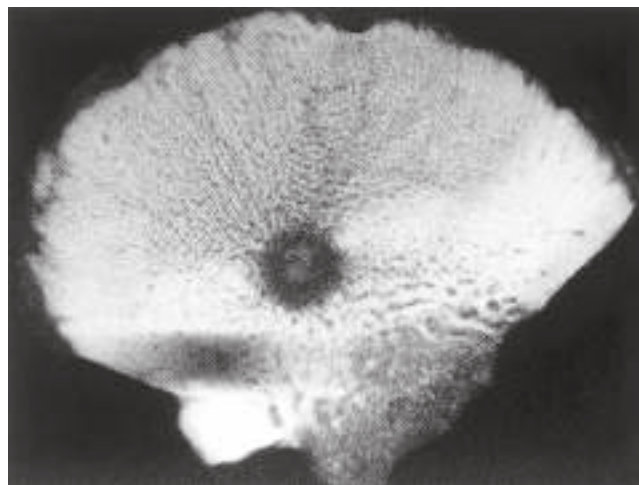


Figure 26.22 Specimen radiograph of intracortical osteoid osteoma. The round radiolucency with central mineralization represents the lesion and is surrounded by abundant reactive bone that has massively thickened the cortex.

infarcts, and prior radiation (sometimes called secondary osteosarcomas). Men are affected slightly more often than women (1.6:1). Although any bone can be involved, tumors usually arise in the metaphyseal region of the long bones; almost 50% are near the knee in the distal femur or proximal tibia.

Osteosarcomas often present with pain, sometimes due to pathologic fractures. Radiographically, the enlarging tumor forms a destructive, mixed lytic and blastic mass with infiltrative margins (Fig. 26.23). The tumor frequently breaks through the cortex and lifts the periosteum, inducing reactive periosteal bone formation. The triangular shadow between the cortex and raised periosteal ends, known radiographically as a *Codman triangle*, indicates an aggressive tumor.

Pathogenesis

The peak incidence of osteosarcoma is during the adolescent growth spurt. The tumor occurs most frequently in the growth plate of rapidly growing bones, where increased proliferation may predispose to mutations that drive oncogenesis. Approximately 70% of osteosarcomas have acquired genetic abnormalities including chromosomal aberrations. These are usually associated with mutations in well-known tumor suppressor genes and oncogenes (Chapter 7), including the following:

- *RB* mutations are present in up to 70% of sporadic osteosarcomas; germline *RB* mutations confer a 1000-fold increased risk of osteosarcoma.
- *TP53* is mutated in the germline of individuals with Li-Fraumeni syndrome, who have a greatly increased incidence of osteosarcoma. Abnormalities that interfere with p53 function are common in sporadic osteosarcomas.
- *CDKN2A* (also known as *INK4a*), which encodes two tumor suppressors, p16 and p14, is inactivated in many osteosarcomas.
- *MDM2* and *CDK4*, which inhibit p53 and *RB* function, respectively, are overexpressed in many low-grade osteosarcomas, often through chromosomal amplification of region 12q13–q15.



Figure 26.23 Distal femoral osteosarcoma with prominent bone formation extending into the soft tissues. The periosteum, which has been lifted, has laid down a proximal triangular shell of reactive bone known as a Codman triangle (arrow).

MORPHOLOGY

Osteosarcomas are bulky, gritty, gray-white tumors that often contain hemorrhage and cystic degeneration (Fig. 26.24). The tumor frequently destroys the surrounding cortices to produce soft tissue masses, spread extensively in the medullary canal, and replace hematopoietic marrow. Tumors infrequently penetrate the epiphyseal plate or enter the joint, where they may grow along tendoligamentous structures or through the attachment site of the joint capsule.

Osteosarcomas demonstrate pleomorphism, large hyperchromatic nuclei, bizarre tumor giant cells, and abundant mitoses including abnormal (e.g., tripolar) forms. Extensive necrosis and intravascular invasion are also common. Diagnosis of osteosarcoma requires the presence of malignant tumor cells producing unmineralized osteoid or mineralized bone (Fig. 26.25), which is typically fine and lacelike, but can also form broad sheets or primitive trabeculae. Neoplastic cells can also produce cartilage; if abundant, such tumors are classified as **chondroblastic osteosarcoma**.

Clinical Features

Based on their known natural history, all osteosarcoma patients are assumed to have occult metastases at the time of diagnosis. As a result, treatment generally includes neoadjuvant chemotherapy, surgery, and postoperative adjuvant chemotherapy. Chemotherapy has greatly improved osteosarcoma prognosis, with 5-year survival reaching 70%



Figure 26.24 Osteosarcoma of the proximal tibia. The tan-white tumor fills most of the medullary cavity of the metaphysis and proximal diaphysis. It has infiltrated through the cortex, lifted the periosteum, and formed soft tissue masses on both sides of the bone.

in individuals without overt metastases at initial diagnosis. Osteosarcoma metastasizes hematogenously to the lungs, bones, brain, and other sites. The outcome for those with clinically evident metastases, recurrent disease, or secondary osteosarcoma remains guarded, with a 5-year survival rate less than 20%.

Cartilage-Forming Tumors

Cartilaginous tumors account for the majority of benign and malignant primary bone tumors. They are characterized by the formation of hyaline or myxoid cartilage; fibrocartilage and elastic cartilage are rare. Benign cartilage tumors are much more common than malignant lesions.

Osteochondroma

Osteochondroma, or exostosis, is the most common benign bone tumor. It is attached to the skeleton by a bony stalk

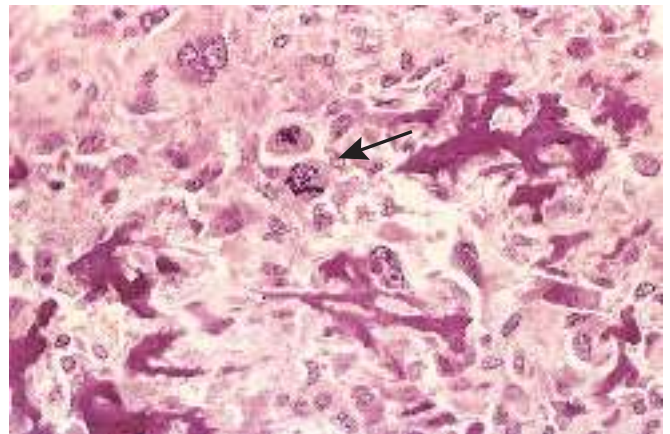


Figure 26.25 Fine, lacelike pattern of neoplastic bone produced by anaplastic malignant tumor cells in an osteosarcoma. Note the abnormal mitotic figures (arrow).

capped by cartilage. About 85% of osteochondromas are solitary and sporadic, with the remainder occurring as part of autosomal dominant *multiple hereditary exostosis syndrome*. Solitary osteochondromas are usually diagnosed in late adolescence and early adulthood, but multiple osteochondromas become apparent during childhood. Men are affected three times more often than women. Osteochondromas develop only in bones of endochondral origin. The most common site is the metaphysis near the growth plate of long tubular bones, especially near the knee, followed by the bones of the pelvis, scapula, and ribs, where they tend to have short stalks. Osteochondromas present as slow-growing masses, which can be painful if they impinge on a nerve or if the stalk is fractured. In many cases, they are detected incidentally. In multiple hereditary exostosis syndrome, the underlying bones may be bowed and shortened, reflecting an associated disturbance in epiphyseal growth.

Pathogenesis

Hereditary exostoses are associated with germline loss-of-function mutations in either the *EXT1* or the *EXT2* gene and subsequent loss of the remaining wild-type allele in chondrocytes of the growth plate. Reduced expression of *EXT1* or *EXT2* has also been observed in sporadic osteochondromas. These genes encode enzymes that synthesize heparan sulfate glycosaminoglycans. The reduced or abnormal glycosaminoglycans may prevent normal diffusion of Indian hedgehog (Ihh), a local regulator of cartilage growth, thereby disrupting Hedgehog signaling and chondrocyte differentiation.

MORPHOLOGY

Osteochondromas are sessile or pedunculated and range in size from 1 to 20 cm. The cap is composed of benign hyaline cartilage

(Fig. 26.26) and is covered by perichondrium. The cartilage has the histologic appearance of a disorganized growth plate and undergoes endochondral ossification, with newly made bone forming the inner portion of the head and stalk. The cortex of the stalk merges with the cortex of the host bone, so that the medullary cavity of the osteochondroma is in continuity with that of the bone from which it arises.

Clinical Features

Osteochondromas usually stop expanding at the time of growth plate closure and, when symptomatic, are cured by simple excision. Secondary chondrosarcoma occurs rarely, usually in tumors associated with multiple hereditary exostosis.

Chondroma

Chondroma is a benign tumor of hyaline cartilage that occurs in bones of endochondral origin. Tumors can arise within the medullary cavity, where they are termed *enchondromas*, or on the bone surface, where they are called *juxtacortical chondromas*. Enchondromas are the most common intraosseous cartilage tumor; they are typically solitary metaphyseal lesions of tubular bones of the hands and feet. Radiographically, enchondromas display a circumscribed lucency with central irregular calcifications, a sclerotic rim, and an intact cortex (Fig. 26.27). *Ollier disease* and *Maffucci syndrome* are nonhereditary disorders characterized by multiple enchondromas. Maffucci syndrome is distinguished by the presence of spindle cell hemangiomas and other noncartilage neoplasms.

Enchondromas are most often diagnosed between 20 and 50 years of age. When they involve large bones, enchondromas are usually asymptomatic and detected incidentally.

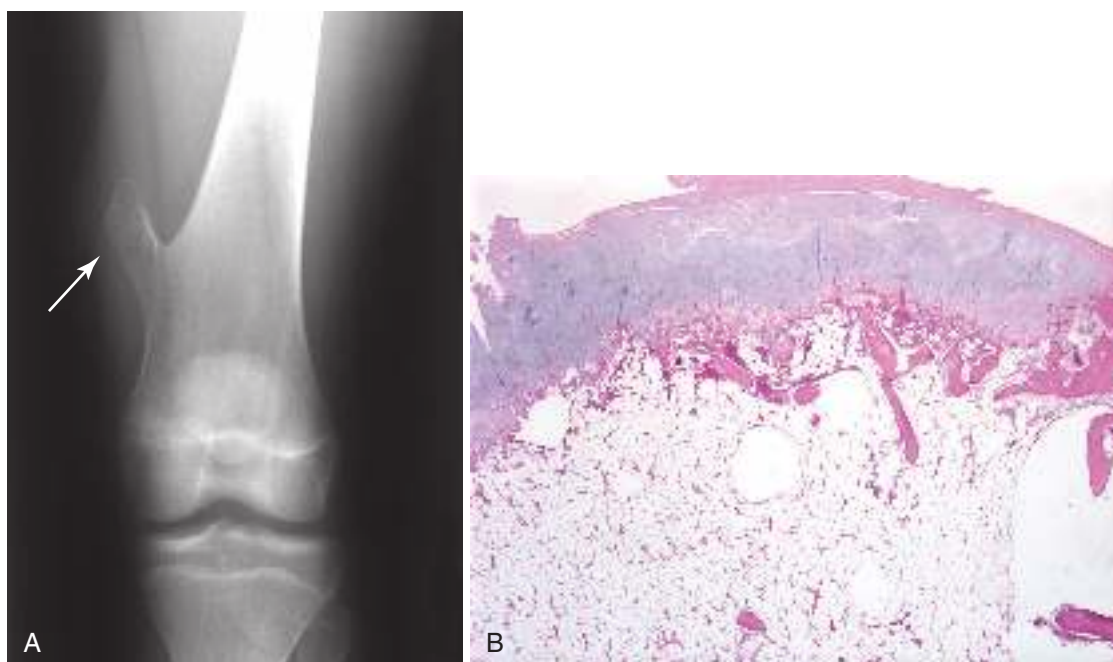


Figure 26.26 Osteochondroma. (A) Radiograph of an osteochondroma arising from the distal femur (arrow). (B) The cartilage cap has the histologic appearance of disorganized growth plate-like cartilage.



Figure 26.27 Enchondroma of the proximal phalanx. The radiolucent nodule of cartilage with central calcification thins but does not penetrate the cortex.

Occasionally, they can be painful or cause pathologic fractures. In enchondromatosis, the tumors may be numerous and large, producing severe deformities.

Pathogenesis

Heterozygous mutations in the *IDH1* and *IDH2* genes are present within enchondromas. Individuals with enchondroma syndromes are genetically mosaic, harboring *IDH* mutations in a subset of otherwise normal cells throughout their bodies. Similarly, *IDH* mutations are present in only a subset of tumor cells in both syndromic and sporadic enchondromas. This unusual situation may be explained by the functional consequences of *IDH* mutations, which cause the encoded proteins, two isoforms of the enzyme isocitrate dehydrogenase, to acquire a new enzymatic activity that leads to the synthesis of 2-hydroxyglutarate. This “oncometabolite” interferes with regulation of DNA methylation (Chapter 7). It is hypothesized that 2-hydroxyglutarate produced by *IDH*-mutated cells diffuses into neighboring cells with normal *IDH* genes, thereby causing oncogenic epigenetic changes in genetically normal neighbors, a phenomenon referred to as transformation by association.

MORPHOLOGY

Enchondromas are well-circumscribed nodules that are usually smaller than 3 cm, gray-blue, and translucent. Histologically, enchondromas are composed of hyaline cartilage containing cytologically benign chondrocytes (Fig. 26.28). Peripheral endochondral ossification may occur, and the center can calcify and infarct. The enchondromas in Ollier disease and Maffucci syndrome are sometimes more cellular than sporadic enchondromas and can exhibit cytologic atypia, making them difficult to distinguish from chondrosarcomas.

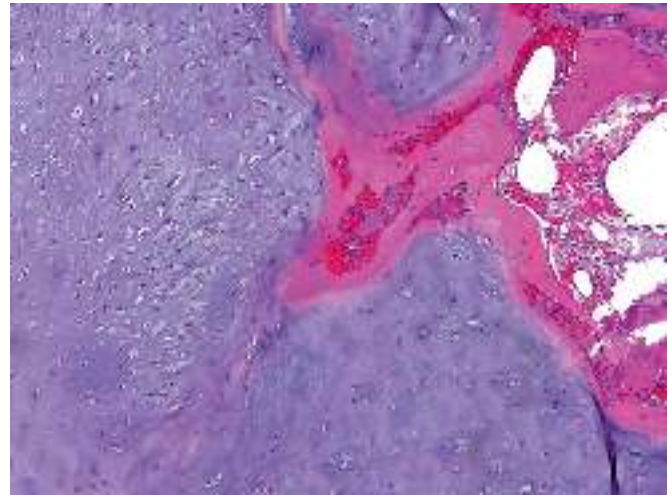


Figure 26.28 Enchondroma composed of a nodule of hyaline cartilage encased by a thin layer of reactive bone.

Clinical Features

The growth potential of chondromas is limited. Treatment depends on the clinical situation and is usually observation or curettage. Solitary chondromas rarely undergo sarcomatous transformation, but those associated with enchondromatosis do so more frequently. Individuals with Maffucci syndrome develop multiple enchondromas and spindle cell hemangiomas and are also at risk of developing other malignancies, including brain gliomas, another type of cancer associated with *IDH* gene mutations (Chapter 28).

Chondrosarcoma

Chondrosarcomas are malignant cartilage-producing tumors. They are subclassified histologically as conventional, clear cell, dedifferentiated, and mesenchymal variants. After osteosarcoma, chondrosarcoma is the second most common malignant matrix-producing tumor of bone. Individuals with chondrosarcoma are usually in their 40s or older, and men are affected twice as frequently as women; most are of the conventional histologic type. The clear cell and mesenchymal variants occur in younger people, in their teens or 20s. Chondrosarcomas commonly arise in the axial skeleton, especially the pelvis, shoulder, and ribs, but, unlike benign enchondroma, distal extremities are rarely involved. On imaging, the calcified matrix of chondrosarcomas appears as foci of flocculent densities. Slow-growing, low-grade tumors cause reactive cortical thickening, whereas aggressive high-grade neoplasms destroy the cortex and form soft tissue masses. The clear cell variant is unique in that it originates in the epiphyses of long tubular bones. About 15% of conventional chondrosarcomas are secondary and arise from enchondromas or osteochondromas.

Pathogenesis

Although chondrosarcomas are genetically heterogeneous, a few recurrent driver mutations are recognized. Chondrosarcomas arising in multiple osteochondroma syndrome exhibit mutations in the *EXT* genes, while both chondromatosis-related and sporadic chondrosarcomas may have *IDH1* or *IDH2* mutations. Silencing of the *CDKN2A* tumor suppressor

locus by DNA methylation is relatively common in sporadic tumors.

MORPHOLOGY

Conventional chondrosarcomas are large bulky tumors made up of nodules of glistening gray-white, translucent cartilage, but the matrix is often gelatinous or myxoid and can ooze from the cut surface (Fig. 26.29A). Spotty calcifications are typically present, and central necrosis may create cystic spaces. Tumor spreads through the cortex and into surrounding muscle and fat. Histologically, neoplastic cartilage infiltrates the marrow space and surrounds preexisting bony trabeculae (Fig. 26.29B). Chondrosarcomas vary in cellularity, degree of cytologic atypia, and mitotic activity and, on this basis, are graded from 1 to 3 (Fig. 26.29C).

Dedifferentiated chondrosarcoma is a low-grade chondrosarcoma with a second, high-grade component that does not produce cartilage. **Clear cell chondrosarcoma** contains sheets of large, malignant chondrocytes that have abundant clear cytoplasm, numerous osteoclast-type giant cells, and intralesional bone formation. **Mesenchymal chondrosarcoma** is composed of islands of well-differentiated hyaline cartilage surrounded by sheets of primitive appearing small round cells.

Clinical Features

Chondrosarcomas present as painful, progressively enlarging masses. The histologic grade correlates directly with biologic behavior. Most conventional chondrosarcomas are grade 1 tumors, which only rarely metastasize and have 5-year survival rates of 80% to 90%. In contrast, 70% of grade 3 tumors spread hematogenously, especially to the lungs, and 5-year survival is only 43%. Treatment of conventional chondrosarcoma is wide surgical excision, but mesenchymal and dedifferentiated tumors require excision and adjuvant chemotherapy because of their more aggressive clinical course.

Tumors of Unknown Origin

Ewing Sarcoma

Ewing sarcoma is a malignant bone tumor characterized by primitive round cells without obvious differentiation.

Ewing sarcomas account for approximately 6% to 10% of primary malignant bone tumors and follow osteosarcoma as the second most common group of bone sarcomas in children. Approximately 80% of patients are younger than 20 years of age. Boys are affected slightly more often than girls, and there is a striking predilection for Caucasians; individuals of African or Asian descent are rarely afflicted.

Ewing sarcoma usually arises in the diaphysis of long tubular bones, especially the femur and the flat bones of the pelvis and presents as a painful, enlarging mass. The affected site is frequently tender, warm, and swollen, and there may be systemic findings that mimic infection, including fever, elevated sedimentation rate, anemia, and leukocytosis. Radiographs show a destructive lytic tumor with permeative, or moth-eaten, margins that extend into surrounding soft tissues. The characteristic periosteal reaction produces layers of reactive bone deposited in an onion-skin fashion.

Pathogenesis

Greater than 90% of Ewing sarcomas contain a balanced translocation involving the *EWSR1* gene on chromosome 22; in a large majority of tumors, the other partner is the *FLI1* gene on chromosome 11, creating an *EWSR1/FLI1* fusion gene. This gene encodes a chimeric EWS/FLI1 protein that binds to chromatin and dysregulates transcription, leading to uncontrolled growth and abnormal differentiation through uncertain mechanisms. The cell of origin is not certain, but mesenchymal stem cells and primitive neuroectodermal cells are most likely.

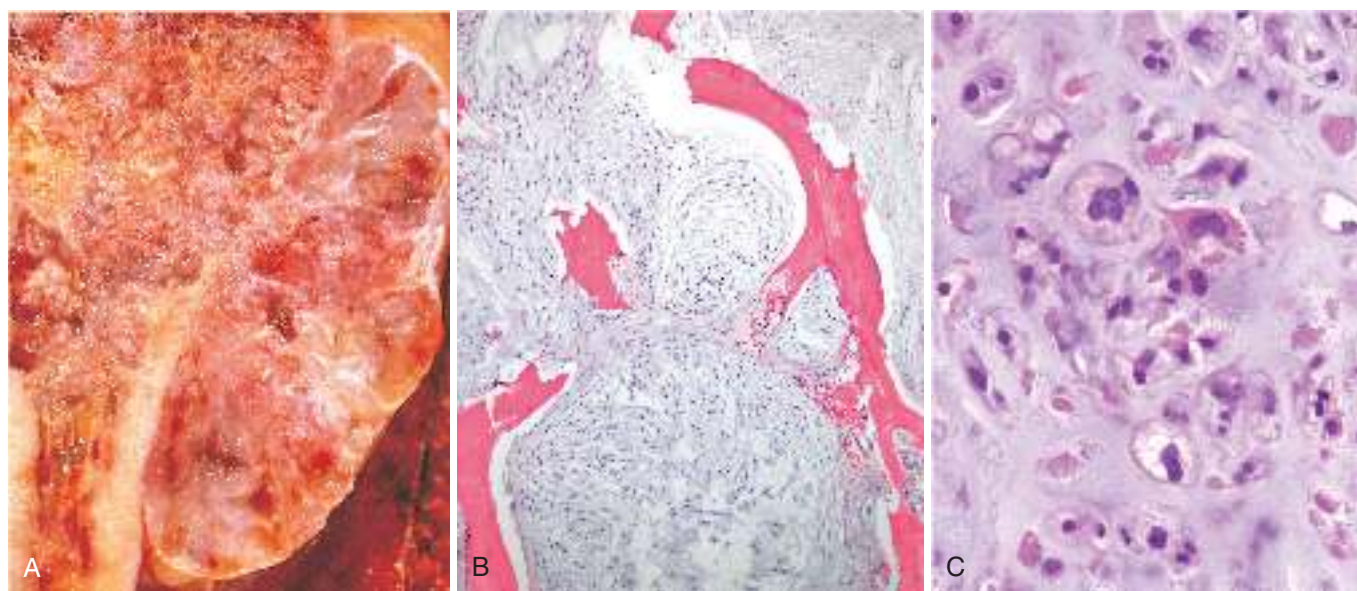


Figure 26.29 Chondrosarcoma. (A) Nodules of hyaline and myxoid cartilage permeate the medullary cavity, grow through the cortex, and form a relatively well-circumscribed soft tissue mass. (B) Conventional chondrosarcoma appearing as a hypercellular cartilaginous mass with entrapped normal bone trabeculae. (C) Anaplastic chondrocytes amid hyaline cartilage matrix in a grade 3 chondrosarcoma.

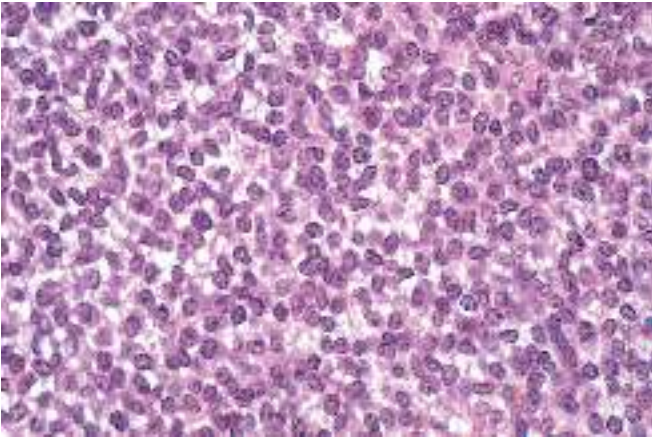


Figure 26.30 Ewing sarcoma composed of sheets of uniform, small round cells with scant clear cytoplasm.

MORPHOLOGY

Ewing sarcoma usually arises in the medullary cavity and invades the cortex, periosteum, and soft tissue. The tumor is soft, tan-white, and frequently contains areas of hemorrhage and necrosis. It is composed of sheets of uniform small, round cells with scant cytoplasm that may be clear due to abundant glycogen (Fig. 26.30). When present, **Homer-Wright rosettes**, which are rounded cell clusters with a central fibrillary core, indicate neuroectodermal differentiation. Fibrous septae may also be present, but there is little stroma. Geographic necrosis may be prominent.

Clinical Features

Although Ewing sarcoma is aggressive, neoadjuvant chemotherapy followed by surgical excision and adjuvant chemotherapy, with or without irradiation, has achieved 75% 5-year survival and 50% long-term cure rates. Chemotherapy-induced necrosis is a positive prognostic indicator.

Giant Cell Tumor

Giant cell tumor is characterized by the presence of numerous multinucleated osteoclast-type giant cells, and is also called *osteoclastoma*. Although giant cell tumors are benign, they can be locally aggressive. This uncommon tumor usually arises in the third through fifth decades of life.

Pathogenesis

Most cells within giant cell tumors are non-neoplastic osteoclasts and their precursors. The neoplastic cells are primitive osteoblast precursors that express high levels of RANKL, which in turn promotes the proliferation of osteoclast precursors and their differentiation into mature osteoclasts. The absence of normal feedback between osteoblasts and osteoclasts results in localized, but highly destructive, bone resorption. The neoplastic cells have acquired mutations in the gene encoding histone 3.3, a chromatin packaging protein; precisely how this leads to tumor formation is unknown.

Giant cell tumors develop within the epiphysis and may extend into the metaphysis. The majority are near the knee, involving the distal femur or proximal tibia, but virtually any bone can be involved. Because they typically arise near



Figure 26.31 Giant cell tumor of the proximal fibula is predominantly lytic and expansile with destruction of the cortex. A pathologic fracture is also present.

joints, giant cell tumors may cause arthritis-like symptoms. They can also present with pathologic fractures.

MORPHOLOGY

Giant cell tumors often destroy the overlying cortex, resulting in a bulging soft tissue mass delineated by a thin shell of reactive bone (Fig. 26.31). Grossly, the tumors are large, red-brown masses that frequently undergo cystic degeneration. Histologically, the tumors consist of uniform oval mononuclear tumor cells and abundant osteoclast-type giant cells with 100 or more nuclei (Fig. 26.32). Necrosis and mitotic activity may be prominent. Although reactive bone may be present, especially at the periphery, tumor cells do not synthesize bone or cartilage.

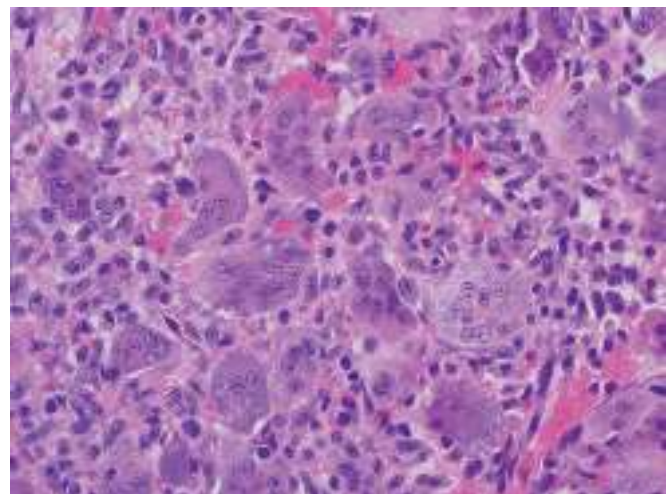


Figure 26.32 Benign giant cell tumor illustrating an abundance of multinucleated giant cells with background mononuclear stromal cells.

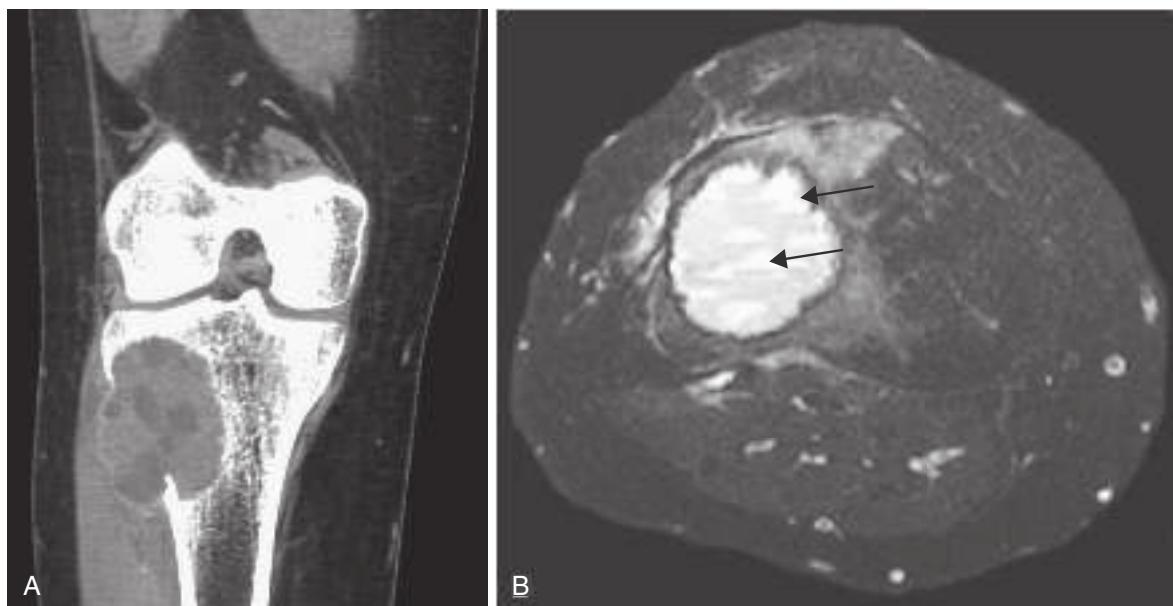


Figure 26.33 (A) Coronal computed axial tomography scan showing an eccentric aneurysmal bone cyst of the tibia. The soft tissue component is delineated by a thin rim of reactive subperiosteal bone. (B) Axial magnetic resonance image demonstrating characteristic fluid levels (arrows).

Clinical Features

Giant cell tumors are treated by curettage, but 40% to 60% recur locally. Although up to 4% metastasize to the lungs, these can regress spontaneously and are seldom fatal. The RANKL inhibitor denosumab has shown promise as an adjuvant therapy.

Aneurysmal Bone Cyst

Aneurysmal bone cyst (ABC) is characterized by multiloculated blood-filled spaces. All age groups are affected, but most cases present in adolescence. ABC develops most frequently in the femur, tibia, and vertebral body posterior elements.

Radiographically, ABC is usually an expansile, well-circumscribed lytic lesion with well-defined margins (Fig. 26.33A). Most show central lysis and a thin sclerotic “eggshell” of reactive bone at the periphery. Supporting trabeculae may create a “soap bubble” appearance on plain films and internal septae with air-fluid levels by computed tomography and magnetic resonance imaging (Fig. 26.33B). These findings are not specific, as similar radiographic (and histologic) changes can also occur as a reaction to trauma and in association with other bone tumors.

Pathogenesis

In nearly 70% of cases of ABC, chromosome 17p13 rearrangements are present within the plump spindle cells, but not in multinucleated giant cells, inflammatory cells, endothelial cells, or osteoblasts. The chromosomal rearrangement results in fusion of the *USP6* coding region with the promoters of genes that are highly expressed in osteoblasts, such as *CDH11*. *USP6* encodes a ubiquitin-specific protease that regulates the activity of the transcription factor NF- κ B, which in turn up-regulates expression of proteins, such as matrix metalloproteases, that lead to cystic bone resorption.

MORPHOLOGY

ABCs consist of multiple blood-filled cystic spaces separated by thin, tan-white septa (Fig. 26.34). The septa are composed of plump spindle cells, multinucleated osteoclast-like giant cells, and reactive woven bone lined by osteoblasts. Bone deposition typically follows the contours of fibrous septa. Approximately one-third of cases contain an unusual densely calcified, basophilic metaplastic matrix referred to as “blue bone.” Necrosis is uncommon unless a pathologic fracture is present.

Clinical Features

ABC presents with localized pain and swelling that may result in a limp, vertebral lesions, and nerve compression.

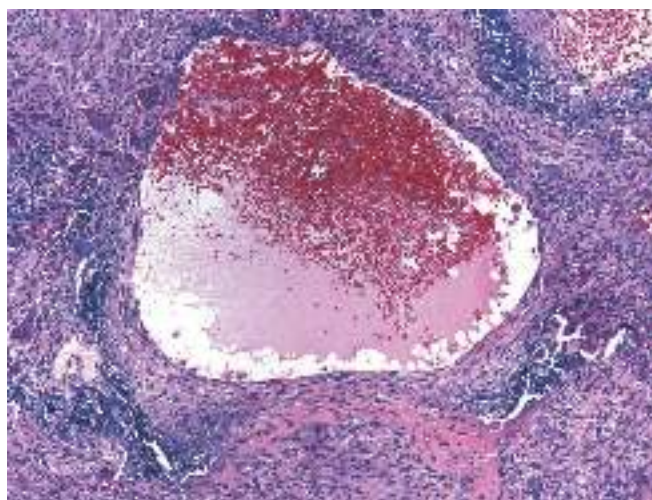


Figure 26.34 Aneurysmal bone cyst with blood-filled cystic space surrounded by a fibrous wall containing proliferating fibroblasts, reactive woven bone, and osteoclast-type giant cells.

Despite being benign, ABC is locally aggressive. Treatment of ABC is by curettage or excision. Recurrence occurs in 10% to 50% of cases.

Lesions Simulating Primary Neoplasms

Fibrous Cortical Defect and Nonossifying Fibroma

Fibrous cortical defects, also known as metaphyseal fibrous defects, are common developmental abnormalities in which fibrous connective tissue replaces bone. These lesions are present in up to 50% of children older than 2 years of age and typically present as an incidental finding in adolescents. The vast majority arise eccentrically in the metaphysis of the distal femur and proximal tibia; almost half are bilateral or multiple. Most are less than 0.5 cm in diameter, but those that grow to 5 or 6 cm are classified as *nonossifying fibromas*.

MORPHOLOGY

Both fibrous cortical defects and nonossifying fibromas are small, sharply demarcated radiolucent masses surrounded by a thin rim of sclerosis (Fig. 26.35). They are grossly yellow-brown and consist histologically of bland fibroblasts that are frequently arranged in a storiform (pinwheel) pattern and macrophages that can take the form of clustered cells with foamy cytoplasm or multinucleated giant cells (Fig. 26.36). Hemosiderin is commonly present.



Figure 26.35 Nonossifying fibroma of the distal tibial metaphysis producing an eccentric, lobulated radiolucency surrounded by a sclerotic margin.

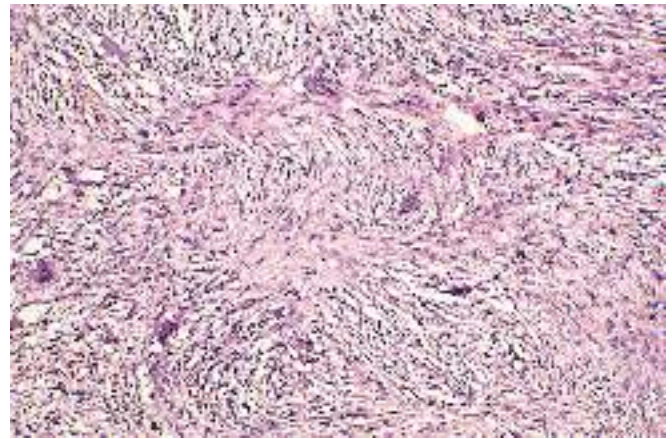


Figure 26.36 Storiform pattern created by benign spindle cells with scattered osteoclast-type giant cells characteristic of a fibrous cortical defect and nonossifying fibroma.

Clinical Features

Fibrous cortical defects are asymptomatic and are detected incidentally on radiographic studies. The radiographic findings are sufficiently specific that biopsy is rarely necessary. Most fibrous cortical defects have limited growth potential and undergo spontaneous resolution over time as they are replaced by normal cortical bone. The few that progressively enlarge into nonossifying fibromas may present with pathologic fracture or require biopsy to exclude other types of tumors. They are treated with curettage and may require bone-grafting for proper healing.

Fibrous Dysplasia

Fibrous dysplasia is a benign tumor that has been likened to localized developmental arrest; all of the components of normal bone are present, but they do not differentiate into mature structures. The lesions arise during skeletal development and appear in several distinctive but sometimes overlapping clinical patterns:

- *Monostotic*: involvement of a single bone
- *Polyostotic*: involvement of multiple bones
- *Mazabraud syndrome*: fibrous dysplasia (usually polyostotic) and soft tissue myxomas
- *McCune-Albright syndrome*: polyostotic disease, associated with café-au-lait skin pigmentations and endocrine abnormalities, especially precocious puberty

Pathogenesis

All forms of fibrous dysplasia result from somatic gain-of-function mutations in *GNAS1*, which encodes the stimulatory α subunit of G_s and is also mutated in pituitary adenomas (Chapter 24). The resulting constitutively active G_s protein promotes cellular proliferation and disturbs osteoblast differentiation. These mutations occur early in embryogenesis, and affected individuals are genetic mosaics. The phenotype depends on (1) the stage of embryogenesis when the mutation is acquired and (2) the fate of the cell harboring the mutation. At one extreme, a mutation during early embryogenesis produces McCune-Albright syndrome. In contrast, a mutation in an osteoblast precursor, during or after formation of the skeleton, results in monostotic fibrous dysplasia.

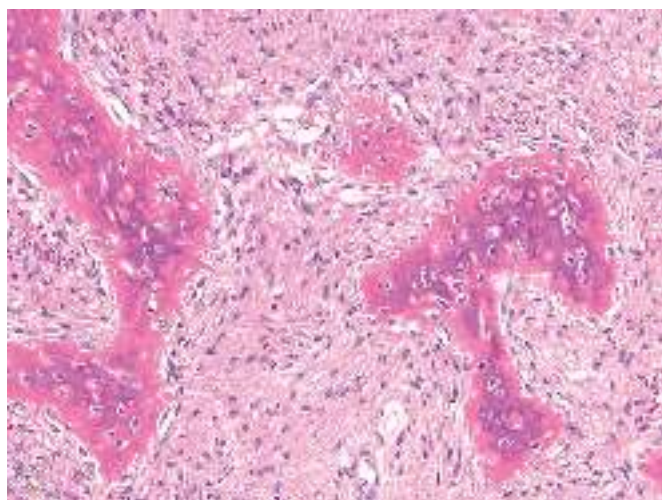


Figure 26.37 Fibrous dysplasia composed of curvilinear trabeculae of woven bone that lack conspicuous osteoblastic rimming and arise in a background of fibrous tissue.

MORPHOLOGY

The lesions of fibrous dysplasia are intramedullary lytic lesions that may expand and cause bowing and cortical thinning. Periosteal reaction usually is absent. Lesional tissue is tan-white and gritty on gross examination and is composed of curvilinear trabeculae of woven bone surrounded by a moderately cellular fibroblastic proliferation without prominent osteoblastic rimming (Fig. 26.37). Nodules of hyaline cartilage with the appearance of disorganized growth plates are present in approximately 20% of cases. Cystic degeneration, hemorrhage, and foamy macrophages are also common.

Clinical Features

Monostotic fibrous dysplasia occurs in early adolescence and often stops enlarging at the time of growth plate closure. The femur, tibia, ribs, jawbones, and calvarium are most commonly affected. The lesion is frequently asymptomatic and usually discovered incidentally but may cause pain, discrepancies in limb length, and pathologic fracture. Growth of the lesions may be reactivated during pregnancy. Symptomatic lesions are treated by curettage, but recurrence is common.

Polyostotic fibrous dysplasia manifests at a slightly earlier age than the monostotic type and may continue to cause problems into adulthood. The femur, skull, and tibia are most frequently affected. Craniofacial involvement is present in 50% of those who have a moderate number of bones affected and in 100% of those with extensive skeletal disease. Involvement of the shoulder and pelvic girdles results in progressive disease and may produce crippling deformities and fractures that require multiple corrective orthopedic surgical procedures. Bisphosphonates can be used to reduce the severity of the bone pain. A rare complication, usually in the setting of polyostotic involvement, is malignant transformation of a lesion into a sarcoma.

Mazabraud syndrome presents with skeletal features of polyostotic fibrous dysplasia in childhood followed by the appearance of intramuscular myxomas in adulthood, often in the same anatomic region as existing fibrous dysplasia. Although benign, myxomas can cause local compression symptoms or limb deformity but are cured by surgical excision.

The most common clinical presentation of McCune-Albright syndrome is precocious sexual development, primarily in girls. The syndrome can include other endocrinopathies such as hyperthyroidism, pituitary adenomas that secrete growth hormone, and primary adrenal hyperplasia. Bone lesions are often unilateral, and skin lesions, when present, are limited to the same side of the body. The cutaneous lesions are classically large macules, dark to café-au-lait in color, with irregular serpiginous borders. Skeletal manifestations are managed as for polyostotic fibrous dysplasia, while the endocrinopathies are treated medically (e.g., with aromatase inhibitors for precocious puberty).

Metastatic Tumors

Metastatic tumors are the most common form of skeletal malignancy, greatly outnumbering primary bone cancers.

The pathways of spread to bone include (1) direct extension, (2) lymphatic or hematogenous dissemination, and (3) intraspinal seeding (via the Batson plexus of veins). Any cancer can spread to bone, but in adults more than 75% of skeletal metastases originate from cancers of the prostate, breast, kidney, and lung. In children, neuroblastoma, Wilms tumor, osteosarcoma, Ewing sarcoma, and rhabdomyosarcoma are the most common metastases to bone.

Skeletal metastases are typically multifocal. However, carcinomas of the kidney and thyroid may present with solitary lesions. Most metastases involve the axial skeleton (vertebral column, pelvis, ribs, skull, and sternum) in which the marrow has a rich capillary network.

The radiographic appearance of metastases may be *lytic* (bone destroying), *blastic* (bone forming), or *mixed* lytic and blastic. Some cancers are associated with predominantly one pattern. For example, prostatic adenocarcinoma is predominantly blastic, whereas carcinomas of the kidney, lung, and gastrointestinal tract and malignant melanoma produce lytic lesions. Bidirectional interactions between metastatic cancer cells and native bone cells account for changes in the bone matrix. Tumor cells do not directly resorb bone in lytic lesions, but they secrete substances such as prostaglandins, cytokines, and PTHrP that up-regulate RANKL on osteoblasts and stromal cells, thereby stimulating osteoclast activity. Conversely, tumor cell growth is supported by the release of matrix-bound growth factors (e.g., TGF- β , IGF-1, and FGF) as bone is resorbed. Sclerotic metastases may result from tumor cells secreting WNT proteins that stimulate osteoblastic bone formation.

The presence of bone metastases portends a poor prognosis because it indicates wide dissemination of the cancer. Goals of therapy are symptomatic relief and prevention of further spread. When treated, options include systemic chemotherapy or immunotherapy, localized radiation, and bisphosphonates. Surgery may be necessary to stabilize pathologic fractures, particularly when metastases involve the spine.

KEY CONCEPTS

BONE TUMORS AND TUMOR-LIKE LESIONS

The majority of bone tumors are classified according to the normal cell they resemble or matrix produced; the remainder are grouped by clinicopathologic features. Most primary bone tumors are benign. Metastases, especially adenocarcinomas, are more common than primary bone neoplasms.

Major categories of primary bone tumors include the following:

- Bone forming: Osteoblastoma and osteoid osteoma consist of benign osteoblasts that synthesize osteoid. Osteosarcoma is a tumor of malignant osteoblasts with an aggressive clinical course that predominantly involves adolescents.

- Cartilage forming: Osteochondroma is a polypoid exostosis with a cartilage cap. Syndromic forms are most often associated with mutations in *EXT* genes. Chondromas are benign intramedullary tumors that produce hyaline cartilage and usually arise in the digits. Chondrosarcomas are malignant tumors of cartilage involving the axial skeleton in adults.
- Ewing sarcoma is an aggressive malignant small round cell tumor associated with t(11;22).
- Fibrous cortical defect and fibrous dysplasia are unusual developmental abnormalities; the latter is caused by somatic gain-of-function mutations in the *GNAS1* gene during embryogenesis.

Joints

Joints allow movement while providing mechanical stability. They are classified as solid (nonsynovial) and cavitated (synovial). Solid joints, or synarthroses, provide structural integrity, lack a joint space, and allow only minimal movement. They are grouped according to the type of connective tissue (fibrous tissue or cartilage) that bridges the ends of the bones. Fibrous synarthroses include the cranial sutures and the bonds between roots of teeth and the jawbones. Immobile cartilaginous joints, or synchondroses, include symphyses (manubriosternalis and pubic). Synovial joints, in contrast, have a joint space that allows for a wide range of motion. Situated between the ends of bones formed via endochondral ossification, they are strengthened by a dense fibrous capsule reinforced by ligaments and muscles. The boundary of the joint space consists of the synovial membrane, which is firmly anchored to the underlying capsule and does not cover the articular surface. The synovium is smooth except near the osseous insertion, where it is thrown into numerous villous folds. Synovial membranes are lined by two types of cells arranged in one to four cell deep layers. Type A synoviocytes are specialized macrophages with phagocytic activity. Type B synoviocytes are similar to fibroblasts and synthesize hyaluronic acid and various proteins. The synovial lining lacks a basement membrane, which allows for efficient exchange of nutrients, waste, and gases between blood and synovial fluid. Synovial fluid is therefore a hyaluronic acid-rich plasma filtrate that acts as a viscous lubricant and provides nutrition for the articular hyaline cartilage.

Hyaline cartilage is a unique connective tissue ideally suited to serve as an elastic shock absorber and wear-resistant surface. Hyaline cartilage is composed of water (70%), type II collagen (10%), proteoglycans (8%), and chondrocytes. The collagen fibers enable resistance to tensile stresses and transmit vertical loads; water and proteoglycans limit compression and friction. Hyaline cartilage lacks blood vessels, lymphatics, and nerves. Chondrocytes synthesize and enzymatically digest matrix, with the half-life of the different components ranging from weeks (proteoglycans) to years (type II collagen). Chondrocytes secrete degradative enzymes in inactive forms and enrich the matrix with enzyme inhibitors. Diseases that destroy articular cartilage do so by activating the degradative enzymes and decreasing the production of their inhibitors, leading to matrix breakdown. Cytokines such as IL-1 and TNF are released by chondrocytes, synoviocytes, fibroblasts, and inflammatory cells and trigger degradative processes. Articular cartilage destruction by indigenous cells contributes to pathogenesis of many joint diseases.

ARTHRITIS

Arthritis refers to inflammation of joints. The clinically most important forms of arthritis are osteoarthritis and rheumatoid arthritis, which differ in pathogenesis and clinical and pathologic manifestations (Table 26.5). Other types of arthritis

Table 26.5 Comparative Features of Osteoarthritis and Rheumatoid Arthritis

	Osteoarthritis	Rheumatoid Arthritis
Primary pathogenic abnormality	Mechanical injury to articular cartilage	Autoimmunity
Role of inflammation	May be secondary; inflammatory mediators exacerbate cartilage damage	Primary: cartilage destruction is caused by T cells and antibodies reactive with joint antigens
Joints involved	Primarily weight bearing (knees, hips)	Often begins with small joints of fingers; progression leads to multiple joints involved
Pathology	Cartilage degeneration and fragmentation, bone spurs, subchondral cysts; minimal inflammation	Inflammatory pannus invading and destroying cartilage; severe chronic inflammation; joint fusion (ankylosis)
Serum antibodies	None	Various, including ACPA, rheumatoid factor
Involvement of other organs	No	Yes (lungs, heart, other organs)

ACPA, Anti-citrullinated peptide antibody.

are caused by immune reactions, infections, and crystal deposition.

Osteoarthritis

Osteoarthritis (OA), also called degenerative joint disease, is characterized by cartilage degeneration that results in structural and functional failure of synovial joints. It is the most common type of joint disease. Although the term osteoarthritis implies an inflammatory disease, it is considered to be primarily a degenerative disease of cartilage.

In most instances, OA appears insidiously, without apparent initiating cause, as an aging phenomenon (*idiopathic* or *primary* OA). In these cases, the disease usually affects few joints (oligoarticular) but may be generalized. In about 5% of cases, OA appears in younger individuals with predisposing conditions, such as a joint deformity, a previous joint injury, or an underlying systemic disease such as diabetes, ochronosis, hemochromatosis, or marked obesity that places joints at risk. In these settings, the disease is called *secondary* OA. Gender has some influence on distribution. The knees and hands are more commonly affected in women and the hips in men. The association with aging is strong; the prevalence of OA increases exponentially beyond 50 years of age, and about 40% of people older than 70 years of age are affected.

Pathogenesis

The lesions of OA stem from **degeneration of the articular cartilage and disordered repair**. Biomechanical stress is the principal pathogenic mechanism, but genetic factors, including polymorphisms in genes encoding components of the matrix and signaling molecules, may predispose to chondrocyte injury that causes matrix alteration (Fig. 26.38). Chondrocytes proliferate and continuously synthesize proteoglycans, but disease develops when degradation exceeds synthesis. This leads to changes in proteoglycan composition as the disease progresses. Chondrocytes also secrete matrix metalloproteases (MMPs) that degrade the type II collagen network. Cytokines and diffusible factors from chondrocytes and synovial cells, particularly TGF- β (which induces the production of MMPs), TNF, prostaglandins, and nitric oxide, are also implicated in OA, and chronic, low-level inflammation contributes to disease progression. Advanced disease is characterized by chondrocyte loss and severe matrix degradation.

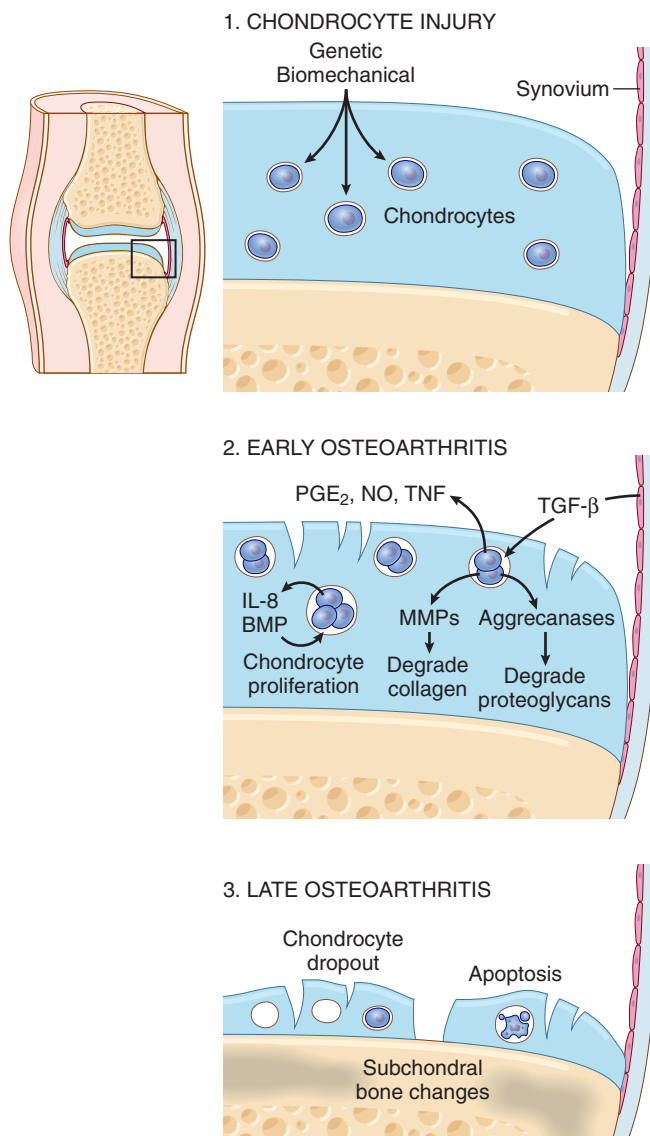


Figure 26.38 Schematic view of osteoarthritis (OA). The process is initiated by biomechanical cartilage injury (1) that can be accelerated in genetically predisposed individuals and results in changes to the extracellular matrix. (2) Although chondrocytes may proliferate and attempt to repair damaged matrix, continued degradation exceeds repair in early OA. (3) Late OA is evidenced by loss of both matrix and chondrocytes with subchondral bone damage.

MORPHOLOGY

Chondrocytes proliferate and form clusters in early stages of OA. Concurrently, matrix water content increases, proteoglycan concentration decreases, and horizontally arranged collagen type II fibers in the superficial zone are cleaved. These processes result in fissures and clefts that create a granular and soft articular surface (Fig. 26.39A). As chondrocytes die, full-thickness portions of cartilage are sloughed into the joint, forming **loose bodies**. Exposed subchondral bone becomes the new articular surface, which is burnished by friction with the opposing surface, giving it the appearance of polished ivory (**bone eburnation**) (Fig. 26.39B). Underlying articular bone undergoes rebuttoning and sclerosis, and develops small fractures, creating gaps that allow

synovial fluid to be forced into the subchondral regions. As the loculated fluid collection increases in size, fibrous-walled cysts form. Mushroom-shaped bony outgrowths (**osteophytes**) develop at the margins of the articular surface and are capped by fibrocartilage and hyaline cartilage that gradually ossify. The synovium is usually only mildly congested and fibrotic and contains few inflammatory cells.

Clinical Features

Individuals with primary osteoarthritis are usually asymptomatic until they are in their 50s. If a young person has significant manifestations of OA, a search for some underlying cause should be made. Characteristic symptoms include deep, achy pain that worsens with use, morning stiffness, crepitus, and limitation of range of movement.

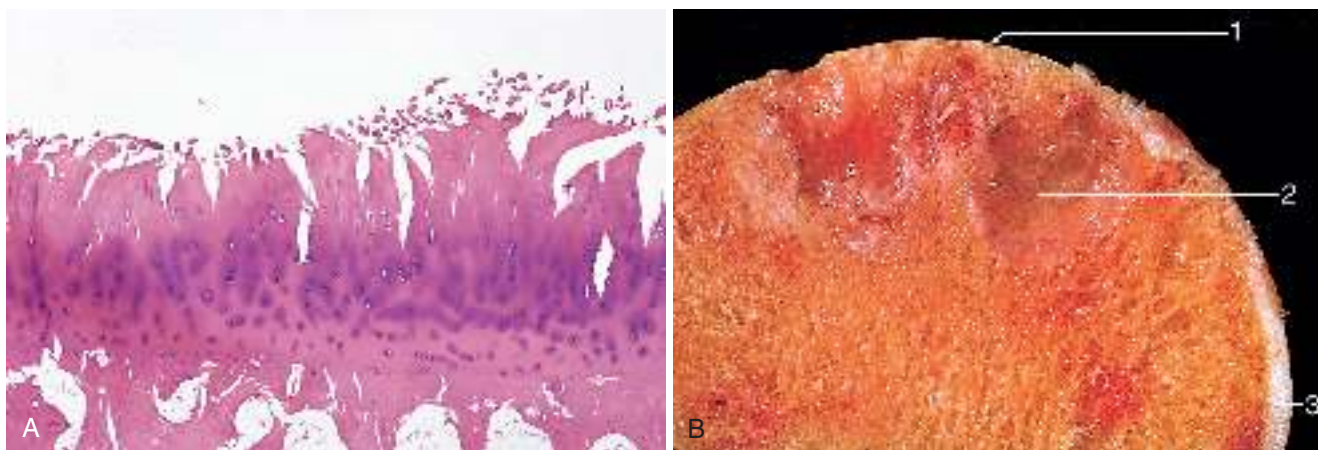


Figure 26.39 Osteoarthritis. (A) Histologic demonstration of the characteristic fibrillation of articular cartilage. (B) Eburnated articular surface exposing subchondral bone (1), subchondral cysts (2), and residual articular cartilage (3).

Impingement on spinal foramina by osteophytes results in cervical and lumbar nerve root compression and radicular pain, muscle spasms, muscle atrophy, and neurologic deficits. Typically, only one or a few joints are involved, except in the uncommon generalized variant. The joints commonly involved include the hips (Fig. 26.40), knees, lower lumbar and cervical vertebrae, proximal and distal interphalangeal joints of the fingers, first carpometacarpal joints, and first tarsometatarsal joints. *Heberden nodes*, prominent osteophytes at the distal interphalangeal joints, are common in women. Wrists, elbows, and shoulders are usually spared. Joint deformities can develop over time, but unlike rheumatoid arthritis (RA; discussed next), fusion does not occur (Fig. 26.41). The level of disease severity

detected radiographically does not correlate well with pain and disability. There are no treatments to prevent or halt progression of primary OA, and therapies include pain management, NSAIDs to reduce inflammation, intraarticular corticosteroids, activity modification, and, for severe cases, arthroplasty.

Rheumatoid Arthritis

Rheumatoid arthritis (RA) is a chronic autoimmune disorder that principally attacks the joints, producing a nonsuppurative proliferative and inflammatory synovitis. RA often progresses to articular cartilage destruction and, in some cases, joint fusion (*ankylosis*). Extraarticular lesions may involve skin, heart, blood vessels, and lungs, leading to clinical manifestations that overlap with other autoimmune disorders including systemic lupus erythematosus and scleroderma. The prevalence of RA in the United States is approximately 1%. The incidence peaks in the second to fourth decades. RA is three times more common in women than in men.

Pathogenesis

The autoimmune response in RA is initiated by CD4+ helper T cells. As in other autoimmune diseases, genetic predisposition and environmental factors contribute to the development, progression, and persistence of disease. The pathologic changes in RA are mediated by antibodies against self antigens and inflammation induced by cytokines, predominantly those secreted by CD4+ T cells (Fig. 26.42).

The activated CD4+ T cells release inflammatory mediators that stimulate other inflammatory cells, leading to tissue injury. Although many cytokine mediators can be isolated from inflamed joints, the most important ones include:

- IFN- γ from Th1 cells, which activates macrophages and resident synovial cells.
- IL-17 from Th17 cells that recruits neutrophils and monocytes.
- RANKL, which is expressed on activated T cells and stimulates bone resorption.
- TNF and IL-1 from macrophages stimulate resident synovial cells to secrete proteases that destroy hyaline cartilage.



Figure 26.40 Severe osteoarthritis of the hip. The joint space is narrowed, and there is subchondral sclerosis with scattered oval radiolucent cysts and peripheral osteophyte lipping (arrows).

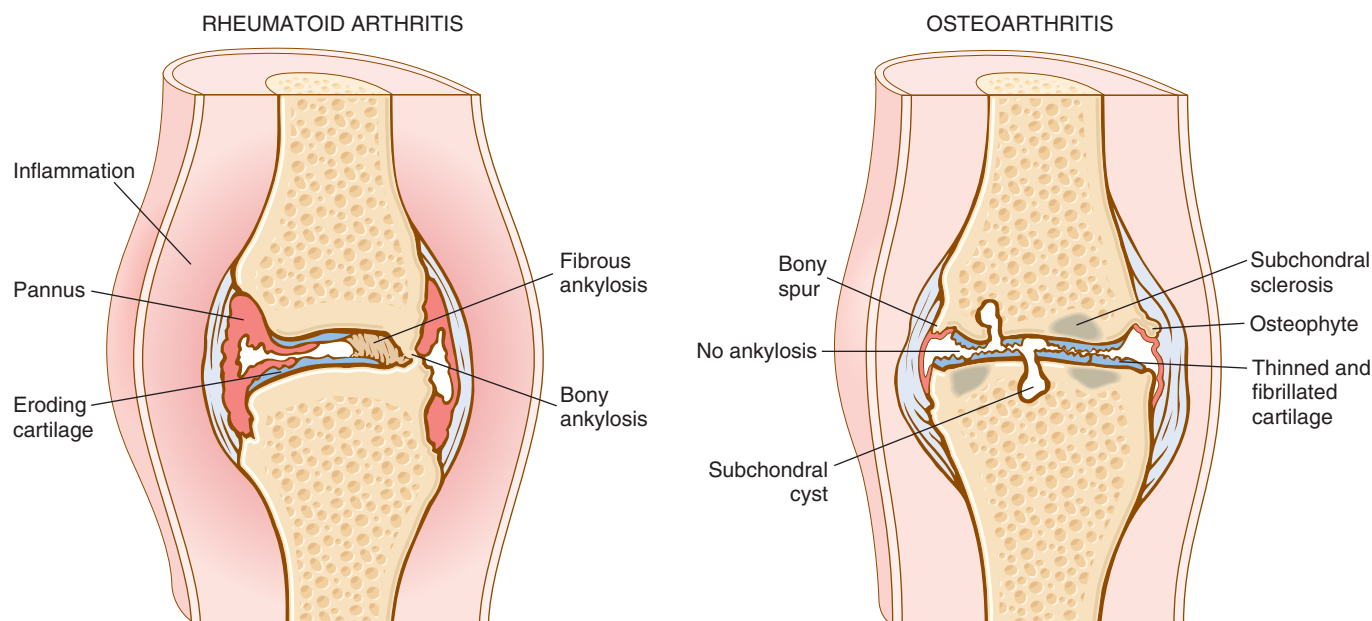


Figure 26.41 Comparison of the morphologic features of rheumatoid arthritis and osteoarthritis.

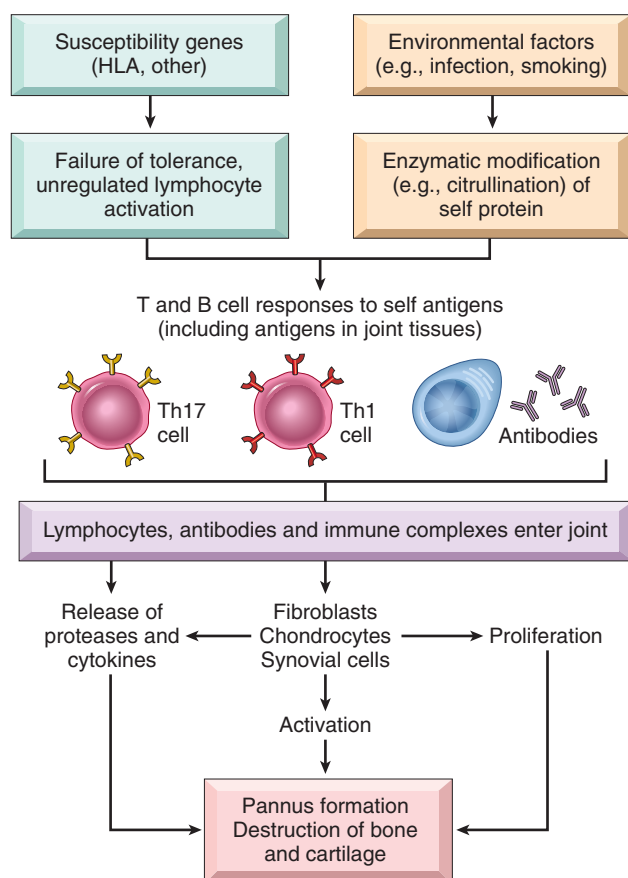


Figure 26.42 Major processes involved in the pathogenesis of rheumatoid arthritis.

Of these, TNF has been most firmly implicated in RA pathogenesis, and anti-TNF biologics have revolutionized its treatment (see later).

In RA, the synovium often contains germinal centers with secondary follicles and abundant plasma cells. Some of these plasma cells secrete antibodies that recognize self antigens. Many of these autoantibodies, which are produced in lymphoid organs as well as synovium, are specific for citrullinated peptides (CCPs) in which arginine residues are post-translationally converted to citrulline. The modified epitopes are found in several proteins found in joints, including fibrinogen, type II collagen, α -enolase, and vimentin. *Anti-citrullinated peptide antibodies* (ACPAs) are diagnostic markers that can be detected in serum of up to 70% of RA patients. Some data suggest that, in combination with T cells reactive to citrullinated proteins, ACPAs drive disease persistence. IgM and IgA autoantibodies that bind to IgG Fc regions are present in 80% of individuals. These autoantibodies, collectively referred to as *rheumatoid factor*, may be deposited in joints as immune complexes, but are not present in all patients and can be detected in some individuals without RA.

It is estimated that 50% of the risk of developing RA is related to inherited genetic susceptibility. The *HLA-DR4* allele is associated with ACPA-positive RA. Evidence suggests that an epitope on a citrullinated protein, vinculin, mimics an epitope on many microbes and can be presented by the class II HLA-DR4 molecule.

Environmental factors that promote autoimmunity are involved, but as in many autoimmune diseases, they have not been clearly defined. For example, infection (including periodontitis) and smoking may promote citrullination of self proteins, creating new epitopes that trigger autoantibody production.

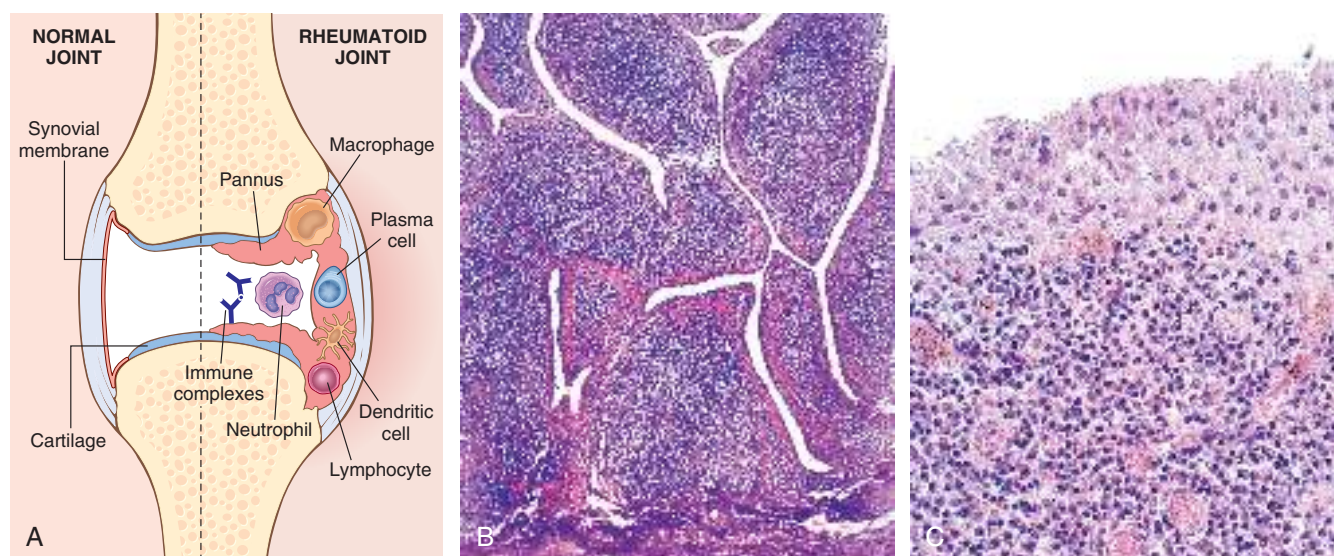


Figure 26.43 Rheumatoid arthritis. (A) Schematic view of the joint lesion. (B) Low magnification reveals marked synovial hypertrophy with formation of villi. (C) At higher magnification, subsynovial tissue containing a dense lymphoid aggregate. (A, Modified from Feldmann M: Development of anti-TNF therapy for rheumatoid arthritis, *Nat Rev Immunol* 2(5):364–371, 2002.)

MORPHOLOGY

RA typically affects small joints of the hands and feet. The synovium becomes grossly edematous, thickened, and hyperplastic, transforming its smooth contour to one covered by delicate and bulbous villi (Fig. 26.43A–B). The characteristic histologic features include (1) synovial cell hyperplasia and proliferation; (2) dense inflammatory infiltrates (frequently forming lymphoid follicles) of CD4+ helper T cells, B cells, plasma cells, dendritic cells, and macrophages (see Fig. 26.43); (3) increased vascularity due to angiogenesis; (4) fibrinopurulent exudate on the synovial and joint surfaces; and (5) osteoclastic activity in subchondral bone, allowing the inflamed synovium to penetrate the bone and cause periarticular erosions and subchondral cysts. Together, these changes produce the **pannus**, a mass of edematous synovium, inflammatory cells, granulation tissue, and fibroblasts that grows over and causes erosion of articular cartilage. In time, after the cartilage has been destroyed, the pannus bridges the apposing bones to form a **fibrous ankylosis**, which eventually ossifies and results in bone fusion, or bony ankylosis.

Rheumatoid nodules are an infrequent manifestation of RA and typically occur in the subcutaneous tissue of the forearm, elbows, occiput, and lumbosacral area. These small masses are firm, nontender, and round to oval. Microscopically, they resemble necrotizing granulomas with a central zone of fibrinoid necrosis surrounded by a prominent rim of activated macrophages and numerous lymphocytes and plasma cells (Fig. 26.44). Severe disease may be associated with **leukocytoclastic vasculitis** (Chapter 11), an acute necrotizing vasculitis of small and large arteries that may involve pleura, pericardium, or lung and evolve into a chronic fibrosing process. Leukocytoclastic vasculitis produces purpura, cutaneous ulcers, and nail bed infarction. Ocular changes such as uveitis and keratoconjunctivitis (similar to Sjögren syndrome, Chapter 6) may also occur.

Clinical Features

RA may be distinguished from other forms of polyarticular inflammatory arthritis serologically by ACPA detection and radiographically by characteristic changes (see later). It begins with malaise, fatigue, and generalized musculoskeletal pain in about half of patients; joint involvement develops after weeks to months. The pattern of joint involvement varies, but it is generally symmetrical and affects small joints before larger ones. Symptoms usually develop in the hands and feet, followed in decreasing frequency by the wrists, ankles, elbows, and knees. In the hands, the metacarpophalangeal and proximal interphalangeal joints are involved, in contrast to OA (see earlier).

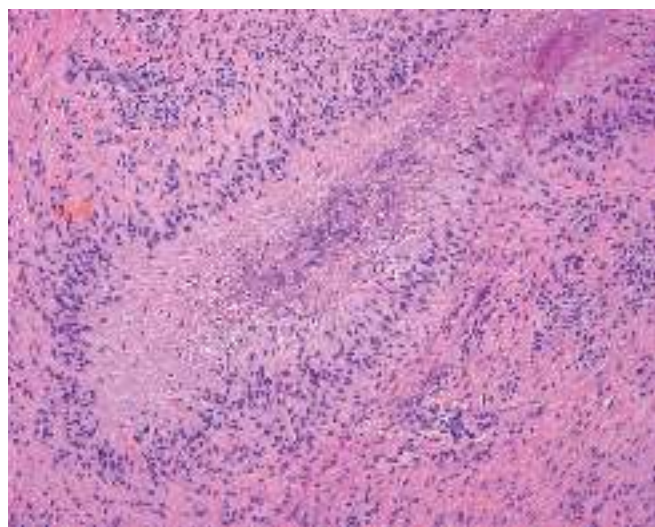


Figure 26.44 Rheumatoid nodule composed of central necrosis rimmed by palisaded macrophages.

Involved joints are swollen, warm, and painful. Unlike OA, morning joint stiffness does not subside with activity. The typical patient has progressive joint enlargement and decreasing range of motion during a chronic waxing and waning course. In a minority of individuals, especially those lacking ACPA and rheumatoid factor, disease may stabilize or even regress. A small number of patients (about 10%) develop acute-onset RA, with severe symptoms and polyarticular involvement that develops over several days.

Inflammation in tendons, ligaments, and occasionally adjacent skeletal muscle frequently accompanies the arthritis and produces characteristic radial deviation of the wrist, ulnar deviation of the fingers, and flexion-hyperextension of the fingers (swan-neck and boutonnière deformities). Radiographic hallmarks include joint effusions and juxta-articular osteopenia with erosions and narrowing of the joint space and loss of articular cartilage (Fig. 26.45).

RA treatment is aimed at relieving pain and inflammation and slowing or arresting joint destruction. Therapies include corticosteroids, other immunosuppressants such as methotrexate, and, most notably, TNF antagonists, which are effective in many patients. Unfortunately, individuals must be maintained on anti-TNF therapy to avoid disease flares.



Figure 26.45 Rheumatoid arthritis of the hand. Characteristic features include diffuse osteopenia, periarticular bony erosions, and marked loss of the joint spaces of the carpal, metacarpal, phalangeal, and interphalangeal joints.

This long-term treatment with anti-TNF agents predisposes individuals to infection with opportunistic organisms such as *M. tuberculosis*. Other biologic agents that interfere with T- and B-lymphocyte responses have also been approved for therapeutic use.

Juvenile Idiopathic Arthritis

Juvenile idiopathic arthritis (JIA) is a heterogeneous group of disorders of unknown cause that present with arthritis before 16 years of age and persist for at least 6 weeks. In the United States, 30,000 to 50,000 individuals are affected. Relative to RA, JIA is more commonly associated with oligoarthritis, systemic disease, and involvement of large joints. Antinuclear antibody (ANA) seropositivity is typical, but rheumatoid nodules are usually absent. JIA pathogenesis is undefined, but, it shares features with adult RA, including associations with certain HLA and PTPN22 gene variants and studies that link its pathogenesis to involvement of Th1 and Th17 cells and inflammatory mediators such as IL-1, IL-17, TNF, and IFN- γ .

Current subclassification of JIA is based on clinical (e.g., oligoarticular, systemic) and laboratory (ANA, rheumatoid factor titers) features. Some subgroups (e.g., systemic, polyarticular rheumatoid factor-positive; enthesitis-related, which refers to involvement of sites of ligament and cartilage insertion into bone [entheses]; oligoarticular) are similar, while others (e.g., polyarticular rheumatoid factor-negative; psoriatic) are highly heterogeneous. Treatment of all subgroups is similar to adult RA, and there has been some success using a systemic IL-6 receptor antibody. Long-term JIA prognosis is variable; many patients have chronic disease, but only about 10% develop serious functional disability.

Seronegative Spondyloarthropathies

Spondyloarthropathies are a heterogeneous group of disorders unified by the following features:

- Absence of rheumatoid factor
- Pathologic changes in the ligamentous attachments (i.e. entheses) rather than synovium
- Sacroiliac joint involvement
- Association with HLA-B27
- Bony proliferation leading to ankylosis

The manifestations are immune mediated and triggered by T-cell responses, presumably against undefined microbial antigens, that cross-react with musculoskeletal system components.

Ankylosing Spondylitis

Ankylosing spondylitis causes destruction of articular cartilage and bony ankyloses, especially of the sacroiliac and vertebral apophyseal joints between tuberosities and processes. The disease presents as lower back pain and spinal immobility, usually in the second and third decades of life. Peripheral joints, such as the hips, knees, and shoulders, are involved in at least one-third of cases. Approximately 90% of patients are HLA-B27 positive. The role of HLA-B27 is unknown; it is presumably related to the ability of this MHC variant to present one or more antigens that

somehow trigger the disease, but neither the antigen nor the pathogenic immune cell is known.

Reactive Arthritis

Reactive arthritis is defined by the triad of arthritis, nongonococcal urethritis or cervicitis, and conjunctivitis following an infection. More recently, the definition has been broadened to include mono- or oligoarticular arthritis that occurs days to weeks after genitourinary (*Chlamydia*) or gastrointestinal (*Shigella*, *Salmonella*, *Yersinia*, *Campylobacter*, and *Clostridioides difficile*) infections. HIV-positive patients may also be affected; the reactive arthritis in these individuals is thought to be caused by pathogens other than HIV. HLA-B27 is common in patients with reactive arthritis. The disease most often presents in young adults, with a global prevalence of approximately 30 per 100,000 adults. Arthritic episodes usually wax and wane for about 6 months, but nearly 50% of patients have recurrent arthritis, tendonitis, and lumbosacral pain. When active disease persists for more than 6 months, the term chronic reactive arthritis is used. Spondyloarthritis can occur in these patients, and spine involvement can be indistinguishable from ankylosing spondylitis.

Patients initially experience an acute-onset asymmetric oligoarthritis, often affecting the lower extremities, especially the knee. Inflammatory low back pain is a common accompanying symptom but is rarely the only symptom. Enthesitis occurs frequently and often involves the insertions of the Achilles tendon and plantar fascia on the calcaneus, resulting in swelling of the heel. *Dactylitis*, or inflammation of the digits, is present in a minority of patients and typically presents as sausage-like digits, joint stiffness, and low back pain. Synovitis of a digital tendon sheath produces “sausage” finger or toe. This may progress to ossification at tendoligamentous insertion sites and development of calcaneal spurs and bony outgrowths in those with chronic reactive arthritis. Extraarticular involvement includes conjunctivitis and uveitis; dysuria, pelvic pain, urethritis, balanitis, cervicitis, mucosal ulcers, skin rashes, and other cutaneous manifestations (e.g., psoriasis-like nail changes); and cardiac valvular disease (e.g., aortic insufficiency).

Infectious Arthritis

Microorganisms of all types can seed joints during hematogenous dissemination. Articular structures can also become infected by direct inoculation or contiguous spread from a soft tissue abscess or focus of osteomyelitis. Because cartilage, unlike bone, has a limited regenerative capacity, the rapid joint destruction that may ensue can result in permanent deformities.

Suppurative Arthritis

Bacterial infections that cause acute suppurative arthritis usually enter the joints from distant sites by hematogenous spread. In neonates, contiguous spread from underlying epiphyseal osteomyelitis is relatively common. *H. influenzae* arthritis predominates in children younger than 2 years of age. *S. aureus* is the main causative agent in older children and adults, while gonococcus is prevalent during late adolescence and young adulthood. Individuals with sickle cell disease are prone to *Salmonella* infection at any age.

Except for gonococcal arthritis, which is seen mainly in women, joint infections are equally common in males and females. Individuals with deficiencies of components of the complement membrane attack complex (C5–C9) are especially susceptible to disseminated gonococcal infections and arthritis. Other predisposing conditions include immunodeficiencies (congenital and acquired), debilitating illness, joint trauma, chronic arthritis of any cause, and intravenous drug use.

The classic presentation is the sudden development of an acutely painful and swollen joint with a restricted range of motion. Fever, leukocytosis, and elevated sedimentation rate are common. In disseminated gonococcal infection, the symptoms are often subacute, and the infection usually involves only a single joint, most commonly the knee, hip, shoulder, elbow, wrist, or sternoclavicular joints. Axial joint involvement is most frequent in drug users. Joint aspiration is diagnostic if it yields purulent fluid in which the infectious organism can be identified. Prompt recognition and effective antimicrobial therapy can prevent joint destruction.

Mycobacterial Arthritis

Mycobacterial arthritis is a chronic progressive monoarticular infection caused by *M. tuberculosis*. It primarily occurs in adults as a complication of adjacent osteomyelitis or after hematogenous dissemination from a visceral (usually pulmonary) site of infection. Onset is insidious and associated with gradually increasing pain. Systemic symptoms may or may not be present. Mycobacterial seeding of the joint induces formation of confluent granulomas with caseous necrosis. The synovium may grow as a pannus over the articular cartilage and erode the bone along the joint margins. Chronic disease results in fibrous ankylosis and obliteration of the joint space. Weight-bearing joints, especially the hips, knees, and ankles, in descending order of frequency, are most commonly affected.

Viral Arthritis

Arthritis can occur with a variety of viral infections. The most common are alphavirus, parvovirus B19, rubella, Epstein-Barr virus, and hepatitis B and C viruses. The manifestations range from acute to subacute arthritis. Joint symptoms may be caused by direct infection of the joint by the virus, as in rubella and some alphavirus infections, or by an autoimmune reaction triggered by the infection.

Lyme Arthritis

Lyme arthritis is caused by the spirochete *Borrelia burgdorferi*, which is transmitted by deer ticks of the *Ixodes ricinus* complex. It is the leading arthropod-borne disease in the United States. It most often occurs in New England, the mid-Atlantic states, and the upper Midwest, but its geographic distribution is expanding. In the classic form, Lyme disease involves multiple organ systems through three clinical phases (Chapter 8). The initial skin infection (*early localized stage*) is followed within days or weeks by dissemination of the organism to other cutaneous sites, cranial nerves, heart, and meninges (*early disseminated stage*). If left untreated, arthritis (*late disseminated stage*) occurs months after infection.

Currently, arthritis occurs in less than 10% of cases of *Borrelia* infection because most individuals are treated and

cured at an earlier stage. However, if left untreated, up to 80% of individuals develop a migratory arthritis (*Lyme arthritis*) lasting weeks to months. Large joints, especially the knees, shoulders, elbows, and ankles, in descending order of frequency, are usually involved. Initial attacks last for a few weeks to months, and are usually limited to one or two joints at a time. Arthritis may subsequently develop at new sites. Spirochetes are only identified in 25% of those with arthritis, but serologic detection of anti-*Borrelia* antibodies is diagnostic. Histologically, infected synovium exhibits chronic synovitis, synoviocyte hyperplasia, fibrin deposition, mononuclear cell infiltrates (especially CD4+ T cells), and *obliterative endarteritis*. In severe cases, the histopathology mimics rheumatoid arthritis.

Lyme disease treatment relies on antibiotics active against *Borrelia* and results in cure rates of 90%. Although an effective vaccine was briefly available, it was withdrawn from the market as a result of negative media reports, coverage of only the most common North American species, lack of safety data in children, and other concerns.

Chronic, antibiotic-refractory arthritis can develop in the late disseminated stage of Lyme disease. In many of these cases, *Borrelia* cannot be detected in the joint fluid, even by polymerase chain reaction. Some have hypothesized that this antibiotic-refractory arthritis is an autoimmune disease triggered by immune responses to *Borrelia* outer surface protein A. Chronic disease is also associated with nonspecific symptoms such as fatigue and cognitive complaints collectively termed *posttreatment Lyme disease syndrome*.

Crystal-Induced Arthritis

Articular crystal deposits are associated with a variety of acute and chronic joint disorders. Endogenous crystals include monosodium urate (*gout*), calcium pyrophosphate dihydrate (*pseudogout*), and basic calcium phosphate. Exogenous crystals, such as the biomaterials used in prosthetic joints, can also induce arthritis as they accumulate with wear. Endogenous and exogenous crystals produce disease by triggering inflammatory reactions that destroy cartilage.

Gout

Gout is marked by transient attacks of acute arthritis initiated by monosodium urate crystals deposited within and around joints. Whether gout is primary or secondary to another underlying disease (Table 26.6), the common feature is excessive uric acid in the tissues and body fluids. In the primary form (90% of cases), gout is the major manifestation of the disease; the cause is unknown.

Pathogenesis

Hyperuricemia (plasma urate above 6.8 mg/dL) is necessary, but not sufficient, for development of gout. Elevated uric acid can result from overproduction, reduced excretion, or both (Table 26.6). Uric acid levels are determined by several factors:

- **Uric acid production.** Urate synthesis is the end product of purine catabolism. Purines are themselves the product of two interlinked pathways: the *de novo* pathway, in which purine nucleotides are synthesized from nonpurine precursors, and the salvage pathways, in which nucleotides

Table 26.6 Classification of Gout

Clinical Category	Uric Acid Production	Uric Acid Excretion
Primary Gout (90%)		
Unknown enzyme defects (85%–90%)	Normal ↑	↓ Normal
Known enzyme defects (e.g., partial HGPRT deficiency)	↑	Normal
Secondary Gout (10%)		
Increased nucleic acid turnover (e.g., leukemia)	↑↑	↑
Chronic renal disease	Normal	↓
Congenital (e.g., Lesch-Nyhan syndrome, HGPRT deficiency)	↑↑	↑

HGPRT, Hypoxanthine guanine phosphoribosyl transferase.

are synthesized from free purine bases in the diet and those generated by purine nucleotide catabolism.

- **Uric acid excretion.** In the kidney, uric acid is filtered by the glomerulus, but is almost completely resorbed by the proximal tubule. The small fraction of total uric acid in the urine is the result of secretion in the distal nephron.

Asymptomatic hyperuricemia appears around puberty in males and after menopause in females. In primary gout, hyperuricemia is usually due to reduced excretion. The mechanism is not known, but genome-wide association studies have identified polymorphisms in genes including *URAT1*, *GLUT9*, and *KCNQ1*, which are involved in transport and homeostasis of urate and other ions. A small fraction of primary gout cases are caused by uric acid overproduction due to identifiable enzymatic defects. For example, partial deficiency of hypoxanthine guanine phosphoribosyl transferase (HGPRT) interrupts the salvage pathway, so purine metabolites cannot be salvaged and are, instead, degraded into uric acid. Complete absence of HGPRT also results in hyperuricemia, but the significant neurologic manifestations of this condition (*Lesch-Nyhan syndrome*) dominate the clinical picture, which is therefore classified as secondary gout. Secondary gout can also be caused by increased production (e.g., rapid cell lysis during chemotherapy for leukemia) or decreased excretion (chronic renal disease).

Inflammation is triggered by precipitation of monosodium urate crystals into the joints, which result in the production of cytokines that recruit leukocytes (Fig. 26.46). Macrophages and neutrophils phagocytose the urate crystals. This activates the inflammasome (Chapter 3) and leads to the secretion of cytokines, including IL-1, which promote accumulation of more neutrophils and macrophages within the joint. A vicious cycle develops as newly recruited inflammatory cells release cytokines, free radicals, proteases, and arachidonic acid metabolites to recruit even more leukocytes. Complement activation by the alternative pathway may also contribute to leukocyte recruitment. Rupture of phagolysosomes induced by ingested urate crystals can cause further release of proteases and inflammatory mediators. The resulting acute arthritis typically remits spontaneously in days to weeks. Repeated attacks of acute arthritis lead eventually to the formation of *tophi*, aggregates of urate crystals and inflammatory tissue in the inflamed synovial membrane

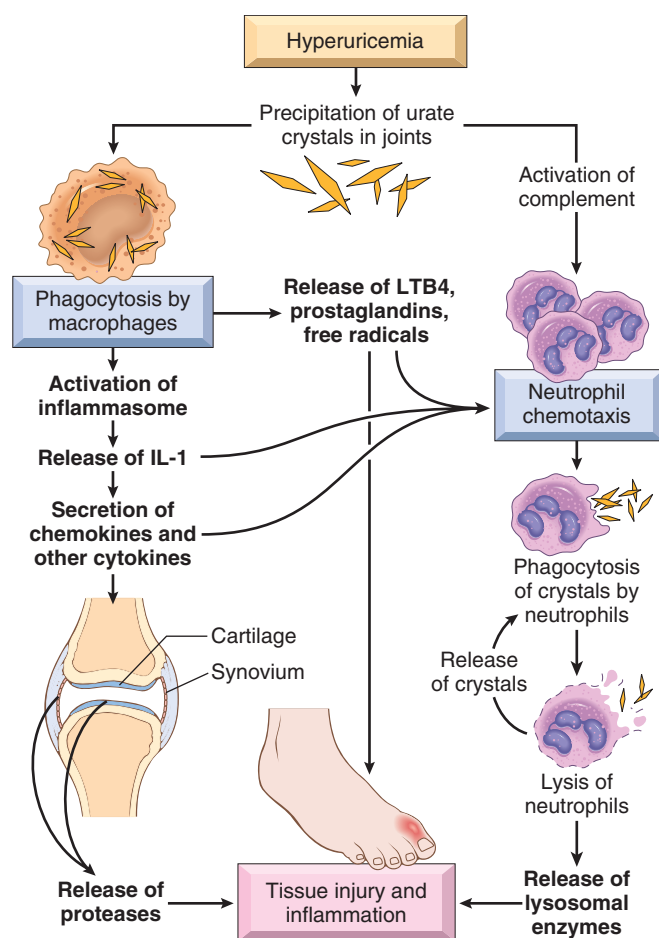


Figure 26.46 Pathogenesis of acute gouty arthritis. Urate crystals are phagocytosed by macrophages and stimulate the production of various inflammatory mediators that elicit the inflammation characteristic of gout. Note that IL-1 stimulates the production of chemokines and other cytokines from a variety of cells. *LTB4*, Leukotriene B₄.

and periarticular tissue as well as severe cartilage damage that compromises joint function.

Only about 10% of individuals with hyperuricemia develop gout. Other factors linked to disease include the following:

- Age of the individual and duration of the hyperuricemia, as gout usually appears after 20 to 30 years of hyperuricemia
- Male sex
- Genetic predisposition, including X-linked *HGPRT* abnormalities and primary gout, which has a multigenic pattern of inheritance
- Alcohol consumption
- Obesity
- Drugs (e.g., thiazides) that reduce urate excretion

MORPHOLOGY

The distinctive patterns of gout are (1) acute arthritis, (2) chronic tophaceous arthritis, (3) accumulation of tophi in extraarticular sites, and (4) urate (gouty) nephropathy.

Acute arthritis is characterized by a dense neutrophilic infiltrate that permeates the synovium and synovial fluid. Slender,

needle-shaped urate crystals are arranged in small clusters in the synovium and are frequently found in the cytoplasm of neutrophils within aspirated joint fluid (Fig. 26.47A). The edematous and congested synovium also contains scattered lymphocytes, plasma cells, and macrophages. The acute attack remits when the episode of crystallization abates and crystals are solubilized.

Chronic tophaceous arthritis evolves from repetitive precipitation of urate crystals during acute attacks. Urate encrusts the articular surface and forms visible deposits in the synovium, which becomes hyperplastic, fibrotic, and thickened by inflammatory cells. The resulting pannus destroys the underlying cartilage and leads to juxta-articular bone erosions. In severe cases, fibrous or bony ankylosis ensues, resulting in loss of joint function.

Tophi are the pathognomonic hallmark of gout. They are formed by large aggregations of urate crystals surrounded by an intense inflammatory reaction of foreign-body giant cells. (Fig. 26.47B–C). Tophi may appear in the articular cartilage, ligaments, tendons, and bursae. Less frequently they occur in soft tissues (earlobes, fingertips) or kidneys. Superficial tophi can ulcerate through the overlying skin.

Urate nephropathy refers to the renal complications caused by urate crystals or tophi in the renal medullary interstitium or tubules. Complications include uric acid nephrolithiasis and pyelonephritis, particularly when urinary obstruction occurs.

Clinical Features

Primary gout initially presents as acute arthritis with sudden-onset, excruciating joint pain, localized hyperemia and warmth, and, occasionally, mild fever; constitutional symptoms are uncommon. Most first attacks are monoarticular, and 50% occur in the first metatarsophalangeal joint of the big toe. Untreated, acute gouty arthritis may last for hours to weeks, but gradually there is complete resolution followed by a symptom-free interval, known as the intercritical period. Some individuals never have another attack, but most experience a second acute episode within months to a few years. In the absence of appropriate therapy, the attacks recur at shorter intervals and frequently become polyarticular, involving other parts of the foot and less commonly the upper extremity. Chronic tophaceous gout develops about 10 years after the initial acute attack and is characterized by juxta-articular bone erosion caused by osteoclastic bone resorption and loss of the joint space.

Nonpharmacologic treatments for gout include lifestyle modifications, such as weight loss in obese individuals; dietary changes to reduce purine intake, particularly of animal and seafood proteins; reduced alcohol and sugar-sweetened beverage consumption; and regular exercise. Pharmacologic therapies include uricosuric drugs (e.g., probenecid), xanthine oxidase inhibitors (e.g., allopurinol), urate oxidases (uricases), NSAIDs, and colchicine. Finally, gout can be precipitated by serum urate-increasing drugs taken for comorbidities; their replacement with alternatives may be helpful. Generally, gout does not shorten the life span, but it does impact quality of life.

Calcium Pyrophosphate Crystal Deposition Disease (Pseudogout)

Calcium pyrophosphate crystal deposition disease (CPPD), also known as *pseudogout*, usually occurs in individuals older than 50 years of age and becomes more common

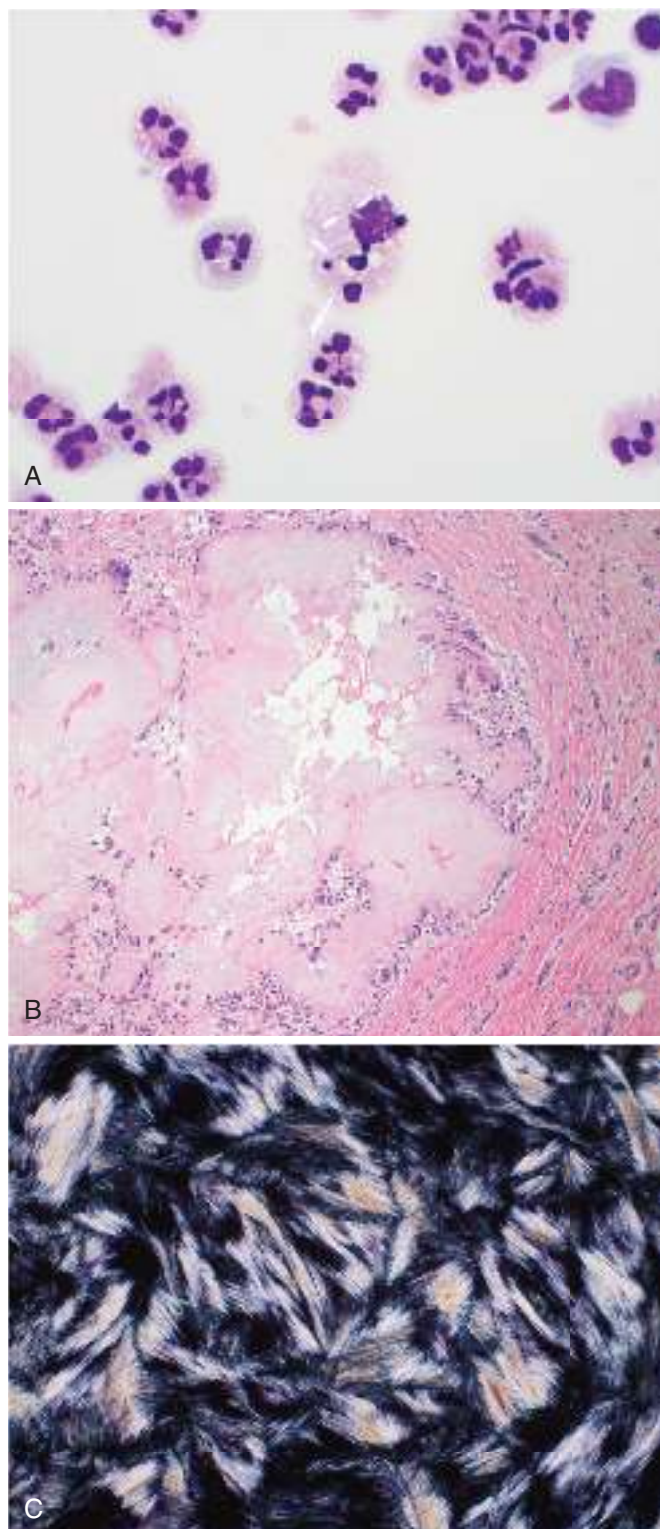


Figure 26.47 Gout. (A) Joint aspirate from acute gout arthritis demonstrates polarizable needle shape urate crystals in neutrophils. (B) Gouty tophus—an aggregate of dissolved urate crystals is surrounded by reactive fibroblasts, mononuclear inflammatory cells, and giant cells. (C) Needle-shaped urate crystals are easily identified under polarized light.

with increasing age; up to 60% of those 85 years of age or older are affected. There is no sex or racial predisposition. CPPD is divided into sporadic (idiopathic), hereditary, and secondary types. An autosomal dominant variant is caused by germline gain-of-function mutations in *ANKH*, an inorganic pyrophosphate transporter, that result in crystal deposition and osteoarthritis at relatively early ages. These mutations are distinct from those associated with craniometaphyseal dysplasia. Various disorders, including previous joint damage, hyperparathyroidism, hemochromatosis, hypomagnesemia, hypothyroidism, ochronosis, and diabetes predispose to secondary CPPD.

Pathogenesis

The basis for crystal formation is not known, but studies suggest that degradation of articular cartilage proteoglycans, which normally inhibit mineralization, allows crystallization around chondrocytes. As in gout, inflammation is caused by activation of the inflammasome in macrophages (see Fig. 26.46).

MORPHOLOGY

The crystals first develop in the articular cartilage, menisci, and intervertebral discs; as the deposits enlarge they may rupture and seed the joint. The crystals form chalky, white friable deposits, which are seen histologically in stained preparations as oval blue-purple aggregates (Fig. 26.48A). Individual crystals are rhomboid, 0.5 to 5 μm in greatest dimension (Fig. 26.48B), and birefringent. Inflammation is usually milder than in gout.

Clinical Features

CPPD is frequently asymptomatic. However, it may produce acute, subacute, or chronic arthritis that can be confused with osteoarthritis or rheumatoid arthritis clinically. Joint involvement may last from several days to weeks and may be monoarticular or polyarticular; the knees, followed by the wrists, elbows, shoulders, and ankles, are most commonly affected. Chondrocalcinosis, radiographically evident calcification of hyaline or fibrocartilage, is often present but is not synonymous with CPPD. Ultimately, approximately 50% of affected individuals experience significant joint damage. Therapy is supportive to minimize symptoms. There is no known treatment that prevents or slows crystal formation.

KEY CONCEPTS

ARTHRITIS

- Osteoarthritis (degenerative joint disease), the most common disease of joints, is a degenerative disorder of articular cartilage in which matrix breakdown exceeds synthesis. Inflammation is minimal and typically secondary, but locally produced cytokines may contribute to progression of joint degeneration.
- Rheumatoid arthritis is a chronic autoimmune inflammatory disease that affects mainly small joints but can be systemic. RA is caused by a cellular and humoral immune response against self antigens, particularly citrullinated proteins. TNF plays a central role, and anti-TNF biologics are effective.
- Seronegative spondyloarthropathies are a heterogeneous group of autoimmune arthritides that preferentially involve the

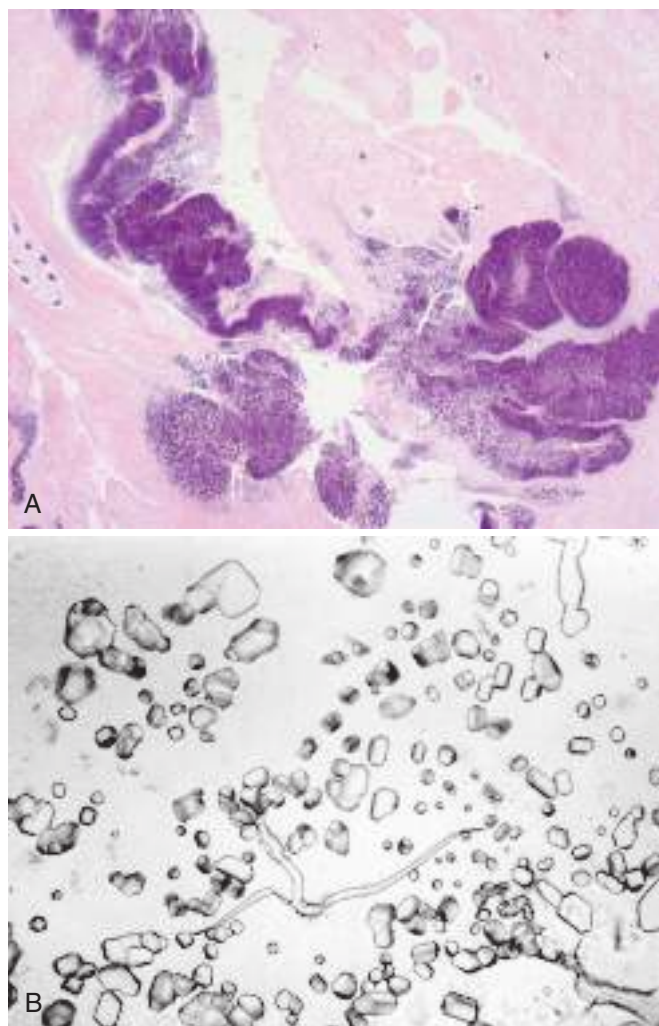


Figure 26.48 Pseudogout. (A) Deposits are present in cartilage and consist of amorphous basophilic material. (B) Smear preparation of calcium pyrophosphate crystals.

sacroiliac, vertebral joints and entheses and are associated with HLA-B27. Reactive arthritis is an uncommon immune-mediated disorder that develops after resolution of genitourinary or gastrointestinal infection.

- Suppurative arthritis describes direct infection of a joint space by bacterial organisms.
- Lyme disease is a systemic infection by *B. burgdorferi* that manifests, in part, as an infectious arthritis, possibly with an autoimmune component in chronic stages.
- Gout and pseudogout result from inflammatory responses to precipitated urate or calcium pyrophosphate, respectively.

JOINT TUMORS AND TUMOR-LIKE CONDITIONS

Reactive tumor-like lesions, such as ganglion cysts, synovial cysts, and osteochondral loose bodies, commonly involve joints and tendon sheaths. They usually result from trauma or degenerative processes and are much more common than neoplasms. Primary neoplasms are rare, usually benign, and tend to recapitulate the cells and tissue types (synovial

membrane, fat, blood vessels, fibrous tissue, and cartilage) native to joints and related structures. The rare malignant tumors are discussed with soft tissue tumors.

Ganglion and Synovial Cysts

Ganglion cysts are small, up to 1.5 cm, and nearly always located near a joint capsule or tendon sheath. A common location is around the joints of the wrist, where the cysts appear as firm, fluctuant, pea-sized translucent nodules. Ganglion cysts develop due to cystic or myxoid connective tissue degeneration; hence the cyst wall lacks a cell lining. Lesions may be multilocular and can enlarge through coalescence and degeneration of adjacent connective tissue. The fluid that fills ganglion cysts is similar to synovial fluid, but there is no communication with the joint space. Despite the name, the lesion is unrelated to ganglia of the nervous system.

Herniation of synovium through a joint capsule or massive enlargement of a bursa may produce a *synovial cyst*. A well-recognized example is the popliteal synovial cyst, or *Baker cyst*, associated with rheumatoid arthritis. The cyst lining resembles the synovium, and both cyst and synovium may be hyperplastic. Cyst fluid often contains inflammatory cells and fibrin.

Tenosynovial Giant Cell Tumor

Tenosynovial giant cell tumor is a benign neoplasm that develops in the synovial lining of joints, tendon sheaths, and bursae. Tenosynovial giant cell tumor can be diffuse (previously known as *pigmented villonodular synovitis*) or localized. The diffuse type tends to involve large joints, while the localized type usually occurs as a discrete nodule attached to a tendon sheath, commonly in the hand. Both variants are most often diagnosed in the 20s to 40s and affect the sexes equally.

Pathogenesis

Both diffuse and localized tenosynovial giant cell tumors harbor a reciprocal somatic chromosomal translocation, t(1;2)(p13;q37), resulting in fusion of the type VI collagen α -3 promoter upstream of the coding sequence of the M-CSF gene. As a result, the tumor cells overexpress M-CSF, which, through autocrine and paracrine effects, stimulates proliferation of macrophages in a manner similar to giant cell tumor of bone (described earlier).

MORPHOLOGY

Tenosynovial giant cell tumors are red-brown to orange-yellow. In diffuse tumors, the normally smooth joint synovium is converted into a tangled mat by red-brown folds, fingerlike projections, and nodules (Fig. 26.49A); localized tumors are well circumscribed. The neoplastic cells, which are a minority of the cells in the mass, are polygonal, moderately sized, and resemble synoviocytes (Fig. 26.49B). In the diffuse variant, they spread along the surface and infiltrate the subsynovial tissue. A solid aggregate may be attached to the synovium by a pedicle. Both diffuse and localized types are densely infiltrated by macrophages that may contain hemosiderin or foamy lipids and can be multinucleated. Older lesions may become fibrotic.

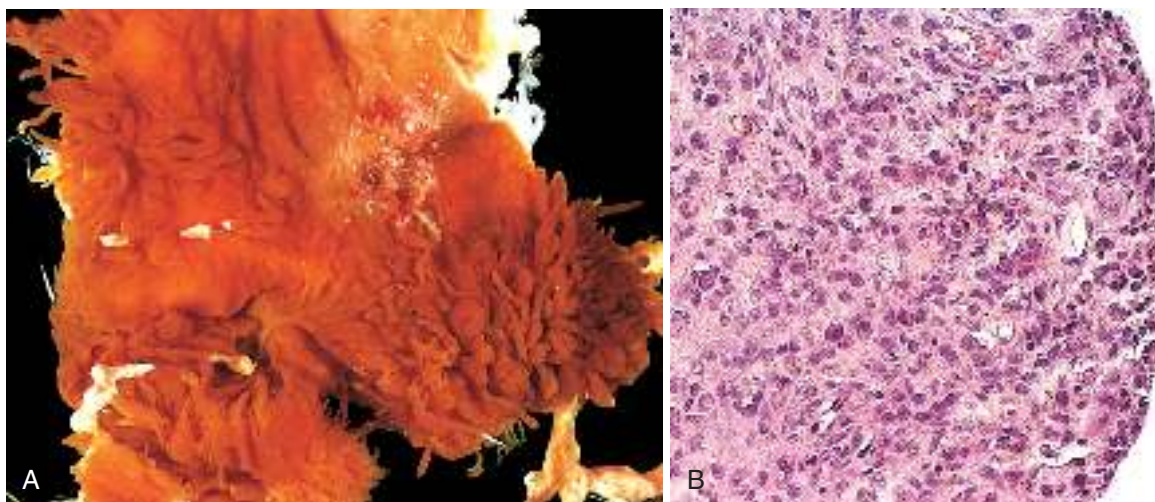


Figure 26.49 Tenosynovial giant cell tumor; diffuse type. (A) Excised synovium with fronds and nodules typical of diffuse tenosynovial giant cell tumor. The color and texture explain the old name of pigmented villonodular synovitis. (B) Sheets of proliferating cells in tenosynovial giant cell tumor bulging the synovial lining.

Clinical Features

Diffuse tenosynovial giant cell tumors present in the knee in 80% of cases. Affected individuals complain of pain, locking, and recurrent swelling, similar to monoarticular arthritis. Sometimes a palpable mass is appreciated. Aggressive tumors erode into adjacent bones and soft tissues, causing confusion with other types of neoplasms. Recurrence

is common. Localized tenosynovial giant cell tumors present as a solitary, slow-growing, painless mass that frequently involves the tendon sheaths along the wrists and fingers. Bone erosion and recurrence are less common than in the diffuse type. Surgical excision is the mainstay of treatment. Clinical trials using antagonists of M-CSF signaling have had promising results.

Soft Tissue Tumors

By convention, soft tissue refers to non-epithelial tissue excluding the skeleton, joints, central nervous system, and hematopoietic and lymphoid tissues. Although nonneoplastic conditions can involve soft tissue, they are seldom confined to this compartment, so the area of soft tissue pathology is restricted to neoplasms. With the exception of skeletal muscle neoplasms, benign soft tissue tumors are 100-fold more frequent than their malignant counterparts, the sarcomas. In the United States, the incidence of soft tissue sarcomas is approximately 12,000 per year, which is less than 1% of all cancers. Sarcomas, however, cause 2% of all cancer mortality, reflecting their aggressive behavior and resistance to chemotherapy. Most soft tissue tumors arise in the extremities, particularly the thigh. Approximately 15% arise in children; the incidence increases with age.

Pathogenesis

Most sarcomas are sporadic and have no known predisposing cause. A small minority are associated with germline mutations in tumor suppressor genes that are associated with well-described syndromes (neurofibromatosis 1, Gardner syndrome, Li-Fraumeni syndrome, Osler-Weber Rendu syndrome). Others are linked to known environmental exposures such as radiation, burns, or toxins.

Unlike carcinomas and certain hematologic malignancies, which often arise from well-recognized precursor lesions,

sarcoma precursors are undefined. Some sarcomas recapitulate a recognizable mesenchymal lineage (e.g. skeletal muscle), but they are thought to arise from pluripotent mesenchymal stem cells, which acquire somatic “driver” mutations in oncogenes and tumor suppressor genes. The genetics of tumorigenesis are heterogeneous, but some generalizations can be made based on karyotypic complexity:

- *Simple karyotype* (20%): Like many leukemias and lymphomas, sarcomas are occasionally euploid tumors with a single or limited number of chromosomal changes (Table 26.7) that occur early in tumorigenesis and are specific enough to serve as diagnostic markers. Tumors with these features most commonly arise in younger individuals and tend to have a monomorphic microscopic appearance. Examples include Ewing sarcoma (discussed earlier) and synovial sarcoma. In some cases, the oncogenic effect of chromosomal rearrangements is reasonably well understood (see Table 26.7). In others, the mechanisms are unknown. Oncogenic tumor-specific fusion proteins represent potential targets for therapy.
- *Complex karyotype* (80%): These tumors are usually aneuploid or polyploid and demonstrate multiple chromosomal gains and losses, a feature that suggests an underlying abnormality producing genomic instability. Examples include leiomyosarcomas and undifferentiated pleomorphic sarcoma. Such tumors are more common in adults and tend to be microscopically diverse, with a

Table 26.7 Chromosomal Abnormalities in Soft Tissue Tumors

Tumor	Cytogenetic Abnormality	Gene Fusion	Proposed Function
Ewing sarcoma	t(11;22)(q24;q12)	<i>EWS-FLI1</i>	Disordered protein with multiple functions, including aberrant transcription, cell cycle regulation, RNA splicing, and telomerase
Extraskelatal myxoid chondrosarcoma	t(21;22)(q22;q12)	<i>EWS-ERG</i>	
Desmoplastic small round-cell tumor	t(9;22)(q22;q12)	<i>EWS-CHN</i>	
Clear-cell sarcoma	t(11;22)(p13;q12)	<i>EWS-WT1</i>	Arrests adipocytic differentiation
Liposarcoma—myxoid type	t(12;22)(q13;q12)	<i>EWS-ATF1</i>	
Synovial sarcoma	t(12;16)(q13;p11)	<i>FUS-DDIT3</i>	Chimeric transcription factors, interrupts cell cycle control
	t(x;18)(p11;q11)	<i>SS18-SSX1</i> <i>SS18-SSX2</i> <i>SS18-SSX4</i>	
Rhabdomyosarcoma—alveolar type	t(2;13)(q35;q14)	<i>PAX3-FOXO1</i>	
	t(1;13)(p36;q14)	<i>PAX7-FOXO1</i>	Chimeric transcription factors, disrupts skeletal muscle differentiation
Dermatofibrosarcoma protuberans	t(17;22)(q22;q15)	<i>COL1A1-PDGFB</i>	Promoter driven overexpression of PDGF- β , autocrine stimulation
Alveolar soft-part sarcoma	t(X;17)(p11.2;q25)	<i>TFE3-ASPL</i>	Unknown
Infantile fibrosarcoma	t(12;15)(p13;q23)	<i>ETV6-NTRK3</i>	Chimeric tyrosine kinase leads to constitutively active Ras/MAPK pathway
Nodular fasciitis	t(22;17)	<i>MYH9-USP6</i>	Increased Wnt/ β -catenin signaling

range of cell sizes and shapes within a single tumor (i.e., they are pleomorphic).

Classification of soft tissue tumors continues to evolve as new molecular genetic abnormalities are identified. Clinically, soft tissue tumors range from benign, self-limited lesions that require minimal treatment to intermediate-grade, locally aggressive tumors with limited metastatic risk to highly aggressive malignancies with significant metastatic risk and mortality. All highly aggressive malignancies are classified as *sarcomas*, but this term is used less consistently among locally aggressive, nonmetastasizing tumors. Pathologic classification integrates morphology (e.g., muscle differentiation), immunohistochemistry, and molecular diagnostics (Table 26.8). In addition to accurate diagnosis, grade (degree of differentiation) and stage (size and depth) are important prognostic indicators. The next section will consider representative soft tissue tumors.

TUMORS OF ADIPOSE TISSUE

Lipoma

Lipoma, a benign tumor of fat, is the most common soft tissue tumor in adults. The conventional lipoma is the most common subtype, from which rare variants are distinguished according to characteristic morphologic and genetic features.

MORPHOLOGY

The conventional lipoma is a well-encapsulated mass of mature adipocytes. It usually arises in the subcutis of the proximal extremities and trunk during middle adulthood. Infrequently, lipomas are large, intramuscular, and poorly circumscribed. **Lipomatosis** occurs when multifocal lipomas involve a limb. Most lipomas are soft, mobile, painless, and cured by simple excision.

Liposarcoma

Liposarcomas, malignant tumors of adipose tissue, are the most common sarcomas of adulthood. These tumors typically develop in deep soft tissues of the proximal extremities and retroperitoneum of individuals in the sixth and seventh decades of life.

Pathogenesis

The three distinct subtypes of liposarcoma, well-differentiated, myxoid, and pleomorphic, have different genetic aberrations. Well-differentiated liposarcomas harbor amplifications of chromosome region 12q13–q15, which includes the p53 inhibitor *MDM2*. In myxoid liposarcoma, a fusion gene generated by a t(12;16) translocation arrests adipocyte differentiation, leading to unregulated proliferation of primitive cells. Pleomorphic liposarcomas have complex karyotypes without reproducible genetic abnormalities.

MORPHOLOGY

Liposarcomas are histologically divided into **well-differentiated**, **myxoid**, and **pleomorphic** subtypes. Well-differentiated liposarcomas are composed of mature adipocytes with scattered atypical spindle cells (Fig. 26.50A). Myxoid liposarcoma is characterized by abundant basophilic extracellular matrix, arborizing capillaries, and primitive cells at various stages of adipocyte differentiation resembling fetal fat (Fig. 26.50B). Pleomorphic liposarcomas are characterized by sheets of anaplastic cells with bizarre nuclei admixed with variable numbers of immature adipocytes, termed lipoblasts.

Clinical Features

Liposarcomas recur locally, and often repeatedly, unless adequately excised. Well-differentiated tumors can be indolent, while pleomorphic liposarcomas are aggressive and frequently metastasize; the behavior of myxoid tumors is intermediate between these extremes.

Table 26.8 Clinical Features of Soft Tissue Tumors

Category	Behavior	Tumor Type	Common Locations	Age (years)	Morphology
Adipose	Benign	Lipoma	Superficial extremity, trunk	40–60	Mature adipose tissue
	Malignant	Well-differentiated liposarcoma	Deep extremity, retroperitoneum	50–60	Adipose tissue with scattered atypical spindle cells
		Myxoid liposarcoma	Thigh, leg	30s	Myxoid matrix, “chicken wire” vessels, round cells, lipoblasts
Fibrous	Benign	Nodular fasciitis	Arm, forearm	20–30	Tissue culture growth, extravasated erythrocytes,
		Deep fibromatosis	Abdominal wall	30–40	Dense collagen, long, unidirectional fascicles
Skeletal muscle	Benign	Rhabdomyoma	Head and neck	0–60	Polygonal rhabdomyoblasts, “spider” cells
	Malignant	Alveolar rhabdomyosarcoma	Extremities, sinuses	5–15	Uniform round discohesive cells between septae
		Embryonal rhabdomyosarcoma	Genitourinary tract	1–5	Primitive spindle cells, “strap” cells
Smooth muscle	Benign	Leiomyoma	Extremity	20s	Uniform, plump eosinophilic cells in fascicles
	Malignant	Leiomyosarcoma	Thigh, retroperitoneum	40–60	Pleomorphic eosinophilic cells
Vascular	Benign	Hemangioma	Head and neck	0–10	Circumscribed mass of capillary or venous channels
	Malignant	Angiosarcoma	Skin, deep lower extremity	50–80	Infiltrating capillary channels
Nerve sheath	Benign	Schwannoma	Head and neck	20–50	Encapsulated, fibrillar stroma, nuclear palisading
		Neurofibroma	Wide, cutaneous, subcutis	10–20+	Myxoid, ropy collagen, loose fascicles, mast cells
	Malignant	Malignant peripheral nerve sheath tumor	Extremities, shoulder girdle	20–50	Tight fascicles, atypia, mitotic activity, necrosis
Uncertain histotype	Benign	Solitary fibrous tumor	Pelvis, pleura	20–70	Branching ectatic vessels
	Malignant	Synovial sarcoma	Thigh, leg	15–40	Tight fascicles of uniform basophilic spindle cells, pseudoglandular structures
		Undifferentiated pleomorphic sarcoma	Thigh	40–70	High-grade anaplastic polygonal, round, or spindle cells, bizarre nuclei, atypical mitoses, necrosis

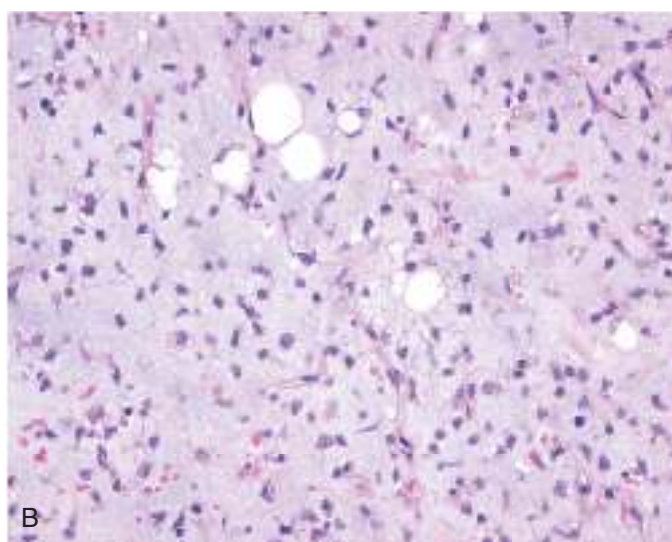
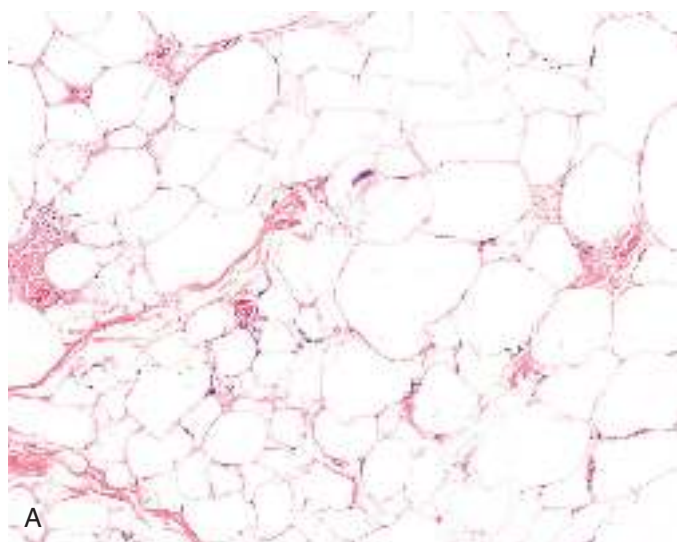


Figure 26.50 Liposarcoma. (A) the well-differentiated subtype consists of mature adipocytes and scattered spindle cells with hyperchromatic nuclei. (B) Myxoid liposarcoma with abundant basophilic ground substance and a rich capillary network with scattered immature adipocytes and more primitive round to stellate cells.

FIBROUS TUMORS

Nodular Fasciitis

Nodular fasciitis is a self-limited fibroblastic and myofibroblastic proliferation that occurs in young adults. The tumors occur most frequently on the forearm, chest, and back and grow rapidly over weeks to months. Due to a history of trauma in 10% to 50% of cases, this lesion was previously considered a reactive lesion. It is, however, a clonal proliferation that harbors a t(17;22) translocation that produces a *MYH9-USP6* fusion. The defect that prevents neoplastic cells from becoming malignant has not been defined. It is, however, interesting to note that aneurysmal bone cyst, a benign neoplasm previously considered reactive, also contains a *USP6* fusion gene, albeit to partners other than *MYH9*. Nodular fasciitis typically regresses spontaneously and, if excised, rarely recurs.

MORPHOLOGY

Nodular fasciitis arises in the deep dermis, subcutis, fascia, or muscle. Grossly the non-encapsulated lesions are well-circumscribed or slightly infiltrative and less than 3 cm in diameter. Histologically, nodular fasciitis is composed of plump, immature-appearing fibroblasts and myofibroblasts containing elongated nuclei with punctate nucleoli. Mitoses are frequent, but atypical forms are notably absent (Fig. 26.51). A gradient, termed zonation, transitions from hypercellular regions with myxoid stroma to hypocellular areas with fibrous stroma. Storiform or fascicular patterns are common in cellular areas. Metaplastic bone, cystic areas, ganglion-like cells, prominent vessels, extravasated red cells, and infiltrating lymphocytes are common.

Fibromatoses

Superficial Fibromatosis

Superficial fibromatosis is a benign growth that can cause local deformity but has an innocuous clinical course. It affects males more frequently than females. These nodular,

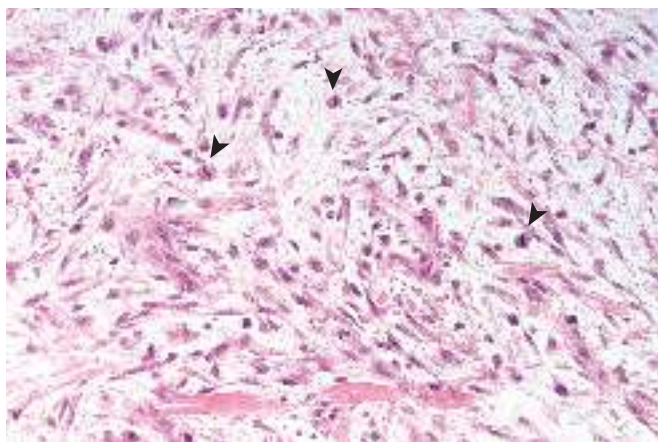


Figure 26.51 Nodular fasciitis with plump, randomly oriented spindle cells surrounded by myxoid stroma. Note the mitotic activity (arrowheads) and extravasated red blood cells.

proliferative lesions are composed of plump spindle cells arranged in poorly defined broad bundles or long, sweeping fascicles surrounded by abundant dense collagen. Clinical subtypes include Dupuytren contracture, Ledderhose disease, and Peyronie disease.

- *Dupuytren contracture*, or *palmar fibromatosis*, is an irregular or nodular thickening of the palmar fascia that can be unilateral or bilateral. Incidence increases with age. Attachment to overlying skin can cause puckering or dimpling and a slowly progressive flexion contracture.
- *Ledderhose disease*, or *plantar fibromatosis*, occurs in boys from under 10 years of age into adolescence. It is unilateral and does not cause contractures but can be associated with palmar and penile fibromatosis.
- *Peyronie disease*, or *penile fibromatosis*, is a palpable induration or mass on the dorsolateral aspect of the penis. Eventually, it may cause abnormal curvature of the shaft and constriction of the urethra.

Palmar and plantar fibromatoses progress in about 50% of cases. The remainder stabilize and do not progress; some resolve spontaneously. Nevertheless, recurrence is common, even after excision.

Deep Fibromatosis (Desmoid Tumors)

Deep fibromatoses, also called desmoid tumors, are large, infiltrative masses that frequently recur but do not metastasize. They arise most frequently in the teens to 30s, predominantly in women. Abdominal fibromatosis generally arises in the musculoaponeurotic structures of the anterior abdominal wall, but tumors can also arise in the limb girdles or mesentery. Deep fibromatoses contain mutations in the *APC* or *CTNNB1* (β -catenin) genes, both of which lead to increased Wnt signaling. The majority of tumors harbor sporadic *CTNNB1* mutations, but individuals with familial adenomatous polyposis, who have germline *APC* mutations, are predisposed to deep fibromatosis (Gardner syndrome, Chapter 17).

MORPHOLOGY

Fibromatoses are gray-white, firm, poorly demarcated masses varying from 1 to 15 cm in diameter. They are rubbery and tough and infiltrate surrounding muscle, nerve, and fat. Bland fibroblasts arranged in long parallel fascicles amid dense collagen are characteristic (Fig. 26.52). The resulting histologic appearance can resemble a scar.

Clinical Features

In addition to possibly being disfiguring or disabling, deep-seated fibromatosis can be painful. Because they are infiltrative, complete excision of deep fibromatosis can be difficult. Recent efforts have concentrated on medical or radiation therapy as alternatives to surgery.

SKELETAL MUSCLE TUMORS

In contrast to tumors of other cell types, skeletal muscle neoplasms are almost all malignant. The benign variant,

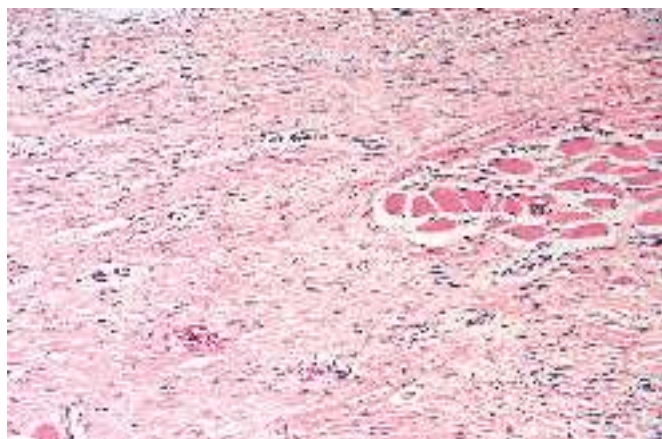


Figure 26.52 Deep fibromatosis infiltrating skeletal muscle.

rhabdomyoma, is frequently associated with tuberous sclerosis (Chapter 28).

Rhabdomyosarcoma

Rhabdomyosarcoma is a malignant mesenchymal tumor with skeletal muscle differentiation. Four subtypes are recognized: alveolar (20%), embryonal (50%), pleomorphic (20%), and spindle cell/sclerosing (10%). Alveolar and embryonal subtypes of rhabdomyosarcoma are the most common soft tissue sarcomas of childhood and adolescence, usually appearing before 20 years of age. Pleomorphic rhabdomyosarcoma occurs in adults, while the spindle cell/sclerosing type affects all ages. Pediatric rhabdomyosarcomas often arise in the sinuses, head and neck, and genitourinary tract; these locations do not normally contain much skeletal muscle, underscoring the hypothesis that sarcomas do not arise from mature, terminally differentiated cells. Alveolar rhabdomyosarcoma frequently contains fusions of the *FOXO1* gene to either *PAX3* or *PAX7* due to t(2;13) or t(1;13) translocations, respectively. *PAX3* is a transcription factor that initiates skeletal muscle differentiation; the chimeric *PAX3-FOXO1* fusion protein interferes with differentiation, a mechanism similar to many of transcription factor fusion proteins found in acute leukemia. The remaining subtypes of rhabdomyosarcoma are genetically heterogeneous.

MORPHOLOGY

Embryonal rhabdomyosarcoma is a soft, gray, infiltrative mass. The tumor cells recapitulate skeletal muscle at various stages of differentiation and include sheets of primitive round and spindled cells in myxoid stroma (Fig. 26.53A). Rhabdomyoblasts with visible cross-striations may be present. **Sarcoma botryoides** (Chapter 22) is an embryonal rhabdomyosarcoma variant that develops in the walls of hollow, mucosal-lined structures, such as the nasopharynx, common bile duct, bladder, and vagina.

Alveolar rhabdomyosarcoma is traversed by a network of fibrous septae that divide the cells into clusters or aggregates, creating a crude resemblance to pulmonary alveoli. Cells in the center of the aggregates are only minimally cohesive, while those

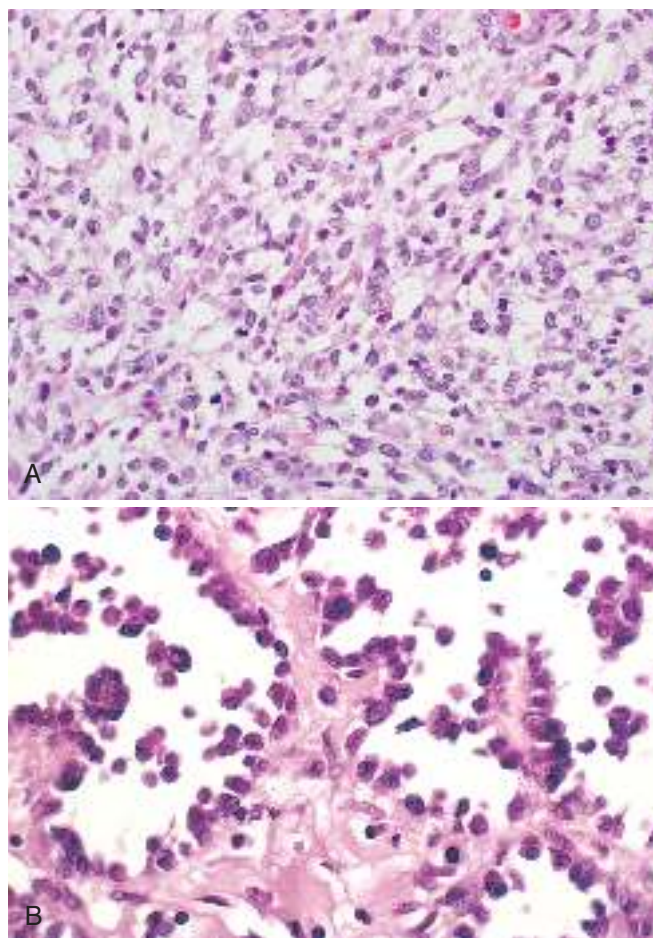


Figure 26.53 Rhabdomyosarcoma. (A) Embryonal subtype composed of malignant cells ranging from primitive and round to densely eosinophilic with skeletal muscle differentiation. (B) Alveolar rhabdomyosarcoma with numerous spaces lined by discohesive, uniform round tumor cells.

at the periphery adhere to the septae. Tumor cells are uniform and round with little cytoplasm. Cross-striations are uncommon (Fig. 26.53B).

Pleomorphic rhabdomyosarcoma is composed of large, sometimes multinucleated, bizarre eosinophilic tumor cells that can overlap histologically with other pleomorphic sarcomas. Immunohistochemistry (e.g., myogenin staining) is usually necessary to confirm rhabdomyoblastic differentiation.

Spindle cell/sclerosing rhabdomyosarcomas consist of fusiform cells with vesicular chromatin arranged in long fascicles or in a storiform pattern. Rhabdomyoblasts are occasionally present. Dense, collagenous, sclerotic stroma may be more common in adults.

Clinical Features

Rhabdomyosarcomas are aggressive and usually treated by surgical resection and chemotherapy, with or without radiation therapy. Histologic type and location are correlated with survival; sarcoma botryoides has the most favorable prognosis, while pleomorphic rhabdomyosarcoma is often fatal.

SMOOTH MUSCLE TUMORS

Leiomyoma

Leiomyomas, benign smooth muscle tumors, often arise in the uterus and are the most common neoplasm in women (Chapter 22). Depending on their number, size, and location, uterine leiomyomas may cause a variety of symptoms including infertility and menorrhagia. *Pilar leiomyomas* arise from cutaneous erector pili muscles and, rarely, develop in deep soft tissues or the gastrointestinal tract. A germline loss-of-function mutation in the fumarate hydratase gene on chromosome 1q42.3 leads to multiple cutaneous leiomyomas, uterine leiomyomas, and renal cell carcinoma. Fumarate hydratase is a Krebs cycle enzyme, and this association is another example of the link between metabolic abnormalities and neoplasia.

Soft tissue leiomyomas are usually 1 to 2 cm and are composed of fascicles of densely eosinophilic spindle cells that often intersect each other at right angles. Tumor cells have blunt-ended, elongated nuclei with minimal atypia and few mitotic figures. Solitary lesions are easily cured. However, multiple tumors may be so numerous that surgical removal is impractical.

Leiomyosarcoma

Leiomyosarcoma, a malignant tumor of smooth muscle, most often develops in the deep soft tissues of the extremities and the retroperitoneum. It accounts for 10% to 20% of soft tissue sarcomas. They occur in adults and afflict women more frequently than men. A particularly deadly form arises from the great vessels, often the inferior vena cava. Leiomyosarcomas have underlying defects in genomic stability leading to complex genotypes.

MORPHOLOGY

Leiomyosarcomas present as painless firm masses. Retroperitoneal tumors may be large and bulky and cause abdominal symptoms. Histologically, they consist of eosinophilic spindle cells with blunt-ended, hyperchromatic nuclei arranged in interweaving fascicles. Immunohistochemical detection of smooth muscle proteins, including smooth muscle actin, desmin, and caldesmon, can aid in diagnosis.

Clinical Features

Leiomyosarcoma treatment depends on tumor size, location, and grade. Superficial or cutaneous leiomyosarcomas are usually small and have a good prognosis, while those involving retroperitoneum are large, frequently unresectable, and often fatal as a result of local extension and metastatic spread, especially to the lungs.

TUMORS OF UNCERTAIN ORIGIN

Although many soft tissue tumors can be assigned to recognizable histologic types, a large proportion do not recapitulate known mesenchymal lineages. This group

includes tumors with simple or complex karyotypes; an example of each is described here.

Synovial Sarcoma

The first described cases of synovial sarcoma arose in soft tissues near the knee joint and led to speculation of a relationship with synovium. The proved to be mistaken, as **synovial sarcoma can present in locations that lack synovium, and its morphologic features are inconsistent with an origin from synoviocytes.** Synovial sarcomas account for approximately 10% of all soft tissue sarcomas and typically occur in people in their 20s to 40s. Individuals often present with a deep-seated mass that has been present for several years. A characteristic chromosomal translocation $t(x;18)(p11;q11)$ that produces fusion of the *SS18* gene to one of three *SSX* genes is present in most synovial sarcomas. The fusions encode chimeric proteins that interfere with normal chromatin remodeling, the repositioning of nucleosomes on chromatin that influences gene expression.

MORPHOLOGY

Synovial sarcomas can be monophasic or biphasic. Monophasic synovial sarcomas consist of uniform spindle cells with scant cytoplasm and dense chromatin arranged in short, tightly packed fascicles. The biphasic type also contains glandlike structures composed of cuboidal to columnar epithelioid cells (Fig. 26.54). Immunohistochemistry is helpful because tumor cells, especially in the biphasic type, are positive for epithelial antigens (e.g., keratins), thereby differentiating them from most other sarcomas.

Clinical Features

Synovial sarcomas are treated aggressively with limb-sparing surgery and chemotherapy. The 5-year survival varies from 25% to 62% and is related to tumor stage and patient age. Common sites of metastases are the lung and, unusually for sarcomas, regional lymph nodes.

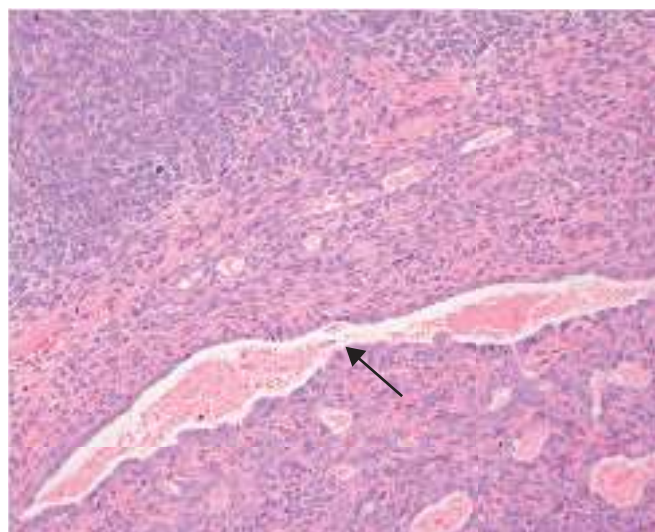


Figure 26.54 Synovial sarcoma revealing the classic biphasic spindle cell and glandlike (arrow) histologic appearance.

Undifferentiated Pleomorphic Sarcoma

Undifferentiated pleomorphic sarcomas include malignant mesenchymal tumors with high-grade, pleomorphic cells that cannot be classified into another category by histomorphology, immunophenotype, and genetics. Most arise in the deep soft tissues of the extremities, especially the thigh of middle-aged or older adults. Most undifferentiated pleomorphic sarcomas are aneuploid with multiple structural and numerical chromosomal changes.

MORPHOLOGY

Undifferentiated pleomorphic sarcomas are usually large, gray-white fleshy masses and can grow to 20 cm, depending on the anatomic compartment. Necrosis and hemorrhage common. Microscopically, they are extremely pleomorphic and composed of sheets of large, anaplastic spindled to polygonal cells with hyperchromatic irregular, sometimes bizarre nuclei (Fig. 26.55). Mitotic figures, including atypical asymmetric forms, are abundant. By definition, tumor cells lack both differentiation along recognized lineages and characteristic genetic defects.

Clinical Features

Undifferentiated pleomorphic sarcomas are aggressive malignancies that are treated with surgery and adjuvant chemotherapy and/or radiation. Despite this, prognosis is generally poor. Metastases occur in 30% to 50% of cases.

KEY CONCEPTS

SOFT TISSUE TUMORS

- Soft tissue tumors are malignant mesenchymal lesions that are distinct from tumors of epithelial, skeletal, central nervous system, hematopoietic, or lymphoid tissues.
- Soft tissue tumors likely arise from pluripotent mesenchymal stem cells rather than mature cells.

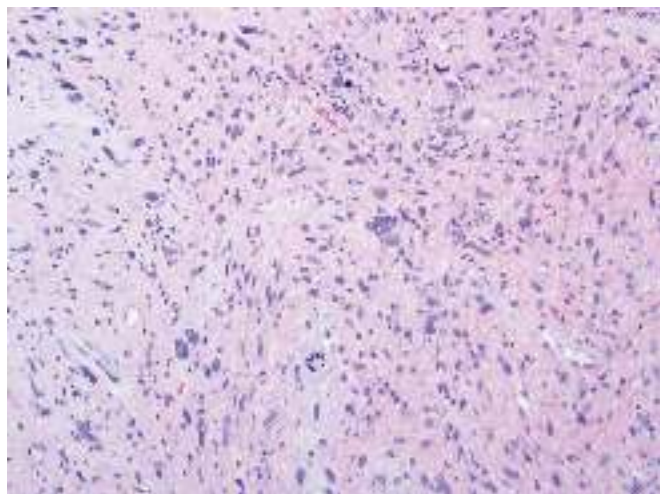


Figure 26.55 Undifferentiated pleomorphic sarcoma revealing anaplastic spindled to polygonal cells.

- Tumors that recapitulate a mature mesenchymal tissue (e.g., smooth muscle) can be subdivided into benign and malignant forms.
- Some tumors are composed of cells for which there is no normal counterpart (e.g., synovial sarcoma and undifferentiated pleomorphic sarcoma).
- Sarcomas with simple karyotypes demonstrate reproducible, chromosomal, and molecular abnormalities that contribute to pathogenesis and have diagnostic utility.
- Most adult sarcomas are genetically heterogeneous with complex karyotypes, histologically pleomorphic, and associated with a poor prognosis.

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Peripheral Nerves and Skeletal Muscles

27

Peter Pytel • Douglas C. Anthony

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Neuromuscular diseases are a complex group of disorders that typically present with weakness, muscle pain, or sensory deficits and may be inherited or acquired. They can be grouped according to anatomy, disease course, and pathogenesis. Physicians keep all these characteristics in mind when evaluating a patient with neuromuscular symptoms. This chapter uses an anatomic approach, grouping neuromuscular disorders into those that preferentially affect the peripheral nerves, the neuromuscular junction, or the skeletal muscles. A discussion of neoplasms that arise from peripheral nerves ends the chapter. Conditions that can produce similar clinical symptoms but are caused by disorders of the central nervous system (CNS) are discussed in Chapter 28.

DISEASES OF PERIPHERAL NERVES

The two components of peripheral nerves involved in impulse transmission are axons and myelin sheaths made by Schwann cells. Injury to either of these components may result in a peripheral neuropathy. Before discussing the pathology of these disorders, a brief review of peripheral nerve structure and function is in order. *Somatic motor function*

is carried out by the motor unit, which consists of (1) a lower motor neuron located in the anterior horn of the spinal cord or in the brainstem; (2) an axon that travels in a nerve to a target; (3) the neuromuscular junctions; and (4) multiple innervated myofibers (muscle fibers). *Somatic sensory function* depends on (1) the distal nerve endings, which may contain specialized structures that register specific sensory modalities; (2) a distal axon segment that travels as part of a peripheral nerve to the dorsal root ganglion; and (3) a proximal axon segment that synapses on neurons in the spinal cord or brainstem. *Autonomic nerve fibers*, which transmit all visceral motor and sensory functions, outnumber somatic fibers in the peripheral nervous system, but signs and symptoms related to their involvement are generally not prominent features of peripheral neuropathies, with a few important exceptions (e.g., in some cases of diabetic neuropathy, discussed later).

Specific sensations (pain, temperature, touch) and motor signals are each conveyed by axons that can be distinguished, in part, based on their diameter. Axonal diameters are in turn correlated with the thickness of their myelin sheaths and with their conduction speeds:

- *Thin unmyelinated fibers* mediate autonomic functions as well as certain pain and temperature sensations and have

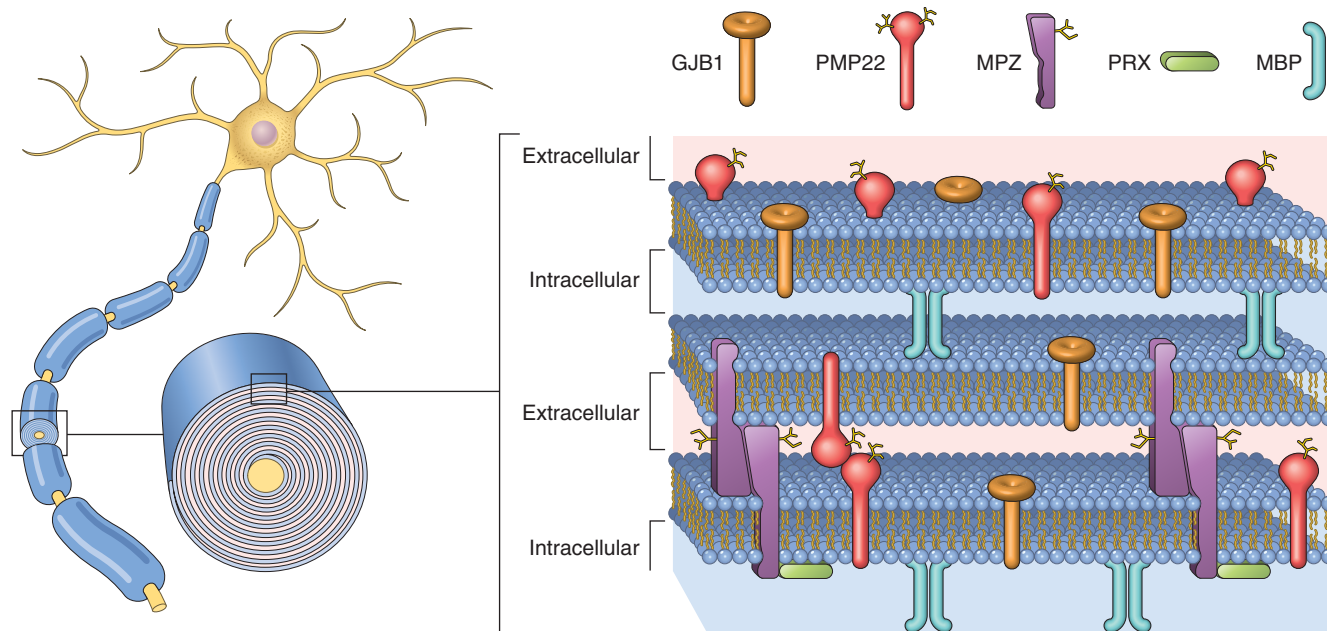


Figure 27.1 Relationship between lipid bilayers and associated proteins in myelin within internodes. Myelin basic protein (MBP) is an intracellular protein that has a role in myelin compaction. Mutant forms of myelin protein zero (MPZ), peripheral myelin protein 22 (PMP22), gap junction protein, beta 1 (GJB1), and periaxin (PRX) cause some forms of Charcot-Marie-Tooth disease, a hereditary demyelinating neuropathy.

the slowest conduction speeds due to their lack of myelin and small axonal diameter.

- *Large diameter axons* with thick myelin sheaths transmit light touch and motor signals and have fast conduction speeds.

In the case of myelinated axons, individual Schwann cells make exactly one myelin sheath that wraps around a single axon to create a myelinated segment called an *internode*. Internodes are separated by unmyelinated gaps referred to as *nodes of Ranvier*, which are uniformly spaced along the length of the axon. A number of specialized proteins are essential for normal assembly and function of myelin within internodes (Fig. 27.1). Unmyelinated axons are also intimately associated with Schwann cells but in a different arrangement in which one cell surrounds segments of multiple axons.

Most peripheral nerves carry out both motor and sensory functions and thus contain axons of varying diameter and myelin thickness. **The axons are bundled together by three major connective tissue components: the epineurium, which encloses the entire nerve; the perineurium, a multilayered concentric connective tissue sheath that groups subsets of axons into fascicles; and the endoneurium, which surrounds individual nerve fibers.**

General Types of Peripheral Nerve Injury

Axonal Neuropathies

Axons are the primary target of the damage in this large group of peripheral neuropathies (Fig. 27.2). The morphologic hallmarks of axonal neuropathies can be produced experimentally by cutting a peripheral nerve, which results in a prototypical pattern of injury described as *Wallerian degeneration*. Portions of axons that are distal to the point of transection are disconnected from the cell bodies (*perikarya*)

and degenerate. Within a day of injury, the distal axons begin to fragment, and the associated myelin sheaths unravel (Fig. 27.3) and disintegrate into spherical structures (*myelin ovoids*). Macrophages are recruited and participate in the removal of axonal and myelin debris. Regeneration starts at the site of transection with the formation of a growth cone and the outgrowth of new branches from the stump of the proximal axon. Schwann cells and their associated basement membranes guide the sprouting axons, which grow at about 1 mm per day, toward their distal target. Continuous pruning of the sprouting axons removes misguided branches. The Schwann cells create new myelin sheaths around the regenerating axons, but these myelin internodes tend to be thinner and shorter than in the original ones. The regeneration is successful only if the two transected ends remain closely approximated. A failure of the outgrowing axons to find their distal target can produce a “pseudotumor” termed *traumatic neuroma*—a nonneoplastic haphazard proliferation of axonal processes and associated Schwann cells that results in a painful nodule (Fig. 27.4).

The changes observed following experimental nerve transections only partially resemble those seen in various axonal neuropathies. One key difference is that in these neuropathies (unlike nerve transection) damage occurs over an extended period of time. As a result, degenerating and regenerating axons coexist in a single biopsy. With time, damage tends to outpace repair, resulting in progressive loss of axons. In cases of toxic and metabolic insults, axons often degenerate in a length-dependent fashion with the longest axons being most susceptible, resulting in a “*dying-back*” type of pattern of progression. The electrophysiologic hallmark of axonal neuropathies is a reduction in signal amplitude owing to the dropout of axons from affected peripheral nerves, with relative preservation of conduction velocity.

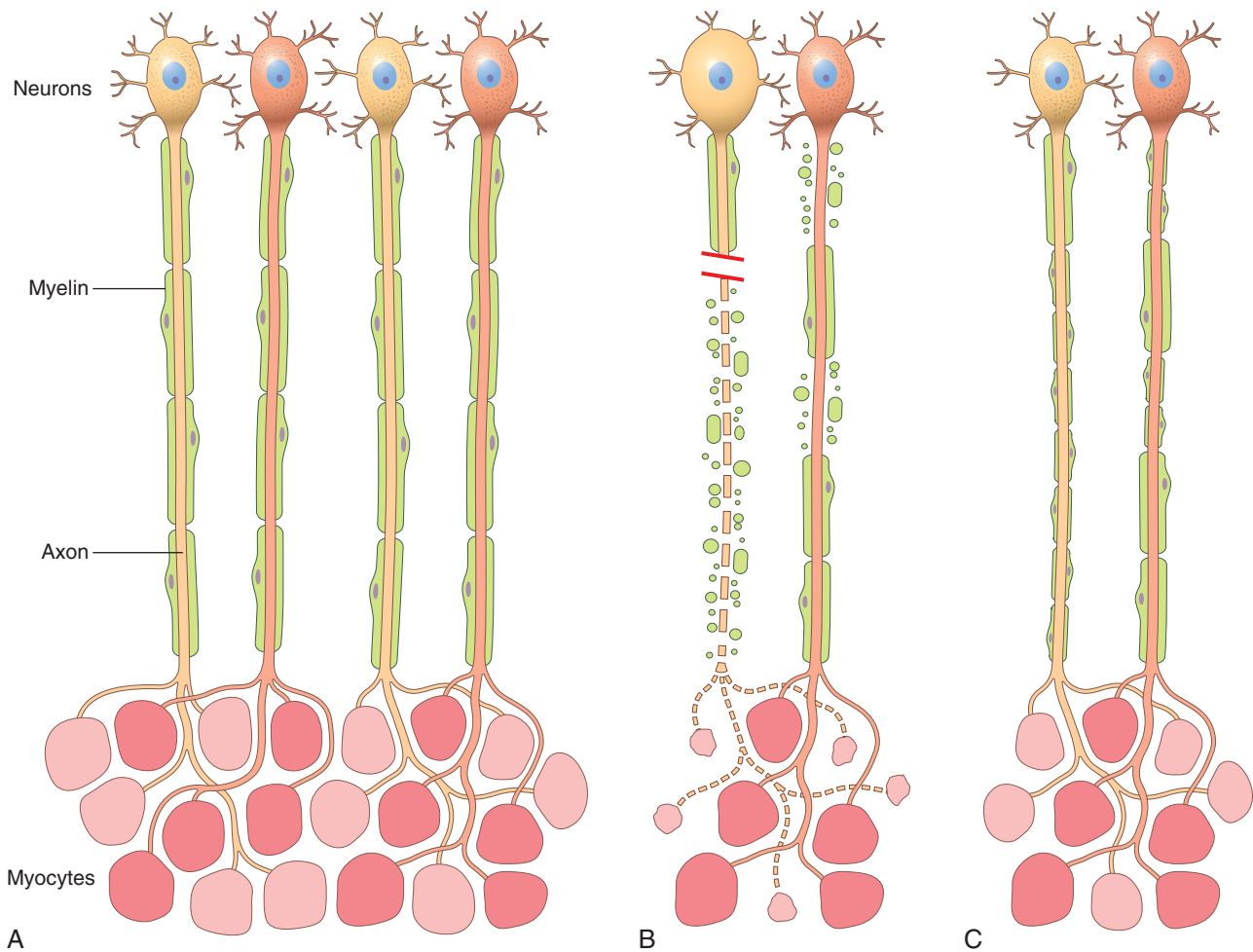


Figure 27.2 Patterns of peripheral nerve damage. (A) In normal motor units, type I and type II myofibers are arranged in a “checkerboard” distribution, and the internodes along the motor axons are uniform in thickness and length. (B) Acute axonal injury (*left axon*) results in degeneration of the distal axon and its associated myelin sheath, with atrophy of denervated myofibers. In contrast, acute demyelinating disease (*right axon*) produces random segmental degeneration of individual myelin internodes, while sparing the axons. (C) Regeneration of axons after injury (*left axon*) allows reinnervation of myofibers. The regenerated axon is myelinated by proliferating Schwann cells, but the new internodes are shorter and the myelin sheaths are thinner than the original ones. Remission of demyelinating disease (*right axon*) allows remyelination to take place, but the new internodes are shorter and have thinner myelin sheaths than flanking normal undamaged internodes. See [Fig. 27.7](#) for comparison with reinnervation.

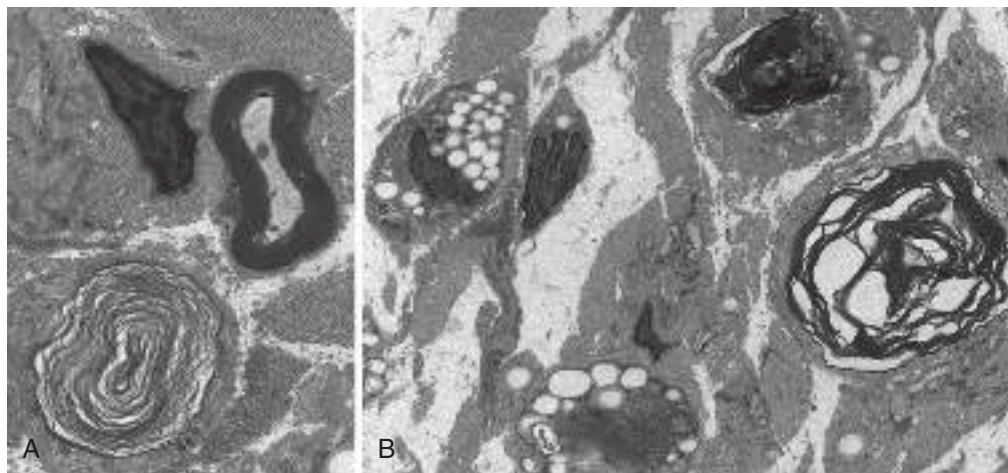


Figure 27.3 Electron micrographs illustrating features of axonal degeneration. (A) Degenerating myelin with loosened myelin layers is seen in the degenerating axon in the lower left corner, to be contrasted with a normal myelin sheath with tightly packed myelin and intact axon in the upper right corner. (B) In addition to an unraveling myelin sheath, several cells contain lipid droplets (seen as vacuoles) derived from degenerating myelin.

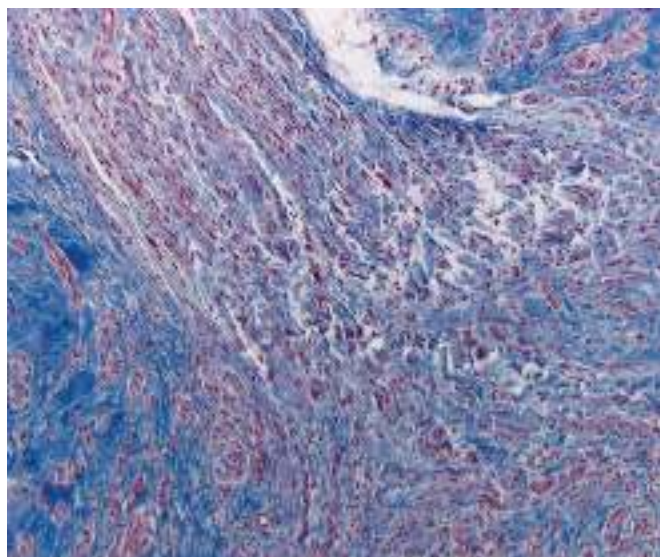


Figure 27.4 Trichrome-stained section of a traumatic neuroma showing the transition from normal nerve containing a parallel arrangement of axons (upper left corner) to a haphazard swirl of red-stained axons associated with admixture of Schwann cells and blue-staining connective tissue.

Demyelinating Neuropathies

In these disorders, Schwann cells with their myelin sheaths are the primary targets of damage (see Fig. 27.2), whereas axons are relatively preserved. This definition is similar to that of demyelinating diseases that affect the CNS (Chapter 28). Individual myelin sheaths degenerate in a seemingly random pattern, resulting in discontinuous damage of myelin segments. In response to this damage, Schwann cells or Schwann cell precursors proliferate and initiate repair through the formation of new myelin sheaths, but these again tend to be shorter and thinner than the original ones. The electrophysiologic hallmark of demyelinating neuropathies is slowed nerve conduction velocity, reflective of the loss of myelin.

Neuronopathies

Neuronopathies result from destruction of neurons, leading to secondary degeneration of axonal processes. Infections like herpes zoster and toxins like platinum compounds are examples of insults that may lead to neuronopathies. Because the damage is at the level of the neuronal cell body, peripheral nerve dysfunction caused by neuronopathies is equally likely to affect proximal and distal parts of the body (unlike peripheral axonopathies, which preferentially affect the distal extremities).

Anatomic Patterns of Peripheral Neuropathies

Peripheral neuropathies can be separated into several groups according to the anatomic distribution of involvement and the associated neurologic deficits. This approach can be helpful clinically, since each pattern has a different set of potential underlying causes. These anatomic patterns of injury are as follows:

- *Mononeuropathies* affect a single nerve and result in deficits in a restricted distribution dictated by nerve anatomy.

Trauma, entrapment, and infections are common causes of mononeuropathy.

- *Polyneuropathies* are characterized by involvement of multiple nerves, usually in a symmetric fashion. In most cases axons are affected in a length-dependent fashion leading to deficits that start in the feet and ascend with disease progression. The hands often start to show involvement by the time deficits extend to the level of the knee, resulting in a characteristic “stocking and glove” distribution of sensory deficits.
- *Mononeuritis multiplex* describes a disease process that damages individual nerves in a haphazard fashion. An affected patient might have a right wrist drop from involvement of the right radial nerve and a left foot drop from peroneal nerve damage. Vasculitis is a common cause of this pattern of injury.
- *Polyradiculoneuropathies* affect nerve roots as well as peripheral nerves, leading to diffuse symmetric symptoms in proximal and distal parts of the body.

Specific Peripheral Neuropathies

Patients with peripheral neuropathy often complain of numbness, painful “pins and needles” sensations, and weakness, most often in the distal portions of the extremities. Many different types of disease processes can damage peripheral nerves, including inflammatory diseases, infections, metabolic changes, toxic injury, trauma, paraneoplastic disease, and inherited gene defects.

Inflammatory Neuropathies

Guillain-Barré Syndrome (Acute Inflammatory Demyelinating Polyneuropathy)

Guillain-Barré syndrome is an immunologically mediated demyelinating peripheral neuropathy that may lead to life-threatening respiratory paralysis. The overall annual incidence is approximately 1 case per 100,000 persons. The disease is characterized clinically by weakness beginning in the distal limbs that rapidly advances to affect proximal muscle function (“ascending paralysis”). Histologic features are inflammation and demyelination of spinal nerve roots and peripheral nerves (radiculoneuropathy).

Pathogenesis

In most cases, Guillain-Barré syndrome is thought to be an acute-onset immune-mediated demyelinating neuropathy. Approximately two-thirds of cases are preceded by an acute, influenza-like illness from which the affected individual has recovered by the time the neuropathy becomes symptomatic. Infections with *Campylobacter jejuni*, cytomegalovirus, Epstein-Barr virus, and *Mycoplasma pneumoniae* as well as prior vaccination have significant epidemiologic associations with Guillain-Barré syndrome. No infectious agent has been demonstrated in affected nerves, and an immunologic reaction is favored as the underlying cause. A similar inflammatory disease of peripheral nerves can be reproduced in experimental animals by immunization with a peripheral nerve myelin protein. A T-cell-mediated immune response ensues, accompanied by segmental demyelination induced by the actions of activated macrophages. Transfer of these T cells to a naive animal results in comparable lesions.

Moreover, lymphocytes from individuals with Guillain-Barré syndrome have been shown to produce demyelination in tissue cultures of myelinated nerve fibers. Circulating antibodies that cross-react with components of peripheral nerves may also play a role.

MORPHOLOGY

The dominant finding in sections stained with hematoxylin and eosin is **inflammation of peripheral nerves**, manifested as perivascular and endoneurial infiltration by lymphocytes, macrophages, and a few plasma cells. Segmental demyelination affecting peripheral nerves is the most prominent lesion, but damage to axons is also seen, particularly when the disease is severe. Electron microscopy has identified an early effect on myelin sheaths. The cytoplasmic processes of macrophages penetrate the basement membrane of Schwann cells, particularly in the vicinity of the nodes of Ranvier, and extend between the myelin lamellae, stripping the myelin sheath from the axon. Ultimately, the remnants of the myelin sheath are engulfed by the macrophages. Inflammation and demyelination can be widespread in the peripheral nervous system but are typically most prominent proximally, close to the nerve roots.

Clinical Features

The clinical picture is dominated by ascending paralysis and areflexia. Deep tendon reflexes disappear early in the process. Sensory involvement, including loss of pain sensation, is often present but is usually not a prominent feature. Nerve conduction velocities are slowed because of multifocal destruction of myelin segments in many axons within a nerve. Cerebrospinal fluid (CSF) protein levels are elevated due to inflammation and altered permeability of the microcirculation within the spinal roots as they traverse the subarachnoid space. Inflammatory cells, on the other hand, remain confined to the roots; therefore, there is little or no CSF pleocytosis. Many patients spend weeks in the intensive care unit (ICU) before recovering normal function. With improved supportive respiratory care, cardiovascular monitoring, and prophylaxis against deep venous thrombosis, the mortality rate has fallen. Plasmapheresis and intravenous immunoglobulin therapy hasten recovery, apparently because these remove pathogenic antibodies and suppress immune function, respectively. However, 2% to 5% of affected patients die of respiratory paralysis, autonomic instability, cardiac arrest, or related complications, and up to 20% of hospitalized survivors suffer long-term disability.

Chronic Inflammatory Demyelinating Poly(radiculo)neuropathy

This is the most common chronic acquired inflammatory peripheral neuropathy, characterized by symmetric mixed sensorimotor polyneuropathy that persists for 2 months or more. By definition, signs and symptoms must be present for at least 2 months, but often the disease evolves over years, usually with relapses and remissions. Typically there is a symmetric, mixed sensorimotor polyneuropathy, but some patients may present with predominantly sensory or motor impairment. Clinical remissions can often be achieved with intravenous immunoglobulin, or with other immunosuppressive therapies, such as plasmapheresis, glucocorticoids,

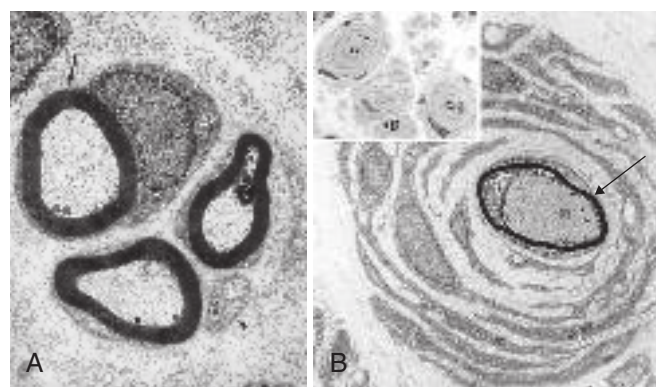


Figure 27.5 Onion bulb neuropathy. (A) shows normal myelinated axons (one that includes the nucleus of the associated Schwann cells is marked with an arrow) and unmyelinated axons (arrowhead). Compared with the normal ultrastructure of axons in a nerve (A), an “onion bulb” (B) is composed of a thinly myelinated axon (arrow) surrounded by multiple concentrically arranged Schwann cells. *Inset*, Light microscopic appearance of an onion bulb neuropathy, characterized by “onion bulbs” surrounding axons. (B, Courtesy G. Richard Dickersin, MD, from *Diagnostic Electron Microscopy: A Text Atlas*, New York, 2000, Igaku-Shoin Medical Publishers, p 984.)

and cytotoxic agents directed against T cells or B cells. The time course and the response to steroids distinguish chronic inflammatory demyelinating polyradiculoneuropathy from Guillain-Barré syndrome.

Pathogenesis

T cells as well as antibodies are implicated in the inflammatory process. Molecules expressed at the Schwann cell-axon junction and in noncompact areas of myelin appear to be the target of the immune response. Complement-fixing immunoglobulin G (IgG) and IgM can be found on the myelin sheaths, and the deposition of these opsonins leads to recruitment of macrophages that strip myelin from axons. Sural nerve biopsies show evidence of recurrent demyelination and remyelination associated with proliferation of Schwann cells. When excessive, this proliferation leads to the formation of so-called *onion bulbs*—structures in which multiple layers of Schwann cells wrap around an axon like the layers of an onion (Fig. 27.5).

Neuropathy Associated With Systemic Autoimmune Diseases

Systemic autoimmune diseases like rheumatoid arthritis, Sjögren syndrome, or systemic lupus erythematosus may be associated with peripheral neuropathies that often take the form of distal sensory or sensorimotor polyneuropathies. These neuropathies are distinct from vasculitic peripheral neuropathies, which can arise as secondary manifestations of these same diseases.

Neuropathy Associated With Vasculitis

Vasculitis is a noninfectious inflammation of blood vessels that can involve and damage peripheral nerves. About one-third of patients with vasculitis, depending on the type of vasculitis, have peripheral nerve involvement, and neuropathy may be the presenting feature. Vasculitis often presents as mononeuritis multiplex, but mononeuritis and polyneuropathy are also encountered.

When peripheral neuropathy occurs in systemic vasculitis, it is most common in the MPO-ANCA-associated vasculitides (microscopic polyangiitis and eosinophilic granulomatosis with polyangiitis/Churg-Strauss syndrome) and polyarteritis nodosa and much less common in PR3-ANCA-associated granulomatosis with polyangiitis. The most common form of vasculitis associated with peripheral neuropathy is a localized form referred to as *nonsystemic vasculitic neuropathy*, which is not associated with any anti-neutrophil cytoplasmic antibodies (ANCA). Regardless of the type of vasculitis, peripheral nerves involved by vasculitis typically show patchy axonal degeneration and loss, with some fascicles being more severely affected than others. Perivascular inflammatory infiltrates are often present. Identification of blood vessels with characteristic forms of acute or chronic damage (Chapter 11) helps establish the diagnosis.

Infectious Neuropathies

Many infectious processes affect peripheral nerves. Among these, leprosy, diphtheria, and varicella-zoster cause relatively specific pathologic changes in nerves that are the focus here. Each of these disorders is also discussed in more detail in Chapter 8.

Leprosy (Hansen Disease)

Peripheral nerves are involved in both lepromatous and tuberculoid leprosy (discussed in Chapter 8).

- In *lepromatous leprosy*, Schwann cells are invaded by *Mycobacterium leprae*, which proliferate and eventually infect other cells. There is evidence of segmental demyelination and remyelination and loss of both myelinated and unmyelinated axons. As the infection advances, endoneurial fibrosis and multilayered thickening of the perineurial sheaths occur. Affected individuals develop a symmetric polyneuropathy that is most severe in the relatively cool distal extremities and in the face because lower temperatures favor mycobacterial growth. The infection prominently involves pain fibers, and the resulting loss of sensation contributes to injury, since the patient is rendered unaware of injurious stimuli and damaged tissues. Thus, large traumatic ulcers may develop.
- *Tuberculoid leprosy* is characterized by an active cell-mediated immune response to *M. leprae* that usually manifests as dermal nodules containing granulomatous inflammation. The inflammation injures cutaneous nerves in the vicinity; axons, Schwann cells, and myelin are lost; and there is fibrosis of the perineurium and endoneurium. In tuberculoid leprosy, affected individuals have much more localized nerve involvement.

Lyme Disease

Lyme disease causes various neurologic manifestations in the second and third stages of the disease. These include polyradiculoneuropathy and unilateral or bilateral facial nerve palsies.

HIV/AIDS

Patients infected with human immunodeficiency virus (HIV) develop several patterns of peripheral neuropathy that are poorly understood, but all appear to be related in some way to immune dysregulation. Early-stage HIV infection can be associated with mononeuritis multiplex

and demyelinating disorders that may resemble Guillain-Barré syndrome or chronic inflammatory demyelinating polyradiculoneuropathy. More commonly, later stages of HIV infection are associated with a distal sensory neuropathy that is often painful.

Diphtheria

Peripheral nerve dysfunction results from the effects of the diphtheria exotoxin. It produces an acute peripheral neuropathy associated with prominent bulbar and respiratory muscle dysfunction, which can lead to death or long-term disability. The mechanism of action of diphtheria toxin is described in Chapter 8. Diphtheria is most commonly found in low income countries and is a continuing medical problem because of incomplete immunization or waning immunity in adults.

Varicella-Zoster Virus

Varicella-zoster is one of the most common viral infections of the peripheral nervous system. Following chickenpox, a latent infection persists within neurons of sensory ganglia. If the virus is reactivated, sometimes many years later, it may be transported along the sensory nerves to the skin. Here it infects keratinocytes, leading to a **painful, vesicular skin eruption in a distribution that follows sensory dermatomes (shingles)**. Most common is the involvement of thoracic or trigeminal nerve dermatomes. The factors underlying reactivation of the virus are not fully understood, but decreased cell-mediated immunity is suspected to play a role. In a small proportion of patients, weakness is also apparent in the same distributions. Affected ganglia show neuronal death, usually accompanied by abundant mononuclear inflammatory cell infiltrates; focal necrosis and hemorrhage may also be found. Peripheral nerves show degeneration of the axons that belong to the dead sensory neurons. Focal destruction of the large motor neurons of the anterior horns or cranial nerve motor nuclei may be seen at the corresponding levels. Intranuclear inclusions generally are not found in the peripheral nervous system.

Metabolic, Hormonal, and Nutritional Neuropathies

Diabetes

Diabetes is the most common cause of peripheral neuropathy. The prevalence of this complication depends on the duration of the disease; up to 50% of patients with diabetes overall and up to 80% of those who have had the disease for more than 15 years have clinical evidence of peripheral neuropathy. Patients with both type 1 and type 2 diabetes are affected (Chapter 24). Several distinct clinicopathologic patterns of diabetes-related peripheral neuropathy are recognized (described later), but the most common by far is an ascending distal symmetric sensorimotor polyneuropathy.

Pathogenesis

The mechanism of diabetic neuropathy is complex and not completely resolved; both metabolic and secondary vascular changes are believed to contribute to the damage of axons and Schwann cells. Hyperglycemia causes the nonenzymatic glycosylation of proteins, lipids, and nucleic acids. The resulting advanced glycosylation end-products (AGEs)

may interfere with normal protein function and activate inflammatory signaling through the receptor for the AGE. Excess glucose within cells is reduced to sorbitol, a process that depletes NADPH and increases intracellular osmolality. These and other metabolic disturbances may predispose peripheral nerves to injury by reactive oxygen species. In addition, the vascular injuries that occur in chronic diabetes due to hyperlipidemia and other metabolic alterations may cause ischemic damage of the nerves.

MORPHOLOGY

In individuals with a distal symmetric sensorimotor neuropathy, the predominant pathologic finding is an axonal neuropathy. Nerve biopsies show reduced numbers of axons. Variable degrees of ongoing axonal damage, marked by degenerating myelin sheaths and regenerative axonal clusters, may be present. Endoneurial arterioles show thickening, hyalinization, and intense periodic acid–Schiff positivity of their walls and extensive reduplication of basement membranes (Fig. 27.6).

Clinical Features

Distal symmetric diabetic polyneuropathy typically presents with sensory symptoms like numbness, loss of pain sensation, difficulty with balance, and paresthesias or dysesthesias. Paresthesias or dysesthesias (abnormal painful sense of touch) are so-called “positive” symptoms—painful sensations that result from abnormal discharges of damaged nerves. Neuropathy leads to considerable morbidity, in particular an increased susceptibility to foot and ankle fractures and chronic skin ulcers, which may eventually lead to amputations.

Another manifestation is dysfunction of the autonomic nervous system; this affects 20% to 40% of individuals with diabetes mellitus, nearly always in association with a distal sensorimotor neuropathy. Diabetic autonomic neuropathy has protean manifestations, including postural hypotension, incomplete emptying of the bladder (resulting in recurrent infections), and sexual dysfunction. Some affected individuals, especially older adults with a long history of diabetes, develop a peripheral neuropathy that manifests with asymmetric presentations, including *mononeuropathy*,

cranial neuropathy, and *radiculoplexus neuropathy*. The last-mentioned is a devastatingly painful acute disorder that presents in the distribution of the brachial or lumbosacral nerve plexus. It is often monophasic and can improve over several months. These asymmetric manifestations may be caused by microvascular disease.

Other Metabolic, Hormonal, and Nutritional Neuropathies

A diverse group of metabolic, hormonal, and nutritional disorders are associated with peripheral neuropathy, including the following:

- **Uremic neuropathy.** Most individuals with renal failure have a peripheral neuropathy. Typically this is a distal, symmetric neuropathy that may be asymptomatic or may be associated with muscle cramps, distal dysesthesias, and diminished deep tendon reflexes. In these patients, axonal degeneration is the primary event; occasionally there is secondary demyelination. Regeneration and recovery are common after dialysis.
- **Thyroid dysfunction.** Hypothyroidism can lead to compression mononeuropathies such as carpal tunnel syndrome or cause a distal symmetric predominantly sensory polyneuropathy. In rare cases, hyperthyroidism is associated with a neuropathy resembling Guillain-Barré syndrome.
- **Vitamin B₁₂ (cyanocobalamin) deficiency** classically results in subacute combined degeneration with damage to long tracts in the spinal cord (Chapter 28) and also peripheral nerves.
- **Deficiencies of vitamin B₁ (thiamine), vitamin B₆ (pyridoxine), folate, copper, and zinc** all have been associated with peripheral neuropathy.

Toxic Neuropathies

Peripheral neuropathies may appear after exposure to industrial or environmental chemicals, biologic toxins, or therapeutic drugs. Important causes of toxic peripheral nerve damage include alcohol (independent of associated nutritional deficiencies), heavy metals (lead, mercury, arsenic, and thallium), and organic solvents. Various medications can cause toxic nerve damage, but the most notorious are chemotherapeutic agents. These include vinca alkaloids and taxanes, microtubule inhibitors that interfere with axonal transport, and cisplatin, which may cause a neuronopathy.

Neuropathies Associated With Malignancy

Neuropathies associated with cancers may stem from local effects, complications of therapy, paraneoplastic effects, or (in the case of B-cell tumors) tumor-derived immunoglobulins.

- **Direct infiltration or compression of peripheral nerves** by tumors is a common cause of mononeuropathy and may be a presenting symptom of cancer. These neuropathies include brachial plexopathy from neoplasms of the apex of the lung, obturator palsy from pelvic malignant neoplasms, and cranial nerve palsies from intracranial tumors or tumors of the base of the skull. A polyradiculopathy involving the lower extremity may develop when the cauda equina is involved by meningeal carcinomatosis.
- **Paraneoplastic neuropathies.** These can occur at any time during the patient's course, but often precede the diagnosis of the underlying tumor. Sensorimotor neuropathy is the

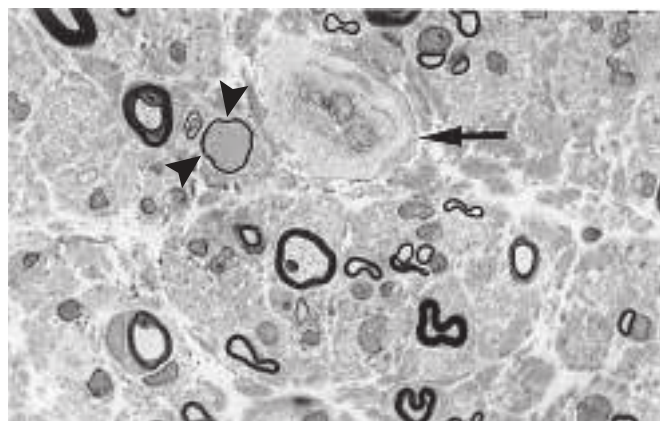


Figure 27.6 Diabetic neuropathy with marked loss of myelinated fibers, a thin myelinated fiber (arrowheads), and thickening of endoneurial vessel wall (arrow).

most common paraneoplastic form, but a pure sensory neuronopathy, chronic inflammatory demyelinating polyneuropathy, plexopathy, and autonomic neuropathy may also be seen. Paraneoplastic sensory neuronopathy is most commonly associated with small cell lung cancer. Antibodies that recognize proteins expressed by cancer cells and normal neurons (for example, anti-Hu antibodies) are often present, but the damage appears to be mediated by a CD8+ cytotoxic T-cell attack on dorsal root ganglion cells. Sensory symptoms usually start distally in an asymmetric and multifocal pattern. Other patients with anti-CV2 autoantibodies (which recognize CRMP5, an intracellular signaling protein) tend to present with a painful axonal asymmetric sensorimotor neuropathy.

- *Neuropathies associated with monoclonal gammopathies.* Neoplastic B cells may secrete monoclonal immunoglobulins or immunoglobulin fragments (so-called paraproteins) that damage nerves. For example, tumors that secrete IgM immunoglobulin may be associated with a demyelinating peripheral neuropathy. In most cases, the pathogenic IgM paraprotein is thought to bind directly to myelin-associated antigens such as myelin-associated glycoprotein. Deposition of IgM can be seen ultrastructurally between the membrane layers of the myelin sheath. IgG or IgA paraproteins may also be associated with peripheral neuropathy. One distinctive presentation is *POEMS syndrome* (polyneuropathy, organomegaly, endocrinopathy, monoclonal gammopathy, and skin changes), in which patients develop a demyelinating neuropathy associated with deposition of paraprotein between noncompacted myelin lamellae. Finally, excess immunoglobulin light chain may deposit as amyloid (Chapter 6), which can lead to peripheral neuropathy due to vascular insufficiency or a direct toxic effect.

Neuropathies Caused by Physical Forces

Peripheral nerves are commonly injured by trauma or entrapment.

- *Lacerations* result from cutting injuries and from sharp fragments of fractured bone, both of which may sever a nerve.
- *Avulsion* of a nerve may occur when tension is applied, often to one of the limbs.
- *Compression neuropathy (entrapment neuropathy)* occurs when a peripheral nerve is chronically subjected to increased pressure, often within an anatomic compartment. Carpal tunnel syndrome, the most common entrapment neuropathy, results from compression of the median nerve at the level of the wrist within the compartment delimited by the transverse carpal ligament. Women are more commonly affected than men, and the problem is frequently bilateral. The disorder may be observed in association with many conditions including tissue edema, pregnancy, inflammatory arthritis, hypothyroidism, amyloidosis (especially related to β_2 -microglobulin deposition in individuals on renal dialysis), acromegaly, diabetes mellitus, and excessive repetitive motions of the wrist. Symptoms are limited to dysfunction of the median nerve and typically include numbness and paresthesias of the tips of the thumb and first two digits. Other nerves prone to compression neuropathies include the ulnar nerve at the level of the elbow, the peroneal nerve at

the level of the knee, and the radial nerve in the upper arm; the latter occurs from sleeping with the arm in an awkward position ("Saturday night palsy"). Another form of compression neuropathy is found in the foot, affecting the interdigital nerve at intermetatarsal sites. This problem, which occurs more often in women than in men, leads to foot pain (metatarsalgia) and is associated with a histologic lesion called a *Morton neuroma*, which is marked by perineural fibrosis.

Inherited Peripheral Neuropathies

Inherited peripheral neuropathies are a group of genetically diverse disorders with overlapping clinical phenotypes that often present in adults. Because of the delayed onset, the possibility of an inherited neuropathy has to be considered in the differential diagnosis for any patient who presents with a peripheral neuropathy. The major types of inherited peripheral neuropathies include (1) hereditary motor and sensory neuropathies, sometimes known as *Charcot-Marie-Tooth* (CMT) disease, described below, which is a genetically heterogeneous disorder caused by mutations in genes that encode proteins involved in the structure and function of either the peripheral nerve axon or the myelin sheath; (2) hereditary motor neuropathies; (3) hereditary sensory neuropathies, with or without autonomic neuropathy; and (4) other inherited conditions causing neuropathy, including familial amyloidosis and inherited metabolic diseases.

Historically, these diseases were classified based on their inheritance pattern and clinical features. Now there is a continuously growing list of altered genes that are linked to these diseases. The complexity of the genetics of inherited neuropathies is no doubt a reflection of the complicated homeostatic mechanisms that sustain normal peripheral nerve function. There is no simple unifying concept tying all the implicated genes together; some of the implicated genes are illustrated in [Fig. 27.1](#). Below, some of the more common and distinctive types of inherited peripheral neuropathies are described in brief.

Hereditary Motor and Sensory Neuropathies/Charcot-Marie-Tooth Disease

These are by far the most common inherited peripheral neuropathies, affecting up to 1 in 2500 people. The initial description of these disorders, based on clinical features, was deceptively simple—an inherited disease associated with distal muscle atrophy, sensory loss, and foot deformities. It is now appreciated that this clinical phenotype can result from mutations in more than 50 different genes. Current systems classify hereditary motor and sensory neuropathies based on the mode of inheritance and the pattern of injury (e.g., axonal or demyelinating). Demyelinating forms of CMT disease are associated with morphologic features of demyelination and remyelination including Schwann cell hyperplasia and onion bulb formation, which may be so severe that the involved nerve is palpably enlarged. Listed are the four most common types, which together account for over 90% of patients with CMT disease:

- *CMT1 encompasses a group of autosomal dominant demyelinating neuropathies* that collectively are the most common subtype of hereditary motor and sensory neuropathy. *CMT1A* accounts for some 55% of genetically defined CMT. It is caused by a duplication of a region on

chromosome 17 that includes the peripheral myelin protein 22 (*PMP22*) gene. The disease usually presents in the second decade of life as a slowly progressive distal demyelinating motor and sensory neuropathy. *CMT1B* is also a demyelinating neuropathy caused by mutations in the myelin protein zero gene (*MPZ*) and accounts for about 9% of genetically defined cases of CMT.

- *CMTX* encompasses X-linked forms of CMT disease. *CMT1X* is the most common of these, accounting for 15% of genetically defined cases of CMT. It is linked to mutations in the *GJB1* gene, which encodes connexin32, a gap junction component that is expressed in Schwann cells.
- *CMT2* includes autosomal dominant neuropathies associated with axonal rather than demyelinating injury. *CMT2A* is the most common subtype, accounting for 4% of all CMT disease. It is caused by mutations in the *MFN2* gene, which is required for normal mitochondrial fusion. The phenotype is typically severe, with disease onset in early childhood.

Hereditary Sensory Neuropathies With or Without Autonomic Neuropathy

This is a diverse group of diseases marked by loss of sensation and variable autonomic disturbances, but with preserved motor strength. Loss of pain and temperature sensation is the most common symptom. The inability to sense pain leads to traumatic injury of the hands and feet. These are typically axonal neuropathies.

Familial Amyloid Polyneuropathies

These are hereditary disorders characterized by amyloid deposition within peripheral nerves. Most are caused by germline mutations of the transthyretin gene. The mutated transthyretin protein, which is normally involved in serum binding and transport of thyroid hormone, is prone to deposit as amyloid fibrils in a number of tissues including peripheral nerve (Chapter 6). The clinical presentation is similar to that of hereditary sensory and autonomic neuropathies.

KEY CONCEPTS

PERIPHERAL NEUROPATHIES

- Anatomic patterns include mononeuropathy, mononeuritis multiplex, polyneuropathy, and polyradiculoneuropathy.
- Damage may occur primarily in Schwann cells (demyelinating neuropathy), axons (axonal neuropathy), or central neurons (neuronopathy); mixed patterns of injury occur.
- Inflammatory disease, infections, metabolic changes, toxic injury, trauma, paraneoplastic disorders, and inherited gene defects all can cause peripheral neuropathy.
- Diabetes mellitus is the most common cause of peripheral neuropathy, which most often presents as a distal symmetric neuropathy.
- Guillain-Barré syndrome and chronic inflammatory demyelinating polyradiculoneuropathy are the major acute and chronic acquired demyelinating peripheral neuropathies, respectively.
- Inherited peripheral neuropathies are genetically and phenotypically diverse disorders that often present in adulthood and may be marked by sensory, motor, or autonomic dysfunction, alone or in combination.

DISEASES OF THE NEUROMUSCULAR JUNCTION

The neuromuscular junction is a complex specialized structure located at the interface of motor nerve axons and skeletal muscle that serves to control muscle contraction. Neuromuscular junctions are found midway along the length of myofibers. Here, the distal ends of peripheral motor nerves branch into small processes that terminate in bulbous synaptic boutons. Upon depolarization, these presynaptic nerve terminals release acetylcholine (ACh) into the synaptic cleft, the space separating the nerve endings from the myofiber membrane (referred to as the *sarcolemma*). The postsynaptic sarcolemma is characterized by complex infoldings and exhibits distinct specializations with localized clustering of ACh receptors. These receptors are responsible for the initiation of signals leading to muscle contraction.

Regardless of cause, patients with disorders that impair the function of neuromuscular junctions complain of painless weakness and fatigue. Autoantibodies that recognize key neuromuscular junction proteins are the most common cause of disrupted neuromuscular transmission, as found in *myasthenia gravis* (literally, “grave weakness”). Understandably, inherited defects in specialized neuromuscular junction proteins are also associated with myasthenic syndromes. Disorders caused by toxins that alter neuromuscular transmission are rarely encountered, but had an important role historically in elucidating how the neuromuscular junction functions.

Antibody-Mediated Diseases of the Neuromuscular Junction

Myasthenia Gravis

Myasthenia gravis is an autoimmune disease that is usually associated with autoantibodies directed against ACh receptors. It has a prevalence of 150 to 200 per 1 million and shows a bimodal age distribution. In young adults, the female-to-male ratio is 2:1, but in older adults there is a male predominance.

Pathogenesis

About 85% of patients have autoantibodies against postsynaptic ACh receptors, while most of the remaining patients have antibodies against the sarcolemmal protein muscle-specific receptor tyrosine kinase. These autoantibodies appear to be pathogenic, as the disease can be passively transferred to animals with serum from affected individuals, and therapeutic maneuvers that decrease autoantibody levels are associated with a reduction in symptoms.

The mechanism of action of the various autoantibodies appears to differ. Anti-ACh receptor antibodies are thought to lead to the aggregation and degradation of the receptors, as well as to damage of the postsynaptic membrane through complement fixation. As a result, postsynaptic membranes show alterations in morphology and are depleted of ACh receptors. This limits the ability of myofibers to respond to ACh. Autoantibodies directed against muscle-specific receptor tyrosine kinase do not fix complement. Instead, these antibodies seem to interfere with the trafficking and

clustering of ACh receptor within the sarcolemmal membrane, the net effect again being decreased ACh receptor function.

There is a strong association between pathogenic anti-ACh receptor autoantibodies and thymic abnormalities. Approximately 10% of patients with myasthenia gravis have a *thymoma*, a tumor of thymic epithelial cells (Chapter 13). An additional 30% of patients (and particularly young patients) have a different thymic abnormality called thymic hyperplasia. This peculiar condition is marked by the appearance of B-cell follicles in the thymus. The thymus normally contains small numbers of myoid cells, stromal cells that express skeletal muscle antigens. It is hypothesized that both thymoma and thymic hyperplasia disrupt normal thymic function in a manner that promotes autoimmunity against ACh receptors expressed on thymic myoid cells. In contrast, thymic abnormalities are usually absent in cases of myasthenia gravis that occur in older patients; the basis for the development of autoantibodies in such cases is unknown.

Clinical Features

Patients with anti-ACh receptor antibodies typically present with fluctuating weakness that worsens with exertion and often over the course of the day. Diplopia and ptosis due to involvement of extraocular muscles are common and distinguish myasthenia gravis from myopathies, in which involvement of extraocular muscles is unusual. In some patients, symptoms are confined to ocular muscles, while others develop generalized weakness that can be so severe as to require mechanical ventilation. Cases with antibodies against muscle-specific receptor tyrosine kinase differ from typical cases by exhibiting more focal muscle involvement (neck, shoulder, facial, respiratory, and bulbar muscles).

Diagnosis is based on clinical history, physical findings, identification of autoantibodies, and electrophysiologic studies. Electrophysiologic studies reveal a decrement in muscle response with repeated stimulation, a characteristic of this disorder. Overall mortality has dropped from over 30% in the 1950s to less than 5% with current therapies. Acetylcholinesterase inhibitors that increase the half-life of ACh are the first line of treatment. Other treatments, such as plasmapheresis and immunosuppressive drugs (e.g., glucocorticoids, cyclosporine, rituximab), can bring symptoms under control by decreasing autoantibody production. Thymectomy is often effective in patients with thymoma, but is of uncertain benefit in those with thymic hyperplasia or lacking thymic abnormalities.

Lambert-Eaton Myasthenic Syndrome

Lambert-Eaton myasthenic syndrome is an autoimmune disorder caused by antibodies that block ACh release by inhibiting a presynaptic calcium channel. In contrast to myasthenia gravis, rapid repetitive stimulation increases muscle response. Muscle strength is augmented after a few seconds of muscle activity. Patients typically present with weakness of their extremities. In about half of cases there is an underlying malignancy, most often neuroendocrine carcinoma of the lung. Symptoms may precede the diagnosis of cancer, sometimes by years. It is thought that the stimulus for autoantibody formation in paraneoplastic cases may be the expression of the same calcium channel in the neoplastic

cells. Patients without cancer often have other autoimmune diseases, such as vitiligo or thyroid disease. Treatment consists of drugs that increase ACh release by depolarizing synaptic membranes and immunosuppressive agents, such as those used to treat myasthenia gravis.

Congenital Myasthenic Syndromes

These rare disorders, which most commonly have an autosomal recessive mode of inheritance, are marked by varying degrees of muscle weakness. Causative mutations have been identified in genes encoding several different presynaptic, synaptic, or postsynaptic proteins. The most common of these are loss-of-function mutations in the gene encoding the ϵ -subunit of the ACh receptor. Another group of mutations affect proteins that are important in normal clustering of ACh receptors on postsynaptic membranes. Many patients with congenital myasthenic syndromes present in the perinatal period with poor muscle tone, external eye muscle weakness, and breathing difficulties, but others have milder forms of the disease and may not come to clinical attention until adolescence or adulthood. The clinical presentation, response to drugs such as acetylcholinesterase inhibitors, and prognosis depend largely on the underlying mutation.

Disorders Caused by Toxins

Botulism is caused by exposure to a neurotoxin (botulinum toxin, popularly known as Botox) that is produced by the anaerobic gram-positive organism *Clostridium botulinum*. Botox acts by blocking the release of ACh from presynaptic neurons (Chapter 8). Curare is the common name for plant-derived muscle relaxants that block ACh receptors, resulting in flaccid paralysis. It was initially discovered and used as poison on arrow tips by indigenous people in the Amazon rain forest. At one time it was used as a muscle relaxant during certain forms of surgery, but has now been supplanted by other related drugs with a similar mechanism of action.

KEY CONCEPTS

DISEASES OF THE NEUROMUSCULAR JUNCTION

- Disorders of neuromuscular junctions present with painless weakness.
- Myasthenia gravis and Lambert-Eaton myasthenic syndrome, the most common forms, are both immune mediated, being caused by antibodies to postsynaptic ACh receptors and presynaptic calcium channels, respectively.
- Myasthenia gravis is often associated with thymic hyperplasia or thymoma, frequently involves ocular muscles, and is marked by fluctuating weakness that worsens with exertion.
- Lambert-Eaton myasthenic syndrome presents with weakness in the extremities that improves with repetitive stimulation and is often a paraneoplastic disorder associated with lung cancer.
- Genetic defects in neuromuscular junction proteins give rise to congenital myasthenic syndromes.
- Biologic toxins such as Botox can block neuromuscular transmission by blocking the release of ACh from presynaptic neurons.

DISEASES OF SKELETAL MUSCLE

Skeletal muscle has unique structural, cellular, and molecular characteristics and accordingly unique patterns of injury and repair. During embryogenesis, skeletal muscle develops through the fusion of mononucleated precursor cells (myoblasts) into multinucleated myotubes. These subsequently mature into myofibers (muscle fibers) of varying length that contain thousands of nuclei. In adult tissues, these myofibers are arranged in fascicles, each associated with a small pool of tissue stem cells referred to as satellite cells, which can contribute to muscle regeneration following injury (outlined later). Myofibers are of two main functional types, type I and type II (Table 27.1), which are admixed in a checkerboard pattern in normal skeletal muscle. Fiber type is determined by signals received from innervating motor neurons, and as a result all fibers that are part of a motor unit are of the same type.

Skeletal Muscle Atrophy

Skeletal muscle atrophy is a common feature of many disorders. Loss of innervation, disuse, cachexia, old age, and primary myopathies all can produce muscle atrophy and, if the atrophy is severe, loss of muscle mass. Although muscle atrophy may occur under many conditions, certain patterns of atrophy are suggestive of specific underlying etiologies:

- *Clusters or groups of atrophic fibers* are seen in neurogenic disease (Fig. 27.7).
- *Perifascicular atrophy* is seen in dermatomyositis (see later).
- *Type II fiber atrophy* with sparing of type I fibers is seen with prolonged corticosteroid therapy or disuse.

Table 27.1 Muscle Fiber Types

	Type I	Type II
Action	Sustained force	Fast movement
Activity type	Aerobic exercise	Anaerobic exercise
Power produced	Low	High
Resistance to fatigue	High	Low
Lipid content	High	Low
Glycogen content	Low	High
Energy metabolism	Low glycolytic capacity, high oxidative capacity	High glycolytic capacity, low oxidative capacity
Mitochondrial density	High	Low
Enzyme activity	NADH-TR, dark staining ATPase at pH 4.3, dark staining ATPase at pH 9.4, light staining	NADH-TR, light staining ATPase at pH 4.3, light staining ATPase at pH 9.4, dark staining
Myosin heavy chain gene expressed	MYH7	MYH2, MYH4, MYH1
Color	Red (high myoglobin content)	Pale red/tan (low myoglobin content)
Prototype	Soleus (pigeon)	Pectoral (pigeon)

ATPase, Adenosine triphosphatase; NADH-TR, nicotinamide adenine dinucleotide, reduced form, tetrazolium reductase.

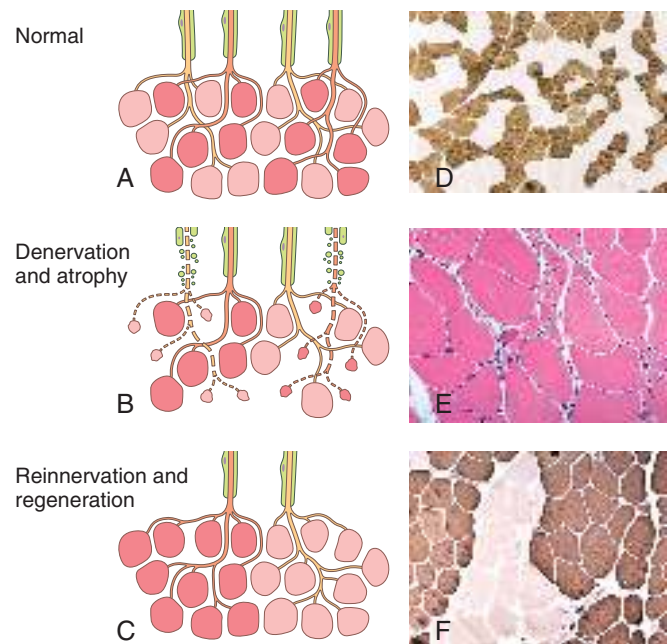


Figure 27.7 (A) This diagrammatic representation of four normal motor units shows a normal checkerboard-type admixture of light- and dark-stained fibers of opposite type. (B) Damage to innervating axons leads to loss of trophic input and atrophy of myofibers. (C) Reinnervation of myofibers can lead to a switch in fiber type and segregation of fibers of like type. As illustrated here, reinnervation is also often associated with an increase in motor unit size, with more myofibers innervated by an individual axon. (D) Normal muscle has a checkerboard-type distribution of type I (light) and type II (dark) fibers on this adenosine triphosphatase reaction (pH 9.4) corresponding to findings in diagram in A. (E) Clustered flattened, "angulated" atrophic fibers (*group atrophy*) are a typical finding associated with disrupted innervation. (F) With ongoing denervation and reinnervation, large clusters of fibers appear that all share the same fiber type (*type grouping*), corresponding to diagram in C.

Neurogenic and Myopathic Changes in Skeletal Muscle

Disorders of skeletal muscles may be caused by direct injury to myofibers (myopathic injury) or by disruption of muscle innervation (neurogenic injury). Each type is described next.

Neurogenic injuries lead to *fiber type grouping* and *grouped atrophy* (see Fig. 27.7), both of which stem from the disruption of muscle innervation. The key to understanding these abnormalities is to recognize that muscle fiber type is determined by the innervating motor neuron. Following denervation, myofibers undergo atrophy, often assuming a flattened, angulated shape. Reinnervation restores fiber size and shape, but may make a denervated myofiber part of a different motor unit, in turn leading to a change in fiber type. In the face of ongoing axonal or neuronal damage and dropout, residual motor axons may innervate increasingly larger numbers of myofibers, leading to enlargement of motor units, each comprised of a single type of muscle fiber (fiber type grouping). These large motor units are also susceptible to grouped atrophy if the innervating axon is damaged.

In contrast, most primary myopathic injuries are associated with a distinct set of morphologic changes that include the following:

- *Segmental myofiber degeneration and regeneration* is seen when only part of a myofiber undergoes necrosis. Degeneration is associated with release of cytoplasmic enzymes such as creatine kinase into the blood, making these useful markers of muscle damage. The sarcomeres and other components of the damaged myofiber segment are engulfed by macrophages in a process termed myophagocytosis. Removal of the degenerated debris sets the stage for regeneration. Fusion of activated satellite cells to damaged myofibers is an important step for regeneration. Eventually, new sarcomeres are generated, and the continuity of the original myofiber is restored. Regenerating myofibers are rich in RNA and therefore blue (basophilic) in hematoxylin and eosin-stained sections. They have enlarged nuclei with prominent nucleoli that are often randomly distributed in the cytoplasm, instead of occupying their normal subsarcolemmal location. Depending on the nature of the primary insult, atrophic myofibers may also be seen. Regeneration can restore normal muscle following an acute, transient injury, but in chronic disease states regeneration often fails to keep pace with damage. In this setting, muscles often show endomysial fibrosis (collagen deposition), dropout of myofibers, and fatty replacement.
- *Myofiber hypertrophy* can be seen as a physiologic adaptation to exercise or in association with certain chronic myopathic conditions.
- *Cytoplasmic inclusions* in the form of vacuoles, aggregates of proteins, or clustered organelles are characteristic of several primary forms of myopathy.

Inflammatory Myopathies

Historically, dermatomyositis, polymyositis, and inclusion body myositis have been considered the three main primary inflammatory myopathies:

- *Dermatomyositis* as a disease that affects skin as well as skeletal muscles. The histologic hallmark, discussed below, is perifascicular atrophy.
- *Inclusion body myositis*, as a slowly progressive disease associated with distinct inclusions termed “rimmed vacuoles” with some similarities to neurodegenerative diseases of the CNS.
- *Polymyositis*, to some extent a diagnosis of exclusion, as a T cell-mediated autoimmune disease affecting skeletal muscles that lacks the features of dermatomyositis or inclusion body myositis.

In addition to the triad mentioned above, other entities have been described and updated classifications have been proposed, but the old historical triad is still used by many physicians. In a more granular classification system, many cases traditionally viewed as polymyositis are now regarded as immune-mediated necrotizing myopathy (IMNM), or as a connective tissue disease-associated myositis. With that, polymyositis is a less common diagnosis than in the past. The systemic connective tissue disorders that can affect skeletal muscle include systemic lupus erythematosus, systemic sclerosis, and sarcoidosis (Chapter 6). Rarely certain

infectious agents can cause inflammation of skeletal muscle (Chapter 8).

Dermatomyositis

Dermatomyositis is a systemic autoimmune disease that typically presents with proximal muscle weakness and skin changes.

Pathogenesis

Dermatomyositis is an immunologic disease in which damage to small blood vessels contributes to muscle injury.

The vasculopathic changes can be seen as *telangiectasias* (dilated capillary loops) in the nail folds, eyelids, and gums and as dropout of capillary vessels in skeletal muscle. Biopsies of muscle and skin may show deposition of the complement membrane attack complex (C5b-9) within capillary beds. An inflammatory signature enriched for genes that are upregulated by type I interferons is seen in muscle and in leukocytes. The prominence of this signature appears to correlate with disease activity. Direct immunologic injury to the muscle fibers may also play a role. Various autoantibodies are detected by serologic studies, and B lymphocytes as well as plasma cells are part of the inflammatory infiltrate that is seen in muscles. Certain autoantibodies detected in the circulation tend to be associated with specific clinical features:

- *Anti-Mi2 antibodies* (directed against a helicase implicated in nucleosome remodeling) show a strong association with prominent Gottron papules and heliotrope rash (described later).
- *Anti-Jo1 antibodies* (directed against the enzyme histidyl t-RNA synthetase) are associated with interstitial lung disease, nonerosive arthritis, and a skin rash described as “mechanic’s hands.” In some classification schemes myopathies with anti-Jo1 antibodies are described separately as “anti-synthetase syndrome associated myositis.”
- *Anti-P155/P140 antibodies* (directed against several transcriptional regulators) are associated with paraneoplastic and juvenile cases of dermatomyositis.

A direct link between these autoantibodies and disease pathogenesis has not yet been established.

MORPHOLOGY

Muscle biopsies of affected patients show infiltrates of mononuclear inflammatory cells that tend to be most pronounced in the perimysial connective tissue and around blood vessels. Sometimes there is a distinctive pattern in which myofiber atrophy is accentuated at the edges of the fascicles—*perifascicular atrophy* (Fig. 27.8B). Segmental fiber necrosis and regeneration may also be seen. Immunohistochemical studies may identify an infiltrate rich in CD4+ T-helper cells and the deposition of C5b-9 in capillary vessels. Electron microscopic studies may show tubuloreticular endothelial cell inclusions, a feature of a number of inflammatory disorders that are linked to a type I interferon response.

Clinical Features

Muscle weakness is slow in onset, symmetric, and often accompanied by myalgias. It typically affects the proximal muscles first. As a result, tasks such as getting up from a

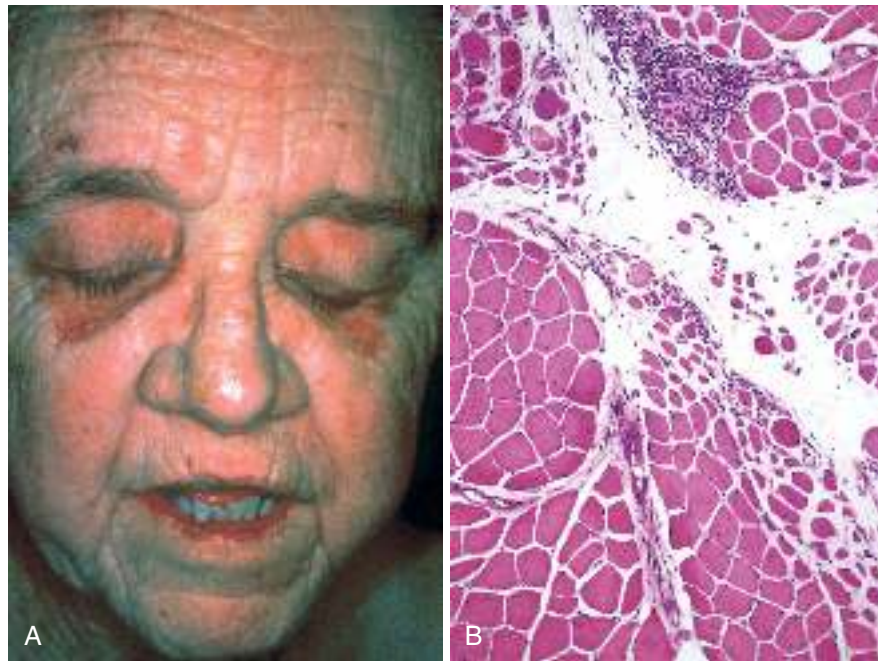


Figure 27.8 (A) Dermatomyositis. Note the heliotrope rash affecting the eyelids. (B) Dermatomyositis. The histologic appearance of muscle shows perifascicular atrophy of muscle fibers and inflammation. (Courtesy Dr. Dennis Burns, Department of Pathology, University of Texas Southwestern Medical School, Dallas, Tex.)

chair and climbing steps become increasingly difficult. Fine movements controlled by distal muscles are affected only late in the disease. Associated myopathic changes on electrophysiologic studies and elevation in serum creatine kinase levels are reflective of muscle damage. Various rashes are described in dermatomyositis, but the most characteristic ones are a lilac-colored discoloration of the upper eyelids (*heliotrope rash*) associated with periorbital edema (Fig. 27.8A) and a scaling erythematous eruption or dusky red patches over the knuckles, elbows, and knees (*Gotttron papules*). Dysphagia resulting from involvement of oropharyngeal and esophageal muscles occurs in one-third of the affected individuals, and another 10% of patients have interstitial lung disease, which can sometimes be rapidly progressive and lead to death. Cardiac involvement is common, but rarely leads to cardiac failure.

Dermatomyositis may occur in adults or in children. The average age of onset of juvenile dermatomyositis is 7 years, whereas adult cases tend to present from the fourth to sixth decade of life. Dermatomyositis is the most common inflammatory myopathy in children. Compared to adult disease, childhood disease is more likely to be associated with calcinosis and lipodystrophy and less likely to be associated with myositis-specific antibodies, cardiac involvement, interstitial lung disease, or an underlying malignancy. As might be expected based on these differences, the overall prognosis is better in children than in adults. From 15% to 24% of adult patients have an associated malignancy, and in such patients, dermatomyositis may be viewed as a paraneoplastic disorder. A dermatomyositis-like picture with myositis and rash has also been described following treatment of cancer with check-point inhibitors (Chapter 7).

Immune-Mediated Necrotizing Myopathy (IMNM)

IMNM represents an autoimmune disease that is often associated with distinct autoantibodies (sometimes also referred to as necrotizing autoimmune myopathy). Affected patients present with subacute muscle weakness that is typically associated with significantly increased creatine kinase levels. Muscle biopsies show fairly prominent myofiber necrosis and regeneration in most cases, while inflammatory cell infiltrates are usually absent or minimal despite the autoimmune nature of the disease. In many patients, IMNM is associated with autoantibodies against HMG-CoA reductase. Their formation is often attributed to prior statin exposure, but some patients develop these autoantibodies without prior use of these medications. In some patients IMNM has been linked to antibodies to the signal recognition particle

Polymyositis

Polymyositis is an adult-onset inflammatory myopathy that shares myalgia and weakness with dermatomyositis but lacks its distinctive cutaneous features and is therefore to some degree a diagnosis of exclusion. Descriptions of other entities have made cases of polymyositis less common as outlined above.

Pathogenesis

The pathogenesis of polymyositis is uncertain, but it is believed to have an immunologic basis. CD8+ cytotoxic T cells are a prominent part of the inflammatory infiltrate in affected muscle, and it is hypothesized that these cells are the mediators of tissue damage. Unlike dermatomyositis, vascular injury does not have a major role in the pathogenesis of polymyositis.

MORPHOLOGY

Mononuclear inflammatory cell infiltrates are present, but in contrast to dermatomyositis, these are usually endomysial in location. Sometimes myofibers with otherwise normal morphology appear to be invaded by mononuclear inflammatory cells, predominantly CD8+ T cells. Degenerating necrotic, regenerating, and atrophic myofibers are typically found in a random or patchy distribution. The perifascicular pattern of atrophy that is characteristic of dermatomyositis is absent.

Inclusion Body Myositis

Inclusion body myositis is a disease of late adulthood that typically affects patients older than 50 years and is the most common inflammatory myopathy in patients older than age 65 years. Most affected individuals present with slowly progressive muscle weakness that tends to be most severe in the quadriceps and the distal upper extremity muscles. Dysphagia from esophageal and pharyngeal muscle involvement is not uncommon. Laboratory studies usually show modestly elevated creatine kinase levels; most myositis-associated autoantibodies are absent, although an antibody to cytosolic 5'-nucleotidase 1A (cN1A) has recently been described. It is present in about half of the patients and is a useful marker of the disease.

MORPHOLOGY

Inclusion body myositis has a number of features that are similar to those found in polymyositis, including:

- Patchy, often endomysial mononuclear inflammatory cell infiltrates rich in CD8+ T cells
- Increased sarcolemmal expression of major histocompatibility complex class I antigens
- Focal invasion of normal-appearing myofibers by inflammatory cells
- Admixed degenerating and regenerating myofibers

Other associated changes, however, are more typical or even specific for inclusion body myositis, as follows:

- Abnormal cytoplasmic inclusions described as “rimmed vacuoles” (Fig. 27.9)
- Tubofilamentous inclusions in myofibers, seen by electron microscopy
- Cytoplasmic inclusions containing proteins typically associated with neurodegenerative diseases, like beta-amyloid, TDP-43, and ubiquitin
- Endomysial fibrosis and fatty replacement, reflective of a chronic disease course

Whether inclusion body myositis is indeed an inflammatory condition or a degenerative process with secondary inflammatory changes remains an unresolved question. It has certain features in common with polymyositis, as discussed earlier. On the other hand, it shares some features with neurodegenerative diseases, such as the presence of abnormal protein aggregates. Furthermore, there are several familial inclusion body myopathies that are also associated with chronic myopathic changes and rimmed vacuoles. These typically lack any associated inflammation—hence the designation *inclusion body “myopathy”* rather than “myositis.” Inflammatory myopathies are treated with immunosuppressive drugs such as steroids, azathioprine, and in some

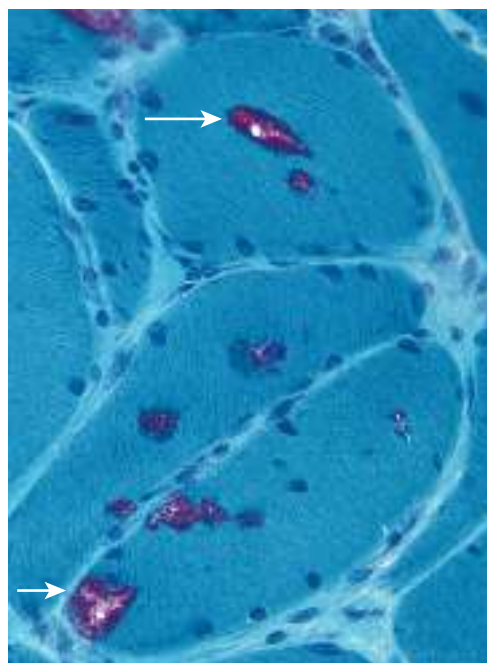


Figure 27.9 Inclusion body myositis, showing myofibers containing rimmed vacuoles—inclusions with reddish granular rimming (arrows). Modified Gomori trichrome stain.

cases intravenous immunoglobulins (IVIG). Such treatments have improved the prognosis of dermatomyositis and polymyositis, but inclusion body myositis responds poorly. This is another feature that argues against an inflammatory or immune origin of inclusion body myositis.

Toxic Myopathies

Toxic myopathies can be caused by prescription or recreational drugs or by certain hormonal imbalances. Among prescription drugs, *statins* are among the leading culprits. Statins are cholesterol-lowering drugs that are widely used to reduce the risks of acute ischemic cardiac events and stroke. **Myopathy is the most common complication of statins** (e.g., atorvastatin, simvastatin, pravastatin). Pure toxic injury is in part linked to dose and statin type and has to be distinguished from IMNM, which is often caused by statin-induced autoantibodies.

Chloroquine and hydroxychloroquine were originally used as antimalarial agents and are currently given as long-term therapy to some patients with systemic autoimmune diseases. These drugs interfere with normal lysosomal function and can cause a drug-induced lysosomal storage myopathy that presents with slowly progressive muscle weakness. The muscle tissue shows myopathic changes including vacuolation that predominantly affects type I fibers. Ultrastructural studies identify aggregates of whorled, lamellar membranous structures, including curvilinear bodies that mimic those seen in ceroid lipofuscinoses (Chapter 28). Cardiac muscle can also be affected by these drugs and can exhibit similar pathologic changes.

ICU myopathy, also known as myosin-deficient myopathy, is a neuromuscular disorder seen in patients during the course of treatment for critical illness (usually in an ICU) especially with corticosteroid therapy. There may be relatively selective degradation of sarcomeric myosin thick

filaments, producing profound weakness that can complicate the clinical course (e.g., by interfering with the weaning of a patient from a mechanical respirator).

Thyroid dysfunction can lead to several types of myopathy. *Thyrotoxic myopathy* presents most commonly as an acute or chronic proximal muscle weakness that may precede other signs of hyperthyroidism. Such patients may also present with exophthalmic ophthalmoplegia, characterized by swelling of the eyelids, edema of the conjunctiva, and diplopia. Hypothyroidism can cause cramping or aching of muscles, and decreased movement. Reflexes may be slowed. Findings in skeletal muscle include fiber atrophy, an increased number of abnormally localized nuclei, glycogen aggregates, and (occasionally) deposition of mucopolysaccharides in connective tissue.

Alcohol can also be myopathic. Most notably, binge drinking may produce an acute toxic syndrome of rhabdomyolysis, myoglobinuria, and renal failure. The affected individual may complain of acute myalgias that are generalized or confined to a single muscle group.

Inherited Diseases of Skeletal Muscle

Inherited mutations are responsible for a diverse collection of disorders marked by defects in skeletal muscle. In some of these disorders, skeletal muscle is the main site of disease,

but in others multiple organs are involved. Of the other organs involved, the heart is of particular importance, since cardiac involvement is common and often life limiting.

Historically, inherited myopathies have been subdivided into several broad categories based on inheritance pattern, anatomic pattern of muscle involvement, onset age, clinical course, and underlying pathogenesis. These categories should largely be understood as illustrative of key concepts. On closer inspection, there are exceptions and there is significant overlap between these disease categories. Included in the group of inherited muscular disorders are the following.

Congenital myopathies (Table 27.2) typically present in infancy with muscle defects that tend to be static or to even improve over time. They are often associated with distinct structural abnormalities of the muscle.

Muscular dystrophies, characterized by progressive muscle damage that typically comes to attention after infancy.

Congenital muscular dystrophies, by contrast, tend to present in infancy and are often associated with developmental abnormalities of the CNS as well as progressive muscle damage. Congenital muscular dystrophies include two important groups:

- *Conditions with defects in extracellular matrix surrounding myofibers.* These are exemplified by Ullrich congenital muscular dystrophy (UCMD) and merosin deficiency. In the former the causative mutations involve one of

Table 27.2 Congenital Myopathies

Disease and Inheritance	Gene and Locus	Clinical Findings	Pathologic Findings
Central core disease; autosomal dominant	Ryanodine receptor-1 (<i>RYR1</i>) gene; 19q13.2	Early-onset hypotonia and weakness; “floppy infant”; associated skeletal abnormalities like scoliosis, hip dislocation, or foot deformities; some <i>RYR1</i> mutations cause central core disease, some cause malignant hyperthermia, and some cause both	Cytoplasmic cores represent demarcated central zones in which the normal arrangement of sarcomeres is disrupted and mitochondria are decreased in number
Nemaline myopathy (NEM)	AD NEM1— α -tropomyosin 3 (<i>TPM3</i>) gene; 1q21.3 AR NEM2—nebulin (<i>NEB</i>) gene; 2q23.3 AR NEM3— α -actin-1 (<i>ACTA1</i>) gene; 1q42.13 AD NEM4—tropomyosin-2 (<i>TPM2</i>) gene; 9p13.3 AR NEM5—troponin T1 (<i>TNNT1</i>) gene; 19q13.42 AR NEM7—cofilin-2 (<i>CFL2</i>) gene; 14q13.1	Childhood weakness; some with more severe weakness, hypotonia at birth (“floppy infant”)	Aggregates of spindle-shaped particles (<i>nemaline rods</i>); occur predominantly in type I fibers; derived from Z-band material (α -actinin) and best seen on modified Gomori stain or by electron microscopy
Centronuclear (myotubular) myopathy	XL—myotubularin (<i>MTM1</i>) gene; Xq28 AD—dynammin-2 (<i>DNM2</i>) gene (and others); 19p13.2 AR—amphiphysin-2 (<i>BIN1</i>) gene; 2q14.3	Severe congenital hypotonia, “floppy infant,” and poor prognosis in X-linked form (“myotubular myopathy”) Childhood onset or young adult onset with other variants with weakness and hypotonia	Many fibers contain nuclei in the geometric center of the myofiber; central nuclei are more common in type I fibers, which are small in diameter, but can occur in both fiber types
Congenital fiber type disproportion	Selenoprotein 1 (<i>SELENON</i>) gene; 1p36.11 α -actin-1 (<i>ACTA1</i>) gene; 1q42.13 Tropomyosin 3 (<i>TPM3</i>) gene; 1q21.3	Hypotonia, weakness, failure to thrive, facial and respiratory weakness, contractures Wide phenotypic spectrum Mutations in <i>SELENON</i> are also associated with protein aggregate myopathy and rigid spine muscular dystrophy; mutations in <i>ACTA1</i> are also associated with nemaline myopathy and protein aggregate myopathy; mutations in <i>TPM3</i> are also associated with nemaline myopathy	Predominance and atrophy of type I fibers (not specific)

AD, Autosomal dominant; AR, autosomal recessive; XL, X-linked.

three collagen VI alpha genes; in the case of merosin deficiency, the gene encoding merosin is disrupted. UCMD is characterized by hypotonia, proximal contractures, and distal hyperextensibility. A morphologic hallmark is mismatched expression of normally co-localized matrix proteins perlecan and collagen VI.

- *Conditions with abnormalities in receptors for extracellular matrix.* In this group are diseases that disrupt the post-translational modification of alpha-dystroglycan (Fig. 27.10) by O-linked glycosylation. Mutations of alpha-dystroglycan itself result in fetal demise, but defects in its post-translational modification result in milder forms of dystroglycan deficiency. Alpha-dystroglycan expression is important for CNS and eye development. Severe cases exhibit features of congenital muscular dystrophy as well as developmental defects of the CNS and eyes that cause seizures, intellectual disability, and blindness. Milder forms may only cause skeletal muscle disease. Some of these mutations are also linked to a presentation described as limb-girdle muscular dystrophy (see later).

The following section focuses on the most common and best understood forms of inherited myopathies.

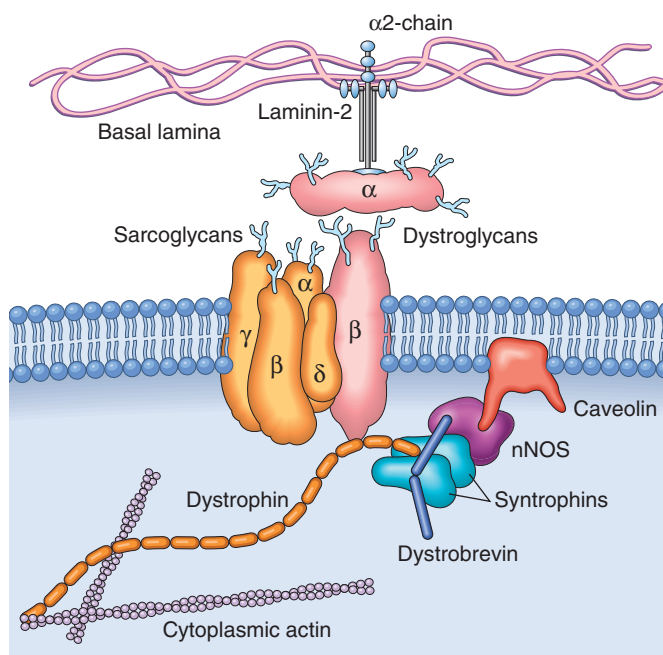


Figure 27.10 Relationship between the cell membrane (sarcolemma) and the sarcolemmal associated proteins. Dystrophin, an intracellular protein, forms an interface between the cytoskeletal proteins and a group of transmembrane proteins, the dystroglycans and the sarcoglycans. These transmembrane proteins have interactions with the extracellular matrix, including the laminin proteins. Dystrophin also interacts with dystrobrevin and the syntrophins, which form a link with neuronal-type nitric oxide synthetase (nNOS) and caveolin. Mutations in dystrophin are associated with the X-linked muscular dystrophies; mutations in caveolin and the sarcoglycan proteins are associated with the limb-girdle muscular dystrophies, which can be autosomal dominant or recessive disorders; and mutations in the α_2 -laminin (merosin) are associated with autosomal recessive congenital muscular dystrophy. Sugar moieties on the dystroglycans are affected by disruption of enzymes responsible for this post-translational modification. Mutations in genes that encode these glycosylation enzymes can cause myopathy as discussed below. These moieties are depicted as light blue branching structure.

Muscular Dystrophies

Muscular dystrophies include many inherited disorders of skeletal muscle that have in common progressive muscle damage that typically manifests between childhood and adulthood. As mentioned earlier, with the exception of congenital muscular dystrophies, these diseases do not present in infancy. While our focus is on X-linked muscular dystrophies, other forms in which the disease pathogenesis is reasonably well understood are also briefly discussed.

X-Linked Muscular Dystrophy With Dystrophin Mutation Including Duchenne and Becker Muscular Dystrophy

The most common muscular dystrophies are X-linked and stem from mutations that disrupt the function of a large structural protein called dystrophin. As a result, these diseases are sometimes referred to as *dystrophinopathies*. The most common early-onset form is referred to as *Duchenne muscular dystrophy*. It has an incidence of 1 per 3500 live male births and has a severe progressive phenotype. *Becker muscular dystrophy* is a second relatively common dystrophinopathy that is characterized by later disease onset and a milder phenotype. Other rare dystrophinopathies may present with isolated cardiomyopathy, asymptomatic elevations of creatine kinase levels, or exercise intolerance because of myalgias and cramps. As with many X-linked diseases, female carriers of dystrophin mutations may be mildly symptomatic due to unfavorable X-chromosome inactivation.

Pathogenesis

Duchenne and Becker muscular dystrophy are caused by loss-of-function mutations in the dystrophin gene on the X chromosome. *Dystrophin* is one of the largest human genes, spanning 2.3 million base pairs and composed of 79 exons. The encoded protein, dystrophin, is a key component of the dystrophin glycoprotein complex (see Fig. 27.10). This complex spans the plasma membrane and serves as a link between the cytoskeleton inside the myofiber and the basement membrane outside of the cell. By doing so, dystrophin is thought to provide mechanical stability to the myofiber and its cell membrane during muscle contraction. Defects in the complex may lead to small membrane tears that permit influx of calcium, triggering events that result in myofiber degeneration. In addition to its mechanical function, dystrophin may have a role in signaling pathways; for example, its carboxy terminus interacts with nitric oxide synthase, which generates NO.

Identification and characterization of specific *dystrophin* mutations has provided an explanation for some of the phenotypic variation in patients with dystrophinopathies. Duchenne muscular dystrophy is typically associated with deletions or frameshift mutations that result in total absence of dystrophin. In contrast, the mutations in Becker muscular dystrophy typically permit the synthesis of a truncated version of dystrophin, which presumably retains partial function.

MORPHOLOGY

The basic morphologic changes that muscular dystrophies elicit in skeletal muscle tissue may differ in severity but do not

discriminate between different forms of dystrophy. The changes in Duchenne muscular dystrophy serve as an example. This disease is marked by chronic muscle damage that outpaces the capacity for repair (Fig. 27.11). Muscle biopsies in young boys show ongoing damage in the form of segmental myofiber degeneration and regeneration associated with an admixture of atrophic myofibers. The fascicular architecture is preserved at this stage of the disease, and there is usually no inflammation except for the presence of myophagocytosis. As the disease progresses, muscle tissue is replaced by collagen and fat cells ("fatty replacement" or "fatty infiltration"). The remaining myofibers at this point in the course show prominent variation in size, from small atrophic fibers to large hypertrophied fibers. This remodeling distorts the fascicular architecture of the muscle, which becomes markedly abnormal over time. **Immunohistochemical studies for dystrophin show absence of the normal sarcolemmal staining pattern in Duchenne muscular dystrophy and reduced staining in Becker muscular dystrophy.**

Clinical Features

Boys with Duchenne muscular dystrophy appear normal at birth. Very early motor milestones are met, but walking is often delayed. The first indications of muscle weakness are clumsiness and inability to keep up with peers. Weakness begins in the pelvic girdle muscles and then extends to the shoulder girdle. Enlargement of the muscles of the lower leg associated with weakness, termed pseudohypertrophy, is often present. The mean age of wheelchair dependence is around 9.5 years. Patients develop joint contractures, scoliosis, worsening respiratory reserve, and sleep hypoventilation.

Dystrophin is also expressed in the heart and the CNS. Dystrophin deficiency in cardiac muscle often leads to the

development of cardiomyopathy and arrhythmias, particularly in older patients. Cognitive impairment and learning disabilities, presumably due to a functional role for dystrophin in the brain, is also common and sometimes produces frank intellectual disability. Despite supportive care, the mean age of death for patients with Duchenne muscular dystrophy is 25 to 30 years of age, with most patients succumbing to respiratory insufficiency, pulmonary infection, or heart failure. This is in contrast to Becker muscular dystrophy, which typically presents in later childhood, adolescence, or adult life; has slower progression; and may have a near-normal life expectancy.

The diagnosis is based on the history, physical examination, and laboratory studies. Serum creatine kinase is markedly elevated during the first decade of life due to ongoing muscle damage, and then falls as the disease progresses and muscle mass is lost. The detection of a dystrophin mutation offers definitive diagnosis.

Treatment of patients with dystrophinopathies is challenging. Current treatment consists primarily of supportive care. Definitive therapy requires restoration of dystrophin levels in skeletal and cardiac muscle fibers. Work in this area is emboldened by the recognition that expression of some dystrophin protein (as in patients with Becker muscular dystrophy) is sufficient to substantially ameliorate the disease phenotype. One approach involves the expression of antisense RNAs that alter RNA splicing so as to cause "skipping" of exons containing deleterious mutations, thus permitting the expression of a truncated, but partially functional, dystrophin protein. A second strategy is exploring the use of drugs that promote ribosomal "read-through" of stop codons, another ploy that may enable the expression of some dystrophin protein. Both of these approaches are mutation-specific and thus need to be tailored to individual

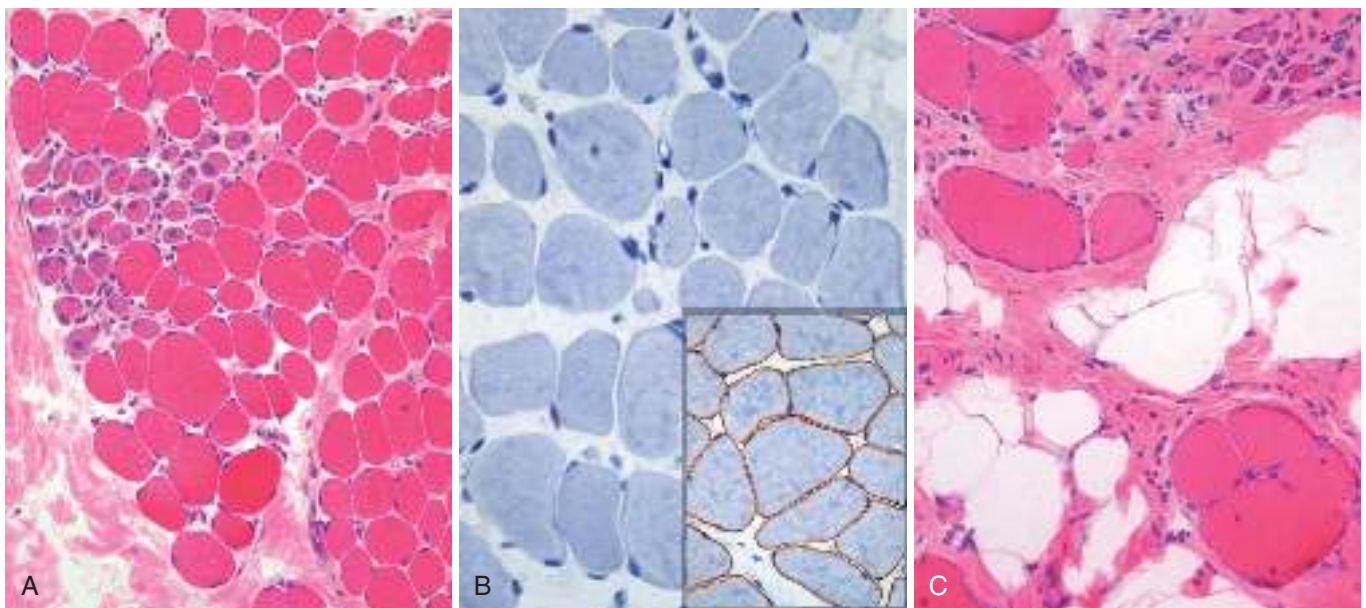


Figure 27.11 Duchenne muscular dystrophy. Histologic images of muscle biopsy specimens from two brothers. (A and B) Specimens from a 3-year-old boy. (C) Specimen from his 9-year-old brother. (A) At a younger age, fascicular muscle architecture is maintained, but myofibers show variation in size. Additionally, there is a cluster of basophilic regenerating myofibers (left side) and slight endomysial fibrosis, seen as focal pink-staining connective tissue between myofibers. (B) Immunohistochemical staining shows complete absence of membrane-associated dystrophin, seen as a brown stain in normal muscle (inset). (C) Biopsy specimen from the older brother illustrates disease progression, which is marked by extensive variation in myofiber size, fatty replacement, and endomysial fibrosis.

patients. Gene therapy (introduction of a normal dystrophin gene) is being investigated, but gene delivery to skeletal muscle cells remains a daunting hurdle.

Limb-Girdle Muscular Dystrophy

Limb-girdle muscular dystrophies are a heterogeneous group of at least 8 autosomal dominant and 23 autosomal recessive entities. Their overall incidence is 1 in 25,000 to 50,000 individuals. As indicated by the name, all forms are characterized by muscle weakness that preferentially involves proximal muscle groups. Both age of onset and disease severity are highly variable. The causative mutations involve genes that participate in diverse cellular functions, making it difficult to discern a unifying mechanism of disease pathogenesis. Based on current knowledge, the implicated genes can be grouped according to function as follows:

- *Genes encoding structural components (sarcoglycans) of the dystrophin glycoprotein complex*
- *Genes encoding enzymes that are responsible for glycosylation of α -dystroglycan, a component of the dystrophin glycoprotein complex (see Fig. 27.10)*
- *Genes encoding proteins that associate with the Z-disks of sarcomeres*
- *Genes encoding proteins involved in vesicle trafficking and cell signaling*
- *Genes that seemingly stand alone, such as CAPN3 encoding the protease calpain 3 and lamin A/C (which is also mutated in some patients with Emery-Dreifuss muscular dystrophy [EMD]; see below)*

Myotonic Dystrophy

Myotonic dystrophy is an autosomal dominant multisystem disorder associated with skeletal muscle weakness, cataracts, endocrinopathy, and cardiomyopathy. It affects about 1 in 10,000 individuals. Myotonia, a sustained involuntary contraction of muscles, is a key feature of the disease. Rarely, patients present with “congenital myotonia,” marked by severe manifestations in infancy.

Pathogenesis

The disease is caused by expansions of CTG triplet repeats in the 3′-noncoding region of the myotonic dystrophy protein kinase (*DMPK*) gene, but precisely how this genetic aberration produces the disease phenotype is unknown. The correlation between the length of expansion and disease severity is variable compared to some other triplet repeat expansion disorders like Huntington disease (Chapter 28). Experimental studies suggest that the skeletal muscle phenotype stems from a “toxic” gain-of-function caused by the triplet repeat expansion. Specifically, the expanded CUG-repeats in the *DMPK* mRNA transcript appear to bind and sequester proteins like muscleblind-like splicing regulator 1, which have an important role in RNA splicing. The resulting disruption of normal splicing events leads to missplicing of other RNA transcripts, including the transcript for a chloride channel called *CLC1*. It is believed that the resulting deficiency of *CLC1* is responsible for the characteristic myotonia. In support of this scenario, one rare form of congenital myotonia is caused by germline loss-of-function mutations in *CLC1*, indicating that *CLC1* is required for normal muscle relaxation.

Emery-Dreifuss Muscular Dystrophy (EMD)

EMD is caused by mutations in genes that encode nuclear lamin proteins. Clinically, it is marked by a triad consisting of slowly progressive humeroperoneal weakness; cardiomyopathy associated with conduction defects; and early contractures of the Achilles tendon, spine, and elbows. The X-linked form (EMD1) and the autosomal form (EMD2) are caused by mutations in the genes encoding *emerin* and *lamin A/C*, respectively, both of which localize to the inner face of the nuclear membrane. It is hypothesized that these proteins help maintain the shape and mechanical stability of the nucleus during muscle contraction. They may also influence gene expression by affecting chromatin organization in the nucleus. How defects in these proteins produce the observed phenotypes is unknown.

Fascioscapulohumeral Dystrophy

Fascioscapulohumeral dystrophy is associated with a characteristic pattern of muscle involvement that includes prominent weakness of facial muscles and muscles of the shoulder girdle. It is an autosomal dominant disease affecting about 1 in 20,000 individuals.

Pathogenesis

The pathogenesis of fascioscapulohumeral dystrophy is complex and only partly understood. It results from various mechanisms that culminate in the (over)expression of a retrogene called *DUX4* that is located in a region of subtelomeric repeats on the long arm of chromosome 4. *DUX4* encodes a transcription factor, suggesting that the disease ultimately results from the overexpression of *DUX4* target genes.

Diseases of Lipid or Glycogen Metabolism

Many inborn errors of lipid or glycogen metabolism affect skeletal muscle. These disorders tend to produce one of two general patterns of muscle dysfunction: In some, patients become symptomatic only with exercise or fasting, which may produce severe muscle cramping and pain or even extensive muscle necrosis (*rhabdomyolysis*). In others, there is slowly progressive damage of muscle, without episodic manifestations. Listed below are some examples:

- *Carnitine palmitoyltransferase II deficiency* is the most common disorder of lipid metabolism to cause episodic muscle damage with exercise or fasting. The defect in this disorder impairs the transport of free fatty acids into mitochondria.
- *Myophosphorylase deficiency (McArdle disease)* is one of the more common glycogen storage diseases affecting skeletal muscle; it also results in episodic muscle damage with exercise.
- *Acid maltase deficiency* results in impaired lysosomal conversion of glycogen to glucose, causing glycogen to accumulate within lysosomes. Severe deficiency results in the generalized glycogenosis of infancy, *Pompe disease* (Chapter 5). Milder deficiency can cause a progressive adult-onset myopathy that preferentially involves the respiratory and truncal muscles. Enzyme replacement therapy has been used to treat some affected patients.

Mitochondrial Myopathies

Mitochondrial diseases are complex systemic conditions that can involve many organ systems, including skeletal muscle. The genetics of these disorders are varied and unusually complex (discussed later), but many of the causative mutations appear to impair the ability of mitochondria to generate adenosine triphosphate (ATP). As a result, these diseases tend to affect skeletal muscles and other tissues rich in cell types with high ATP requirements, particularly cardiac muscle cells and neurons.

Skeletal muscle involvement can manifest as weakness, elevations in serum creatine kinase levels, or rhabdomyolysis. Although the anatomic pattern of muscle weakness is variable, involvement of extraocular eye muscles is common and can be a clue to the diagnosis. Indeed, *chronic progressive external ophthalmoplegia* is a common feature of mitochondrial disorders and may occur as an isolated phenomenon or a part of a multisystem syndrome. The reason that extraocular eye muscles are particularly sensitive to mitochondrial disease is uncertain, but it may be that these muscles have exceptionally high requirements for ATP. In line with this idea, extraocular eye muscles have the most mitochondria per mass of any of the body's muscles.

Mitochondrial proteins and tRNAs may be encoded by either the nuclear genome or the mitochondrial genome (mtDNA). While mutations in nuclear mitochondrial genes follow Mendelian inheritance patterns, mutations in mtDNA are maternally inherited, since all of the mitochondria in the embryo are contributed by the oocyte (Chapter 5). In addition, unlike nuclear DNA, which is present in only two copies and is evenly distributed from a mother cell to daughter cells during cell division, each cell contains thousands of mtDNA copies, which are distributed in a random fashion to daughter cells at the time of cell division. It is believed that disease results only when a certain threshold of mutated mtDNA copies is exceeded within a substantial fraction of "at-risk" cells (e.g., skeletal muscle cells) in a tissue.

MORPHOLOGY

The most consistent pathologic change in skeletal muscle is abnormal aggregates of mitochondria that are seen preferentially in the subsarcolemmal area of affected myofibers, producing an appearance that is referred to as "ragged red fibers" (Fig. 27.12). By electron microscopy, morphologically abnormal mitochondria may be seen. Loss of particular mitochondrial enzyme activities characterizes some mitochondrial diseases and may be appreciated by histochemical staining for cytochrome oxidase. Some mitochondrial diseases lack morphologic changes and can be diagnosed only through enzymatic assays or genetic analyses.

Clinical Features

Due to the complexity of mitochondrial genetics, genotype/phenotype relationships in mitochondrial disorders are not straightforward. For example, a single point mutation in the mitochondrial leucine tRNA gene may produce isolated chronic progressive external ophthalmoplegia in one patient and in another, a much more severe phenotype, *mitochondrial encephalomyopathy with lactic acidosis and stroke-like episodes*.

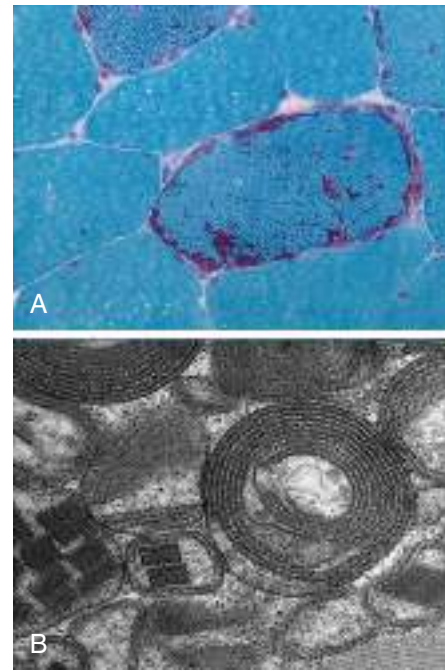


Figure 27.12 (A) Ragged red fiber with increased reddish granular subsarcolemmal staining reflective of abnormal aggregation of mitochondria. (B) Electron micrograph showing morphologically abnormal mitochondria with concentric membranous rings (so-called "phonograph records") and rhomboid paracrystalline inclusions (lower left side).

Similarly, deletions in mtDNA may lead to either isolated ophthalmoplegia or to *Kearns-Sayre syndrome*, characterized by ophthalmoplegia, pigmentary degeneration of the retina, and complete heart block. *Myoclonic epilepsy with ragged red fibers* and *Leber hereditary optic neuropathy* are other examples of mitochondrial disease caused by point mutations in mtDNA. Many mitochondrial disorders, such as subacute necrotizing encephalopathy (*Leigh syndrome*), are remarkably heterogeneous genetically and may be caused by mutations in either mtDNA or the nuclear genome. In the case of Leigh syndrome, causative mutations have been identified in more than 30 different genes, the common feature being that all of the affected genes encode proteins with essential roles in mitochondrial metabolism.

Spinal Muscular Atrophy and Differential Diagnosis of a Hypotonic Infant

Spinal muscular atrophy is a neuropathic disorder in which loss of motor neurons leads to muscle weakness and atrophy. Infants with neurologic or neuromuscular disease may present with generalized hypotonia ("floppy infant"). The differential diagnosis of infantile hypotonia includes primary diseases of skeletal muscle (e.g., congenital myasthenic syndrome, congenital myotonia, congenital myopathies, and congenital muscular dystrophies); abnormalities of the brain (e.g., encephalopathy); and neuronopathies, of which spinal muscular atrophy is a prototypic example.

Spinal muscular atrophy is an autosomal recessive disorder with an incidence of 1 in 6000 births and is caused by loss-of-function mutations in the *SMN1* (survival of motor neuron-1) gene. The function of the gene is uncertain—the

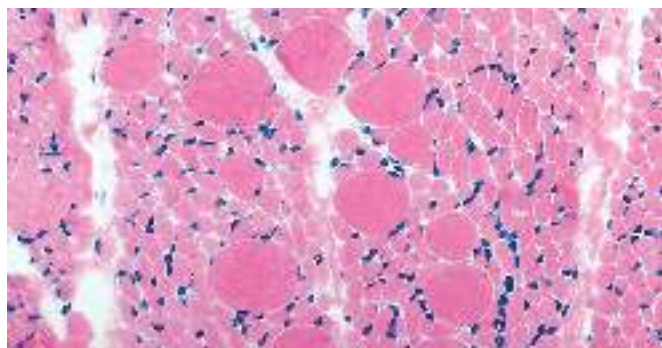


Figure 27.13 Spinal muscular atrophy with only rare hypertrophied myofibers admixed with numerous atrophic rounded myofibers. The larger fibers are those that are innervated and have undergone compensatory hypertrophy.

encoded protein may have a role in RNA splicing—but SMN1 deficiency has a dramatic effect on motor neuron survival, sometimes leading to loss of motor neurons in utero. The resulting denervation of skeletal muscle may lead to characteristic morphologic changes consisting of large zones of severely atrophic myofibers mixed with scattered normal-sized or hypertrophied myofibers, found individually or in small groups (Fig. 27.13). These normal or hypertrophied fibers are those that retain innervation from remaining motor neurons.

Ion Channel Myopathies (Channelopathies)

Channelopathies are a group of inherited diseases caused by mutations affecting the function of ion channel proteins. Most channelopathies are autosomal dominant disorders with variable penetrance. Depending on the channel that is affected, clinical manifestations may include epilepsy, migraine, movement disorders with cerebellar dysfunction, peripheral nerve disease, and muscle disease.

Different ion channel myopathies may cause decreased or increased excitability resulting in hypotonia or hypertonia. Disorders associated with hypotonia can be further subclassified based on whether symptomatic patients have elevated, depressed, or normal serum potassium levels and are called *hyperkalemic*, *hypokalemic*, or *normokalemic periodic paralysis*, respectively. Examples of gene mutations that are associated with muscle dysfunction are the following:

- **KCNJ2:** Mutations affecting this potassium channel cause *Andersen-Tawil syndrome*, an autosomal disorder associated with periodic paralysis, cardiac arrhythmias, and skeletal abnormalities.
- **SCN4A:** Mutations affecting this sodium channel cause several autosomal disorders with presentations ranging from myotonia to periodic paralysis.
- **CACNA1S:** Missense mutations in this protein, a subunit of a muscle calcium channel, are the most common cause of hypokalemic paralysis.
- **CLC1:** Mutations affecting this chloride channel cause myotonia congenita. As already discussed, *CLC1* expression is decreased in myotonic dystrophy.
- **RYR1:** Mutations in the *RYR1* gene disrupt the function of the ryanodine receptor, which regulates calcium release from the sarcoplasmic reticulum. *RYR1* mutations are linked to a congenital myopathy (central core disease)

and to *malignant hyperthermia*. The latter is characterized by a hypermetabolic state (tachycardia, tachypnea, muscle spasms, and later hyperpyrexia) that can be triggered by anesthetics, most commonly halogenated inhalational agents and succinylcholine. Upon exposure to anesthetic, the mutated receptor allows increased efflux of calcium from the sarcoplasmic reticulum, leading to tetany and excessive heat production.

KEY CONCEPTS

DISORDERS OF SKELETAL MUSCLE

- Altered muscle function may stem from neurogenic or primary myopathic processes.
- Myopathic disorders are often marked by degeneration and regeneration of myofibers.
- The three traditional inflammatory myopathies are polymyositis, dermatomyositis, and inclusion body myositis, but other forms like immune-mediated necrotizing myopathy (IMNMs) are now recognized.
 - Inclusion body myositis is a chronic progressive disease of older patients associated with rimmed vacuoles.
 - Dermatomyositis occurs in children and adults, the latter frequently as a paraneoplastic disorder. Immune damage to small blood vessels and perifascicular atrophy are common features.
 - Polymyositis is an adult-onset myopathy caused by CD8+ T cells. This disease is less common than originally thought because many patients are reclassified as having IMNM, systemic connective tissue disease, or even early inclusion body myositis.
- Muscular dystrophies and congenital myopathies result from genetic mutations that disrupt the function of proteins that are important for various aspects of muscle development, function, and regeneration. Some of these diseases present in infancy, others in adulthood. They may be relentlessly progressive or cause relatively static deficits.
- Myopathy can result from toxic injury or be the result of metabolic diseases including those of lipid metabolism, glycogen metabolism, and mitochondria.

PERIPHERAL NERVE SHEATH TUMORS

The vast majority of benign and malignant neoplasms of peripheral nerve sheaths are composed of cells that show evidence of Schwann cell differentiation. These include the three common types: schwannoma, neurofibroma, and malignant peripheral nerve sheath tumor (MPNST). Other rare tumors arising from nerves may show evidence of perineurial cell differentiation. There is an abrupt transition between myelination by oligodendrocytes (central myelin) and myelination by Schwann cells (peripheral myelin) that occurs as nerves extend out from the substance of the brain. Thus, peripheral nerve tumors sometimes arise within the dura as well as along the distal course of peripheral nerves.

Peripheral nerve sheath tumors have several unique features. Firstly, their association with relatively common familial tumor syndromes, including neurofibromatosis type 1 (NF1), neurofibromatosis type 2 (NF2), and schwannomatosis.

Secondly, MPNSTs seen in the context of NF1 are thought to arise through malignant transformation of preexisting benign plexiform neurofibromas. Although malignant transformation of a preexisting benign lesion is a common origin for certain carcinomas (e.g., colon cancer), it is unusual in soft tissue tumors. Tumors with skeletal muscle differentiation are discussed in Chapter 26.

Schwannomas

These are benign tumors that exhibit Schwann cell differentiation and often arise directly from peripheral nerves. Schwannomas are a component of NF2, and even sporadic schwannomas are commonly associated with inactivating mutations in the *NF2* gene on chromosome 22. Loss of expression of the *NF2* gene product, merlin, is a consistent finding in all schwannomas. Through interactions affecting the actin cytoskeleton, merlin participates in the regulation of several key signaling pathways that are involved in control of cell shape, cell growth, and the attachment of cells to one another (cell adhesion).

MORPHOLOGY

Schwannomas are well-circumscribed, encapsulated masses that are attached loosely to the associated nerve without invading it. As a result they can often be resected without sacrificing nerve function. Grossly, these tumors form firm, gray masses. Microscopically, they are comprised of an admixture of dense and loose areas referred to as **Antoni A** and **Antoni B** areas, respectively (Fig. 27.14A). The dense eosinophilic Antoni A areas often contain spindle cells arranged into cellular intersecting fascicles. Palisading of nuclei is common. Resulting structures with central “nuclear-free zones” ramified by palisading nuclei are termed **Verocay bodies** (Fig. 27.14B). In the loose, hypocellular **Antoni B** areas the spindle cells are spread apart by a prominent myxoid extracellular matrix that may be associated with microcyst formation. Schwann cells are characterized by the presence of a spindled elongated nucleus with a wavy or buckled shape. Electron microscopy shows basement membrane deposits encasing single cells and collagen fibers. Because the lesion displaces the nerve of origin as it grows, axons are largely excluded from the tumor, although they may become entrapped in the capsule. The Schwann cell origin of these tumors is borne out by their uniform immunoreactivity for S-100. A variety of degenerative changes may be found in schwannomas, including nuclear pleomorphism, xanthomatous change, vascular hyalinization, cystic change, and necrosis. Some large mitotically active schwannomas lacking Antoni B areas can mimic a sarcoma. Schwannomas may recur locally if incompletely resected, but malignant transformation is extremely rare (in contrast to plexiform neurofibromas, discussed later).

Clinical Features

Most schwannomas cause symptoms by local compression of the involved nerve or adjacent structures (e.g., brainstem or spinal cord). Within the cranial vault, most schwannomas occur at the cerebellopontine angle, where they are attached to the vestibular branch of the eighth nerve. Affected individuals often present with tinnitus and hearing loss; the tumor is commonly referred to as an *acoustic neuroma*—a double misnomer, since the tumor neither arises

from the acoustic portion of the nerve nor is it a neuroma. Elsewhere within the dura, sensory nerves are preferentially involved, including branches of the trigeminal nerve and dorsal roots. When extradural, schwannomas can arise in association with large nerve trunks or as soft tissue lesions without an identifiable associated nerve. Surgical removal is curative.

Neurofibromas

Neurofibromas are benign nerve sheath tumors that are more heterogeneous in composition than schwannomas. **The neoplastic Schwann cells are admixed with perineurial-like cells, fibroblasts, mast cells, and CD34+ spindle cells.** Neurofibromas may be either sporadic or NF1-associated. Different types of neurofibroma can be distinguished depending on their growth pattern.

- *Superficial cutaneous neurofibromas* often present as pedunculated nodules that can be seen isolated (if sporadic) or multiple (if NF1-associated).
- *Diffuse neurofibromas* often present as a large plaque-like elevation of skin and are typically NF1-associated.
- *Plexiform neurofibromas* can be found in deep or superficial locations in association with nerve roots or large nerves and are uniformly NF1-associated.

Pathogenesis

Only the Schwann cells in neurofibromas show complete loss of the *NF1* gene product, neurofibromin, indicating that these are the neoplastic cells. You will recall from Chapter 7 that neurofibromin is a tumor suppressor that inhibits RAS activity by stimulating the activity of a guanosine triphosphatase (GTPase). Loss of the GTPase activity causes RAS to be trapped in GTP bound active state. Haploinsufficiency for the *NF1* gene in other associated cells may also contribute to the growth of NF1-associated tumors. For example, there is evidence that *NF1*-haploinsufficient mast cells are hypersensitive to KIT ligand produced by Schwann cells and in response secrete factors that stimulate Schwann cell growth. This form of tumor/stromal cell cross-talk may be targetable with inhibitors of the KIT receptor tyrosine kinase. Other studies suggest that plexiform neurofibromas and dermal neurofibromas arise from different neural crest-derived precursor cells. Malignant transformation of neurofibroma to MPNST is usually observed in the plexiform variant but is sometimes also seen in the diffuse type. The overall incidence of MPNST in NF1 patients is about 5% to 10%, but patients with large numbers of plexiform neurofibromas and large deletions in the *NF1* gene are at higher risk.

MORPHOLOGY

Localized cutaneous neurofibroma. These are small, well-delineated but unencapsulated nodular lesions that arise in the dermis and subcutaneous fat. They have relatively low cellularity and contain bland Schwann cells admixed with stromal cells such as mast cells, perineurial cells, CD34+ spindle cells, and fibroblasts. Adnexal structures are sometimes entrapped at the edges of the lesion. The stroma of these tumors contains loose collagen.

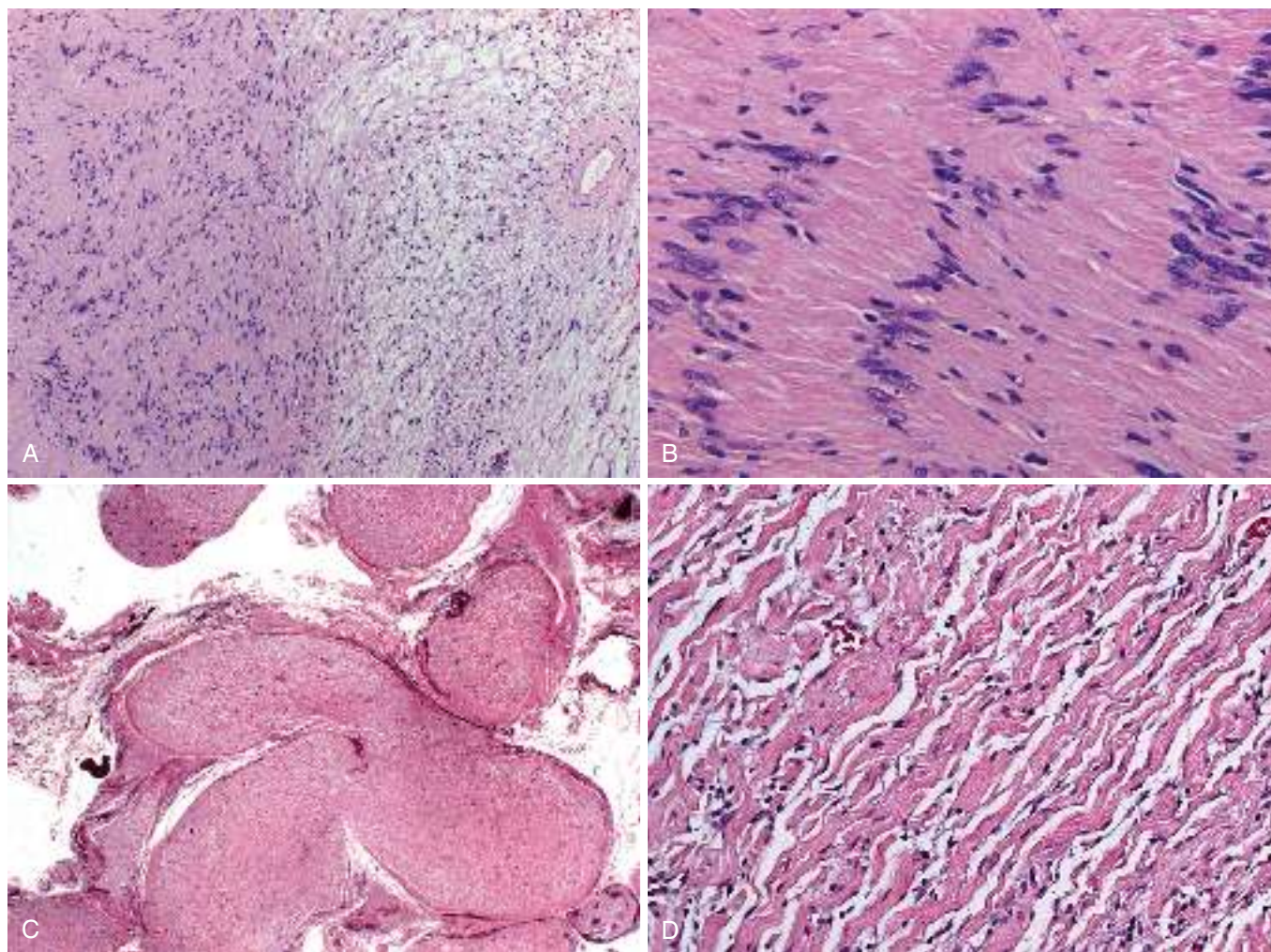


Figure 27.14 Schwannoma and plexiform neurofibroma. (A and B) Schwannoma. (A) Schwannomas often contain dense eosinophilic Antoni A areas (*left*) and loose, pale Antoni B areas (*right*), as well as hyalinized blood vessels (*right*). (B) Antoni A area with the tumor cell nuclei aligned in palisading rows leaving anuclear zones and resulting in the formation of structures termed *Verocay bodies*. (C and D) Plexiform neurofibroma. (C) Multiple nerve fascicles are expanded by infiltrating tumor cells. (D) At high power, bland spindle cells are admixed with wavy collagen bundles resembling carrot shavings.

Diffuse neurofibroma. This tumor has morphologic features similar to those seen in localized cutaneous neurofibromas but exhibits a distinctly different growth pattern. The tumor diffusely infiltrates the dermis and subcutaneous connective tissue, entrapping fat and appendage structures and producing a plaque-like appearance. Some of these neurofibromas can grow to large sizes. Focal collections of cells mimicking the appearance of Meissner corpuscles (so-called **pseudo-Meissner corpuscles** or **tactile-like bodies**) are an associated feature.

Plexiform neurofibroma. These tumors grow within and expand nerve fascicles (Fig. 27.14C), entrapping associated axons. The external perineurial layer of the nerve is preserved, giving individual nodules an encapsulated appearance. The expanded, ropy thickening of multiple nerve fascicles results in what is sometimes referred to as a “bag of worms” appearance. The tumor has cellular composition similar to that of other neurofibromas. The extracellular matrix varies from loose and myxoid to more collagenous and fibrous. Often the collagen is seen in bundles likened to “shredded carrot” (Fig. 27.14D).

Malignant Peripheral Nerve Sheath Tumors

Most MPNSTs (approximately 85%) are high-grade tumors, but low-grade variants are recognized. About half arise in NF1 patients and are assumed to result from malignant transformation of a plexiform neurofibroma. Sporadic cases may arise *de novo*. Most are associated with larger peripheral nerves in the chest, abdomen, pelvis, neck, or limb-girdle. MPNSTs exhibit complex chromosomal aberrations including chromosome gains, losses, and rearrangements. The molecular alterations driving malignant transformation of a neurofibroma to MPNST are still poorly understood.

MORPHOLOGY

The lesions are poorly defined tumor masses that frequently infiltrate along the axis of the parent nerve and invade adjacent soft tissues. A wide range of histologic appearance can be encountered. Typical cases show a fasciculated arrangement of

spindle cells. At low power the tumor often appears “marbleized” due to variations in cellularity. Mitoses, necrosis, and nuclear anaplasia are common. An interesting phenomenon observed in MPNST is described as “divergent differentiation.” This term refers to the presence of focal areas that exhibit other lines of differentiation, including glandular, cartilaginous, osseous, or rhabdomyoblastic morphology. A tumor exhibiting the latter is referred to as **Triton tumor**. Due to the poorly differentiated nature of MPNST, the distinction from an undifferentiated sarcoma may not be straightforward. Helpful clues include a diagnosis of NF1 in the affected patient and a clearly demonstrated anatomic relationship to a nerve or to a preexisting neurofibroma.

Neurofibromatosis Type 1 and Type 2

Neurofibromatosis Type 1

This is a common autosomal dominant disorder with a frequency of 1 in 3000. It is a systemic disease associated with nonneoplastic manifestations and with a variety of tumors, including neurofibromas of all types, MPNSTs, gliomas of the optic nerve, other glial tumors and hamartomatous lesions, and pheochromocytomas. Other features include intellectual disability or seizures, skeletal defects, pigmented nodules of the iris (*Lisch nodules*), and cutaneous hyperpigmented macules (*café au lait spots*). The disease is caused by loss-of-function mutations in the *NF1* gene, located at 17q11.2, which encodes the tumor suppressor neurofibromin. The neoplastic cells in NF1-related tumors lack neurofibromin due to biallelic defects in the *NF1* gene. As has been mentioned earlier, NF-1 protein has GTPase activity that restrains RAS function. In the absence of NF-1, RAS remains trapped in its active state.

The disease has a high penetrance but variable expressivity. Some patients exhibit only subtle features, while others show disease that is restricted to certain parts of the body, a distribution that is attributable to mosaicism. An unfortunate subset has severe disease. Large chromosomal deletions that span *NF1* and extend to involve adjacent genes tend to be associated with more severe phenotypes.

Neurofibromatosis Type 2

This is an autosomal dominant disorder resulting in a range of tumors, most commonly bilateral eighth nerve schwannomas, multiple meningiomas, and ependymomas of the spinal cord. Many individuals with NF2 also have nonneoplastic lesions, which include nodular ingrowth of Schwann cells into the spinal cord (schwannosis), meningoangiomas (a proliferation of meningeal cells and blood vessels that grows into the brain), and glial hamartia (microscopic nodular collections of glial cells at abnormal locations, often in the superficial and deep layers of cerebral cortex). This disorder is much less common than NF1, having a frequency of 1 in 40,000 to 50,000. Certain other rare familial syndromes are also associated with multiple schwannomas, such as schwannomatosis and Carney complex.

The *NF2* gene is located on chromosome 22q12 and is also commonly mutated in sporadic meningiomas and schwannomas. The *NF2* gene product, merlin, is a cytoskeletal protein that participates in the regulation of several key signaling pathways that are involved in control

of cell shape, cell growth, and the attachment of cells to one another (cell adhesion). There is some correlation between the type of mutation and clinical symptoms, with nonsense and frameshift mutations causing more severe phenotypes than missense mutations.

KEY CONCEPTS

PERIPHERAL NERVE SHEATH TUMORS

- The three common peripheral nerve sheath tumors—schwannoma, neurofibroma, and MPNST—all likely arise from cells of Schwann cell lineage.
- Schwannomas are encapsulated benign tumors that can be associated with NF2.
- Neurofibromas are benign peripheral nerve sheath tumors sometimes associated with NF1 that can be subtyped as localized cutaneous, diffuse, or plexiform.
- MPNSTs can be de novo sporadic neoplasms or NF1-associated tumors arising through malignant transformation of a (plexiform) neurofibroma.

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The Central Nervous System

28

Marta Margeta • Arie Perry

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The principal functional unit of the central nervous system (CNS) is the *neuron*. Neurons of different types and in different locations have distinct properties, including diverse functional roles, patterns of synaptic connections, neurotransmitters used, and metabolic requirements, which vary with electrical activity. A set of neurons, not necessarily clustered together in a region of the brain, may thus show *selective vulnerability* to particular insults because they share one or more of these properties. Because brain functions are anatomically compartmentalized, the pattern of clinical signs and symptoms that follow injury depend as much on the region of brain that is affected as on the pathologic process. Mature neurons are incapable of cell division, so destruction of even a small number of neurons essential for a specific function may produce a neurologic deficit.

In addition to neurons, the CNS contains other cells, such as *astrocytes* and *oligodendrocytes*, which make up the *glia*. Different cellular components of the CNS are affected by distinct unique neurologic disorders and also respond to common insults (e.g., ischemia, infection) in a manner that is distinct from other tissues. We start our discussion of diseases of the CNS with an overview of the patterns of injury of different cells and the reactions of these cells to various insults.

CELLULAR PATHOLOGY OF THE CENTRAL NERVOUS SYSTEM

Neurons and glia of the CNS display a range of functional and morphologic changes following injury. Understanding these patterns can provide information about the mechanism of cellular injury and type of disease.

Reactions of Neurons to Injury

Neuronal injury may be an acute process, often as a consequence of depletion of oxygen or glucose or trauma, or may develop over a period of years, as occurs in degenerative disorders of the brain. Neurons are highly metabolically active and require a continuous supply of oxygen and glucose to meet their metabolic needs. Moreover, since neurons are postmitotic cells that are incapable of proliferation, they must be maintained throughout life. Perhaps because of their lengthy lifespan and high metabolic activity, neurons are unusually susceptible to the accumulation of misfolded proteins, which trigger the unfolded protein response (Chapter 2) and appear to be central to the pathogenesis of many degenerative disorders of the CNS.

MORPHOLOGY

Acute neuronal injury (“red neurons”) refers to changes seen following acute CNS hypoxia/ischemia, severe hypoglycemia, and other acute insults, and are the earliest morphologic markers of neuronal cell death (see Fig. 28.15A). “Red neurons” are evident by 6 to 12 hours after an irreversible hypoxic/ischemic insult. Typical features include shrinkage of the cell body, pyknosis of the nucleus, disappearance of the nucleolus, loss of Nissl substance, and intense cytoplasmic eosinophilia.

Subacute and chronic neuronal injury (“degeneration”)

refers to neuronal death that occurs as a result of a progressive disease of months to years duration, as is seen in slowly evolving neurodegenerative diseases such as amyotrophic lateral sclerosis and Alzheimer disease. Prior to death, neurons often suffer a loss of synapses (sites of interneuronal communication), which may stem from aberrations in synaptic pruning (a process responsible for normal brain development and plasticity). Loss of synapses is followed by cell death (often selectively involving functionally related groups of neurons) and reactive gliosis. At an early stage, the cell loss is difficult to appreciate; the associated reactive glial changes are often the best indicator of neuronal injury. In many of these diseases, the predominant mechanism of cell death appears to be apoptosis.

Axonal reaction is a change observed in the cell body during regeneration of the axon; it is best seen in anterior horn cells of the spinal cord when motor axons are cut or seriously damaged. The increase in protein synthesis that occurs in response to the injury is reflected in enlargement and rounding up of the cell body, peripheral displacement of the nucleus, enlargement of the nucleolus, and dispersion of Nissl substance from the center to the periphery of the cell (central chromatolysis).

Neuronal damage may be associated with a wide range of subcellular alterations in the neuronal organelles and cytoskeleton.

Neuronal inclusions may occur as a manifestation of aging, and consist of intracytoplasmic accumulations of complex lipids (lipofuscin), proteins, or carbohydrates. Abnormal cytoplasmic accumulations of lipids and other substances also occur with certain inborn errors of metabolism, genetic disorders caused by mutations that lead to the loss of specific enzyme activities (Chapter 5). Viral infection can lead to the appearance of viral inclusions, e.g., in herpetic infection (Cowdry A or B bodies), rabies (Negri body), and cytomegalovirus infection.

Some degenerative diseases of the CNS are associated with neuronal intracytoplasmic inclusions, such as neurofibrillary tangles of Alzheimer disease and Lewy bodies of Parkinson disease; others cause abnormal vacuolization of the perikaryon and neuronal cell processes in the neuropil (Creutzfeldt-Jakob disease).

Wallerian degeneration refers to degeneration of axons after disruption of nerve fibers (see Chapter 27).

Reactions of Astrocytes to Injury

Gliosis is the most important histopathologic marker of CNS injury, regardless of etiology, and is characterized by hypertrophy and hyperplasia of astrocytes. The astrocyte derives its name from its star-shaped appearance. These cells have multipolar, extensively branched cytoplasmic processes that emanate from the cell body and express glial fibrillary acidic protein (GFAP), a cell type-specific intermediate filament. Astrocytes act as metabolic buffers and detoxifiers within the brain. Additionally, through the foot processes (which surround capillaries or extend to the subpial and subependymal zones), they contribute to barrier functions by controlling the flow of macromolecules between the blood, the cerebrospinal fluid (CSF), and the brain.

MORPHOLOGY

In gliosis, the nuclei of astrocytes (which are typically oval with evenly dispersed, pale chromatin) enlarge, become vesicular, and

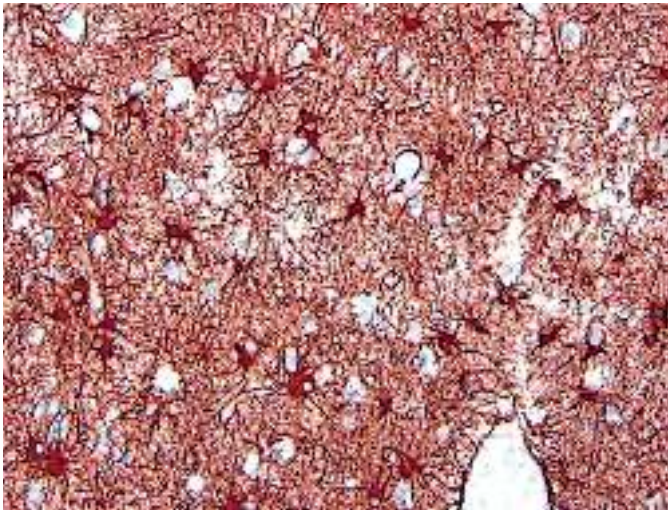


Figure 28.1 Reactive astrocytes. Immunohistochemical staining for glial fibrillary acidic protein (brown) highlights the stellate astrocytic processes from which the cell's name is derived.

occasionally develop prominent nucleoli. The previously indistinct cytoplasm becomes bright pink due to increased expression of GFAP, and the cells develop numerous stout, ramifying processes; these cells are called **reactive or gemistocytic astrocytes** (Fig. 28.1). There are at least two subtypes of reactive astrocytes; although morphologically indistinguishable, these cells form in response to different injurious stimuli, have distinct gene expression patterns, and have different effects on neuronal health and survival, with one subtype promoting CNS injury and the other contributing to CNS repair.

Acute astrocytic injury, as occurs in hypoxia, hypoglycemia, and toxic injuries, is manifested by cellular swelling, as in other cells (Chapter 2). The **Alzheimer type II astrocyte** (unrelated to the disease but first described by the same neuroscientist) is a gray matter cell with a large (two to three times normal) nucleus, pale-staining central chromatin, an intranuclear glycogen droplet, and a prominent nuclear membrane and nucleolus. Alzheimer type II astrocytes are mainly seen in individuals with hyperammonemia due to chronic liver disease, Wilson disease, or hereditary metabolic disorders of the urea cycle.

Other types of astrocyte injury lead to the formation of cytoplasmic inclusion bodies. **Rosenthal fibers** are thick, elongated, brightly eosinophilic, irregular structures that occur within astrocytic processes and contain two heat-shock proteins (α B-crystallin and hsp27) as well as ubiquitin. Rosenthal fibers are typically found in regions of long-standing gliosis; they are also characteristic of one type of glial tumor, pilocytic astrocytoma (see Fig. 28.49C). In Alexander disease, a leukodystrophy associated with mutations in the gene encoding GFAP, abundant Rosenthal fibers are found in periventricular, perivascular, and subpial locations. More commonly seen are **corpora amylacea**, or polyglucosan bodies. These are round, faintly basophilic, periodic acid-Schiff (PAS)-positive, concentrically lamellated structures of 5 to 50 μ m in diameter that are located wherever there are astrocytic end processes, especially in the subpial and perivascular zones. They consist primarily of glycosaminoglycan polymers, as well as heat-shock proteins and ubiquitin. Corpora amylacea occur in increasing numbers with advancing age and are thought to represent a

degenerative change in the astrocyte. **Lafora bodies**, which are seen in the cytoplasm of neurons (as well as hepatocytes, myocytes, and other cells) in a variant of myoclonic epilepsy, have a similar structure and biochemical composition.

Reactions of Microglia to Injury

Microglia are phagocytic cells derived early in embryonic development from the yolk sac or fetal liver that serve as the resident macrophages of the CNS and share many surface markers with bone marrow-derived peripheral monocytes/macrophages. At rest, microglia are tiled (i.e., cover non-overlapping territories) and have highly branched, amoeboid processes. During development, microglia prune unused synaptic connections, most likely through a phagocytic process that may involve the complement system; aberrant reactivation of this developmental process has recently been implicated in a number of different brain diseases, including schizophrenia, encephalitis, Alzheimer disease, and frontotemporal dementia. Microglia respond to injury by (1) proliferating; (2) developing elongated nuclei; (3) forming aggregates around small foci of tissue necrosis (*microglial nodules*); or (4) congregating around cell bodies of dying neurons (*neuronophagia*). In addition to resident microglia, blood-derived macrophages may also be present in inflammatory foci.

Reactions of Other Glial Cells to Injury

Oligodendrocytes are cells that wrap their cytoplasmic processes around axons and form myelin. Each oligodendrocyte myelinates numerous internodes on multiple axons, in contrast to the myelinating Schwann cell in peripheral nerves, which only myelinates the internode of a single axon. Injury or apoptosis of oligodendrocytes is a feature of acquired demyelinating disorders and leukodystrophies. Oligodendroglial nuclei may harbor viral inclusions in progressive multifocal leukoencephalopathy (PML). Glial cytoplasmic inclusions, primarily composed of α -synuclein, are found in oligodendrocytes in multiple system atrophy (MSA).

Ependymal cells, the ciliated columnar epithelial cells lining the ventricles, do not show specific reaction patterns. When there is inflammation or marked dilation of the ventricular system, disruption of the ependymal lining is paired with proliferation of subependymal astrocytes to produce small irregularities on the ventricular surfaces (*ependymal granulations*). Certain infectious agents, particularly CMV, may produce extensive ependymal injury, with viral inclusions in ependymal cells. However, neither oligodendrocytes nor ependymal cells mediate significant responses to most forms of injury in the CNS.

KEY CONCEPTS

CELLULAR PATHOLOGY OF THE CENTRAL NERVOUS SYSTEM

- Each cellular component of the nervous system responds to injury in a distinctive way.

- Neuronal injury commonly results in cell death, either by apoptosis or necrosis. Loss of neurons or synapses is often difficult to detect morphologically, but is a major contributor to neurologic dysfunction.
- Astrocytes respond to injury through apparent hypertrophy (increased cytoplasm due to accumulation of the intermediate filament protein GFAP) and hyperplasia; this response can be beneficial or harmful, depending on the type of injurious stimulus and the exact nature of astrocytic response.
- Microglia, the resident monocyte-lineage population of the CNS, proliferate and accumulate in response to injury. Aberrant activation of microglia- and complement-mediated synaptic pruning plays an important role in the pathogenesis of many different neurologic disorders.

CEREBRAL EDEMA, HYDROCEPHALUS, RAISED INTRACRANIAL PRESSURE, AND HERNIATION

The brain and the spinal cord are encased and protected by the rigid skull, dural reflections, and the bony spinal canal. The pressure within the cranial cavity may rise in any one of three commonly observed clinical settings: generalized brain edema, increased CSF volume, and focally expanding mass lesions. Depending on the degree and rapidity of the pressure increase and the nature of the underlying lesion, the consequences range from subtle neurologic deficits to death.

Cerebral Edema

Cerebral edema (more precisely, brain parenchymal edema) is the result of increased fluid leakage from blood vessels and injury to various cells of the CNS. There are two main pathways of edema formation in the brain.

- *Vasogenic edema* is an increase in extracellular fluid caused by blood-brain barrier disruption and increased vascular permeability, allowing fluid to shift from the intravascular compartment to the intercellular spaces of the brain. The paucity of lymphatics greatly impairs the resorption of excess extracellular fluid. Vasogenic edema may be either localized, e.g., adjacent to inflammation or neoplasms, or generalized, as may occur following a global ischemic injury.
- *Cytotoxic edema* is an increase in intracellular fluid secondary to neuronal, glial, or endothelial cell membrane injury, as might be encountered with a generalized hypoxic/ischemic insult or with a metabolic derangement that prevents maintenance of the normal membrane ionic gradients.

In practice, conditions associated with generalized edema often have elements of both vasogenic and cytotoxic edema. In generalized edema, the gyri are flattened, the intervening sulci are narrowed, and the ventricular cavities are compressed. As the brain expands, herniation may occur.



Figure 28.2 Hydrocephalus. Dilated lateral ventricles seen in a coronal section through the midthalamus.

Hydrocephalus

Hydrocephalus is the accumulation of excessive CSF within the ventricular system (Fig. 28.2). The choroid plexus within the ventricular system produces CSF, which normally circulates through the ventricular system and enters the cisterna magna at the base of the brainstem through the foramina of Luschka and Magendie. Subarachnoid CSF bathes the superior cerebral convexities and is absorbed by the arachnoid granulations. Most cases of hydrocephalus are a consequence of impaired flow and resorption of CSF; overproduction is a rare cause that can accompany tumors of the choroid plexus. In all forms, the increased volume of CSF expands the ventricles and can elevate the intracranial pressure.

When hydrocephalus develops in infancy before closure of the cranial sutures, the head enlarges. By contrast, after the sutures close, hydrocephalus is associated with expansion of the ventricles and increased intracranial pressure, without a change in head circumference. If the ventricular system is only focally obstructed, due to a mass in the third ventricle or to aqueductal stenosis, it is called *noncommunicating (obstructive) hydrocephalus*. In *communicating hydrocephalus*, the ventricular system remains in continuity with the subarachnoid space and there is enlargement of the entire ventricular system; causes include overproduction of CSF from a choroid plexus tumor and arachnoid fibrosis following meningitis. The term *hydrocephalus ex vacuo* refers to a compensatory increase in ventricular volume secondary to a loss of brain parenchyma.

Raised Intracranial Pressure and Herniation

Herniation is the displacement of brain tissue past rigid dural folds (the falx and tentorium) or through openings in the skull because of increased intracranial pressure. As the volume of the brain increases, CSF is displaced, leading to increasing pressure within the cranial cavity; herniation occurs when this pressure exceeds the brain's limited capacity to accommodate the increased intracranial pressure. Brain herniation is mostly caused by mass effects, either diffuse

(generalized brain edema) or focal (tumors, abscesses, or hemorrhages). Elevated intracranial pressure may also compress the vasculature and reduce perfusion of the brain, causing ischemic injury and further exacerbating cerebral edema.

MORPHOLOGY

The brain may herniate through different openings, and if the expansion is sufficiently severe, herniation may occur simultaneously in several locations (Fig. 28.3):

- **Subfalcine (cingulate) herniation** occurs when unilateral or asymmetric expansion of a cerebral hemisphere displaces the cingulate gyrus under the falx. This may lead to compression of the anterior cerebral artery and its branches, resulting in secondary infarcts.
- **Transtentorial (uncal, mesial temporal) herniation** occurs when the medial aspect of the temporal lobe is compressed against the free margin of the tentorium. With increasing displacement of the temporal lobe, the third cranial nerve is compromised, resulting in pupillary dilation and impaired ocular movements on the side of the lesion. The posterior cerebral artery may also be compressed, resulting in an infarct of its territory (which includes the primary visual cortex). When the extent of herniation is large enough, the contralateral cerebral peduncle may be compressed, resulting in hemiparesis ipsilateral to the side of the herniation. Progression of transtentorial herniation is often accompanied by secondary hemorrhagic lesions in the midbrain and pons, termed **Duret hemorrhages** (Fig. 28.4). These linear or flame-shaped lesions usually occur in the midline and paramedian regions and are believed to be

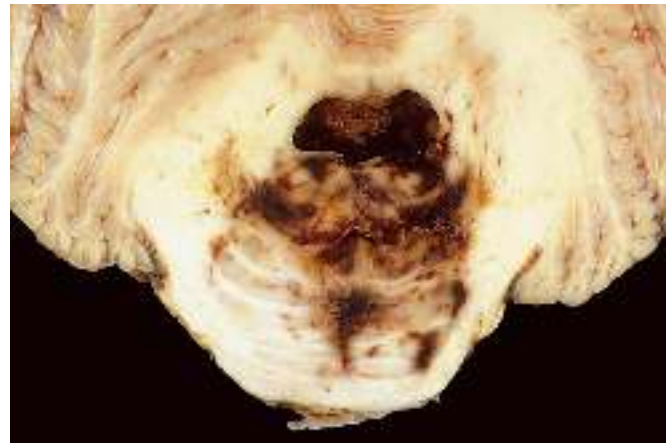


Figure 28.4 Duret hemorrhage. As mass effect displaces the brain downward, there is disruption of the vessels that enter the pons along the midline, leading to hemorrhage.

due to distortion or tearing of penetrating veins and arteries supplying the upper brainstem.

- **Tonsillar herniation** refers to displacement of the cerebellar tonsils through the foramen magnum. This pattern of herniation is life-threatening because it causes brainstem compression and compromises vital respiratory and cardiac centers in the medulla.

KEY CONCEPTS

CEREBRAL EDEMA, HYDROCEPHALUS, RAISED INTRACRANIAL PRESSURE, AND HERNIATION

- Cerebral edema is the accumulation of excess fluid within the brain parenchyma. Hydrocephalus is an increase in CSF volume within all or part of the ventricular system.
- Increase in the intracranial volume (due to increased CSF volume, parenchymal edema, hemorrhage, or tumor) raises intracranial pressure.
- Increased intracranial pressure may result in cerebral herniation, decreased perfusion, and secondary infarction of the affected areas.

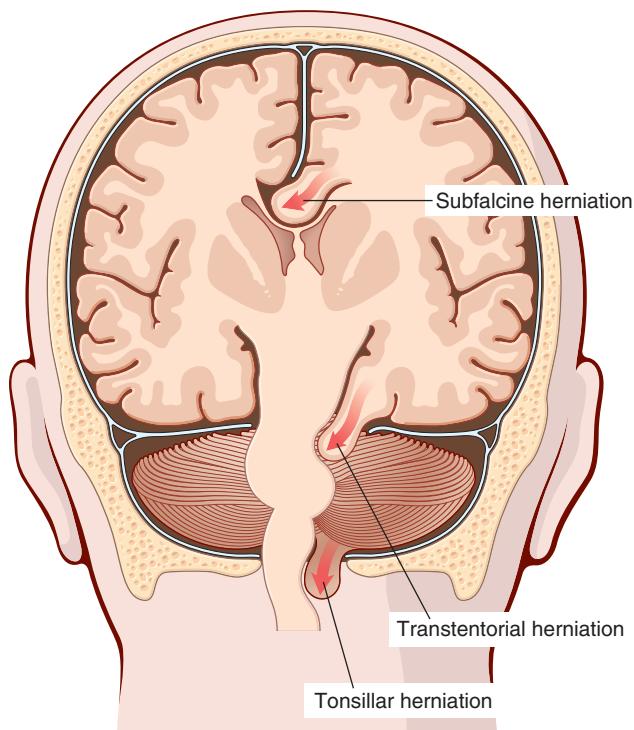


Figure 28.3 Major herniation syndromes of the brain: subfalcine, transtentorial, and tonsillar.

MALFORMATIONS AND DEVELOPMENTAL DISORDERS

Although the pathogenesis and etiology of many CNS malformations remain unknown, both genetic and environmental influences appear to be involved. Genomic sequencing has begun to uncover a range of genetic variants that are associated with certain malformations. The causal relationship between these genetic alterations and the pathogenesis of the malformations is the subject of active research. Besides genetic factors, many toxic compounds and infectious agents also have teratogenic effects and may cause brain malformations.

Neural Tube Defects

Neural tube defects are midline malformations that involve some combination of neural tissue, meninges, and overlying

bone or soft tissue; collectively, they are the most common CNS malformations. Two distinct pathogenic mechanisms are contributory: (1) failure of neural tube closure, in which secondary mesenchymal tissue defects stem from aberrant skeletal modeling around the malformed tube (e.g., anencephaly and myelomeningocele); and (2) primary bony defects that are caused by abnormal axial mesoderm development and lead to secondary CNS abnormalities (e.g., encephalocele, meningocele, and spina bifida).

MORPHOLOGY

- **Anencephaly** is a malformation of the anterior end of the neural tube that leads to absence of most of the brain and calvarium. Forebrain development is disrupted at approximately 28 days of gestation, and all that remains in its place is the **area cerebrovasculosa**, a flattened remnant of disorganized brain tissue with admixed ependyma, choroid plexus, and meningotheelial cells. The posterior fossa structures may be spared, depending on the extent of the skull deficit; descending tracts associated with disrupted structures are, as expected, absent.
- **Myelomeningocele** (or meningocele) refers to extension of CNS tissue through a defect in the vertebral column; the term **meningocele** applies when there is only a meningeal extrusion. Myelomeningoceles occur most commonly in the lumbosacral region (Fig. 28.5). Affected individuals have motor and sensory deficits in the lower extremities as well as disturbances of bowel and bladder control. They are often complicated by superimposed infection of the cord due to defective barrier function of the thin, overlying skin.



Figure 28.5 Myelomeningocele. Both meninges and spinal cord parenchyma are included in the cystlike structure visible just above the buttocks.

- **Encephalocele** refers to an extrusion of malformed brain tissue through a midline defect in the cranium. It most often occurs in the occiput, although nasofrontal variants involving the orbit, ethmoid, or cribriform plate (sometimes misleadingly referred to as a “nasal glioma”) also are seen.
- **Spinal dysraphism** or **spina bifida** (the most common neural tube defects) may be an asymptomatic bony defect (spina bifida occulta) or a severe malformation with a flattened, disorganized segment of spinal cord, associated with an overlying meningeal outpouching.

Clinical Features

The frequency of neural tube defects varies widely among different ethnic groups, with the overall recurrence rate for a neural tube defect in subsequent pregnancies estimated at 4% to 5%. Folate deficiency during the first several weeks of gestation is a well-established risk factor; differences in rates of neural tube defects among populations can be attributed in part to polymorphisms in enzymes involved in folic acid metabolism. Folate supplementation can lower the risk of neural tube defects, but because neural tube closure is normally complete by day 28 of embryonic development (before most pregnancies are recognized), it must be given to women throughout their reproductive years to be fully effective. Precisely how folate deficiency increases the risk is uncertain; effects on DNA methylation (an important epigenetic mode of gene regulation) are suspected.

Forebrain Anomalies

Abnormalities in the generation and migration of neurons result in malformations of the forebrain that may be focal or involve entire structures. The pool of proliferating precursor cells in the developing brain lies in the germinal matrix adjacent to the ventricular system; the total number of neurons is determined by the fraction of proliferating cells that undergo transition into migrating cells with each cell cycle. The migration of neurons from the germinal matrix to the cerebral cortex follows two paths: radial migration, for progenitor cells destined to become excitatory neurons; and tangential migration, for those that will become inhibitory interneurons.

A range of different anatomic malformation patterns has been defined. Recently, it has become clear that many of these patterns are caused by mutations in genes that are required for proper cerebral development. Changes may be seen in the complexity of the brain surface (either too few or too many gyri present), the organization of the brain into normal lobes, the structure of the cerebral cortex, or the distribution of neurons within the brain.

- The volume of brain may be abnormally large (*megalocephaly*) or abnormally small (*microcephaly*). Microcephaly, by far the more common of the two, is typically accompanied by a small head circumference. It is associated with a number of conditions, including chromosome abnormalities, fetal alcohol syndrome, and viral infection acquired in utero (e.g., with human immunodeficiency virus 1 [HIV-1] or Zika virus). It is postulated that the underlying anomaly is a reduction in the number of

neurons that reach the neocortex, which leads to a simplification of gyral folding, a mechanism supported by experimental results in mouse models.

- **Lissencephaly** is a malformation characterized by reduction in the number of gyri, which in the extreme case may show no gyral pattern (*agyria*). Two general patterns are observed, a smooth-surface form (type 1), and a rough- or cobblestone-surface form (type 2). In general, type 1 forms are associated with mutations that disrupt mechanisms involved in cell migration, such as mutations in the cytoskeletal “motor” proteins that drive migration of neuroblasts. In contrast, type 2 lissencephaly is most commonly associated with genetic alterations that disrupt the “stop signal” for migration. This signal depends on a set of specifically glycosylated proteins, and mutations in the enzymes that place sugars onto these proteins are the most common causes of this form of lissencephaly.
- **Polymicrogyria** is characterized by numerous small, irregularly formed cerebral convolutions with shallow sulci. The cerebral cortex is composed of four or fewer layers (instead of the normal six layers), with fusion of the molecular layers between gyri. Polymicrogyria can be induced by localized tissue injury toward the end of neuronal migration, although genetically determined forms, which are typically bilateral and symmetric, are also recognized.
- **Neuronal heterotopias** are a group of migrational disorders defined by collections of subcortical neurons in inappropriate locations along the pathway of migration. As might be expected, one location is along the ventricular surface—as though the cells never left their place of birth. Periventricular nodular heterotopias can be caused by mutations in the gene encoding filamin A, an actin-binding protein responsible for assembly of complex meshworks of filaments. This gene is on the X chromosome, and the mutant allele causes male lethality; in females, the process of X inactivation separates neurons into those with a normal allele (in the correct location) and those with the mutant allele (in the heterotopia). Another microtubule-associated protein, doublecortin (DCX), is also encoded by a gene on the X chromosome; mutations in this gene result in lissencephaly in males and in subcortical bandlike heterotopias in females (this parallel layer of gray matter imparts the impression of a “double cortex”). Nodular subcortical heterotopias may also be encountered.
- **Holoprosencephaly** is a spectrum of malformations characterized by incomplete separation of the cerebral hemispheres across the midline. Severe forms manifest midline facial abnormalities, including cyclopia; less severe variants (*arrhinencephaly*) show absence of the olfactory cranial nerves and related structures. Intrauterine diagnosis of severe forms by ultrasonography is now possible. Holoprosencephaly is associated with trisomy 13 as well as other genetic syndromes. Mutations in genes that encode components of the sonic hedgehog signaling pathway may also produce holoprosencephaly.
- **Agenesis of the corpus callosum**, a relatively common malformation, is the absence of the white matter bundles that carry cortical projections from one hemisphere to the other (Fig. 28.6). Radiologic imaging studies show misshapen lateral ventricles (“bat-wing” deformity); on coronal whole-mount sections of the brain, “Probst



Figure 28.6 Agenesis of the corpus callosum. The midsagittal view of the left hemisphere shows the lack of a corpus callosum and cingulate gyrus.

bundles” of anteroposteriorly oriented white matter can be demonstrated. Agenesis of the corpus callosum is sometimes associated with intellectual disability but may also be found in normal individuals. It may be sporadic or familial and can be present in isolation or in association with a range of other malformations.

Posterior Fossa Anomalies

A distinct set of malformations primarily affect the brainstem and the cerebellum, which often show dramatic changes in size and shape. These may be accompanied by morphologic changes in other regions of the brain.

MORPHOLOGY

- **Arnold-Chiari malformation** (Chiari type II malformation) consists of a small posterior fossa, a misshapen midline cerebellum with downward extension of vermis through the foramen magnum (Fig. 28.7), and, almost invariably, hydrocephalus and a lumbar myelomeningocele. Other associated changes may include caudal displacement of the medulla, malformation of the tectum, aqueductal stenosis, cerebral heterotopias, and hydromyelia (see later).
- **Chiari type I malformation** is a less severe disorder in which low-lying cerebellar tonsils extend down into the vertebral canal. It may be a silent abnormality or may become symptomatic because of impaired CSF flow and medullary compression; if present, these symptoms can usually be corrected by neurosurgical intervention.
- **Dandy-Walker malformation** is characterized by an enlarged posterior fossa. The cerebellar vermis is absent or present only in rudimentary form in its anterior portion; in its place is a large midline cyst that is lined by ependyma and is contiguous with leptomeninges on its outer surface. This cyst represents the expanded, roofless fourth ventricle in the absence of a normally formed vermis. Dysplasias of brainstem nuclei are commonly found in association with Dandy-Walker malformation.



Figure 28.7 Arnold-Chiari malformation. Midsagittal section showing small posterior fossa contents, downward displacement of the cerebellar vermis, and deformity of the medulla (arrows indicate the approximate level of the foramen magnum).

- **Joubert syndrome** and related disorders share hypoplasia of the cerebellar vermis with apparent elongation of the superior cerebellar peduncles and an altered shape of the brainstem; together these changes give rise to the “molar tooth sign” on imaging. This group of malformations has been found to be caused by diverse mutations affecting genes that encode components of the primary (nonmotile) cilium.

Syringomyelia and Hydromyelia

These disorders are characterized by expansion of the ependyma-lined central canal of the cord (*hydromyelia*) or by the formation of a fluid-filled cleftlike cavity in the inner portion of the cord (*syringomyelia*, *syrinx*) that may extend into the brainstem (*syringobulbia*).

Syringomyelia may be associated with a Chiari malformation; it may also occur in association with intraspinal tumors or following traumatic injury. In general, the histologic appearance is similar in all of these conditions, with destruction of the adjacent gray and white matter, surrounded by dense reactive gliosis. The disease generally becomes manifest in the second or third decade of life. The distinctive symptoms and signs of a syrinx are the isolated loss of pain and temperature sensation in the upper extremities because of disruption of the crossing anterior spinal commissural fibers of the spinal cord.

KEY CONCEPTS

MALFORMATIONS AND DEVELOPMENTAL DISORDERS

- Malformations may be associated with single-gene mutations, larger-scale genetic alterations, or exogenous factors.

- Overall, the earlier in development a malformation occurs, the more severe the morphologic and functional phenotype.
- Neural tube defects are caused by the failure of the tube to close or by skeletal bony defects that lead to secondary tube abnormalities; they range from incidental findings to severe malformations.
- Cortical development depends on proper orchestration of progenitor cell proliferation in the germinal matrix and migration of progenitors upward into the developing cortex. Disruption of these processes can alter the size, shape, and organization of the brain.
- Malformations involving the posterior fossa are typically distinct from those that affect the cerebral hemispheres.

PERINATAL BRAIN INJURY

Brain injury occurring in the perinatal period is an important cause of childhood-onset neurologic disability. Injuries that occur early in gestation may destroy brain tissue without eliciting the reactive changes observed in adult brain and, therefore, may be difficult to distinguish from malformations. Different patterns of injury may occur.

- The term *cerebral palsy* refers to a nonprogressive neurologic motor deficit characterized by combinations of spasticity, dystonia, ataxia/athetosis, and paresis, attributable to brain injury occurring during the prenatal and perinatal periods. Signs and symptoms may not be apparent at birth and only declare themselves later, as development proceeds. Postmortem examinations of children with cerebral palsy have shown a wide range of neuropathologic findings, including destructive lesions traced to remote events that may have caused hemorrhage and infarction.
- In premature infants, there is an increased risk of *germinal matrix hemorrhage*, often near the junction between the developing thalamus and caudate nucleus. Hemorrhages may remain small and localized or extend into the ventricular system and subarachnoid space, sometimes leading to hydrocephalus and death in severe cases.
- Infarcts may occur in the supratentorial periventricular white matter (*periventricular leukomalacia*), especially in premature infants; they take the form of chalky yellow plaques that consist of discrete white matter necrosis, often with dystrophic calcification. Ultimately, the infarcted areas develop into large cystic lesions (Fig. 28.8); when damage is extensive and involves both gray and white matter, the condition is termed *multicystic encephalopathy*.
- In perinatal ischemic lesions of the cerebral cortex, the depths of sulci bear the brunt of injury and result in mushroom-shaped gyri with thinned-out, gliotic stalks (*ulegyria*). The basal ganglia and thalamus may also suffer ischemic injury, with patchy neuronal loss and reactive gliosis. Later, aberrant and irregular myelinization gives rise to a marble-like appearance of the deep nuclei (*status marmoratus*). Because the lesions are in the caudate, putamen, and thalamus, movement disorders such as choreoathetosis are common clinical sequelae.



Figure 28.8 Chronic stage of periventricular leukomalacia. Large cystic spaces in the periventricular white matter (seen in both hemispheres of this brain) are the long-term sequelae of a severe prenatal or perinatal ischemic injury.

TRAUMA

The anatomic location of the lesion and the limited capacity of the brain for functional repair are major determinants of the consequences of CNS trauma. Injury of several cubic centimeters of brain parenchyma may be clinically silent (e.g., in the frontal lobe), severely disabling (e.g., in the spinal cord), or fatal (e.g., in the brainstem).

The physical forces associated with head injury may result in skull fractures, parenchymal injury, and vascular injury; all three can coexist. The magnitude and distribution of a traumatic brain lesion depends on the shape of the object causing the trauma, the force of impact, and whether the head is in motion at the time of injury. A blow to the head may be penetrating or blunt; it may cause either an open or a closed injury.

Skull Fractures

A fracture in which bone is displaced into the cranial cavity by a distance greater than the thickness of the bone is called a *displaced skull fracture*. The thickness of the cranial bones varies; therefore, their resistance to fracture differs greatly. Also, the relative incidence of fractures among skull bones is related to the pattern of falls. For example, when an individual falls while awake (such as might occur when stepping off a ladder), the site of impact is often the occipital portion of the skull; in contrast, a fall that follows loss of consciousness (as might follow a syncopal attack) can result in either frontal or occipital impact. Symptoms referable to the lower cranial nerves or the cervicomedullary region, and the presence of orbital or mastoid hematomas distant from the point of impact, raise the suspicion of a basal skull fracture, which typically follows impact to the occiput or sides of the head; CSF leakage from the nose or ear and infection (meningitis) may follow.

Parenchymal Injuries

The structural consequences and clinical manifestations of the parenchymal injury depend on the nature and severity of the causative head trauma and range from mild (concussion) to severe (contusions and lacerations).

Concussion

Concussion is a clinical syndrome of altered consciousness secondary to head injury, typically brought about by a change in the momentum of the head (e.g., following impact with a rigid object). The characteristic clinical picture includes the sudden onset of transient neurologic dysfunction, including loss of consciousness, temporary respiratory arrest, and loss of reflexes. Although neurologic recovery is usually complete, amnesia for the event often persists. The pathogenesis of the disruption of neurologic function is unknown, but likely involves dysregulation of the reticular activating system in the brainstem. Postconcussive neuropsychiatric syndromes, typically associated with repetitive injuries, are well recognized, and there is increasing evidence that significant cognitive impairment can emerge along with distinct pathologic findings termed chronic traumatic encephalopathy (discussed later).

Direct Parenchymal Injury

Contusions and *lacerations* are brain injuries caused by transmission of kinetic energy to the brain. A contusion is analogous to the familiar bruise caused by blunt trauma, and a laceration is an injury caused by penetration of an object and tearing of tissue. As with any other organ, a blow to the surface of the brain, transmitted through the skull, leads to rapid tissue displacement, disruption of vascular channels, and subsequent hemorrhage, tissue injury, and edema. Hemorrhage can extend into the subarachnoid space from these lesions. The crests of gyri are most susceptible because this is where the direct force is greatest. The most common locations for contusions correspond to the most frequent sites of direct impact and to regions of the brain that overlie a rough and irregular inner skull surface, such as the frontal lobes along the orbital ridges and the temporal lobes. Contusions are less frequent over the occipital lobes, brainstem, and cerebellum, but may be seen when there is an adjacent skull fracture (fracture contusions).

A person who suffers a blow to the head may develop a contusion at the point of contact (a *coup* injury) or on the brain surface diametrically opposite to it (a *contrecoup* injury). Their macroscopic and microscopic appearances are indistinguishable, and the distinction between them is based on identification of the point of impact. In general, if the head is immobile at the time of trauma, only a coup injury is found. If the head is mobile, both coup and contrecoup lesions may be found, though the latter predominate and are thought to develop when the brain strikes the opposite inner surface of the skull after sudden deceleration.

Sudden impacts that result in violent posterior or lateral hyperextension of the neck (as occurs when a pedestrian is struck from the rear by a vehicle) may avulse the pons from the medulla or the medulla from the cervical cord, causing instant death.

MORPHOLOGY

When seen on cross-section, contusions are wedge shaped, with the broad base lying along the surface at the point of impact (Fig. 28.9A). The appearance of contusions is similar regardless of the source of the trauma. In the earliest stages, there is edema and hemorrhage, which is often pericapillary. During the next few hours, the extravasation of blood extends throughout the involved tissue, across the width of the cerebral cortex, and into the white matter and subarachnoid space. Morphologic evidence of neuronal injury (pyknosis of the nucleus, eosinophilia of the cytoplasm, and disintegration of the cell) takes 12 to 24 hours to appear, although functional deficits generally occur earlier. Axonal swellings develop along the full length of the damaged neurons. The inflammatory response to the injured tissue follows its usual course, with the appearance of sparse neutrophils followed by abundant macrophages. Old traumatic lesions on the surface of the brain have a characteristic gross appearance. They are depressed, retracted, yellowish brown patches involving the crests of gyri, most commonly those that are located at the sites of contrecoup injuries (inferior frontal cortex, temporal and occipital poles); these lesions, called **plaque jaune** (Fig. 28.9B), can become epileptic foci.

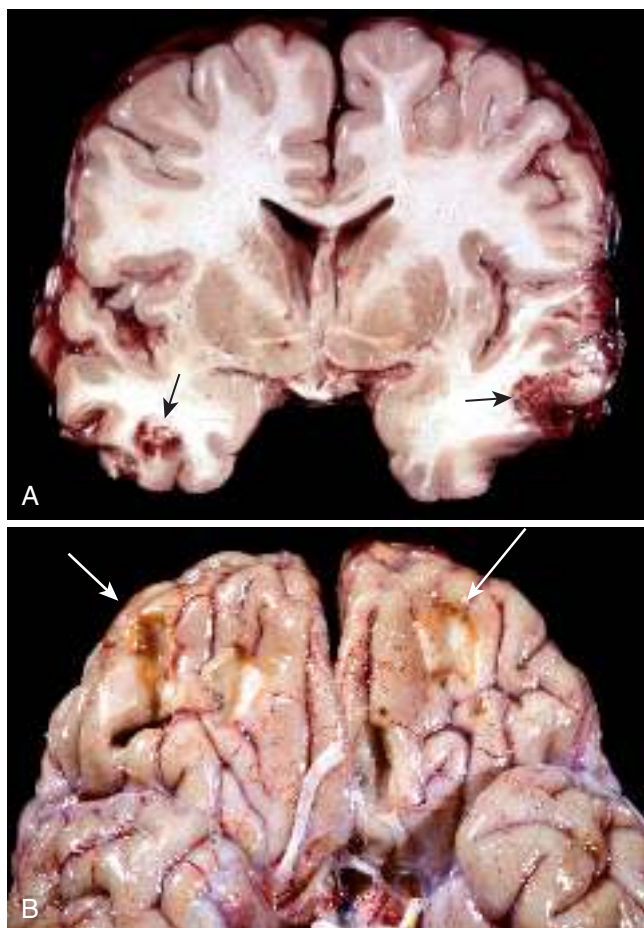


Figure 28.9 (A) Acute contusions are present in both temporal lobes, with areas of hemorrhage and tissue disruption (arrows). (B) Remote contusions (arrows) are present on the inferior frontal surface of this brain and have a yellow color (plaque jaune) that reflects hemosiderin accumulation.

More extensive hemorrhagic regions of brain trauma give rise to larger cavitated lesions, which resemble remote infarcts. In old contusions, gliosis and residual hemosiderin-laden macrophages predominate.

Diffuse Axonal Injury

Traumatic brain injuries may also damage the deep white matter regions, cerebral peduncles, superior colliculi, and deep reticular formation in the brainstem. The microscopic findings include axonal swelling, indicative of *diffuse axonal injury*, and focal hemorrhagic lesions. As many as 50% of individuals who develop coma shortly after trauma, even without cerebral contusions, are believed to have diffuse axonal injury. Axons are injured directly by mechanical forces, with subsequent alterations in axoplasmic flow. Such injuries may occur from marked changes in angular acceleration (e.g., from blast injuries), even in the absence of physical impacts involving the skull.

MORPHOLOGY

Diffuse axonal injury is characterized by widespread, often asymmetric axonal swellings that appear within hours of the injury and may persist for much longer. The swelling is best demonstrated with silver impregnation techniques or with immunoperoxidase stains for axonally transported proteins, such as amyloid precursor protein and α -synuclein. Later, increased numbers of microglia are seen in damaged areas of the cerebral cortex, and subsequently there is degeneration of the involved fiber tracts.

Traumatic Vascular Injury

Vascular injury is a frequent component of CNS trauma. It results from disruption of the vessel wall and leads to hemorrhage in different anatomic sites (Table 28.1). Depending on the position of the ruptured vessel, hemorrhage may occur in the epidural, subdural, subarachnoid, and intraparenchymal compartments, sometimes in combination (Fig. 28.10). Epidural and subdural hemorrhages rarely occur outside of the setting of trauma; in some settings, such as coagulopathy or significant cerebral atrophy, subdural hemorrhages can follow even minor trauma. Subarachnoid hemorrhage almost always accompanies parenchymal trauma, but can also develop spontaneously secondary to vascular anomalies (discussed later).

Epidural Hematoma

The dura is fused with the periosteum and is supplied by a number of dural arteries. These arteries, most notably the middle meningeal artery, are vulnerable to traumatic injury. In adults, this most often occurs with temporal skull fractures in which the fracture crosses the course of the vessel. In children, in whom the skull is deformable, a temporary displacement of the skull bones leading to laceration of a vessel can occur in the absence of a skull fracture.

Once a vessel has been torn, the extravasation of blood under arterial pressure can cause the dura to separate from the periosteum, creating a space (Fig. 28.11). The expanding hematoma compresses the underlying brain. When blood accumulates slowly, patients may experience a lucid period

Table 28.1 Patterns of Hemorrhage in the Central Nervous System

Location	Etiology	Additional Features
Epidural space	Trauma	Usually associated with skull fracture (in adults); rapidly evolving neurologic symptoms (often after a short lucid period) that require intervention
Subdural space	Trauma	May follow minor trauma; slowly evolving neurologic symptoms, often with a delay from the time of injury
Subarachnoid space	Trauma Vascular abnormality (arteriovenous malformation or aneurysm)	Typically associated with underlying parenchymal injury Sudden onset of severe headache, often with rapid neurologic deterioration; secondary injury may emerge and is associated with vasospasm
Intraparenchymal space	Trauma (contusions) Ischemia (hemorrhagic conversion of an ischemic infarct) Cerebral amyloid angiopathy Hypertension Tumors (primary or metastatic)	Selective involvement of the crests of gyri, where the brain is in contact with the inner surface of the skull (frontal and temporal tips, orbitofrontal surface) Petechial hemorrhages in an area of previously ischemic brain, usually following the cortical ribbon “Lobar” hemorrhage, involving subcortical white matter and often with extension into the subarachnoid space Centered in the deep white matter; thalamus, basal ganglia, or brainstem; may extend into the ventricular system Associated with high-grade gliomas or certain metastases (melanoma, choriocarcinoma, renal cell carcinoma)

before the onset of neurologic signs. A symptomatic epidural hematoma is a neurosurgical emergency; without prompt diagnosis and drainage, fatal brain herniation may occur within a few hours.

Subdural Hematoma

The dura is composed of two layers—an external collagenous layer and an inner more cellular layer containing fibroblasts. Bridging veins travel from the convexities of the cerebral hemispheres through the subarachnoid space and dura to empty into the dural sinuses. The brain is suspended in CSF, but the venous sinuses are fixed relative to the dura; as a result, traumatic displacement of the brain can tear the veins at the point where they penetrate the dura. The extravasated blood dissects through the two layers of the dura, producing a subdural hematoma. In older individuals

with brain atrophy, the bridging veins are stretched, hence the increasing incidence of subdural hematoma with aging. Infants are also very susceptible to subdural hematomas because their bridging veins are thin-walled.

MORPHOLOGY

Grossly, **acute subdural hematomas** appear as a collection of freshly clotted blood along the brain surface, without extension into the depths of sulci (Fig. 28.12). The underlying brain is flattened, and the subarachnoid space is often clear. Usually, venous bleeding is self-limited and the resulting hematoma is broken down and organized over time; this most often occurs in the following sequence:

- Lysis of the clot (about 1 week)
- Growth of fibroblasts from the dural surface into the hematoma (2 weeks)

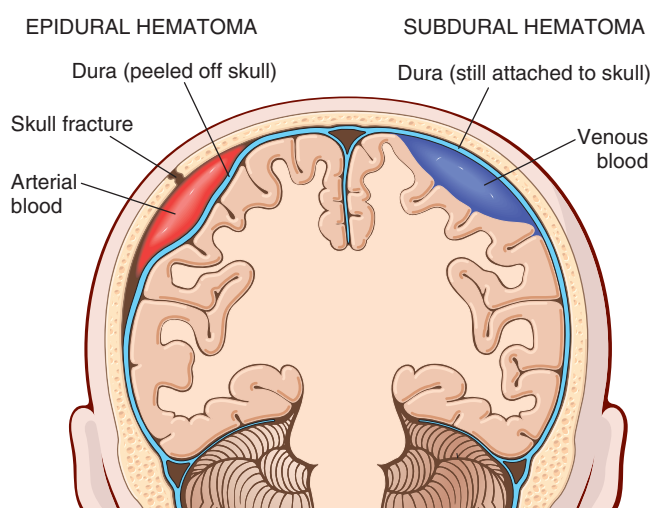


Figure 28.10 Epidural hematoma (left) in which rupture of a meningeal artery, usually associated with a skull fracture, leads to accumulation of arterial blood between the dura and the skull. In a subdural hematoma (right), damage to bridging veins between the brain and the superior sagittal sinus leads to the accumulation of blood between the dura and the arachnoid.

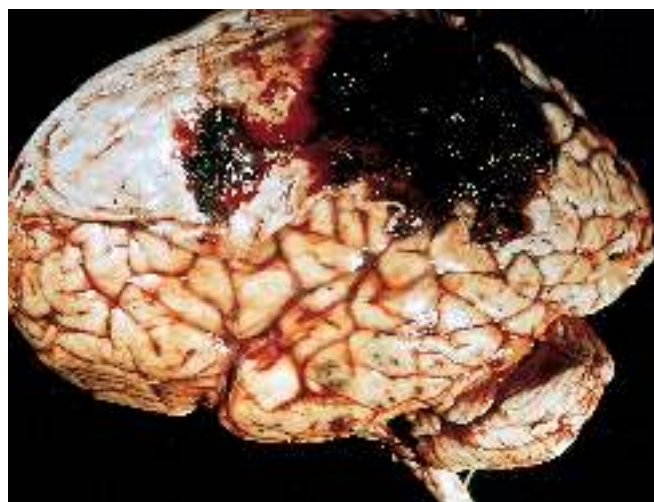


Figure 28.11 Epidural hematoma covering a portion of the dura. Also present are multiple small contusions in the temporal lobe. (Courtesy the late Dr. Raymond D. Adams, Massachusetts General Hospital, Boston, Mass.)

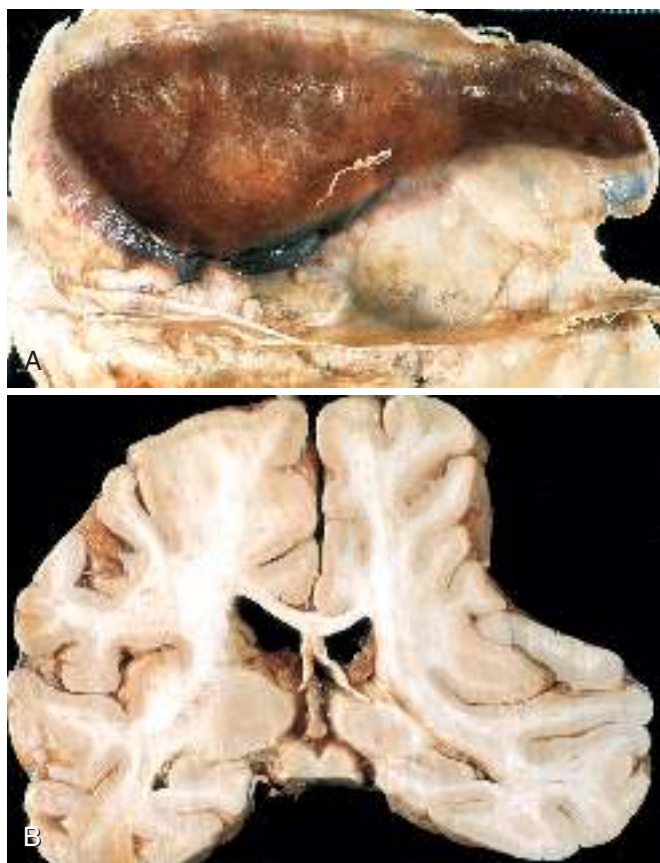


Figure 28.12 (A) Large organizing subdural hematoma attached to the dura. (B) Coronal section of the brain showing atrophy of the hemisphere compressed by the subdural hematoma shown in A.

- Early development of hyalinized connective tissue (1 to 3 months)

Typically, the organized hematoma is firmly attached to the inner surface of the dura by ingrowing fibrous tissue and is free of the underlying arachnoid, which does not contribute to healing. The lesion can eventually retract as the granulation tissue matures until only a thin layer of reactive connective tissue remains ("subdural membranes"). In other cases, however, multiple recurrent episodes of bleeding occur (**chronic subdural hematoma**), presumably from the thin-walled vessels of the granulation tissue.

Clinical Features

Symptomatic subdural hematomas most often manifest within 48 hours of injury. They are most common over the lateral aspects of the cerebral hemispheres and are bilateral in about 10% of cases. Neurologic signs are attributable to the pressure exerted on the adjacent brain. There may be focal signs, but often the clinical manifestations are nonlocalizing and include headache and confusion. Slowly progressive neurologic deterioration is typical, but acute decompensation may also occur. The treatment of subdural hematomas is to remove the blood and associated organizing tissue. The risk of repeat bleeding is greatest in the first few months after the initial hemorrhage.

Sequelae of Brain Trauma

A broad range of neurologic syndromes may become manifest months or years after brain trauma of any cause. These have gained increasing notice in the context of litigation involving issues of compensation for those in the civilian work force, professional athletes, and the military services.

- *Posttraumatic hydrocephalus* is largely due to obstruction of CSF resorption from hemorrhage into the subarachnoid space.
- *Chronic traumatic encephalopathy* (CTE, previously referred to as "dementia pugilistica") is a dementing illness that develops after repeated head trauma. Affected brains are atrophic, with enlarged ventricles, and show accumulation of tau-containing neurofibrillary tangles in a characteristic pattern involving gyral depths and perivascular regions in the frontal and temporal lobe cortices. Although concussion is thought of as having no structural consequences, it is clear that repeated events are antecedents to CTE; however, it remains uncertain which factors determine whether encephalopathy will ultimately develop (the number, frequency, and/or severity of individual traumatic events, or some combination of these).
- Other important sequelae of brain trauma include post-traumatic epilepsy, risk of infection, and psychiatric disorders.

Spinal Cord Injury

The spinal cord is vulnerable to trauma from its skeletal encasement. Most injuries that damage the cord are associated with the transient or permanent displacement of the vertebral column. The level of cord injury determines the extent of the neurologic manifestations: lesions involving the thoracic vertebrae or below can lead to paraplegia; cervical lesions result in quadriplegia; those above C4 can, in addition, lead to respiratory compromise from paralysis of the diaphragm. Damage at the region of impact to descending and ascending white matter tracts isolates the distal spinal cord from the rest of the brain. This interruption, paired with the localized gray matter damage at the level of the impact, is the principal cause of neurologic deficits.

MORPHOLOGY

The histologic features of traumatic injury of the spinal cord are similar to those found at other sites in the CNS. At the level of injury the acute phase consists of hemorrhage, necrosis, and axonal swelling in the surrounding white matter. The lesion tapers above and below the level of injury. In time, central areas of neuronal destruction become cystic and gliotic; cord sections above and below the lesion show secondary ascending and descending Wallerian degeneration, respectively, involving the long white-matter tracts at the site of trauma.

KEY CONCEPTS

Perinatal Brain Injury

- Timing of injury is critical, with earlier events resulting in greater damage and deficits.

- Cerebral palsy is the term used for nonprogressive deficits associated with injury during the prenatal and perinatal periods.

Trauma

- Physical injury to the brain can occur when the inside of the skull comes into forceful contact with the brain.
- In blunt trauma, if the head is mobile there may be brain injury both at the original point of contact (coup injury) and on the opposite side of the brain (contrecoup injury).
- Parenchymal injuries take the form of contusions, with hemorrhage extending into the subarachnoid space.
- Rapid displacement of the head and brain can tear axons (diffuse axonal injury), often immediately causing severe, irreversible neurologic deficits.
- Traumatic tearing of blood vessels leads to epidural or subdural hematoma.

CEREBROVASCULAR DISEASE

Cerebrovascular disease—injury to the brain as a consequence of altered blood flow—can be grouped into ischemic and hemorrhagic etiologies, with tissue infarction the ultimate consequence of both. “Stroke” is the clinical designation applied to these conditions and is defined as neurologic signs and symptoms that can be explained by a vascular mechanism, have an acute onset, and persist beyond 24 hours. (If symptoms disappear within 24 hours, the event is termed “transient ischemic event.”) Cerebrovascular disease is the third leading cause of death (after heart disease and cancer) in the United States, and the most prevalent cause of morbidity and mortality from neurologic disease. Stroke stems from two major mechanisms:

- *Ischemia and/or hypoxia* resulting from impairment of blood supply and oxygenation of CNS tissue. This can be either a global or focal process, with the clinical manifestations determined by the region of brain affected. In the brain, embolism is a more common cause of vascular occlusion than thrombosis.
- *Hemorrhage* resulting from rupture of CNS vessels. Common etiologies include hypertension and vascular anomalies (aneurysms and malformations; see [Table 28.1](#)).

Hypoxia and Ischemia

Although the brain accounts for only 1% to 2% of body weight, it receives approximately 15% of the resting cardiac output and accounts for 20% of the body’s oxygen consumption. Cerebral blood flow remains relatively constant over a wide range of blood pressure and intracranial pressure because of autoregulation of vascular resistance. The brain is strictly dependent on aerobic metabolism to meet its constant energy demands, and it may be deprived of oxygen by either hypoxemia (low blood oxygen content) or ischemia (inadequate blood flow). Inadequate blood flow may result from a reduction in perfusion pressure (as in hypotension), small- or large-vessel obstruction, or both.

When blood flow to a portion of the brain is reduced, the survival of the tissue at risk depends on the presence of collateral circulation, the duration of ischemia, and the magnitude and rapidity of the reduction of flow. These factors determine, in turn, the precise anatomic site and size of the lesion and, consequently, the clinical deficit.

The general biochemical changes in cells resulting from ischemia are discussed in Chapter 2. In addition to processes shared with ischemia in other parts of the body, ischemia in the CNS can result in inappropriate release of excitatory amino acid neurotransmitters such as glutamate, which can damage neurons by allowing excessive influx of calcium ions through N-methyl-D-aspartate (NMDA)-type glutamate receptors; this phenomenon is termed *excitotoxicity*. In the region of transition between necrotic tissue and the normal brain, there is an area of “at-risk” brain, referred to as the *penumbra*; this region can be rescued from cell death in many animal models with a variety of anti-apoptotic interventions, implying that ischemic neurons may die by apoptosis as well as necrosis.

Focal Cerebral Ischemia

Focal cerebral ischemia follows reduction or cessation of blood flow to a localized area of the brain due to partial or complete arterial obstruction. When the ischemia is sustained, infarction follows in the territory of the compromised vessel. The size, location, and shape of the infarct and the extent of tissue damage that results are influenced by the duration of the ischemia and the adequacy of collateral flow. The major source of collateral flow is the circle of Willis (supplemented by external carotid-ophthalmic artery collaterals). Inconstant collateral leptomeningeal vessels from the surface of the brain may also supply the distal branches of the anterior, middle, and posterior cerebral arteries through cortical-leptomeningeal anastomoses; in contrast, there is little if any collateral flow for the deep penetrating vessels of the thalamus, basal ganglia, and deep white matter.

Occlusive vascular disease of severity sufficient to lead to cerebral infarction may be due to embolization from a distant source, in situ thrombosis, or various vasculitides; the pathology of these conditions is also discussed in Chapters 4 and 11.

- *Embolism* to the brain occurs from a variety of sources. Cardiac mural thrombi are among the most common culprits; myocardial infarct, valvular disease, and atrial fibrillation are important predisposing factors. Next in frequency are thromboemboli originating from arteries, most often atheromatous plaques within the carotids. Other sources of emboli include paradoxical thromboemboli, particularly in children with cardiac anomalies; thromboemboli associated with cardiac surgery; and emboli of other types (tumor, fat, or air). The territory supplied by the middle cerebral artery—the direct extension of the internal carotid artery—is most frequently affected by embolic infarction; the incidence is about equal in the two hemispheres. Emboli tend to lodge where blood vessels branch or in areas of preexisting luminal stenosis. “Shower embolization,” as in fat embolism, may occur after fractures; affected individuals manifest generalized cerebral dysfunction with disturbances of higher

cortical function and consciousness, often without localizing signs. Widespread hemorrhagic lesions involving the white matter are characteristic of embolization of bone marrow after trauma.

- *Thrombotic occlusion of the cerebral arteries* is most commonly caused by acute change of vulnerable atherosclerotic plaques, as in coronary artery disease (Chapter 12). The most common sites are the carotid bifurcation, the origin of the middle cerebral artery, and either end of the basilar artery. Thrombi cause progressive narrowing of the lumen, may be accompanied by anterograde extension, and may progress to fragmentation and distal embolization. Atherosclerotic cerebrovascular disease is frequently associated with systemic diseases such as hypertension and diabetes.
- *Inflammatory processes* that involve blood vessels may also lead to luminal narrowing, occlusion, and, hence, cerebral infarcts. Although infectious vasculitis of small and large vessels occurs with syphilis and tuberculosis, it is now more common in the setting of immunosuppression and opportunistic infection (e.g., aspergillosis). *Polyarteritis nodosa* and other noninfectious vasculitides may involve cerebral vessels and cause single or multiple infarcts throughout the brain. *Primary angiitis of the CNS* can also develop in the absence of systemic vasculitis.
- Other conditions that may cause thrombosis (and intracranial hemorrhage) include hypercoagulable states, dissecting aneurysm of extracranial arteries in the neck that supply the brain, and drug abuse (amphetamines, heroin, cocaine).

Brain infarcts are subdivided into two broad groups based on the presence of secondary hemorrhage. Because the brain has end-organ circulation with limited collateral supply, occlusive brain infarcts generally start as nonhemorrhagic (pale/anemic; Fig. 28.13A); clinically, these nonhemorrhagic infarcts are called ischemic, a confusing term as every infarct, not just this type, is caused by tissue ischemia. Secondary hemorrhage can occur from ischemia-reperfusion injury following spontaneous or therapeutic dissolution or fragmentation of the intravascular occlusive material. This process (termed *secondary hemorrhagic transformation* and leading to

a hemorrhagic infarct) develops if the causative ischemic event lasts long enough to damage small blood vessels in the affected area; the resulting reperfusion hemorrhages are largely petechial in nature, but may be multiple or even confluent (see Fig. 28.13B). The clinical management of patients with nonhemorrhagic and hemorrhagic infarcts differs greatly, although the underlying causes are the same (for instance, thrombolytic therapy is contraindicated in a patient with brain hemorrhage of any etiology).

MORPHOLOGY

Both the gross and microscopic appearance of a **nonhemorrhagic infarct** changes over time. Grossly, there is little change in appearance during the first 6 hours of irreversible injury. By 48 hours, however, the tissue becomes pale, soft, and swollen, and the gray-white matter junction becomes indistinct. From 2 to 10 days, the brain becomes gelatinous and friable, and the previously ill-defined boundary between normal and infarcted tissue becomes more distinct as edema resolves in the viable adjacent tissue. From 10 days to 3 weeks, the tissue liquefies, eventually leaving a fluid-filled cavity that continues to expand until all of the dead tissue has been removed (Fig. 28.14).

Microscopically, the tissue reaction evolves along the following sequence:

- **Acute infarct** (Fig. 28.15A). After the first 6 to 12 hours, neurons in the affected area show eosinophilic neuronal necrosis (increased eosinophilia of the cytoplasm followed by nuclear pyknosis and karyorrhexis; “dead red neurons”); both cytotoxic and vasogenic edema are present. There is loss of the usual tinctorial characteristics of white- and gray-matter structures. Endothelial and glial cells, mainly astrocytes, swell, and myelinated fibers begin to disintegrate. Up to 48 hours, neutrophilic emigration progressively increases and then falls off (but is never as prominent as in myocardial infarction).
- **Subacute (evolving) infarct** (Fig. 28.15B). Phagocytic cells, derived from circulating monocytes and activated microglia, are evident at 48 to 72 hours and become the predominant cell type in the ensuing 2 to 3 weeks. The macrophages become stuffed with the products of myelin breakdown or blood and may persist in the lesion for months to years. Reactive astrocytes

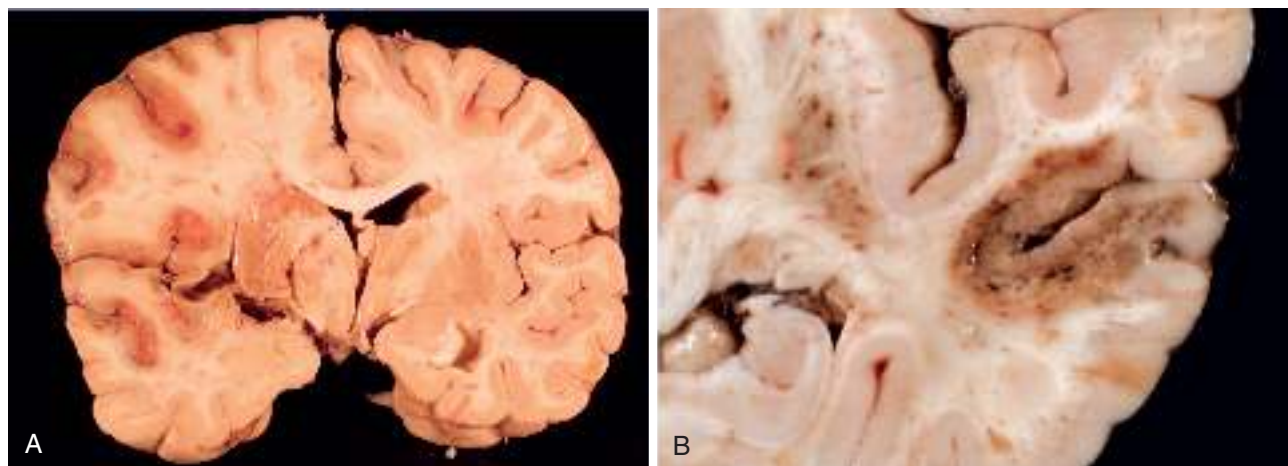


Figure 28.13 (A) An ischemic infarct involves the territory of the middle cerebral artery, including the striatum, on the left side of this brain. (B) A hemorrhagic infarct with punctate hemorrhages, consistent with ischemia-reperfusion injury, is present in the temporal lobe.

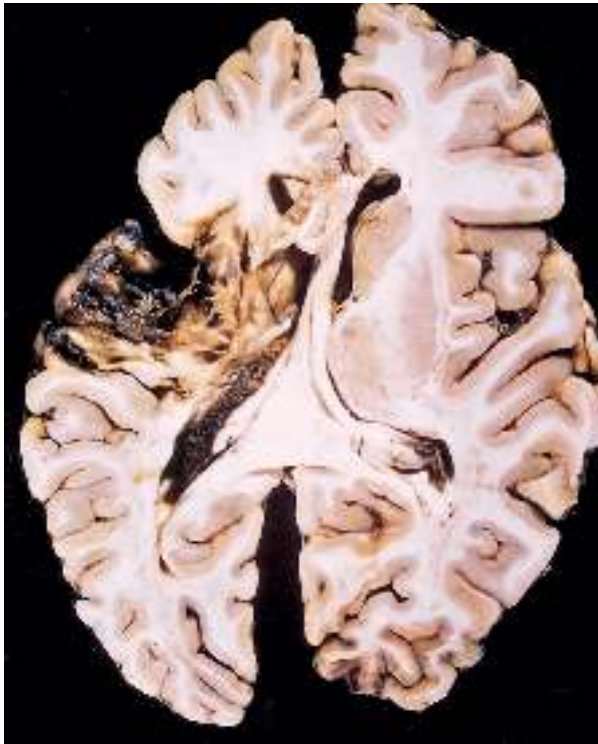


Figure 28.14 Old cystic infarct showing cavitation from loss of brain parenchyma.

and newly formed vessels can be seen at the periphery of the lesion as early as 1 week after the insult. As the process of liquefaction and phagocytosis proceeds, astrocytes at the edges of the lesion progressively enlarge, divide, and develop a prominent network of cytoplasmic extensions.

- **Healed infarct (Fig. 28.15C).** After several months, the astrocytic response recedes, leaving behind a dense meshwork of glial fibers admixed with new capillaries and some perivascular connective tissue. In the cerebral cortex, the cavity is separated from the meninges and subarachnoid space by a gliotic layer of tissue, which is derived from the molecular layer of the cortex. The pia and arachnoid are not affected. Infarcts undergo these reactive and reparative stages from the edges inward; thus, different areas of a lesion may appear chronologically divergent, revealing the natural progression of the response.

The features and temporal evolution of **hemorrhagic infarctions** parallel ischemic infarctions, with the addition of blood extravasation and resorption. In individuals receiving anticoagulant treatment, hemorrhagic infarcts may be associated with extensive intracerebral hematomas. Venous infarcts are often hemorrhagic and may occur after thrombotic occlusion of the superior sagittal sinus or other sinuses, or after occlusion of the deep cerebral veins. Neoplasms, localized infections, and other conditions leading to a hypercoagulable state increase the risk for venous thrombosis.

Lacunar Infarcts

Hypertension affects the deep penetrating arteries and arterioles that supply the basal ganglia and hemispheric white matter as well as the brainstem; these cerebral vessels develop *arteriolosclerosis*, also known as small vessel disease because it affects arteries 40 to 900 μm in diameter, described in more detail later. If this disease process progresses to thrombosis and complete vessel occlusion, the end-result is development of small cavitary infarcts known as *lacunes* or *lacunar infarcts* (Fig. 28.16). These lakelike spaces, arbitrarily defined as less than 15 mm wide, can be single or multiple and involve the putamen, globus pallidus, thalamus, internal capsule, deep white matter, caudate nucleus, and pons, in descending order of frequency. On microscopic

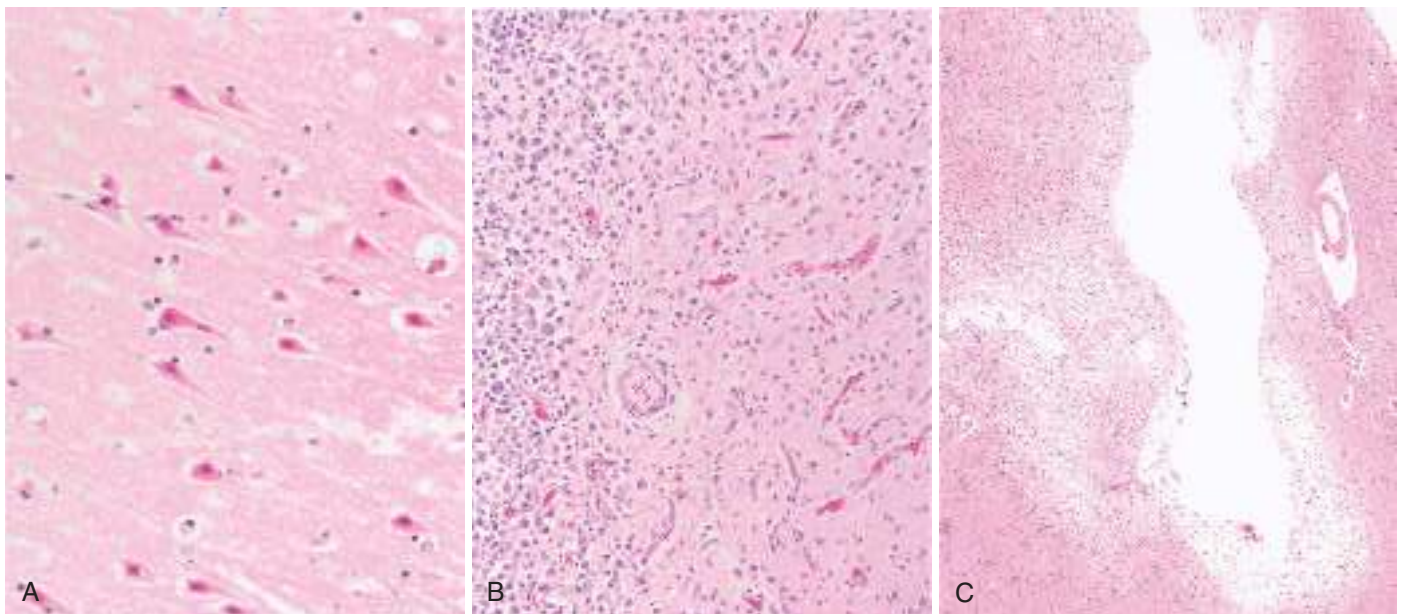


Figure 28.15 Cerebral infarcts. (A) Acute ischemic injury causes diffuse eosinophilia of neurons, which are beginning to shrink. (B) After about 10 days, the lesion is characterized by the presence of foamy macrophages (best seen on the left) and adjacent reactive gliosis with neovascularization (on the right). (C) Remote small infarct is seen as an area of tissue loss surrounded by residual gliosis.

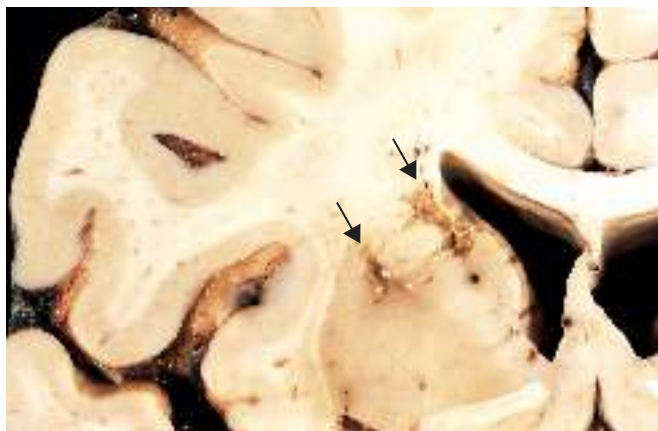


Figure 28.16 Lacunar infarcts in the caudate and putamen (arrows).

examination, there is tissue loss surrounded by gliosis. Depending on their location in the CNS, lacunar infarcts can be clinically silent or cause severe neurologic impairment. Affected vessels may also be associated with widening of the perivascular spaces without tissue infarction (*état criblé*).

Clinical Features

Deficits produced by infarction are determined by the anatomic distribution of the damage, rather than the underlying cause. Neurologic symptoms referable to the area of injury often develop rapidly, over minutes, and may continue to evolve over hours. There can be improvement in severity of symptoms that is associated with reversal of injury in the ischemic penumbra and resolution of associated local edema. In general, there is often slow improvement during a period of months. Because strokes are frequently associated with cardiovascular disease, many of the genetic and lifestyle risk factors are shared. Early diagnosis is of critical importance, as rapid treatment of nonhemorrhagic strokes with thrombolytic agents can often limit or entirely prevent the development of permanent neurologic deficits.

Global Cerebral Hypoxia/Ischemia

Global cerebral hypoxia or ischemia occurs when there is a generalized reduction of cerebral perfusion (as in cardiac arrest, shock, and severe hypotension) or decreased oxygen carrying capacity of the blood (e.g., in carbon monoxide poisoning). The clinical outcome varies with the severity and length of the insult. In mild cases, there may only be a transient postischemic confusional state, followed by complete recovery. Nevertheless, irreversible CNS damage may occur in individuals who experience global hypoxic and/or ischemic insults (*diffuse hypoxic/ischemic encephalopathy*). Among CNS cells, there is a hierarchy of sensitivity to hypoxia/ischemia: neurons are the most sensitive, although glial cells (oligodendrocytes and astrocytes) are also vulnerable. The most sensitive neurons in the brain are pyramidal neurons in the hippocampus (especially area CA1, also referred to as Sommer sector), cerebellar Purkinje cells, and pyramidal neurons in the cerebral cortex (especially layers III and V); the molecular mechanisms underlying this selective vulnerability are not understood. With severe global cerebral hypoxia/ischemia, widespread neuronal death occurs, irrespective of regional vulnerability; patients who survive this injury

often remain in a persistent vegetative state. Other patients meet the current clinical criteria for “brain death,” including evidence of irreversible diffuse cortical injury (isoelectric, or “flat,” electroencephalogram), brainstem damage (such as absent reflexes and respiratory drive), and absent cerebral perfusion. When individuals with this pervasive form of injury are maintained on mechanical ventilation, the brain gradually undergoes widespread liquefaction, producing so-called “respirator brain.”

Border zone (“watershed”) infarcts occur in the regions of the brain or spinal cord that lie at the most distal reaches of the arterial blood supply (i.e., the border zones between arterial territories). In the cerebral hemispheres, the border zone between the anterior and the middle cerebral artery distributions is at greatest risk. Damage to this region, located a few centimeters lateral to the interhemispheric fissure, results in a cortical wedge-shaped infarct that usually shows secondary hemorrhagic transformation (Fig. 28.17) and is often bilateral. Border zone infarcts usually develop after severe hypotensive episodes and are most commonly seen in patients resuscitated after cardiac arrest.

MORPHOLOGY

In the setting of global ischemia, the brain becomes edematous and swollen, producing widening of the gyri and narrowing of the sulci. The cut surface shows poor demarcation between gray and white matter. The microscopic features of irreversible ischemic injury evolve over time and mimic the changes seen in infarcts; the distinction between global and focal ischemic injury is based not on the nature of cellular pathology but on the overall pattern of brain involvement. **Early changes**, occurring 6 to 12 hours



Figure 28.17 Classic watershed infarct with secondary hemorrhagic transformation (arrow); boundary between anterior and middle cerebral artery circulations.

after the insult, are seen in neurons (dead red neurons described earlier); similar acute changes occur somewhat later in astrocytes and oligodendroglia. **Subacute changes**, occurring at 24 hours to 2 weeks, include tissue necrosis, influx of macrophages, vascular proliferation, and reactive gliosis. **Repair**, robust after approximately 2 weeks, is characterized by removal of necrotic tissue, loss of normal CNS architecture, and gliosis. In the cerebral neocortex, the neuronal loss and gliosis are uneven, with preservation of some layers and destruction of others, producing a pattern of injury termed **laminar necrosis**.

Intracranial Hemorrhage

Hemorrhages may occur at any site within the cranium—outside the brain or within it (intraparenchymal). Hemorrhages in the epidural or subdural space are typically associated with trauma and were discussed earlier; hemorrhages within the brain parenchyma and in the subarachnoid space, in contrast, are more often a manifestation of underlying cerebrovascular disease and are discussed in the following sections (see Table 28.1).

Intraparenchymal Hemorrhage

Rupture of a small intraparenchymal vessel can result in a primary hemorrhage within the brain, often associated with sudden onset of neurologic symptoms (stroke); this should not be confused with the secondary hemorrhagic transformation of an occlusive infarct (described earlier). Spontaneous (nontraumatic) intraparenchymal hemorrhages occur most commonly in middle to late adult life, with a peak incidence at about 60 years of age. Hemorrhages in the basal ganglia and thalamus are commonly designated “ganglionic hemorrhages,” whereas those that occur in the lobes of the cerebral hemispheres are called “lobar hemorrhages”; the two major causes of these patterns of hemorrhage are hypertension and cerebral amyloid angiopathy, respectively. In addition, other local and systemic factors may cause or contribute to nontraumatic hemorrhage, including systemic coagulation disorders, neoplasms, vasculitis, aneurysms, and vascular malformations.

Hypertension is the risk factor most commonly associated with deep brain parenchymal hemorrhages, accounting for more than 50% of clinically significant hemorrhages and for roughly 15% of deaths among individuals with chronic hypertension. *Hypertensive intraparenchymal hemorrhage* may originate in the putamen (50% to 60% of cases), thalamus, pons, cerebellar hemispheres (rarely), and other regions of the brain (Fig. 28.18A). Hypertension leads to a number of vessel wall abnormalities, including accelerated atherosclerosis in larger arteries, hyaline arteriosclerosis in smaller arteries, and (in severe cases) proliferative changes and frank necrosis of arterioles. Arteriolar walls affected by hyaline change (Fig. 28.18B) are thickened but more vulnerable to rupture than normal vessels; these changes are most prominent in the basal ganglia and the subcortical white matter. As described earlier, if small arteries affected by hyaline arteriosclerosis do not rupture but are occluded, the result is lacunar infarction.

Cerebral amyloid angiopathy (CAA) is the risk factor most commonly associated with lobar hemorrhages (Fig. 28.18C). In CAA, amyloidogenic peptides, usually the same

ones found in Alzheimer disease ($A\beta$; see later), are deposited in the walls of medium- and small-caliber meningeal, cortical, and cerebellar vessels; involved vessels are rigid, and as a result fail to collapse during tissue processing and sectioning. Although similar to hyaline arteriosclerosis on routine hematoxylin and eosin (H&E) stain, the hyaline material in CAA consists of β amyloid rather than collagen (Fig. 28.18D) and is primarily seen in the leptomeningeal and cortical (rather than basal ganglia and white matter) vessels. Amyloid deposition can weaken the vessel wall and lead to hemorrhage; as a result, many individuals with CAA have evidence of numerous small hemorrhages within the brain (“microbleeds”) that can be visualized by various imaging methods. As with Alzheimer disease (discussed later), there is a relationship between polymorphisms in the gene that encodes apolipoprotein E (ApoE) and risk of disease; specifically, the presence of either an $\epsilon 2$ or $\epsilon 4$ allele increases the risk of bleeding. Autosomal dominant forms of CAA are associated with certain mutations in the *APP* gene, which encodes the precursor for the $A\beta$ peptides that are prone to deposit as amyloid.

Other forms of hereditary small-vessel diseases of the CNS have been identified. *Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy* (CADASIL) is an autosomal dominant disorder caused by mutations in the *NOTCH3* gene that lead to misfolding of the extracellular domain of the NOTCH3 receptor. The disease is characterized clinically by recurrent small vessel strokes (usually infarcts, less often hemorrhages) and dementia. Imaging studies show that the first detectable changes are in white matter, usually presenting around 35 years of age and then progressing further over time. Other forms of the heritable small vessel disease include a disorder associated with mutations in the gene for COL4A1, a component of the vascular basement membrane.

MORPHOLOGY

Acute primary intraparenchymal hemorrhages are characterized by a central core of clotted blood that compresses the adjacent parenchyma; this compression leads to secondary infarction of the affected brain tissue, with anoxic neuronal and glial changes as well as edema. Eventually the edema resolves, hemosiderin- and lipid-laden macrophages appear, and proliferation of reactive astrocytes is seen at the periphery of the lesion; the cellular events then follow the same time course that is observed after cerebral infarction. Old hemorrhages show areas of parenchymal cavitory destruction with a rim of brownish discoloration.

Clinical Features

Intracerebral hemorrhage can be clinically devastating if it involves a large part of the brain or extends into the ventricular system. When hemorrhage affects smaller regions, it is either clinically silent or evolves like an infarct; over weeks or months, there is a gradual removal of the hematoma, sometimes with considerable clinical improvement. Again, the location of the hemorrhage determines the clinical manifestations.

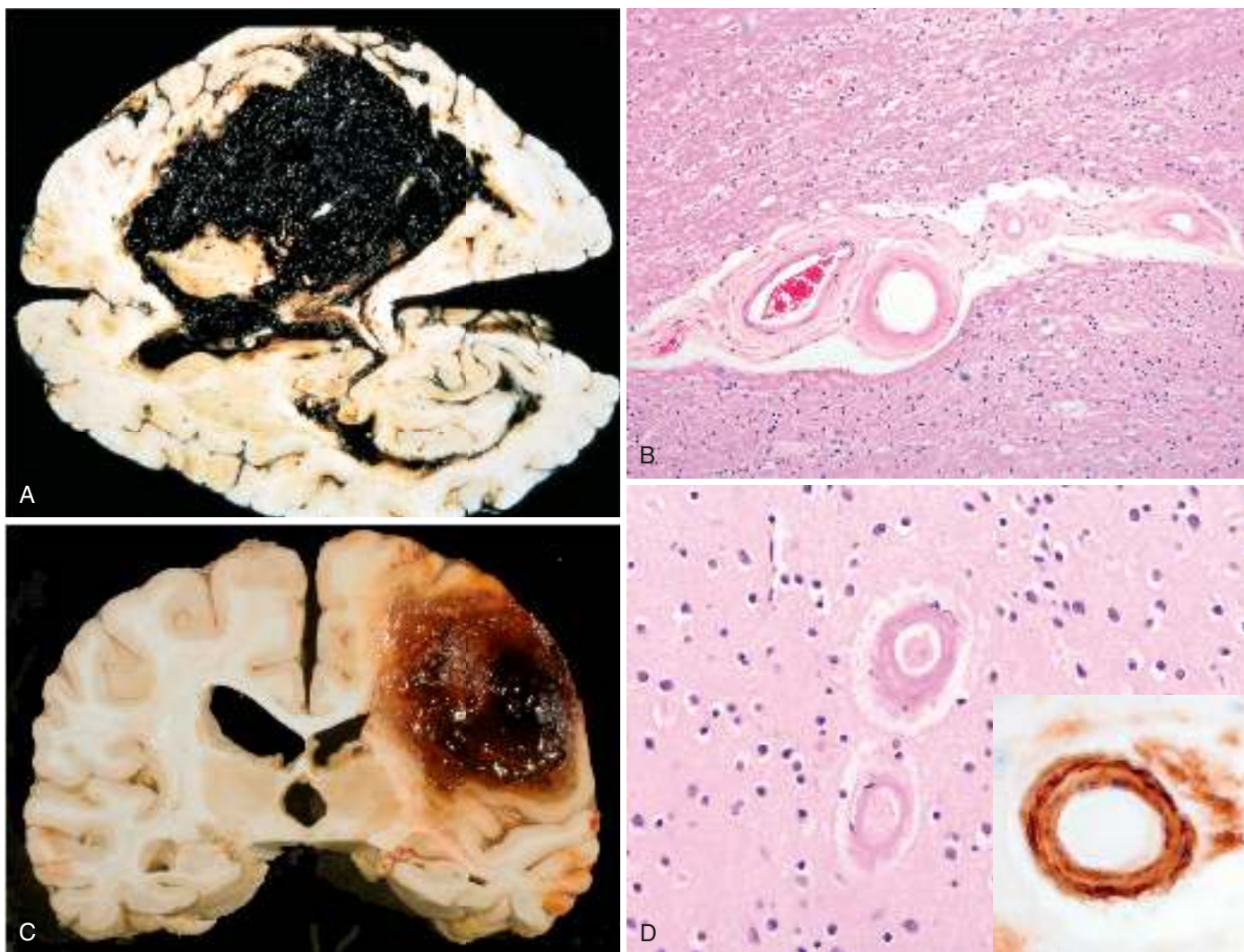


Figure 28.18 (A) Massive hypertensive ganglionic hemorrhage rupturing into a lateral ventricle. (B) Hyaline arteriosclerosis (fibrosis and thickening of the arteriolar walls) develops in the basal ganglia and subcortical white matter of patients with long-standing hypertension; it is a risk factor for hypertensive hemorrhages as well as lacunar infarcts. (C) Large lobar hemorrhage due to cerebral amyloid angiopathy; it focally dissects into the subarachnoid space. (D) Amyloid deposition in cortical arterioles in cerebral amyloid angiopathy; inset, immunohistochemical staining for A β highlights the deposited material in the vessel wall. (C, Courtesy Dr. Dimitri Agamanolis, <http://neuropathology-web.org>.)

Subarachnoid Hemorrhage and Ruptured Saccular Aneurysm

The most frequent cause of spontaneous subarachnoid hemorrhage is rupture of a saccular (“berry”) aneurysm in a cerebral artery. (As noted earlier, brain trauma is the most common cause of subarachnoid hemorrhage overall.) Nontraumatic subarachnoid hemorrhage may also result from rupture of a primary intracerebral hemorrhage into the ventricular system, vascular malformation, hematologic disturbances, and tumors.

Saccular aneurysm is the most common type of intracranial aneurysm; other aneurysm types include atherosclerotic (fusiform; mostly of the basilar artery), mycotic, traumatic, and dissecting. These latter three, like saccular aneurysms, are most often found in the anterior circulation, but more often cause cerebral infarction rather than subarachnoid hemorrhage.

Saccular aneurysms are found in about 2% of the population according to recent data from community-based radiologic studies. About 90% of saccular aneurysms are found near major arterial branch points in the anterior circulation (Fig. 28.19); multiple aneurysms exist in 20% to 30% of cases based on autopsy series.

Pathogenesis

Although the etiology of saccular aneurysms remains obscure, the structural abnormality of the involved vessel (absence of smooth muscle and intimal elastic lamina) suggests that they are developmental anomalies. The majority occur sporadically, but genetic factors may be important in their pathogenesis because there is an increased incidence of aneurysms in first-degree relatives of those affected. There is also an increased incidence in individuals with certain Mendelian disorders (e.g., autosomal dominant polycystic kidney disease, Ehlers-Danlos syndrome type

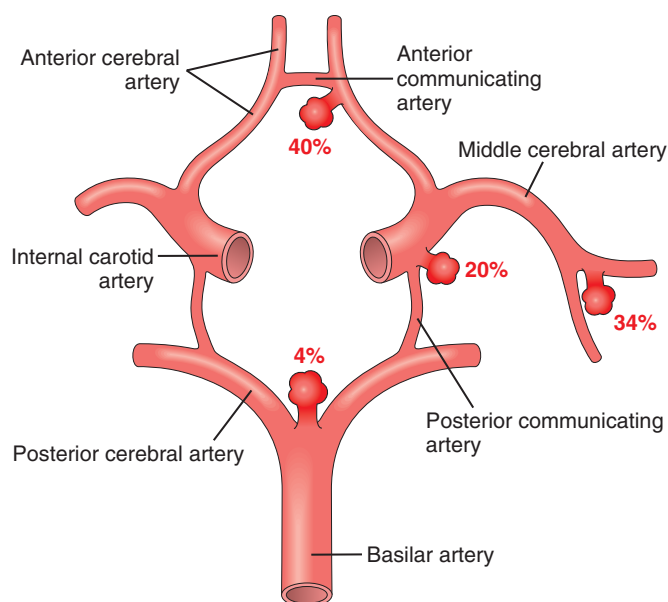


Figure 28.19 Common sites of saccular (berry) aneurysms in the circle of Willis.

IV, neurofibromatosis type 1 [NF1], and Marfan syndrome), fibromuscular dysplasia of extracranial arteries, and coarctation of the aorta. Other predisposing factors include cigarette smoking and hypertension (estimated to be present in about half of affected individuals). Although they are sometimes referred to as “congenital,” the aneurysms are not present at birth but develop over time because of an underlying defect in the media of the vessel wall.

MORPHOLOGY

An unruptured saccular aneurysm is a thin-walled outpouching, usually at an arterial branch point along the circle of Willis or a

major vessel just beyond. Saccular aneurysms measure from a few millimeters to 2 or 3 cm in diameter and have a bright red, shiny surface and a thin, translucent wall (Fig. 28.20). Atheromatous plaques, calcification, or thrombi may be found in the wall or lumen of the aneurysm. Sometimes there is evidence of prior hemorrhage, in the form of brownish discoloration of the adjacent brain and meninges. The neck of the aneurysm may be wide or narrow. Rupture usually occurs at the apex of the sac and leads to extravasation of blood into the subarachnoid space, the substance of the brain, or both. The arterial wall adjacent to the neck of the aneurysm often shows some intimal thickening and attenuation of the media. Smooth muscle and intimal elastic lamina do not extend into the neck and are absent from the aneurysm sac itself, which is made up of thickened hyalinized intima and a covering of adventitia.

Clinical Features

Rupture of an aneurysm leading to subarachnoid hemorrhage is most frequent in the fifth decade and is slightly more frequent in women. Overall, aneurysms rupture at a rate of 1.3% per year, but the risk is higher for larger aneurysms; for example, aneurysms greater than 10 mm in diameter have a roughly 50% risk of bleeding per year. Rupture may occur at any time, but in about one-third of cases it is associated with acute increases in intracranial pressure, such as with straining at stool or sexual orgasm. Blood under arterial pressure is forced into the subarachnoid space, and affected individuals are stricken with a sudden, excruciating headache (“the worst headache I’ve ever had”) and rapidly lose consciousness. Between 25% and 50% of patients die with the first rupture, but patients who survive often improve and recover consciousness in minutes. Repeat bleeding is common in survivors and unpredictable in timing. With each episode of bleeding, the prognosis is worse.

The clinical consequences of blood in the subarachnoid space can be separated into acute events (occurring within

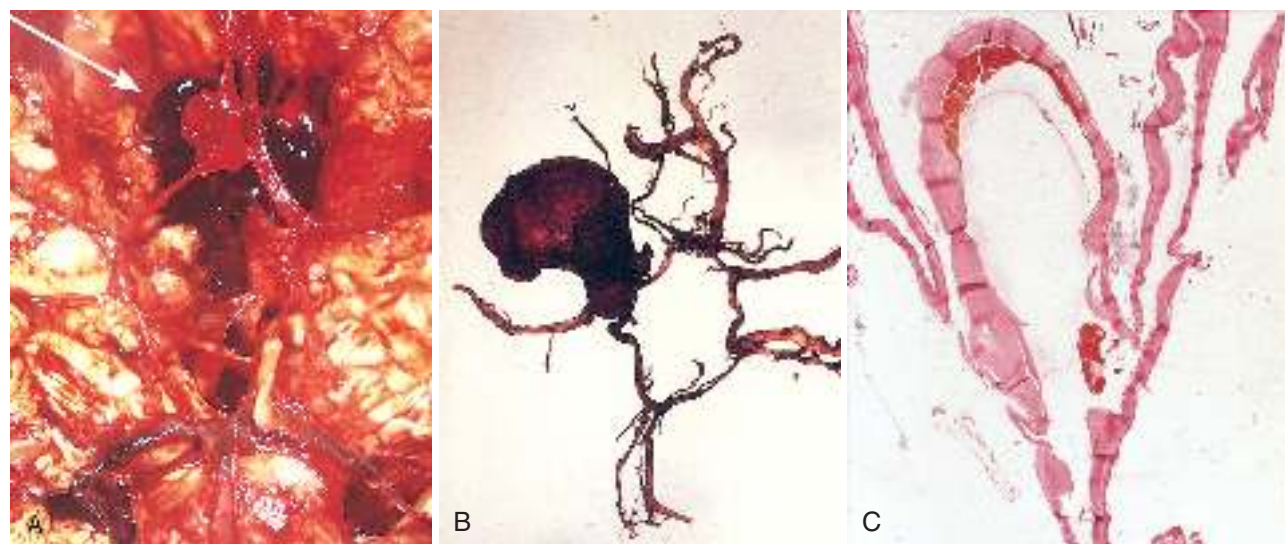


Figure 28.20 (A) View of the base of the brain, dissected to show the circle of Willis with an aneurysm of the anterior cerebral artery (arrow). (B) Dissected circle of Willis to show large aneurysm. (C) Section through a saccular aneurysm showing the hyalinized fibrous vessel wall (hematoxylin and eosin).

hours to days after the hemorrhage) and late sequelae associated with the healing process. In the first few days after a subarachnoid hemorrhage, regardless of the etiology, there is an increased risk of additional ischemic injury from vasospasm affecting vessels bathed in the extravasated blood. This problem is of greatest significance in cases of basal subarachnoid hemorrhage, in which vasospasm can involve major vessels of the circle of Willis. Various mediators have been proposed to have a role in this process, including endothelins, nitric oxide, and arachidonic acid metabolites. In the healing phase of subarachnoid hemorrhage, meningeal fibrosis and scarring occur, sometimes leading to obstruction of CSF flow as well as interruption of the normal pathways of CSF resorption.

Vascular Malformations

Vascular malformations of the brain are classified into four principal groups: arteriovenous malformations, cavernous malformations, capillary telangiectasias, and venous angiomas. Of these, the first two are the types associated with risk of hemorrhage and development of neurologic symptoms. Although lacking morphologic features that typify neoplasms, arteriovenous malformations are frequently associated with activating somatic mutations in the *KRAS* oncogene within the endothelial cells that line the malformed vessels, suggesting that dysregulated RAS signaling has a central role in their pathogenesis.

MORPHOLOGY

Arteriovenous malformations (tangled networks of wormlike vascular channels with high blood flow due to prominent, pulsatile arteriovenous shunting) may involve vessels in the subarachnoid space, the brain, or both (Fig. 28.21). They are composed of greatly enlarged blood vessels separated by gliotic tissue, often with evidence of prior hemorrhage. Some vessels can be recognized as arteries with duplication and fragmentation of the internal elastic lamina, while others show marked thickening or partial replacement of the media by hyalinized connective tissue.

Cavernous malformations consist of distended, loosely organized vascular channels arranged back to back with collagenized

walls of variable thickness; there is usually no brain parenchyma between vessels in this type of malformation. They occur most often in the cerebellum, pons, and subcortical regions, in decreasing order of frequency, and are “low-flow” channels that do not participate in arteriovenous shunting. Foci of old hemorrhage, infarction, and calcification frequently surround the abnormal vessels.

Clinical Features

Arteriovenous malformations are the most common clinically significant vascular malformation. Males are affected twice as frequently as females. The lesion often presents between 10 and 30 years of age as a seizure disorder, an intracerebral hemorrhage, or a subarachnoid hemorrhage. The most common site is the territory of the middle cerebral artery, particularly its posterior branches. Large arteriovenous malformations occurring in the newborn period can lead to congestive heart failure because of shunt effects, especially if the malformation involves the vein of Galen. Cavernous malformations are unique among this group of disorders in that familial forms are relatively common. Multiplicity of lesions is a hallmark of familial cases, which are inherited as a highly penetrant autosomal dominant trait.

Vascular Dementia

Individuals who, over the course of many months and years, suffer multiple, bilateral, gray matter (cortex, thalamus, basal ganglia) and white matter (centrum semiovale) infarcts may develop a distinctive clinical syndrome characterized by dementia, gait abnormalities, and pseudobulbar signs, often with superimposed focal neurologic deficits. The syndrome, generally referred to as *vascular dementia*, is caused by multifocal vascular disease of several types, including (1) cerebral atherosclerosis, (2) vessel thrombosis or embolization from carotid vessels or from the heart, and (3) cerebral arteriosclerosis from chronic hypertension. When the pattern of injury preferentially involves large areas of the subcortical white matter with myelin and axon loss, the disorder is referred to as *Binswanger disease* (*subcortical white matter dementia*); this distribution of vascular white-matter injury must be distinguished clinically and radiologically from other diseases that affect the hemispherical white matter. In addition, many individuals with neurodegenerative diseases resulting in cognitive impairment or dementia also have evidence of cerebrovascular disease. The presence of significant cerebrovascular disease increases risk of neurologic impairment for a given level of lesions associated with the degenerative diseases, suggesting that it is an independent contributing factor to disruption of normal brain function.

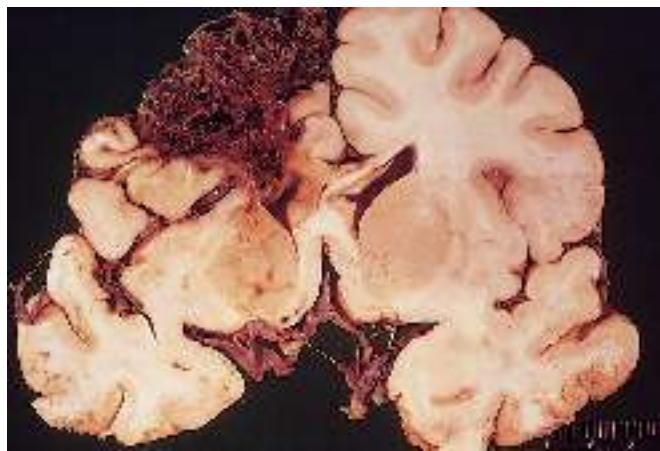


Figure 28.21 Large arteriovenous malformation in the left cerebral hemisphere.

KEY CONCEPTS

CEREBROVASCULAR DISEASES

- Stroke is the clinical term for acute-onset neurologic deficits that last longer than 24 hours and that are a consequence of vessel occlusion or vessel rupture.
- Cerebral infarction follows loss of blood supply and can be focal, widespread, or restricted to regions with the least robust vascular supply (boundary zones).

- Large artery infarcts are most commonly embolic; with subsequent dissolution of the embolus and reperfusion, a nonhemorrhagic infarct can become hemorrhagic.
- Primary intraparenchymal hemorrhages lead to infarction of the adjacent brain parenchyma and are typically either ganglionic (most commonly due to hypertension) or lobar (most commonly due to CAA).
- Hyaline arteriolosclerosis (small vessel disease), which is caused by a long-standing hypertension, can lead to either lumen occlusion (resulting in a lacunar infarct) or wall rupture (resulting in a ganglionic hemorrhage).
- Spontaneous subarachnoid hemorrhage is usually caused by a structural vascular abnormality, such as an aneurysm or arteriovenous malformation.

INFECTIONS

Infection may damage the nervous system directly through injury of neurons or glia by the infectious agent, or indirectly through microbial toxins, the destructive effects of the inflammatory response, or as a result of immune-mediated mechanisms. There are four principal routes by which microbes enter the nervous system.

- *Hematogenous spread* is the most common; infectious agents ordinarily gain access through the arterial circulation, but retrograde venous spread can occur via anastomoses with veins of the face.
- *Direct implantation* of microorganisms is most often traumatic, but can be sometimes associated with congenital malformations (e.g., meningocele) that provide ready access for microorganisms.
- *Local extension* can originate from infected adjacent structures, such as air sinuses, teeth, skull, or vertebrae.
- Viruses also may be transported along the *peripheral nervous system*, as occurs with rabies and herpes zoster viruses.

General aspects of the pathology of infectious agents are discussed in Chapter 8; distinctive forms of CNS infections are described herein (Table 28.2).

Acute Meningitis

Meningitis is an inflammatory process of the leptomeninges and CSF within the subarachnoid space, usually caused by an infection. *Meningoencephalitis* refers to inflammation of the meninges and brain parenchyma. Although infections are the most common causes of meningitis and meningoencephalitis, this reaction may also occur in response to a nonbacterial irritant introduced into the subarachnoid space (*chemical meningitis*) or in the context of a systemic autoimmune disease. Based on the etiology and clinical evolution of the illness, infectious meningitis is broadly classified into *acute pyogenic* (usually bacterial), *aseptic* (usually acute or subacute viral), and *chronic* (usually tuberculous, spirochetal, or cryptococcal). Each type is accompanied by characteristic changes in the CSF.

Table 28.2 Common Central Nervous System Infections

Type of Infection	Clinical Syndrome	Common Causative Organisms
Bacterial Infections		
Meningitis	Acute pyogenic meningitis	<i>Escherichia coli</i> or group B streptococci (infants) <i>Neisseria meningitidis</i> (young adults) <i>Streptococcus pneumoniae</i> or <i>Listeria monocytogenes</i> (older adults)
	Chronic meningitis	<i>Mycobacterium tuberculosis</i>
Localized infections	Abscess	Streptococci and staphylococci
	Empyema	Polymicrobial (staphylococci, anaerobic gram-negative)
Viral Infections		
Meningitis	Acute aseptic meningitis	Enteroviruses Influenza species Lymphocytic choriomeningitis virus
Encephalitis	Encephalitic syndromes	Herpes simplex (HSV-1, HSV-2) Cytomegalovirus Human immunodeficiency virus JC polyomavirus (progressive multifocal leukoencephalopathy)
	Arthropod-borne encephalitis	West Nile virus Eastern equine encephalitis virus Western equine encephalitis virus St. Louis encephalitis virus La Crosse encephalitis virus Venezuelan equine encephalitis virus Japanese encephalitis virus Tick-borne encephalitis virus
Brainstem and spinal cord syndromes	Rhombencephalitis Acute flaccid myelitis/ poliomyelitis	Rabies Poliovirus West Nile virus Enterovirus D68
Spirochetes and Fungi		
Meningoencephalitic syndromes	Neurosyphilis Lyme disease (neuroborreliosis) Fungal meningitis	<i>Treponema pallidum</i> <i>Borrelia burgdorferi</i> <i>Cryptococcus neoformans</i> <i>Candida albicans</i>
Protozoa and Metazoa		
Encephalitis	Amebic encephalitis	<i>Balamuthia</i> and <i>Acanthamoeba</i> species <i>Naegleria</i> species
Localized infections	Toxoplasmosis Cysticercosis	<i>Toxoplasma gondii</i> <i>Taenia solium</i>

Acute Pyogenic (Bacterial) Meningitis

Distinctive microorganisms cause acute pyogenic meningitis in various age groups: *Escherichia coli* and the group B streptococci in neonates; *Streptococcus pneumoniae* and *Listeria monocytogenes* in the elderly; and *Neisseria meningitidis* in adolescents and young adults, with clusters of cases raising public health concerns. The introduction of immunization against *Haemophilus influenzae* has markedly reduced the incidence of this infection in the developed world, particularly among infants (who used to be at high risk).

Affected individuals typically show systemic signs of infection superimposed on symptoms related to meningeal irritation and neurologic impairment, including headache, photophobia, irritability, clouding of consciousness, and neck stiffness. A spinal tap yields cloudy or frankly purulent CSF with as many as 90,000 neutrophils per cubic millimeter, increased CSF pressure and increased protein concentration, and markedly reduced glucose content. Untreated pyogenic meningitis can be fatal, while effective treatment with antibiotics markedly reduces mortality. One feared complication is the *Waterhouse-Friderichsen syndrome*, which results from meningitis-associated septicemia and hemorrhagic infarction of the adrenal glands (Chapter 24). It occurs most often with meningococcal and pneumococcal meningitis. In the immunosuppressed individual, purulent meningitis may be caused by several other infectious agents, such as *Klebsiella* or anaerobic organisms, and may have an atypical clinical course and uncharacteristic CSF findings, rendering timely diagnosis more difficult.

MORPHOLOGY

In acute meningitis, an exudate is evident within the leptomeninges over the surface of the brain (Fig. 28.22). The meningeal vessels are engorged and stand out prominently. The anatomic distribution of the exudate varies; in *H. influenzae* meningitis, for example, it is usually basal, whereas in pneumococcal meningitis it is often densest over the cerebral convexities near the sagittal sinus. From



Figure 28.22 Pyogenic meningitis. A thick layer of suppurative exudate covers the brain surface and thickens the leptomeninges.

the areas of greatest accumulation, tracts of pus follow along blood vessels on the surface of the brain. When the meningitis is fulminant, the inflammation may extend to the ventricles, producing ventriculitis. The ventricles may also be the portal for CSF involvement by blood-borne infections because the choroid plexus lacks a blood-brain barrier.

On microscopic examination, neutrophils fill the subarachnoid space in severely affected areas and are found predominantly around the leptomeningeal blood vessels in less severe cases. Particularly in untreated meningitis, Gram stain reveals variable numbers of bacteria. In fulminant meningitis, the inflammatory cells infiltrate the walls of the leptomeningeal veins and may extend focally into the substance of the brain (cerebritis); secondary vasculitis and venous thrombosis may lead to hemorrhagic cerebral infarction.

Leptomeningeal fibrosis may follow pyogenic meningitis and cause hydrocephalus. Particularly in pneumococcal meningitis, large quantities of the capsular polysaccharide of the organism produce a gelatinous exudate that promotes arachnoid fibrosis, a condition referred to as *chronic adhesive arachnoiditis*.

Acute Aseptic (Viral) Meningitis

Aseptic meningitis is a clinical term used for an absence of organisms by bacterial culture in a patient with manifestations of meningitis, including meningeal irritation, fever, and alterations of consciousness of relatively acute onset. The disease is generally of viral etiology, but may be bacterial, rickettsial, or autoimmune in origin. The clinical course is less fulminant than that of pyogenic meningitis, and the CSF findings also differ; in aseptic meningitis, there is a lymphocytic pleocytosis, the protein elevation is only moderate, and the glucose content is nearly always normal. The viral aseptic meningitides are usually self-limited and are treated symptomatically. Remarkably, the etiologic agent is identified in only a minority of cases; however, this may change through use of more sensitive and specific detection techniques (such as next-generation sequencing), which are under development. When pathogens are identified, enteroviruses are the most common etiology, accounting for 80% of cases. The spectrum of pathogens varies seasonally and geographically. An aseptic meningitis-like picture may also develop subsequent to rupture of an epidermoid cyst into the subarachnoid space or the introduction of a chemical irritant (chemical meningitis). The CSF is sterile in these cases, and there is pleocytosis with neutrophils and an increased protein concentration, but the sugar content is usually normal.

Acute Focal Suppurative Infections

Focal suppurative infections are typically caused by pyogenic bacteria or fungi, and can arise in several different compartments including the brain parenchyma, subdural space, and extradural space.

Brain Abscess

A brain abscess is a localized focus of necrosis of brain tissue with accompanying inflammation, usually caused by a bacterial infection. Brain abscesses may arise by direct implantation of organisms, local extension from adjacent

foci (mastoiditis, paranasal sinusitis), or hematogenous spread (usually from a primary site in the heart, lungs, or bones of the extremities, or secondary to bacteremia from dental procedures). Predisposing conditions include acute bacterial endocarditis, which may give rise to multiple brain abscesses; congenital heart disease with right-to-left shunting and loss of pulmonary filtration of organisms; chronic pulmonary sepsis, as in bronchiectasis; and systemic disease with immunosuppression. Streptococci and staphylococci are the most common offending organisms identified in nonimmunosuppressed patients.

MORPHOLOGY

Abscesses are discrete lesions with central liquefactive necrosis surrounded by brain swelling (Fig. 28.23). At the outer margin of the necrotic lesion, there is exuberant granulation tissue with neovascularization. The newly formed vessels are abnormally permeable, accounting for marked vasogenic edema in the adjacent brain tissue. In well-established lesions, a collagenous capsule is produced by fibroblasts derived from the walls of blood vessels. Outside the fibrous capsule is a zone of reactive gliosis containing numerous gemistocytic astrocytes.

Clinical Features

Cerebral abscesses are destructive lesions, and patients often present with progressive focal neurologic deficits; signs and symptoms related to increased intracranial pressure may also develop. Typically, the CSF has a high white cell count and an increased protein concentration, but the glucose content is normal. The source of infection may be apparent or may be traced to a small distant focus that is not symptomatic. The increased intracranial pressure can lead to fatal herniation; other complications include abscess rupture with ventriculitis or meningitis, and venous sinus thrombosis. With surgery and antibiotic treatment, the otherwise high mortality rate can be reduced to less than 10%.

Subdural Empyema

Bacterial, and rarely fungal, infections of the skull bones or air sinuses can spread to the subdural space, producing a subdural empyema. Although the underlying arachnoid and

subarachnoid spaces are usually not affected, a large subdural empyema with mass effect and/or thrombophlebitis of the bridging veins can cause venous occlusion and infarction of the brain. In addition to symptoms referable to the source of the infection, most patients are febrile and have headache and neck stiffness. The CSF profile is similar to that seen in brain abscesses because both are parameningeal infectious processes. If untreated, focal neurologic signs, lethargy, and coma may develop. With prompt diagnosis and treatment, including surgical drainage, resolution and full recovery is possible, the only residuum being a thickened dura.

Extradural Abscess

Extradural abscess, commonly associated with osteomyelitis, often arises from an adjacent focus of infection, such as sinusitis, or following a surgical procedure. When the process occurs in the spinal epidural space, it may cause spinal cord compression and constitute a neurosurgical emergency.

Chronic Bacterial Meningoencephalitis

Chronic bacterial infection of the meninges and the brain may be caused by *Mycobacterium tuberculosis*, *Treponema pallidum*, and *Borrelia* species.

Tuberculosis

Tuberculosis of the CNS may be part of active disease elsewhere in the body, or appear in isolation following seeding from silent lesions elsewhere, usually the lungs. It may involve the meninges or the brain.

MORPHOLOGY

The most common pattern of tuberculous involvement is a diffuse **meningoencephalitis**. The subarachnoid space contains a gelatinous or fibrinous exudate that characteristically involves the base of the brain, effacing the cisterns and encasing cranial nerves. There may be discrete, white areas of inflammation scattered over the leptomeninges. On microscopic examination, involved areas contain a mixed inflammatory infiltrate with lymphocytes, plasma cells, and macrophages. Florid cases show well-formed granulomas with caseous necrosis and giant cells. Arteries running through the subarachnoid space may show obliterative endarteritis and marked intimal thickening. Organisms can often be seen with acid-fast stains. The infectious process may spread to the choroid plexus and ependymal surface, traveling through the CSF. In long-standing cases, a dense, fibrous adhesive arachnoiditis may develop, most conspicuous around the base of the brain. Hydrocephalus may result.

CNS involvement may also take the form of one or more well-circumscribed intraparenchymal masses (**tuberculomas**), which may be associated with meningitis. A tuberculoma may be as large as several centimeters in diameter, causing significant mass effect. These granulomas usually have central caseous necrosis; calcification may occur in inactive lesions.

Clinical Features

Patients with tuberculous meningitis usually have headache, malaise, mental confusion, and vomiting. The CSF typically shows a pleocytosis made up of mononuclear cells or a

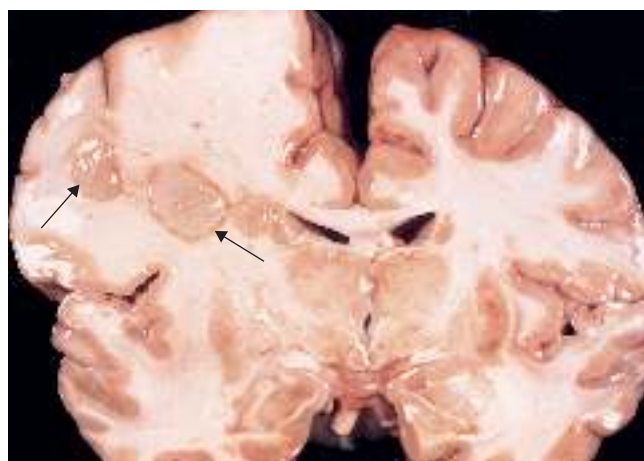


Figure 28.23 Cerebral abscesses (arrows).

mixture of neutrophils and mononuclear cells, an elevated protein concentration (often strikingly so), and a moderately reduced or normal glucose. The most serious complications of chronic tuberculous meningitis are arachnoid fibrosis producing hydrocephalus and obliterative endarteritis producing arterial occlusion and brain infarction. When the process involves the spinal cord subarachnoid space, nerve roots may also be affected. Tuberculomas produce symptoms typical of space-occupying brain lesions and must be distinguished from CNS tumors.

CNS tuberculosis in patients with acquired immunodeficiency syndrome (AIDS) is pathologically similar, but there may be less host reaction than in immunocompetent individuals.

Neurosyphilis

Neurosyphilis is a manifestation of the tertiary stage of syphilis and occurs in only about 10% of individuals with untreated infection. The major patterns of CNS involvement are meningovascular neurosyphilis, parietic neurosyphilis, and tabes dorsalis. Affected individuals often show an incomplete or mixed picture, most commonly the combination of tabes dorsalis and parietic disease (taboparesis). Because of impaired cell-mediated immunity, individuals infected with HIV are at increased risk for neurosyphilis, particularly acute syphilitic meningitis or meningovascular disease; the rate of disease progression and severity are also accelerated.

MORPHOLOGY

Neurosyphilis presents in several distinct forms.

- **Meningovascular neurosyphilis** is chronic meningitis involving the base of the brain and more variably the cerebral convexities and spinal leptomeninges. In addition, there may be an associated obliterative endarteritis (Heubner arteritis) accompanied by a distinctive perivascular inflammatory reaction rich in plasma cells and lymphocytes. Cerebral gummas (plasma cell-rich mass lesions) may also occur in the meninges and extend into the parenchyma.
- **Paretic neurosyphilis** is caused by invasion of the brain by *T. pallidum* and manifests as insidious but progressive cognitive impairment associated with mood alterations (including delusions of grandeur) that terminate in severe dementia (**general paresis of the insane**). Parenchymal damage of the cerebral cortex is particularly common in the frontal lobe, but also occurs in other areas of the isocortex. The lesions are characterized by loss of neurons, proliferation of microglia, gliosis, and iron deposits; the latter are demonstrable with the Prussian blue stain perivascularly and in the neuropil, and are presumably the sequelae of small bleeds stemming from microvascular damage. The spirochetes can, at times, be demonstrated in tissue sections.
- **Tabes dorsalis** is the result of damage to the sensory axons in the dorsal roots. This causes impaired joint position sense and ataxia (locomotor ataxia); loss of pain sensation, leading to skin and joint damage (Charcot joints); other sensory disturbances, particularly the characteristic “lightning pains”; and absence of deep tendon reflexes. On microscopic examination, there is loss of both axons and myelin in the dorsal roots, with corresponding pallor and atrophy in the dorsal columns of the spinal cord. Organisms are not demonstrable in the cord lesions.

Neuroborreliosis (Lyme Disease)

Lyme disease is caused by the spirochete *Borrelia burgdorferi*, which is transmitted by various species of *Ixodes* tick (Chapter 8). Involvement of the nervous system is referred to as *neuroborreliosis*. Neurologic symptoms are highly variable and include aseptic meningitis, facial nerve palsies and other polyneuropathies, as well as encephalopathy. The rare cases that have come to autopsy have shown a focal proliferation of microglial cells in the brain as well as scattered extracellular organisms.

Viral Meningoencephalitis

Viral encephalitis is a parenchymal infection of the brain almost invariably associated with meningeal inflammation (meningoencephalitis) and sometimes with simultaneous involvement of the spinal cord (encephalomyelitis).

Some viruses have a propensity to infect the nervous system. Such neural tropism takes several forms: some infect specific cell types (e.g., oligodendrocytes), while others preferentially involve particular areas of the brain (e.g., medial temporal lobes or the limbic system). Latency is an important phase of several viral infections of the CNS (e.g., herpes zoster and progressive multifocal leukoencephalopathy). Systemic viral infections in the absence of direct evidence of viral penetration into the CNS may be followed by an immune-mediated disease, such as perivenous demyelination. Viral infection of the fetus may cause congenital malformations, as occurs with rubella and Zika virus. A slowly progressive degenerative disease syndrome may follow many years after a viral illness; an example is postencephalitic parkinsonism after the 1918 viral influenza pandemic.

Arthropod-Borne Viral Encephalitis

Arboviruses are an important cause of epidemic encephalitis, especially in tropical regions of the world, and are capable of causing serious morbidity and high mortality. In the Western hemisphere, the most important arboviruses are Eastern and Western equine, West Nile, Venezuelan, St. Louis, and La Crosse; elsewhere in the world, pathogenic arboviruses include Japanese B (Far East), Murray Valley (Australia and New Guinea), and tick-borne (Russia and Eastern Europe).

All of these viruses have animal hosts and insect vectors. Clinically, affected individuals develop generalized neurologic deficits, such as seizures, confusion, delirium, and stupor or coma, as well as focal signs, such as reflex asymmetry and ocular palsies. Involvement of the spinal cord in West Nile encephalitis can lead to a polio-like syndrome with paralysis. In general, the CSF is usually colorless, with slightly elevated pressure, an elevated protein level, and a normal glucose. Initially the CSF exhibits a neutrophilic pleocytosis, but this rapidly converts to a lymphocytosis.

MORPHOLOGY

The encephalitides caused by various arboviruses produce similar histopathologic changes that differ only in severity and extent. Characteristically, there is a meningoencephalitis marked by the perivascular accumulation of lymphocytes (and sometimes

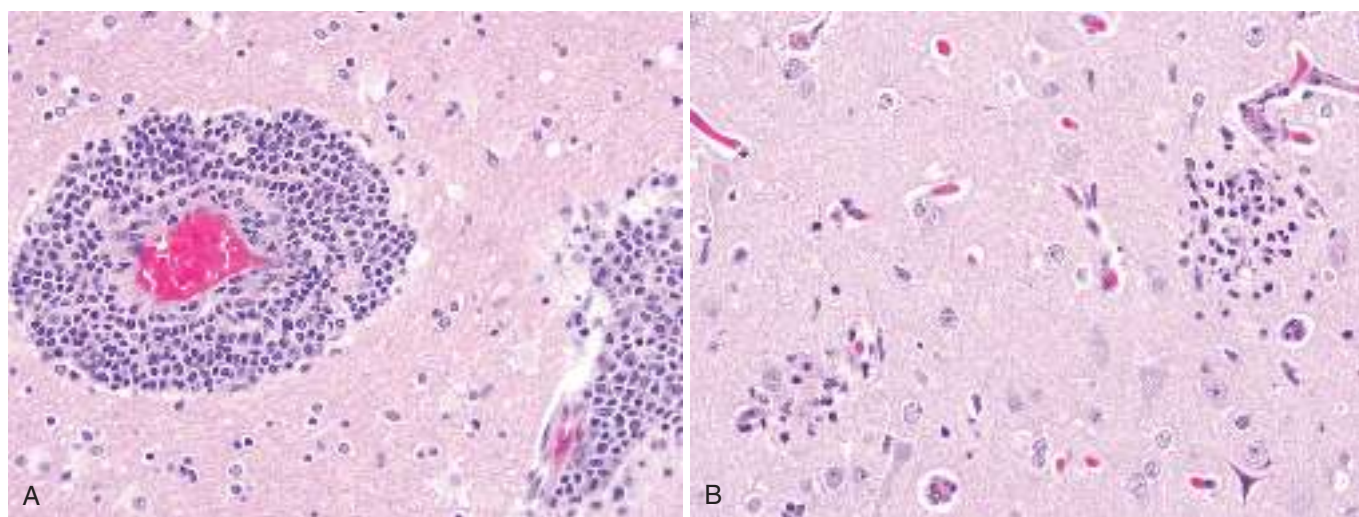


Figure 28.24 Characteristic findings of viral encephalitis include perivascular cuffs of lymphocytes (A) and microglial nodules (B).

neutrophils) (Fig. 28.24A). Multiple foci of necrosis of gray and white matter are found; in particular, there is evidence of single-cell neuronal necrosis with phagocytosis of the debris (**neuronal phagocytosis**). Microglial cells form small aggregates, called **microglial nodules** (see Fig. 28.24B). In severe cases, there may be a necrotizing vasculitis with associated focal hemorrhages. Although some viruses declare their presence by formation of intracellular inclusions, in tissue samples the causative virus is most often identified by a combination of ultrastructural, immunohistochemical, and molecular methods.

Herpes Simplex Virus Type 1

Herpes simplex virus type 1 (HSV-1) encephalitis occurs most commonly in children and young adults; only about 10% of affected individuals have a history of prior herpetic infection. The typical presenting symptoms are alterations in mood, memory, and behavior. Antiviral agents now provide

effective treatment in many cases, with a significant reduction in the mortality rate. In some individuals, HSV-1 encephalitis follows a subacute course with clinical manifestations (weakness, lethargy, ataxia, seizures) that evolve during a more protracted period (4 to 6 weeks).

MORPHOLOGY

This encephalitis starts in and most severely involves the inferior and medial regions of the temporal lobes and the orbital gyri of the frontal lobes (Fig. 28.25A). The infection is necrotizing and often hemorrhagic in the most severely affected regions. Perivascular inflammatory infiltrates are usually present, and Cowdry type A intranuclear viral inclusions may be found in both neurons and glia (Fig. 28.25B). In individuals with slowly evolving HSV-1 encephalitis, there is more diffuse involvement of the brain.

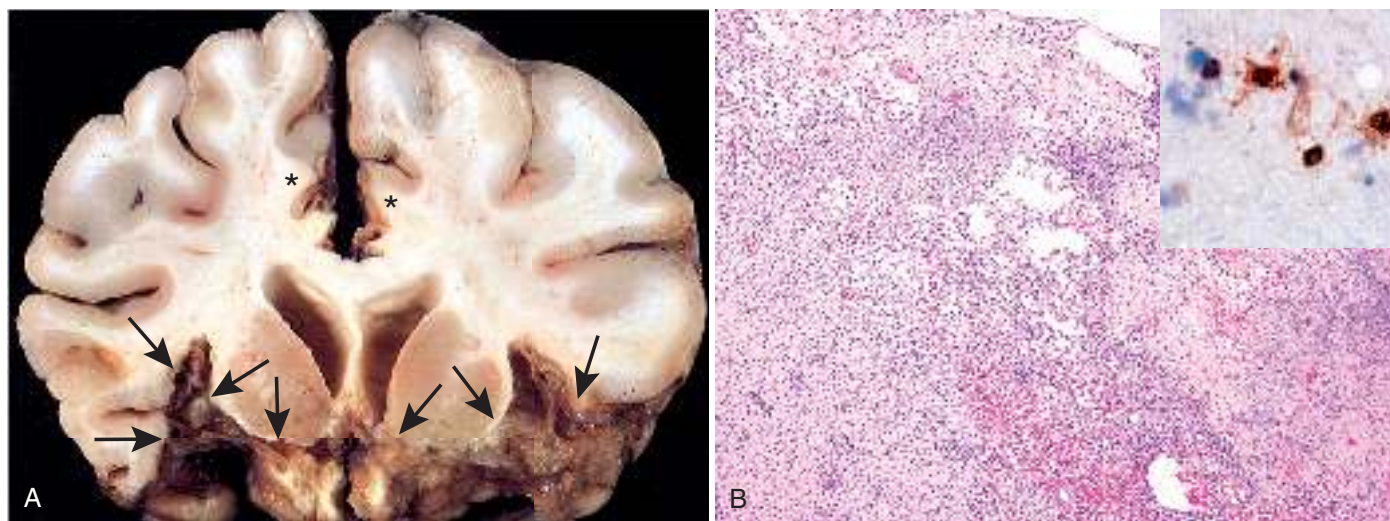


Figure 28.25 (A) Herpes encephalitis showing extensive destruction of inferior frontal and anterior temporal lobes (arrows) and the cingulate gyri (asterisks). (B) Necrotizing inflammatory process characterizes acute herpes encephalitis; nuclear viral inclusions can be highlighted by immunostaining (inset). (A, Courtesy Dr. T.W. Smith, University of Massachusetts Medical School, Worcester, Mass.)

Herpes Simplex Virus Type 2

Herpes simplex virus type 2 (HSV-2) can infect the nervous system. In adults it causes meningitis, but as many as 50% of neonates born by vaginal delivery to women with active primary HSV genital infections acquire the infection during passage through the birth canal and develop severe encephalitis. In individuals with active HIV infection, HSV-2 may cause an acute hemorrhagic and necrotizing encephalitis.

Varicella-Zoster Virus

Primary infection with varicella-zoster virus causes one of the childhood exanthems (*chickenpox*), without evidence of neurologic involvement. Following the cutaneous infection, the virus enters a latent phase within sensory neurons of the dorsal root or trigeminal ganglia. Reactivation of infection in adults (*shingles* or *herpes zoster*) usually manifests as a painful, vesicular skin eruption confined to one or several dermatomes. Herpes zoster is typically self-limited, but there may be a postherpetic neuralgia syndrome (particularly after 60 years of age) characterized by persistent pain, sometimes induced by stimuli that are otherwise not painful. Vaccination is now available to prevent these complications of varicella-zoster virus reactivation.

Cytomegalovirus

CMV infection of the nervous system occurs in fetuses and immunosuppressed individuals. The outcome of infection in utero is periventricular necrosis that produces severe brain destruction followed later by microcephaly and periventricular calcification. CMV is a common opportunistic viral pathogen in individuals with AIDS, with CNS involvement also occurring in this setting.

MORPHOLOGY

In the immunosuppressed individual, CMV most commonly causes subacute encephalitis, which may be associated with CMV inclusion-bearing cells (see Fig. 8.12). The infection tends to localize in the subependymal regions of the brain, where it results in a severe hemorrhagic necrotizing ventriculoencephalitis and choroid plexitis. The virus can also attack the lower spinal cord and roots, producing a painful radiculoneuritis. Any cell in the CNS (neurons, glia, ependyma, or endothelium) may be infected. Prominent enlarged cells with intranuclear and intracytoplasmic inclusions can be readily identified by conventional light microscopy; CMV infection is confirmed by immunohistochemistry.

Poliomyelitis

Although paralytic poliomyelitis has been eradicated by vaccination in many parts of the world, there are still a few countries where it remains a serious problem. In addition, a cluster of cases of acute flaccid myelitis (a polio-like syndrome) has recently occurred in the United States and seems to have been caused by a novel strain of enterovirus D68 (which usually causes only a mild respiratory infection). In nonimmunized individuals, poliovirus infection causes a subclinical or mild gastroenteritis, similar to that caused by other members of the picornavirus group of enteroviruses. In a small fraction of the vulnerable population, however, the virus secondarily invades the nervous system.

MORPHOLOGY

Acute cases show mononuclear cell perivascular cuffs and neuronophagia of the **anterior horn motor neurons of the spinal cord**. The inflammatory reaction is usually confined to the anterior horns but may extend into the posterior horns, and the damage is occasionally severe enough to produce cavitation. Poliovirus RNA has been detected in anterior horn cell motor neurons; the cranial motor nuclei are sometimes involved as well. Post-mortem examination in long-term survivors of symptomatic poliomyelitis shows loss of neurons and gliosis in the affected anterior horns of the spinal cord, atrophy of the anterior (motor) spinal roots, and neurogenic atrophy of denervated muscle.

Clinical Features

CNS infection manifests initially with meningeal irritation and a CSF picture consistent with aseptic meningitis; the disease may progress no further or may advance to involve the spinal cord. When the disease affects the motor neurons of the spinal cord, it produces a flaccid paralysis associated with muscle wasting and hyporeflexia in the corresponding region of the body—the permanent neurologic residue of poliomyelitis. Because of the destruction of motor neurons, paresis or paralysis follows; when the diaphragm and intercostal muscles are affected, severe respiratory compromise and even death may occur. A myocarditis sometimes complicates the acute infection. *Postpolio syndrome* can develop in patients 25 to 35 years after resolution of the initial illness. It is characterized by progressive weakness associated with decreased muscle mass and pain, and has been attributed to superimposed loss of remaining motor neurons, without any convincing evidence of viral re-activation.

Rabies

Rabies is severe encephalitis transmitted to humans by the bite of a rabid animal, usually a dog or various wild mammals that are natural reservoirs. Exposure to certain species of bats, even without a known bite, can also lead to rabies.

MORPHOLOGY

External examination of the brain shows intense edema and vascular congestion. Microscopically, there is widespread neuronal degeneration and an inflammatory reaction that is most severe in the brainstem; the basal ganglia, spinal cord, and dorsal root ganglia may also be involved. **Negri bodies**, the pathognomonic microscopic finding, are cytoplasmic, round to oval, eosinophilic inclusions found in pyramidal neurons of the hippocampus and Purkinje cells of the cerebellum, sites usually devoid of inflammation (Fig. 28.26). Rabies virus can be detected within Negri bodies by ultrastructural and immunohistochemical methods.

Clinical Features

Because the virus enters the CNS by ascending along the peripheral nerves from the wound site, the incubation period (usually between 1 and 3 months) depends on the distance between the wound and the brain. The disease begins with nonspecific symptoms such as malaise, headache, and fever, but the conjunction of these symptoms with local paresthesias around the wound is diagnostic. As the infection advances,

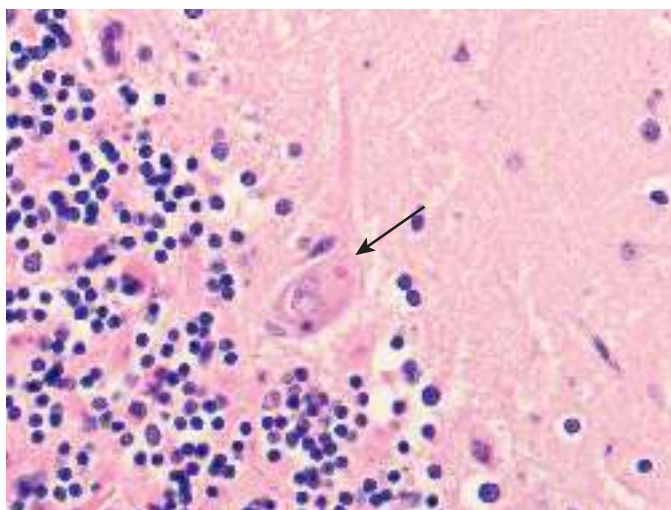


Figure 28.26 The diagnostic histologic finding in rabies is the eosinophilic Negri body, as seen here in a Purkinje cell (arrow).

the affected individual exhibits extraordinary CNS excitability; the slightest touch is painful and produces violent motor responses or even convulsions. Contracture of the pharyngeal musculature on swallowing produces foaming at the mouth, which may create an aversion to swallowing even water (hydrophobia). There are signs of meningeal irritation and, as the disease progresses, flaccid paralysis. Alternating periods of mania and stupor progress to coma and eventually death ensues from respiratory failure.

Human Immunodeficiency Virus

In the period before the availability of effective antiretroviral therapy, neuropathologic changes were demonstrated at postmortem examination in as many as 80% to 90% of cases of AIDS. These changes stem from direct effects of the virus on the nervous system, opportunistic infections, and primary CNS lymphoma, most commonly an EBV-positive B-cell tumor. There has been a decrease in the frequency of these secondary effects of HIV infection thanks to the efficacy of multidrug antiretroviral therapy.

HIV aseptic meningitis occurs within 1 to 2 weeks of seroconversion in about 10% of patients; antibodies to HIV can be demonstrated, and the virus can be isolated from the CSF. The neuropathologic studies of the early, acute phases of HIV invasion of the CNS have shown mild lymphocytic meningitis, perivascular inflammation, and some myelin loss. Among CNS cell types, only microglia express both the CD4 coreceptor and the chemokine receptors (CCR5 or CXCR4) that are required in combination for efficient infection by HIV. During the chronic phase, HIV encephalitis is commonly found when symptomatic individuals come to autopsy.

An *immune reconstitution inflammatory syndrome* (IRIS) has been identified in patients with AIDS after effective treatment; the syndrome is recognized as a paradoxical deterioration after starting therapy, and consists of an exuberant “reconstituted” inflammatory response while on antiretroviral therapy (Chapter 6). In the CNS of patients with opportunistic infections, IRIS has caused paradoxical exacerbation of symptoms; neuropathologic studies confirm intense inflammation with an influx of CD8⁺ lymphocytes.

MORPHOLOGY

HIV encephalitis is a chronic inflammatory reaction associated with widely distributed **microglial nodules**, often containing macrophage-derived **multinucleated giant cells** (Fig. 28.27); foci of tissue necrosis and reactive gliosis are sometimes seen together with these lesions. Some of the microglial nodules are found near small blood vessels, which show abnormally prominent endothelial cells and perivascular foamy or pigment-laden macrophages. These changes are especially prominent in the subcortical white matter, diencephalon, and brainstem. In some cases, there is also a disorder of white matter characterized by multifocal or diffuse areas of myelin pallor, axonal swelling, and gliosis. HIV can be detected in CD4⁺ microglia and mononuclear or multinucleated macrophages.

Clinical Features

Cognitive changes, some mild and others florid enough to be termed *HIV-associated dementia*, appear to have persisted in the era of effective anti-HIV treatment regimens; collectively, they are termed *HIV-associated neurocognitive disorders* (HAND). Rather than having a specific pathologic lesion as its correlate, HAND is most closely related to inflammatory activation of microglial and perivascular macrophages, some of which are infected by HIV. A wide range of possible mechanisms for neuronal dysfunction and injury have been proposed, including the actions of inflammatory cytokines, a cascade of toxic effects of HIV-derived proteins, neurotoxic effects of anti-HIV therapy, accelerated aging, and aberrant synaptic pruning; in all probability, most (if not all) of these mechanisms contribute to the pathogenesis of this disorder.

Progressive Multifocal Leukoencephalopathy

PML is an encephalitis caused by the JC polyomavirus; because the virus preferentially infects oligodendrocytes, demyelination is its principal pathologic effect. The disease occurs almost exclusively in immunosuppressed individuals in various clinical settings, including chronic lymphoproliferative or myeloproliferative illnesses, immunosuppressive chemotherapy

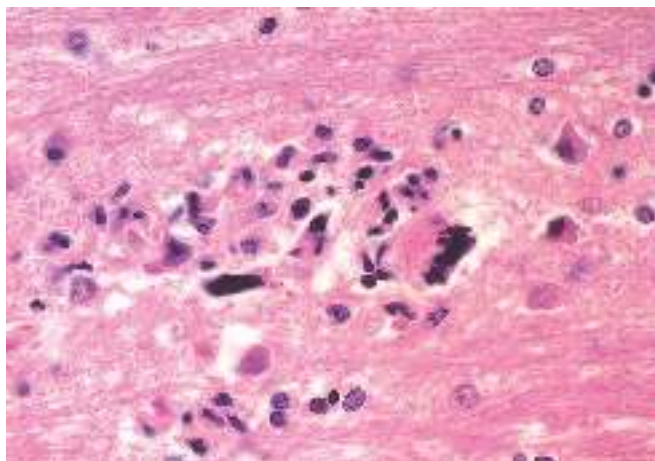


Figure 28.27 HIV encephalitis. Note the microglial nodule and multinucleated giant cells.

(including monoclonal antibody therapy targeting certain integrins), granulomatous diseases, and AIDS.

Although most people have serologic evidence of exposure to JC virus by 14 years of age, primary infection is asymptomatic; PML results from the reactivation of virus in the setting of immunosuppression. Clinically, affected individuals develop focal and relentlessly progressive neurologic symptoms and signs, with imaging studies showing extensive, often multifocal, cerebral or cerebellar white matter lesions.

MORPHOLOGY

The lesions consist of patches of irregular, ill-defined white matter injury that range in size from millimeters to large confluent regions (Fig. 28.28A). Microscopically, a PML lesion shows an area of demyelination, most often in a subcortical location, in the center of which are sheets of lipid-laden macrophages and a reduced number of axons (Fig. 28.28B). Greatly enlarged oligodendrocyte nuclei containing glassy amphophilic viral inclusions (Fig. 28.28B, inset), which can be identified by immunohistochemistry, are typically found at the lesion edge. Bizarre giant astrocytes with one to several irregular, hyperchromatic nuclei are intermixed with more typical reactive astrocytes. Infection of granule cell neurons in the cerebellum has been demonstrated in rare instances.

Fungal Meningoencephalitis

Fungal infections of the CNS are encountered primarily in immunocompromised individuals. The brain is usually involved following widespread hematogenous dissemination of fungi; the most frequent offenders are *Candida albicans*, *Mucor* species, *Aspergillus fumigatus*, and *Cryptococcus neoformans*. In endemic areas, pathogens such as *Histoplasma capsulatum*, *Coccidioides immitis*, and *Blastomyces dermatitidis* may involve the CNS after a primary pulmonary or cutaneous infection; again, this often follows immunosuppression. Although most fungi reach the brain by hematogenous

dissemination, direct extension may also occur, particularly in mucormycosis in the setting of diabetes mellitus.

The three main forms of injury in CNS fungal infection are chronic meningitis, vasculitis, and parenchymal invasion. Vasculitis is most frequently seen with *Mucormycosis* and *Aspergillosis*, both of which directly invade blood vessel walls, but it occasionally occurs with other infections such as candidiasis. The resultant vascular thrombosis produces infarction that is often strikingly hemorrhagic.

Parenchymal infection, usually in the form of granulomas or abscesses, can occur with most of the fungi and often coexists with meningitis. The most commonly encountered fungi that invade the brain are *Candida* and *Cryptococcus*. Candidiasis usually produces multiple microabscesses, with or without granuloma formation; in contrast, parenchymal aggregates of *Cryptococcal* organisms are typically found within expanded perivascular (Virchow-Robin) spaces and are associated with minimal to no inflammation or gliosis.

Cryptococcal meningitis, a common opportunistic infection in the setting of AIDS, may be fulminant and fatal in as little as 2 weeks or indolent, evolving over months or years. The CSF may contain few cells but usually has a high concentration of protein. The mucoid-encapsulated yeasts can be visualized in the CSF with special stains or detected indirectly using assays for cryptococcal antigens (see Fig. 8.43).

Most cases of cryptococcal infection in immunosuppressed individuals are caused by *C. neoformans*. Recently, a second species, *C. gattii*, has been recognized and appears more likely to cause disease in immunocompetent individuals. Some case studies suggest that *C. gattii* involvement of the CNS is more likely to take the form of mass lesions (“cryptococcomas”) and to present with symptoms related to increased CNS pressure due to mass effects.

Other Infectious Diseases of the Nervous System

Protozoal diseases (including malaria, toxoplasmosis, amebiasis, and trypanosomiasis), rickettsial infections (e.g.,

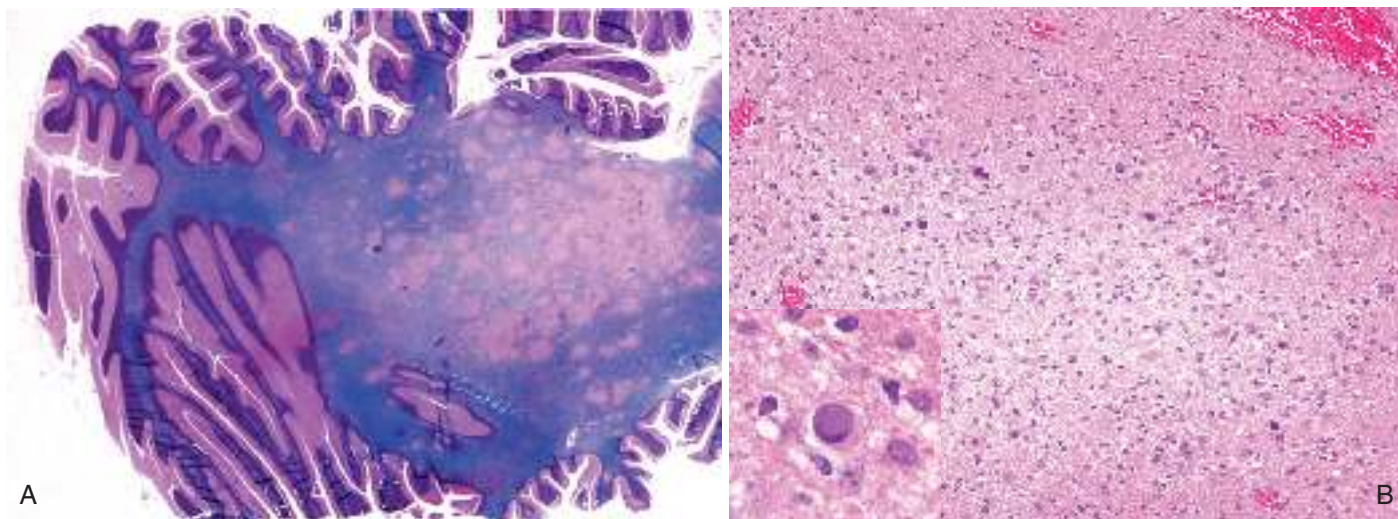


Figure 28.28 (A) Progressive multifocal leukoencephalopathy. Section stained for myelin showing irregular, poorly defined areas of demyelination, which become confluent in places. (B) Microscopically, the lesions consists of areas of demyelination. Inset, Enlarged oligodendrocyte nucleus represents the effect of viral infection.

typhus and Rocky Mountain spotted fever), and metazoal diseases (especially cysticercosis and echinococcosis) may also involve the CNS and are discussed in Chapter 8.

- **Cerebral toxoplasmosis** is an opportunistic infection commonly found in the setting of HIV-associated immunosuppression. The clinical symptoms of infection of the brain by *Toxoplasma gondii* are subacute, evolving during a 1- or 2-week period, and may be both focal and diffuse. Computed tomography and magnetic resonance imaging studies may show multiple ring-enhancing lesions; however, this radiographic appearance is not specific, as CNS lymphoma, tuberculosis, and fungal infections produce similar findings. In nonimmunosuppressed hosts, the impact of toxoplasmosis on the brain is most often seen when primary maternal infection occurs early in pregnancy. Such infections often spread to the brain of the developing fetus and cause severe damage in the form of multifocal necrotizing lesions that may calcify. With early diagnosis, cerebral toxoplasmosis is often treatable with antibiotics, which may also be administered empirically if this infection is in the differential diagnosis.
- **Cerebral amebiasis.** A rapidly fatal necrotizing encephalitis results from infection with *Naegleria* species, and a chronic granulomatous meningoencephalitis has been associated with infection with *Acanthamoeba*. The amoebae may be difficult to distinguish morphologically from activated macrophages (Fig. 28.29). Methenamine silver or PAS stains are helpful in visualizing the organisms, although definitive identification ultimately depends on immunofluorescence studies, culture, and molecular methods.
- **Cerebral malaria.** Cerebral malaria is a complication of infection by *Plasmodium falciparum*, the species with the highest mortality. Most likely the result of sticking of infected red cells to inflamed vascular endothelium, cerebral involvement by malaria is accompanied by reduced cerebral blood flow and results in ataxia, seizures, and coma in the acute phase, and leads to long-term cognitive deficits in up to 20% of affected children (Chapter 8).

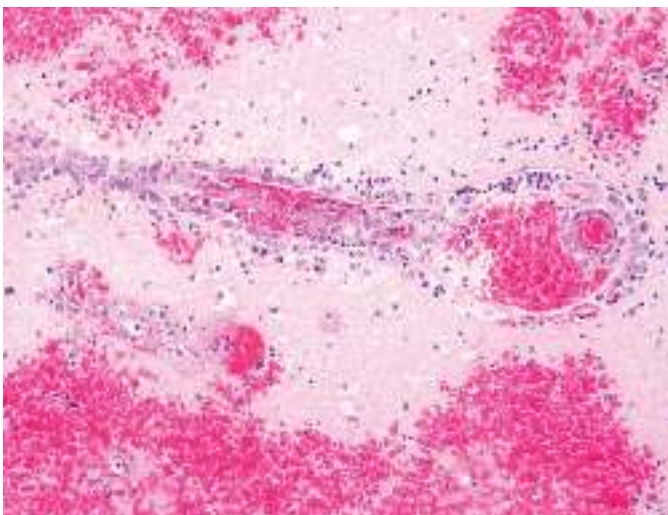


Figure 28.29 Necrotizing amebic meningoencephalitis; in this image, the organisms are largely perivascular. (*Acanthamoeba* resemble macrophages on routine stains, but are slightly smaller and less granular.)

MORPHOLOGY

Toxoplasmosis of the CNS produces brain abscesses, which are found most often in the cerebral cortex (near the gray-white junction) and deep gray nuclei, less often in the cerebellum and brainstem, and rarely in the spinal cord (Fig. 28.30A). Acute lesions exhibit central necrosis, petechial hemorrhages surrounded by acute and chronic inflammation, macrophage infiltration, and vascular proliferation. Both free tachyzoites and encysted bradyzoites may be found at the periphery of the necrotic foci. The organisms are often seen on routine H&E or Giemsa stains, but are more easily recognized by immunohistochemical methods (Fig. 28.30B). The blood vessels near these lesions often show intimal proliferation or frank vasculitis with fibrinoid necrosis and thrombosis.

KEY CONCEPTS

INFECTIONS

- Pathogens from viruses through parasites can infect the brain. Different pathogens use distinct routes to reach the brain and cause different patterns of disease.
- Routes of access of organisms to the brain include: hematogenous spread (e.g., abscess formation in the setting of endocarditis), direct extension (following trauma or with extension from the sinuses), and retrograde transport along nerves (e.g., rabies).
- Bacterial infections may cause meningitis, cerebral abscesses, or a chronic meningoencephalitis. The distribution of pathogens is influenced by various host factors, such as age and level of immune function.
- Viral infections can cause meningitis or meningoencephalitis. Some viruses have characteristic patterns of infection (HSV-1 in the temporal lobes, polio in the anterior horn).
- HIV can directly cause meningoencephalitis, or indirectly affect the brain by increasing the risk of opportunistic infections (toxoplasmosis, CMV) and EBV-positive CNS lymphoma.

DEMYELINATING DISEASES

Demyelinating diseases of the CNS are acquired conditions characterized by preferential damage to myelin with relative preservation of axons. The clinical deficits, at least initially, are due to the effect of myelin loss on the transmission of electrical impulses along axons. The natural history of demyelinating diseases is determined, in part, by the limited capacity of the CNS to regenerate normal myelin and by the degree of secondary damage to axons that occurs as the disease runs its course.

Several pathologic processes can cause loss of myelin. These include immune-mediated destruction of myelin (as in multiple sclerosis [MS]) and infections (as in PML, described earlier). In addition, inherited disorders may affect synthesis or turnover of myelin components; these are termed *leukodystrophies* and are discussed with metabolic disorders.

Multiple Sclerosis (MS)

MS is an autoimmune demyelinating disorder characterized by distinct episodes of neurologic deficits that are

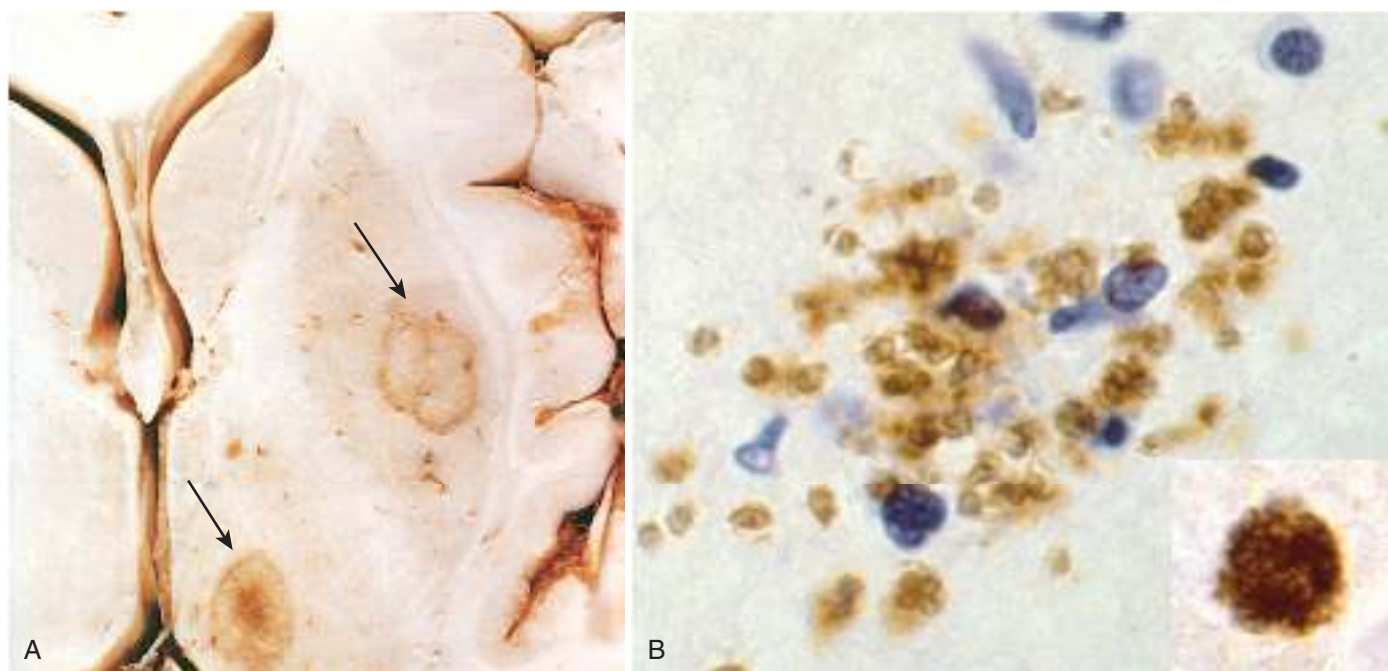


Figure 28.30 (A) Toxoplasma abscesses in the putamen and thalamus (arrows). (B) Free tachyzoites demonstrated by immunostaining; inset, *Toxoplasma* pseudocyst with bradyzoites highlighted by immunostaining.

separated in time and are attributable to patchy white matter lesions that are separated in space. It is the most common of the demyelinating disorders, having a prevalence of approximately 1 per 1000 persons in most of the United States and Europe. The disease may become clinically apparent at any age, although onset in childhood or after 50 years of age is relatively rare. Women are affected twice as often as are men. In most individuals with MS, the clinical course takes the form of relapsing and remitting episodes of variable duration (weeks to months to years) marked by neurologic defects, followed by gradual and partial recovery of neurologic function. The frequency of relapses tends to decrease over time, but there is a steady neurologic deterioration in most affected individuals.

Pathogenesis

MS is caused by an autoimmune response directed against components of the myelin sheath. As in other autoimmune disorders, the pathogenesis of this disease involves both genetic and environmental factors (Chapter 6). The incidence of MS is 15-fold higher when the disease is present in a first-degree relative and roughly 150-fold higher with an affected monozygotic twin. Despite a series of well-powered studies, only a portion of the genetic basis of the disease has been explained. There is a strong association with a DR haplotype of the major histocompatibility complex. Genome-wide association studies have identified additional associations with the IL-2 and IL-7 receptor genes, and subsequently with a number of other genes encoding proteins involved in the immune response, including cytokines and their receptors, costimulatory molecules, and cytoplasmic signaling molecules; many of these loci are associated with other autoimmune diseases. These genetic studies have not explained why the clinical course for individuals with MS is so variable.

Environmental factors are also important. For example, there is geographic variation in the prevalence of MS, with a higher number of cases diagnosed away from the equator; it has been proposed that this latitude dependence may be related to a low level of vitamin D (an immune system modulator) in people who are not exposed to sunlight during winter.

Immune mechanisms that underlie the destruction of myelin are the focus of much investigation. The available evidence indicates that the disease is initiated by Th1 and Th17 T cells that react against myelin antigens and secrete cytokines. Th1 cells secrete IFN- γ , which activates macrophages, while Th17 cells promote the recruitment of leukocytes (Chapter 6). The demyelination is caused by these activated leukocytes and their injurious products. The infiltrate in plaques and surrounding regions of the brain consists of T cells (mainly CD4+, some CD8+) and macrophages. How the autoimmune reaction is initiated is not understood; a role of viral infection (e.g., EBV) has been proposed but remains controversial.

Based on the growing understanding of the pathogenesis of MS, therapies are being developed that modulate or inhibit T-cell responses and block the recruitment of T cells to the brain. A potential contribution of humoral immunity has also long been suspected, based on the early observation of oligoclonal bands of immunoglobulin in CSF; the demonstration that treatment with agents that deplete B cells decreases the incidence of demyelinating lesions in patients with MS lends support to this idea.

MORPHOLOGY

MS is a white matter disease that is best appreciated in sections of the brain and spinal cord. In the fresh state, the lesions are

firmer than the surrounding white matter (“**sclerosis**”) and appear as well circumscribed, somewhat depressed, glassy, gray-tan, irregularly shaped **plaques** (Fig. 28.31A). The area of demyelination often has sharply defined borders, a feature best appreciated with stains for myelin (Fig. 28.31B). The size of lesions varies considerably, from small foci that are only recognizable microscopically to large confluent plaques. Plaques commonly occur adjacent to the lateral ventricles and are also frequent in the corpus callosum, optic nerves and chiasm, brainstem, ascending and descending fiber tracts, cerebellum, and spinal cord. Plaques can extend into gray matter because myelinated fibers are present there as well.

Microscopically, in an **active plaque**, there is ongoing myelin breakdown associated with abundant foamy macrophages (Fig. 28.32A); lymphocytes are also present, mostly as perivascular cuffs, especially at the outer edge of the lesion. Active lesions are often centered on small veins; myelin is usually completely absent (Fig. 28.32B), but axons are relatively preserved (Fig. 28.32C). In time, astrocytes undergo reactive changes. As lesions become quiescent, the inflammatory cells slowly disappear. Within **inactive plaques**, there is no macrophage-rich infiltrate, little to no myelin is found, and there is a reduction in the number of oligodendrocyte nuclei; instead, reactive gliosis is prominent. Axons in old gliotic plaques are usually greatly diminished in number.

In some MS plaques (shadow plaques), the white matter pallor is less severe, and abnormally thinned-out myelin sheaths can be demonstrated, especially at the outer edges. This phenomenon is most commonly interpreted as evidence of partial and incomplete remyelination by surviving oligodendrocytes. Abnormally myelinated fibers have also been observed at the edges of typical plaques. Although these histologic findings suggest a limited potential for remyelination in the CNS, the remaining axons within most MS plaques remain unmyelinated.

Clinical Features

Although MS lesions can occur anywhere in the CNS and consequently may induce a wide range of clinical manifestations, certain patterns of neurologic symptoms and signs are more common. Unilateral visual impairment due to involvement of the optic nerve (*optic neuritis*) is a frequent initial manifestation of MS. However, only a minority of individuals (10% to 50%, depending on the population studied) with an episode of optic neuritis go on to develop MS (which requires multiple episodes to support the diagnosis). Involvement of the brainstem produces cranial nerve signs (ataxia, nystagmus, and internuclear ophthalmoplegia from interruption of the fibers of the medial longitudinal fasciculus). Spinal cord lesions give rise to motor and sensory impairment of trunk and limbs, spasticity, and loss of bladder control.

Examination of the CSF in individuals with MS shows a mildly elevated protein level and in one-third of cases a moderate pleocytosis. IgG levels in the CSF are increased, and oligoclonal IgG bands are usually observed on immunoelectrophoresis; these are indicative of the presence of a small number of activated B-cell clones in the CNS, which are postulated to be self-reactive. Magnetic resonance imaging has taken on a prominent role in assessing disease progression; these studies, along with autopsy and clinical findings, indicate that some plaques may be clinically silent even in otherwise symptomatic patients. Treatment consists of several types of immunosuppressive or immunomodulatory

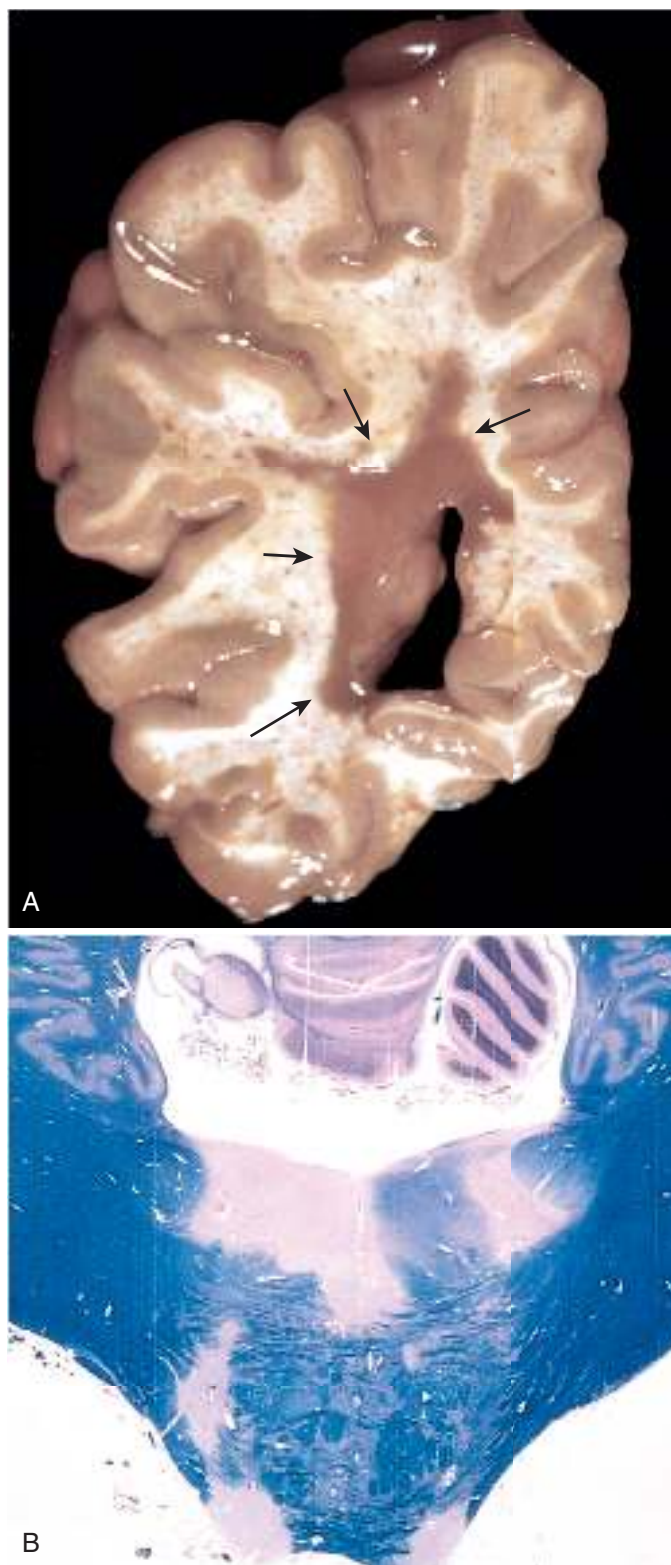


Figure 28.31 Multiple sclerosis. (A) Section of fresh brain shows gray-brown plaque around occipital horn of the lateral ventricle (arrows). (B) Regions of demyelination (MS plaques) around the fourth ventricle lack the normal blue staining of myelin (Luxol fast blue periodic acid–Schiff stain).

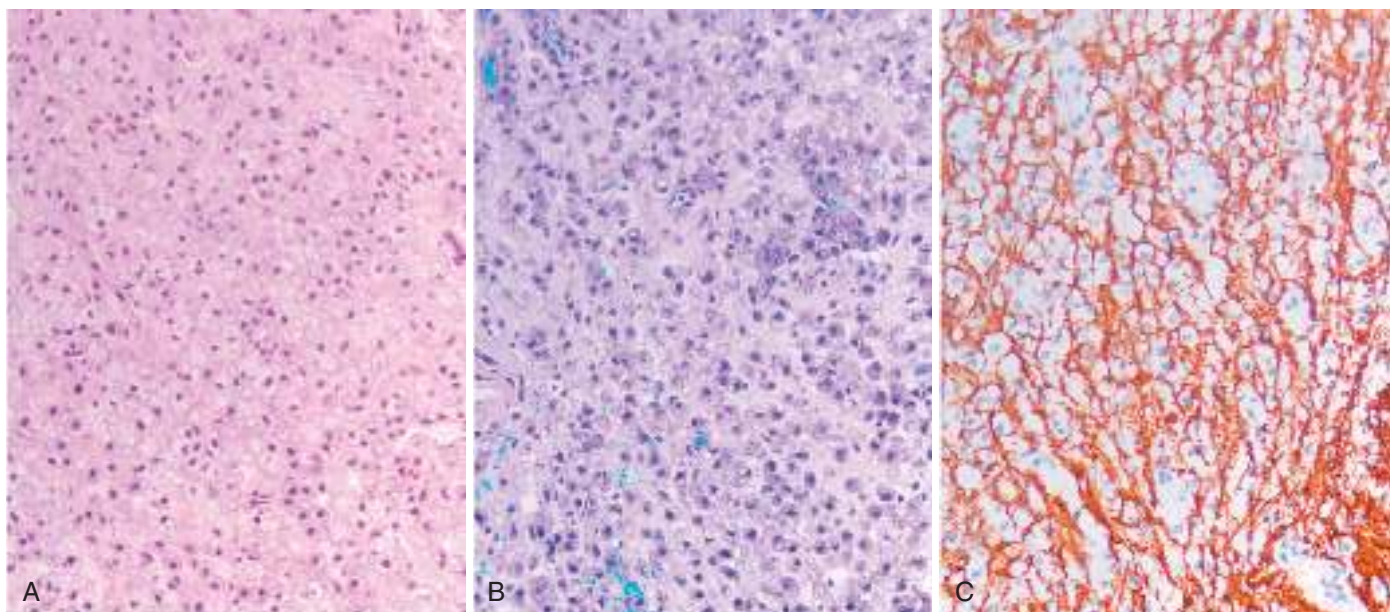


Figure 28.32 Multiple sclerosis. (A) Active demyelinating plaques appear very cellular due to the presence of numerous lipid-laden macrophages. (B) The same lesion stained with the Luxol fast blue periodic acid-Schiff stain shows a complete lack of myelin. (C) Relative preservation of axons is seen on the neurofilament immunostain (brown).

agents, which may slow the progression of the disease but are not curative.

Neuromyelitis Optica

Neuromyelitis optica (NMO) is a syndrome with synchronous (or near-synchronous) bilateral optic neuritis and spinal cord demyelination. Once considered a variant of MS, it is clearly a distinct disorder. NMO has an even greater skewing toward women than MS, is more commonly associated with poor recovery from the first attack, and is often characterized by the presence of aquaporin-4 antibodies, which appear to be pathogenic. Areas of demyelination in NMO show loss of aquaporin-4, the major water channel of astrocytes. White cells (often including neutrophils) are common in the CSF. Within the damaged areas of white matter, there is typically necrosis, an inflammatory infiltrate including neutrophils and eosinophils, and vascular immunoglobulin and complement deposition. Acute attacks are treated with glucocorticoids or plasma exchange, while long-term treatments include agents that decrease antibody titers (e.g., by depleting B cells) or that inhibit complement (e.g., inhibitors of the C5-C9 membrane attack complex).

Acute Disseminated Encephalomyelitis and Acute Necrotizing Hemorrhagic Encephalomyelitis

Acute disseminated encephalomyelitis is a diffuse, monophasic demyelinating disease that follows a viral infection or, rarely, a viral immunization. Symptoms typically develop 1 or 2 weeks after the antecedent event and include headache, lethargy, and coma rather than focal findings, as seen in MS; in contrast to MS, all brain lesions appear similar, consistent with the clinically monophasic nature of the disorder. The clinical course is rapid, and as many as 20% of those affected die; the remaining patients usually recover completely.

Acute necrotizing hemorrhagic encephalomyelitis (also known as acute hemorrhagic leukoencephalitis or Weston Hurst disease) is a fulminant syndrome of CNS demyelination, typically affecting young adults and children. The illness is almost invariably preceded by a recent episode of upper respiratory infection, most often of unknown cause. This disorder shows histologic similarities with acute disseminated encephalomyelitis; however, the damage is more severe. The disease is fatal in many patients, with significant deficits present in most survivors.

The lesions of acute disseminated encephalomyelitis are similar to those induced by immunization of animals with myelin components or with early rabies vaccines that had been prepared from brains of infected animals. This has led to the suggestion that acute disseminated encephalomyelitis is an acute autoimmune reaction to myelin and that acute necrotizing hemorrhagic encephalomyelitis is a hyperacute variant, but inciting antigens have yet to be identified.

Central Pontine Myelinolysis

Central pontine myelinolysis is an acute disorder characterized by loss of myelin in the base of the pons and portions of the pontine tegmentum, typically in a roughly symmetric pattern. It most commonly arises 2 to 6 days after rapid correction of hyponatremia, although it can also be associated with other severe electrolyte disturbances or osmolar imbalances, and is clinically known as osmotic demyelination syndrome. It appears that rapid increases in osmolality damage oligodendrocytes through uncertain mechanisms. Inflammation is absent from the lesions, and neurons and axons are well preserved. Because of the synchronous onset of damage, all lesions appear to be at the same stage of myelin loss and reaction. Although originally described in the pons, extra-pontine lesions with similar appearance and apparent etiology may also occur.

Although it can involve most parts of the brain, periventricular and subpial regions are spared, and it is extremely rare for the process to extend below the pontomedullary junction. The clinical presentation is a rapidly evolving quadriplegia, which may be fatal or lead to severe long-term deficits, including the “locked-in” syndrome, in which patients are fully conscious yet unresponsive. It is imperative that hyponatremia be corrected slowly and carefully to prevent this tragic complication.

KEY CONCEPTS

DEMYELINATING DISEASES

- Because of the critical role of myelin in nerve conduction, diseases of myelin can lead to widespread and severe neurologic deficits.
- Demyelinating diseases show evidence of breakdown and destruction of previously normal myelin, often by inflammatory processes. Secondary injury to axons typically emerges over time.
- MS, an autoimmune demyelinating disease primarily affecting young adults, is the most common disorder of myelin. It often pursues a relapsing-remitting course, with eventual progressive accumulation of neurologic deficits that is thought to reflect secondary axon loss.
- Less common forms of immune-mediated demyelination generally follow infections and have an acute, monophasic clinical course.

NEURODEGENERATIVE DISEASES

Neurodegenerative diseases are disorders characterized by the progressive loss of particular groups of neurons, which often have shared functions. Thus, the different diseases tend to involve particular neural systems and have relatively stereotypic presenting signs and symptoms.

The pathologic process that is common across most of the neurodegenerative diseases is the accumulation of protein aggregates (hence the occasional use of the term “proteinopathy”). Protein aggregates may arise because of mutations that alter the affected protein’s conformation or disrupt the pathways that are involved in the processing or clearance of an otherwise normal protein. In other situations, there may be a subtle imbalance between protein synthesis and clearance (from genetic, environmental, or stochastic factors) that allows gradual accumulation of proteins.

Regardless of how they arise, the protein aggregates typically are resistant to degradation and show aberrant localization within neurons. The current evidence suggests that large (i.e., microscopically visible) protein aggregates are not toxic to cells; their formation appears to be an adaptive response that enables cells to sequester smaller (oligomeric) aggregates of the same proteins, which are directly toxic to neurons. However, as more and more protein is shunted into the aggregates, the normal function of the protein may also be lost, and this may also contribute to cell injury. Other recent evidence suggests that protein aggregates are capable of behaving like prions (see later); that is, aggregates derived from one cell may be taken up by another and provoke additional protein aggregation.

The data supporting this concept are largely derived from experimental animal studies, but studies of patients who died with Alzheimer or Parkinson disease are consistent with the idea that these diseases spread from one site in the brain to another. However, only classic prion diseases have been shown to be truly transmissible.

The protein aggregates are recognized histologically as inclusions, which serve as diagnostic hallmarks. The basis for aggregation varies from one disease to another. It may be directly related to an intrinsic feature of a mutated protein (e.g., expanded polyglutamine repeats in Huntington disease [HD]), an intrinsic feature of a peptide derived from a larger precursor protein (e.g., A β in Alzheimer disease [AD]), or an unexplained alteration of a normal cellular protein (e.g., α -synuclein in sporadic Parkinson disease [PD]).

Neurodegenerative diseases vary with respect to the anatomic localization of involved areas and their specific cellular abnormalities (e.g., tangles, plaques, Lewy bodies). Accordingly, they can be classified using two different approaches:

- *Symptomatic/anatomic*: based on the anatomic regions that are most affected, which is typically reflected in the clinical symptoms (e.g., neocortical involvement results in cognitive impairment and dementia)
- *Pathologic*: based on the types of inclusions or abnormal structures observed (e.g., diseases with inclusions containing tau or containing synuclein)

Nevertheless, within the spectrum of degenerative diseases there is remarkable overlap in terms of characteristic neurologic deficits, functional/anatomic distribution of lesions, and cellular pathology (Tables 28.3 and 28.4). For the sake of simplicity, we will follow the time-honored classification based on the original description of these diseases.

Prion Diseases

Prion diseases are rapidly progressive neurodegenerative disorders caused by aggregation and intercellular spread of a misfolded prion protein (PrP); they may be sporadic, familial, or transmitted. Prion diseases include Creutzfeldt-Jakob disease (CJD), Gerstmann-Sträussler-Scheinker syndrome, fatal familial insomnia, and kuru in humans; scrapie in sheep and goats; mink-transmissible encephalopathy; chronic wasting disease of deer and elk; and bovine spongiform encephalopathy. All of these diseases are characterized morphologically by “spongiform change” caused by intracellular vacuoles in neurons and glia, and clinically by a rapidly progressive dementia.

Pathogenesis

Prion diseases are conceptually important because they exemplify degenerative disorders that are caused by “spreading” of misfolded proteins, a remarkable phenomenon that allows a pathogenic protein to acquire some of the characteristics of an infectious organism. Normal PrP is a 30-kD cytoplasmic protein of unknown function. Disease occurs when PrP undergoes a conformational change from its normal α -helix-containing isoform (PrP^C) to an abnormal β -pleated sheet isoform, usually termed PrP^{Sc} (for scrapie) (Fig. 28.33); associated with this conformational change, PrP acquires resistance to digestion with proteases, such as

Table 28.3 Features of the Major Neurodegenerative Diseases

Disease	Clinical Pattern	Inclusions	Genetic Causes
Prion diseases	Rapidly progressive dementia	Kuru plaques and diffuse PrP ^{sc} deposits	PrP
Alzheimer disease (AD)	Dementia	A β (plaques) and tau (tangles)	APP, PS-1, PS-2, trisomy 21
Frontotemporal lobar degeneration (FTLD)	Behavioral changes, language disturbance	tau TDP-43 FUS	tau TDP-43, progranulin, C9orf72 FUS
Parkinson disease (PD)	Hypokinetic movement disorder with or without dementia	α -synuclein tau α -synuclein or none	α -synuclein (mutations or amplification) LRRK2 DJ-1, PINK1, parkin
Progressive supranuclear palsy (PSP)	Parkinsonism with abnormal eye movements	tau	tau
Corticobasal degeneration (CBD)	Parkinsonism with asymmetric movement disorder	tau	
Multiple system atrophy (MSA)	Parkinsonism, cerebellar ataxia, autonomic failure	α -synuclein	
Huntington disease (HD)	Hyperkinetic movement disorder	Huntington (polyglutamine)	Htt
Spinocerebellar ataxias (SCA-1, 2, 3, 6, 7, 17 and DRPLA)	Cerebellar ataxia	Various proteins (polyglutamine containing)	Multiple loci
Amyotrophic lateral sclerosis (ALS)	Weakness with upper and lower motor neurons signs	SOD1 TDP-43 FUS	SOD1 TDP-43, C9orf72 FUS
Spinal bulbar muscular atrophy (SBMA)	Lower motor neuron weakness, diminished androgen	Androgen receptor (polyglutamine containing)	Androgen receptor

proteinase K. Accumulation of PrP^{sc} in neural tissue seems to be the cause of the pathologic changes in these diseases, but how this material induces the development of cytoplasmic vacuoles and eventual neuronal death is still unknown. Immunostaining for PrP after partial digestion with proteinase K allows detection of PrP^{sc}, which is diagnostic.

The conformational change resulting in PrP^{sc} may occur spontaneously at an extremely low rate (resulting in sporadic cases of CJD) or at a higher rate if various mutations are present in PrP^c, such as occurs in familial forms of CJD, Gerstmann-Sträussler-Scheinker syndrome, and fatal familial

insomnia. Independent of the means by which it originates, PrP^{sc} then facilitates, in a cooperative fashion, the conversion of other PrP^c molecules to PrP^{sc} molecules; it is this propagation of PrP^{sc} that accounts for the transmissible variants of prion diseases, which include iatrogenic CJD, variant CJD, and kuru. The suggestion that, at least within an individual, there may be cell-to-cell spread of disease-associated protein aggregates provides a link between prion diseases and other neurodegenerative disorders such as Alzheimer and Parkinson disease.

Creutzfeldt-Jakob Disease (CJD)

The most common prion disease, CJD is a rare disorder that manifests clinically as a rapidly progressive dementia. The sporadic form of CJD has an annual incidence of approximately 1 per 1,000,000 people and accounts for about 90% of CJD cases; familial forms are caused by mutations in *PRNP*, the gene that encodes PrP. The disease has a peak incidence in the seventh decade. There are also well-established cases of iatrogenic transmission, notably by corneal or dural transplantation, deep implantation of electrodes in the brain, and administration of contaminated preparations of cadaveric human growth hormone. The onset is marked by subtle changes in memory and behavior followed by a rapidly progressive dementia, often associated with pronounced involuntary jerking muscle contractions on sudden stimulation (startle myoclonus). Signs of cerebellar dysfunction, usually manifested as ataxia, are present in a minority of affected individuals. The disease is uniformly fatal; the average survival is only 7 months after the onset of symptoms. A few patients have lived for several years, and these long-surviving cases show extensive atrophy of involved gray matter.

Table 28.4 Relationship Between Proteins and Neurodegenerative Diseases

Protein	Diseases With Inclusions
A β	Alzheimer disease
tau	Alzheimer disease Frontotemporal lobar degeneration Parkinson disease (with <i>LRRK2</i> mutations) Progressive supranuclear palsy Corticobasal degeneration Chronic traumatic encephalopathy
TDP-43	Frontotemporal lobar degeneration Amyotrophic lateral sclerosis
FUS	Frontotemporal lobar degeneration Amyotrophic lateral sclerosis
α -synuclein	Parkinson disease Multiple system atrophy
Polyglutamine aggregates (different protein in each disease)	Huntington disease Some forms of spinocerebellar ataxia Spinal bulbar muscular atrophy

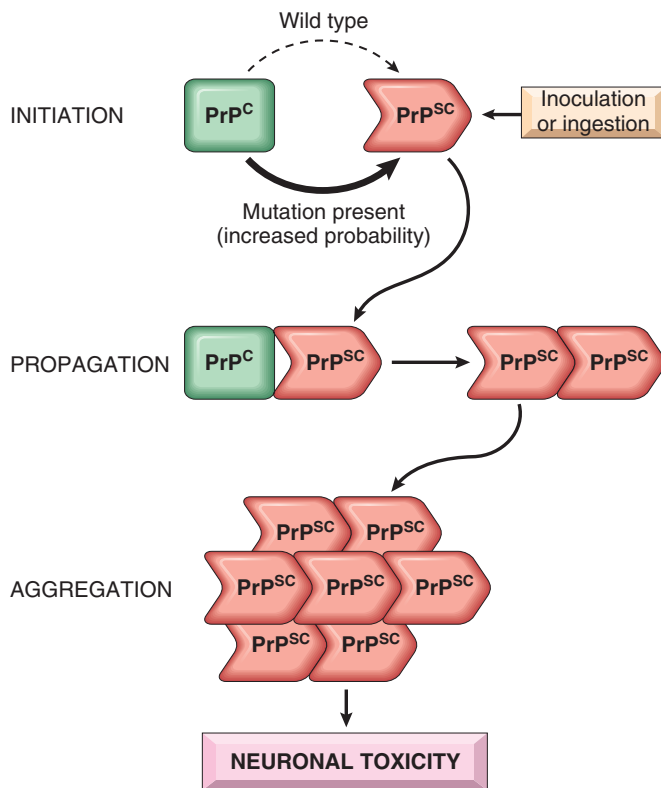


Figure 28.33 Pathogenesis of prion disease. α -helical PrP^{C} may spontaneously shift to the β -sheet PrP^{Sc} conformation, an event that occurs at a much higher rate in familial diseases associated with germline PrP mutations. PrP^{Sc} may also be acquired from exogenous sources, such as contaminated food, medical instrumentation, or medicines. Once present, PrP^{Sc} converts additional molecules of PrP^{C} into PrP^{Sc} through physical interaction, eventually leading to the formation of pathogenic PrP^{Sc} aggregates.

Variant Creutzfeldt-Jakob Disease

Starting in 1995, a new CJD-like illness came to medical attention in the United Kingdom. This illness was different from classical CJD in several important respects: the disease affected young adults, behavioral disorders figured prominently in the early stages of the disease, and the neurologic syndrome progressed more slowly than in individuals with other forms of CJD. Multiple lines of evidence indicated that the new variant of CJD (vCJD) was linked to exposure to bovine spongiform encephalopathy, either through consumption of contaminated foods or via transfusion of blood from patients in the asymptomatic/preclinical stage of vCJD. Recognition of this link led to public health measures that have been effective in limiting the spread of vCJD; the incidence in the United Kingdom peaked in 2000 and dropped precipitously since that time.

MORPHOLOGY

The progression of the dementia in CJD is usually so rapid that there is little if any grossly evident brain atrophy. The pathognomonic finding is a **spongiform** transformation of the cerebral cortex and deep gray matter structures (caudate, putamen); this multifocal process results in the uneven formation of small and apparently empty microscopic vacuoles of varying sizes within

the neuropil and sometimes in the perikaryon of neurons (Fig. 28.34A). In advanced cases, there is severe neuronal loss, reactive gliosis, and sometimes expansion of the vacuolated areas into cystlike spaces (“status spongiosus”). Inflammation is notably absent. Electron microscopy shows the vacuoles to be membrane-bound and located within the cytoplasm of neuronal processes. **Kuru plaques** are extracellular deposits of aggregated abnormal PrP . They are Congo red- and PAS-positive and usually occur in the cerebellum (Fig. 28.34B), but are abundant in the cerebral cortex in cases of vCJD (Fig. 28.34C). In all forms of prion disease, immunohistochemical staining demonstrates the presence of proteinase K-resistant PrP^{Sc} in tissue.

Alzheimer Disease (AD)

AD is the most common cause of dementia in older adults, with an increasing incidence as a function of age. The disease usually becomes clinically apparent as insidious impairment of higher cognitive functions. As the disease progresses, deficits in memory, visuospatial orientation, judgment, personality, and language gradually emerge; over a course of 5 to 10 years, the affected individual becomes profoundly disabled, mute, and immobile. Patients rarely become symptomatic before 50 years of age; the incidence of the disease increases with age, and the prevalence roughly doubles every 5 years, starting from a level of 1% for the 60- to 64-year-old population and reaching 40% or more for the 85- to 89-year-old cohort. This progressive increase in incidence with increasing age has given rise to major medical, social, and economic concerns in countries with aging populations. About 5% to 10% of cases are familial; these have provided important insight into the pathogenesis of the more common sporadic form of the disease. Although pathologic examination of brain tissue obtained at autopsy remains necessary for definitive diagnosis of AD, the combination of clinical assessment and current radiologic methods allows accurate premortem diagnosis in 80% to 90% of cases.

Pathogenesis

The fundamental abnormality in AD is the accumulation of two proteins ($\text{A}\beta$ and tau) in specific brain regions, likely as a result of excessive production and defective removal (Fig. 28.35). The two pathologic hallmarks of AD, particularly evident in the end stages of the illness, are *amyloid plaques* and *neurofibrillary tangles*. Plaques are deposits of aggregated $\text{A}\beta$ peptides in the neuropil, while tangles are aggregates of the microtubule binding protein tau, which develop intracellularly and then persist extracellularly after neuronal death. Both plaques and tangles appear to contribute to the neural dysfunction, and the interplay between the processes that lead to the accumulation of these two types of abnormal protein aggregates is a critically important aspect of AD pathogenesis that has yet to be fully unraveled.

Several lines of evidence strongly support a model in which **$\text{A}\beta$ generation is the critical initiating event for the development of AD.** First, there are diseases in which tau deposits appear, such as frontotemporal lobar degenerations, progressive supranuclear palsy, and corticobasal degeneration (discussed later), but $\text{A}\beta$ deposits do not ensue, and full-blown AD does not develop. This suggests that having

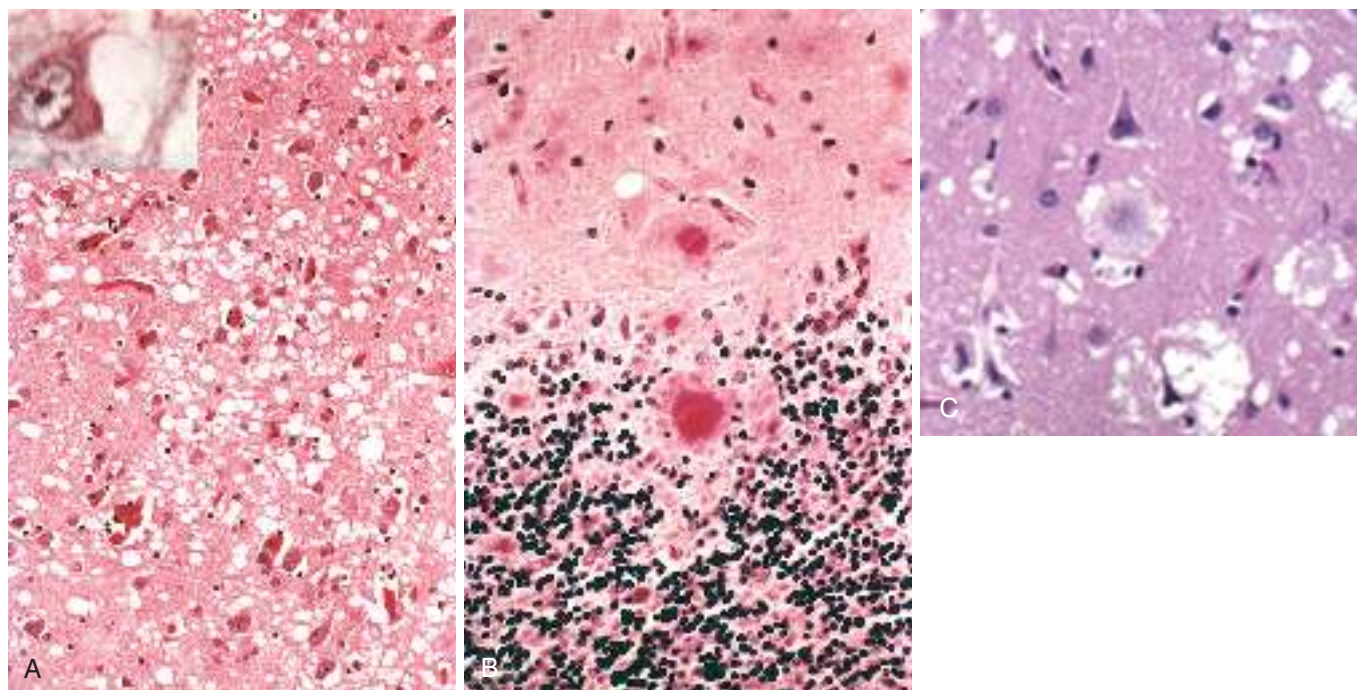


Figure 28.34 Prion disease. (A) Spongiform change in the cerebral cortex. *Inset*, High magnification of neuron with vacuoles. (B) Cerebellar cortex showing kuru plaques (periodic acid–Schiff stain) that consist of aggregated PrP^{Sc}. (C) Cortical kuru plaques surrounded by spongiform change in variant Creutzfeldt-Jakob disease.

abnormal deposits of tau in the brain is not a sufficient stimulus to elicit deposition of A β . Additionally, multiple lines of genetic evidence (discussed later) point to the likely importance of altered A β metabolism. In contrast, mutations in the *MAPT* gene that encodes tau do not give rise to AD but rather cause frontotemporal lobar degeneration (discussed later).

The pathogenesis of AD thus involves A β , tau, and several other genetic and host factors, as follows:

- **Role of A β .** APP is a cell surface protein with a single transmembrane domain that may function as a receptor, possibly for prion protein (PrP^{Sc}) among other ligands. The A β portion of the protein extends from the extracellular region into the transmembrane domain (see Fig. 28.35). Processing of APP begins with cleavage in the extracellular domain, followed by an intramembranous cleavage. There are two potential extracellular sites of cleavage, which may be carried out by two different classes of proteases, α -secretase and β -secretase. If the first cut occurs at the α -secretase site, then A β is not generated (the non-amyloidogenic pathway); this cleavage event mostly occurs at the cell surface because proteases with α -secretase activity are involved in the shedding of surface proteins. Surface APP may also be endocytosed into vesicles, where it can be cleaved by β -secretase, which cuts at a slightly more N-terminal site within APP (the amyloidogenic pathway). Following cleavage of APP at either of these sites, the γ -secretase complex performs an intramembranous cleavage. Cleavage by α -secretase and β -secretase releases A β 42, which is prone to aggregation and amyloid formation. The γ -secretase complex—containing presenilin-1 or presenilin-2, nicastrin, PEN2, and APh1—is also responsible for processing of Notch receptors and many other membrane proteins.

Once generated, A β 42 is highly prone to aggregation—first into small oligomers (which appear to be the toxic form responsible for neuronal dysfunction), and eventually into large aggregates and fibrils that can be visualized by microscopy.

Familial forms of AD support the central role of A β generation as a critical step for initiation of AD pathogenesis. The gene encoding APP, on chromosome 21, lies in the Down syndrome region; AD pathology contributes to the cognitive impairment of patients with this chromosomal abnormality, with appearance of histopathologic AD findings in the second and third decades followed by neurologic decline about 20 years later. A similar gene dosage effect is produced by localized chromosome 21 duplications that span the *APP* locus in some patients with familial AD. Point mutations in *APP* are another cause of familial AD. Some mutations lie near the β -secretase and γ -secretase cleavage sites, and others sit in the A β sequence and increase its propensity to aggregate. The two loci identified as causes of the majority of early-onset familial AD (*PSEN1* on chromosome 14 and *PSEN2* on chromosome 1) encode presenilin-1 and presenilin-2. These mutations lead to a gain of function, such that the γ -secretase complex generates increased amounts of A β , particularly A β 42.

- **Role of tau.** Because neurofibrillary tangles contain the tau protein, there has been much interest in the role of this protein in AD. Tau is a microtubule-associated protein present in axons in association with the microtubular network. With the development of tangles in AD, it shifts to a somatic-dendritic distribution, becomes hyperphosphorylated, and loses the ability to bind to microtubules. The formation of tangles is an important component of AD, and the increased tangle burden in the brain over

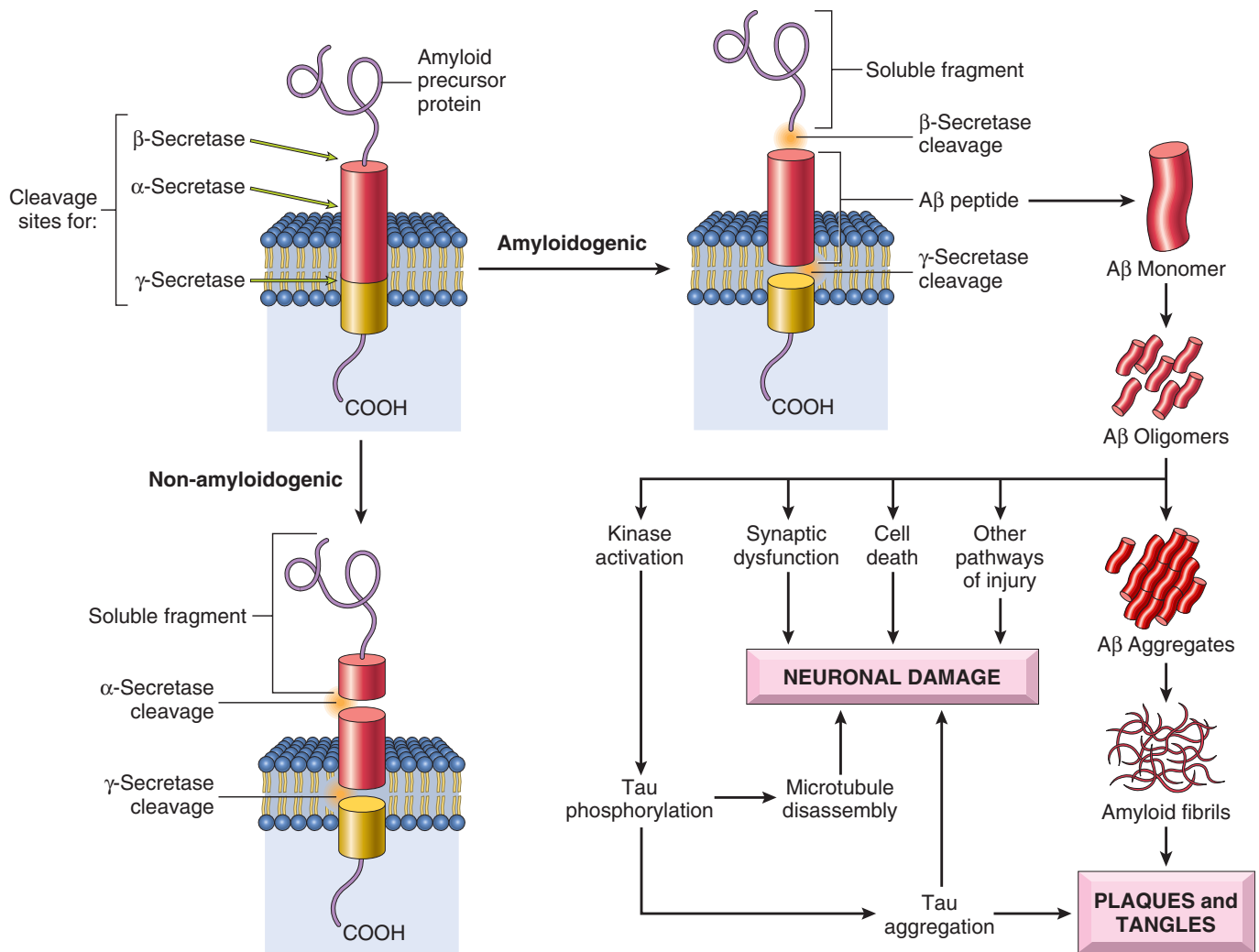


Figure 28.35 Protein aggregation in Alzheimer disease. Amyloid precursor protein cleavage by α -secretase and γ -secretase produces a harmless soluble peptide, whereas amyloid precursor protein cleavage by β -amyloid-converting enzyme and γ -secretase releases $A\beta$ peptides, which form pathogenic aggregates and contribute to the characteristic plaques and tangles of Alzheimer disease.

the course of the illness eventually appears to become independent of the $A\beta$. How tangles injure neurons remains poorly understood, but two possible pathways have been suggested, which are not mutually exclusive: (1) the aggregates of tau protein elicit a stress response; and (2) the loss of tau protein destabilizes microtubules.

- **Other genetic risk factors.** The genetic locus on chromosome 19 that encodes apolipoprotein E (ApoE) has a strong influence on the risk of developing AD. Three alleles exist ($\epsilon 2$, $\epsilon 3$, and $\epsilon 4$) based on two amino acid polymorphisms. The dosage of the $\epsilon 4$ allele increases the risk of AD and lowers the age of onset of the disease, such that individuals with the $\epsilon 4$ allele are overrepresented in populations of patients with AD. This ApoE isoform promotes $A\beta$ generation and deposition, but also appears to exacerbate $A\beta$ -independent, tau-mediated neurodegeneration. Overall, this locus has been estimated to convey about one-fourth of the risk for development of late-onset AD. Genome-wide association studies have identified several other loci that contribute to the risk of AD; the connection between these genetic loci and the pathogenesis of AD remains to be explored.

- **Role of inflammation.** Both small aggregates and larger deposits of $A\beta$ elicit an inflammatory response from microglia and astrocytes. This response probably assists in the clearance of the aggregated peptide, but it may also stimulate the secretion of mediators that cause damage. Additional consequences of the activation of these inflammatory cascades may include alterations in tau phosphorylation, oxidative injury to neurons, and aberrant pruning of synapses.
- **Basis for cognitive impairment.** Although there remains disagreement regarding the best correlate of dementia in individuals with AD, it is clear that the presence of a large burden of plaques and tangles is highly associated with severe cognitive dysfunction. The number of neurofibrillary tangles correlates better with the degree of dementia than does the number of neuritic plaques. Biochemical markers that have been correlated with the degree of dementia include loss of choline acetyltransferase, loss of synaptophysin immunoreactivity, and amyloid burden.
- **Biomarkers.** Among the more important recent developments in the understanding of AD is the discovery of

possible biomarkers, which draw on the understanding of the biologic processes discussed earlier. It is now possible to demonstrate A β deposition in the brain through imaging methods that rely on 18F-labeled amyloid-binding compounds; this approach can identify asymptomatic patients who are in the early stages of AD. Additional evidence of neuronal degeneration associated with AD-related pathologic processes includes the presence of increased phosphorylated tau and reduced A β in the CSF. Together, these biomarkers have allowed for the identification of preclinical stages of AD, well in advance of the development of dementia or other signs and symptoms; this in turn has enabled the focus of pharmacologic trials to shift toward individuals in the earliest stages of the illness, in whom it is hoped interventions will slow or prevent disease progression and limit disability.

MORPHOLOGY

Grossly, the brain shows variable **cortical atrophy** marked by gyral narrowing and sulcal widening that is most pronounced in the frontal, temporal, and parietal lobes (Fig. 28.36). With significant atrophy, there is compensatory ventricular enlargement (hydrocephalus ex vacuo) secondary to reduced brain volume. Structures of the medial temporal lobe, including the hippocampus, entorhinal cortex, and amygdala, are involved early in the disease course and are usually severely atrophied in the later stages.

The major microscopic abnormalities of AD are **neuritic (senile) plaques** and **neurofibrillary tangles**. There is progressive, eventually severe, neuronal loss and reactive gliosis in the same regions that bear the burden of plaques and tangles.

Neuritic plaques are focal, spherical collections of dilated, tortuous, axonal or dendritic processes (dystrophic neurites) often around a central amyloid core, which may be surrounded by a clear halo (Fig. 28.37A). Dystrophic neurites contain tau aggregates that are biochemically similar to neurofibrillary tangles.



Figure 28.36 Alzheimer disease with cortical atrophy most evident on the right, where meninges have been removed. (Courtesy the late Dr. E.P. Richardson, Jr., Massachusetts General Hospital, Boston, Mass.)

Neuritic plaques range in size from 20 to 200 μ m in diameter; microglial cells and reactive astrocytes are present at their periphery. Plaques are found in the hippocampus, amygdala, and neocortex, although there is usually relative sparing of primary motor and sensory cortices (this also applies to neurofibrillary tangles). The amyloid core, which can be stained by Congo red or a beta amyloid immunostain, contains several abnormal proteins. The dominant component of the amyloid plaque core is A β , a peptide derived by proteolytic cleavage of APP (Figs. 28.37B and 28.35); other proteins, including components of the complement cascade, proinflammatory cytokines, α_1 -antichymotrypsin, and apolipoproteins, are present in lesser abundance. In some cases, there is deposition of A β peptides in the absence of the surrounding dystrophic neurites. These lesions, termed **diffuse plaques**, are found mainly in superficial portions of the cerebral cortex, the basal ganglia, and the cerebellar cortex. Based on studies of individuals with trisomy 21, diffuse plaques are believed to be an early stage of plaque development. Although neuritic plaques contain both A β_{40} and A β_{42} , diffuse plaques are predominantly made up of A β_{42} .

Neurofibrillary tangles are tau-containing bundles of filaments in the cytoplasm of the neurons that displace or encircle the nucleus. In pyramidal neurons, they often have an elongated “flame” shape; in rounder cells, the basket weave of fibers around the nucleus takes on a rounded contour (“globose” tangles). Neurofibrillary tangles are visible as basophilic fibrillary structures with H&E staining (Fig. 28.37C) but are demonstrated much more clearly by silver (Bielschowsky) staining (Fig. 28.37D) and with tau immunohistochemistry (Fig. 28.37E). They are commonly found in cortical neurons, especially in the entorhinal cortex, as well as in other sites such as pyramidal cells of the hippocampus, the amygdala, the basal forebrain, and the raphe nuclei. Neurofibrillary tangles are insoluble and apparently resistant to clearance in vivo, thus remaining visible in tissue sections as “ghost” or “tombstone” tangles long after the death of the parent neuron. Ultrastructurally, neurofibrillary tangles are composed predominantly of paired helical filaments along with some straight filaments that appear to have a similar composition. Aggregated tau is also present in dystrophic neurites that form the outer portions of neuritic plaques and in axons coursing through the affected gray matter as neuropil threads. In contrast to neuritic and diffuse plaques, tangles are found in other neurodegenerative diseases and are thus not specific for AD.

In addition to the diagnostic features of plaques and tangles, several other pathologic findings are seen in the setting of AD. **Cerebral amyloid angiopathy (CAA)** (Fig. 28.18D) is an almost invariable accompaniment of AD; however, it can also be found in brains of individuals without AD. In contrast to the amyloid deposited in neuritic and diffuse plaques, which mainly consists of A β_{42} , the amyloid in CAA is predominantly comprised of A β_{40} .

Although abundant burdens of plaques and tangles characterize the end stage of AD, in which affected individuals are fully demented, it is clear that these histologic changes appear well in advance of clinical symptoms. To provide a correlation between neuropathologic findings and clinical symptomatology, the most recent recommendations for describing these lesions consider all deposition of A β in brain parenchyma to be a form of AD neuropathologic change. The scheme then generates a histopathologic score based on the distribution of A β deposits, plaques, and tangles, which is used to predict the likelihood of an individual being cognitively impaired that is based on population studies.

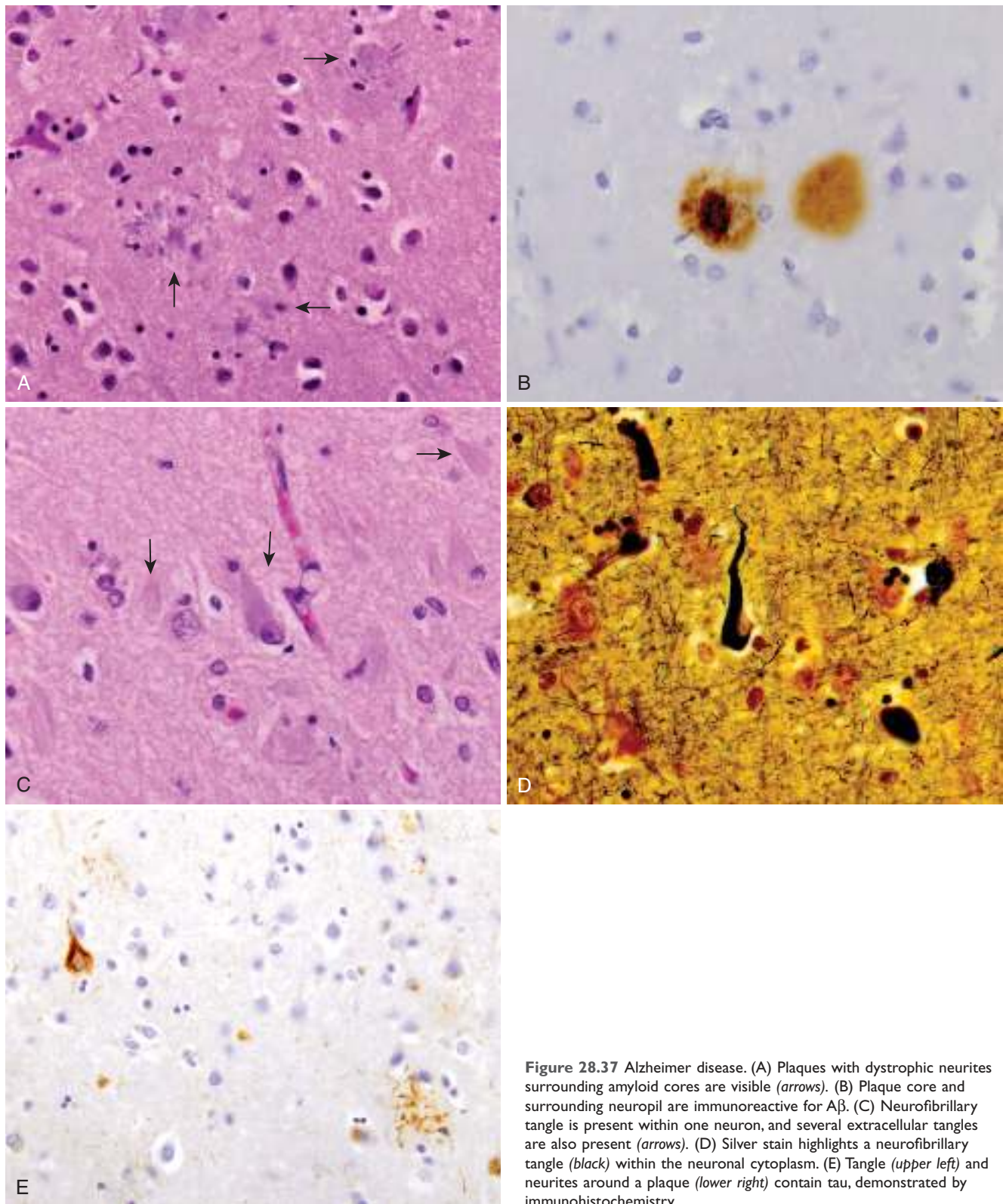


Figure 28.37 Alzheimer disease. (A) Plaques with dystrophic neurites surrounding amyloid cores are visible (*arrows*). (B) Plaque core and surrounding neuropil are immunoreactive for A β . (C) Neurofibrillary tangle is present within one neuron, and several extracellular tangles are also present (*arrows*). (D) Silver stain highlights a neurofibrillary tangle (*black*) within the neuronal cytoplasm. (E) Tangle (*upper left*) and neurites around a plaque (*lower right*) contain tau, demonstrated by immunohistochemistry.

Clinical Features

The progression of AD is slow but relentless, with a symptomatic course often running more than 10 years. Initial symptoms are forgetfulness and other memory disturbances; with progression, other symptoms emerge, including language deficits, loss of mathematical skills, and loss of learned motor skills. In the final stages, affected individuals may become incontinent, mute, and unable to walk; intercurrent disease, often pneumonia, is usually the terminal event. Current clinical trials are focused on treating subjects in early, preclinical stages of the illness, using strategies that include clearing A β from the brain through immunologic approaches, disrupting the generation of A β with pharmacologic agents that target secretases, as well as approaches aimed at preventing alterations in tau.

Frontotemporal Lobar Degenerations (FTLDs)

FTLDs are a heterogeneous group of disorders associated with focal degeneration of frontal and/or temporal lobes. They are clinically distinguished from AD by the pattern of involvement, with alterations in personality, behavior, and language (aphasias) preceding memory loss; however, as with other neurodegenerative diseases, postmortem pathologic evaluation is required for definitive diagnosis. Global dementia occurs with progressive disease, and a subset of patients develops extrapyramidal motor loss. Several clinical variants have been described based on whether the behavioral change or aphasias dominate, but they have overlapping features. FTLDs are one of the more common causes of early-onset dementia and occur at the same frequency as AD in those younger than 65 years of age. Commonly referred to in the clinical setting as *frontotemporal dementia* (FTD), the preferred pathologic terminology highlights the anatomic pattern of involvement (lobar degeneration) rather than the clinical symptom (dementia).

Like most neurodegenerative diseases, FTLD is associated with cellular inclusions comprised of specific proteins; the two most common patterns are those with tau inclusions (FTLD-tau) and those with TDP-43 inclusions (FTLD-TDP). Within each of these groups, there are heritable and sporadic forms. There is no fixed relationship between the clinical subtypes of FTLD and the type of neuronal inclusions.

FTLD-tau

These are forms of FTLD in which the affected cortical regions demonstrate progressive neuronal loss and reactive gliosis, along with the presence of cytoplasmic tau inclusions in neurons. Unlike Alzheimer disease, which is characterized by the combination of A β and tau deposition, FTLD-tau shows only tau aggregation and accumulation. In some cases FTLD-tau inclusions resemble the tangles seen in AD, and in other forms of the disease there are smooth contoured inclusions (Pick bodies).

Pathogenesis

FTLD-tau may be associated with mutations in *MAPT*, the gene that encodes tau, or it may arise sporadically. As mentioned earlier, tau interacts with microtubules; its binding to microtubules may be regulated by phosphorylation. There

is an inverse relationship between the degree of tau phosphorylation and its ability to bind microtubules. Tau, particularly when phosphorylated, has a propensity to aggregate. Interestingly, tau exists as a complex series of isoforms that are encoded by different mRNA splice variants; the balance between these isoforms appears to be critical for normal tau function in neurons, and disturbances in isoform ratio may also provoke tau aggregation.

Two different types of tau mutations have been described. Some missense point mutations effect tau phosphorylation, tipping the balance from microtubule binding to aggregation. Other mutations include point mutations that affect splicing; many of these are intronic and alter the loop-stem structures recognized by the spliceosome. The resulting change in isoform ratio is thought to lead to neuronal dysfunction and, as discussed earlier, may also enhance tau aggregation.

It remains unclear how abnormal tau injures neurons, although there appears to be both a loss-of-function component, as aggregation depletes neurons of tau, and a toxic gain-of-function component due to presence of aberrantly hyperphosphorylated aggregated protein.

MORPHOLOGY

There is atrophy of frontal and temporal lobes to variable extent and severity. The pattern of atrophy can often be predicted from the clinical symptoms. The atrophic regions of cortex are marked by neuronal loss, gliosis, and the presence of tau-containing neurofibrillary tangles (Fig. 28.38A). These tangles may contain a variety of tau isoforms. Nigral degeneration may occur. Inclusions can also be found in glial cells in some forms of the disease.

In **Pick disease** (a subtype of FTD-tau), the brain invariably shows a pronounced and frequently asymmetric atrophy of the frontal and temporal lobes, with conspicuous sparing of the posterior two-thirds of the superior temporal gyrus and only rare involvement of the parietal and occipital lobes. The atrophy can be severe, reducing gyri to a wafer-thin ("knife-edge") appearance. The neuronal loss is most severe in the outer three layers of the cortex. Some of the surviving neurons show a characteristic swelling (Pick cells), and others contain **Pick bodies**, cytoplasmic, round to oval, filamentous inclusions that stain strongly with silver methods (Fig. 28.38B).

FTLD-TDP

Some individuals with clinically diagnosed FTD and macroscopic changes of relatively localized cortical atrophy (the "lobar degeneration" of the term) have inclusions that do not contain tau and instead contain TDP-43, an RNA-binding protein. Individuals with FTLD-TDP typically present with either behavioral problems or language deficits, features similar to FTLD-tau.

Pathogenesis

Mutations in three different genes have been found in the inherited forms of FTLD-TDP.

- The most common familial form of FTLD-TDP is the result of an expansion of a hexanucleotide repeat in the 5' region of *C9orf72* (a gene encoding a protein of unknown function); the spectrum of disease associated with *C9orf72* expansion also includes amyotrophic lateral sclerosis

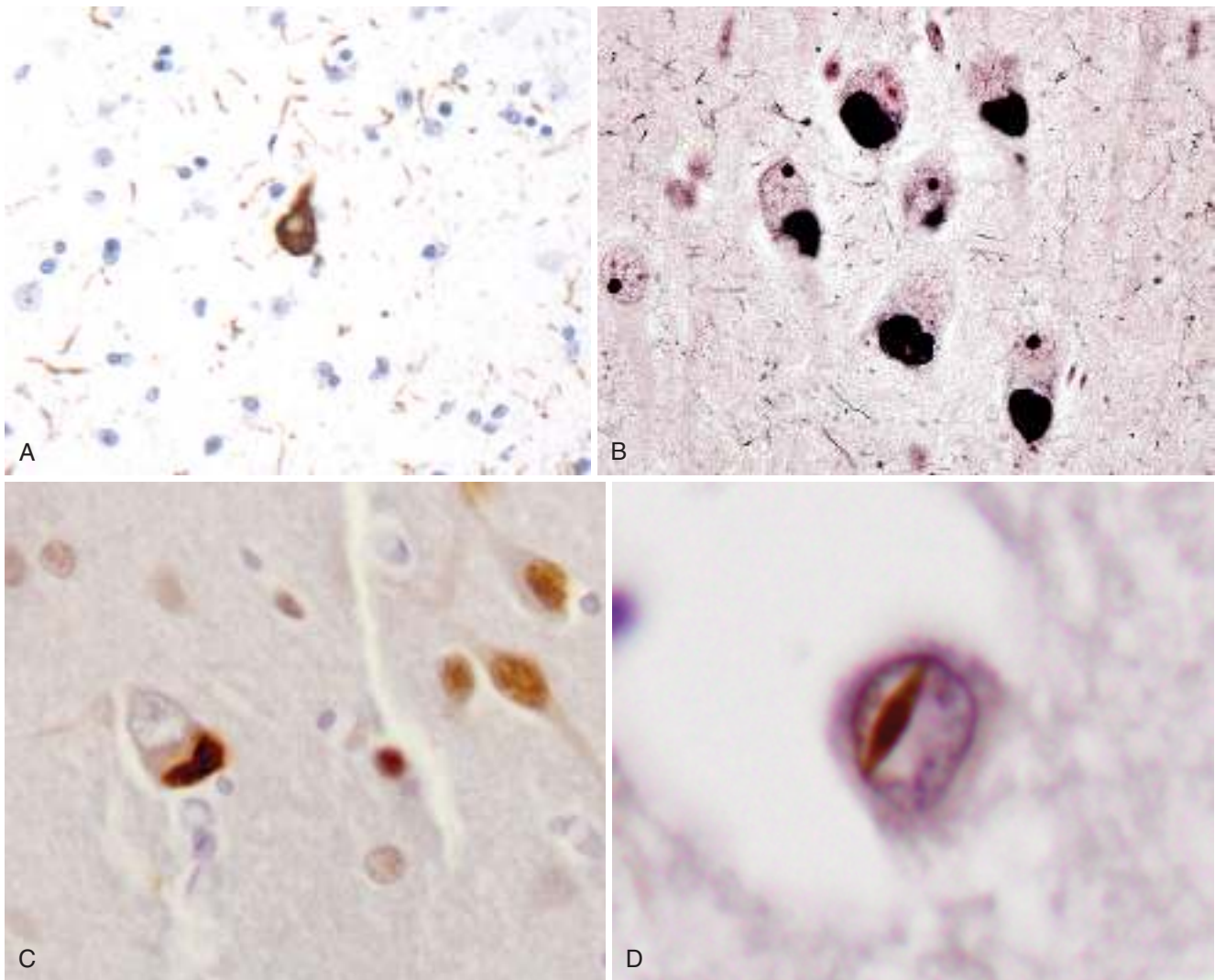


Figure 28.38 Frontotemporal lobar degenerations (FTLDs). (A) FTLD-tau. A tangle is present along with numerous tau-containing neurites. (B) Pick disease. Pick bodies are round, homogeneous neuronal cytoplasmic inclusions that stain intensely with silver stains. (C) FTLD-TDP. Cytoplasmic inclusions containing TDP-43 are seen in association with loss of normal nuclear immunoreactivity. (D) FTLD-TDP. With progranulin mutations, the TDP-43-containing inclusions are commonly intranuclear.

(discussed later). How the repeat expansion results in formation of aggregates of TDP-43 is a mystery at present. Interestingly, the brains from *C9orf72*-associated FTLD and amyotrophic lateral sclerosis cases also contain ubiquitin-positive inclusions that consist of dipeptide repeats generated by aberrant translation of the expanded repeats (in individuals without repeat expansion, this region of the gene contains regulatory sequences and is not translated).

- Mutations in the *TARDBP* gene (encoding TDP-43) are less common and also occur in some familial cases of amyotrophic lateral sclerosis. TDP-43 is an RNA-binding protein with roles in RNA processing and the formation of stress granules, RNA-protein aggregates of uncertain function. Both loss of function and toxic gain of function may contribute to this form of FTD.
- A third form of FTLD-TDP is the result of mutations in the *GRN* gene, which encodes a protein called progranulin;

in contrast to *TARDBP* and *C9orf72*, mutations in *GRN* have not been linked to amyotrophic lateral sclerosis. Progranulin mutations cause loss of function, and the disease is believed to stem from deficient progranulin activity. Progranulin is a secreted protein expressed in glia and neurons that is cleaved into multiple small peptides; these peptides have been implicated in regulation of inflammation in the brain, but the link between this activity and the accumulation of TDP-43-containing inclusions in FTLD is currently obscure.

There are forms of FTLD in which there are neither tau- nor TDP-containing inclusions. Although these are infrequent, the underlying genetic causes show overlap with the pathways involved in other forms of FTLD-TDP. Mutations in the *FUS* (fused in sarcoma) gene may cause either FTLD or ALS, and *FUS* is another RNA-binding protein that may be involved in formation of stress granules.

MORPHOLOGY

The appearance is similar to the other forms of FTLT, with variable atrophy of frontal and temporal lobes accompanied by neuronal loss and gliosis. Normally, TDP-43 is found diffusely in the nucleus but in FTLT-TDP, TDP-43 is found in inclusions (Fig. 28.38C) that may be in the cell body (neuronal cytoplasmic inclusions), in the nucleus (neuronal intranuclear inclusions), or in neurites (TDP-43 threads). In the inclusions, TDP-43 is phosphorylated and ubiquitinated. Inclusions are most abundant in the frontal and temporal cortex, the striatum, and the dentate gyrus of the hippocampus. There is a strong correlation between the presence of needle-like intranuclear inclusions and progranulin mutations (Fig. 28.38D).

There are also forms of FTLT in which there are neither tau- nor TDP-containing inclusions. Although these are infrequent, the underlying genetic causes show connections with the pathways involved in other forms of FTLT-TDP. Mutations in the *FUS* (fused in sarcoma) gene may cause either FTLT or amyotrophic lateral sclerosis, and *FUS* is another RNA-binding protein that may be involved in formation of stress granules.

Parkinson Disease (PD)

PD is a neurodegenerative disease marked by a hypokinetic movement disorder that is caused by loss of dopaminergic neurons from the substantia nigra. The clinical syndrome of *parkinsonism* combines diminished facial expression (often termed masked facies), stooped posture, slowing of voluntary movement, festinating gait (progressively shortened, accelerated steps), rigidity, and a “pill-rolling” tremor. This type of motor disturbance is seen in a number of conditions that are associated with damage to the nigrostriatal dopaminergic system. Although PD is the most common cause of parkinsonism, similar symptoms can be induced by dopaminergic antagonists or by toxins that selectively damage the dopaminergic system. Also discussed later are other rare diseases that have parkinsonism as part of the clinical presentation.

The clinical diagnosis of PD is based on the presence of the triad of parkinsonism—tremor, rigidity, and bradykinesia—in the absence of a toxic or other known underlying etiology. This clinical impression is confirmed by symptomatic response to L-DOPA replacement therapy. Although the diagnosis of PD is based in large part on the presence of the motor symptoms, which reflect the decreased dopaminergic innervation of the striatum, the disease is not restricted to dopaminergic neurons or to the basal ganglia; in fact, degeneration of the substantia nigra (which results in the motor symptoms) represents a mid-stage in a progressive disease that begins lower in the brainstem and eventually progresses to involve the cerebral cortex, leading to cognitive impairment (see “[Dementia with Lewy Bodies](#),” later in this chapter).

The dopaminergic neurons of the substantia nigra project to the striatum, and their degeneration in PD is associated with a reduction in striatal dopamine content. The severity of the motor syndrome is proportional to the dopamine deficiency, which can, at least in part, be corrected by

replacement therapy with L-DOPA (the immediate precursor of dopamine). However, treatment does not reverse the structural changes or arrest the progress of the disease; with progression, drug therapy becomes less effective, and symptoms more difficult to manage. Deep brain stimulation has emerged over the past decade as a therapy for the motor symptoms of PD. In addition, the well-characterized neural and biochemical deficits in PD have also provided a rationale for therapeutic trials of neural transplantation and gene therapy.

An acute parkinsonian syndrome and destruction of neurons in the substantia nigra follows exposure to MPTP (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine), discovered as a contaminant in illicitly synthesized batches of the opioid meperidine. This toxin has been used to generate animal models of PD, which are used to test new therapies. Epidemiologic evidence has also suggested that pesticide exposure is a risk factor for PD, while caffeine and nicotine appear to be protective.

Pathogenesis

PD is associated with protein accumulation and aggregation, mitochondrial abnormalities, and neuronal loss in the substantia nigra and elsewhere in the brain. Although most PD is sporadic, identification of genetic causes helped shed light on its pathogenesis.

- The first mutations identified as a cause of autosomal dominant PD involved *SNCA*, a gene that encodes α -synuclein, an abundant lipid-binding protein normally localized to synapses. This protein is a major component of the *Lewy body*, which is the diagnostic hallmark of PD. Mutations in *SNCA* are rare; they include point mutations and amplifications of the region of chromosome 4q21 that contains the gene. The occurrence of disease caused by increases in gene copy number implies a gene dosage effect and suggests that polymorphisms in the α -synuclein promoter influence the risk of PD. Like A β in AD, α -synuclein forms aggregates; of these, small oligomers appear to be the most toxic to neurons. There is also evidence that aggregates can be released from one neuron and taken up by another, suggesting a prion-like pattern of spread within the brain. Consistent with this idea, α -synuclein-containing aggregates (Lewy bodies and Lewy neurites) are first seen in the medulla and then in contiguous areas of the brain, ascending through the brainstem and extending into limbic structures and finally the neocortex. Remarkably, emerging evidence suggests that α -synuclein aggregates are initially formed in the neurons of the enteric nervous system and spread from the gut to the medulla via the vagus nerve; this “gut-brain” hypothesis of PD pathogenesis provides a possible explanation for association of PD with exposure to pesticides and other environmental toxins.
- *Mitochondrial dysfunction* has been implicated as a contributing factor based on autosomal recessive forms of PD that are caused by mutations in genes that encode the proteins DJ-1, PINK1, and parkin. DJ-1 has multiple cellular roles, including acting as a transcriptional regulator, but in settings of oxidative stress, it can relocate to the mitochondria and have cytoprotective effects. PINK1 is a kinase that localizes to the outer membrane of dysfunctional mitochondria, where it recruits and phosphorylates

parkin. This modification activates parkin, an E3 ubiquitin ligase that targets a number of substrates for proteasomal destruction. Under normal circumstances, the combination of PINK1 and parkin results in clearance of dysfunctional mitochondria through mitophagy; this process is impaired by defects in PINK1 or parkin. Intriguingly, levels of mitochondrial complex I, a component of the oxidative phosphorylation cascade, are reduced in the brains of patients with sporadic PD.

- Heterozygous mutations in the lysosomal enzyme glucocerebrosidase are the most important risk factor for development of PD, accounting for ~5% of PD cases (homozygous mutations in this enzyme cause Gaucher disease, a lysosomal storage disease described in more detail in Chapter 5); this genetic link suggests that dysfunction of the lysosome-autophagy pathway contributes to PD pathogenesis, most likely through abnormal metabolism of α -synuclein, activation of the unfolded protein response, and increased inflammation.
- Mutations in the gene encoding *LRRK2* (leucine-rich repeat kinase 2) are the most common cause of autosomal dominant PD and are also found in some late-onset, apparently sporadic cases of the disease. *LRRK2* is a cytoplasmic kinase; several of the pathogenic mutations increase its kinase activity, suggesting that gains in *LRRK2* function—either hyperphosphorylation of normal targets or emergence of novel targets—might contribute to the development of PD. However, the molecular mechanism linking *LRRK2* dysfunction and α -synuclein accumulation has not yet been elucidated.

MORPHOLOGY

A characteristic gross finding in PD is **pallor of the substantia nigra** (compare Fig. 28.39A and B) and locus ceruleus, which is due to loss of the pigmented, catecholaminergic neurons in these nuclei. **Lewy bodies** (Fig. 28.39C) are usually found in some of the remaining neurons; these are single or multiple, cytoplasmic, eosinophilic, round to elongated inclusions that often have a dense core surrounded by a pale halo. Ultrastructurally, Lewy bodies are composed of fine filaments, densely packed in the core but loose at the rim; these filaments are composed of α -synuclein. Lewy bodies can also be found in the cholinergic neurons of the basal nucleus of Meynert, as well as in other brainstem nuclei including the locus ceruleus and the dorsal motor nucleus of the vagus. Areas of neuronal loss also typically show gliosis. Lewy neurites are dystrophic processes that contain aggregated α -synuclein.

Dementia With Lewy Bodies

About 10% to 15% of individuals with PD develop dementia, particularly with advancing age. Characteristic features of this disorder include a fluctuating course, visual hallucinations, and prominent frontal signs. Although some affected individuals have pathologic evidence of Alzheimer disease (or, less frequently, other degenerative diseases associated with cognitive changes) in combination with PD, in others the most prominent histologic correlate is the presence of widespread Lewy bodies in neurons of the cortex and brainstem. In some cases, dementia with Lewy bodies represents an advanced stage of PD in which protein aggregates

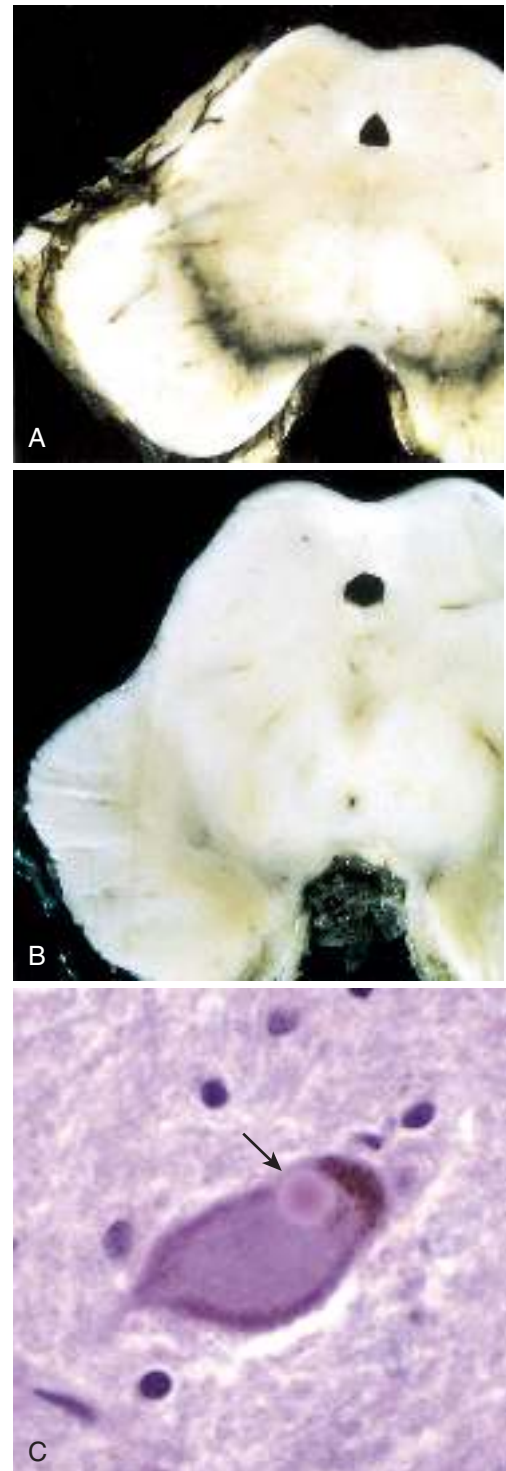


Figure 28.39 Parkinson disease. (A) Normal substantia nigra. (B) Depigmented substantia nigra in idiopathic Parkinson disease. (C) Lewy body in a substantia nigra neuron, staining bright pink (arrow).

appear to have “spread,” possibly through propagation of misfolded proteins, to neurons in the cerebral cortex. In other individuals, however, the dementia is the first presenting symptom, and the motor symptoms appear later, suggesting a descending rather than ascending pattern of disease spread.

Cortical Lewy bodies are less distinctive than those in the brainstem, but are also composed predominantly of α -synuclein. Immunohistochemical staining for α -synuclein also reveals the presence of abnormal neurites that contain aggregated protein—called *Lewy neurites* even though Lewy never saw them! In this setting, the gross pathologic findings typically include depigmentation of the substantia nigra and locus ceruleus, paired with relative preservation of the cortex, hippocampus, and amygdala. The burden of cortical Lewy bodies is usually extremely low, and the mechanism by which this disease wreaks havoc on cognitive functioning is not clear. There is evidence that the burden of oligomeric α -synuclein in cortical neurons is more important in disease causation than the Lewy bodies.

Atypical Parkinsonian Syndromes

As discussed earlier, the clinical syndrome of parkinsonism, with bradykinesia and rigidity, reflects dysfunction of the dopaminergic nigrostriatal pathway. In addition to the forms of PD already discussed, there are a variety of disorders that include parkinsonism as a component of the symptoms. These diseases, in general, are minimally responsive to treatment with L-DOPA; they are also distinguished from PD through the presence of additional signs and symptoms. For these reasons, they are considered to be *atypical parkinsonian syndromes* or *Parkinson-plus syndromes*. These disorders include progressive supranuclear palsy and corticobasal degeneration, both of which are tauopathies, and multisystem atrophy, a synucleinopathy.

Progressive Supranuclear Palsy (PSP)

PSP is a tauopathy in which affected individuals develop progressive truncal rigidity, disequilibrium with frequent falls, and difficulty with voluntary eye movements. Other common symptoms include nuchal dystonia, pseudobulbar palsy, and a mild progressive dementia. The onset is usually between the fifth and seventh decades, with males affected approximately twice as frequently as females. The disease is often fatal within 5 to 7 years of onset.

Although the pathologic hallmark of PSP is the presence of tau-containing inclusions in neurons and glia, causative mutations in the tau gene have been identified in only a few cases. However, risk for sporadic PSP is linked to single nucleotide polymorphisms that map near the tau gene locus. Other risk alleles have been identified through genome-wide association studies, but how they influence development of PSP is unclear.

Corticobasal Degeneration (CBD)

CBD is a progressive tauopathy that is most often characterized by extrapyramidal rigidity, asymmetric motor disturbances (jerking movements of limbs), and impaired higher cortical function (typically in the form of apraxia). As with PSP, cognitive decline may occur, typically later in the illness. The same tau variant linked to PSP is also highly associated with CBD. Overall, CBD and PSP share many clinical and pathologic features; in general, with PSP there is a greater burden of tau-containing lesions in brainstem and deep gray matter, while in CBD the balance is shifted toward involvement of the cerebral cortex.

MORPHOLOGY

In PSP, there is widespread neuronal loss in the globus pallidus, subthalamic nucleus, substantia nigra, colliculi, periaqueductal gray matter, and dentate nucleus of the cerebellum; globose fibrillary tangles are found in these affected regions, in neurons as well as in glia. In CBD, the brain shows cortical atrophy, mainly of the motor, premotor, and anterior parietal lobes. In affected regions of cortex, there is severe loss of neurons, gliosis, and “**ballooned**” neurons. Tau immunoreactivity has been found in astrocytes (“tufted astrocytes”), oligodendrocytes (“coiled bodies”), basal ganglionic neurons, and, variably, cortical neurons. Clusters of tau-positive processes around astrocytes (“astrocytic plaques”) and the presence of tau-positive threads in gray and white matter may be the most specific pathologic findings of CBD. The substantia nigra and locus ceruleus show neuronal loss, ballooned neurons, and tangles.

Multiple System Atrophy (MSA)

MSA is a sporadic disorder that affects several functional systems in the brain and is characterized by α -synuclein inclusions in the cytoplasm of oligodendrocytes. In contrast to other degenerative diseases, which primarily affect neurons, the histopathologic changes in MSA are primarily observed in glial cells and are associated with degeneration of white matter tracts. In addition, there is accompanying neuronal degeneration, but typically without the presence of inclusions. The “multiple” in the term *multiple system atrophy* refers to three distinct neuroanatomic circuits that are commonly involved: the striatonigral circuit (leading to parkinsonism), the olivopontocerebellar circuit (leading to ataxia), and the autonomic nervous system including the central elements (leading to autonomic dysfunction, with orthostatic hypotension as a prominent component). In a given individual, one of these components may predominate at the onset of the illness, but typically the other systems become affected as MSA progresses.

Pathogenesis

As in PD, α -synuclein is the major component of the inclusions. MSA is a sporadic disease, and no causative mutations in the gene encoding α -synuclein have been identified; nonetheless, polymorphisms near this gene appear to confer increased risk. The relationship between glial cytoplasmic inclusions and disease is supported by the observation that the burden of inclusions increases as the disease progresses, although inclusions eventually disappear as cells die in the final stages. It appears that glial cytoplasmic inclusions can occur in the absence of neuronal loss, suggesting that they are the primary pathologic event; for example, glial cytoplasmic inclusions are consistently observed in the white matter projecting to and from the motor cortex. The origin of the α -synuclein in oligodendrocytes remains perplexing because this is a neuronal protein associated with synaptic vesicles. Several studies have shown that there is no up-regulation of α -synuclein expression in white matter or in oligodendrocytes in MSA. It has been suggested that oligodendrocytes may acquire α -synuclein aggregates secondarily from injured or dying neurons. When α -synuclein is present in oligodendrocytes, they become more sensitive

to oxidative stress and show impaired interaction with the extracellular matrix.

MORPHOLOGY

The pathologic findings in MSA match the clinical presentation in any particular case. In cerebellar forms, there is atrophy of the cerebellum, including the cerebellar peduncles, pons (especially the base, Fig. 28.40A), and medulla (especially the inferior olive); in parkinsonian forms, the atrophy involves the substantia nigra and striatum (especially the putamen). Autonomic symptoms are related to cell loss from the catecholaminergic nuclei of the medulla and the intermediolateral cell column of the spinal cord. Atrophic brain regions show evidence of neuronal loss as well as variable numbers of neuronal cytoplasmic and nuclear inclusions.

The diagnostic glial cytoplasmic inclusions were originally demonstrated in oligodendrocytes with silver impregnation methods and contain α -synuclein as well as ubiquitin (see Fig. 28.40B). Similar inclusions may also be found in the cytoplasm of neurons, in neuronal and glial nuclei, and in axons.

Huntington Disease (HD)

HD is an autosomal dominant disease caused by degeneration of striatal neurons and characterized by a progressive movement disorder and dementia. Jerky, hyperkinetic, sometimes dystonic movements involving all parts of the body (chorea) are characteristic; affected individuals may later develop bradykinesia and rigidity. The disease is relentlessly progressive and uniformly fatal, with an average course of about 15 years.

Pathogenesis

HD is a prototypic polyglutamine trinucleotide repeat expansion disease (Chapter 5). The gene for HD, *HTT*, located on chromosome 4p16.3, encodes a 348-kD protein known as *huntingtin*. In the first exon of the gene, there is a stretch of CAG repeats that encodes a polyglutamine region near the N terminus of the protein. Normal *HTT* genes contain 6 to 35 copies of the repeat; when the number of repeats is increased beyond this level, it is associated with disease. There is an inverse relationship between repeat number and age of onset, such that longer repeats tend to be associated with earlier onset. Repeat expansions occur during spermatogenesis, so that paternal transmission is associated with earlier onset in the next generation, a phenomenon termed *anticipation*. In contrast to many other degenerative diseases, there is no sporadic form of HD. New mutations are uncommon; most apparently sporadic cases are explained by nonpaternity, the death of a parent before the disease develops, or an unaffected father with a mild repeat expansion that enlarges to a pathogenic size during spermatogenesis.

The biologic function of normal huntingtin remains unknown, but it appears that the expansion of the polyglutamine region bestows a toxic gain-of-function on huntingtin. For this reason, various approaches to silencing expression of the mutant allele are being investigated as potential therapies. It is interesting to note that although huntingtin is expressed in all tissues of the body, the deleterious effects of mutant huntingtin occur only in selected parts of the central nervous system.

Although development of intranuclear inclusions containing huntingtin is a pathologic hallmark of HD, this process is not directly involved in cellular injury and, based on in vitro evidence, actually appears to be neuroprotective (among the

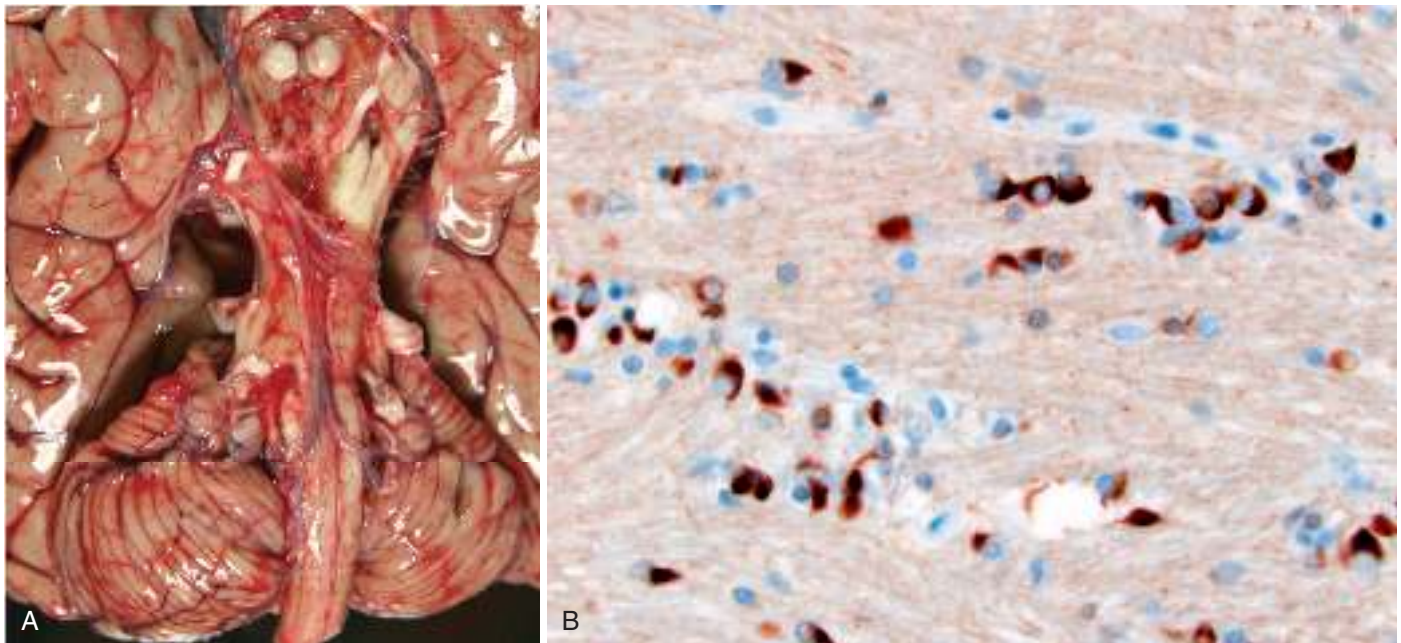


Figure 28.40 Multiple system atrophy (MSA). (A) Severe atrophy of the basis pontis in a case of MSA-C. (B) Inclusions in oligodendrocytes contain α -synuclein.

striatal neurons overexpressing mutant huntingtin, those that form inclusions survive the longest). Although the mechanism for this neuroprotective effect is not clear, it is generally thought that it stems from the sequestration of neurotoxic forms of mutated huntingtin. Transcriptional dysregulation has been implicated in HD, based on the observation that mutant forms of huntingtin bind various transcriptional regulators. Some of the transcription factors that interact with mutant huntingtin are involved in mitochondrial biogenesis and protection against oxidative injury, and their reduced activity may result in increased susceptibility to oxidative stress. Other alterations that may contribute to the pathogenesis of HD include reduced expression of the growth factor brain-derived neurotrophic factor (BDNF), and disruption of proteasomal and autophagic degradation pathways. Interestingly, emerging evidence suggests that aggregated huntingtin can be transferred between cells, another example of possible prion-like spread of a pathogenic protein.

MORPHOLOGY

The brain is small and shows striking **atrophy of the caudate nucleus and the putamen**, components of the dorsal striatum; the globus pallidus may atrophy secondarily, and the lateral and third ventricles are dilated (Fig. 28.41A). Atrophy is frequently also seen in the frontal lobe, less often in the parietal lobe, and occasionally throughout the entire cortex. On microscopic examination, there is profound loss of striatal neurons; the most marked changes are found in the caudate nucleus, especially in the tail and in portions nearer the ventricle. Pathologic changes develop in a medial-to-lateral direction in the caudate and from dorsal to ventral in the putamen. Medium-sized, spiny neurons that use γ -aminobutyric acid as their neurotransmitter, along with enkephalin, dynorphin, and substance P, are especially affected; fibrillary gliosis is more extensive than in the usual reaction to neuronal loss. There is a direct relationship between the degree of degeneration in the striatum and the severity of clinical symptoms. Protein aggregates containing huntingtin can be found in neurons in the striatum and cerebral cortex (see Fig. 28.41, inset).

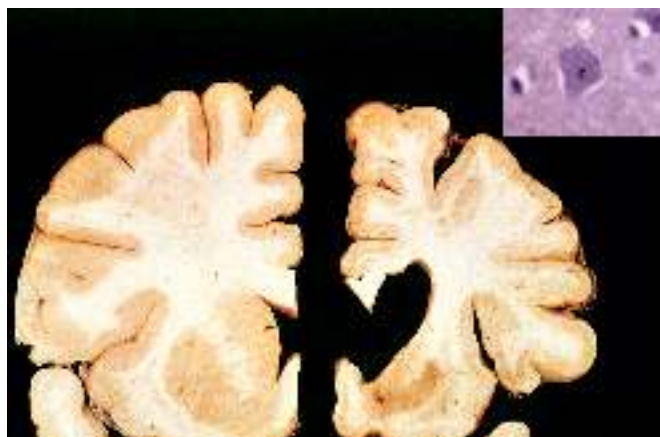


Figure 28.41 Huntington disease. Normal hemisphere on the left compared with the hemisphere with Huntington disease on the right showing atrophy of the striatum and ventricular dilation. Inset, Intranuclear inclusions in neurons are highlighted by an immunohistochemical stain for ubiquitin. (Courtesy Dr. J-P Vonsattel, Columbia University, NY.)

Clinical Features

The loss of striatal neurons, which function to dampen motor activity, results in increased motor output, often manifested as choreoathetosis. The cognitive changes associated with the disease are probably related to the neuronal loss in cerebral cortex. The age at onset is most commonly in the fourth and fifth decades and is related to the length of the CAG repeat in the *HTT* gene. Motor symptoms often precede the cognitive impairment. The movement disorder of HD is choreiform, with increased and involuntary jerky movements of all parts of the body; writhing movements of the extremities are typical. Early symptoms of higher cortical dysfunction include forgetfulness as well as cognitive and affective dysfunction, with progression to a severe dementia.

The most common immediate cause of death in HD patients is pneumonia, typically occurring at an advanced stage of the disease. Tragically, HD patients resort to suicide at a rate that is approximately twice that of the general population. Given the ability to screen for disease-causing mutations, one might assume that genetic screening of individuals at risk would be routine; however, this is a situation in which the ability to detect the likelihood of disease has surpassed any possible treatment. In the absence of effective therapy and given the devastating nature of the disease, the ethics of screening is debatable.

Spinocerebellar Degenerations

The shared feature of this group of degenerative diseases is that they cause neuronal loss and dysfunction in the cerebellum, the spinal cord, and peripheral nerves. In other regards, they are quite heterogeneous, with differences in genetics, age at onset, and signs and symptoms. Genetic analysis has impacted the classification of these diseases, but has yet to lead to clearer insight into their pathogenesis or to effective treatments.

The term *spinocerebellar ataxia* (SCA) is applied to a group of autosomal dominant inherited spinocerebellar degenerations. We will also briefly discuss Friedreich ataxia and ataxia-telangiectasia, two of the more common autosomal recessive disorders that are characterized by spinocerebellar degeneration. Finally, there is a small set of hereditary disorders characterized by episodes of ataxia or other symptoms of isolated cerebellar dysfunction, which are mostly associated with mutations in genes for ion channel subunits.

Spinocerebellar Ataxias (SCA)

This group of spinocerebellar degenerations is remarkably heterogeneous; current classifications that rely on a combination of genetic, clinical, and pathologic findings include 35 distinct subtypes. Here we focus on the common varieties of pathogenic mutations that are found in SCA.

Pathogenesis

The genes responsible for over half of the numerous SCAs have been identified. Three distinct types of pathogenic mutations are recognized:

- *Polyglutamine diseases* linked to expansion of a CAG repeat, similar to HD. Seven types of SCA are caused by this type of mutation, and in each the mutated protein

accumulates in neurons as nuclear inclusions, suggesting the involvement of pathogenic mechanisms previously discussed for HD. The group of polyglutamine diseases includes SCA1, SCA2, SCA3 (also known as *Machado-Joseph disease*), SCA6, SCA7 (relatively unique in that it includes visual impairment), SCA17, and dentatorubropallidoluysian atrophy.

- *Expansion of noncoding repeats*, similar to myotonic dystrophy. This mechanism is involved in five forms of SCA (SCA8, SCA10, SCA12, SCA31, and SCA36); in most, the genetic locus underlying the disease has been defined, and each involves a different locus. The pathogenic connection between expansion of these repeats and disease is obscure.
- *Other mutations*. The remaining SCAs are associated with deletions, insertions, and/or point mutations in a variety of genes that encode proteins involved in signal transduction (e.g., receptors, ion channels, kinases).

Friedreich Ataxia

Friedreich ataxia is an autosomal recessive disease characterized by progressive ataxia, spasticity, weakness, sensory neuropathy, and cardiomyopathy. It generally begins in the first decade of life with gait ataxia, followed by hand clumsiness and dysarthria. Deep tendon reflexes are depressed or absent, but an extensor plantar reflex is typically present. Joint position and vibratory sense are impaired, and there is sometimes loss of pain, temperature sensation, and light touch; many affected individuals develop pes cavus and kyphoscoliosis. Most patients become wheelchair-bound within about 5 years of onset, and life expectancy is typically limited to 40 or 50 years of age. The accompanying cardiomyopathy is associated with a high incidence of arrhythmias and congestive heart failure, which contribute to the deaths of most affected individuals. Concomitant diabetes is found in up to 25% of patients.

Friedreich ataxia is caused by expansion of a GAA trinucleotide repeat in the first intron of a gene on chromosome 9q13 that encodes frataxin, a protein found in the mitochondrial inner membrane, where it is involved in assembly of iron-sulfur cluster enzymes of mitochondrial complexes I and II. Affected individuals have extremely low levels of this protein, and the severity of the disease course may correlate better with the loss of frataxin than with the size of the GAA-repeat expansion. With reduced frataxin, there is decreased mitochondrial oxidative phosphorylation (similar to the defect in mitochondrial encephalomyopathies) as well as increased free iron; the presence of free iron within the mitochondria may contribute to oxidative stress. Most cases of Friedreich ataxia are associated with GAA repeat expansion in both alleles, but sometimes one allele has a repeat expansion and the other harbors a point mutation. Both of these patterns of inheritance are consistent with the disease stemming from loss of frataxin function.

Ataxia-Telangiectasia

Ataxia-telangiectasia (Chapter 7) is an autosomal recessive disorder characterized by an ataxic-dyskinetic syndrome beginning in early childhood, with the subsequent development of telangiectasias in the conjunctiva and skin, along with immunodeficiency. The ataxia-telangiectasia mutated (*ATM*) gene on chromosome 11q22-q23 encodes

a kinase with a critical role in orchestrating the cellular response to double-stranded DNA breaks. In addition to this critical cellular role, signals mediated through ATM also regulate apoptosis, maintenance of telomeres, mitochondrial homeostasis, response to oxidative stress, and the ubiquitin-proteasomal degradation system. It remains unclear which of these pathways contribute to the degenerative phenotype observed in the setting of loss of ATM protein in neurons. Intriguingly, there are comparable patterns of neurodegeneration associated with other diseases in which repair of single-stranded DNA breaks is disrupted.

Clinical Features

The disease is relentlessly progressive, with death early in the second decade. Common initial symptoms are recurrent sinopulmonary infections and unsteadiness in walking; later on, speech is noted to become dysarthric, and eye movement abnormalities develop. Many affected individuals develop lymphoid neoplasms, which are most often T-cell leukemias.

Amyotrophic Lateral Sclerosis (ALS)

ALS is a progressive disorder in which there is loss of upper motor neurons in the cerebral cortex and lower motor neurons in the spinal cord and brainstem. Loss of these neurons results in denervation of muscles, producing weakness that becomes profound as the disease progresses. The disease has an overall incidence of about 2 per 100,000 people, affects men slightly more frequently than women, and commonly emerges in the fifth decade or later. Sporadic ALS is more common than familial ALS, which may account for up to 20% of cases.

Pathogenesis

Both sporadic and familial ALS are associated with degeneration of upper and lower motor neurons, often in association with toxic protein accumulation. Close to two dozen genetic loci have been identified as causing familial ALS, with nearly all being autosomal dominant disorders.

One of the earliest discovered hereditary forms of ALS has mutations in the gene encoding copper-zinc superoxide dismutase (*SOD1*) on chromosome 21; this variant accounts for about 20% of familial cases. Initially, identification of *SOD1* mutations suggested that neuronal injury in ALS might reflect an impaired capacity to detoxify free radicals, but it is now believed that mutations lead to an adverse gain-of-function phenotype associated with mutant *SOD1* protein. It appears that mutated *SOD1* protein misfolds and forms aggregates (which can include wild-type protein) that cause cellular injury through a variety of postulated mechanisms, including dysregulation of proteasomal and autophagic catabolism, disruption of axonal transport and mitochondrial function, and sequestration of other proteins within the aggregates. Accumulation of protein aggregates can eventually trigger the unfolded protein response, with subsequent initiation of apoptosis. The overall importance of protein degradation pathways is reinforced by discovery of a range of uncommon mutations in genes implicated in protein degradation that are also associated with familial ALS.

The most common mutation that gives rise to ALS and FTLN simultaneously is an expansion of a hexanucleotide

repeat in the 5'-untranslated region of a transcript of unknown function, *C9orf72*. This mutation is estimated to be the basis of up to 40% of familial ALS and a smaller fraction of what appear to be sporadic cases of ALS. Non-AUG-initiated translation (in all three reading frames) can occur from these expanded repeats, and neuronal deposits of the derived peptides have been found in the setting of the mutation; whether these novel peptide aggregates contribute to cellular injury remains obscure.

Other genetic loci that cause ALS and FTLN encode stress granule proteins with RNA-binding capacity, such as TDP-43 and FUS. The underlying link between alterations in RNA-binding proteins and manifestations of motor neuron disease remains unclear; possibly, nuclear depletion of TDP-43 results in inappropriate processing of some RNAs, while the aggregation of the protein in the cytoplasm activates the unfolded protein response common to many of the proteinopathies.

MORPHOLOGY

The anterior roots of the spinal cord are thin (Fig. 28.42A) due to loss of lower motor neuron axons, and the precentral motor gyrus in the cortex may be atrophic in especially severe cases. There is a reduction in the number of anterior horn neurons throughout the length of the spinal cord, associated with reactive gliosis. Similar findings are seen in the hypoglossal, ambiguus, and motor trigeminal cranial nerve nuclei. Remaining neurons often contain PAS-positive cytoplasmic inclusions called Bunina bodies (which appear to be remnants of autophagic vacuoles) and TDP-43-positive cytoplasmic inclusions (which are seen in sporadic cases and familial cases caused by mutations in *C9orf72* and the TDP-43 gene [*TARDBP*], but are absent in *SOD1*-mutant and *FUS*-mutant cases). Skeletal muscles innervated by the degenerated lower motor neurons show neurogenic atrophy. Loss of the upper motor neurons leads to degeneration of the corticospinal tracts, resulting in volume loss and absence of myelinated fibers, which may be particularly evident at the lower segmental levels (Fig. 28.42B).

Clinical Features

Early symptoms of ALS include asymmetric weakness of the hands, manifested as dropping of objects and difficulty performing fine-motor tasks, and cramping and spasticity of the arms and legs. As the disease progresses, muscle strength and bulk diminish, and involuntary contractions of individual motor units, termed fasciculations, occur. The disease eventually involves the respiratory muscles, leading to recurrent bouts of pneumonia. Although most affected individuals have a combination of upper and lower motor neuron involvement, other patterns are observed. The term *progressive muscular atrophy* applies to uncommon cases in which lower motor neuron involvement predominates, while *primary lateral sclerosis* refers to cases with mostly upper motor neuron involvement. In some affected individuals, degeneration of the lower brainstem cranial motor nuclei occurs early and progresses rapidly, a pattern referred to as *progressive bulbar palsy* or *bulbar ALS*. In these individuals, abnormalities of swallowing and speech dominate, and the clinical course is inexorable during a 1- or 2-year period. When bulbar involvement is less severe, about half

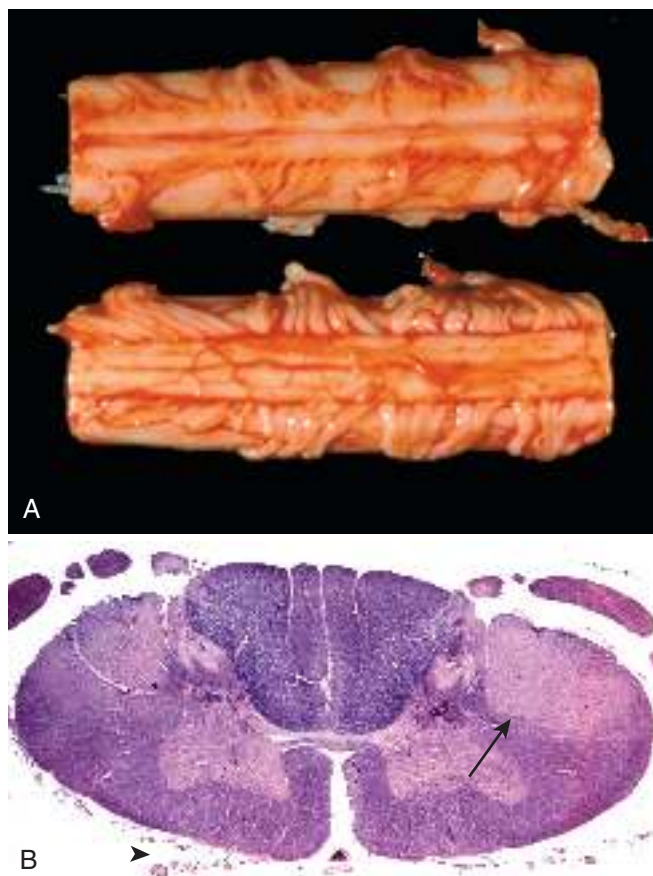


Figure 28.42 Amyotrophic lateral sclerosis. (A) Segment of spinal cord viewed from anterior (upper) and posterior (lower) surfaces showing attenuation of anterior (motor) roots compared with posterior (sensory) roots. (B) Spinal cord showing loss of myelinated fibers (lack of stain) in corticospinal tracts (best seen on the right side of this specimen; arrow) as well as degeneration of anterior roots (arrowhead).

of affected individuals are alive 2 years after diagnosis. The motor neurons innervating extra-ocular muscles are among the last to be involved in ALS; with long survival, usually associated with ventilator support, even this form of motor output fails. Familial cases develop symptoms earlier than most sporadic cases, but the clinical course is comparable.

Although ALS is considered a disease of the motor system, a significant fraction of affected individuals have evidence of more widespread cerebral cortical disease. The clinical presentation of cerebral disease is usually a frontotemporal dementia, with pathologic findings most often matching those of FTLN associated with TDP-43 inclusions. The mechanistic link between these two processes is further strengthened by the presence of TDP-43-containing inclusions in many cases of ALS as well as overlap of genetic alterations in ALS and FTLN.

Other Motor Neuron Diseases

In addition to ALS, motor neurons are the primary target in a few other neurodegenerative diseases with distinct etiology and pathogenesis.

Spinal and Bulbar Muscular Atrophy (Kennedy Disease)

This X-linked polyglutamine repeat-expansion disease is characterized by distal limb amyotrophy and bulbar signs, such as atrophy and fasciculations of the tongue and dysphagia, that are associated with degeneration of lower motor neurons in the spinal cord and brainstem. The expanded repeat occurs in the first exon of the androgen receptor and results in androgen insensitivity, gynecomastia, testicular atrophy, and oligospermia. The basis for the selective motor neuron involvement is unclear, but as in other polyglutamine expansion diseases such as HD and some forms of spinocerebellar atrophy, there are intranuclear inclusions that contain the involved protein. Progression of the disease is slow, and most affected individuals remain ambulatory until late in the disease course. Life span is normal.

Spinal Muscular Atrophy (SMA)

SMA includes a group of genetically linked disorders of childhood characterized by marked loss of lower motor neurons that results in progressive weakness. The most severe form (SMA type I, Werdnig-Hoffmann disease) has onset during the first year of life, and is usually fatal by the age of 2 years. Other forms, with later onset, have more gradual courses; in SMA type III (Kugelberg-Welander disease), motor disability usually emerges during later childhood and adolescence.

The severity of SMA is related to loss of a protein (termed SMN) that is involved in the assembly of the spliceosome. Most SMN protein comes from mRNA transcripts derived from the *SMN1* gene on chromosome 5q. There is an adjacent *SMN2* gene that differs by a few base pairs; these changes alter mRNA splicing and increase the production of SMN protein that is unstable and rapidly degraded. *SMN2* is an example of a gene that shows copy number variation, such that normal individuals may have one or multiple copies of *SMN2* on each chromosome 5. All forms of SMA are caused by loss-of-function mutations in *SMN1* (usually deletions), with the differences in clinical phenotype being determined by the number of *SMN2* gene copies. A novel gene therapy for SMA is now available that uses an adeno-associated virus to express SMN in motor neurons, albeit at a cost of over \$2 million. Genetic therapies that alter the splicing of *SMN2*-derived transcripts have also been approved for treatment of this devastating disease.

KEY CONCEPTS

NEURODEGENERATIVE DISEASES

- Neurodegenerative diseases are characterized by progressive neuronal loss involving specific neuronal circuits and brain regions. Most of these diseases are associated with accumulation of abnormal protein aggregates, typically in the form of cellular inclusions. The disease phenotype reflects the patterns of brain involvement rather than the type of inclusion.
- These diseases can be grouped by clinical presentation into dementias, hypokinetic movement disorders, hyperkinetic movement disorders, cerebellar ataxias, and motor neuron diseases.
- Prion diseases may be sporadic, familial, or transmissible. Regardless of the cause, the disease is driven by the conversion of a normal cellular protein (PrP^c) into an abnormal conformation

(PrP^{sc}), with the acquisition of distinct characteristics including relative resistance to protease digestion, self-propagation, and the ability to spread. The most common clinical presentation is that of rapidly progressive dementia.

- Among dementias, Alzheimer disease (with plaques of A β and tangles of tau) is the most common; other predominantly dementing diseases include the various forms of frontotemporal lobar degeneration (usually with either TDP-43- or tau-containing inclusions) and dementia with Lewy bodies (with α -synuclein inclusions).
- Among the hypokinetic movement disorders, Parkinson disease is the most common (again with α -synuclein-containing inclusions); other diseases with parkinsonism include progressive supranuclear palsy and corticobasal degeneration (both forms of tauopathy).
- Amyotrophic lateral sclerosis is the most common form of motor neuron disease, with diverse genetic causes as well as sporadic forms.

GENETIC METABOLIC DISEASES

Genetic disorders that disrupt metabolic processes in neurons and glia typically present early in life and are often relentlessly progressive. Some of these disorders manifest in the immediate postnatal period, while others emerge later in life. Overall, metabolic diseases with earlier onset tend to have a more rapid and aggressive clinical course.

This heterogeneous group of disorders can be organized based on the organelles or the type of cell that is affected, and by the anatomic distribution of disease, as follows:

- *Neuronal storage diseases* are predominantly autosomal recessive disorders caused by the deficiency of an enzyme or trafficking protein that is required for the catabolism of sphingolipids (including the gangliosides), mucopolysaccharides, or mucolipids.
- *Leukodystrophies* are mostly autosomal recessive disorders caused by mutations in genes encoding enzymes involved in myelin synthesis or catabolism.
- *Mitochondrial encephalomyopathies* are disorders of oxidative phosphorylation, often affecting multiple tissues including skeletal muscle (Chapter 27). These disorders may be caused by mutations in the mitochondrial or the nuclear genomes.

Neuronal Storage Diseases

These disorders are characterized by the accumulation of storage material within neurons, typically followed by neuronal death. The neurologic manifestations that result from this neuronal dysfunction or death are most often seizures and generalized loss of neurologic function. Although many of these disorders are associated with deficits of specific enzymes, others appear to be caused by defects in protein or lipid trafficking within neurons. This is a large class of disorders, with genetic heterogeneity even within clinically homogeneous entities. A few diseases in this category (Tay-Sachs and Niemann-Pick diseases, mucopolysaccharidoses, and others) are described in more detail in Chapter 5.

Leukodystrophies

These disorders are caused by mutations of genes whose products are involved in the generation, turnover, or maintenance of myelin. Normal brain function depends as much on the connections between neurons as it does on the integrity of the neurons themselves; therefore, the progressive and cumulative damage to myelinated axons in leukodystrophies has devastating consequences. Several clinical features separate these dysmyelinating leukodystrophies from demyelinating diseases such as MS: the leukodystrophies typically present with an insidious and progressive loss of cerebral function, often at younger ages, and are associated with diffuse and symmetric changes on imaging studies. Although many of the leukodystrophies are caused by single enzyme defects resulting in altered metabolism of myelin-associated lipids, a variety of other genetic alterations can lead to white matter diseases. The following examples are representative of this spectrum of disorders.

- *Krabbe disease* is an autosomal recessive leukodystrophy resulting from a deficiency of galactocerebroside β -galactosidase, the enzyme required for the catabolism of galactocerebroside to ceramide and galactose. As a consequence of the impaired catabolism of galactocerebroside in the brain, an alternative catabolic pathway shunts galactocerebroside to galactosylsphingosine; elevated levels of this compound are toxic to both oligodendrocytes and astrocytes. The clinical course is rapidly progressive, with onset of symptoms (dominated by motor signs such as stiffness and weakness) between 3 and 6 months of age; survival beyond 2 years of age is uncommon. The brain shows loss of myelin and oligodendrocytes (Fig. 28.43); a similar process affects peripheral nerves. Neurons and axons are relatively spared. A unique and diagnostic feature of Krabbe disease is the aggregation of engorged macrophages (*globoid cells*) in the brain parenchyma and around blood vessels (Fig. 28.43, inset).
- *Metachromatic leukodystrophy* is an autosomal recessive disease that results from a deficiency of the lysosomal enzyme, arylsulfatase A. This enzyme, present in a variety of tissues, cleaves the sulfate from sulfate-containing lipids (sulfatides) as the first step in their degradation. The enzyme deficiency thus leads to an accumulation of the sulfatides, especially cerebroside sulfate, that are toxic to white matter. The most striking histologic finding is demyelination with resulting gliosis. Macrophages with vacuolated cytoplasm are scattered throughout the white matter. The membrane-bound vacuoles contain complex crystalloid structures composed of sulfatides; when bound to certain dyes such as toluidine blue, sulfatides shift the absorbance spectrum of the dye, a property called *metachromasia*. Similar metachromatic material can be detected in peripheral nerves and in urine, the latter being a sensitive method of establishing the diagnosis.
- *Adrenoleukodystrophy* is an X-linked recessive disease caused by loss-of-function mutations in *ABCD1*, which encodes an ATP-binding cassette transporter protein that is required for transport of fatty acids into peroxisomes. In the typical form, young boys present with behavioral changes and adrenal insufficiency. The loss of *ABCD1* leads to an inability to catabolize very-long-chain fatty

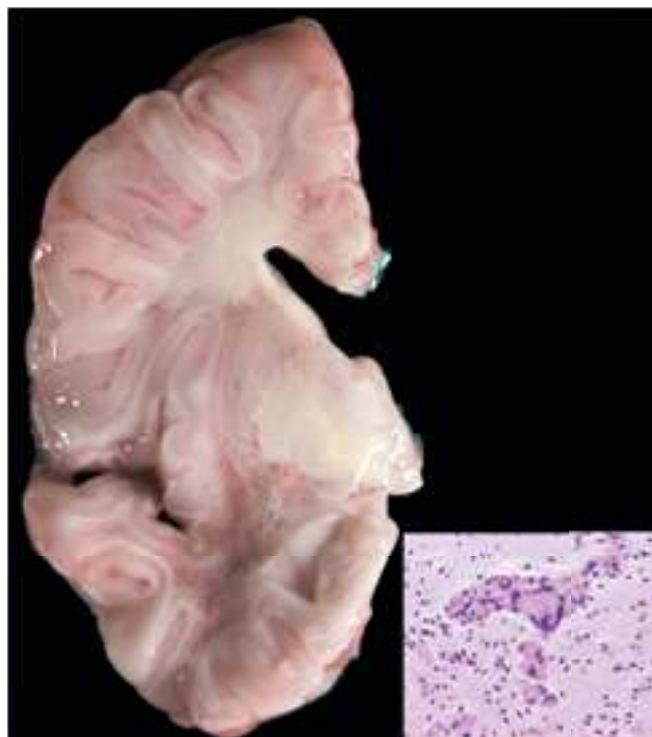


Figure 28.43 Krabbe disease. Much of the white matter is gray/yellow because of the loss of myelin. Inset, “Globoid” cells are the hallmark of the disease.

acids within peroxisomes, resulting in elevated serum levels of these lipids. The symptoms result from progressive CNS, peripheral nerve, and adrenal disease. In the white matter, myelin loss is accompanied by gliosis and extensive lymphocytic infiltration. Adrenal cortical atrophy explains the hypocortisolism.

A wide range of other leukodystrophies are recognized, some with defects in lipid metabolism. Several other mechanisms of white matter injury have been identified, including pathogenic mutations in genes that encode proteins required for myelin formation (*Pelizaeus-Merzbacher disease*), the intermediate filament protein GFAP (*Alexander disease*), or the translation initiation factor eIF2B (*vanishing white matter leukoencephalopathy*).

Mitochondrial Encephalomyopathies

Disorders of energy generation can cause a range of neurologic disease, often in association with abnormalities in other tissues. Although many of the inherited disorders of mitochondrial oxidative phosphorylation present as muscle diseases (Chapter 27), the critical dependence of neurons on oxidative phosphorylation for generation of ATP is reflected in frequent CNS involvement.

The mitochondrial genome, which is entirely inherited from the mother, encodes only 13 proteins, 22 tRNAs, and two rRNAs. The remaining proteins that are required for mitochondrial structure and function are encoded by the nuclear genome. Thus, some mitochondrial disorders show maternal transmission because the affected genes lie in the mitochondrial genome, but many others do not. There is a

complex genotype-phenotype relationship in these disorders: the same mutation may manifest as different phenotypes, and the same phenotype may be caused by different mutations.

A critical mitochondrial disease concept is *heteroplasmy*, defined as the presence of both wild type and mutated mitochondrial genomes within different populations of mitochondria in single cells. Because mitochondria are distributed in a more or less random fashion when “mother cells” divide, daughter cells may inherit different proportions of functional and dysfunctional mitochondria, leading to wide cell-to-cell variation in disease expression. In general, mitochondrial disorders in the CNS selectively target neurons; the disruption of energy generation is often reflected in elevated tissue lactate levels, which can be demonstrated by spectroscopic imaging methods. At the histologic level, there can be loss of enzymatic activity of cytochrome C oxidase, which can be assessed with special histochemical stains.

Some of the better recognized mitochondrial encephalopathies include the following entities:

- *Mitochondrial encephalomyopathy, lactic acidosis, and strokelike episodes (MELAS)* is characterized by recurrent episodes of acute neurologic dysfunction, cognitive changes, muscle weakness, and lactic acidosis. The strokelike episodes are often associated with reversible deficits that do not correspond to specific vascular territories. Pathologically, areas of infarction are observed, sometimes with vascular proliferation and focal calcification. The most common mutation observed in MELAS is in the gene encoding mitochondrial tRNA-leucine (*MTTL1*).
- *Myoclonic epilepsy with ragged red fibers (MERRF)* is a maternally transmitted disease in which affected individuals have myoclonus (a seizure disorder) and a myopathy that is characterized by ragged red fibers on muscle biopsy (Chapter 27). Ataxia, associated with neuronal loss from the cerebellar system, is also common. Most cases of MERRF are associated with mutations in mitochondrial tRNA genes.
- *Leigh syndrome* is a disease of infancy characterized by lactic acidemia, arrest of psychomotor development, feeding problems, seizures, extraocular palsies, and weakness with hypotonia; death usually occurs within 1 to 2 years. On histologic examination, there are multifocal regions of destruction of brain tissue associated with a spongiform appearance and vascular proliferation. Brainstem nuclei, the thalamus, and the hypothalamus are typically involved, usually in a symmetric manner. A wide spectrum of nuclear and mitochondrial DNA mutations has been identified.

KEY CONCEPTS

GENETIC METABOLIC DISEASES

- Mutations that disrupt metabolic or synthetic pathways can affect the nervous system. These pathways can involve general cellular processes or those that are relatively specific to the nervous system.
- Diseases with earlier onset are typically more severe in the degree of damage and pace of illness.
- Neuronal storage diseases are usually autosomal recessive disorders. The characteristic finding is accumulation of material within neurons, along with evidence of neuronal death.

- Leukodystrophies are also typically autosomal recessive disorders associated with disruption of the synthesis or turnover of myelin components.
- Mitochondrial encephalomyopathies are a pleiotropic set of disorders that involves neurons and tissues outside of the nervous system. These can be caused by mutations in the nuclear or mitochondrial genomes.

TOXIC AND ACQUIRED METABOLIC DISEASES

Toxic and acquired metabolic diseases are relatively common causes of neurologic illnesses. These diseases are discussed in Chapter 9; only aspects that are relevant to CNS pathology are presented here.

Vitamin Deficiencies

Neural tissue has a very high metabolic demand and is thus mainly affected by deficiencies of B vitamins, which play a key role in cellular metabolism; two distinct clinicopathologic syndromes are recognized.

Thiamine (Vitamin B₁) Deficiency

The classical neurologic disease resulting from thiamine deficiency is called the *Wernicke-Korsakoff syndrome* and consists of two related manifestations. *Wernicke encephalopathy* is characterized by acute psychosis and ophthalmoplegia. These symptoms are reversible when treated with thiamine. However, if unrecognized and untreated, they may be followed by a prolonged and largely irreversible condition (*Korsakoff syndrome*) that is marked by disturbances of short-term memory and confabulation. The syndrome is particularly common in the setting of chronic alcoholism, but it may also be encountered in individuals with thiamine deficiency resulting from gastric disorders, including carcinoma, chronic gastritis, or persistent vomiting.

MORPHOLOGY

In Wernicke encephalopathy, there are foci of hemorrhage and necrosis in the mamillary bodies and the walls of the third and fourth ventricles. Early lesions show dilated capillaries with prominent endothelial cells; subsequently, the capillaries become leaky, producing hemorrhagic areas. With time, there is infiltration of macrophages and development of a cystic space containing hemosiderin-laden macrophages. These chronic lesions predominate in individuals with Korsakoff syndrome. Lesions in the dorsomedial nucleus of the thalamus seem to correlate best with memory disturbance and confabulation.

Vitamin B₁₂ Deficiency

Subacute combined degeneration of the spinal cord is caused by deficiency of vitamin B₁₂ and is marked by degeneration of the ascending and descending spinal tracts. The lesions result from a defect in myelin formation; the mechanism of this defect is not known. Symptoms may present over a few weeks, initially with bilaterally symmetrical numbness,

tingling, and slight ataxia in the lower extremities, which may progress to include spastic weakness of the lower extremities. Complete paraplegia may occur, usually only later in the course. With prompt vitamin replacement therapy, clinical improvement occurs; however, once complete paraplegia has developed, recovery is poor. On microscopic examination, there is swelling of myelin layers, producing vacuoles in the affected tracts; with time, axons degenerate as well. In the early stages of the disease, the mid-thoracic level of the spinal cord is preferentially affected, from which the process may extend proximally and distally.

Neurologic Sequelae of Metabolic Disturbances

Hypoglycemia

Because the brain requires glucose and oxygen for its energy production, the cellular effects of diminished glucose resemble those of oxygen deprivation, described earlier. Some regions of the brain are more sensitive to hypoglycemia than others. Glucose deprivation initially leads to selective injury to large cortical pyramidal neurons, which, if severe, may result in pseudolaminar necrosis of the cortex, predominantly involving deep layers. The Sommer sector (area CA1) of the hippocampus is also vulnerable, as are cerebellar Purkinje cells, albeit to a lesser extent than with hypoxia. If the level and duration of hypoglycemia are of sufficient severity, there may be widespread injury to many areas of the brain.

Hyperglycemia

Hyperglycemia, most commonly associated with inadequately controlled diabetes mellitus and associated with either ketoacidosis or hyperosmolar coma, does not elicit significant morphologic changes in the brain. The affected individual becomes dehydrated and develops confusion, stupor, and eventually coma. The fluid depletion must be corrected gradually; otherwise, severe cerebral edema may follow.

Hepatic Encephalopathy

The encephalopathy found in the setting of impaired liver function (Chapter 18) is accompanied by a glial response within the CNS; critical mediators appear to include elevated ammonia levels as well as proinflammatory cytokines. Astrocytes with enlarged nuclei and minimal cytoplasm, known as Alzheimer type II cells, appear in the cerebral cortex, basal ganglia, and other subcortical gray matter regions.

Toxic Disorders

Cellular and tissue injury from toxic agents is discussed in Chapter 9. Aspects of several important toxic disorders that are of unique neurologic importance are discussed here.

Carbon Monoxide

Many of the pathologic findings that follow acute carbon monoxide exposure are the result of impaired oxygen-carrying capacity of hemoglobin; in addition, CO can interact with the heme of cytochrome C oxidase, inhibiting tissue utilization of oxygen by blocking electron transport in the mitochondria. Selective injury of the neurons in layers III and V of the cerebral cortex, Sommer sector of the hippocampus, and Purkinje cells is characteristic. A unique pattern

of bilateral globus pallidus necrosis may also occur. Demyelination of white matter tracts may be a later event.

Ethanol

The effects of acute ethanol intoxication are reversible, but chronic alcohol abuse is associated with a variety of neurologic sequelae, including Wernicke-Korsakoff syndrome from thiamine deficiency (see earlier). The toxic effects of chronic alcohol intake may be either direct or secondary to nutritional deficits. Cerebellar dysfunction occurs in about 1% of chronic alcoholics, and is associated with a clinical syndrome of truncal ataxia, unsteady gait, and nystagmus. The histologic changes are atrophy and loss of granule cells, predominantly in the superior anterior vermis (Fig. 28.44). In advanced cases, there is loss of Purkinje cells and proliferation of the adjacent astrocytes (*Bergmann gliosis*) between the depleted granular cell layer and the molecular layer of the cerebellum. The fetal alcohol syndrome is discussed in Chapter 10.

Radiation

Exposure of the brain to radiation can occur accidentally or as part of therapeutic regimens for tumors. As discussed in Chapter 9, exposure to very high doses of radiation (>10 Gy) can cause intractable nausea, confusion, convulsions, and rapid onset of coma, followed by death. Delayed effects of radiation can also present with rapidly evolving symptoms, including headaches, nausea, vomiting, and papilledema, which may appear months to years after irradiation.

The pathologic findings consist of large areas of coagulative necrosis, primarily in white matter, with all tissue elements within the area affected. This is accompanied by marked edema and gliosis in the surrounding tissue, along with vascular fibrinoid necrosis and sclerosis. The combination of radiation and methotrexate, administered either concurrently or sequentially, can act synergistically to cause tissue injury; in this setting, the pattern of injury is similar in appearance to that caused by radiation alone. Axons and cell bodies in the vicinity of the radiation-induced lesions



Figure 28.44 Alcoholic cerebellar degeneration. The anterior and superior portion of the vermis (upper portion of figure) is atrophic, with widened spaces between the folia.

undergo dystrophic mineralization. Radiation can also induce tumors, which usually develop years after radiation therapy and include sarcomas, gliomas, and meningiomas.

KEY CONCEPTS

TOXIC AND ACQUIRED METABOLIC DISEASES

- Certain vitamin deficiencies result in neurologic disease as well as systemic disorders, with thiamine and vitamin B₁₂ deficiencies being the most common.
- The metabolic demand of the CNS makes it highly susceptible to injury from hypoglycemia and carbon monoxide poisoning.
- Chronic alcohol exposure may result in several forms of brain injury; atrophy of the superior anterior cerebellar vermis is particularly characteristic.
- Metabolic disturbances may disrupt brain function, often without detectable morphologic changes.

TUMORS

The annual incidence of tumors of the CNS ranges from 10 to 17 per 100,000 persons for intracranial tumors and 1 to 2 per 100,000 persons for intraspinal tumors; the majority of these are primary tumors. Tumors of the CNS account for nearly 20% of all cancers of childhood. Seventy percent of childhood CNS tumors arise in the posterior fossa; a comparable number of tumors in adults arise within the cerebral hemispheres above the tentorium.

Although pathologists have developed classification schemes that distinguish between benign and malignant lesions on histologic grounds, the clinical course of a patient with a brain tumor is strongly influenced by patterns of growth and location. Thus, even benign tumors and low-grade malignancies can lead to serious clinical deficits and may prove fatal. Because of the capacity of diffuse gliomas to widely infiltrate the brain parenchyma, such tumors are not amenable to complete surgical resection without compromising neurologic function. Also, any CNS neoplasm, regardless of histologic grade or classification, may be lethal if situated in a critical brain region; for example, a benign posterior fossa meningioma may cause cardiorespiratory arrest by compressing vital centers in the medulla. Even the most highly malignant gliomas rarely metastasize outside the CNS. Some malignant pediatric tumors spread through the CSF when they encroach on the subarachnoid space, resulting in cerebrospinal dissemination far from the original tumor site.

Classification of tumors is one of the arts of pathology, drawing on long-recognized histologic and biologic features and newer molecular analyses. Treatment protocols and clinical trials are currently based on the 2016 World Health Organization (WHO) classification, which segregates tumors into one of four grades according to their biologic behavior, ranging from grade I to grade IV. Like most cancers, malignant brain tumors tend to progress and become more aggressive with time due to clonal evolution, a change in behavior that is often reflected in a change to a higher tumor grade (which, in this case, also means a new tumor name).

The major classes of primary brain tumors to be considered here include gliomas, neuronal/glioneuronal tumors,

embryonal tumors, and a few less common categories. In addition, meningeal tumors and familial tumor syndromes will be covered.

Gliomas

Gliomas, the most common group of primary brain tumors, include *astrocytomas*, *oligodendrogliomas*, and *ependymomas*. These tumor types have characteristic histologic features that form the basis for the classification. However, as is true of many hematologic malignancies (Chapter 13), it appears that these tumors are derived from a multipotent progenitor cell that preferentially differentiates down a particular cellular lineage. Many CNS tumors have a predilection for specific anatomic regions within the brain and tend to occur within particular age groups.

Astrocytoma

The two major categories of astrocytic tumors are the **diffusely infiltrating astrocytomas (WHO grade II to IV)** and the **more localized astrocytomas, the most common example of which is the pilocytic astrocytoma, WHO grade I**. Astrocytomas may occur from the first decade of life onward and may be found anywhere along the neuroaxis from the cerebral hemispheres to the spinal cord.

Infiltrating Astrocytomas (WHO Grades II to IV)

Infiltrating astrocytoma and glioblastoma (the synonym for “grade IV astrocytoma”) account for about 80% of primary brain tumors in adults. Usually found in the cerebral hemispheres, they may also occur in the cerebellum, brainstem, or spinal cord, most often in the fourth through sixth decades. The most common presenting signs and symptoms are seizures, headaches, and focal neurologic deficits related to the anatomic site of involvement. The degree of histologic differentiation of infiltrating astrocytomas correlates well with clinical outcome; tumors range from *diffuse astrocytoma* (grade II) to *anaplastic astrocytoma* (grade III) and *glioblastoma* (grade IV), and are further stratified based on mutations of the isocitrate dehydrogenase genes (*IDH1* or *IDH2*) into *IDH-mutant* and *IDH-wild-type* forms, the former associated with considerably better prognosis than the latter (Table 28.5). There are no WHO grade I infiltrating astrocytomas because they are considered malignant by definition.

Pathogenesis

Glioblastoma tends to occur in two different clinical settings—most commonly as a new onset disease, typically in older individuals (primary glioblastoma), and less frequently in younger patients due to progression of a lower-grade astrocytoma (secondary glioblastoma). Secondary glioblastomas and their lower-grade precursors are associated with driver mutations of *IDH1*, or (less often) its homologue *IDH2* (see Chapter 7 for a discussion of these oncometabolites and their role in tumorigenesis). In contrast, most primary glioblastomas are *IDH-wild type*, WHO grade IV; they harbor other common genetic alterations, most commonly gains of chromosome 7, loss of chromosome 10, *TERT*-promoter mutations, and *EGFR* gene amplification. These and other less common genetic and genomic alterations contribute to the acquisition of cancer hallmarks (Chapter 7). For example, most gliomas have genetic aberrations that

Table 28.5 Features of Diffuse Gliomas

	Low-Grade or Anaplastic Astrocytoma (WHO Grade II or III)	Low-Grade or Anaplastic Oligodendroglioma (WHO Grade II or III)	Glioblastoma, IDH-mut (WHO Grade IV)	Glioblastoma, IDH-wt (WHO Grade IV)
IDH status	Mutant	Mutant	Mutant	Wild-type
Other genetics	<i>TP53</i> -mut, <i>ATRX</i> -mut	1p/19q-codeleted	<i>TP53</i> -mut, <i>ATRX</i> -mut	+7/-10, <i>pTERT</i> -mut, <i>EGFR</i> -amp
Primary or secondary	Usually primary	Usually primary	Usually secondary	Usually primary
Typical morphology	Nuclear atypia; mitoses in grade III	Round nuclei, clear haloes; mitoses, MVP, and/or necrosis in grade III	Nuclear atypia, mitoses, MVP, necrosis	Nuclear atypia, mitoses, MVP, necrosis
Prognosis	OS: 5 to 15 years	OS: 10 to 20 years	OS: 2 to 4 years	OS: 6 months to 2 years

amp, Amplification; *mut*, mutant; *MVP*, microvascular proliferation; OS, overall survival time after onset of disease (average); *p*, promoter wt, wild-type; +, whole-chromosome gain; -, whole-chromosome loss.

lead to evasion of senescence (either telomerase mutations or mutations that lead to alternative lengthening of telomeres); escape from normal growth controls (biallelic *CDKN2A* deletion); activation of growth factor signaling pathways (*EGFR* or *PDGFR* gene amplification); and resistance to apoptosis (*TP53* mutation).

MORPHOLOGY

Diffuse astrocytomas are poorly defined, gray, infiltrative tumors that expand and distort involved brain (Fig. 28.45A). These tumors range in size from a few centimeters to lesions that replace nearly the entire brain. The cut surface of the tumor may be either firm or soft and gelatinous; cystic degeneration may be seen. The tumor may appear well demarcated, but is more commonly ill-defined due to infiltration beyond the perceived margins.

Microscopically, these tumors are hypercellular compared to normal white matter and feature enlarged, elongated or irregular, hyperchromatic nuclei embedded within a fibrillar background

(Fig. 28.45B) that is often GFAP-immunoreactive. An immunostain for the IDH1 R132H mutant protein highlights the tumor cells in up to 90% of cases (Fig. 28.45B, inset); *IDH1* and *IDH2* sequencing is required to identify less common pathogenic IDH mutations, and is done when the immunostain is negative. Individual tumor cells infiltrate brain tissue some distance away from the main lesion.

Anaplastic astrocytomas have a similar appearance, but are more densely cellular and have readily detectable mitotic activity.

In **glioblastomas**, variation in the appearance of the tumor from region to region is characteristic; some areas are firm and gray-white, while others are soft and yellow due to necrosis or red due to hemorrhage and hypervascularity (Fig. 28.46). The histologic appearance is similar to anaplastic astrocytoma with the additional features of necrosis and/or microvascular proliferation. Necrosis in glioblastoma often occurs in a serpentine pattern with tumor hypercellularity along the edges of the necrotic regions, a histologic pattern referred to as palisading (Fig. 28.47). The microvascular cell proliferation produces tufts of cells that pile

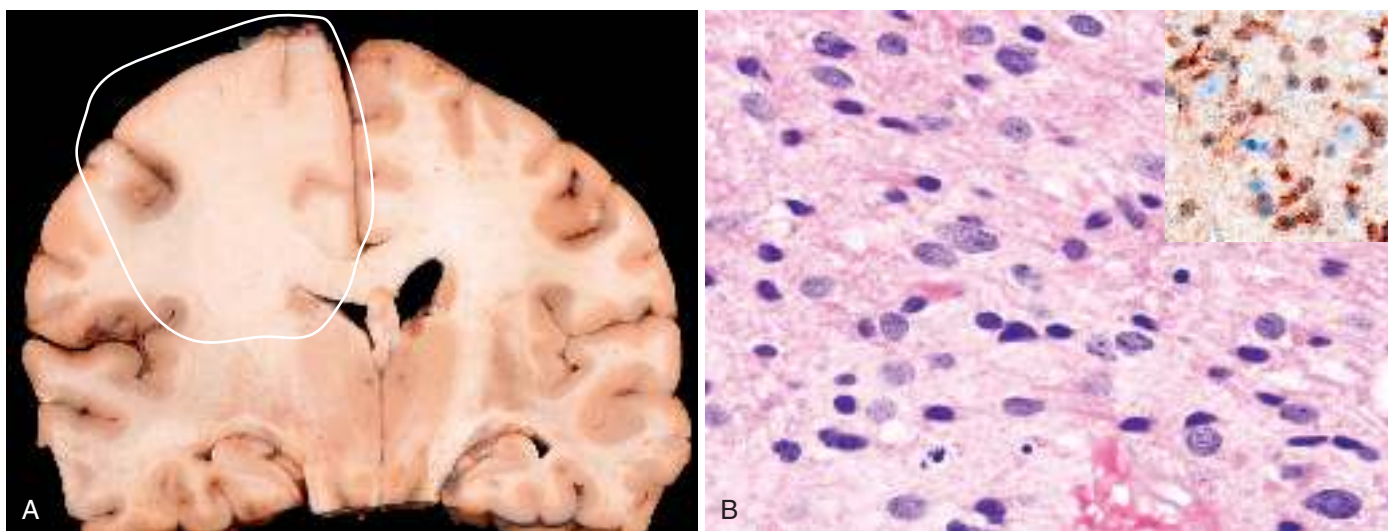


Figure 28.45 Diffuse astrocytoma. (A) On coronal section at autopsy, the left frontal white matter is expanded, and there is blurring of the corticomedullary junction due to infiltrative tumor (circled region). (B) This histologic section from the white matter shows enlarged, irregular, hyperchromatic nuclei that appear embedded within the native fibrillar matrix of the brain; the smaller round and oval nuclei are native oligodendrocytes and reactive astrocytes, respectively. *Inset*, An immunostain for IDH1 R132H is positive for mutant protein in tumor cells, some of which surround the larger immunonegative cortical neurons ("perineuronal satellitosis").

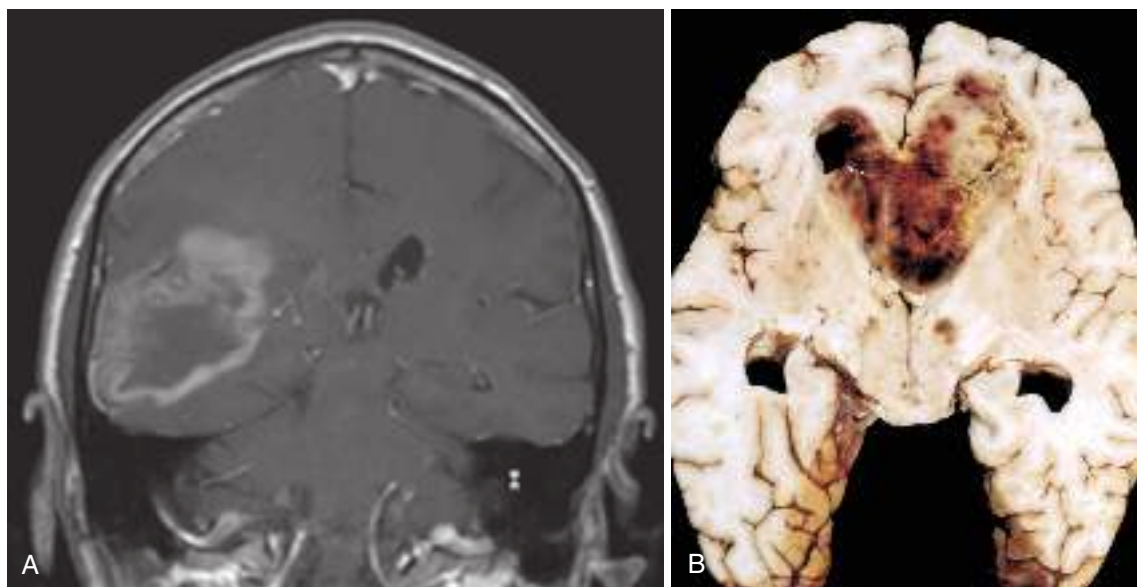


Figure 28.46 (A) Contrast T1-weighted coronal magnetic resonance image shows a large mass in the right temporal lobe with “ring” enhancement. (B) Glioblastoma appearing as a necrotic, hemorrhagic, infiltrating mass.

up and bulge into the lumen of small blood vessels (Fig. 28.47 inset); VEGF, produced by malignant astrocytes in response to hypoxia, contributes to this distinctive vascular change. Because histologic features can be extremely variable from one region to another, small biopsy specimens may not be representative, and the tumor may be undergraded.

Clinical Features

The presenting symptoms of infiltrating astrocytomas depend, in part, on the location and growth rate of the tumor. WHO grade II astrocytomas may remain stable or progress slowly; mean survival exceeds 5 years. Eventually, however, clinical deterioration occurs due to the emergence of a more rapidly growing subclone of higher histologic

grade. Radiologic studies show a non-enhancing mass and adjacent “edema,” which often includes a tumor infiltrate. High-grade astrocytomas have abnormal vessels that are “leaky,” with an abnormally permeable blood-brain barrier, and therefore demonstrate contrast enhancement, which is often ringlike in glioblastomas (Fig. 28.46A). The prognosis for individuals with IDH-wild-type glioblastoma (~90% of glioblastomas) is very poor, although the use of newer chemotherapeutic agents has provided some benefit. Epigenetic silencing of the promoter for the gene encoding the DNA repair enzyme MGMT predicts responsiveness to DNA alkylating drugs (e.g., temozolomide), as would be expected given that MGMT is critical for the repair of chemotherapy-induced DNA modifications. With current treatment, which consists of resection followed by radiation therapy and chemotherapy, the mean length of survival after diagnosis has increased to 15 months; 25% of such patients are alive after 2 years. Survival is substantially shorter in older patients and in patients with lower performance status or with large unresectable lesions. As stated earlier, the IDH-mutant glioblastomas are also aggressive but have a better prognosis overall, with an average survival of 2 to 3 years.

Oligodendroglioma

The other major subtype of infiltrating glioma is comprised of cells that resemble oligodendrocytes. When corrected for tumor grade, the oligodendrogliomas have the best prognosis among glial tumors; as with their astrocytic counterparts, they are now defined using morphologic and genetic features. These tumors constitute 5% to 15% of gliomas and are most common in the fourth and fifth decades. Patients may have had several years of neurologic complaints, often including seizures. The lesions are found mostly in the cerebral hemispheres and have a predilection for white matter.

Pathogenesis

Oligodendroglioma is molecularly defined by an IDH mutation in combination with whole-arm chromosomal codeletion

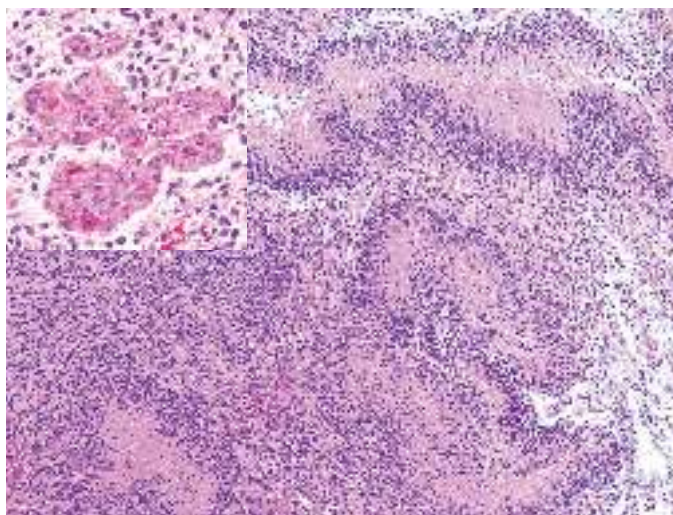


Figure 28.47 Glioblastoma. Serpiginous foci of palisading necrosis (tumor nuclei lined up around the red nucleate zones of necrosis). Inset, Microvascular proliferation.

of 1p and 19q. Additional genetic alterations occur with progression to anaplastic oligodendroglioma; the more common of these include losses of chromosome 9p, which result in deletion of the *CDKN2A* tumor suppressor gene. In contrast to high-grade astrocytic tumors, *EGFR* gene amplification is not seen. The vast majority of oligodendrogliomas feature the same *TERT*-promoter mutations as those encountered in glioblastoma.

MORPHOLOGY

Oligodendrogliomas are gelatinous, gray masses, often with cysts, focal hemorrhage, and calcification. The tumors are composed of sheets of regular cells with spherical nuclei containing finely granular chromatin (similar to normal oligodendrocytes) surrounded by a clear halo of vacuolated cytoplasm (Fig. 28.48). The tumor typically contains a delicate network of anastomosing capillaries, resembling “chicken wire.” Calcification, present in as many as 90% of these tumors, ranges from microscopic foci to massive depositions. Tumor cells that infiltrate the cerebral cortex often collect around neurons (perineuronal satellitosis). Mitotic activity and proliferation indices are low in low-grade (WHO grade II) oligodendroglioma.

Anaplastic oligodendrogliomas (WHO grade III) are characterized by a higher cell density, nuclear anaplasia, increased mitotic activity, and occasionally necrosis. Such cases may arise *de novo*, but more often progress from WHO grade II oligodendrogliomas.

Clinical Features

In general, individuals with oligodendrogliomas have a better prognosis than do those with astrocytomas. Current treatment with surgery, chemotherapy, and radiation therapy has yielded average survivals of 15 to 20 and 10 to 15 years for WHO grades II and III tumors, respectively. Progression from low- to higher-grade lesions occurs, typically over a period of 5 or more years.

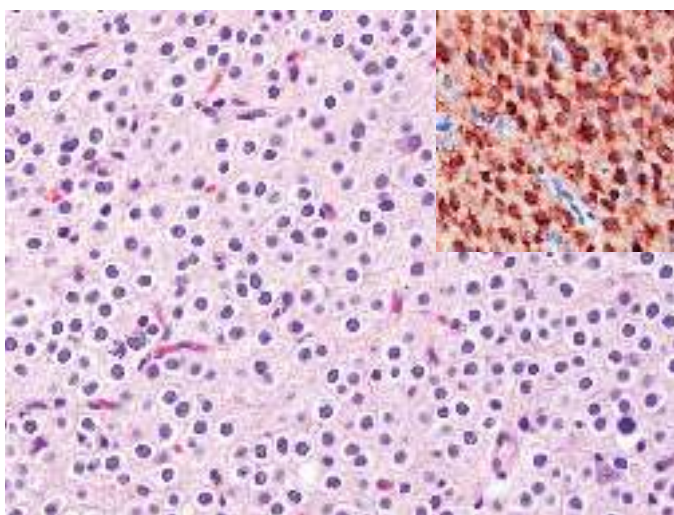


Figure 28.48 Oligodendroglioma. Tumor nuclei are round, with cleared cytoplasm forming “halos” and vasculature composed of thin-walled capillaries. Similar to diffuse astrocytoma (see Fig. 28.46B), tumor cells are usually positive for IDH1 R132H-mutant protein (inset).

Pilocytic Astrocytoma

Pilocytic astrocytomas (WHO grade I) are distinguished from infiltrative astrocytomas by their more discrete borders, unique histopathologic and molecular characteristics, and relatively benign behavior. They typically occur in children and young adults and are usually located in the cerebellum, but they may also appear in the region around the third ventricle, optic nerves, spinal cord, and occasionally in the cerebral hemispheres. The histologic separation of these tumors from other astrocytomas is supported by their relative circumscription, unique histopathologic features, and distinct genetic alterations of the MAP kinase pathway, the most common being a *KIAA1549-BRAF* gene fusion/duplication. Pilocytic astrocytomas grow very slowly and can often be treated by resection alone. Symptomatic recurrence and incompletely resected lesions may be treated with repeat surgery, chemotherapy, or targeted therapies.

MORPHOLOGY

Pilocytic astrocytoma is often radiologically and grossly well demarcated and cystic (Fig. 28.49A), with a contrast-enhancing mural nodule. Histologically, it differs from infiltrative astrocytoma of any grade by showing only limited brain invasion, with an often biphasic architecture combining loose “microcystic” and compact, densely fibrillar areas (Fig. 28.49B). The tumor is composed of bipolar cells with long, thin “hairlike” processes that are GFAP-positive and form a dense fibrillary meshwork; Rosenthal fibers (eosinophilic corkscrew-shaped inclusions) and eosinophilic granular bodies (mulberry-like inclusions) are characteristic findings (Fig. 28.49C). In contrast to diffuse gliomas, the presence of microvascular proliferation or necrosis does not imply an unfavorable prognosis.

Ependymoma

Ependymomas are tumors that most often arise in proximity to the ependyma-lined ventricular system, including the oft-obliterated central canal of the spinal cord. In the first two decades of life, they typically occur near the fourth ventricle and constitute 5% to 10% of the primary brain tumors in this age group; less commonly, they are found in the cerebral hemispheres, where some lack an obvious connection to the ventricles. In adults, the spinal cord is the most common location; tumors in this site are more frequent in the setting of neurofibromatosis type 2 (NF2).

Pathogenesis

Given the association of spinal ependymomas with NF2, it is not surprising that the *NF2* gene on chromosome 22 is commonly mutated in ependymomas arising in the spinal cord but not at other sites. Ependymomas do not share the genetic alterations that are found in infiltrating gliomas. Despite their similar morphologic appearance, ependymomas occurring at different sites (supratentorial, posterior fossa, spinal) tend to be associated with distinct sets of driver mutations.

MORPHOLOGY

Grossly and on MRI, ependymomas are solid (i.e., noninfiltrative) masses. In the posterior fossa, they typically arise from the floor of

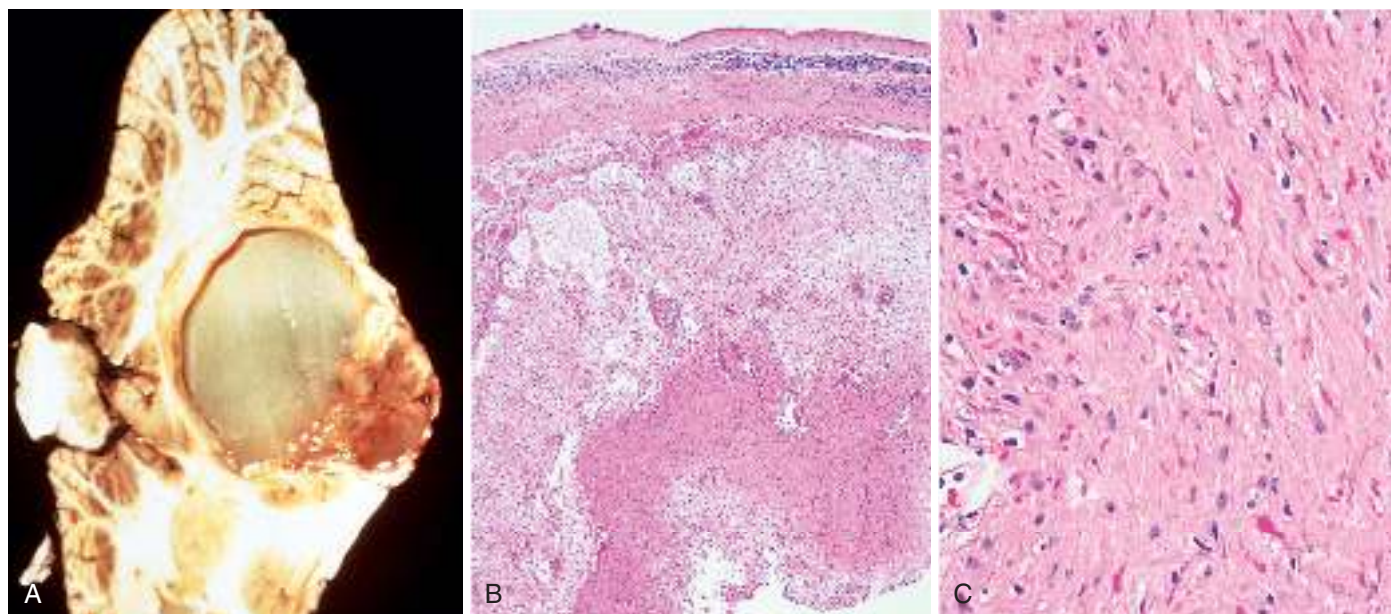


Figure 28.49 Pilocytic astrocytoma. (A) Grossly, this cerebellar tumor forms a mural nodule within a cyst. (B) At low magnification, one can appreciate atrophic cerebellar cortex (*top*), sharp circumscription, and a biphasic pattern with alternating loose (*middle*) and compact (*bottom*) regions of tumor growth. (C) Higher magnification reveals oval to irregular tumor nuclei resembling those of diffuse astrocytoma, but with numerous Rosenthal fibers (brightly eosinophilic corkscrew-shaped inclusions) in the background.

the fourth ventricle (Fig. 28.50A), occasionally growing through the foramina of Magendie or Luschka and extending below the foramen magnum into the cervical spinal canal. Although ependymomas are moderately well demarcated from adjacent brain, the proximity to vital structures often makes complete removal impossible; in the spinal cord, however, total resection is more feasible. Ependymomas are composed of cells with fibrillary processes that contain regular, round to oval nuclei with granular chromatin. Tumor cells may form glandlike round or elongated structures (rosettes, canals) that resemble the embryologic ependymal canal, with long, delicate processes extending into a lumen (Fig. 28.50B); more frequently present are **perivascular pseudorosettes** (Fig. 28.50B) in which there is a perivascular nuclear-free zone consisting

of thin ependymal processes radiating toward the central blood vessel. Although most ependymomas are well differentiated and behave as WHO grade II lesions, anaplastic ependymomas (WHO grade III) feature increased cell density, high mitotic rates, areas of palisading necrosis, and/or microvascular proliferation. However, the tumor grade is less predictive of outcome than either the extent of surgical resection or the molecular subtype.

Two other variants of ependymoma merit brief mention. **Myxopapillary ependymomas** (WHO grade I) are distinct lesions that occur in the filum terminale of the spinal cord. Cuboidal tumor cells are arranged in a variably papillary architecture around mucin-rich fibrovascular cores. Prognosis depends on completeness of surgical resection.

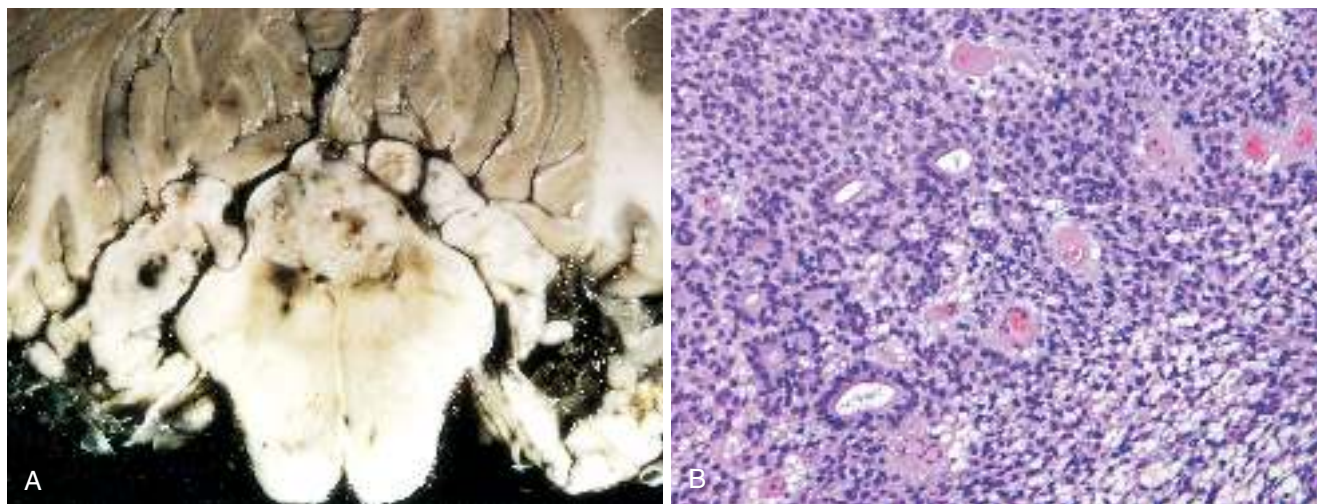


Figure 28.50 Ependymoma. (A) Tumor of the fourth ventricle, distorting, compressing, and infiltrating surrounding structures. (B) The microscopic appearance includes both true rosettes (with a glandlike central lumen) and perivascular pseudorosettes (nuclear-free zone composed of fibrillary processes radiating toward a central blood vessel).

Subependymomas (WHO grade I) are solid, sometimes calcified, slow-growing nodules that are usually found protruding into the lateral or fourth ventricle by growing under the ependyma. They are usually asymptomatic and are incidental findings at autopsy or imaging. However, if they are sufficiently large or strategically located, they can cause obstructive hydrocephalus. Subependymomas have a characteristic microscopic appearance, consisting of clusters of ependymal-appearing nuclei scattered in a dense fibrillar background of glial processes.

Clinical Features

Posterior fossa ependymomas often manifest with hydrocephalus secondary to progressive obstruction of the fourth ventricle. Despite the relationship of ependymomas to the ventricular system, CSF dissemination is uncommon. The clinical outcome for completely resected ependymomas is considerably better than for their subtotally resected counterparts.

Choroid Plexus Tumors

Both benign and malignant choroid plexus tumors are recognized, all of which are rare. *Choroid plexus papillomas* are intraventricular tumors that are most common in children, in whom they tend to occur in the lateral ventricle; in adults, they more frequently involve the fourth ventricle. These papillary growths recapitulate the structure of the normal choroid plexus, wherein epithelioid tumor cells cover fibrovascular stalks. Clinically, choroid plexus papillomas usually present with hydrocephalus due to obstruction of the ventricular system by tumor or overproduction of CSF. The far rarer *choroid plexus carcinomas* resemble adenocarcinoma; these tumors are almost always found in young children, where metastatic carcinoma is not typically encountered.

Neuronal and Glioneuronal Tumors

Far less common than glial tumors are those that exhibit neuronal differentiation. In general, neuronal tumors are more often seen in children and young adults with epilepsy.

- **Gangliogliomas** (WHO grade I) are tumors composed of a mixture of mature neuronal and glial cells. They are typically superficial lesions that present with seizures and are the most common neuronal tumors in the CNS. Most of these tumors are slow growing. When gangliogliomas present with medically refractory epilepsy, surgical resection is usually effective in controlling the seizures. Approximately 20% to 50% of these tumors have an activating mutation in the *BRAF* gene (V600E), which occasionally serves as the basis of targeted therapy. Gangliogliomas are most commonly found in the temporal lobe and often have a cystic component. The neoplastic ganglion cells are irregularly clustered and often have random orientation of processes; dysmorphic and/or binucleate forms are found. The glial component of these lesions most often resembles a pilocytic astrocytoma.
- **Dysembryoplastic neuroepithelial tumor** is a rare, benign (WHO Grade I) tumor often associated with epilepsy. It has a good prognosis following surgical resection, with

both low recurrence rates and favorable seizure control. These lesions are typically located in the superficial temporal lobe, although other sites may be seen; they usually form multiple discrete, mucin-rich, intracortical nodules of small, oligodendrocyte-like cells arranged in columns around central axonal bundles or capillaries. There are also “floating neurons” that sit in basophilic, mucin-rich pools.

Embryonal Neoplasms

A large category of mostly pediatric tumors is described as embryonal, meaning that they appear primitive or undifferentiated, although a limited degree of neuronal and less commonly glial differentiation is often encountered. The most common embryonal neoplasm is the *medulloblastoma*, which accounts for 20% of pediatric brain tumors.

Medulloblastoma

This malignant embryonal tumor occurs predominantly in children and exclusively in the cerebellum (by definition). Although histologically, molecularly, and prognostically distinct subtypes have been identified, all are considered grade IV tumors. Rapid growth may occlude the flow of CSF, leading to hydrocephalus.

Pathogenesis

Molecular subtypes of medulloblastoma have been identified through genomic studies, revealing alterations of signaling pathways involved in normal cerebellar development, such as the sonic hedgehog-patched (SHH) pathway (which is involved in control of normal proliferation of cerebellar granule cells) and the WNT/ β -catenin signaling pathway. Medulloblastomas in which the Wnt pathway is activated are often located around the anterior fourth ventricle or middle cerebellar peduncle, while sonic hedgehog-activated medulloblastomas most often involve the vermis or lateral cerebellum. On the basis of molecular alterations, medulloblastoma can be divided into four or five molecular groups that vary in terms of age of onset, tumor location, and prognosis.

MORPHOLOGY

Medulloblastoma is often well circumscribed, gray, and friable, and may extend to the surface of the cerebellar folia and involve the leptomeninges (Fig. 28.51A). On microscopic examination, the tumor is very densely cellular, with sheets of small primitive-appearing cells (see Fig. 28.51B); individual tumor cells have scant cytoplasm and hyperchromatic nuclei that are frequently elongated or crescent-shaped. Mitoses are abundant, and markers of cellular proliferation, such as Ki-67, are positive in a high percentage of the cells. In the **classic** subtype, tumor cells form **Homer Wright rosettes** (Fig. 28.51B; also seen in neuroblastoma described in Chapter 10) and often express neuronal markers such as synaptophysin; the expression of glial markers (e.g., GFAP) is less common. The **desmoplastic/nodular variant** is characterized by internodular areas of stromal response, marked by collagen and reticulin deposition (i.e., desmoplasia), and “pale islands” or nodules that have more neuropil and show greater neuronal differentiation (Fig. 28.51C). The **large cell/anaplastic variant**

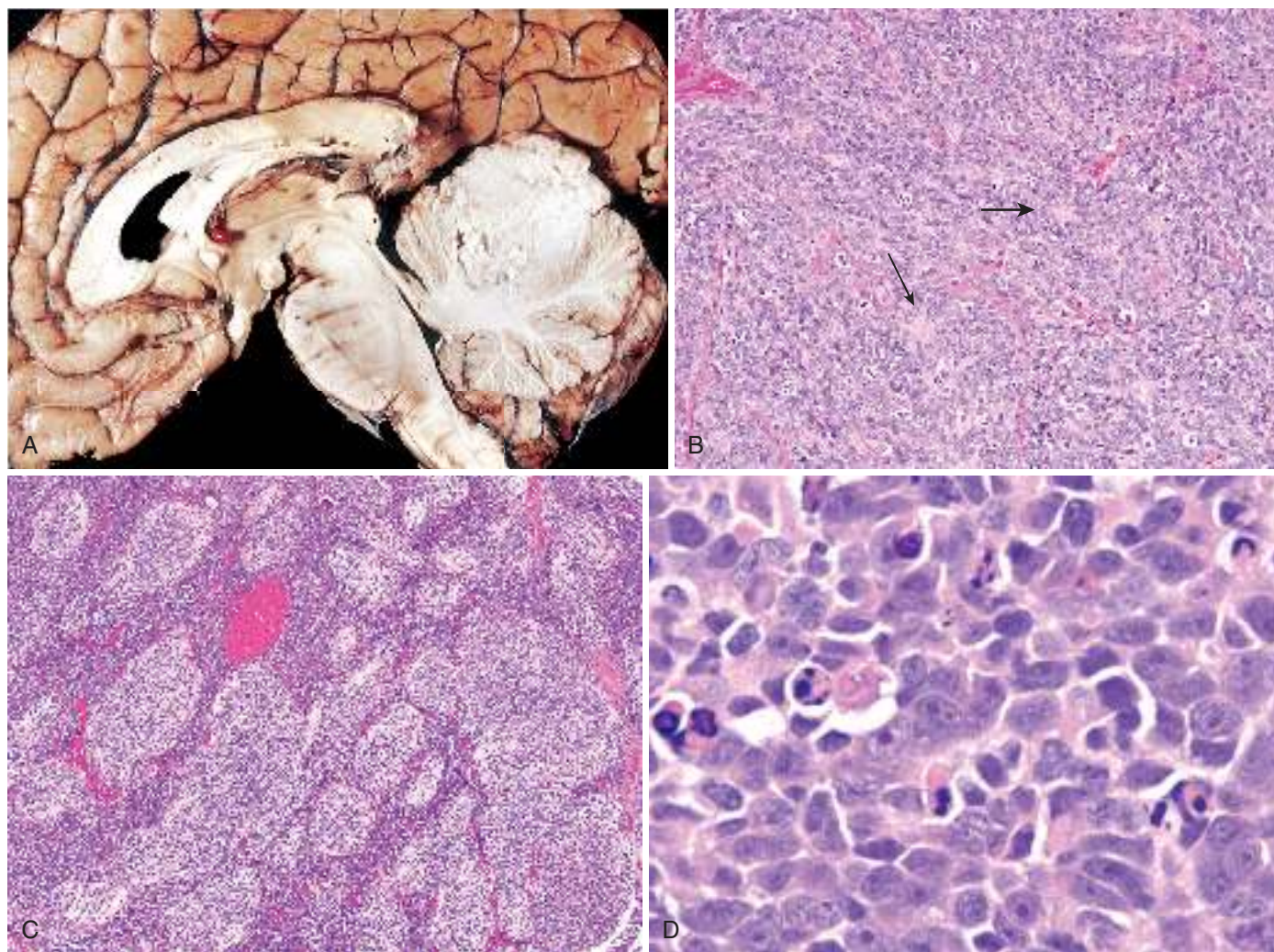


Figure 28.51 Medulloblastoma. (A) Sagittal section of brain showing medulloblastoma replacing part of the superior cerebellar vermis. (B) The microscopic appearance of classic medulloblastoma includes primitive appearing “small blue cells” that form sheets and Homer Wright (neuroblastic) rosettes with central neuropil (arrows). (C) The desmoplastic/nodular variant includes reticulin-rich internodular zones with more primitive appearing cells and “pale nodules” representing centers of partial neuronal differentiation; this variant is nearly always the SHH-activated molecular subtype. (D) The large cell/anaplastic variant features increased cell size, large cherry red nucleoli, and cell wrapping (wherein tumor cells wrap around one another or surround pyknotic nuclei of dying tumor cells).

is characterized by large and irregular vesicular nuclei, prominent nucleoli, cell wrapping (a cell-in-cell appearance), and high mitotic/apoptotic indices (Fig. 28.51D).

Medulloblastomas have a propensity to spread to the sub-arachnoid space. Dissemination through the CSF is a common complication, giving rise to either extensive growth along the surface likened to “icing,” or nodular “drop metastases,” which tend to involve the cauda equina.

Clinical Features

Medulloblastomas are highly malignant, and the prognosis for untreated patients is dismal; however, it is exquisitely radiosensitive. The molecular and histologic subtype dramatically influences prognosis, with WNT-activated medulloblastomas associated with nearly 100% survival at 5 years, whereas more aggressive subtypes fall into the 20% to 30% survival range. To account for these differences, ongoing clinical trials are exploring the option of either less

aggressive or more aggressive treatment regimens based on subtype. Similarly, targeted therapies are being explored for those with actionable molecular alterations.

Other rare embryonal neoplasms that have unique clinicopathologic and genetic features are now being recognized throughout the CNS; their classification continues to evolve.

Primary CNS Lymphoma

Primary CNS lymphoma accounts for 2% of extranodal lymphomas and 1% of intracranial tumors. It is the most common CNS neoplasm in immunosuppressed individuals, including those with AIDS and immunosuppression after transplantation. In non-immunosuppressed populations, the age spectrum is relatively wide, but the frequency increases after 60 years of age.

The term *primary* emphasizes the distinction between these lesions and secondary involvement of the CNS by

lymphoma arising elsewhere in the body (Chapter 13). Primary brain lymphoma is often multifocal and may also involve the eye, yet involvement outside of the CNS (in lymph nodes or bone marrow) is a rare and late complication. Conversely, lymphoma arising outside the CNS rarely involves the brain parenchyma; in this situation, secondary CNS involvement usually affects the meninges or the CSF, the latter sometimes diagnosed by the presence of malignant cells in a lumbar puncture specimen.

The vast majority of primary CNS tumors are diffuse large B-cell lymphomas; they are aggressive and generally have worse outcomes than tumors of comparable histology that occur at non-CNS sites (Chapter 13). In the setting of immunosuppression, the malignant B cells are usually latently infected by Epstein-Barr virus. In those who are not immunosuppressed, primary CNS large B-cell lymphomas often have amplification and overexpression of the *PDL1* gene, which encodes an important immune checkpoint protein that inhibits T-cell responses (Chapter 7). Trials of immune checkpoint inhibitors are ongoing and have produced excellent responses in some patients.

MORPHOLOGY

Primary CNS lymphomas are frequently multiple, soft gray-white masses in the deep subcortical parts of the brain; periventricular spread is common. The malignant cells infiltrate the parenchyma of the brain and accumulate around blood vessels, express B-cell markers such as CD20, and have high proliferative indices. When tumors arise in the setting of immunosuppression, necrosis is often prominent and the Epstein-Barr virus is detectable by in situ hybridization for EBERs, small nuclear RNAs that are encoded by the viral genome.

Meningiomas

Meningiomas are predominantly benign tumors of adults that arise from the meningotheial cells of the arachnoid and are usually attached to the dura. Meningiomas may

be found along any of the external surfaces of the brain as well as within the ventricular system, where they arise from the stromal arachnoid cells of the choroid plexus. Prior radiation therapy to the head and neck, typically decades earlier, is a risk factor for development of meningioma. Other tumors such as metastases, solitary fibrous tumors, and a range of poorly differentiated sarcomas may also grow as dura-based masses.

Pathogenesis

The most common cytogenetic abnormality is loss of chromosome 22, especially the long arm (22q). The deletions include the region of 22q12 that harbors the *NF2* gene, which encodes the protein merlin; as expected, meningiomas are a common lesion in the setting of *NF2* (see later). Of sporadic meningiomas, 50% to 60% harbor mutations in the *NF2* gene; these are most commonly seen in convexity tumors, including those that eventually become higher grade. In meningiomas without *NF2* mutations, the most common mutations occur in *TRAF7*, *KLF4*, *AKT1*, and *SMO*; most such tumors involve the skull base and have a lower risk of malignant progression. Higher-grade meningiomas more often have chromosomal losses involving several chromosomes, *TERT*-promoter mutations, and homozygous *CDKN2A* gene deletions.

MORPHOLOGY

Meningiomas are usually rubbery, rounded, or bosselated dural masses that compress underlying brain but are easily separated from it (Fig. 28.52A). They may also grow **en plaque**, in which the tumor spreads in a sheetlike fashion along the surface of the dura; this form is commonly associated with hyperostotic changes in the adjacent bone due to invasion, although this by itself does not signify malignancy. The lesions range from firm to finely gritty (sandlike), the latter typically due to numerous psammomatous calcifications (“psammos” is Greek for *sand*).

Most meningiomas have a relatively low risk of recurrence or aggressive growth, and so are considered WHO grade I. Many histologic patterns are observed, with most showing no prognostic

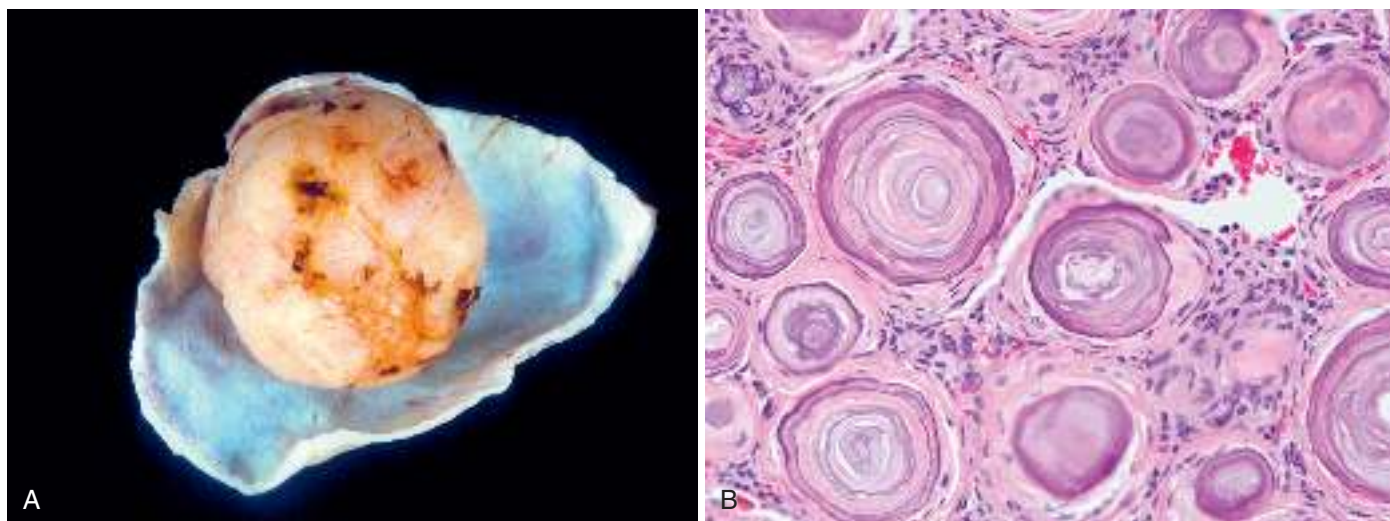


Figure 28.52 (A) Resected meningioma specimen showing rounded contour and dural attachment. (B) Meningioma with a whorled pattern of cell growth and numerous psammoma bodies (calcifications with concentric rings).

significance. The most common include **meningothelial**, containing clusters of epithelioid cells with fuzzy or indiscernible cell membranes; **fibroblastic**, with intersecting fascicles of spindled cells and abundant collagen deposition; **transitional**, with mixed meningothelial and fibroblastic features, often including many whorls (concentric wrapping of tumor cells around one another); and **psammomatous**, with a predominance of psammoma bodies, the latter representing concentric rings of calcification deposited over pre-existing whorls (see Fig. 28.52B).

Atypical meningiomas (WHO grade II) represent about one-fourth of meningiomas and are associated with higher recurrence rates and more aggressive local growth, which may require radiation therapy in addition to surgery. They are distinguished from lower-grade meningiomas by having an increased mitotic index, brain invasion, or other aggressive features. Certain histologic patterns (**clear cell** and **chordoid**) are considered grade II by definition because of their more aggressive behavior.

Anaplastic (malignant) meningioma (WHO grade III) is a highly aggressive malignancy that often resembles a carcinoma or sarcoma, but retains some evidence of meningothelial origin. They account for 1% to 3% of all meningiomas and may arise de novo or from malignant progression of lower-grade meningiomas. Mitotic rates are often markedly elevated. **Papillary** and **rhabdoid** meningiomas are rare WHO grade III variants that also feature high recurrence and mortality rates.

Clinical Features

Meningiomas are usually slow-growing tumors. Patients present either with vague nonlocalizing symptoms or with focal findings referable to compression of underlying brain. Common sites of involvement include the parasagittal aspect of the brain convexity, dura over the lateral convexity, wing of the sphenoid, olfactory groove, sella turcica, and foramen magnum. Meningiomas are uncommon in children and generally show a moderate (3:2) female predominance, although the ratio is 10:1 for spinal meningiomas (which are also commonly psammomatous). Lesions are usually solitary, but when present at multiple sites, especially in association with vestibular schwannoma and/or spinal ependymoma, the possibility of NF2 should be considered. Meningiomas often express a variety of hormone (e.g., progesterone) receptors and may grow more rapidly during pregnancy, only to regress after delivery.

Metastatic Tumors

Metastatic lesions, mostly carcinomas, account for approximately one-fourth to one-half of intracranial tumors in hospitalized patients. The five most common primary sites are lung, breast, skin (melanoma), kidney, and gastrointestinal tract, accounting for about 80% of all metastases combined. Some rare tumors (e.g., choriocarcinoma) have a high likelihood of metastasizing to the brain, whereas other more common tumors (e.g., prostatic adenocarcinoma) almost never do so. The meninges are also a frequent site of involvement by metastatic disease, where they may either form discrete masses or a more disseminated subarachnoid, meningitis-like pattern known as “meningeal carcinomatosis”; the latter is often highly challenging to treat and is associated with both cranial and spinal neuropathies from nerve root

compression. Metastatic tumors present clinically as mass lesions and may occasionally be the first manifestation of systemic cancer. In general, localized treatment of solitary brain metastases improves the quality and duration of the patient’s life.

MORPHOLOGY

Intraparenchymal metastases form sharply demarcated masses, often at the junction of gray and white matter, usually surrounded by a zone of edema. The boundary between tumor and brain parenchyma is usually well-defined microscopically, although melanoma does not always follow this rule and may be more infiltrative. The tumors typically resemble their parent tumor type, but can be very poorly differentiated in some cases; for the roughly 10% that present as a metastasis of unknown primary, immunohistochemistry is often helpful for further subtyping. Meningeal carcinomatosis, with tumor nodules studding the surface of the brain, spinal cord, and intradural nerve roots, is most commonly associated with carcinomas of the lung and breast.

Paraneoplastic Syndromes

In addition to the direct and localized effects produced by CNS metastases, peripheral neoplasms sometimes produce *paraneoplastic syndromes* that involve the peripheral and/or central nervous systems and can precede the clinical recognition of the primary tumor. A variety of paraneoplastic syndromes has been described; a shared underlying mechanism appears to be the development of an immune response against tumor antigens that cross-reacts with antigens in the central or peripheral nervous system. The spectrum of known circulating antibodies and target antigens continues to expand; illustrative CNS examples are as follows:

- *Subacute cerebellar degeneration* is associated with destruction of Purkinje cells, gliosis, and a chronic inflammatory cell infiltrate. One group of affected patients has a circulating PCA-1 antibody (anti-Yo) that recognizes cerebellar Purkinje cells; this antibody occurs predominantly in women with ovarian, uterine, or breast carcinomas.
- *Limbic encephalitis* is characterized by subacute dementia and shows perivascular inflammatory cuffs, microglial nodules, some neuronal loss, and gliosis, all of which are most evident in the anterior and medial portions of the temporal lobe; the microscopic picture resembles that of a viral encephalitis. A comparable process involving the brainstem can be seen in isolation or together with limbic system involvement. Some affected patients have a circulating ANNA-1 antibody (anti-Hu) that recognizes neuronal nuclei in the central and peripheral nervous systems; ANNA-1 is most commonly associated with small cell carcinoma of the lung. Another group of patients has a circulating antibody that recognizes the NMDA receptor and cross-reacts with hippocampal neurons. Originally identified in women with ovarian teratomas, the same clinical syndrome is now also recognized in a small proportion of patients with sporadic encephalitis. A third group of patients has a circulating VGKC-complex antibody that recognizes the voltage-gated potassium channel; this antibody may also be associated with peripheral neuropathy.

- *Eye movement disorders*, most commonly opsoclonus, may occur by themselves or in association with other evidence of cerebellar and brainstem dysfunction. In children, this is most commonly associated with neuroblastoma and is often accompanied by myoclonus.

The peripheral nervous system can also be affected by the following paraneoplastic syndromes:

- *Subacute sensory neuropathy* may be found in association with limbic encephalitis or in isolation. It is marked by loss of sensory neurons and lymphocytic inflammation in the dorsal root ganglia.
- *Lambert-Eaton myasthenic syndrome* is caused by antibodies against the voltage-gated calcium channel in the pre-synaptic elements of the neuromuscular junction. This syndrome can also be seen in the absence of malignancy.

For some paraneoplastic syndromes, there is evidence that immunotherapy (removal of circulating antibodies and immunosuppression) and tumor removal lead to clinical improvement. In general, clinical syndromes associated with plasma membrane-reactive antibodies (e.g., antibodies that recognize VGKC or NMDA receptor) respond to immunotherapy better than those associated with intracellular antigens (e.g., ANNA-1 and PCA-1).

Familial Tumor Syndromes

A number of inherited diseases is associated with increased risk of neoplasms (Chapter 7). In several of these (discussed in the following sections), tumors of the nervous system are a prominent aspect of the disease.

Neurofibromatoses and Schwannomatosis

The two forms of neurofibromatosis, NF1 and NF2, are familial autosomal dominant syndromes characterized by tumors of the peripheral and central nervous systems; in contrast, schwannomatosis involves the peripheral nervous system only and may be either familial or sporadic. NF1 is most common, with a frequency of 1 in 3000, and is characterized by neurofibromas of peripheral nerves, gliomas of the optic nerve, pigmented nodules of the iris (Lisch nodules), and cutaneous hyperpigmented macules (café au lait spots). NF2 has a frequency of 1 in 40,000 to 50,000 and is characterized by the occurrence of bilateral vestibular (cranial nerve VIII) schwannomas, multiple meningiomas, and ependymomas of the cervical spinal cord. Schwannomatosis, roughly as common as NF2, is the most recently recognized syndrome; it is defined by multiple nonvestibular schwannomas, either throughout the body or limited to one region. These disorders are discussed in greater detail in Chapter 27.

Tuberous Sclerosis Complex

Tuberous sclerosis complex (TSC) is an autosomal dominant syndrome that occurs at a frequency of approximately 1 in 6000 births. It is characterized by the development of hamartomas and benign neoplasms involving the brain and other tissues; the most frequent clinical manifestations are seizures, autism, and intellectual disability. Hamartomas within the CNS take the form of cortical tubers and subependymal nodules; subependymal giant cell astrocytomas are benign neoplasms that appear to develop from the

hamartomatous nodules in the same location. Cortical tubers are often epileptogenic, and surgical resection can be beneficial when medical management of the seizures fails. Elsewhere in the body, renal angiomyolipomas, retinal glial hamartomas, pulmonary lymphangiomyomatosis, and cardiac rhabdomyomas develop during childhood and adolescence. Cysts may be found at various sites, including the liver, kidneys, and pancreas. Cutaneous lesions include angiofibromas, localized leathery thickenings (shagreen patches), hypopigmented areas (ash-leaf patches), and subungual fibromas.

One tuberous sclerosis gene (*TSC1*) is found on chromosome 9q34 and encodes a protein known as hamartin; the more commonly mutated gene (*TSC2*) is on 16p13.3 and encodes tuberlin. These two proteins form a complex that inhibits the kinase mTOR, a key regulator of protein synthesis and other aspects of anabolic metabolism. Mutations in *TSC1* or *TSC2* disrupt this control and lead to increased and unregulated mTOR activity. Of note, mTOR controls cell size, and the tumor cells associated with TSC, particularly the subependymal giant cell astrocytomas, are remarkable for having voluminous cytoplasm. Treatment is usually symptomatic, including anticonvulsant therapy for control of seizures; however, treatment with mTOR inhibitors has sometimes resulted in dramatic clinical improvement.

MORPHOLOGY

Cortical hamartomas of TSC are firm areas that, in contrast to the softer adjacent cortex, have been likened to potatoes (hence the appellation “tubers”). The tubers are composed of haphazardly arranged neurons that lack the normal laminar organization. In addition, some dysplastic cells have an appearance intermediate between glia and neurons (large vesicular nuclei with nucleoli, resembling neurons, and abundant eosinophilic cytoplasm resembling gemistocytic astrocytes); they often express intermediate filaments of both neuronal (neurofilament) and glial (GFAP) types. Similar hamartomatous features are present in the subependymal nodules, where the large astrocyte-like cells cluster beneath the ventricular surface. These small waxlike masses that bulge into the ventricular system gave rise to the term **candle-gutterings**. Subependymal giant cell astrocytoma is histologically similar, but larger; it often presents with obstructive hydrocephalus.

Von Hippel-Lindau Disease

Individuals with this autosomal dominant disease develop hemangioblastomas of the CNS; cysts that involve the pancreas, liver, and kidneys; renal cell carcinomas; and pheochromocytomas. Hemangioblastomas are most common in the cerebellum and retina, but may also occur in other CNS locations. The disease frequency is 1 in 30,000 to 40,000.

The gene associated with von Hippel-Lindau disease, called *VHL*, is a tumor-suppressor gene located on chromosome 3p25.3; it encodes a protein (VHL) that is a component of a ubiquitin ligase complex that down-regulates hypoxia-induced factor 1 (HIF-1). HIF-1 is a transcription factor that regulates expression of vascular endothelial growth factor (VEGF), erythropoietin, and other growth factors. VEGF leads to excessive vessel growth, contributing to the development of hemangioblastomas. Erythropoietin overexpression leads to polycythemia in about 10% of hemangioblastoma patients.

HIF also regulates the expression of genes that control cellular metabolism and cell growth, activities that likely contribute to tumor formation.

MORPHOLOGY

Hemangioblastomas are highly vascular neoplasms that form a mural nodule associated with a large fluid-filled cyst. The lesion consists of numerous capillary-size or somewhat larger thin-walled vessels with intervening neoplastic cells that have vacuolated, lipid-rich cytoplasm. The neoplastic stromal cells express erythropoietin, VEGF, and inhibin (a diagnostically useful marker), and are believed to be derived from an early mesenchymal progenitor cell capable of differentiating into both endothelial and hemopoietic cells.

Therapy is directed at the symptomatic neoplasms, including resection of the cerebellar hemangioblastomas and laser therapy for retinal hemangioblastomas.

KEY CONCEPTS

TUMORS

- Tumors of the CNS may arise from the brain coverings (meningiomas), the brain parenchyma (gliomas, neuronal tumors, choroid plexus tumors), and other resident CNS cell populations (primary CNS lymphoma); they can also originate elsewhere in the body (metastases).
- Even low-grade or benign tumors can have poor clinical outcomes, depending on where they occur in the brain.
- Distinct types of tumors affect specific brain regions (e.g., cerebellum for medulloblastoma) and specific age populations (medulloblastoma and pilocytic astrocytoma in pediatric age groups; glioblastoma and lymphoma in older patients).
- Glial tumors are broadly classified into astrocytomas, oligodendrogliomas, and ependymomas. Increasing tumor malignancy is associated with more cytologic anaplasia, increased cell density, necrosis, and mitotic activity. Diagnostic, prognostic, and/or predictive genetic alterations have recently been recognized in some of these tumors.
- Metastatic spread of brain tumors to other regions of the body is rare, but the brain is a common recipient of metastatic disease from systemic malignancies. Carcinomas are most common.

ACKNOWLEDGMENT

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The Eye

29

Robert Folberg

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**The End-Stage Eye: Phthisis
Bulbi 1328**

Although this chapter comes at the end of the book, it is not an afterthought. Vision is a major quality-of-life issue. Before the public awareness of acquired immunodeficiency syndrome (AIDS) and Alzheimer disease, the most feared disease among Americans was cancer, and the second most feared disease was blindness. So great is the fear of blindness that even today, people often tell their physicians, "Doctor, I'd rather be dead than be blind!"

In general, diseases that produce loss of vision do not attract as much attention as do many of the life-threatening conditions described in this book. For example, age-related macular degeneration (AMD) is the most common cause of irreversible visual loss in the United States. Most individuals with AMD do not suffer from a total loss of vision—an immersion into darkness. The histopathology is unimpressive: small scars develop in the macula. But consider the effect of these tiny scars in a retired schoolteacher with AMD. The central portion of her or his vision is lost. The faces of spouse or grandchildren are not visible. He or she cannot read a book or newspaper. Once a model of independence, this teacher can no longer drive a car and must

be chauffeured everywhere. In short, this person is robbed of the common joys that most of us take for granted.

To study the eye, one needs to comprehend all that has come before. For example, the pathology of the eyelids builds on knowledge of dermatopathology (see Chapter 25), and the pathology of the retina and optic nerve extends what was learned in Chapter 28 about the brain and central nervous system. However, the study of ocular pathology does not merely repeat what has been presented thus far. **The eye provides the only site in which a physician can directly visualize a variety of microcirculatory disturbances ranging from arteriosclerosis to angiogenesis in the clinic.** Although there are conditions that are unique to the eye (e.g., cataract and glaucoma), many ocular conditions share similarities with disease processes elsewhere in the body that are modified by the unique structure and function of the eye (Fig. 29.1).

In recent years, the elucidation of the molecular pathogenesis of disease has been translated rapidly to therapeutic applications in the eye. Many blinding conditions, such as corneal neovascularization, diabetic retinopathy, and certain

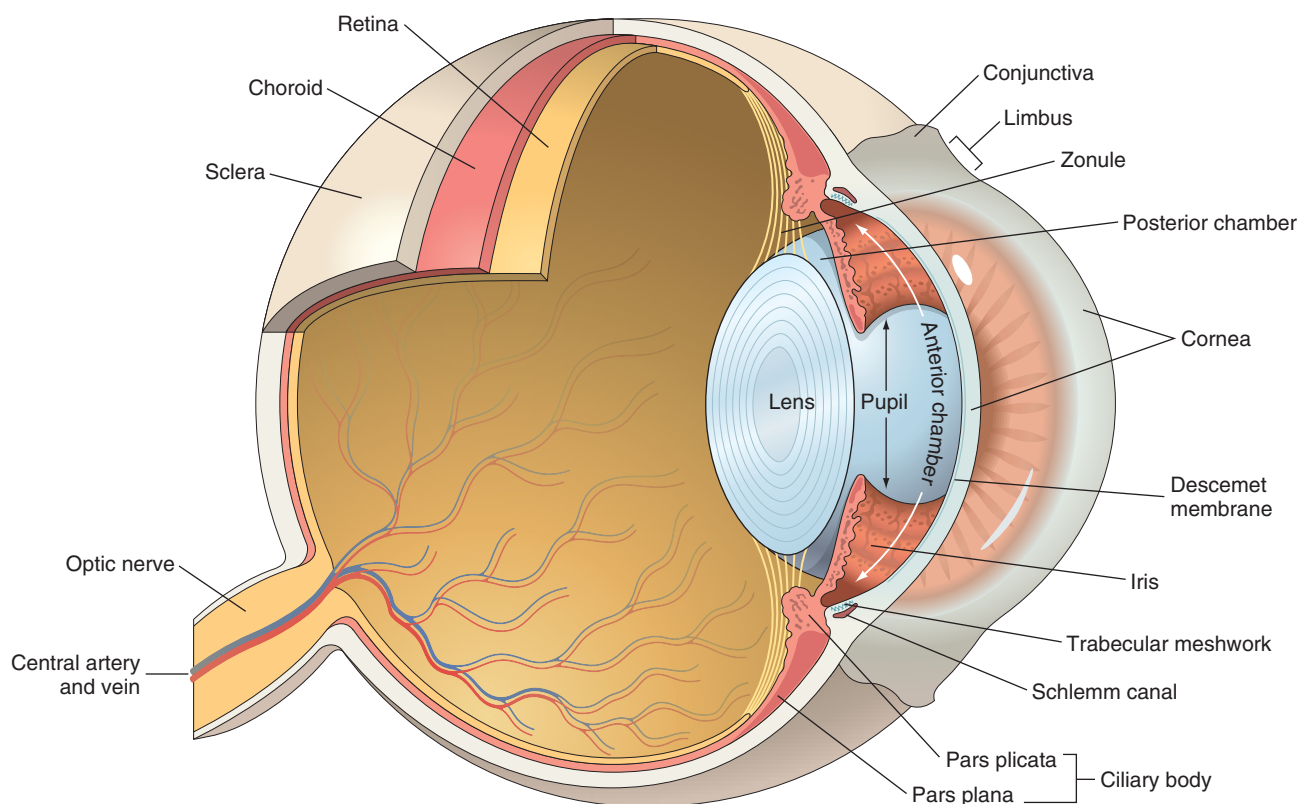


Figure 29.1 Anatomy of the eye.

forms of age-related neovascularization, result from pathologic angiogenesis. Successful treatment of these conditions with vascular endothelial growth factor (VEGF) antagonists has saved vision in patients who might have been blinded just a few years ago.

This chapter is organized on the basis of ocular anatomy. The discussion of each region of the eye begins with anatomic and functional considerations and their impact on the understanding of ocular diseases.

ORBIT

Functional Anatomy and Proptosis

The orbit is a compartment that is closed medially, laterally, and posteriorly. Diseases that increase orbital contents therefore displace the eye forward, a condition known as *proptosis*. Aside from the obvious cosmetic concerns, the proptotic eye might not be covered completely by the eyelids, and the tear film might not be distributed evenly across the cornea. Chronic corneal exposure to air is injurious, leading to pain and predisposing to corneal ulceration and infection. Proptosis may be axial (directly forward) or positional. For example, any enlargement of the lacrimal gland from inflammation (e.g., *sarcoidosis*) or neoplasm (e.g., *lymphoma*, *pleomorphic adenoma*, or *adenoid cystic carcinoma*) produces a proptosis that displaces the eye inferiorly and medially because the lacrimal gland is positioned superotemporally within the orbit.

Masses contained within the cone formed by the horizontal rectus muscles generate axial proptosis: the eye bulges straight forward. The two most common primary tumors of the optic nerve (a tract of the central nervous system), *glioma* and *meningioma*, produce axial proptosis because the optic nerve is positioned within the muscle cone. The orbital contents are subject to the same disease processes that affect other tissues. Representative inflammatory conditions and neoplasms of the orbit are discussed briefly next.

Thyroid Ophthalmopathy (Graves Disease)

In the chapter on endocrine disorders (Chapter 24) it was noted that axial proptosis is an important clinical manifestation of Graves disease. Proptosis is caused by the accumulation of extracellular matrix proteins and variable degrees of fibrosis in the rectus muscles (Fig. 29.2). The development of thyroid ophthalmopathy may, in some cases, be independent of the status of thyroid function.

Other Orbital Inflammatory Conditions

The floor of the orbit is the roof of the maxillary sinus, and the medial wall of the orbit—the lamina papyracea—separates the orbit from the ethmoidal sinuses. As a result, uncontrolled sinus infection may spread to the orbit either as an acute bacterial infection or as a component of a fungal infection. This occurs most commonly in immunosuppressed individuals, in patients with diabetic ketoacidosis, or, rarely, in persons without any predisposition. Systemic conditions such as *granulomatosis with polyangiitis* (Chapter 11) may present first in the orbit and may be confined there for



Figure 29.2 The extraocular muscles are greatly distended in this postmortem dissection of tissues from a patient with thyroid (Graves) ophthalmopathy. Note that the tendons of the muscles are spared. (Courtesy Dr. Ralph C. Eagle, Jr., Wills Eye Hospital, Philadelphia, Pa.)

prolonged periods of time, or alternatively, it may involve the orbit secondarily by extension from the sinuses.

Idiopathic orbital inflammation, also known as orbital inflammatory pseudotumor (Fig. 29.3), is another inflammatory condition affecting the orbit. This condition may be unilateral or bilateral and may affect all orbital tissue elements or may be confined to the lacrimal gland (*sclerosing dacryoadenitis*), extraocular muscles (*orbital myositis*), or Tenon capsule, the fascial layer that wraps around the eye (*posterior scleritis*). IgG4-related disease (Chapter 6) should be excluded before declaring an orbital inflammation to be idiopathic.

MORPHOLOGY

Idiopathic orbital inflammation is characterized histologically by chronic inflammation and variable degrees of fibrosis. The inflammatory infiltrate typically includes lymphocytes and plasma cells and occasionally eosinophils. Germinal centers, when present,

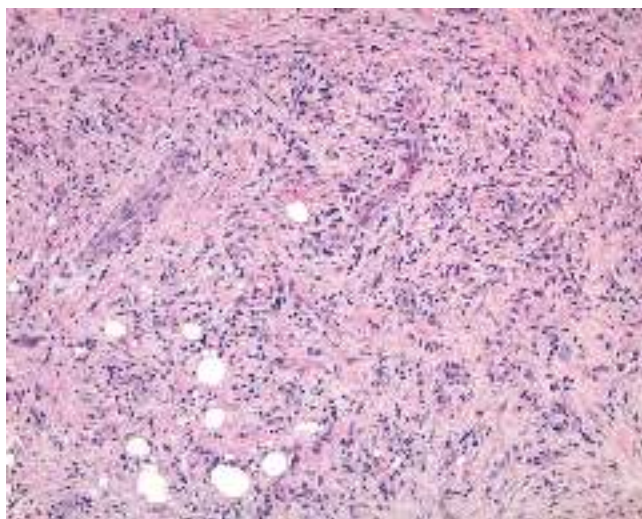


Figure 29.3 In idiopathic orbital inflammation (orbital inflammatory pseudotumor) the orbital fat is replaced by fibrosis. Note the chronic inflammation, accompanied in this case by eosinophils.

raise the suspicion of a reactive lymphoid hyperplasia. Vasculitis may be present, suggesting an underlying systemic condition. The presence of necrosis and degenerating collagen along with vasculitis should raise the suspicion of granulomatosis with polyangiitis. Idiopathic orbital inflammation is typically confined to the orbit but may develop concomitantly with sclerosing inflammation in the retroperitoneum, the mediastinum, and the thyroid, especially as a manifestation of IgG4-related disease.

Neoplasms

The most frequently encountered primary neoplasms of the orbit are vascular in origin: capillary hemangioma of infancy and early childhood and lymphangioma (both of which are unencapsulated) and encapsulated cavernous hemangioma found typically in adults. These are described in other chapters. Only a handful of orbital masses are encapsulated (e.g., pleomorphic adenoma of the lacrimal gland, dermoid cyst, neurilemmoma), and the recognition of encapsulation on imaging studies allows the surgeon to anticipate pathologic findings.

Non-Hodgkin lymphoma, like idiopathic orbital inflammation, can affect the entire orbit or can be confined to compartments of the orbit such as the lacrimal gland. Orbital lymphomas are classified according to the World Health Organization (WHO) classification system (Chapter 13).

Primary orbital malignancies may arise from any of the orbital tissues and are classified according to the scheme used for the parent tissue. For example, the lacrimal gland may be considered a minor salivary gland, and tumors of the lacrimal gland are classified as salivary gland tumors.

Metastases to the orbit may present with distinctive signs and symptoms that point to the origin of the tumor. For example, metastatic prostatic carcinoma may present clinically like idiopathic orbital inflammation; metastatic neuroblastoma and Wilms tumor—richly vascular neoplasms—may produce characteristic periocular ecchymoses. Neoplasms may also invade from the sinuses into the orbit.

KEY CONCEPTS

- Proptosis results from lesions or pathologic changes in tissue that occupy space in the orbit. The orbit is a compartment that is only open anteriorly and is closed in all other dimensions by bone.
- Inflammation in the orbit may develop by extension of local disease in adjacent tissues (e.g., sinusitis) or as a component of systemic disease (e.g., granulomatosis with polyangiitis).
- The most common primary tumors of the orbit are vascular (e.g., capillary and cavernous hemangiomas).

EYELID

Functional Anatomy

The eyelid is composed of skin externally and mucosa (the conjunctiva) on the surface apposed to the eye (Fig. 29.4). In addition to covering and protecting the eye, elements within

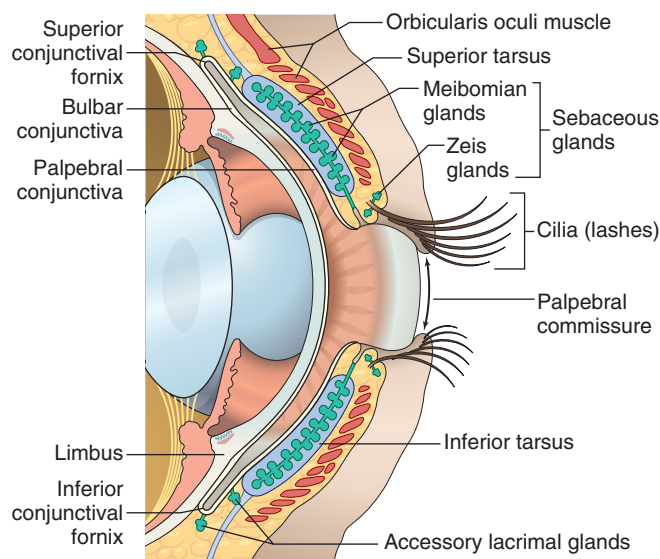


Figure 29.4 Anatomy of the conjunctiva and eyelids.

the eyelid generate critical components of the tear film. If the drainage system of the sebaceous glands is obstructed by chronic inflammation at the eyelid margin (*blepharitis*) or, less commonly, by neoplasm, lipid may extravasate into surrounding tissue and provoke a granulomatous response producing a lipogranuloma, or *chalazion*.

Neoplasms

The most common malignancy of the eyelid is basal cell carcinoma. Surprisingly, primary melanomas of the eyelid skin are extremely rare. Regardless of histogenesis, eyelid neoplasms may distort tissue and prevent the eyelids from closing completely. Because chronic exposure to air damages the cornea, prompt treatment of locally invasive basal cell carcinomas is imperative to preserve vision. Basal cell carcinoma has a distinct predilection for the lower eyelid and the medial canthus.

Sebaceous carcinoma may form a local mass that mimics *chalazion* or may diffusely thicken the eyelid. This neoplasm may also resemble inflammatory processes such as *blepharitis* or *ocular cicatricial pemphigoid* because of a predilection for intraepithelial spread as occurs in Paget disease of the nipple (Chapter 23) or vulva (Chapter 22). Sebaceous carcinoma tends to spread first to the parotid and submandibular nodes. The overall mortality rate can be as high as 22%. Sebaceous carcinoma of the eyelid is less likely to be associated with Muir-Torre syndrome than sebaceous neoplasms developing elsewhere.

MORPHOLOGY

In moderately differentiated or well-differentiated sebaceous carcinoma, vacuolization of the cytoplasm is present and helps in the diagnosis. This cancer may, however, resemble a variety of other malignancies histologically, including basal cell carcinoma; hence establishing the correct diagnosis can be difficult. Pagetoid spread (Fig. 29.5) may mimic Bowenoid actinic keratosis in the

eyelid and carcinoma in situ in the conjunctiva. Sebaceous carcinoma may spread through the conjunctival epithelium and the epidermis to the lacrimal drainage system and the nasopharynx. It may also extend into the lacrimal gland ductules and thereby into the main lacrimal gland.

In individuals with AIDS, *Kaposi sarcoma* may develop in either the eyelid or the conjunctiva. In the eyelid the lesion may appear clinically to have a purple hue because the vascular lesion is embedded in the dermis, but in the thin mucous membrane of the conjunctiva, Kaposi sarcoma appears bright red and may be confused clinically with a subconjunctival hemorrhage.

KEY CONCEPTS

- Basal cell carcinoma is the most common primary malignancy of the eyelid and may be very invasive locally.
- Sebaceous carcinoma of the eyelid, by contrast, may metastasize and is therefore a serious and potentially life-threatening condition.

CONJUNCTIVA

Functional Anatomy

The conjunctiva is divided into zones (see Fig. 29.4), each with distinctive histologic features and responses to disease. The conjunctiva lining the interior of the eyelid, the *palpebral conjunctiva*, is tightly tethered to the tarsus and may respond to inflammation by being thrown into minute papillary folds as may occur in allergic conjunctivitis and bacterial conjunctivitis. The conjunctiva in the *fornix* is a pseudostratified columnar epithelium rich in goblet cells. The fornix also contains accessory lacrimal tissue, and the ductules of the

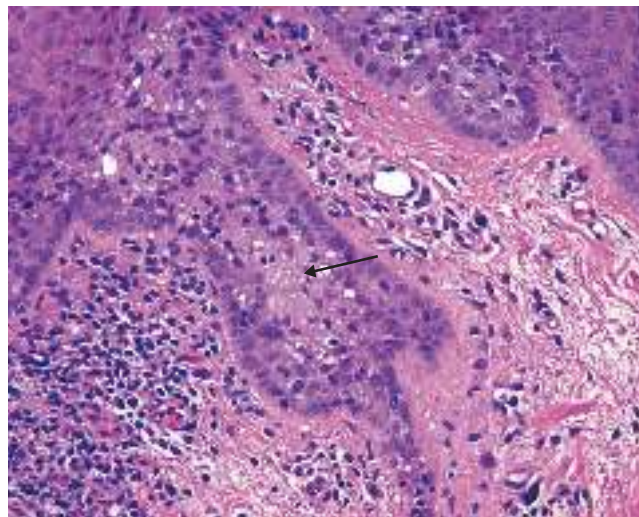


Figure 29.5 Pagetoid spread of sebaceous carcinoma. Neoplastic cells with foamy cytoplasm are present within the epidermis (arrow). Invasive sebaceous carcinoma was identified elsewhere in this biopsy sample.

main lacrimal gland pierce through the conjunctiva in the fornix superiorly and laterally. The lymphoid population of the conjunctiva is most noticeable in the fornix, and *in viral conjunctivitis, lymphoid follicles may enlarge sufficiently to be visualized clinically* by slit-lamp examination. Granulomas associated with systemic sarcoidosis may be detected in the conjunctival fornix, and the yield of granulomas from a nondirected conjunctival biopsy in individuals suspected of having sarcoid may be as high as 50%. Primary lymphoma of the conjunctiva (typically indolent marginal zone B-cell lymphoma) is most likely to develop in the fornix. The *bulbar conjunctiva*—the conjunctiva that covers the surface of the eye—is a nonkeratinizing stratified squamous epithelium. The limbus, the intersection between the sclera and cornea, also marks the transition between conjunctival and corneal epithelium (see Fig. 29.1).

The conjunctiva, like the eyelid, is richly invested with lymphatic channels. Malignant neoplasms arising in the eyelid and conjunctiva tend to spread to regional lymph nodes (parotid and submandibular node groups).

Conjunctival Scarring

Many cases of bacterial or viral conjunctivitis cause redness and itching, but most heal without sequelae. However, infection with *Chlamydia trachomatis* (trachoma) may produce significant conjunctival scarring. Conjunctival scarring is also seen after exposure of the ocular surface to caustic alkalis or as a sequela to ocular cicatricial *pemphigoid* (Chapter 25). A reduction in the number of goblet cells due to conjunctival scarring leads to a decrease in surface mucin, which is essential for the adherence of the aqueous component of tears to the corneal epithelium. Thus, even if the aqueous component of the tear film is adequate, the affected individual will suffer from a dry eye. More commonly, however, dry eye results from a deficiency in the aqueous component of the tear film generated by the accessory lacrimal glands embedded within the eyelid and fornix.

The conjunctiva may be scarred iatrogenically through reaction to drugs or as a consequence of surgery. In other parts of the body, cancer surgery requires excision of the lesion with a margin of normal tissue to ensure complete removal. However, extensive surgical excision of even diseased conjunctiva can remove a large number of goblet cells or compromise lacrimal gland ductules that traverse the conjunctiva. Thus, removal of a conjunctival neoplasm or a precursor lesion may leave the affected individual with a painful dry eye that can compromise vision. Therefore, surgeons often remove only the invasive components of conjunctival neoplasms and treat the intraepithelial components with tissue-sparing modalities such as cryotherapy or topical chemotherapy delivered as eyedrops.

Pinguecula and Pterygium

Both pinguecula and pterygium appear as submucosal elevations on the conjunctiva. They result from actinic damage and are therefore located in the sun-exposed regions of the conjunctiva (i.e., in the fissure between both the upper and the lower eyelids—the interpalpebral fissure). Pterygium typically originates in the conjunctiva

astride the limbus. It is formed by a submucosal growth of *fibrovascular connective tissue that migrates onto the cornea*, dissecting into the plane occupied normally by the Bowman layer. Pterygium does not cross the pupillary axis and, aside from the possible induction of mild astigmatism, does not pose a threat to vision. Although most pterygia are entirely benign, it is worthwhile submitting the excised tissue for pathologic examination because, on occasion, precursors of actinic-induced neoplasms—squamous cell carcinoma and melanoma—are detected in these lesions. Pinguecula, which, like pterygium, appears astride the limbus, is a small, yellowish submucosal elevation.

Neoplasms

Both squamous neoplasms and melanocytic neoplasms and their precursors tend to develop at the limbus. Conjunctival *squamous cell carcinoma* may be preceded by intraepithelial neoplastic changes analogous to those seen in the evolution of cervical squamous cell carcinoma. In the conjunctiva the spectrum of changes from mild dysplasia through carcinoma in situ is designated as *ocular surface squamous neoplasia*. Squamous papillomas and conjunctival intraepithelial neoplasia may be associated with the presence of human papillomavirus types 16 and 18.

Conjunctival nevi are encountered commonly but seldom invade the cornea or appear in the fornix or over the palpebral conjunctiva. Pigmented lesions in these zones of the conjunctiva most likely represent melanomas or melanoma precursors. Compound nevi of the conjunctiva characteristically contain subepithelial cysts lined by surface epithelium (Fig. 29.6A, B). In late childhood or adolescence, conjunctival nevi may acquire an inflammatory component rich in lymphocytes, plasma cells, and eosinophils. The resultant *inflamed juvenile nevus* is completely benign.

Conjunctival melanomas are unilateral neoplasms, typically affecting fair-complexioned individuals in middle age (Fig. 29.6C, D). Most cases of conjunctival melanoma develop through a phase of intraepithelial growth termed *primary acquired melanosis with atypia* or *conjunctival melanocytic intraepithelial neoplasia* (C-MIN). Between 50% and 90% of individuals with incompletely treated primary acquired melanosis with atypia develop conjunctival melanoma. *BRAF* V600 mutations may be identified in nearly 40% of conjunctival melanomas. Perhaps the best treatment of conjunctival melanoma is its prevention through extirpation of its precursor lesion. The lesions tend to spread first to the parotid or submandibular lymph nodes. Approximately 25% of conjunctival melanomas prove to be fatal.

KEY CONCEPTS

- Conjunctival scarring, a consequence of a variety of conditions, may result in painful loss of vision by interfering with the delivery and maintenance of the tear film.
- Many conjunctival neoplasms originate at the limbus, the seat of stem cells of the ocular surface.
- Conjunctival malignancies—especially conjunctival melanomas—tend to spread through the rich lymphatics of the conjunctiva to regional lymph nodes.

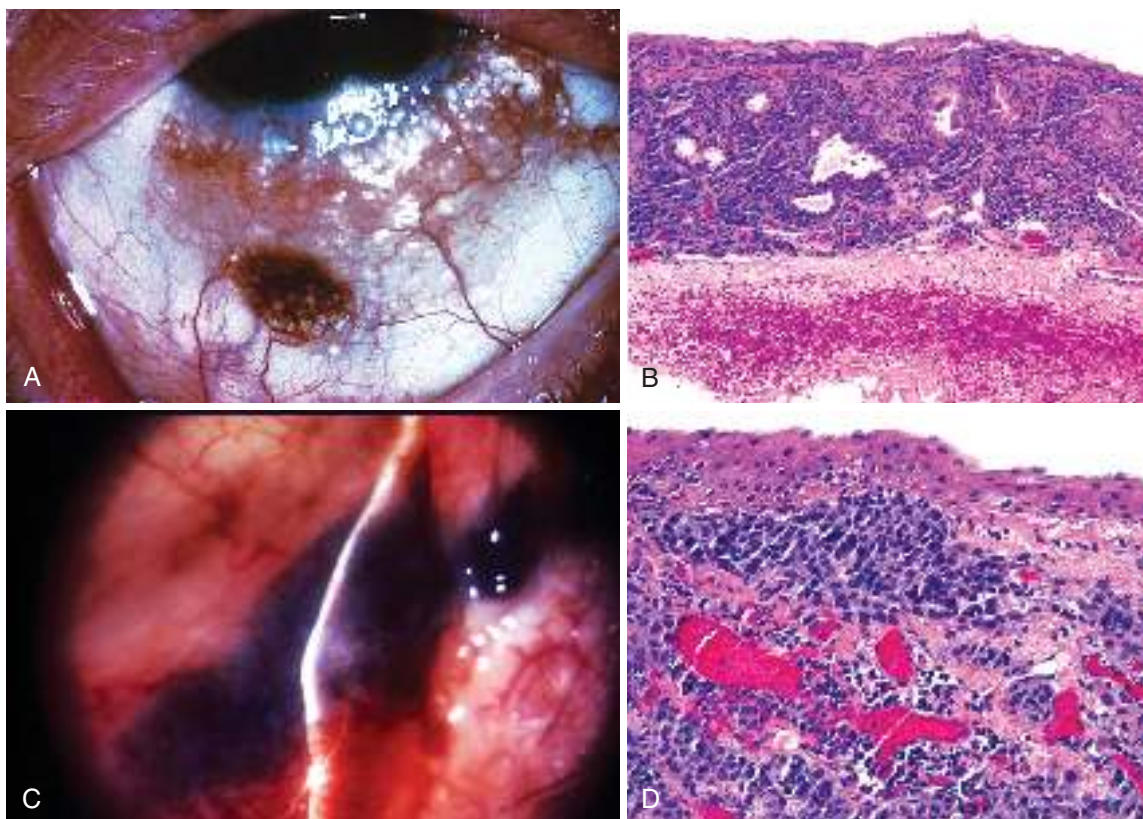


Figure 29.6 (A, B) Cystic compound nevus of the conjunctiva. (C, D) Conjunctival malignant melanoma. In C, note the deflection of the beam of the slit lamp over the surface of the lesion, indicative of invasion. (A, B, From Folberg R, et al: Benign conjunctival melanocytic lesions: clinicopathologic features, *Ophthalmology* 96:436, 1989.)

SCLERA

The sclera consists mainly of collagen and contains few blood vessels and fibroblasts; hence wounds and surgical incisions tend to heal poorly. Immune complex deposits within the sclera, such as in *rheumatoid arthritis*, may produce a necrotizing *scleritis*.

The sclera may appear “blue” in a variety of conditions. Some of these are:

- It may become thin following episodes of *scleritis*, and the normally brown color of the uvea may appear blue clinically because of the optical Tyndall effect.
- Sclera may be thinned in eyes with exceptionally high intraocular pressure and because this zone of scleral ectasia is lined by uveal tissue, the resulting lesion, known as a *staphyloma*, also appears blue.
- The sclera may appear blue in *osteogenesis imperfecta*.
- The sclera may appear blue because of a heavily pigmented congenital nevus of the underlying uvea, a condition known as *congenital melanosis oculi*. When accompanied by periocular cutaneous pigmentation, this condition is known as *nevus of Ota*.

CORNEA

Functional Anatomy

The cornea and its overlying tear film—not the lens—make up the major refractive surface of the eye (Fig. 29.7). Par-enthetically, *myopia* typically develops because the eye is too long for its refractive power, and *hyperopia* results from an eye that is too short. The popularity of procedures such as laser-assisted in situ keratomileusis (LASIK) to sculpt the cornea and change its refractive properties attests to the importance of corneal shape in contributing to the refractive power of the eye.

Anteriorly, the cornea is covered by *epithelium* that rests on a basement membrane. The *Bowman layer*, situated just beneath the epithelial basement membrane, is acellular and forms an efficient barrier against the penetration of malignant cells from the epithelium into the underlying stroma.

The *corneal stroma* lacks blood vessels and lymphatics, a feature that contributes not only to the transparency of the cornea, but also to high rate of success of corneal transplantation. Indeed, nonimmunologic graft failure

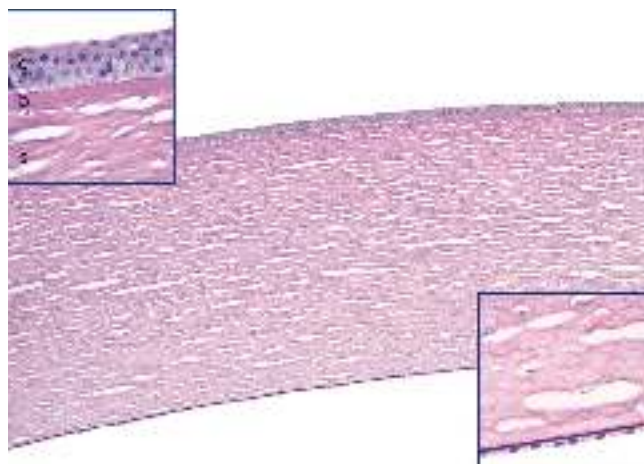


Figure 29.7 Normal corneal microarchitecture. The corneal tissue is stained by periodic acid–Schiff (PAS) to highlight basement membranes. The inset at the upper left is a high magnification of the anterior layers of the cornea: epithelium (e), Bowman layer (b), and stroma (s). A very thin PAS-positive basement membrane separates the epithelium from the Bowman layer. Note that the Bowman layer is acellular. The inset at the lower right is a high magnification of the PAS-positive Descemet membrane and the corneal endothelium. The “holes” in the stroma are artifactual spaces between parallel collagenous stromal lamellae.

(associated with loss of endothelial cells and subsequent corneal edema) is seen more commonly than is immunologic graft rejection. The risk of corneal graft rejection increases with stromal vascularization and inflammation. A precise alignment of collagen in the corneal stroma also contributes to transparency.

Corneal vascularization may accompany chronic corneal edema, inflammation, and scarring. The application of topical VEGF antagonists affords a promising approach to preventing corneal vascularization. Scarring and edema both disrupt the spatial alignment of stromal collagen and contribute to corneal opacification. Scars may result from trauma or inflammation. Normally, the corneal stroma is in a state of relative deturgescence (dehydration), maintained in large part by active pumping of fluid from the stroma back into the anterior chamber by the corneal endothelium.

The corneal *endothelium* is derived from neural crest and is not related to vascular endothelium. It rests on its basement membrane, Descemet membrane. A decrease in endothelial cells or a malfunction of endothelium results in stromal edema, which may be complicated by bullous separation of the epithelium (*bullous keratopathy*). *Descemet membrane* increases in thickness with age. It is the site of copper deposition in the Kayser–Fleischer ring of Wilson disease (Chapter 18).

Keratitis and Ulcers

Various pathogens—bacterial, fungal, viral (especially herpes simplex and herpes zoster), and protozoal (*Acanthamoeba*)—can cause corneal ulceration. In all forms of keratitis, dissolution of the corneal stroma may be accelerated by activation of collagenases within corneal epithelium and stromal fibroblasts (also known as keratocytes). Exudate

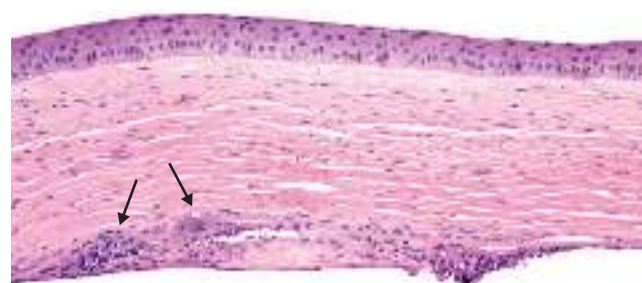


Figure 29.8 Chronic herpes simplex keratitis. The cornea is thin and scarred (note the increased number of fibroblast nuclei). Granulomatous reaction in the Descemet membrane, illustrated in this photomicrograph (arrows), is a histologic hallmark of chronic herpes simplex keratitis.

and cells leaking from iris and ciliary body vessels into the anterior chamber may be visible by slit-lamp examination and may accumulate in sufficient quantity to become visible even by a penlight examination (*hypopyon*). Although the corneal ulcer may be infectious, the hypopyon seldom contains organisms and is an example par excellence of the vascular response to acute inflammation. The specific forms of keratitis may have certain distinctive features. For example, chronic herpes simplex keratitis may be associated with a granulomatous reaction involving the Descemet membrane (Fig. 29.8).

Corneal Degenerations and Dystrophies

Ophthalmologists have traditionally divided many corneal disorders into degenerations and dystrophies. Corneal degenerations may be either unilateral or bilateral and are typically nonfamilial. By contrast, corneal dystrophies are typically bilateral and are hereditary. Corneal dystrophies may affect selective corneal layers or the changes may be distributed throughout multiple layers.

Band Keratopathies

Two types of band keratopathy serve as examples of corneal degenerations. *Calcific band keratopathy* is characterized by deposition of calcium in the Bowman layer. This condition may complicate chronic uveitis, especially in individuals with chronic juvenile rheumatoid arthritis. *Actinic band keratopathy* develops in individuals who are exposed chronically to high levels of ultraviolet light. In this condition, extensive solar elastosis develops in the superficial layers of corneal collagen in the sun-exposed interpalpebral fissure—hence the horizontally distributed band of pathology. Similar to pinguecula, the sun-damaged collagen of the cornea may take on a yellow hue to the point that this condition is sometimes erroneously called “oil-droplet keratopathy.”

Keratoconus

With an incidence of 1 in 2000, *keratoconus* is a fairly common disorder characterized by progressive thinning and ectasia of the cornea without evidence of inflammation or vascularization. Such thinning results in a cornea that has a conical rather than spherical shape. This abnormal shape generates irregular astigmatism that is difficult to correct with spectacles. Rigid contact lenses generate a smooth, spherical

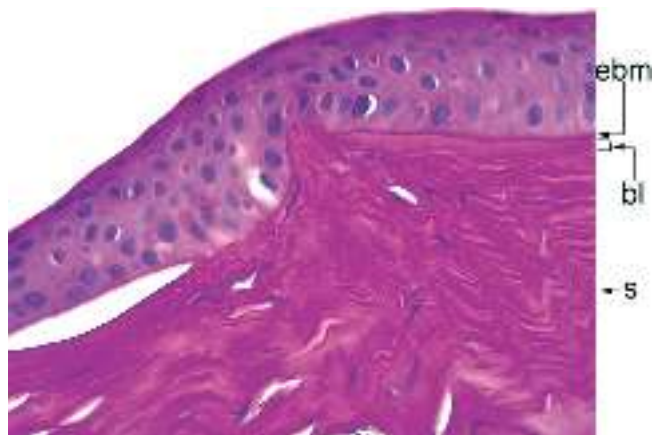


Figure 29.9 Keratoconus. The tissue section is stained by periodic acid–Schiff to highlight the epithelial basement membrane (*ebm*), which is intact; the Bowman layer (*bl*), situated between the epithelial basement membrane; and the stroma (*s*). Following the Bowman layer from the right side of the photomicrograph toward the center, there is a discontinuity, diagnostic of keratoconus. The epithelial separation just to the left of the Bowman layer discontinuity resulted from an episode of corneal hydrops, caused by a break in the Descemet membrane (not shown).

surface to the cornea and may provide refractive relief for individuals with keratoconus. Patients whose vision cannot be corrected with spectacles or contact lenses are excellent candidates for corneal transplantation, which has a high degree of success in this condition. Unlike many types of degeneration, keratoconus is typically bilateral. Keratoconus is associated with Down syndrome, Marfan syndrome, and atopic disorders. Its development may stem from a genetic predisposition superimposed by an environmental insult, such as eye rubbing in response to atopic conditions.

MORPHOLOGY

Thinning of the cornea with breaks in the Bowman layer are the histologic hallmarks of keratoconus (Fig. 29.9). In some patients the Descemet membrane may rupture precipitously, allowing the aqueous humor in the anterior chamber to gain access to the corneal stroma. The sudden effusion of aqueous humor through a gap in the Descemet membrane—corneal **hydrops**—may also cause vision to worsen suddenly. An episode of hydrops may be followed by corneal scarring that can also contribute to visual loss. Acute corneal hydrops can complicate Descemet membrane ruptures that develop secondary to extraordinary elevations of intraocular pressure in *infantile glaucoma* (*Haab striae*) or following the now uncommon obstetric forceps injury to the eye.

Fuchs Endothelial Dystrophy

Fuchs endothelial dystrophy, one of several dystrophies, results from loss of endothelial cells and the resulting edema and thickening of the stroma. It is one of the principal indications for corneal transplantation in the United States. The two major clinical manifestations of Fuchs endothelial dystrophy—*stromal edema* and *bullous keratopathy*—are both



Figure 29.10 Fuchs dystrophy. This tissue section is stained by periodic acid–Schiff to highlight the Descemet membrane, which is thick. Numerous drop-like excrescences—guttata—protrude downward from the Descemet membrane. Endothelial cell nuclei are not seen. Epithelial bullae, not shown in this micrograph, were present, reflecting corneal edema.

related to a primary loss of endothelial cells. Early in the course of the disease, endothelial cells produce drop-like deposits of abnormal basement membrane material (*guttata*) that can be visualized clinically by slit-lamp examination. With disease progression, there is a decrease in the total number of endothelial cells, and the residual cells are incapable of maintaining stromal deturgescence. Consequently the stroma becomes edematous and thickens; it acquires a ground-glass appearance clinically, and vision is blurred (Fig. 29.10). Because of chronic edema, the stroma may eventually become vascularized. On occasion the number of endothelial cells may decrease following cataract surgery even in individuals who do not have early forms of Fuchs dystrophy. This condition, known as *pseudophakic bullous keratopathy*, is also a common indication for corneal transplantation.

KEY CONCEPTS

- The cornea—not the lens—is the major refractive surface of the eye. Keratoconus is an example of a condition that distorts the contour of the cornea and alters this refractive surface, producing an irregular form of astigmatism.
- The normal cornea is avascular, a feature that contributes to transparency and the low incidence of graft rejection after corneal transplantation.
- Inflammations of the cornea may be accompanied by a noninfectious exudative process in the anterior chamber that may organize to distort anterior segment anatomy and contribute to secondary glaucoma and to cataract.
- Corneal dystrophies are generally inherited and degenerations are typically not inherited. Fuchs dystrophy and pseudophakic bullous keratopathy both produce visual loss through the final common pathway of corneal edema, and both of these conditions are leading indications for corneal transplantation in the United States.

ANTERIOR SEGMENT

Functional Anatomy

The anterior chamber is bounded anteriorly by the cornea, laterally by the trabecular meshwork, and posteriorly by the iris (Fig. 29.11). Aqueous humor, formed by the pars plicata of the ciliary body, enters the posterior chamber, bathes the lens, and circulates through the pupil to gain access to the anterior chamber. The posterior chamber lies behind the iris and in front of the lens.

The lens is a closed epithelial system; the basement membrane of the lens epithelium (known as the lens capsule) totally envelops the lens. Thus, the lens epithelium does not exfoliate like the epidermis or a mucosal epithelium. Instead, the lens epithelium and its derivative fibers accumulate within the confines of the lens capsule, thus “infolding.”

With aging, therefore, the size of the lens increases. Neoplasms of the lens have not been described.

Cataract

The term *cataract* describes lenticular opacities that may be **congenital or acquired**. Systemic diseases (e.g., galactosemia, diabetes mellitus, Wilson disease, and atopic dermatitis), drugs (especially corticosteroids), radiation, trauma, and many intraocular disorders are associated with cataract. Age-related cataract typically results from opacification of the lens nucleus (*nuclear sclerosis*). The accumulation of urochrome pigment may render the lens nucleus brown, thus distorting the individual's perception of blue color (the predominance of yellow hues in Rembrandt's paintings later in life might have been a consequence of nuclear sclerotic cataracts). Other physical changes in the lens may generate opacities. For example, the lens cortex may liquefy. Migration

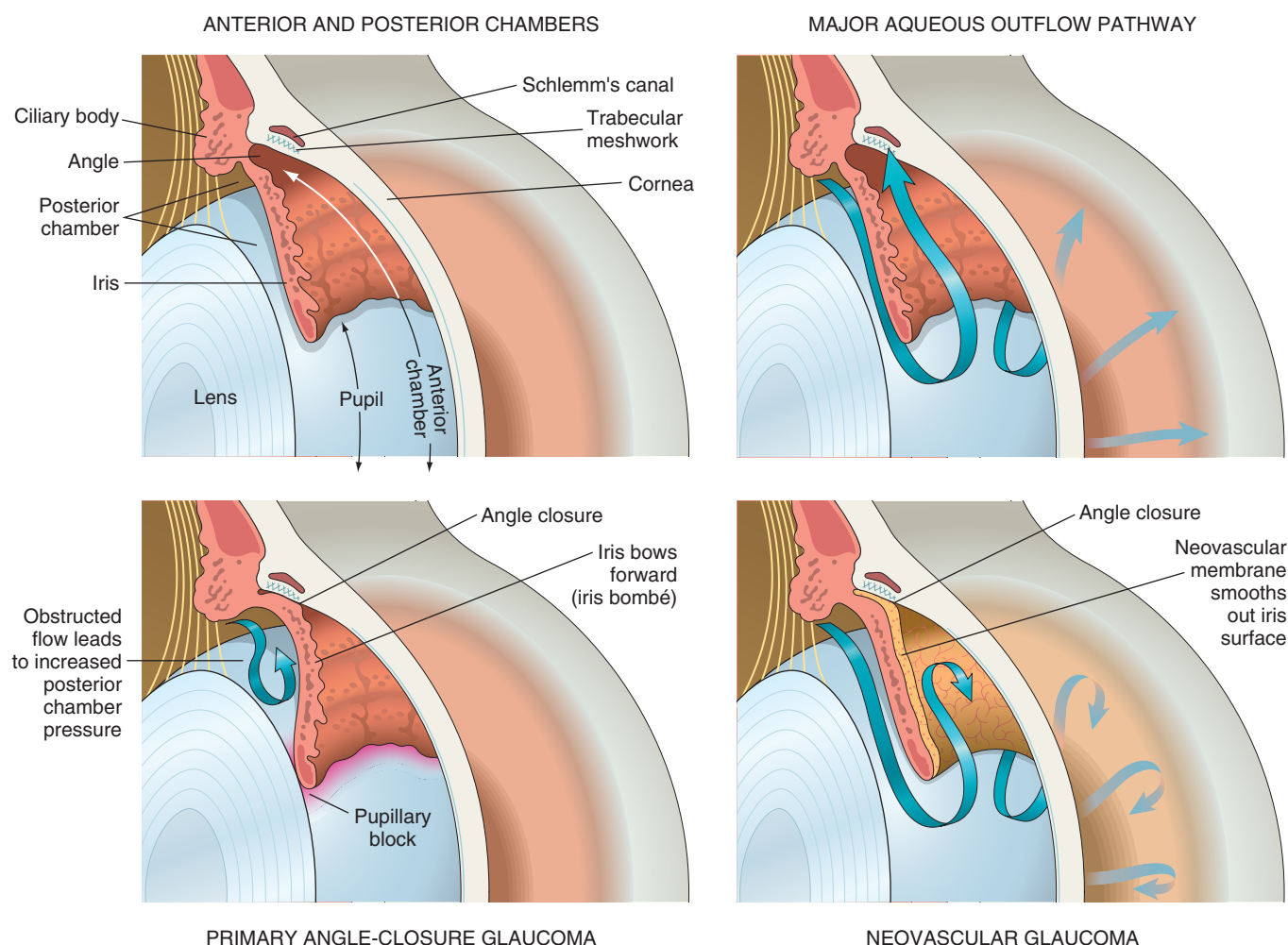


Figure 29.11 Upper left, The normal eye. Note that the surface of the iris is highly textured with crypts and folds. Upper right, The normal flow of aqueous humor. Aqueous humor, produced in the posterior chamber, flows through the pupil into the anterior chamber. The major pathway for the egress of aqueous humor is through the trabecular meshwork, into the Schlemm canal. Minor outflow pathways (uveoscleral and iris, not depicted) contribute to a limited extent to aqueous outflow. Lower left, Primary angle-closure glaucoma. In anatomically predisposed eyes, transient apposition of the iris at the pupillary margin to the lens blocks the passage of aqueous humor from the posterior chamber to the anterior chamber. Pressure builds in the posterior chamber, bowing the iris forward (iris bombé) and occluding the trabecular meshwork. Lower right, A neovascular membrane has grown over the surface of the iris, smoothing the iris folds and crypts. Myofibroblasts within the neovascular membrane cause the membrane to contract and to become apposed to the trabecular meshwork (peripheral anterior synechiae). Outflow of aqueous humor is blocked, and the intraocular pressure becomes elevated.

of the lens epithelium posterior to the lens equator may result in *posterior subcapsular cataract* secondary to enlargement of abnormally positioned lens epithelium. The technique that is most commonly used to remove opacified lenses extracts the lens contents, leaving the lens capsule intact. A prosthetic intraocular lens is typically inserted into the eye.

Anterior Segment and Glaucoma

The term *glaucoma* refers to a collection of diseases characterized by distinctive changes in the visual field and in the cup of the optic nerve. Most of the glaucomas are associated with elevated intraocular pressure, although some individuals with normal intraocular pressure may develop characteristic optic nerve and visual field changes (*normal or low-tension glaucoma*). The relationship between intraocular pressure and optic nerve damage is discussed later under Optic Nerve.

To understand the pathophysiology of glaucoma it is useful to consider the formation and drainage of aqueous humor. As Fig. 29.11 illustrates, aqueous humor is produced in the ciliary body and passes from the posterior chamber through the pupil into the anterior chamber. Although there are multiple pathways for the egress of fluid from the anterior chamber, most of the aqueous humor drains through the trabecular meshwork, situated in the angle formed by the intersection between the corneal periphery and the anterior surface of the iris. With this background, glaucoma can be classified into two major categories.

- In *open-angle glaucoma* the aqueous humor has complete physical access to the trabecular meshwork, and the elevation in intraocular pressure results from an increased resistance to aqueous outflow in the open angle.
- In *angle-closure glaucoma* the peripheral zone of the iris adheres to the trabecular meshwork and physically impedes the egress of aqueous humor from the eye.

Both open-angle and angle-closure glaucoma can be subclassified into primary and secondary types. In *primary open-angle glaucoma*, the most common form of glaucoma, the angle is open, and few changes are apparent structurally.

There are multiple causes of *secondary open-angle glaucoma*. Pseudoexfoliation glaucoma, perhaps the most common form of secondary open-angle glaucoma, is associated with the deposition of fibrillar material of varying composition throughout the anterior segment. In addition to deposition in the anterior chamber, fibrillar material is deposited around blood vessels in connective tissue and in many visceral organs such as liver, kidney, and gallbladder.

Particulate material such as high-molecular-weight lens proteins produced by phacolysis, senescent red cells after trauma (*ghost cell glaucoma*), iris epithelial pigment granules (*pigmentary glaucoma*), and necrotic tumors (*melanolytic glaucoma*) can clog the trabecular meshwork in the presence of an open angle. Elevations in the pressure on the surface of the eye (episcleral venous pressure) in the presence of an open angle also contribute to secondary open-angle glaucoma. This type of glaucoma is associated with surface ocular vascular malformations seen in *Sturge-Weber syndrome* or is a consequence of arterialization of the episcleral veins following a spontaneous or traumatic carotid-cavernous fistula.

Primary angle-closure glaucoma typically develops in eyes with shallow anterior chambers, often found in individuals with hyperopia. Transient apposition of the pupillary margin of the iris to the anterior surface of the lens may result in obstruction to the flow of aqueous humor through the pupillary aperture (*pupillary block*). Continued production of aqueous humor by the ciliary body thus elevates pressure in the posterior chamber and may bow the iris periphery forward (*iris bombé*), apposing it to the trabecular meshwork. These anatomic changes provoke a marked elevation in intraocular pressure (see Fig. 29.11). Since the crystalline lens is avascular and the lens epithelium receives its nutrition from the aqueous humor, unremitting elevation in intraocular pressure in primary angle-closure glaucoma can damage the lens epithelium. This leads to minute anterior subcapsular opacities that are visible by slit-lamp examination. Although the affected individual might have a normal complement of healthy corneal endothelial cells, sustained elevated intraocular pressure can produce corneal edema and bullous keratopathy.

There are many causes of *secondary angle-closure glaucoma*. Contraction of various types of pathologic membranes that form over the surface of the iris can draw the iris over the trabecular meshwork, occluding aqueous outflow. For example, chronic retinal ischemia is associated with the upregulation of VEGF and other proangiogenic factors. The appearance of VEGF in the aqueous humor is thought to induce the development of thin, clinically transparent fibrovascular membranes over the surface of the iris. Contraction of myofibroblastic elements in these membranes leads to occlusion of the trabecular meshwork by the iris: *neovascular glaucoma* (see Fig. 29.11). Necrotic tumors, especially retinoblastomas, can also induce iris neovascularization and glaucoma. Secondary angle-closure glaucoma may be caused by other mechanisms as well; for example, tumors in the ciliary body can mechanically compress the iris onto the trabecular meshwork, closing off the major pathway of aqueous outflow.

Endophthalmitis and Panophthalmitis

In intraocular inflammation, vessels in the ciliary body and iris become leaky, allowing cells and exudate to accumulate in the anterior chamber. These changes can be visualized with a slit lamp; at times the inflammatory cells may adhere to the corneal endothelium, forming clinically visible *keratic precipitates*. The size and shape of these precipitates can provide clues to the underlying cause of the inflammation. For example, aggregates of macrophages on the endothelium in sarcoid produce characteristic “mutton-fat” keratic precipitates.

Just as pleural exudate in acute bronchopneumonia can lead to adhesions between the visceral and parietal pleura, the presence of exudate in the anterior chamber can lead to the formation of adhesions between the iris and the trabecular meshwork or cornea (*anterior synechiae*) or between the iris and the anterior surface of the lens (*posterior synechiae*). Anterior synechiae can lead to elevation in intraocular pressure, which may lead to optic nerve damage. Prolonged contact between the iris and the anterior surface of the lens can deprive lens epithelium of contact with aqueous humor and can induce fibrous metaplasia of the lens epithelium:

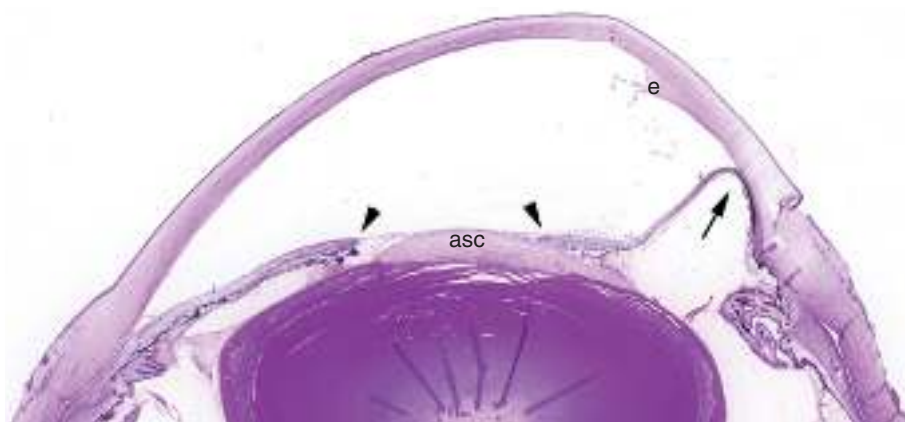


Figure 29.12 Sequelae of anterior segment inflammation. This eye was removed for complications of chronic corneal inflammation (not visible at this magnification). The exudate (e) present in the anterior chamber would have been visualized with a slit lamp as an optical “flare.” The iris is adherent focally to the cornea, obstructing the trabecular meshwork (anterior synechia, arrow), and to the lens (posterior synechia, arrowheads). An anterior subcapsular cataract (asc) has formed. The radial folds in the lens are artifacts.

anterior subcapsular cataract (Fig. 29.12). The pharmacologic induction of pupillary dilation and cycloplegia in individuals with intraocular inflammation is intended in part to prevent the formation of synechia and their sequelae.

Although inflammation confined to the anterior segment is technically intraocular inflammation, the term *endophthalmitis* is reserved for inflammation within the vitreous humor. The retina lines the vitreous cavity, and suppurative inflammation in the vitreous humor is poorly tolerated by the retina; a few hours may be sufficient to cause irreversible retinal injury. Endophthalmitis is classified as *exogenous* (originating in the environment and gaining access to the interior of the eye through a wound) or *endogenous* (delivered to the eye hematogenously). The term *panophthalmitis* is applied to inflammation within the eye that involves the retina, choroid, and sclera and extends into the orbit (Fig. 29.13).

KEY CONCEPTS

- The term cataract describes opacities of the lens that may be congenital or acquired.
- The term glaucoma describes a group of conditions characterized by distinctive changes in the visual field and the size and shape of the optic nerve cup and usually by an elevation in intraocular pressure.
- Glaucoma may develop in the context of either an open or closed angle. Open-angle and angle-closure glaucomas are further subclassified into primary and secondary types.
- Endophthalmitis is a term used to describe inflammation of the interior of the eye involving the vitreous humor, and panophthalmitis is the term used to describe inflammation of the interior of the eye that also extends into the uvea and sclera.
- Endophthalmitis may originate from infection within the body (endogenous endophthalmitis complicating generalized sepsis) or as a complication of corneal infection or a wound, an accidental injury, or a surgical procedure (exogenous endophthalmitis).
- In panophthalmitis, inflammation extends from the interior of the eye into the ocular coats: the retina, choroid, and the sclera.

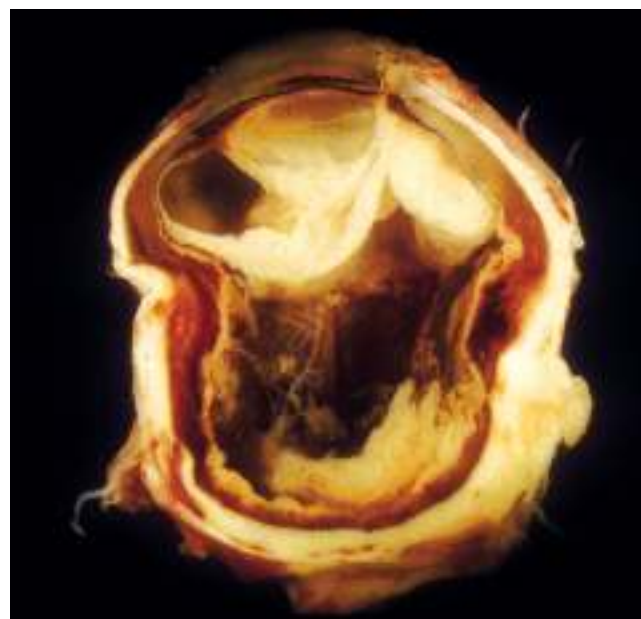


Figure 29.13 Exogenous panophthalmitis. This eye was removed after a foreign body injury. Note the suppurative inflammation behind the lens that is drawn up to the right of the lens to the cornea, the site of the wound. The central portion of the vitreous humor was extracted surgically (by vitrectomy). Note the adhesions to the surface of the eye at the 8 o'clock position, indicating that the intraocular inflammation has spread through the sclera into the orbit: panophthalmitis. (From Folberg R: The eye. In Spencer WH, editor: *Ophthalmic Pathology—An Atlas and Textbook*, ed 4, Philadelphia, 1985, W.B. Saunders.)

UVEA

Together with the iris, the choroid and ciliary body constitute the uvea. The choroid is among the most richly vascularized sites in the body.

Uveitis

The term *uveitis* can be applied to any type of inflammation in one or more of the tissues that compose the uvea. Thus, the iritis that develops after blunt trauma to the eye or that accompanies a corneal ulcer is technically a form of uveitis. However, in clinical practice the term *uveitis* is restricted to a diverse group of chronic diseases that may be either components of a systemic process or localized to the eye. Uveal inflammation may be manifest principally in the anterior segment (e.g., in *juvenile rheumatoid arthritis*) or may affect both the anterior and the posterior segments. The complications of chronic anterior segment inflammation were discussed earlier; the remainder of this discussion therefore focuses on the effects of uveal inflammation on the posterior segment of the eye. As will be described briefly, uveitis is frequently accompanied by retinal pathology. Uveitis may be caused by infectious agents (e.g., *Pneumocystis carinii*), may be idiopathic (e.g., sarcoidosis), or may be autoimmune in origin (sympathetic ophthalmia). Examples are described later.

Granulomatous uveitis is a common complication of sarcoidosis (Chapter 15). In the anterior segment it gives rise to an exudate that evolves into “mutton-fat” keratic precipitates described earlier. In the posterior segment, sarcoid may involve the choroid and retina. Thus, granulomas may be seen in the choroid. Retinal pathology is characterized by perivascular inflammation; this is responsible for the well-known ophthalmoscopic sign of “candle wax drippings.” Conjunctival biopsy can be used to detect granulomatous inflammation and confirm the diagnosis of ocular sarcoid.

Numerous infectious processes can affect the choroid or the retina. Inflammation in one compartment is typically associated with inflammation in the other. Retinal *toxoplasmosis* is usually accompanied by uveitis and even scleritis. Individuals with AIDS may develop cytomegalovirus retinitis and uveal infection such as *Pneumocystis* or mycobacterial choroiditis.

Sympathetic ophthalmia is an example of noninfectious uveitis limited to the eye. This condition is characterized by bilateral granulomatous inflammation typically affecting all components of the uvea: a panuveitis. Sympathetic ophthalmia, which blinded young Louis Braille, may complicate a penetrating injury of the eye. In the injured eye, retinal antigens sequestered from the immune system may gain access to lymphatics in the conjunctiva and thus set up a delayed hypersensitivity reaction that affects not only the injured eye but also the contralateral, noninjured eye. The condition may develop from 2 weeks to many years after injury. Enucleation of a blind eye (which can be the sympathizing eye rather than the directly injured eye) may yield diagnostic findings. It is characterized by diffuse granulomatous inflammation of the uvea (choroid, ciliary body, and iris). Plasma cells are typically absent, but eosinophils may be identified in the infiltrate (Fig. 29.14). Sympathetic ophthalmia is treated by the administration of systemic immunosuppressive agents.

Neoplasms

The most common intraocular malignancy of adults is metastasis to the uvea, typically to the choroid. The

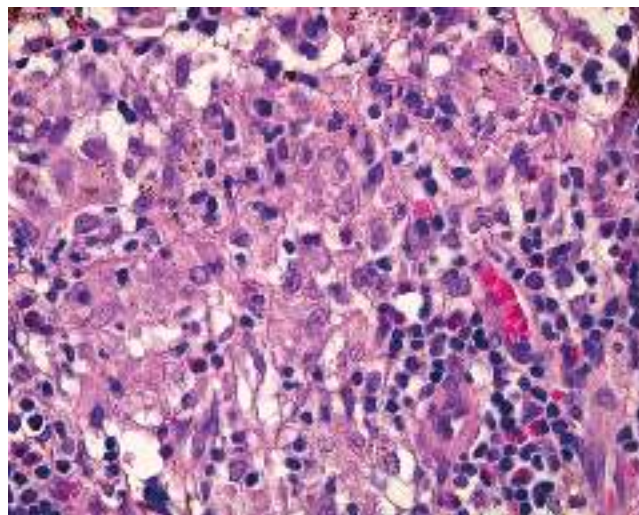


Figure 29.14 Sympathetic ophthalmia. The granulomatous inflammation depicted here was identified diffusely throughout the uvea. The uveal granulomas may contain melanin pigment and may be accompanied by eosinophils.

occurrence of metastases to the eye is associated with an extremely short survival, and treatment of ocular metastases, usually by radiotherapy, is palliative.

Uveal Nevi and Melanomas

Uveal melanoma is the most common primary intraocular malignancy of adults. In the United States, these tumors account for approximately 5% of melanomas and have an age-adjusted incidence of 5.1 per 1 million per year. Uveal nevi, especially choroidal nevi, are rather common, affecting an estimated 5% of the Caucasian population.

Epidemiology and Pathogenesis

Unlike cutaneous melanoma, the occurrence of uveal melanoma has remained stable over many years, and there is no clear link between exposure to ultraviolet light and risk. In line with this observation, sequencing of tumor genomes has revealed that the molecular pathogenesis of uveal melanoma is distinct from that of cutaneous melanoma. Unlike cutaneous and conjunctival melanomas, *BRAF* mutations do not appear to play a role in the pathogenesis of uveal melanomas. The most important oncogenes in uveal melanoma are *GNAQ* and *GNA11*, both of which encode G protein-coupled receptors. Roughly 85% of uveal melanomas harbor a gain-of-function mutation in one of these genes that activates pathways that promote proliferation, such as the mitogen-activated protein kinase (MAPK) pathway (Chapter 7). Notably, uveal nevi are also associated with *GNAQ* and *GNA11* mutations, yet rarely transform to melanoma, indicating that other genetic events are required for the development of uveal melanoma. One such event is loss of chromosome 3, which appears to be selected for because it leads to deletion of *BAP1*, a tumor suppressor gene on chromosome 3 that encodes a deubiquitinating enzyme. BAP1 is a component of protein complexes that place repressive marks on chromatin that lead to gene silencing; thus, uveal melanoma has joined the increasing list of cancers in which epigenetic alterations appear to have a central role in tumor pathogenesis (Chapter

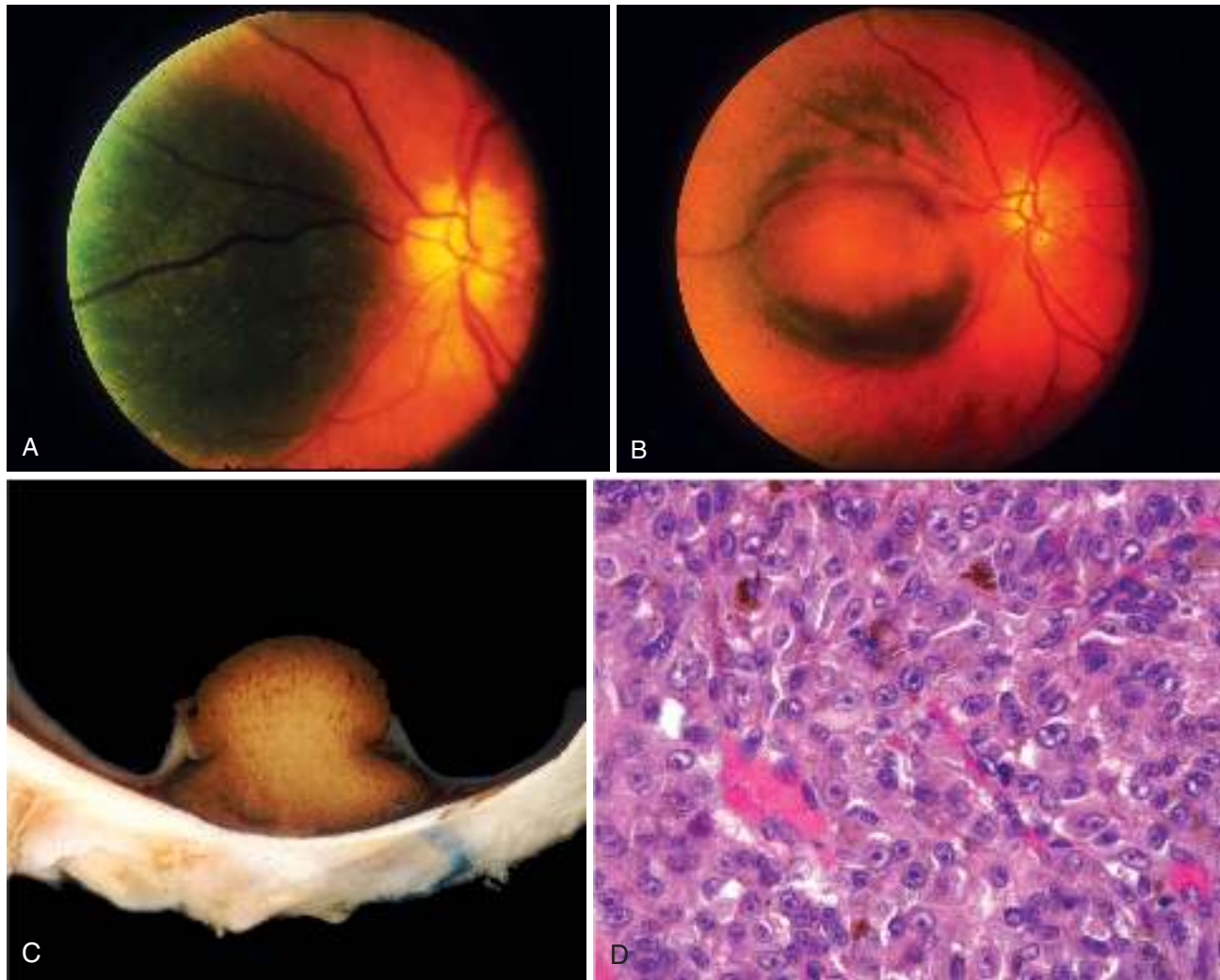


Figure 29.15 Uveal melanoma. (A) Fundus photograph from an individual with a relatively flat pigmented lesion of the choroid near the optic disc. (B) Fundus photograph of the same individual several years later; the tumor has grown and has ruptured through the Bruch membrane. (C) Gross photograph of a choroidal melanoma that has ruptured the Bruch membrane. The overlying retina is detached. (D) Epithelioid melanoma cells associated with an adverse outcome. (A–C, From Folberg R: *Pathology of the Eye—An Interactive CD-ROM Program*, Philadelphia, 1996, Mosby.)

7). Germline mutations in *BAP1* may predispose patients to the development of uveal melanomas, renal cell carcinomas, and malignant mesothelioma among several others.

MORPHOLOGY

Histologically, uveal melanomas may contain two types of cells, spindle and epithelioid, in various proportions (Fig. 29.15). **Spindle cells** are fusiform in shape, whereas **epithelioid cells** are spherical and have greater cytologic atypicity. Like cutaneous melanomas, large numbers of tumor-infiltrating lymphocytes may be seen in some cases. An unusual feature that is commonly seen is the presence of looping slit-like spaces lined by laminin that surround packets of tumor cells. These spaces (which are not blood vessels) connect to blood vessels and serve as extravascular conduits for the transport of plasma and possibly blood. In vitro studies and examination of human tissues suggest that these unusual growth patterns are promoted by tumor cells through a process termed **vasculogenic mimicry**.

Uveal melanomas, with very rare exception, spread exclusively by a hematogenous route (the only exception being the rare case of melanoma that spreads through the sclera and invades the conjunctiva, thereby gaining access to conjunctival lymphatics). Most uveal melanomas spread first to the liver; an excellent example of a tumor-specific tropism for a particular metastatic site.

Clinical Features

Most uveal melanomas are incidental findings or present with visual symptoms, which may be related to retinal detachment or glaucoma. The prognosis of choroidal and ciliary body melanomas is related to (1) size (in contrast to cutaneous melanoma, the lateral extent of the tumor rather than tumor depth is the size dimension related to adverse outcome); (2) cell type (tumors containing epithelioid cells have a worse prognosis than do those containing exclusively spindle cells); (3) and proliferative index. Cytogenetic profiles, especially monosomy 3, and gene expression profiling may

be helpful in stratifying patients into categories with differing risks of developing metastatic disease.

There seems to be no difference in survival between tumors treated by removal of the eye (enucleation) and those receiving eye-sparing radiotherapy, which is the treatment of choice. Melanomas situated exclusively in the iris tend to follow a relatively indolent course, whereas melanomas of the ciliary body and choroid are more aggressive.

Although the 5-year survival rate is approximately 80%, the cumulative melanoma mortality rate is 40% at 10 years, increasing 1% per year thereafter. Metastases may appear “out of the blue” many years after treatment, making uveal melanoma a prime candidate for the investigation of the phenomenon of tumor dormancy. Targeted therapies such as MAPK inhibitors have shown some encouraging responses in clinical trials, but currently there is no effective treatment for metastatic uveal melanoma.

KEY CONCEPTS

- Uveitis is restricted to a diverse group of chronic diseases that may be either components of a systemic process or localized to the eye.
- Sarcoid is an example of a systemic condition that may produce granulomatous uveitis, and sympathetic ophthalmia may produce bilateral granulomatous inflammation as a possible consequence of penetrating injury to one eye.

- The most common intraocular tumor of adults is metastasis to the eye.
- The most common primary intraocular tumor of adults is uveal melanoma.
- Uveal melanoma disseminates hematogenously, and the first evidence of metastasis is typically detected in the liver.
- Uveal melanoma shows marked differences in epidemiologic risk factors and driver mutations compared to cutaneous melanoma.

RETINA AND VITREOUS

Functional Anatomy

The neurosensory retina, like the optic nerve, is an embryologic derivative of the diencephalon. The retina therefore responds to injury by means of gliosis. As in the brain, there are no lymphatics. The architecture of the retina accounts for the ophthalmoscopic appearance of a variety of ocular disorders. Hemorrhages in the nerve fiber layer of the retina are oriented horizontally and appear as streaks or “flames”; the external retinal layers are oriented perpendicular to the retinal surface, and hemorrhages in these outer layers appear as dots (the tips of cylinders). Exudates tend to accumulate in the outer plexiform layer of the retina, especially in the macula (Fig. 29.16).

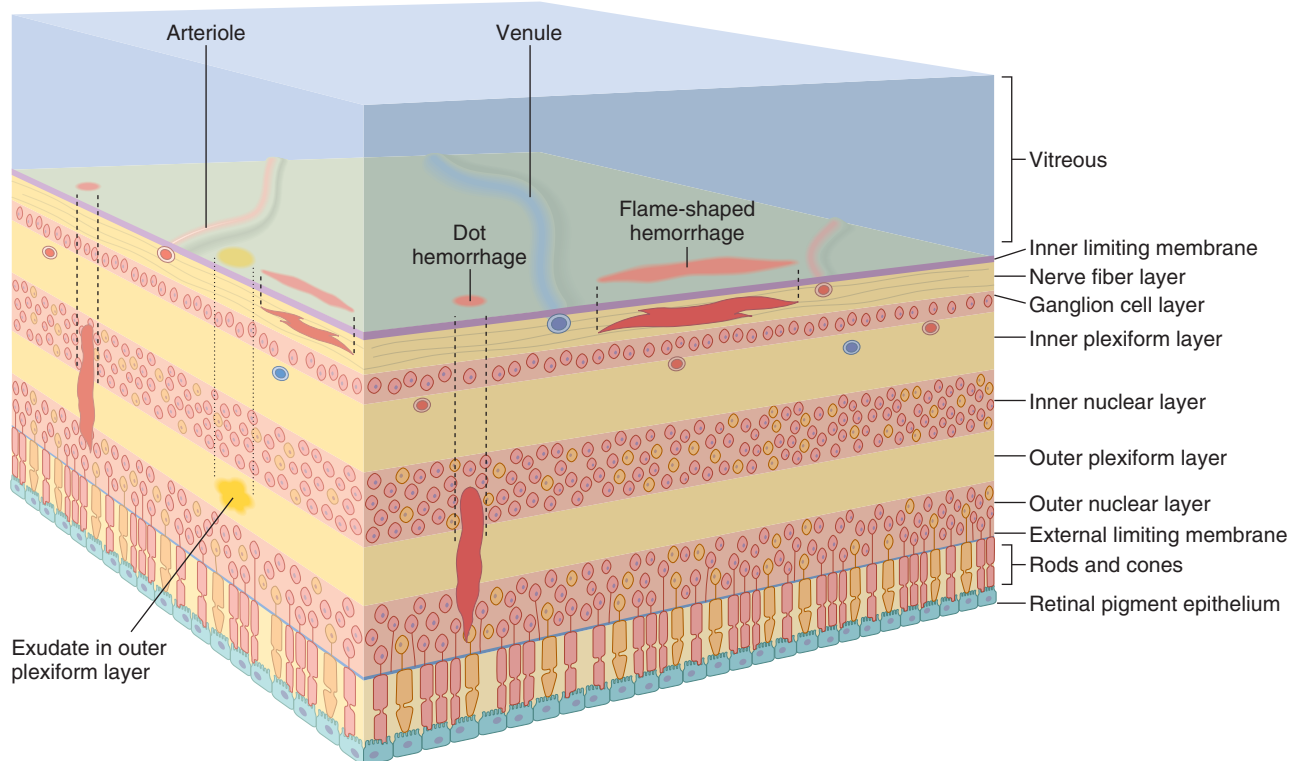
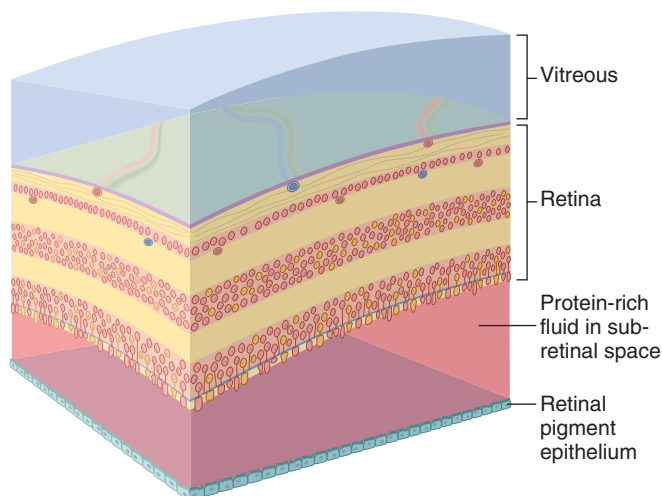
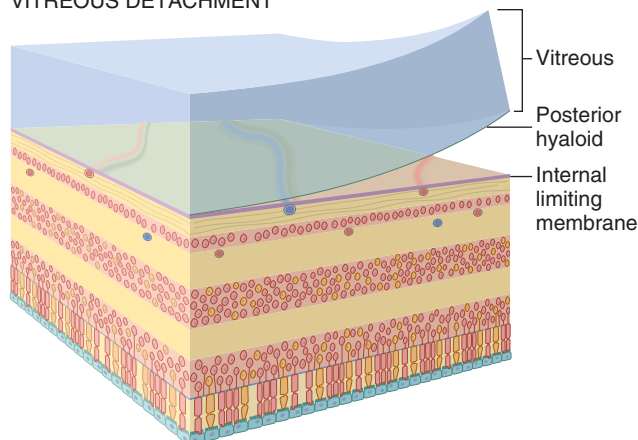


Figure 29.16 Clinicopathologic correlations of retinal hemorrhages and exudates. The location of the hemorrhage within the retina determines its appearance by ophthalmoscopy. The retinal nerve fiber layer is oriented parallel to the internal limiting membrane, and hemorrhages of this layer appear to be flame-shaped ophthalmoscopically. The deeper retinal layers are oriented perpendicular to the internal limiting membrane, and hemorrhages in this location appear as cross-sections of a cylinder, or “dot” hemorrhages. Exudates that originate from leaky retinal vessels accumulate in the outer plexiform layer.

NON-RHEGMATOGENOUS RETINAL DETACHMENT



VITREOUS DETACHMENT



RHEGMATOGENOUS RETINAL DETACHMENT

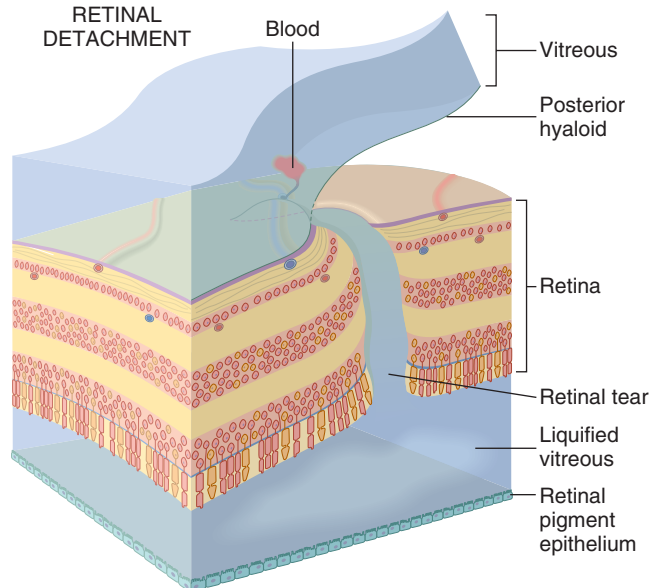


Figure 29.17 Retinal detachment is defined as the separation of the neurosensory retina from the retinal pigment epithelium (RPE). Retinal detachments are classified broadly into nonrhegmatogenous (without a retinal break) and rhegmatogenous (with a retinal break) types. *Top*, In nonrhegmatogenous retinal detachment the subretinal space is filled with protein-rich exudate. Note that the outer segments of the photoreceptors are missing (see Fig. 29.16 for orientation of layers). This indicates a chronic retinal detachment, a finding that can be seen in both nonrhegmatogenous and rhegmatogenous detachments. *Middle*, Posterior vitreous detachment involves the separation of the posterior hyaloid from the internal limiting membrane of the retina and is a normal occurrence in the aging eye. *Bottom*, If during a posterior vitreous detachment the posterior hyaloid does not separate cleanly from the internal limiting membrane of the retina, the vitreous humor will exert traction on the retina, which will be torn at this point. Liquefied vitreous humor seeps through the retinal defect, and the retina is separated from the RPE. The photoreceptor outer segments are intact, illustrating an acute detachment.

The retinal pigment epithelium (RPE), like the retina, is derived embryologically from the primary optic vesicle, an outpouching of the brain. Separation of the neurosensory retina from the RPE defines a *retinal detachment*. The RPE has an important role in the maintenance of the outer segments of the photoreceptors. Disturbances in the RPE-photoreceptor interface are implicated in hereditary retinal degenerations such as *retinitis pigmentosa*.

The adult vitreous humor is avascular. Incomplete regression of fetal vasculature running through the vitreous humor can produce significant pathology as a retrolental mass (*persistent fetal vasculature*). The vitreous humor can be opacified by hemorrhage from trauma or retinal neovascularization. With age the vitreous humor may liquefy and collapse, creating the visual sensation of “floaters.” Also, with aging, the posterior face of the vitreous humor—the posterior hyaloid—may separate from the neurosensory retina (*posterior vitreous detachment*). The relationship between the posterior hyaloid and the neurosensory retina has a key role in the pathogenesis of retinal neovascularization and in some forms of retinal detachment.

Retinal Detachment

Retinal detachment (separation of the neurosensory retina from the RPE) is broadly classified by etiology based on the presence or absence of a break in the retina. *Rhegmatogenous retinal detachment* is associated with a full-thickness retinal defect. Retinal tears may develop after the vitreous collapses structurally, and the posterior hyaloid exerts traction on points of abnormally strong adhesion to the retinal internal limiting membrane. Liquefied vitreous humor then seeps through the tear and gains access to the potential space between the neurosensory retina and the RPE (Fig. 29.17). Reattachment of the retina to the RPE generally requires relief of vitreous traction through indenting of the sclera by surgical procedures. This can be accomplished by the application of strips of silicon to the surface of the eye (scleral buckling) and possibly by removal of vitreous material (vitrectomy). Rhegmatogenous retinal detachment

may be complicated by *proliferative vitreoretinopathy*, the formation of epiretinal or subretinal membranes by retinal glial cells or RPE cells.

Nonrhegmatogenous retinal detachment (retinal detachment without retinal break) may complicate retinal vascular disorders associated with significant exudation and any condition that damages the RPE and permits fluid to leak from the choroidal circulation under the retina. Retinal detachments associated with choroidal tumors and malignant hypertension are examples of nonrhegmatogenous retinal detachment.

Retinal Vascular Disease

Hypertension

Normally, the thin walls of retinal arterioles permit a direct visualization of the circulating blood by ophthalmoscopy. In retinal arteriolosclerosis the thickened arteriolar wall changes the ophthalmic perception of circulating blood: vessels may appear narrowed, and the color of the blood column may change from bright red to copper and to silver depending on the degree of vascular wall thickness (Fig. 29.18A). Retinal arterioles and veins share a common adventitial sheath. Therefore, in pronounced retinal arteriolosclerosis the arteriole may compress the vein at points where both vessels cross (Fig. 29.18B). Venous stasis distal to arteriolar-venous crossing may precipitate occlusions of the retinal vein branches.

In malignant hypertension, vessels in the retina and choroid may be damaged. Damage to choroidal vessels may produce focal choroidal infarcts, seen clinically as *Elschnig*

spots. Damage to the choriocapillaris, the internal layer of the choroidal vasculature, may, in turn, damage the overlying RPE and permit the exudate to accumulate in the potential space between the neurosensory retina and the RPE, thereby producing a retinal detachment. Exudate from damaged retinal arterioles typically accumulates in the outer plexiform layer of the retina (see Fig. 29.18A). The ophthalmoscopic finding of a macular star—a spoke-like arrangement of exudate in the macula in malignant hypertension—results from exudate accumulating in the outer plexiform layer of the macula that is oriented obliquely instead of perpendicular to the retinal surface.

Occlusion of retinal arterioles may produce infarcts of the nerve fiber layer of the retina (axons of the retinal ganglion cell layer populate the nerve fiber layer). Axoplasmic transport in the nerve fiber layer is interrupted at the point of axonal damage, and accumulation of mitochondria at the swollen ends of damaged axons creates the histologic illusion of cells (*cytoid bodies*). Collections of cytoid bodies populate the nerve fiber layer infarct, seen ophthalmoscopically as “cotton-wool spots” (Fig. 29.19). Although nerve fiber layer infarcts are described here in the context of hypertension, they may be detected in a variety of retinal occlusive vasculopathies. For example, retinal nerve fiber layer infarcts may develop in individuals with AIDS due to a retinal vasculopathy that is similar to the brain vasculopathy that may develop in this condition.

Diabetes Mellitus

The eye is profoundly affected by diabetes mellitus. The effects of hyperglycemia on the lens and iris have already

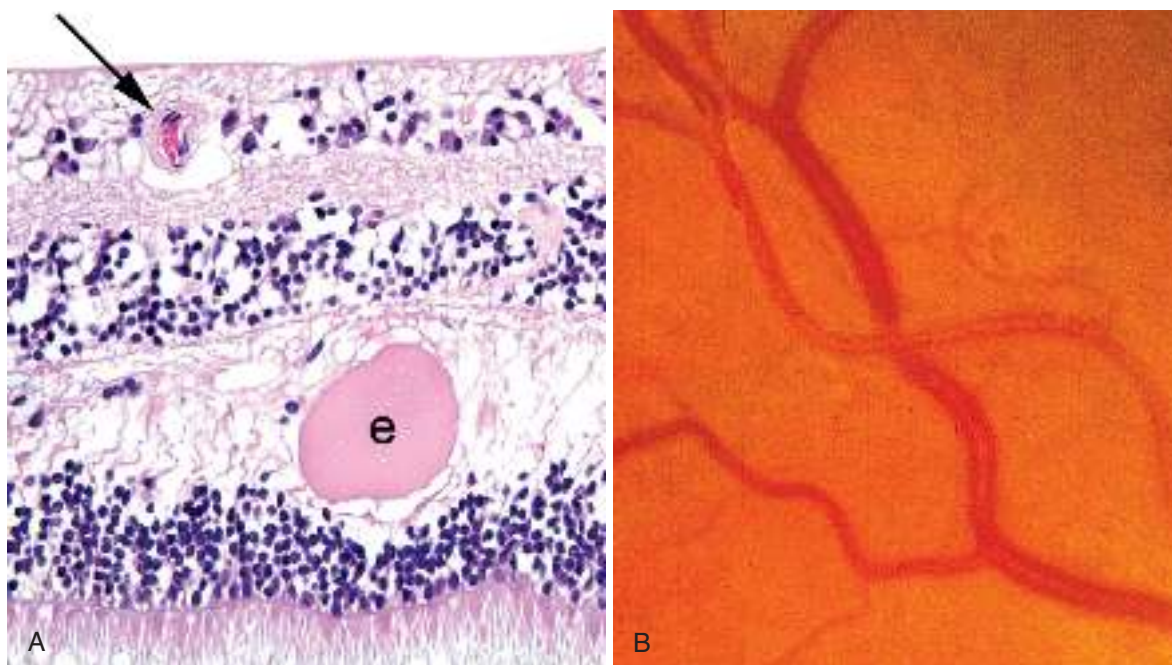


Figure 29.18 The retina in hypertension. (A) The wall of the retinal arteriole (arrow) is thick. Note the exudate (e) in the retinal outer plexiform layer. (B) The fundus in hypertension. The diameter of the arterioles is reduced, and the color of the blood column appears to be less saturated (copper wire-like). If the wall of the vessel were thicker still, the degree of red color would diminish such that the vessels might appear clinically to have a “silver wire” appearance. In this fundus photograph, note that the vein is compressed where the sclerotic arteriole crosses over it. (B, Courtesy Dr. Thomas A. Weingeist, Department of Ophthalmology and Visual Science, University of Iowa, Iowa City, Ia.)

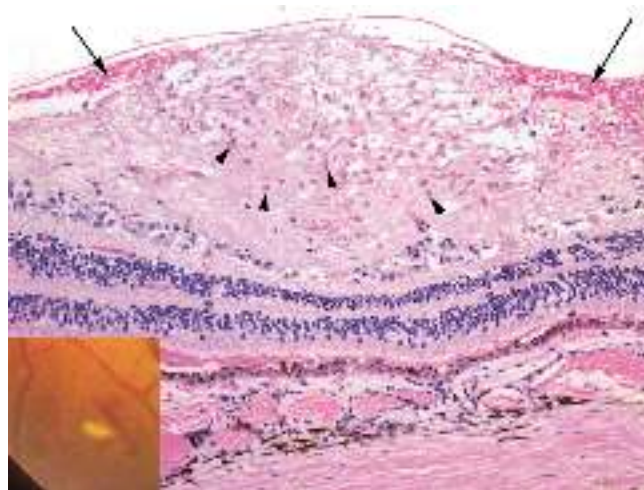


Figure 29.19 Nerve fiber layer infarct. A “cotton-wool spot” is illustrated in the inset, adjacent to a flame-shaped (nerve fiber layer) hemorrhage. The histology of a cotton-wool spot—an infarct of the nerve fiber layer of the retina—is illustrated in the photomicrograph. A focal swelling of the nerve fiber layer is occupied by numerous red to pink cytotoid bodies (arrowheads). Hemorrhage (arrows) surrounding the nerve fiber layer infarct as illustrated here is a variable and inconsistent finding. (Fundus photograph, Courtesy Dr. Thomas A. Weingeist, Department of Ophthalmology and Visual Science, University of Iowa, Iowa City, Ia.)

been mentioned. Thickening of the basement membrane of the epithelium of the pars plicata of the ciliary body is a reliable histologic marker of diabetes mellitus in the eye (Fig. 29.20) and is reminiscent of similar changes in the glomerular mesangium. This discussion focuses on the retinal microangiopathy associated with diabetes mellitus, a prototype for the consideration of other retinal microangiopathies.

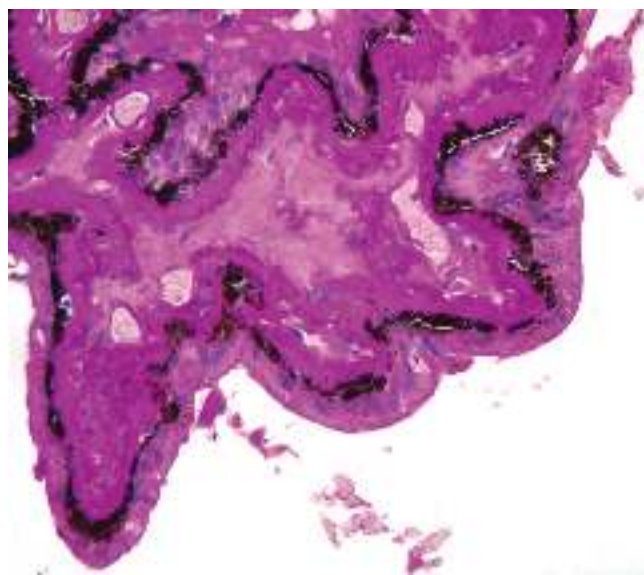


Figure 29.20 The ciliary body in chronic diabetes mellitus, periodic acid–Schiff stain. Note the massive thickening of the basement membrane of the ciliary body epithelia, reminiscent of changes in the mesangium of the renal glomerulus.

MORPHOLOGY

The retinal vasculopathy of diabetes mellitus can be classified into **nonproliferative** and **proliferative diabetic retinopathy**.

Nonproliferative diabetic retinopathy includes a spectrum of changes resulting from structural and functional abnormalities of retinal vessels (i.e., confined beneath the internal limiting membrane of the retina). As with diabetic microangiopathy in general, the **basement membrane of retinal blood vessels is thickened**. In addition, the number of pericytes relative to endothelial cells diminishes. **Microaneurysms** are an important manifestation of diabetic microangiopathy. They are typically smaller than the resolution of direct ophthalmoscopes, and findings customarily described as microaneurysms by ophthalmoscopy may in fact be retinal microhemorrhages. Structural changes in the retinal microcirculation have been associated with a physiologic breakdown in the blood-retinal barrier. Recall that VEGF was initially called vascular permeability factor. Thus, the retinal microcirculation in diabetics may be exceptionally leaky, giving rise to **macular edema**, a common cause of visual loss in these patients. The vascular changes may also produce **exudates** that accumulate in the outer plexiform layer. Although the retinal microcirculation is often hyperpermeable, it is also subject to the effects of micro-occlusion. Both vascular incompetence and vascular micro-occlusions can be visualized clinically after intravenous injection of fluorescein. Nonperfusion of the retina due to the microcirculatory change described earlier is associated with upregulation of VEGF and intraretinal angiogenesis (located beneath the internal limiting membrane of the retina).

Proliferative diabetic retinopathy is defined by the appearance of new vessels sprouting on the surface of either the optic nerve head or the surface of the retina (Fig. 29.21C). The term “retinal neovascularization” is applied only when the newly formed vessels breach the internal limiting membrane of the retina. The quantity and location of retinal neovascularization guide the ophthalmologist in the treatment of proliferative diabetic retinopathy. The web of newly formed vessels is referred to as a neovascular membrane. It is composed of angiogenic vessels with or without a substantial supportive fibrous or glial stroma (Fig. 29.21B).

If the vitreous humor has not detached and the posterior hyaloid is intact, neovascular membranes extend along the potential plane between the retinal internal limiting membrane and the posterior hyaloid. If vitreous humor later separates from the internal limiting membrane of the retina (**posterior vitreous detachment**), there may be massive hemorrhage from the disrupted neovascular membrane. In addition, scarring associated with the organization of the retinal neovascular membrane may wrinkle the retina, disrupting the orientation of retinal photoreceptors and producing visual distortion, and may exert traction on the retina, separating it from the RPE (retinal detachment). **Traction retinal detachment** may begin as a nonrhegmatogenous detachment, but severe traction may tear the retina, producing a traction rhegmatogenous detachment.

Retinal neovascularization may be accompanied by the development of a neovascular membrane on the iris surface, presumably secondary to increased levels of VEGF in the aqueous humor. Contraction of the iris neovascular membrane may lead to adhesions between the iris and trabecular meshwork (anterior synechiae), thus occluding a major pathway for aqueous outflow and thereby contributing to elevation of the intraocular pressure (**neovascular glaucoma**).

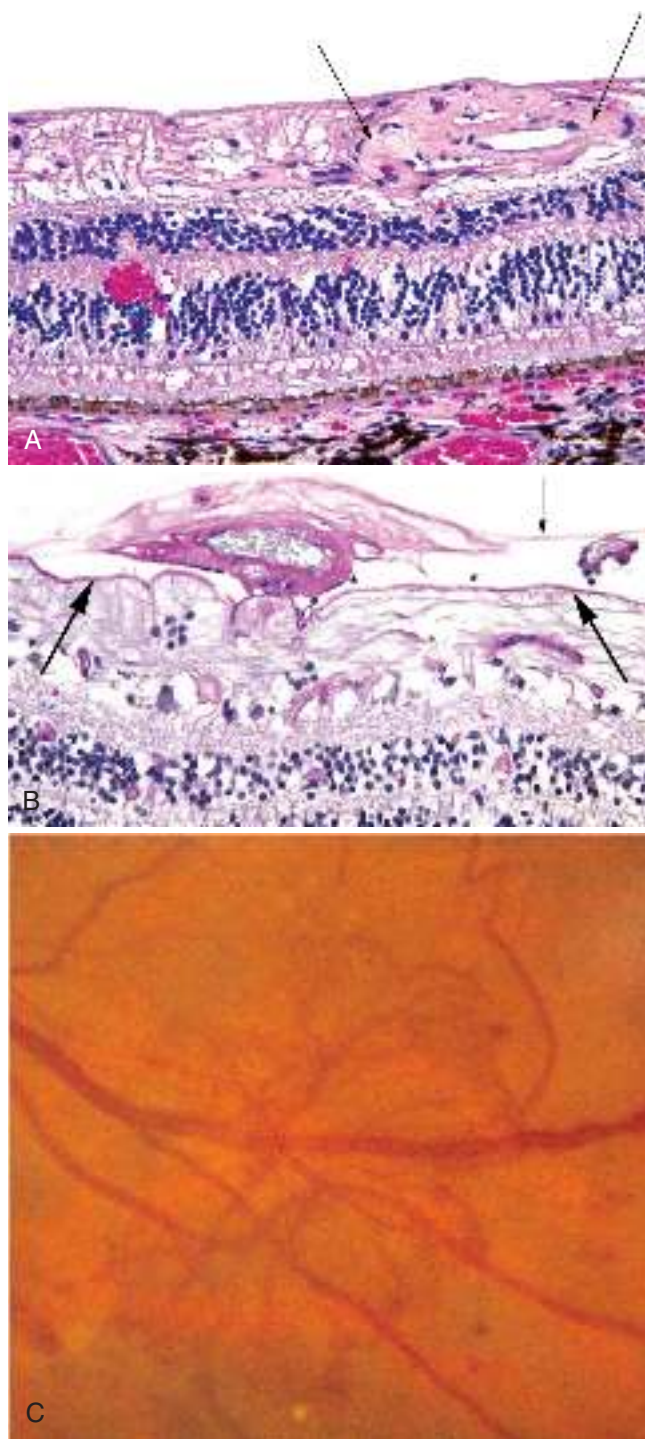


Figure 29.21 The retina in diabetes mellitus (see Fig. 29.16 for a schematic of retinal structure). (A) A tangle of abnormal vessels lies just beneath the internal limiting membrane of the retina on the right half of the photomicrograph (between arrows). This is an example of intraretinal angiogenesis known as intraretinal microangiopathy (IRMA). Note the retinal hemorrhage in the outer plexiform layer in the left half. The ganglion cell layer and the nerve fiber layer—the axons of the ganglion cells—are absent. The rarefied space beneath internal limiting membrane to the left of the focus of IRMA consists largely of elements of retinal glial (Müller) cells. Absence of the ganglion cell and nerve fiber layers is a hallmark of glaucoma. The chronic diabetes mellitus in this individual was complicated by iris neovascularization and secondary angle-closure glaucoma (neovascular glaucoma). (B) In this section stained by periodic acid-Schiff, the internal limiting membrane is indicated by the *thick arrows*, and the posterior hyaloid of the vitreous is indicated by the *thin arrow*. In the potential space between these two landmarks, the vessels to the left of the *thin arrow* are invested with a fibrous-glial stroma and would appear ophthalmoscopically as a white neovascular membrane. The thin-walled vessel to the right of the *thin arrow* is not invested with connective tissue. A posterior vitreous detachment in an eye such as this might exert traction on these new vessels and precipitate a massive vitreous hemorrhage. (C) Ophthalmoscopic view of retinal neovascularization (known clinically as neovascularization “elsewhere” in contrast to neovascularization of the optic disc) creating a neovascular membrane.

Retinopathy of Prematurity (Retrolental Fibroplasia)

At term, the temporal (lateral) aspect of the retinal periphery is incompletely vascularized, whereas the medial aspect is vascularized. In premature or low-birth-weight infants treated with oxygen, immature retinal vessels in the temporal retinal periphery constrict, rendering the retinal tissue distal to this zone ischemic. Retinal ischemia may result in upregulation of proangiogenic factors such as VEGF and lead to retinal angiogenesis. Contraction of the resulting peripheral retinal neovascular membrane may “drag” the temporal aspect of the retina toward the peripheral zone, displacing the macula (situated temporal to the optic nerve) laterally. Neovascular membrane contraction may create sufficient force to cause retinal detachment. The use of VEGF inhibition in this condition is under investigation.

Sickle Retinopathy, Retinal Vasculitis, and Radiation Retinopathy

Retinopathy affecting individuals with sickle hemoglobinopathies (Chapter 14) has been divided into two types that roughly parallel those used for diabetic retinopathy: nonproliferative (intraretinal angiopathic changes) and proliferative (retinal neovascularization). The final common pathway in both types is vascular occlusion. Low oxygen tension within the blood vessels in the retinal periphery results in red cell sickling and microvascular occlusions. In the nonproliferative form (which occurs in individuals with hemoglobin SS and SC genotypes), vascular occlusions are thought to contribute to preretinal, intraretinal, and subretinal hemorrhages. The resolution of these hemorrhages may give rise to several ophthalmoscopically visible changes, known as *salmon patches*, *iridescent spots*, and *black sunburst lesions*.

Ablating nonperfused retina by laser photocoagulation or cryopexy triggers regression of both retinal and iris neovascularization, emphasizing the central role that retinal hypoxia has in these disorders. More recently, the injection of VEGF inhibitors into the vitreous has been used to treat diabetic macular edema and retinal neovascularization, a successful example of how knowledge of the molecular pathogenesis of a condition may evolve into a successful therapeutic strategy.

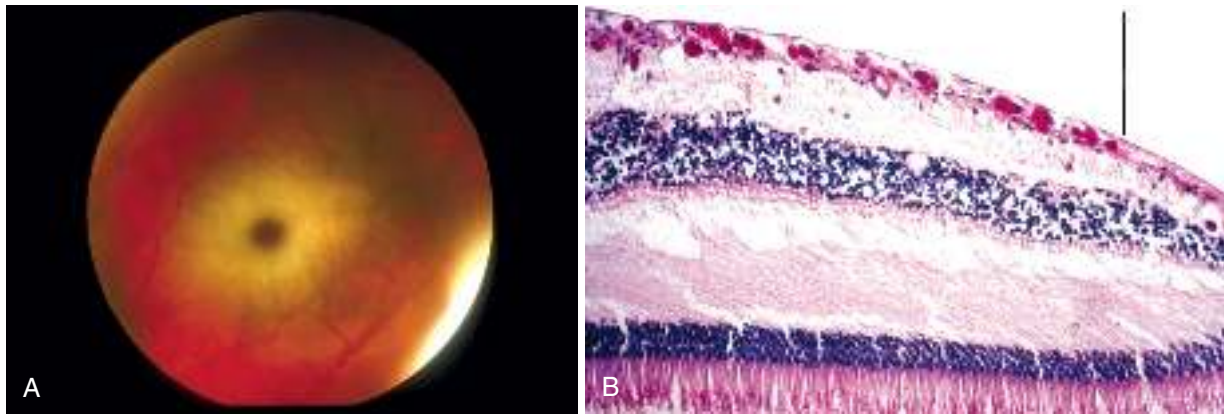


Figure 29.22 Cherry-red spot in Tay-Sachs disease. (A) Fundus photograph of the cherry-red spot in Tay-Sachs disease. (B) Photomicrograph of the macula in an individual with Tay-Sachs disease, stained with periodic acid-Schiff to highlight the accumulation of ganglioside material in the retinal ganglion cells. The presence of ganglion cells filled with gangliosides outside the fovea blocks the transmission of the normal orange-red color of the choroid, but absence of ganglion cells within the fovea (to the right of the vertical bar) permits the normal orange-red color to be visualized, accounting for the so-called cherry-red spot. (A, Courtesy Dr. Thomas A. Weingeist, Department of Ophthalmology and Visual Science, University of Iowa, Iowa City, Ia.; B, from the teaching collection of the Armed Forces Institute of Pathology.)

Organization of pre-retinal hemorrhage may result in retinal traction and retinal detachment. Vascular occlusions may also contribute to angiogenesis secondary to upregulation of both VEGF and basic fibroblast growth factor. This can give rise to areas of florid neovascularization in the periphery of the retina, described clinically as “sea-fans.”

Neovascularization also occurs in a variety of other clinical settings such as peripheral retinal vasculitis and in irradiation used to treat intraocular tumors. The feature common to these conditions is damage to retinal vessels, producing zones of retinal ischemia that trigger retinal angiogenesis and its complications hemorrhage and traction, which, in turn, may cause detachment.

Retinal Artery and Vein Occlusions

The central retinal artery or its branches can be occluded by disorders that affect the vessels in general. For example, the lumen of the central retinal artery can be narrowed significantly by atherosclerosis, thus predisposing to thrombosis. Emboli to the central retinal artery can originate from thrombi in the heart or from ulcerated atheromatous plaques in the carotid arteries. Fragments of atherosclerotic plaques can lodge within the retinal circulation (*Hollenhorst plaques*). Total occlusion of a branch of retinal artery can produce a segmental infarct of the retina. With sudden cessation of blood supply, the retina (an embryologic derivative of brain tissue) swells acutely and becomes optically opaque. By ophthalmoscopy the fundus in the affected area appears white instead of red or orange because the retinal opacity blocks the view of the richly vascular choroid.

Total occlusion of the central retinal artery can produce a diffuse infarct of the retina. Following an acute occlusion, the retina appears relatively opaque by ophthalmoscopy. The fovea and foveola are physiologically thin; therefore, the normal orange-red of the choroid is not only visible but also highlighted by the surrounding opaque retina—the origin of the *cherry-red spot* of the central retinal artery occlusion. Cherry-red spots can also be seen in rare storage diseases such as *Tay-Sachs* and *Niemann-Pick* diseases because of the structural organization of the retina. The storage

material accumulates in retinal ganglion cells: the ganglion cell layer of the macula surrounding the fovea is thick, but there are no ganglion cells in the center of the macula, the fovea. Thus, the fovea is relatively transparent to the underlying choroidal vasculature but is rimmed by relatively opaque retina, the result of storage material accumulating in the perifoveal macular ganglion cells (Fig. 29.22).

Retinal vein occlusion may occur with or without ischemia. In ischemic retinal vein occlusion, VEGF and other proangiogenic factors are upregulated in the retina, leading to neovascularization of the retina and surface of the optic nerve head as well as neovascularization of the iris and subsequent angle-closure glaucoma. Nonischemic retinal vein occlusion may be complicated by hemorrhages, exudates, and macular edema.

Age-Related Macular Degeneration (AMD)

AMD results from damage to the macula, which is required for central vision. It occurs in two forms, dry and wet, that are distinguished by the presence of neovascularization in the wet form and its absence in the dry form. From the name of this disorder, it is clear that advancing age is a risk factor. The cumulative incidence of AMD in individuals 75 years of age and older is 8%, and with increasing longevity AMD is becoming a major health problem.

Atrophic or “dry” AMD is characterized ophthalmoscopically by diffuse or discrete deposits in the Bruch membrane (*drusen*) and geographic atrophy of the RPE. Loss of vision may be severe in these individuals. The oral consumption of zinc and of vitamins with antioxidant properties may be associated with a slowing of the progression to AMD. There is currently no effective treatment for “dry” or atrophic AMD; a regenerative approach to replacing diseased RPE cells with stem cells is under investigation.

Neovascular or “wet” AMD is characterized by *choroidal neovascularization*, defined by the presence of vessels that presumably originate from the choriocapillaris and penetrate through the Bruch membrane beneath the RPE (Fig. 29.23). This neovascular membrane may also penetrate the RPE and

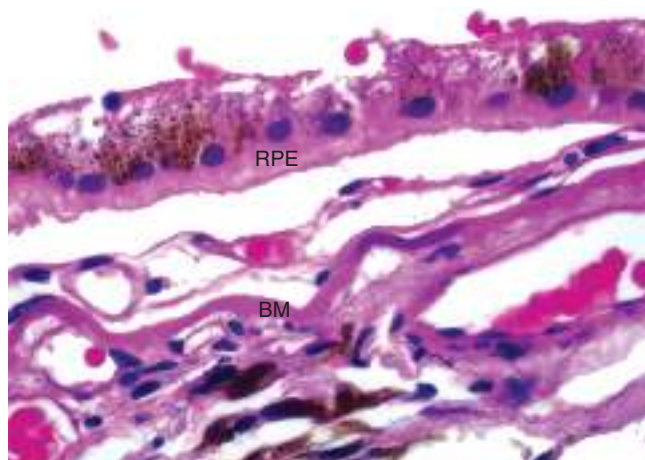


Figure 29.23 “Wet” age-related macular degeneration. A neovascular membrane is positioned between the retinal pigment epithelium (RPE) and Bruch membrane (BM). Note the blue discoloration of BM to the right of the label, indicating focal calcification.

become situated directly beneath the neurosensory retina. The vessels in this membrane may leak, and the exuded blood may be organized by RPE cells into macular scars. Occasionally, these vessels are the source of hemorrhage, leading to the localized suffusion of blood that may be mistaken clinically for an intraocular neoplasm, or give rise to diffuse vitreous hemorrhage. Currently the mainstay of treatment for neovascular AMD is the injection of VEGF antagonists into the vitreous of the affected eye. Photodynamic therapy is also being evaluated for effectiveness.

Choroidal neovascular membranes can develop in diverse conditions that are unrelated to age, such as pathologic myopia (Fuchs spot), following disruption of the Bruch membrane (due to trauma or other causes), or an immunologic response to systemic histoplasmosis (presumed ocular histoplasmosis syndrome).

To understand the pathogenesis of AMD it is important to appreciate the existence of a structural and functional unit composed of the RPE, Bruch membrane (which contains the basement membrane of the RPE), and the innermost layer of the choroidal vasculature, the choriocapillaris. Disturbance in any component of this “unit” affects the health of the overlying photoreceptors, producing visual loss.

Attention is now focused on the roles of several genes, especially *CFH* (complement factor H) and other complement regulatory genes in the pathogenesis of this condition. The complement regulatory gene variants that are associated with AMD all appear to decrease their function, implying that AMD may stem from an excess of complement activity. Environmental exposures such as cigarette smoking may also increase risk of AMD, especially in genetically predisposed individuals.

Other Retinal Degenerations

Retinitis Pigmentosa

Retinitis pigmentosa is an inherited condition resulting from mutations that affect rods and cones or RPE. It can cause varying degrees of visual impairment including, in

some cases, total blindness. The term “retinitis” pigmentosa is an unfortunate relic of the time when these disorders were incorrectly presumed to be inflammatory. They may be inherited as X-linked recessive, autosomal recessive, or autosomal dominant (the age of onset correlates with the inheritance pattern, with autosomal dominant retinitis pigmentosa appearing later in life). Retinitis pigmentosa may be part of a syndrome, such as the *Bardet-Biedl syndrome*, *Usher syndrome*, or *Refsum disease*, or may develop in isolation (nonsyndromic retinitis pigmentosa).

Retinitis pigmentosa is linked to mutations affecting *RHO*, *USH2A*, *RGPR*, and *EYS* genes; they regulate the functions of either the photoreceptor cells or the RPE. Typically, both rods and cones are lost to apoptosis, though in varying proportions. Loss of rods may lead to early night blindness and constricted visual fields. As cones are lost, central visual acuity may be affected. Clinically, retinal atrophy is accompanied by constriction of retinal vessels and optic nerve head atrophy (“waxy pallor” of the optic disc) and the accumulation of retinal pigment around blood vessels, thus accounting for the “pigmentosa” in the disease name. The electroretinogram reveals abnormalities characteristic of this disease.

Retinitis

A variety of pathogens can contribute to the development of infectious retinitis. For example, *Candida* may disseminate to the retina hematogenously, especially in the setting of intravenous drug abuse or in systemic candidemia from other causes. Hematogenous dissemination of pathogens to the retina typically results in multiple retinal abscesses. As was mentioned previously, cytomegalovirus retinitis is an important cause of visual morbidity in immunocompromised individuals, especially those with AIDS.

Retinal Neoplasms

Retinoblastoma

Retinoblastoma is the most common primary intraocular malignancy of children. The molecular genetics of retinoblastoma are discussed in detail in Chapter 7. Although the name retinoblastoma might suggest origin from a primitive retinal cell that is capable of differentiation into both glial and neuronal cells, it is now clear that the cell of origin of retinoblastoma is a neuronal progenitor. Recall that in approximately 40% of cases, retinoblastoma occurs in individuals who inherit a germline mutation of one *RB* allele. Retinoblastoma arises when the retinal progenitor suffers a second, somatic mutation and the *RB* gene function is lost. In the sporadic cases, both *RB* alleles are lost by somatic mutations. Retinoblastomas arising in those with germline mutations are often bilateral. In addition, they may be associated with pinealoblastoma (“trilateral” retinoblastoma), which is associated with a dismal outcome.

MORPHOLOGY

The pathology of retinoblastoma, both hereditary and sporadic types, is identical. Tumors may contain both undifferentiated and

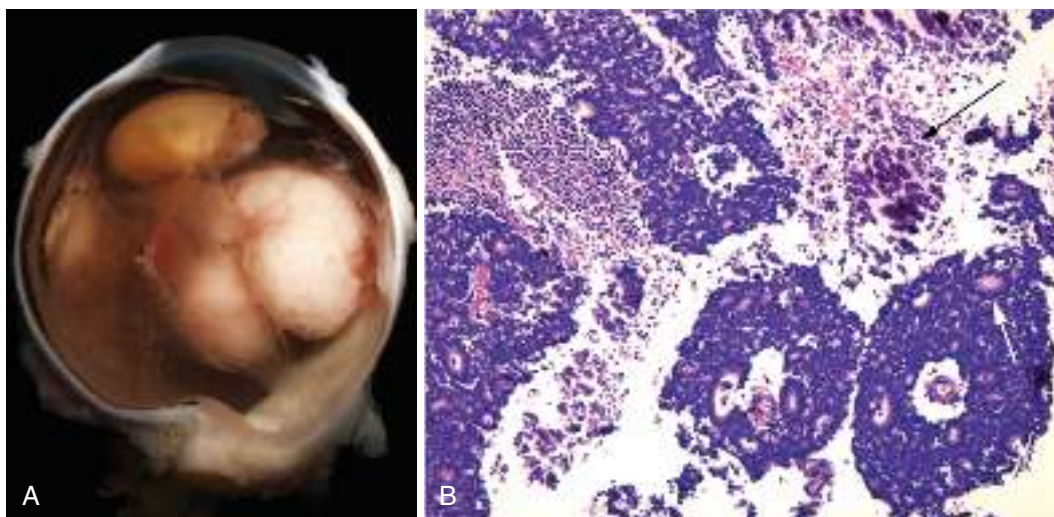


Figure 29.24 Retinoblastoma. (A) Gross photograph of retinoblastoma. (B) Tumor cells appear viable when in proximity to blood vessels, but necrosis is seen as the distance from the vessel increases. Dystrophic calcification (*black arrow*) is present in the zones of tumor necrosis. Flexner-Wintersteiner rosettes—arrangements of a single layer of tumor cells around an apparent “lumen”—are seen throughout the tumor, and one such rosette is indicated by the *white arrow*.

differentiated elements. The former appear as collections of small, round cells with hyperchromatic nuclei. In well-differentiated tumors there are **Flexner-Wintersteiner rosettes** and fleurettes reflecting photoreceptor differentiation. It should be noted, however, that the degree of tumor differentiation does not appear to be associated with the prognosis. As seen in [Fig. 29.24](#), viable tumor cells are found encircling tumor blood vessels with zones of necrosis typically found in relatively avascular areas, illustrating the dependence of retinoblastoma on its blood supply. Focal zones of dystrophic calcification are characteristic of retinoblastoma.

In an effort to preserve vision and eradicate the tumor, many ophthalmic oncologists now attempt to reduce tumor burden by administration of chemotherapy, including selective delivery of the drug to the eye through the ophthalmic artery; after chemoreduction, tumors may be obliterated by laser treatment or cryopexy. Retinoblastoma tends to spread to the brain and bone marrow and seldom disseminates to the lungs. Prognosis is adversely affected by extraocular extension and invasion along the optic nerve and by choroidal invasion. A variant of retinoblastoma—retinocytoma or retinoma—has been reported and appears to be a premalignant lesion. The appearance of retinoblastoma in one eye and retinocytoma in the other eye is characteristic of heritable retinoblastoma.

Retinal Lymphoma

Primary retinal lymphoma is an aggressive tumor that characteristically involves the two retinal layers derived from brain, the neurosensory retina and the RPE. Primary intraocular lymphoma tends to occur in older individuals and may mimic uveitis clinically. Most are diffuse large B-cell lymphomas (Chapter 13). Spread to the brain commonly occurs via the optic nerve. The diagnosis depends on a demonstration of lymphoma cells in vitreous aspirates.

KEY CONCEPTS

- Retinal detachment, a separation of the neurosensory retina from the RPE, may be the consequence of a break in the retina (rhegmatogenous retinal detachment) or may develop without a retinal break because of pathology within or beneath the retina (nonrhegmatogenous retinal detachment).
- The clinical appearance of the retina by ophthalmoscopy can be linked to specific pathologic changes: the change in caliber and color of retinal blood vessels may reflect varying degrees of arteriolosclerosis, and the location of hemorrhages and exudates in the retina is related to their locations within the retinal layers.
- Several major causes of blindness result from pathologic intraocular angiogenesis including proliferative diabetic retinopathy and exudative (wet) age-related macular degeneration, among many other conditions. VEGF antagonists may prevent visual loss in many of these conditions.
- Retinoblastoma is the most common primary intraocular tumor of children.
- Primary retinal lymphoma is an aggressive tumor that often involves the brain as well.

OPTIC NERVE

As a sensory tract of the central nervous system, the optic nerve is surrounded by meninges, and cerebrospinal fluid circulates around the nerve. The pathology of the optic nerve is similar to the pathology of the brain. For example, the most common primary neoplasms of the optic nerve are glioma (typically pilocytic astrocytomas) and meningioma.

Anterior Ischemic Optic Neuropathy

There are striking similarities between stroke and a condition known in ophthalmic terminology as anterior ischemic

optic neuropathy (AION). As used clinically, the term AION includes a spectrum of injuries to the optic nerve varying from ischemia to infarction. Thus, transient partial interruptions in blood flow to the optic nerve can produce episodes of transient loss of vision, whereas total interruption in blood flow can give rise to an optic nerve infarct which may be segmental or total. Zones of relative ischemia may surround segmental infarcts of the optic nerve. Optic nerve function in these poorly perfused but not infarcted zones may recover. The optic nerve does not regenerate, and visual loss from infarction is permanent.

Interruption in the blood supply to the optic nerve can result from inflammation of the vessels that supply the optic nerve or from embolic or thrombotic events. Bilateral total infarcts of the optic nerve resulting in total blindness have been reported in temporal arteritis, adding urgency to the treatment of this condition with high doses of corticosteroids.

Papilledema

Edema of the head of the optic nerve may develop as a consequence of compression of the nerve (as in a primary neoplasm of the optic nerve when swelling of the nerve head produces unilateral disc edema) or from elevations of cerebrospinal fluid pressure surrounding the nerve (resulting typically in bilateral disc edema). The concentric increase in pressure encircling the nerve contributes to venous stasis and also interferes with axoplasmic transport, leading to nerve head swelling. Swelling of the optic nerve head in elevated intracranial pressure is bilateral and is commonly termed *papilledema*. Typically, acute papilledema from increased intracranial pressure is not associated with visual loss. Ophthalmoscopically, the optic nerve head is swollen and hyperemic; by contrast, the optic nerve head in the relatively acute phases of anterior ischemic optic neuropathy appears swollen and pale because of decreased nerve perfusion (Fig. 29.25). In papilledema secondary to increased intracranial pressure, the optic nerve may remain congested for a prolonged period of time.

Glaucomatous Optic Nerve Damage

As discussed, the majority of individuals with glaucoma have elevated intraocular pressure. However, there is a small group that develops the visual field and optic nerve changes typical of glaucoma with normal intraocular pressure—so-called **normal-tension glaucoma**. Conversely, some individuals with elevated intraocular pressure who are followed over long periods of time never develop visual field changes or optic nerve cupping. Therefore, it is clear that there is a spectrum of neuronal susceptibility to the effects of elevated intraocular pressure. Considerable research is now directed toward understanding mechanisms by which the optic nerve axons may be protected from injury.

MORPHOLOGY

Characteristically, there is a diffuse loss of ganglion cells and thinning of the retinal nerve fiber layer (Fig. 29.26), which can be measured by optical coherence tomography. In advanced cases, the optic nerve is both cupped and atrophic, a combination unique

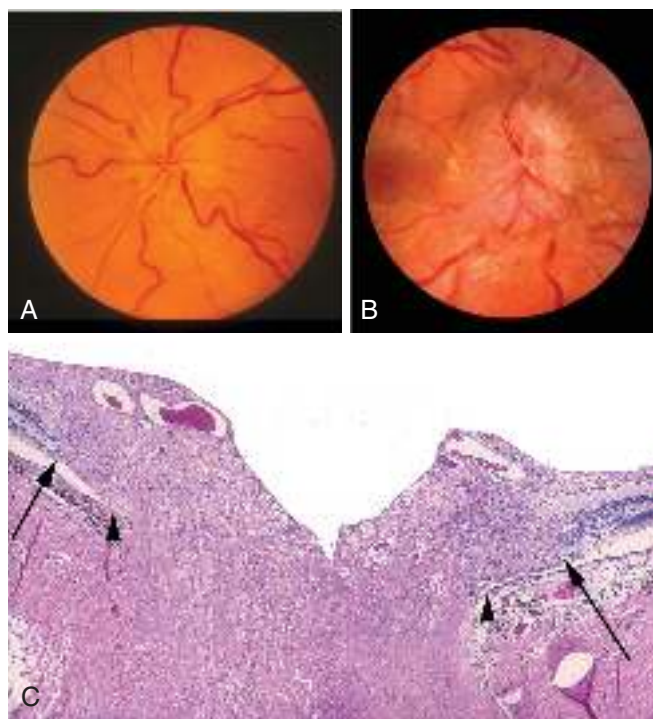


Figure 29.25 The optic nerve in anterior ischemic optic neuropathy (AION) and papilledema. (A) In the acute phases of AION the optic nerve may be swollen, but it is relatively pale because of decreased perfusion. (B) In papilledema secondary to increased intracranial pressure, the optic nerve is typically swollen and hyperemic. (C) Normally, the termination of Bruch membrane (arrowheads) is aligned with the beginning of the neurosensory retina, as indicated by the presence of stratified nuclei (arrows), but in papilledema the optic nerve is swollen, and the retina is displaced laterally. This is the histologic explanation for the blurred margins of the optic nerve head seen clinically in this condition. (A and B, Courtesy Dr. Sohan S. Hayreh, Department of Ophthalmology and Visual Science, University of Iowa, Iowa City, Ia.; C, from the teaching collection of the Armed Forces Institute of Pathology.)

to glaucoma. Elevated intraocular pressure in infants and children can lead to diffuse enlargement of the eye (**buphthalmos**) or enlargement of the cornea (**megalocornea**). After the eye reaches its adult size, prolonged elevation of intraocular pressure can lead to focal thinning of the sclera, and uveal tissue may line ectatic sclera (**staphyloma**).

Other Optic Neuropathies

Optic neuropathy may be inherited or may be secondary to nutritional deficiencies or toxins such as methanol. Individuals may suffer severe visual compromise. If the nerve fibers that originate from the macula are affected, then central visual acuity is lost.

Leber hereditary optic neuropathy results from inheritance of mitochondrial gene mutations (Chapter 5). Since neuronal health is dependent on axoplasmic transport of mitochondria, mitochondrial dysfunctions give rise to neurologic disorders including optic neuropathy. Leber optic neuropathy shows maternal inheritance pattern typical of mitochondrial gene mutations. Males are affected far more commonly than

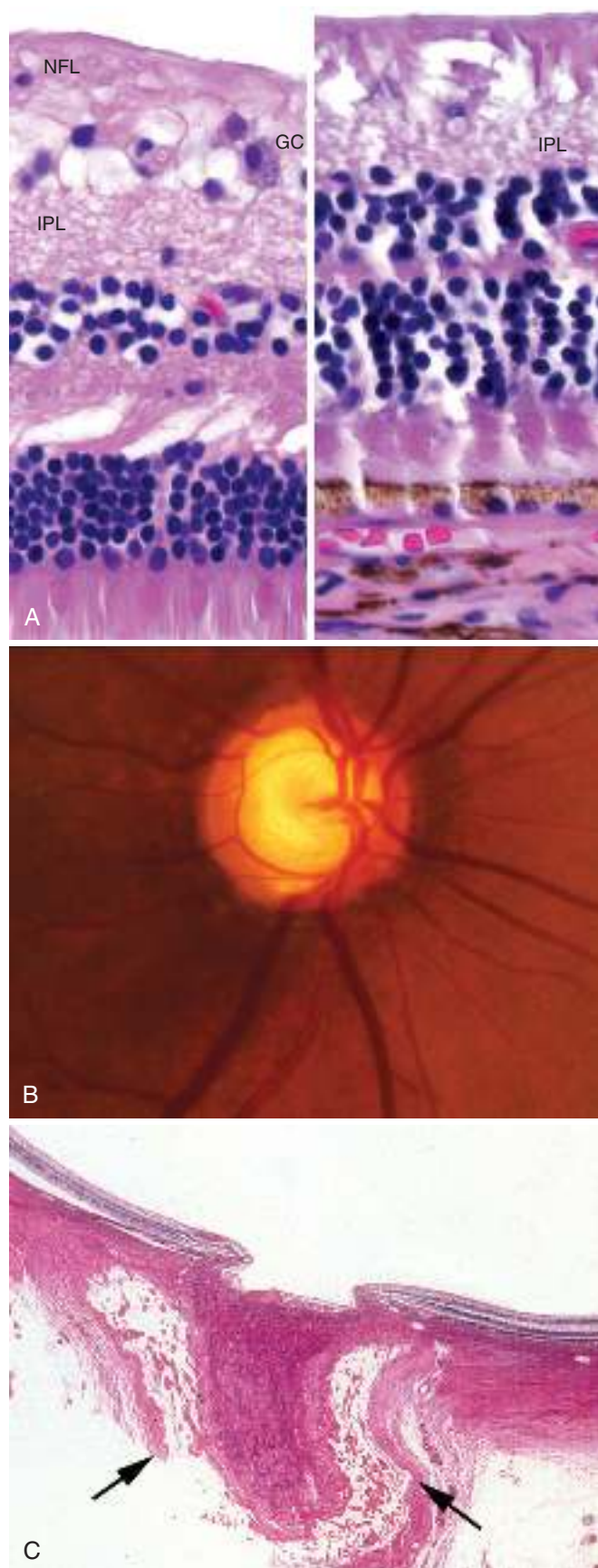


Figure 29.26 The retina and optic nerve in glaucoma. (A) *Left panel*, normal retina; *right panel*, the retina in long-standing glaucoma (same magnification). The full thickness of the glaucomatous retina is captured (*right*), a reflection of the thinning of the retina in glaucoma. In the glaucomatous retina, the areas corresponding to the nerve fiber layer (NFL) and ganglion cell layer (GC) are atrophic; the inner plexiform layer (IPL) is labeled for reference. Note also that the outer nuclear layer of the glaucomatous retina is aligned with the inner nuclear layer of the normal retina due to the thinning of the retina in glaucoma. See Fig. 29.16 for orientation. (B) Glaucomatous optic nerve cupping results in part from loss of retinal ganglion cells, the axons of which populate the optic nerve. (C) The arrows point to the dura of the optic nerve. Notice the wide subdural space, a result of atrophy of the optic nerve. There is a striking degree of cupping on the surface of the nerve as a consequence of long-standing glaucoma.

females. The usual age of onset is between 10 and 30 years. It begins with clouding of vision that may progress to total loss of vision.

Optic Neuritis

Many unrelated conditions have historically been grouped under the heading of optic neuritis. Unfortunately, the term itself suggests optic nerve inflammation, which might not accurately describe the pathophysiologic changes. In common clinical usage the term *optic neuritis* is used to describe a loss of vision secondary to demyelination of the optic nerve. One of the most important causes of optic neuritis is multiple sclerosis (Chapter 28). Indeed, optic neuritis may be the first manifestation of this disease. The 10-year risk of developing multiple sclerosis after the first attack of optic neuritis increases if the affected person has concomitant evidence of brain lesions as detected by magnetic resonance imaging. Individuals with a single episode of optic nerve demyelination may recover vision and remain disease-free.

KEY CONCEPTS

- The term “anterior ischemic optic neuropathy” refers to a spectrum of ischemic injuries to the optic nerve varying from transient ischemia to infarction.
- Bilateral swelling of the optic nerve head known as papilledema may develop as a consequence of elevated cerebrospinal fluid pressure and stasis of axoplasmic transport within the optic nerve. Unilateral optic nerve head swelling may result from compression of the optic nerve such as in primary tumors of the nerve.
- In chronic glaucoma, the optic nerve may atrophy and the cup on the surface of the nerve may enlarge and deepen.
- Optic neuropathy may be inherited (as in Leber hereditary optic neuropathy) or may result from nutritional deficiencies or toxins such as methanol.

THE END-STAGE EYE: PHTHISIS BULBI

Trauma, intraocular inflammation, chronic retinal detachment, and many other conditions can give rise to an eye that is both small (atrophic) and internally disorganized: phthisis bulbi. Congenitally small eyes—hypoplastic or microphthalmic eyes—are generally not disorganized internally. Phthisical eyes typically show the following changes: the presence of exudate or blood between the ciliary body and sclera and the choroid and sclera (*ciliochoroidal effusion*); the presence of a membrane extending across the eye from one aspect of the ciliary body to the other (*cyclitic membrane*); chronic retinal detachment; optic nerve atrophy; the presence of intraocular bone, which is thought by many to originate from osseous metaplasia of the RPE; and a thickened sclera, especially posteriorly. Ciliochoroidal effusion is typically associated with the physiologic state of low intraocular pressure (*hypotony*). The normal pull of the extraocular muscles on a hypotonous eye may render the appearance of the eye as square rather than round.

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